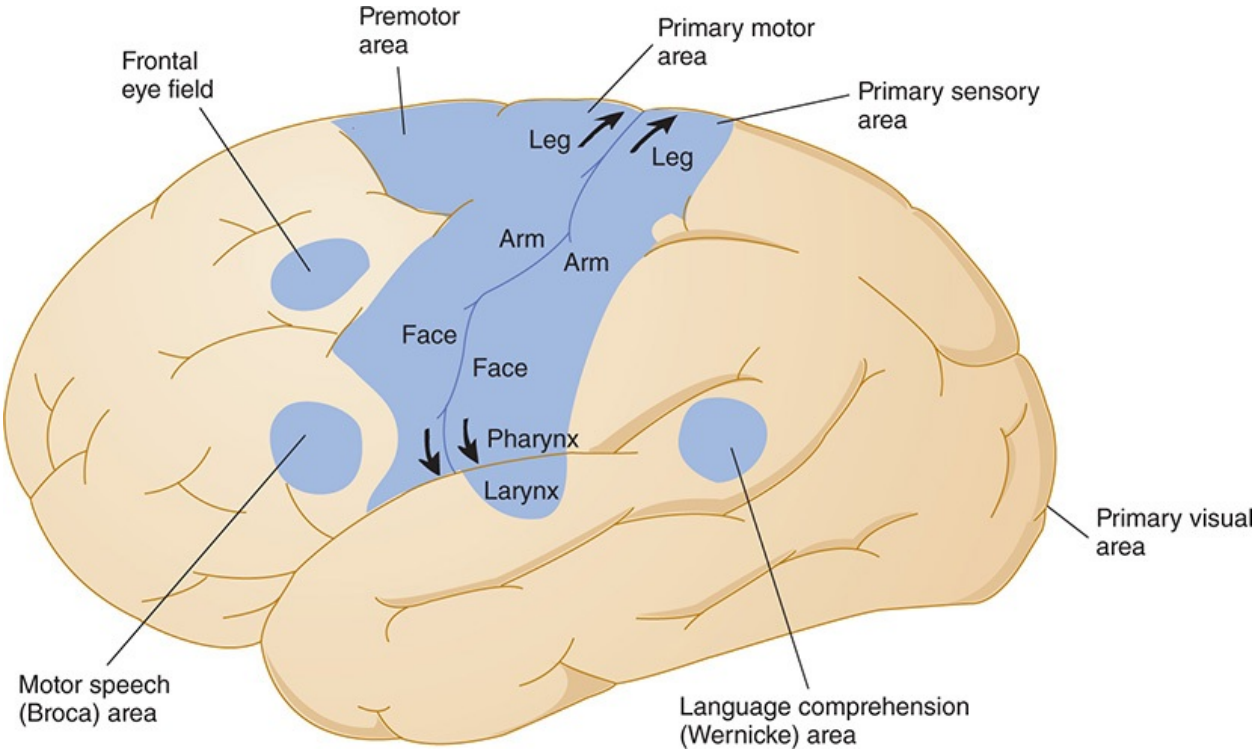


FIGURE 8–8 A lateral view of the left hemisphere showing some principal cortical areas and their functional correlates in the human brain. The primary somatosensory area is in the postcentral gyrus of the parietal lobe, and the primary motor cortex is in the precentral gyrus. (Reproduced with permission from Waxman SG: *Clinical Neuroanatomy*, 26th ed. New York, NY: McGraw-Hill; 2010.)

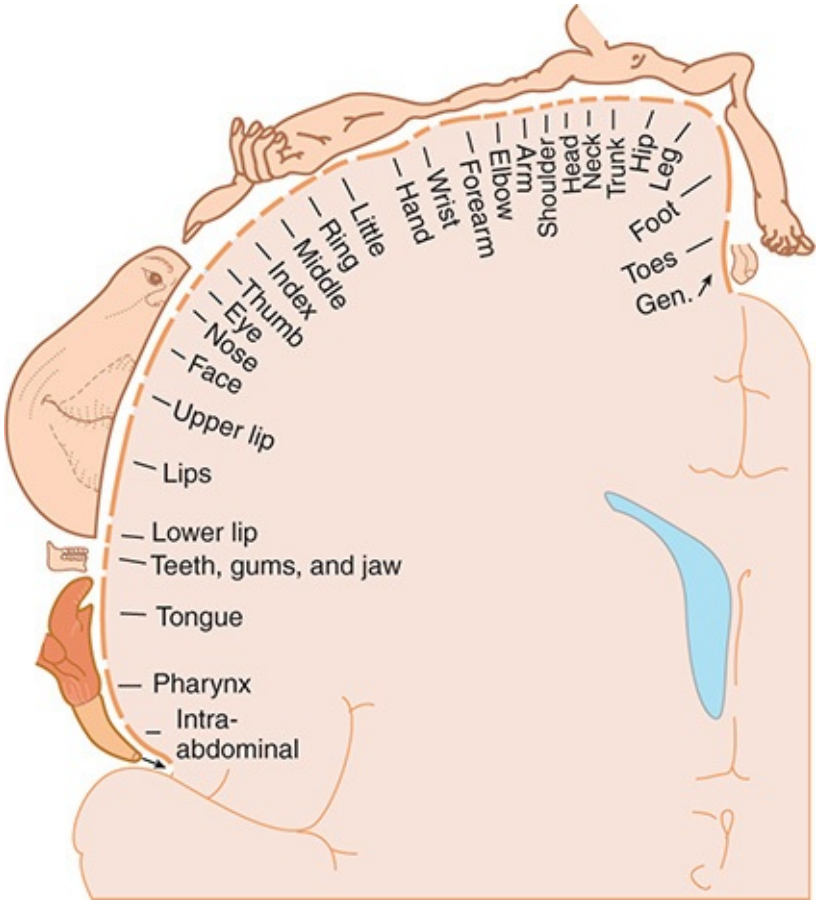


FIGURE 8–9 Sensory homunculus, drawn overlying a coronal section through the postcentral gyrus. The parts of the body are represented in order along the postcentral gyrus, with the legs on top and the head at the foot of the gyrus. The size of the cortical receiving area for impulses from a particular part of the body is proportionate to the use of the part. Gen., genitalia.

In addition to the primary somatosensory cortex, there are two other cortical regions that contribute to the integration of sensory information. The **sensory association area** is in the parietal cortex, and the **secondary somatosensory cortex** is located in the wall of the **lateral fissure** (also called **sylvian fissure**) that separates the temporal from the frontal and parietal lobes. These regions receive input from the primary somatosensory cortex.

Conscious awareness of the positions of the various parts of the body in space depends in part on impulses from sensory receptors in and around the joints. Impulses from these receptors, from touch receptors in the skin and other tissues, and from muscle spindles are synthesized in the cortex into a conscious picture of the position of the body in space (proprioception).

VENTROLATERAL SPINOTHALAMIC TRACT

Fibers from nociceptors and thermoreceptors synapse on neurons in the dorsal horn of the spinal cord. The axons from these dorsal horn neurons cross the midline and ascend in the ventrolateral quadrant of the spinal cord, where they form the ventrolateral spinothalamic pathway (Figure 8–7). Fibers within this tract synapse in the VPL. Some dorsal horn neurons that receive nociceptive input synapse in the reticular formation of the brainstem **(spinoreticular pathway)** and then project to the centrolateral nucleus of the thalamus.

Positron emission tomographic (PET) and functional magnetic resonance imaging (fMRI) studies in healthy humans indicate that pain activates the primary and secondary somatosensory cortex and the cingulate gyrus on the side opposite the stimulus. In addition, the amygdala, frontal lobe, and the insular cortex are activated. These technologies were important in distinguishing two components of pain pathways. The pathway to the primary somatosensory cortex is responsible for the discriminative aspect of pain (pain localization). In contrast, the pathway that includes synapses in the brainstem reticular formation and centrolateral thalamic nucleus projects to the frontal lobe, limbic system, and insular cortex. This pathway mediates the motivational-affective component of pain.

Visceral sensation travels along the same central pathways as somatic sensation in the spinothalamic tracts and thalamic radiations, and the cortical receiving areas for visceral sensation are intermixed with the somatic receiving areas.

CORTICAL PLASTICITY

The extensive neuronal connections described above are not innate and immutable but can be changed relatively rapidly by experience to reflect the use of the represented area. **Clinical Box 8–4** describes remarkable changes in cortical and thalamic organization that occur in response to limb amputation to lead to the phenomenon of **phantom limb pain.**

CLINICAL BOX 8–4

Phantom Limb Pain

In 1551, Ambroise Pare, a military surgeon, wrote “…the patients, long after

the amputation is made, say they still feel pain in the amputated part. Of this they complain strongly, a thing worthy of wonder and almost incredible to people who have not experienced this." This is perhaps the earliest description of **phantom limb pain.** Between 50% and 80% of amputees experience phantom sensations, usually pain, in the region of their amputated limb. Phantom sensations may also occur after the removal of body parts other than the limbs, for example, after amputation of the breast, extraction of a tooth **(phantom tooth pain),** or removal of an eye **(phantom eye syndrome).** Numerous theories have been evoked to explain this phenomenon. The current theory is based on evidence that the brain can reorganize if sensory input is cutoff. The **ventral posterior thalamic nucleus** is one example where this change can occur. In patients who have had their leg amputated, single neuron recordings show that the thalamic region that once received input from the leg and foot now respond to stimulation of the stump (thigh). Others have demonstrated remapping of the somatosensory cortex. For example, in some individuals who have had an arm amputated, stroking different parts of the face can lead to the feeling of being touched in the area of the missing limb.

THERAPEUTIC HIGHLIGHTS

The use of **epidural anesthesia** during the amputation surgery may prevent the acute pain associated with the surgery, thereby reducing the need for opioid therapy in the immediate postoperative period. This anesthetic procedure resulted in a reduced incidence of phantom pain. **Spinal cord stimulation** has been shown to be an effective therapy for phantom pain. Electric current is passed through an electrode that is placed next to the spinal cord to stimulate spinal pathways. This interferes with the impulses ascending to the brain and lessens the pain felt in the phantom limb. Instead, amputees feel a tingling sensation in the phantom limb.

Cortical structures are capable of dramatic reorganization **(plasticity)** after loss of a body part such as a digit; the cortical representation of the neighboring digits spreads into the cortical area that was formerly occupied by the representation of the amputated digit. Cortical connections of sensory units to the cortex have extensive convergence and divergence, with connections that can become weak with disuse and strong with use.

PET scanning in humans also documents plastic changes, sometimes from

one sensory modality to another. Thus, for example, tactile and auditory stimuli increase metabolic activity in the visual cortex in blind individuals. Conversely, deaf individuals respond faster and more accurately than normal individuals to moving stimuli in the visual periphery. Plasticity also occurs in the motor cortex.

EFFECTS OF CNS LESIONS

Clinical Box 8–2 describes some of the deficits noted after damage within the somatosensory pathways. **Clinical Box 8–5** describes the characteristic changes in sensory and motor functions that occur in response to spinal cord hemisection.

Damage to the dorsal columns leads to ipsilateral loss of the ability to detect light touch, vibration, and proprioception from body structures represented caudal to the level of damage. Damage to the ventrolateral spinothalamic pathway leads to contralateral loss of pain and temperature sensation below the level of the lesion. Such spinal damage could occur with a penetrating wound or a tumor.

CLINICAL BOX 8–5

Brown-Séquard Syndrome

A functional hemisection of the spinal cord causes a characteristic and easily recognized clinical picture that reflects damage to ascending sensory (dorsal column pathway, ventrolateral spinothalamic tract) and descending motor (corticospinal tract) pathways, which is called the **Brown-Séquard syndrome.** The lesion to fasciculus gracilus or fasciculus cuneatus leads to ipsilateral loss of discriminative touch, vibration, and proprioception below the level of the lesion. The loss of the spinothalamic tract leads to contralateral loss of pain and temperature sensation beginning one or two segments below the lesion. Damage to the corticospinal tract produces weakness and spasticity in certain muscle groups on the same side of the body. Although a precise spinal hemisection is rare, the syndrome is common because it can be caused by a spinal cord tumor, spinal cord trauma, degenerative disk disease, and ischemia.

THERAPEUTIC HIGHLIGHTS

Spinal or vertebral stabilization is needed when spinal trauma is indicated. Drug

treatments for Brown-Séquard syndrome are based on the etiology and time since onset. High doses of **corticosteroids** have been shown to be of value particularly if administered soon after the onset of such a spinal cord injury. Corticosteroids decrease the inflammation by suppressing polymorphonuclear leukocytes and reverse the increase in capillary permeability. Drugs may be needed to treat spasticity, pain, or other possible complications of the spinal cord injury. Physical therapy is important to maintain strength and joint mobility and improving respiratory function.

Damage to the primary somatosensory cortex does not abolish somatic sensation. Irritation of this region causes **paresthesia** or an abnormal sensation of numbness and tingling on the contralateral side of the body. Destructive lesions impair the ability to localize noxious stimuli in time, space, and intensity. Damage to the cingulate cortex impairs the recognition of the aversive nature of a noxious stimulus.

An infarct in the thalamus can lead to a loss of sensation. **Thalamic pain syndrome** can occur during recovery from a thalamic infarct. The syndrome is characterized by chronic pain on the side of the body contralateral to the stroke.

MODULATION OF PAIN TRANSMISSION

PROCESSING INFORMATION IN THE DORSAL HORN

Transmission in nociceptive pathways can be interrupted by actions within the dorsal horn of the spinal cord at the site of sensory afferent termination. Many people have learned from practical experience that rubbing or shaking an injured area decreases the pain due to the injury. The relief may be due to the simultaneous activation of innocuous cutaneous mechanoreceptors whose afferents emit collaterals that terminate in the dorsal horn. The activity of these cutaneous mechanosensitive afferents may reduce the responsiveness of dorsal horn neurons to their input from nociceptive afferent terminals. This is called the **gate-control mechanism** of pain modulation and it serves as the rationale behind the use of **transcutaneous electrical nerve stimulation (TENS)** for pain relief. This method uses electrodes to activate Aα and Aβ fibers near the injury.

Opioids are a commonly used analgesic that can exert their effects at various

places in the CNS, including in the spinal cord and dorsal root ganglia. **Figure 8–10** shows some of the various modes of action of opioids to decrease nociceptive transmission. There are interneurons in the superficial regions of the dorsal horn that contain endogenous opioid peptides (**enkephalin** and **dynorphin**). These interneurons terminate in the region of the dorsal horn where nociceptive afferents terminate. Opioid receptors are located on the terminals of nociceptive fibers and on dendrites of dorsal horn neurons, allowing for both presynaptic and postsynaptic sites of actions for opioids. Activation of the postsynaptic opioid receptors hyperpolarizes the dorsal horn interneuron by causing an increase in K^+ conductance. Activation of the presynaptic opioid receptors leads to a decrease in Ca^{2+} influx, resulting in a decrease in release of glutamate and substance P. Together these actions reduce the duration of the EPSP in the dorsal horn neuron. Activation of opioid receptors on dorsal root ganglia cell bodies also contributes to reduced transmission from nociceptive afferents.

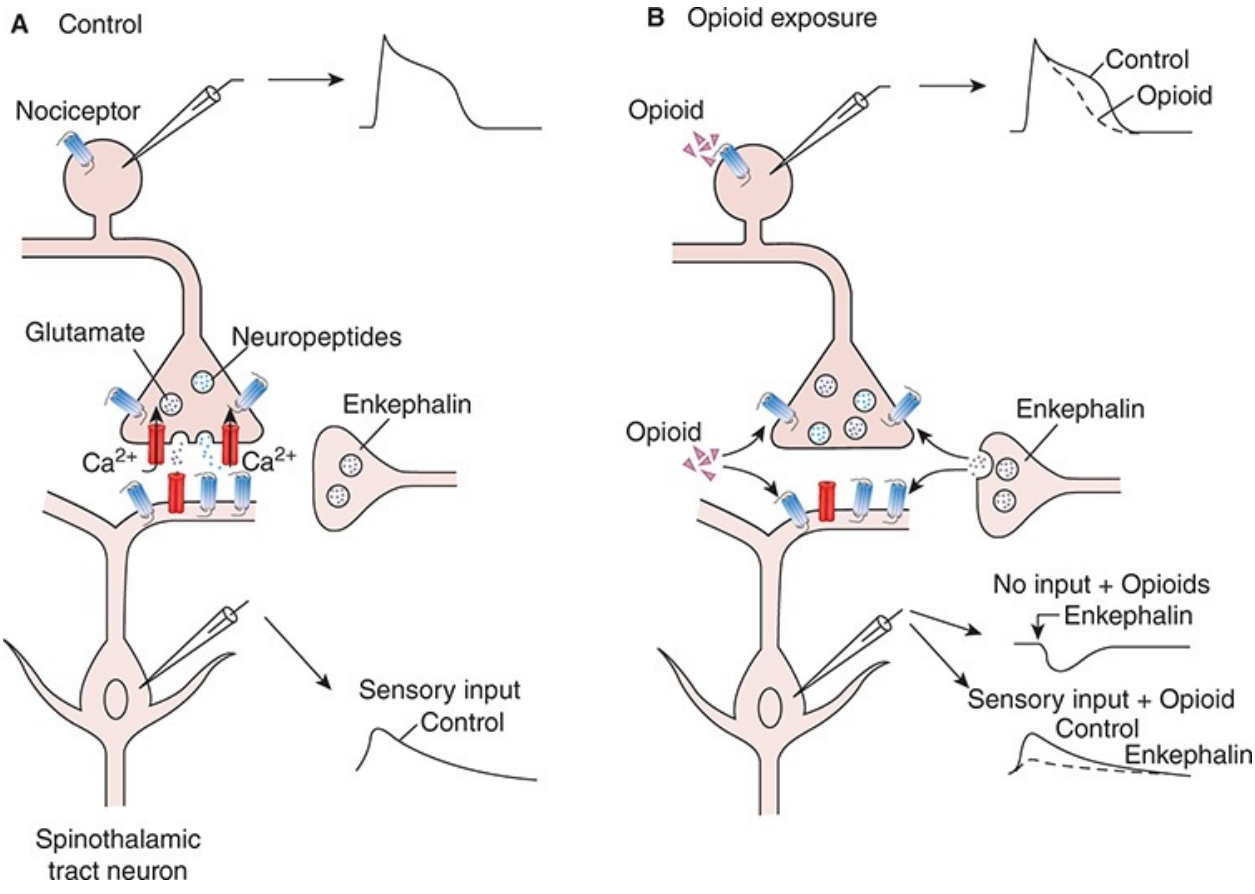


FIGURE 8–10 Actions of opioids to reduce sensory transmission in pain pathways at the level of the dorsal root ganglion (DRG) and spinal cord

dorsal horn region. A) Activation of a nociceptor leads to the release of glutamate and neuropeptides from its nerve terminals that synapse on spinothalamic tract projection neurons. This leads to the depolarization (activation) of spinothalamic tract projection neurons. Enkephalin (ENK)-containing interneurons mediate their effects via opioid receptors on the terminals of nociceptive afferent fibers and on dendrites of dorsal horn neurons to exert both presynaptic and postsynaptic inhibition. **B)** The action of an opioid (eg, morphine) within the DRG is to decrease Ca^{2+} influx leading to a decrease in the duration of the invoked action potential in the nociceptive neuron and a reduction in transmitter release from the nociceptive neuron onto a neuron in the dorsal horn. Opioids also hyperpolarize the membrane of dorsal horn neuron by activation of a K^{+} conductance; opioids also decrease the amplitude of the excitatory postsynaptic potential (EPSP) produced by stimulation of nociceptors.

CLINICAL BOX 8–6

Motivation & Addiction

Neurons in the forebrain **ventral tegmental area** and **nucleus accumbens** are involved in motivated behaviors such as reward, laughter, pleasure, addiction, and fear. These areas are referred to as the brain's **reward center** or **pleasure center.** The **mesocortical dopaminergic neurons** that project from the midbrain to the **nucleus accumbens** and the frontal cortex are also involved. **Addiction,** defined as the repeated compulsive use of a substance despite negative health consequences, can be produced by many different drugs. According to the World Health Organization, over 76 million people worldwide suffer from alcohol abuse, and over 15 million suffer from drug abuse. Not surprisingly, alcohol and drug addiction are associated with the reward system. The best studied addictive drugs are opioids (eg, morphine and heroin); others include cocaine, amphetamine, alcohol, cannabinoids, and nicotine. These drugs affect the brain in different ways, but all have in common the fact that they increase the amount of dopamine available to act on $\mathbf{D_3}$ **receptors** in the nucleus accumbens. Thus, acutely they stimulate the reward system of the brain. Long-term addiction involves the development of **tolerance,** which is the need for increasing amounts of a drug to produce a high. Also, **withdrawal** produces psychological and physical symptoms. One of the characteristics of addiction is the tendency of addicts to relapse after treatment. For opioid addicts, the relapse rate in the first year is about 80%.

Relapse often occurs on exposure to sights, sounds, and situations that were previously associated with drug use. Even a single dose of an addictive drug facilitates release of excitatory neurotransmitters in brain areas concerned with memory. The medial frontal cortex, hippocampus, and amygdala are concerned with memory, and they project via excitatory glutamatergic pathways to the nucleus accumbens. Relatively little is known about the specific brain mechanisms that cause tolerance and dependence. However, the two can be separated. Absence of **β-arrestin-2** blocks tolerance but has no effect on dependence. β-Arrestin-2 is a member of a family of proteins that inhibit heterotrimeric G-proteins by phosphorylating them.

THERAPEUTIC HIGHLIGHTS

Withdrawal symptoms and cravings associated with addiction to opioids can be reversed by treatment with drugs such as **methadone** and **buprenorphine** that act on the same CNS receptors as morphine and heroin. The US Federal Drug Administration has approved the use of three drugs for treatment of alcohol abuse: **naltrexone, acamprosate,** and **disulfiram.** Naltrexone is an opioid receptor antagonist that blocks the reward system and the craving for alcohol. Acamprosate may reduce the withdrawal effects associated with alcohol abuse. Disulfiram causes an accumulation of acetaldehyde by preventing the full degradation of alcohol. This leads to an unpleasant reaction to alcohol ingestion (eg, flushing, nausea, and palpitations). **Topiramate,** a Na^+ channel blocker, is showing promise in clinical trials of alcohol addiction. This is the same drug that is effective in treatment of migraine headaches.

Patients who receive long-term pain management with morphine may become resistant to the drug, requiring progressively higher doses for pain relief. This **acquired tolerance** is different from **addiction,** which refers to a psychological craving. Psychological addiction rarely occurs when morphine is used to treat chronic pain, provided the patient does not have a history of drug abuse. **Clinical Box 8–6** describes mechanisms involved in motivation and addiction.

ROLES OF PERIAQUEDUCTAL GRAY & BRAINSTEM

Another site of action for morphine and endogenous opioid peptides is the

midbrain **periaqueductal gray (PAG).** An injection of opioids into the PAG induces analgesia. The PAG is part of a descending pathway that modulates pain transmission by inhibiting primary afferent transmission in the dorsal horn (Figure 8–6). These PAG neurons project directly to and activate two groups of neurons in the brainstem: serotonergic neurons in the **nucleus raphe magnus** and catecholaminergic neurons in the **rostral ventromedial medulla.** Neurons in these regions project to the dorsal horn of the spinal cord where they release serotonin and norepinephrine, respectively, to inhibit the activity of dorsal horn neurons that receive input from nociceptive afferent fibers (Figure 8–6). This inhibition occurs, at least in part, due to the activation of the dorsal horn enkephalin-containing interneurons. A group of pontine catecholaminergic neurons in the **locus coeruleus** are also elements of this descending pain modulating pathway. These neurons also exert their analgesic effect by the release of norepinephrine in the dorsal horn.

The analgesic effect of **electroacupuncture** may involve the release of endogenous opioids and activation of this descending pain modulatory pathway. Electroacupuncture activates ascending sensory pathways that emit collaterals in the PAG and in the brainstem serotonergic and catecholaminergic regions. The analgesic effect of electroacupuncture is prevented by administration of naloxone, an opioid receptor antagonist.

STRESS-INDUCED ANALGESIA

It is well known that soldiers wounded in the heat of battle often feel no pain until the battle is over. This is an example of **stress-induced analgesia** that can also be exemplified by reduced pain sensitivity when being attacked by a predator or other stressful events. Release of norepinephrine, perhaps from brainstem catecholaminergic neurons, in the amygdala may contribute to this phenomenon. As noted above, the amygdala is a part of the limbic system that is involved in mediating the motivational-affective responses to pain.

The release of endogenous **cannabinoids** such as 2-arachidonoylglycerol (2AG) and anandamide may contribute to stress-induced analgesia. These chemicals can act on two types of G-protein-coupled receptors (CB_1 and CB_2). CB_1 receptors are found in many brain regions, and activation of these receptors accounts for the euphoric actions of cannabinoids. CB_2 receptors are expressed in activated microglia under pathologies that are associated with chronic neuropathic pain (see Clinical Box 8–3). Binding of an agonist to CB_2 receptors

on microglia reduces the inflammatory response and has an analgesic effect. Work is underway to develop selective CB_2 agonists for therapeutic treatment of neuropathic pain.

CHAPTER SUMMARY

- Touch and pressure are sensed by four types of mechanoreceptors that are innervated by myelinated Aα and Aβ sensory afferents. They are rapidly adapting Meissner corpuscles (respond to tapping and slow vibrations), slowly adapting Merkel cells (respond to sustained pressure and touch), slowly adapting Ruffini corpuscles (respond to skin stretch and vibration), and rapidly adapting Pacinian corpuscles (respond to fast vibration and deep pressure).
- Nociceptors and thermoreceptors are located on the free nerve endings of unmyelinated C fibers or lightly myelinated Aδ fibers in hairy and glaborous skin and deep tissues. These nerve endings have various types of receptors that are activated by noxious chemical (eg, TRPV1 and ASIC), mechanical (eg, P2X, P2Y and TRPA1), and thermal (eg, TRPV1) stimuli. In addition, chemical mediators (eg, bradykinin, prostaglandin, serotonin, and histamine) released in response to tissue injury directly activate or sensitize nociceptors.
- The receptor potential is the nonpropagated depolarizing potential recorded in a sensory receptor after an adequate stimulus is applied. As the stimulus intensity is increased, the size of the receptor potential is increased. When it reaches a critical threshold, an action potential is generated in the sensory nerve.
- Converting a receptor stimulus to a recognizable sensation is termed “sensory coding.” All sensory systems code for four elementary attributes of a stimulus: modality (receptor selectivity), location (receptive field), intensity (receptor sensitivity), and duration (receptor adaptation).
- Pain is an unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage, whereas nociception is the unconscious activity induced by a harmful stimulus applied to sense receptors. First pain is mediated by Aδ fibers and causes a sharp, localized sensation. Second pain is mediated by C fibers and causes a dull, intense, diffuse, and unpleasant feeling. Acute pain has a sudden onset, recedes during the healing process, and serves as an important

protective mechanism. Chronic pain is persistent and caused by nerve damage or prolonged inflammation; it is often associated with hyperalgesia (an exaggerated response to a noxious stimulus) and allodynia (a sensation of pain in response to an innocuous stimulus).

- Visceral pain is poorly localized, unpleasant, and associated with nausea and autonomic symptoms. It often radiates (or is referred) to other somatic structures perhaps due to convergence of somatic and visceral nociceptive afferent fibers on the same second-order neurons in the spinal dorsal horn that project to the thalamus and then to the primary somatosensory cortex.
- Discriminative touch, proprioception, and vibratory sensations are relayed via the dorsal column (medial lemniscus) pathway to the VPL in the thalamus and then to the primary somatosensory cortex. Pain and temperature sensations are mediated via the ventrolateral spinothalamic tract, which projects to the VPL and then to cortex. The discriminative aspect of pain results from activation of the primary somatosensory cortex; the motivational-affective component of pain is from activation of the frontal lobe, limbic system, and insular cortex.
- Damage to the dorsal columns leads to ipsilateral loss of the ability to detect light touch, vibration, and proprioception from body structures represented caudal to the level of damage. Damage to the ventrolateral spinothalamic pathway leads to contralateral loss of pain and temperature sensation below the level of the lesion. Damage to the primary somatosensory impairs the ability to localize noxious stimuli in time, space, and intensity. Damage to the cingulate cortex impairs the recognition of the aversive nature of a noxious stimulus.
- Transmission in pain pathways is modulated by endogenous opioids that can act in the PAG, brainstem, spinal cord, and dorsal root ganglia. Descending pain modulating pathways include neurons in the PAG, nucleus raphe magnus, rostral ventromedial medulla, and locus coeruleus.
- New pain therapies focus on synaptic transmission in nociception and peripheral sensory transduction. Capsaicin transdermal patches or creams reduce pain by exhausting the supply of substance P in nerves and by acting on TRPV1 receptors in the skin. Lidocaine and mexiletine are useful in some cases of chronic pain and act by blocking Nav1.8, which is uniquely associated with nociceptive neurons in dorsal root ganglia. Ziconotide, a voltage-gated N-type Ca^{2+} channel blocker, is used for intrathecal analgesia in patients with refractory chronic pain. Gabapentin, an anticonvulsant drug,

is effective in treatment of neuropathic and inflammatory pain by acting on voltage-gated Ca^{2+} channels. Topiramate, a Na^{+} channel blocker, is another anticonvulsant drug that can be used to treat migraines. NMDA receptor antagonists can be co-administered with an opioid to reduce tolerance to an opioid.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A 28-year-old man was seen by a neurologist because he had experienced prolonged episodes of tingling and numbness in his right arm. He underwent a neurologic exam to evaluate his sensory nervous system. Which of the following cutaneous mechanoreceptors is correctly paired with the type of stimulus to which it is most apt to respond?

   A. Pacinian corpuscle and rapid vibration

   B. Meissner corpuscle and skin stretch

   C. Merkel cells and slow vibration

   D. Ruffini corpuscles and sustained pressure

2. An MD/PhD student was recording responses in different cutaneous receptors and noted the following. One receptor was inactive until the skin temperature was increased to 33°C and then its firing rate continued to increase as the skin temperature was gradually raised to 45°C. A second receptor was inactive until the skin temperature reached 46°C. A third receptor was inactive at skin temperatures of 40°C, but then steadily increases its firing rate as skin temperature was lowered to 24°C. For each of these cases, classify the type of receptor and the nonselective cation channel that was possibly activated.

   A. Receptor one is a thermal nociceptor and the channel activated was TRPV1; receptor two is a thermal nociceptor and the channel activated was TRPA1; receptor three was an innocuous cold receptor and the channel activated was TRPV4.

   B. Receptor one is an innocuous warm receptor and the channel activated was TRPV1; receptor two is a thermal nociceptor and the channel activated was TRPM8; receptor three was an innocuous cold receptor and the channel activated was TRPV3.

   C. Receptor one is an innocuous warm receptor and the channel activated was TRPV3; receptor two is a thermal nociceptor and the channel activated was

TRPV1; receptor three was an innocuous cold receptor and the channel activated was TRPM8.

D. Receptor one is a thermal nociceptor and the channel activated was TRPA1; receptor two is a thermal nociceptor and the channel activated was TRPV1; receptor three was an innocuous cold receptor and the channel activated was TRPM8.

3. List the steps involved in the generation of an action potential in a sensory nerve fiber beginning with the stimulation of a Pacinian corpuscle.

A. Light touch is applied to the Pacinian corpuscle, and a receptor potential is generated; as the pressure is increased, the size of the receptor potential is increased; when it reaches 30 mV, an action potential is produced at a point of the sensory nerve within the corpuscle.

B. Light touch is applied to the Pacinian corpuscle, and a receptor potential is generated; as more receptors are brought into the receptive field, the size of the receptor potential increases; when it reaches 30 mV, an action potential is produced at the first node of Ranvier.

C. Sustained pressure is applied to the Pacinian corpuscle, and a receptor potential is generated; as more receptors are activated, the size of the receptor potential increases; when it reaches 10 mV, an action potential is produced at the first node of Ranvier.

D. Rapid vibration is applied to the Pacinian corpuscle, and a graded receptor potential is generated; when the receptor potential reaches 10 mV, an action potential is produced at the first node of Ranvier.

E. Rapid vibration is applied to the Pacinian corpuscle, and a receptor potential is generated; when the receptor potential reaches 10 mV, an action potential is produced in the unmyelinated portion of the sensory fiber.

4. A medical student was doing research in a sensory neurophysiology laboratory. In preparation for his research, the principal investigator of the laboratory asked him to compare the four basic attributes of a stimulus to sensory receptors. The four attributes of sensory coding are

A. modality, adequate threshold, sensitivity, and location.

B. adequate threshold, receptive field, adaptation, and projection.

C. specific energy, adequate threshold, sensation, and duration.

D. sensitization, discrimination, energy, and projection.

E. modality, location, intensity, and duration.

5. A 23-year-old woman fell asleep on the beach while sunbathing. She awoke a

few hours later to find that she had a very bad sunburn. That evening while taking a shower, the lukewarm water (40°C) touching her back caused her to feel pain. What types of receptors were activated by the lukewarm water and why did she experience pain?

A. Thermal nociceptors and nociceptive pain

B. Thermal nociceptors and allodynia

C. Thermal nociceptors and hyperalgesia

D. Innocuous thermal receptors and hyperalgesia

E. Innocuous thermal receptors and allodynia

6. A 32-year-old woman experienced the sudden onset of a severe cramping pain in the abdominal region. She also became nauseated. List some of the common features of visceral pain.

A. It results from activation of nociceptors in the viscera that are innervated by the same fibers as innervate skin, induces rapid shard pain, causes spasms of the visceral muscle, and shows relatively rapid adaptation.

B. It is mediated by Aδ and C fibers in the ventral roots of spinal nerves, radiates to a nearby or distant somatic structure, is accompanied by sweating, and is relayed to the cortex by the spinothalamic tract.

C. It is poorly localized, is accompanied by sweating, radiates to a somatic structure that may be some distance away, and is relayed to the somatosensory cortex via the spinothalamic tract.

D. It requires simultaneous activation of nociceptors within and outside of the viscera, causes spasms of the visceral and skeletal muscle, and is relayed to the cortex by the dorsal column pathway.

E. It is well localized, is accompanied by sweating, radiates to a somatic structure that may be some distance away, and causes spasms of visceral muscle.

7. A ventrolateral cordotomy is performed that produces relief of pain in the right leg. It is effective because it interrupts the

A. left dorsal column.

B. left ventrolateral spinothalamic tract.

C. right ventrolateral spinothalamic tract.

D. right medial lemniscal pathway.

E. a direct projection to the primary somatosensory cortex.

8. A medical student is studying neurons that are part of a descending pain

modulating pathway. What brain region is correctly paired with the neurotransmitters it releases and the location where the neurotransmitter is released?

A. Periaqueductal gray neurons release endorphins in the spinal dorsal horn.

B. Nucleus raphe magnus releases serotonin in the dorsal root ganglion.

C. Locus coeruleus neurons release serotonin in the nucleus raphe magnus.

D. Locus coeruleus neurons release norepinephrine in the spinal dorsal horn.

E. Periaqueductal gray releases dynorphin in the rostral ventromedial medulla.

9. A 47-year-old woman experienced a migraine headache that was not relieved by her current pain medications. The doctor opted to prescribe a drug that targets receptors or ion channels within the pain pathway. Which of the following drugs is used to treat migraines and which receptor or ion channel does it affect?

A. Topiramate and voltage-gated Na^+ channel

B. Ziconotide and voltage-gated N-type Ca^{2+} channel

C. Valproate and TRPV1 receptors

D. Carbamazepine and voltage-gated Na^+ channels

E. Gabapentin and Nav1.8

10. A 40-year-old man loses his right hand in a farm accident. Four years later, he has episodes of severe pain in the missing hand (phantom limb pain). A detailed PET scan study of his cerebral cortex might be expected to show

A. expansion of the right hand area in his right primary somatosensory cortex.

B. expansion of the right hand area in his left primary somatosensory cortex.

C. a metabolically inactive spot where his hand area in his left primary somatosensory cortex would normally be.

D. projection of fibers from neighboring sensory areas into the right hand area of his right primary somatosensory cortex.

E. projection of fibers from neighboring sensory areas into the right hand area of his left primary somatosensory cortex.

11. A 50-year-old woman undergoes a neurologic exam that indicates loss of pain and temperature sensitivity, vibratory sense, and proprioception in the left leg. These symptoms could be explained by

A. a tumor on the right medial lemniscal pathway in the sacral spinal cord.

B. a peripheral neuropathy.

C. a tumor on the left medial lemniscal pathway in the sacral spinal cord.

D. a tumor affecting the right posterior paracentral gyrus.
E. a large tumor in the right lumbar ventrolateral spinal cord.

CHAPTER 9

Smell & Taste

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the structure and function of the neural elements in the olfactory epithelium and olfactory bulb.
- Identify the significance of the family of olfactory receptor genes.
- Explain how odorant receptors are activated and the mechanism by which signal transduction occurs in these receptors.
- Label the components of the pathway by which impulses generated in the olfactory epithelium reach five regions of the olfactory cortex.
- Describe the location and cellular composition of taste buds.
- Name the five major taste modalities and compare the signal transduction mechanisms in the receptors mediating these different taste modalities.
- Label the components of the pathways by which impulses generated in taste receptors reach the gustatory region of the insular cortex.
- Name and discuss abnormalities in odor and taste sensations.

INTRODUCTION

Smell (**olfaction**) and taste (**gustation**) are examples of visceral senses because of their close association with gastrointestinal function. Physiologically, they are related to each other as the flavor of food is a combination of its taste and smell. This is why food may taste “different” if one has a cold that depresses the sense of smell. Smell and taste receptors are **chemoreceptors** that are stimulated by chemical molecules in solution in mucus in the nose (**odorants**) and saliva in the mouth (**tastants**). The sensations of smell and taste likely evolved as protective mechanisms to avoid the intake of potentially harmful substances.

SMELL

OLFACTORY EPITHELIUM

The yellowish pigmented **olfactory epithelium** is a specialized portion of the nasal mucosa that covers an area of 10 cm^2 in the roof of the nasal cavity near the septum in humans **(Figure 9–1)**. The olfactory epithelium is the place in the body where the nervous system is closest to the external world. It contains three types of neurons that are important for olfaction: **olfactory sensory neurons, supporting cells, and basal stem cells**.

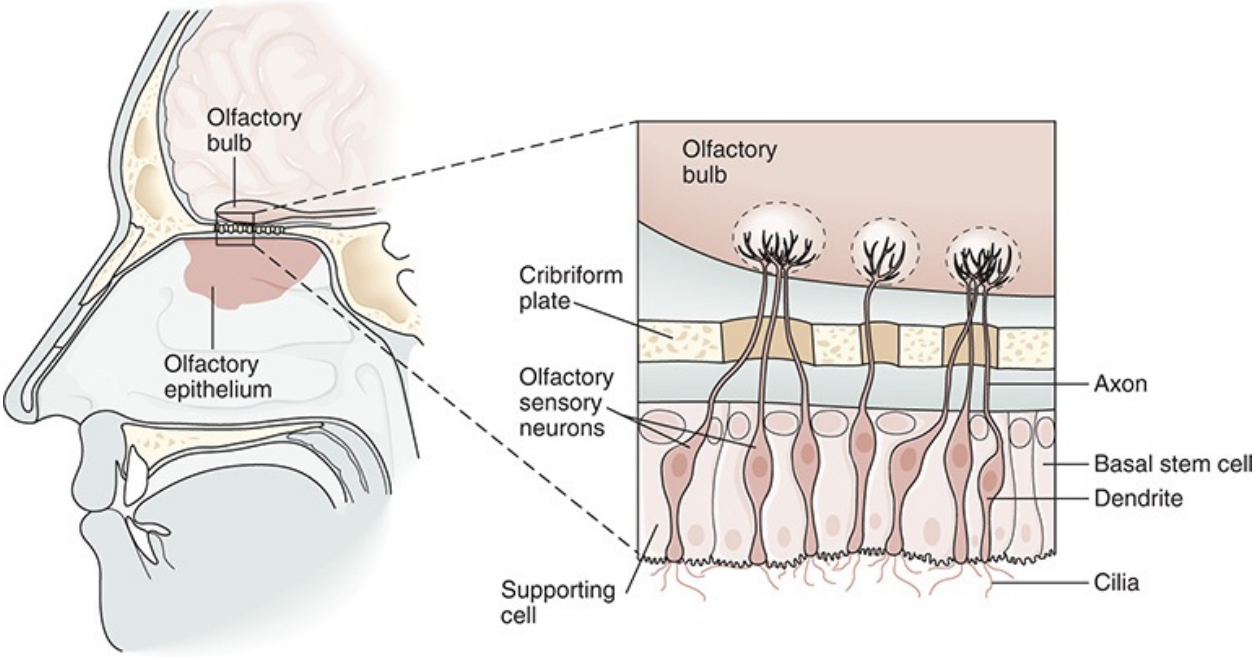


FIGURE 9–1 Structure of the olfactory epithelium. There are three cell types: olfactory sensory neurons, supporting (sustentacular) cells, and basal stem cells

at the base of the epithelium. Each olfactory sensory neuron has a dendrite that projects to the epithelial surface. Numerous cilia protrude into the mucus layer lining the nasal lumen. Odorants bind to specific odorant receptors on the cilia and initiate a cascade of events leading to generation of action potentials in the sensory axon. Each olfactory sensory neuron has a single axon that projects to the olfactory bulb, a small ovoid structure that rests on the cribriform plate of the ethmoid bone. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

The bipolar olfactory sensory neurons (also called olfactory receptor cells) are responsible for olfactory transduction. They have a short, thick dendrite that projects into the nasal cavity where it terminates in a knob containing 6–12 **cilia** that protrude into the thin layer of mucus overlying the epithelium (Figure 9–1). The axons of the olfactory sensory neurons (ie, **olfactory nerve**) pass through the **cribriform plate** of the ethmoid bone to enter the **olfactory bulbs.**

The supporting cells secrete the mucus that provides the appropriate molecular and ionic environment for odor detection in the olfactory epithelium. Odor-producing molecules (odorants) dissolve in the mucus and bind to **odorant receptors** on the cilia of the olfactory sensory neurons. **Odorant- binding proteins** in the mucus may facilitate the diffusion of odorants to and from the odorant receptor. Basal stem cells undergo mitosis to generate new olfactory sensory neurons as needed to replace those damaged by exposure to the environment; olfactory sensory neurons generally survive for only 1–2 months.

ODORANT RECEPTORS & SIGNAL TRANSDUCTION

The olfactory system can discriminate perhaps more than 1 million distinct odors due in part to the existence of many different functional odorant receptors. There are about 1000 olfactory genes in humans, accounting for 3% of the human genome; approximately 400 of these genes function as odorant receptors. The amino acid sequences of odorant receptors are very diverse, but all are **G-protein-coupled receptors (GPCRs)**. When an odorant molecule binds to its receptor, the G-protein subunits (α, β, γ) dissociate **(Figure 9–2)**. The α-subunit activates adenylyl cyclase to catalyze the production of cAMP which acts as a second messenger to open cation channels, increasing the membrane permeability to Na^+, K^-, and Ca^{2+}. The net effect is an inward-directed Ca^{2+}

current which produces the **graded receptor potential.** This then opens Ca^{2+}-activated Cl^- channels, further depolarizing the cell due to the high intracellular Cl^- levels in olfactory sensory neurons. If the stimulus is sufficient for the receptor potential to exceed its threshold, an action potential in the olfactory nerve (1st cranial nerve) is triggered.

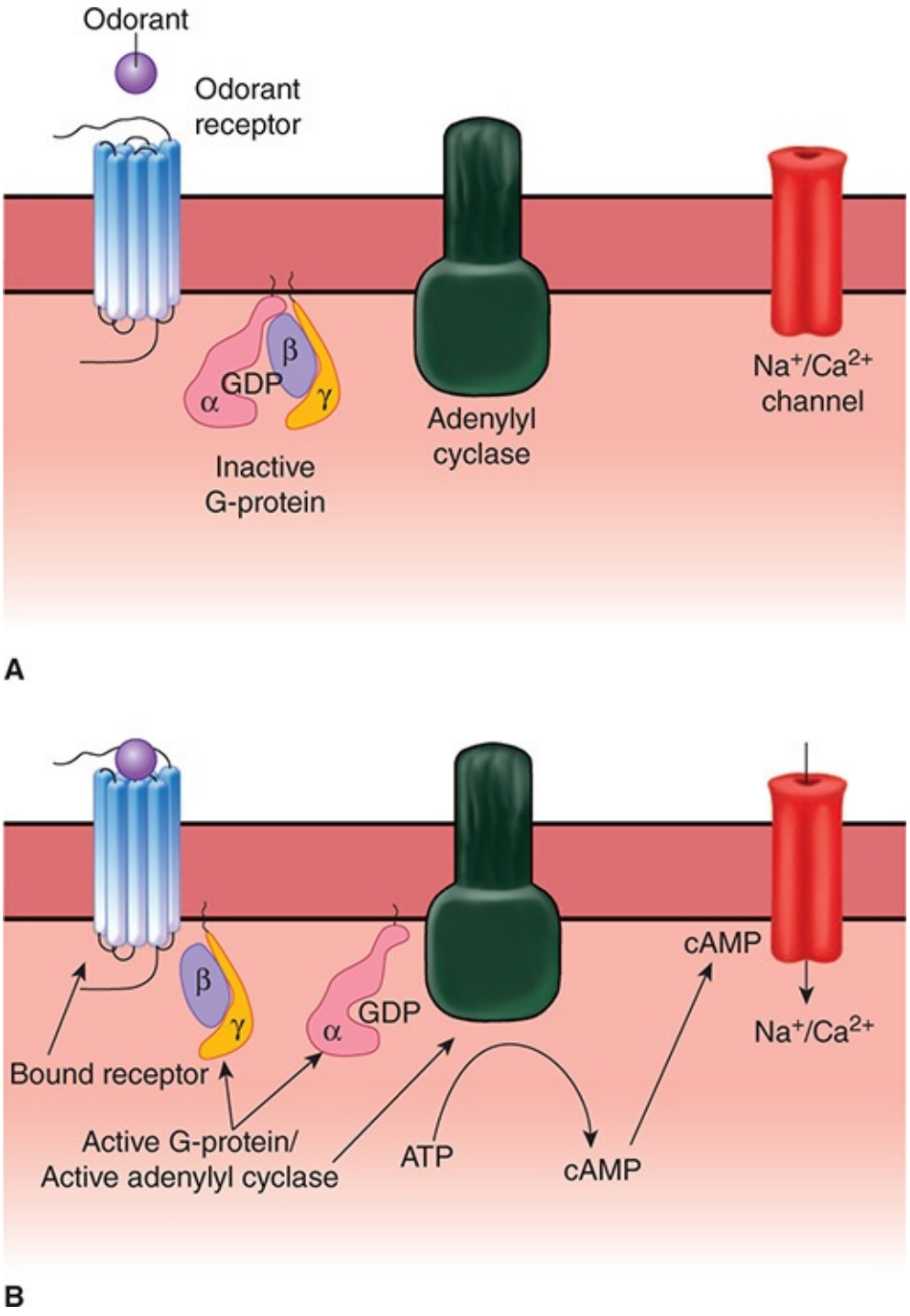


FIGURE 9–2 Signal transduction in an odorant receptor. A) Olfactory receptors are examples of G-protein-coupled receptors; ie, they are associated with three G-protein subunits (α, β, γ). **B)** When an odorant binds to the receptors, the subunits dissociate. The α-subunit of G-proteins activates adenylyl cyclase to catalyze production of cAMP. cAMP acts as a second messenger to open cation channels. Inward diffusion of Na^+ and Ca^{2+} produces depolarization.

OLFACTORY SENSORY PATHWAY

In the olfactory bulb, the axons of the olfactory sensory neurons synapse on the primary dendrites of the **mitral cells** and **tufted cells** **(Figure 9–3)** to form **olfactory glomeruli.** Each olfactory sensory neuron expresses only one of the 400 functional olfactory genes, but each odorant can bind to a large pool of odorant receptors. Each olfactory sensory neuron projects to only one or two glomeruli. This provides a distinct two-dimensional map in the olfactory bulb that is unique to the odorant. The mitral cells with their glomeruli project to different parts of the olfactory cortex. The central olfactory system is able to decode the pattern of receptor-cell activity that signals the identity of the odorant.

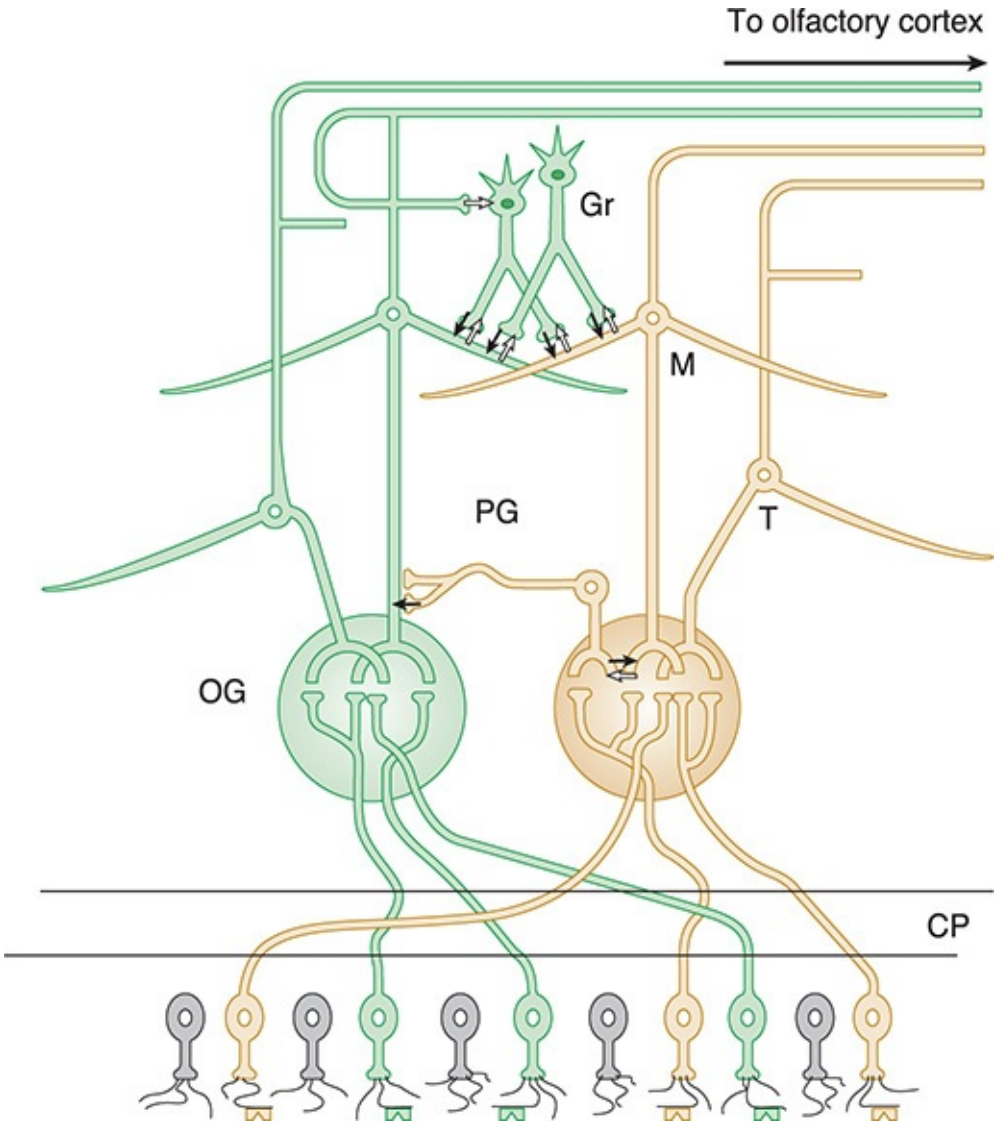


FIGURE 9–3 Basic neural circuits in the olfactory bulb. Olfactory receptor cells with one type of odorant receptor project to one olfactory glomerulus (OG) and olfactory receptor cells with another type of receptor project to a different OG. Solid black arrows signify inhibition via GABA release, and white arrows signify excitatory connections via glutamate release. CP, cribriform plate; Gr, granule cell; M, mitral cell; PG, periglomerular cell; T, tufted cell. (Adapted with permission from Mori K, et al: The olfactory bulb: coding and processing of odor molecular information, Science 1999 Oct 22; 286(5440): 711–715.)

The olfactory bulbs also contain **periglomerular cells,** which are inhibitory neurons connecting one glomerulus to another, and **granule cells,** which have no axons and make reciprocal synapses with the lateral dendrites of the mitral and tufted cells. At these synapses, the mitral or tufted cells excite the granule cell by

releasing **glutamate,** and the granule cells in turn inhibit the mitral or tufted cell by releasing **GABA.** The lateral inhibition mediated by periglomerular and granule cells functions to sharpen and focus olfactory signals.

The axons of the mitral and tufted cells pass posteriorly through the **lateral olfactory stria** to terminate on apical dendrites of pyramidal cells in five regions of the **olfactory cortex: anterior olfactory nucleus, olfactory tubercle, piriform cortex, amygdala,** and **entorhinal cortex** **(Figure 9–4)**. From these regions, information travels directly to the frontal cortex or via the thalamus to the orbitofrontal cortex. Conscious discrimination of odors is dependent on the pathway to the orbitofrontal cortex. The pathway to the amygdala mediates the emotional responses to olfactory stimuli, and the pathway to the entorhinal cortex is concerned with olfactory memories.

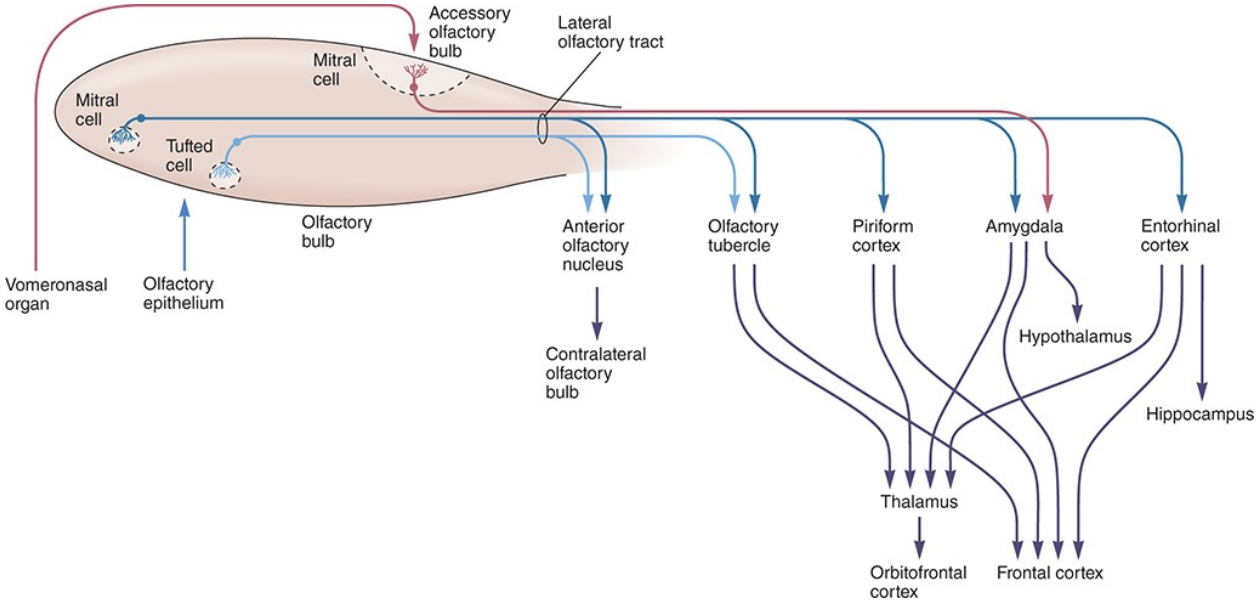


FIGURE 9–4 Diagram of the olfactory pathway. Information is transmitted from the olfactory bulb by axons of mitral and tufted relay neurons in the lateral olfactory tract. Mitral cells project to five regions of the olfactory cortex: anterior olfactory nucleus, olfactory tubercle, piriform cortex, and parts of the amygdala and entorhinal cortex. Tufted cells project to anterior olfactory nucleus and olfactory tubercle; mitral cells in the accessory olfactory bulb project only to the amygdala. Conscious discrimination of odor depends on the neocortex (orbitofrontal and frontal cortices). Emotive aspects of olfaction derive from limbic projections (amygdala and hypothalamus). (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

In some mammals, the nasal cavity contains another patch of olfactory epithelium located along the nasal septum in the **vomeronasal organ** that is concerned with the perception of **pheromones.** Vomeronasal sensory neurons project to the **accessory olfactory bulb** (Figure 9–4) and from there to regions of the amygdala and hypothalamus that are concerned with reproduction and ingestive behavior. The vomeronasal organ has about 100 G-protein-coupled odorant receptors that differ in structure from those in the rest of the olfactory epithelium.

ODOR DETECTION THRESHOLD

Odorants are generally small, containing between 3 and 20 carbon atoms; molecules with the same number of carbon atoms but different structural configurations have different odors. Relatively high water and lipid solubility is characteristic of substances with strong odors. Some common abnormalities in odor detection are described in **Clinical Box 9–1**.

The **odor detection threshold** is the lowest concentration of a chemical that can be detected. Examples of substances detected at very low concentrations include hydrogen sulfide (0.0005 parts per million, ppm), acetic acid (0.016 ppm), kerosene (0.1 ppm), and gasoline (0.3 ppm). Some toxic substances are essentially odorless; they have odor detection thresholds higher than lethal concentrations. For example, carbon dioxide is detected at 74,000 ppm but is lethal at 50,000 ppm. The odor detection threshold for a given odorant is not the same in all individuals.

The sense of smell is said to be more acute in women than in men, and in women it is most acute at the time of ovulation. Although olfactory discrimination is remarkable, determination of differences in the intensity of any given odor is poor. The concentration of an odor-producing substance must be changed by about 30% before a difference can be detected. The comparable visual discrimination threshold is a 1% change in light intensity.

CLINICAL BOX 9–1

Abnormalities in Odor Detection

Anosmia (inability to smell) and **hyposmia** or **hypesthesia** (diminished olfactory sensitivity) can result from simple nasal congestion, nasal polyps, or prolonged use of nasal decongestants. It may also be a sign of a more serious

problem such as damage to the olfactory nerves due to fractures of the cribriform plate or head trauma, tumors (eg, neuroblastomas or meningiomas), and respiratory tract infections. **Congenital anosmia** is a rare disorder in which an individual is born without the ability to smell. Olfactory dysfunction is often one of the earliest clinical symptoms of Alzheimer disease. According to the National Institutes of Health, 1–2% of the North American population under the age of 65 experiences a significant degree of loss of smell. However, 50% of individuals between the ages of 65 and 80 and >75% of those over the age of 80 have an impaired ability to identify smells. Because of the close relationship between taste and smell, anosmia is associated with a reduction in taste sensitivity **(hypogeusia).** Anosmia is generally permanent in cases in which the olfactory nerve or other neural elements in the olfactory neural pathway are damaged. In addition to not being able to experience the enjoyment of pleasant aromas and a full spectrum of tastes, individuals with anosmia are at risk because they are not able to detect the odor from dangers such as gas leaks, fire, and spoiled food. **Hyperosmia** (enhanced olfactory sensitivity) is less common than loss of smell, but pregnant women commonly become oversensitive to smell. **Dysosmia** (distorted sense of smell) can be caused by several disorders including sinus infections, partial damage to the olfactory nerves, and poor dental hygiene. An aura of a disagreeable odor (eg, burning rubber) can occur when an individual experiences an **uncinate seizure** that originates in the medial temporal lobe.

THERAPEUTIC HIGHLIGHTS

Quite often anosmia is a temporary condition due to sinus infection or a common cold, but it can be permanent if caused by nasal polyps or trauma. **Antibiotics** can be prescribed to reduce the inflammation caused by polyps and improve the ability to smell. In some cases, surgery is performed to remove the nasal polyps. Topical **corticosteroids** have also been shown to be effective in reversing the loss of smell due to nasal and sinus diseases.

TASTE

TASTE BUDS

The specialized sense organ for taste (gustation) consists of about 5000 **taste buds** located primarily on the **papillae** of the dorsal surface of the tongue in humans **(Figure 9–5)**. The **fungiform papillae** are rounded structures most numerous near the tip of the tongue; the **circumvallate papillae** are prominent structures arranged in a V on the back of the tongue; the **foliate papillae** are on the posterior edge of the tongue. Each fungiform papilla has up to five taste buds, mostly located at the top of the papilla, while each circumvallate and foliate papilla contain up to 100 taste buds, mostly located along the sides of the papillae. Taste buds are also located in the soft palate, epiglottis, and pharynx.

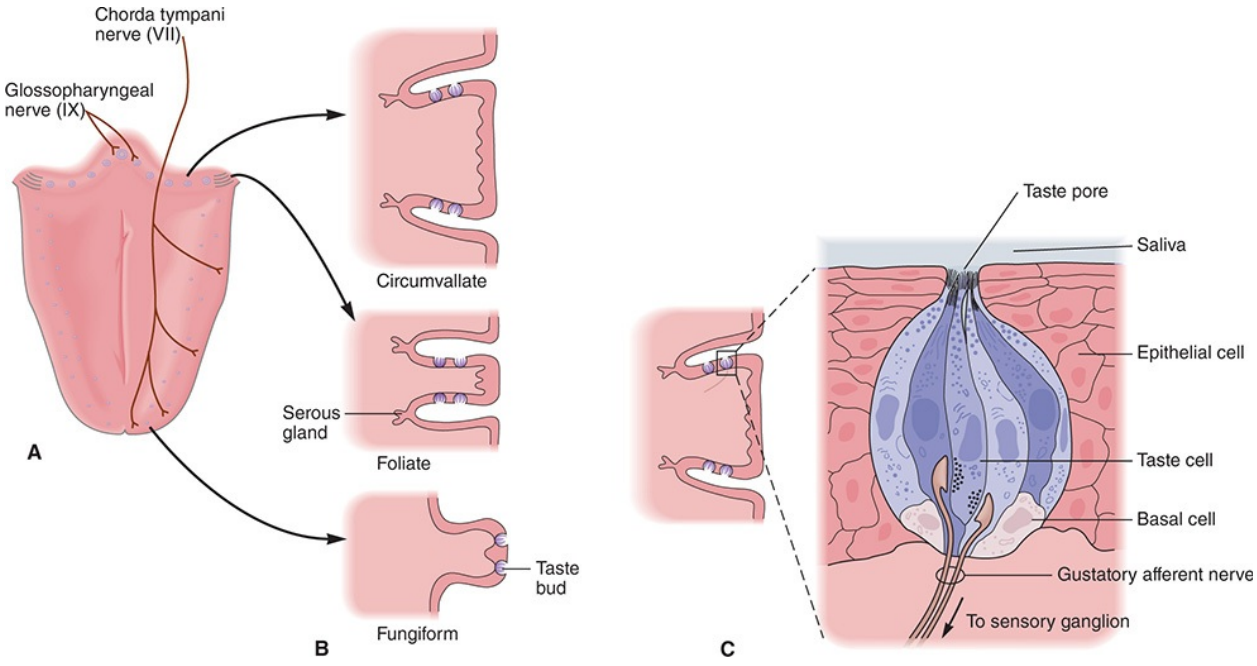


FIGURE 9–5 Taste buds located in papillae of the human tongue. A) Taste buds on the anterior two-thirds of the tongue are innervated by the chorda tympani branch of the facial nerve; those on the posterior one-third of the tongue are innervated by the lingual branch of the glossopharyngeal nerve. **B)** The three major types of papillae (circumvallate, foliate, and fungiform) are located on specific parts of the tongue. **C)** Taste buds are composed of basal stem cells and three types of taste cells (dark, light, and intermediate). Taste cells extend from the base of the taste bud to the taste pore, where microvilli contact tastants dissolved in saliva and mucus. (Modified with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

Each taste bud contains 50–100 **taste receptor cells** and numerous **basal cells**

and **support cells** (Figure 9–5). The taste receptor cells are modified epithelial cells that respond to chemical stimuli or tastants. The apical ends of taste cells have **microvilli** that project into the **taste pore**, a small opening on the dorsal surface of the tongue where tastes cells are exposed to the oral contents. Saliva in the oral cavity acts as a solvent for tastants; after dissolving, the chemical diffuses to the taste receptor sites. Saliva may also function to cleanse the mouth to prepare the taste receptors for a new stimulant.

Each taste bud is innervated by about 50 nerve fibers, and conversely, each nerve fiber receives input from an average of five taste buds. The basal cells arise from the epithelial cells surrounding the taste bud. They differentiate into new taste cells as taste cells survive for only about 10 days. If the sensory nerve is cut, the taste buds it innervates degenerate and eventually disappear.

TASTE PATHWAYS

The sensory nerve fibers from the taste buds on the anterior two-thirds of the tongue travel in the **chorda tympani branch of the facial nerve,** and those from the posterior third of the tongue reach the brainstem via the **glossopharyngeal nerve (Figure 9–6)**. The fibers from areas other than the tongue (eg, pharynx) reach the brain stem via the **vagus nerve.** On each side, the myelinated but relatively slowly conducting taste fibers in these three nerves unite in the gustatory portion of the **nucleus of the tractus solitarius (NTS)** in the medulla oblongata (Figure 9–6). From there, axons of second-order neurons ascend in the ipsilateral medial lemniscus and project directly to the ventral **posteromedial nucleus of the thalamus.** From the thalamus, the axons of the third-order neurons pass to neurons in the **anterior insula** and the frontal operculum in the ipsilateral cerebral cortex. This region is rostral to the face area of the postcentral gyrus, which is probably the area that mediates conscious perception of taste and taste discrimination.

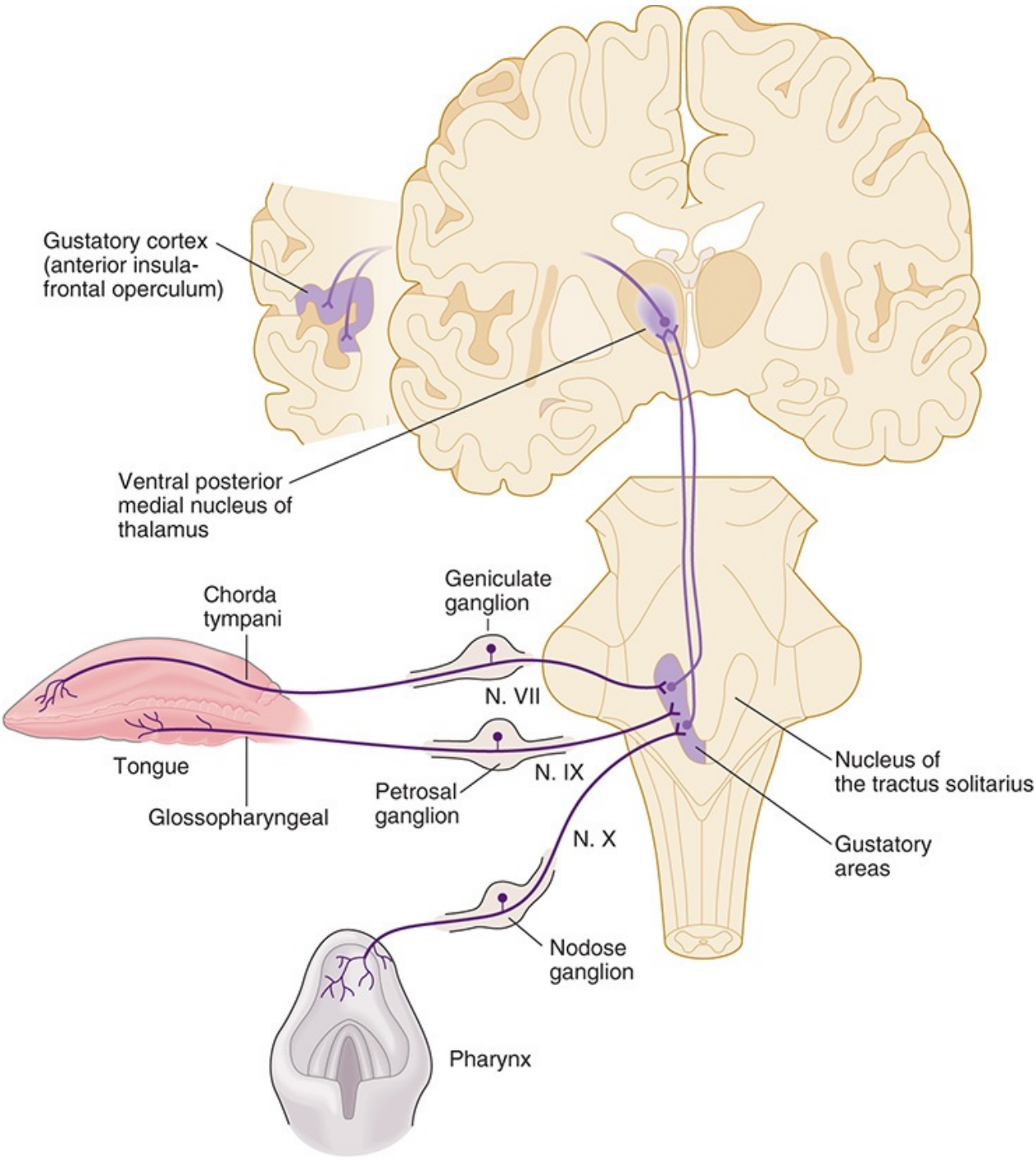


FIGURE 9–6 Diagram of taste pathways. Signals from the taste buds travel via different nerves to gustatory areas of the nucleus of the tractus solitarius, which relays information to the thalamus; the thalamus projects to the gustatory cortex. (Modified with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

Sensory fibers in the trigeminal (5th cranial) nerve also innervate the tongue and contribute to the burning sensation experienced when we eat foods containing capsaicin. Taste buds are surrounded by TRPV1 receptors on trigeminal nociceptive fibers that are activated in response to eating spicy foods.

TASTE MODALITIES, RECEPTORS, & TRANSDUCTION

Humans have five basic taste modalities: **salt, sweet, sour, bitter,** and **umami.** Common stimuli for these sensory modalities are sodium chloride, sucrose, hydrochloric acid, quinine, and monosodium glutamate, respectively. All tastants are sensed from all parts of the tongue and adjacent structures. Afferent nerves to the NTS contain fibers from all types of taste receptors, without any clear localization of types. An individual taste receptor cell may respond to more than one type of tastant. The central nervous system can distinguish the various tastes from one another because each type of taste receptor cell connects to a particular gustatory axon.

Figure 9–7 shows the putative receptors for the five modalities of taste. They include the two major types of receptors: **ligand-gated channels** (ionotropic receptors) and GPCRs (metabotropic receptors). Salt and sour tastes are triggered by activation of ionotropic receptors; sour, bitter, and umami tastes are triggered by activation of metabotropic receptors. Many GPCRs in the human genome are taste receptors (T1Rs, T2Rs families).

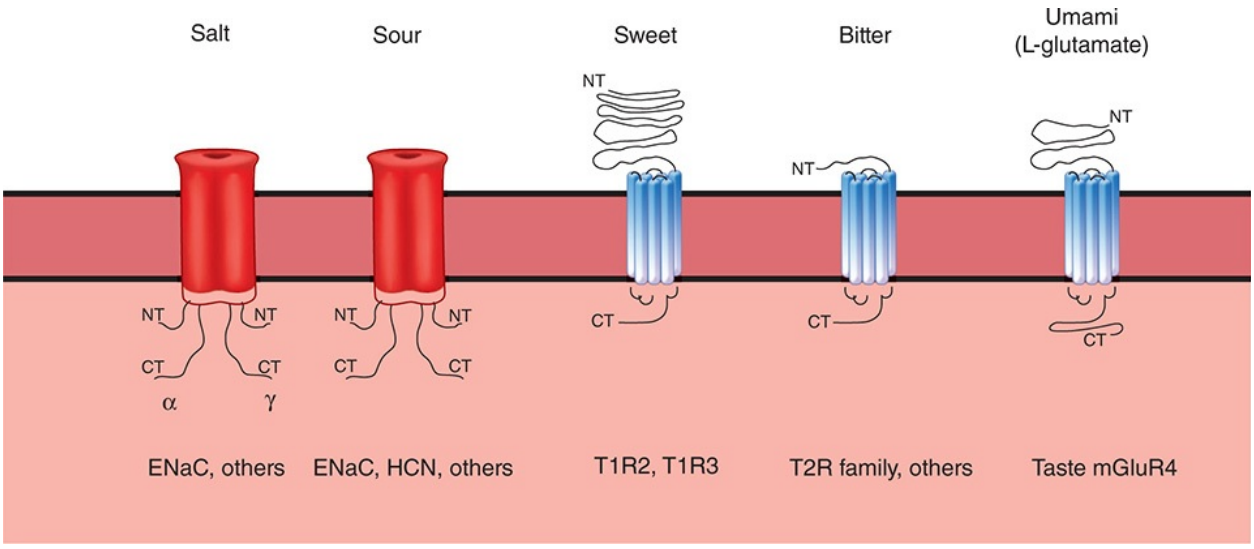


FIGURE 9–7 Signal transduction in taste receptors. Salt and sour tastes are

mediated via the epithelial sodium channel (ENaC). This receptor has two subunits (α and γ), each crossing the membrane twice, resulting in intracellular N and C termini (NT, CT). Salt is sensed following Na^+ movement; sour is mediated by movement of H^+. Sweet, bitter, and umami tastes are sensed via G-protein-coupled receptors that span the membrane seven times and have varying lengths of CT and NT (represented as ribbon structures). Sweet tastes are detected by the T1R2 and T1R3 families; bitter and umami tastes are detected by the T2R family and mGluR4, respectively.

Salt-sensitive taste is mediated by an **epithelial sodium channel (ENaC).** The entry of Na^+ into the salt receptor depolarizes the membrane, generating a receptor potential. The sour taste is triggered by protons (H^+ ions). ENaCs permit the entry of protons and may contribute to the sensation of sour taste. The H^+ ions can also bind to and block a K^+-sensitive channel. The fall in K^+ permeability can depolarize the membrane. Also, a hyperpolarization-activated cyclic nucleotide-gated cation channel (HCN) and other mechanisms may contribute to sour transduction.

Substances that taste sweet are detected by at least two types of GPCRs, T1R2 and T1R3. Sugars taste sweet, but so do compounds such as saccharin that have an entirely different structure. Natural sugars such as sucrose and synthetic sweeteners may act on gustducin via different receptors. Sweet-responsive receptors act via cyclic nucleotides and inositol phosphate metabolism.

Bitter taste is produced by a variety of unrelated compounds. Many of these are poisons, and bitterness serves as a warning to avoid them. Some bitter compounds (eg, quinine) are membrane permeable and bind to and block K^+-selective channels. Many bitter tastants (eg, strychnine) bind to GPCRs (T2R family) that couple to the heterotrimeric G-protein, **gustducin**. Gustducin lowers cAMP and increases the formation of inositol phosphates (IP_3), which releases Ca^{2+} to trigger depolarization.

Umami tastants activate a receptor comprised of T1R1 and T1R3. Umami taste may also involve the activation of a truncated metabotropic glutamate receptor, **mGluR4,** in the taste buds.

TASTE THRESHOLDS & INTENSITY DISCRIMINATION

The ability of humans to discriminate differences in the intensity of tastes is

relatively crude. A 30% change in the concentration of the tastant is necessary before a difference can be detected. **Taste threshold** refers to the minimum concentration at which a substance can be perceived. Table 9–1 shows the threshold concentration of various substances needed for a taste bud to respond. Bitter substances tend to have the lowest threshold. Some toxic substances such as strychnine have a bitter taste at very low concentrations, preventing accidental ingestion of this chemical, which causes fatal convulsions. Some common abnormalities in taste detection are described in Clinical Box 9–2.

TABLE 9–1 Some taste thresholds.

Substance	Taste	Threshold Concentration (μmol/L)
Hydrochloric acid	Sour	100
Sodium chloride	Salt	2000
Strychnine hydrochloride	Bitter	1.6
Glucose	Sweet	80,000
Sucrose	Sweet	10,000
Saccharin	Sweet	23

CLINICAL BOX 9–2

Abnormalities in Taste Detection

Ageusia (absence of the sense of taste) and **hypogeusia** (diminished taste sensitivity) can be caused by damage to the lingual or glossopharyngeal nerve. Neurological disorders such as vestibular schwannoma, Bell palsy, familial dysautonomia, multiple sclerosis, certain infections (eg, primary ameboid meningoencephalopathy), and poor oral hygiene can also cause problems with taste sensitivity. Ageusia can be an adverse side effect of various drugs, including cisplatin and captopril, or vitamin B_3 or zinc deficiencies. Aging and tobacco abuse also contribute to diminished taste. **Dysgeusia** or **parageusia** (unpleasant perception of taste) causes a metallic, salty, foul, or rancid taste. In many cases, dysgeusia is a temporary problem.

Factors contributing to ageusia or hypogeusia can also lead to abnormal taste sensitivity. Taste disturbances can also occur under conditions in which **serotonin (5-HT)** and **norepinephrine (NE)** levels are altered (eg, during anxiety or depression). This implies that these neuromodulators contribute to the determination of **taste thresholds.** Administration of a **5-HT reuptake inhibitor** reduces sensitivity to sucrose (sweet taste) and quinine (bitter taste). In contrast, administration of an **NE reuptake inhibitor** reduces bitter taste and sour thresholds. About 25% of the population has a heightened sensitivity to taste, in particular to bitterness. These individuals are called **supertasters;** this may be due to the presence of an increased number of fungiform papillae on their tongue.

THERAPEUTIC HIGHLIGHTS

Improved oral hygiene and adding zinc supplements to one's diet can correct the inability to taste in some individuals.

CHAPTER SUMMARY

- The olfactory epithelium in the upper portion of the nasal cavity contains three types of cells involved in olfaction: olfactory sensory neurons that are responsible for olfactory transduction, supporting cells that secrete the mucus that provides the appropriate molecular and ionic environment for odor detection, and basal stem cells that generate new olfactory sensory neurons to replace those damaged by exposure to the environment.
- Processing of olfactory information occurs in the olfactory bulb where the axons of olfactory sensory neurons synapse on mitral cells and tufted cells to form olfactory glomeruli. The olfactory bulb also contains inhibitory periglomerular cells and granule cells which make reciprocal synapses with mitral and tufted cells.
- The olfactory system can discriminate perhaps more than 1 million distinct odors due in part to the existence of 400 functional odorant genes (receptors). Each olfactory sensory neuron expresses only one of the 400 functional olfactory genes, but each odorant can bind to a large pool of odorant receptors.
- Odorant receptors are part of a large family of GPCRs. When an odorant

binds to a receptor, the G-protein subunits dissociate and the α-subunit activates adenylyl cyclase to increase production of cAMP which opens cation channels to increase membrane permeability to Na^+, K^-, and Ca^{2+}. Cl^- channels then open to further depolarize olfactory sensory neurons.

- Projections from mitral and tufted cells travel via the lateral olfactory stria directly to five regions of the olfactory cortex: anterior olfactory nucleus, olfactory tubercle, piriform cortex, amygdala, and entorhinal cortex.
- Taste buds are the specialized sense organs for taste and are composed of epithelial taste cells and basal stem cells. Taste buds are located primarily in the mucosa of the walls of papillae of the tongue, but also in the epiglottis, palate, and pharynx.
- The five major taste modalities are salt, sour, bitter, sweet, and umami. Signal transduction mechanisms include passage through ion channels (ENaC for salt and sour), binding to and blocking ion channels (sour), and binding to GPCRs requiring second messenger systems (T1R2, T1R3 for sweet; T2R for bitter; and mGluR4 for umami).
- The afferents from taste buds in the anterior two-thirds of the tongue travel via the facial nerve, those from the posterior one-third of the tongue travel via the glossopharyngeal nerve, and those located elsewhere travel via the vagus nerve. All of the sensory fibers from taste buds synapse in the NTS. From there, axons ascend via the ipsilateral medial lemniscus to the ventral posteromedial nucleus of the thalamus, and onto the anterior insula and frontal operculum in the ipsilateral cerebral cortex.
- Disorders of olfaction include anosmia (inability to smell), hyposmia (diminished olfactory sensitivity), hyperosmia (enhanced olfactory sensitivity), and dysosmia (distorted sense of smell). Causes include damage to the olfactory nerve, tumors, respiratory tract infections, and poor dental hygiene. Disorders of taste include ageusia (absence of the sense of taste), hypogeusia (diminished taste sensitivity), and dysgeusia (unpleasant perception of taste). Causes include damage to the facial or glossopharyngeal nerve, neurological disorders, drugs, vitamin deficiencies, and poor oral hygiene.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A young boy was diagnosed with congenital anosmia, a rare disorder in which an individual is born without the ability to smell. Which parts of the nervous system might be defective in an individual with congenital anosmia to account for the inability to detect odors?

   A. Glossopharyngeal nerve, olfactory bulb, ventral posterior medial nucleus of the thalamus, and anterior insula-frontal operculum

   B. Olfactory sensory neurons, olfactory glomeruli, nucleus of the tractus solitarius, and ventral posterior lateral nucleus of the thalamus

   C. Olfactory nerve, olfactory bulb, medial olfactory tract, and anterior insula-frontal operculum

   D. Olfactory sensory neuron, 1st cranial nerve, olfactory glomeruli, and frontal cortex

   E. Trigeminal nerve, olfactory glomeruli, lateral olfactory tubercle, and entorhinal cortex

2. While working in a laboratory studying the olfactory system, a medical student was intrigued by the fact that a simple sense organ like the human olfactory epithelium can discriminate perhaps more than 1 million distinct odors. What factors may contribute to this phenomenon?

   A. There are 500 types of odorant receptors and over 1000 types of odorant-binding proteins that sequester odorants to enhance sensory discrimination.

   B. Each olfactory sensory neuron expresses a single odorant receptor gene and projects to a particular subset of mitral cells that connect to distinct parts of the olfactory cortex.

   C. Odorants bind to a mixture of GPCR and ion channel receptors on olfactory sensory neurons and the axons of these sensory neurons form anatomically discrete synaptic units called olfactory glomeruli.

   D. Lateral inhibition within olfactory glomeruli sharpen and focus olfactory signals and granule cells within the olfactory glomerulus make specific projections to the postcentral gyrus in the somatosensory cortex.

   E. There are about 5000 types of odorant receptors and each odorant binds to only one of these.

3. As part of a research experience, a medical student was reviewing reports on the effects of exposure to various neurotoxins on odor detection in humans. Which cells in the olfactory system are responsible for the ability to retain the sense of smell despite the fact that toxins can damage elements of the nasal mucosa?

A. Basal cells in the olfactory bulb undergo mitosis to generate new olfactory sensory neurons.

B. Surviving olfactory sensory neurons undergo neuroplasticity and make connections with the mitral and tufted cells that were originally connected to the destroyed sensory neurons.

C. Supporting cells in the olfactory epithelium release neurotrophic factors that stimulate the genesis of new olfactory sensory neurons.

D. Basal cells in the olfactory epithelium regenerate the neurons comprising the 1st cranial nerve.

E. Olfactory sensory neurons in the olfactory bulb are repaired because the surrounding environment contains neurotrophic factors.

4. A 9-year-old boy had frequent episodes of uncontrollable nose bleeds. At the advice of his clinician, he underwent surgery to correct a problem in his nasal septum. A few days after the surgery, he told his mother he could not smell the cinnamon rolls she was baking in the oven. Which of the following sequence of events occurs when an odorant binds to an odorant receptor?

A. Binding of the odorant to a ligand-gated ion channel promotes the influx of Na^+ ions in the olfactory sensory neuron which leads to the induction of an action potential in the olfactory nerve.

B. Binding of the odorant promotes dissociation of G-protein subunits and the γ-subunit activates adenylyl cyclase to increase cGMP which then selectively opens Na^+ channels on the nerve membrane to induce an action potential in the olfactory nerve.

C. When the odorant binds to a combination of ligand-gated ion channels and GPCRs, there is an influx of Ca^{2+} that triggers an influx of Na^+ to induce an action potential in the olfactory sensory neuron.

D. Binding of the odorant to a GPCR on the cilia of the olfactory sensory neurons causes dissociation of the G-protein subunits, the α-subunit activates adenylyl cyclase to increase cAMP which opens cation channels, leading to increased permeability to Na^+, K^-, and Ca^{2+} on the nerve membrane and depolarization of the olfactory nerve.

5. After watching the movie *Christmas Story*, a 10-year-old boy wanted to see if his tongue would really stick to a frozen pole. Much to his surprise, it did. When he pulled his tongue from the pole, he injured the anterior one-third of his tongue. What sensory nerve arises from this portion of the tongue, where are the cell bodies of these sensory neurons, and where does the nerve

terminate?

A. Facial nerve, geniculate ganglion, and gustatory area of the nucleus of the tractus solitarius

B. Vagus nerve, nodose ganglion, and gustatory area of the nucleus ambiguus

C. Chorda tympani branch of the facial nerve, taste buds, and gustatory area of nucleus of the tractus solitarius

D. Glossopharyngeal nerve, petrosal ganglion, and gustatory area of the nucleus ambiguus

E. Glossopharyngeal nerve, taste buds, and caudal portion of the nucleus of the tractus solitarius

6. A 37-year-old woman was diagnosed with multiple sclerosis. One of the potential consequences of this disorder is diminished taste and smell sensitivity. What is the relationship between the sensations of taste and smell?

A. Odorant receptors and taste receptors are both innervated by trigeminal nerves.

B. The afferent fibers from odorant receptors and taste receptors terminate on the same second-order neurons in the brainstem.

C. Odors from food enter our nasal passages at the same time that the taste receptors in our mouth are stimulated by the food, and the two chemosensory systems interact to establish the flavor of what we ingest.

D. Olfaction is closely related to gustation because odorant and taste receptors send signals to adjacent regions of the postcentral gyrus of the cortex which are connected via axon collaterals.

7. A medical student was doing research on gustatory function with a focus on the anatomy and physiology of taste buds. What are the locations and cellular composition of taste buds?

A. Taste buds are located on the sensory endings of the 7th, 9th, and 10th cranial nerves and contain taste receptors, salivary glands, and basal stem cells.

B. Taste buds are located on the dendrites of the 7th, 9th, and 10th cranial nerves and contain papillae, salivary glands, and taste cells.

C. Taste buds are located in the papillae of the tongue and epiglottis and contain tastants, salivary glands, and basal cells.

D. Taste buds are located in the papillae of the tongue and contain tastants, neuroepithelial cells, and the axons of the 7th, 9th, and 10th cranial nerves.

E. Taste buds are located in the papillae of the tongue and contain taste

receptor cells, support cells, and basal cells.

8. A 31-year-old woman is a smoker who has had poor oral hygiene for most of her life. In the past few years she has noticed a reduced sensitivity to the taste of foods which she used to enjoy eating. What types of taste receptors may be damaged if she has difficulty sensing sweet and bitter substances?

   A. Epithelial sodium channel (sweet) and hyperpolarization-activated cyclic nucleotide-gated cation channel (bitter)

   B. Hyperpolarization-activated cyclic nucleotide-gated cation channel (sweet) and epithelial sodium channel (bitter)

   C. T2R family of GPCRs (sweet) and metabotropic glutamate receptor (mGluR4, bitter)

   D. T1R2 and T1R3 family of GPCRs (sweet) and T2R family of GPCRs (bitter)

   E. Metabotropic glutamate receptor (mGluR4, sweet) and epithelial sodium channel (bitter)

CHAPTER 10
Vision

OBJECTIVES

After studying this chapter, you should be able to:

- Identify the function of the various parts of the eye.
- Explain how light rays in the environment are brought to a focus on the retina and the role of accommodation and the pupil light reflex in this process.
- Explain the refractive deficits responsible for hyperopia, myopia, presbyopia, and astigmatism.
- Describe the functional organization of the retina.
- List the sequence of events involved in phototransduction.
- Describe the electrical responses produced by bipolar cells, horizontal cells, amacrine cells, and ganglion cells.
- Define dark adaptation, visual acuity, and age-related macular degeneration.
- Trace the neural pathways that transmit visual information from photoreceptors to the visual cortex.
- Describe the neural pathways involved in color vision and the types of color blindness.
- Predict the visual field deficits that would occur after lesions within specific

parts of the visual pathway.

- Identify the muscles involved in the four types of eye movements and the function of these movements.

INTRODUCTION

The eyes are complex sense organs that have evolved from primitive light-sensitive spots on the surface of invertebrates. They gather information about the environment; the brain interprets this information to form an image of what appears within the field of vision. Within its protective casing, each eye has a layer of photoreceptors that respond to light, a lens system that focuses the light on these receptors, and a system of nerves that conducts impulses from the receptors to the brain. A great deal of work has been done on the neurophysiology of vision; in fact, it may be the most studied and best understood sensory system. This chapter describes the way the components of the visual system operate to set up conscious visual images.

ANATOMY OF THE EYE

The principal structures of the eye are shown in **Figure 10–1**. The outer white protective layer of the eyeball is the **sclera** through which no light can pass. It is modified anteriorly to form the transparent **cornea,** through which light rays enter the eye. The lateral margin of the cornea is contiguous with the **conjunctiva,** a clear mucous membrane that covers the sclera. Just inside the sclera is the **choroid,** a vascular layer that provides oxygen and nutrients to the structures in the eye. The **retina,** the neural tissue containing the **photoreceptors,** lines the posterior two-thirds of the choroid.

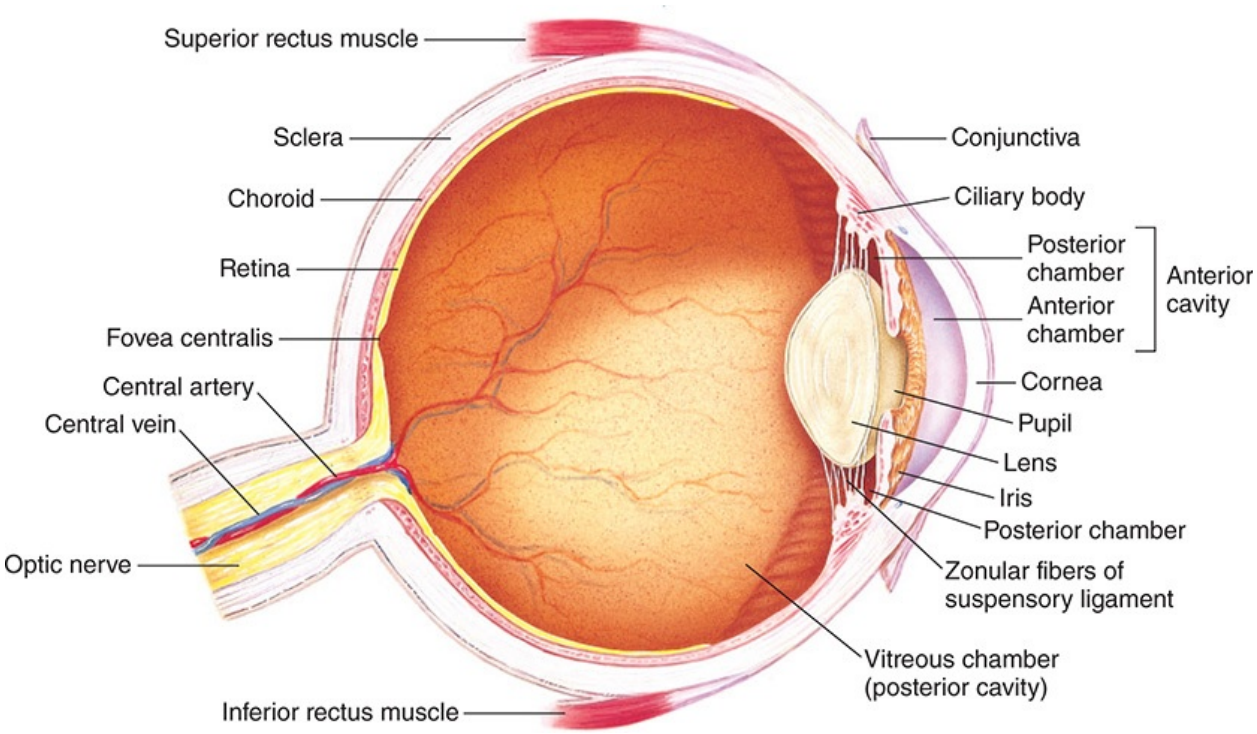


FIGURE 10–1 A schematic of the anatomy of the eye. (Reproduced with permission from Fox SI: *Human Physiology*. New York, NY: McGraw-Hill; 2008.)

The **crystalline lens** is a transparent structure held in place by a circular **lens suspensory ligament (zonule).** The zonule is attached to the **ciliary body** that contains circular muscle fibers and longitudinal muscle fibers that attach near the corneoscleral junction. The pigmented and opaque **iris,** the colored portion of the eye, is in front of the lens. The iris, ciliary body, and choroid are collectively called the **uvea.** The iris contains sphincter muscles that constrict **(miosis)** and radial muscles that dilate **(mydriasis)** the **pupil** that are under the control of parasympathetic and sympathetic nerves, respectively (see Chapter 13). Variations in the diameter of the pupil can produce up to a 16-fold change in the amount of light reaching the retina.

The **aqueous humor** is a clear protein-free liquid that nourishes the cornea and iris; it is produced in the ciliary body by diffusion and active transport from plasma. It flows through the pupil and fills the **anterior chamber** of the eye. It is normally reabsorbed through a network of trabeculae into the **canal of Schlemm,** a venous channel at the junction between the iris and the cornea **(filtration angle).** Obstruction of this outlet leads to increased **intraocular pressure (IOP),** a critical risk factor for glaucoma **(Clinical Box 10–1)**.

The **posterior chamber** is a narrow aqueous-containing space between the iris, zonule, and the lens. The **vitreous chamber** is the space between the lens and the retina that is filled primarily with a clear gelatinous material called the **vitreous (vitreous humor).**

The eye is well protected from injury by the bony walls of the orbit. The cornea is moistened and kept clear by tears that course from the **lacrimal gland** in the upper portion of each orbit across the surface of the eye to empty via the **lacrimal duct** into the nose. Blinking helps keep the cornea moist.

THE IMAGE-FORMING MECHANISM

The eyes convert energy in the visible spectrum into action potentials in the optic nerve. The wavelength of visible light ranges from 397 to 723 nm. The images of objects in the environment are focused on the retina. The light rays striking the retina generate potentials in the photoreceptors. Impulses initiated in the retina are conducted to the cerebral cortex, where they produce the sensation of vision.

PRINCIPLES OF OPTICS

Light rays are bent when they pass from a medium of one density into a medium of a different density, except when they strike perpendicular to the interface **(Figure 10–2)**. The bending of light rays is called **refraction** and is the mechanism that allows one to focus an accurate image onto the retina. Parallel light rays striking a biconvex lens are refracted to a point **(focal point** or **principal focus)** behind the lens. The focal point is on a line passing through the centers of curvature of the lens, the **principal axis.** The distance between the lens and the principal focus is the **focal length.** Light rays from an object that strike a lens more than 6 m (20 ft) away are, for practical purposes, parallel. The rays from an object closer than 6 m are diverging and are therefore brought to a focus farther back on the principal axis than the principal focus. Biconcave lenses cause light rays to diverge.

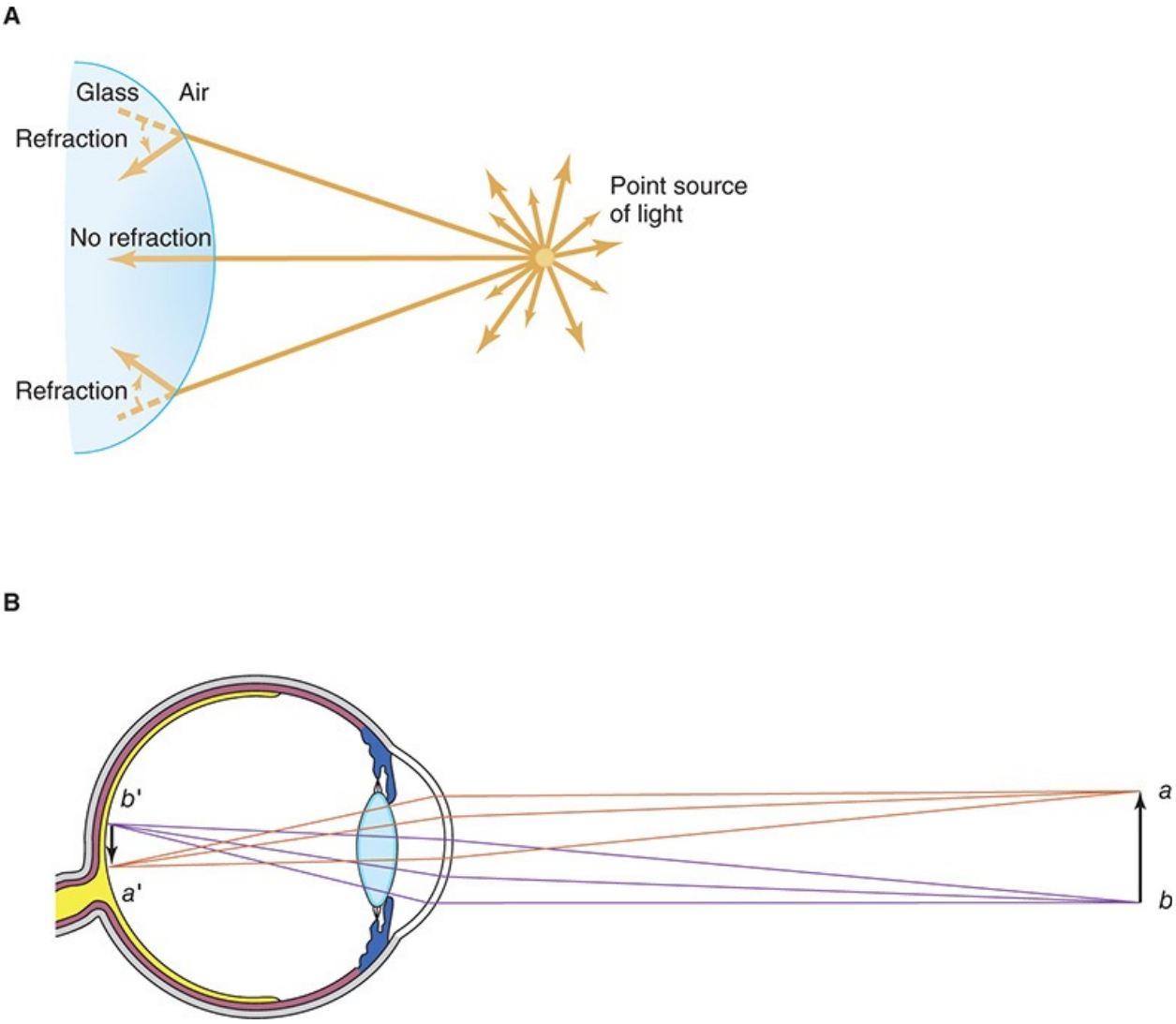


FIGURE 10–2 Focusing point sources of light. A) When diverging light rays enter a dense medium at an angle to its convex surface, refraction bends them inward. **B)** Refraction of light by the lens system. For simplicity, refraction is shown only at the corneal surface (site of greatest refraction) although it also occurs in the lens and elsewhere. Incoming light from *a* (above) and *b* (below) is bent in opposite directions, resulting in *b′* being above *a′* on the retina. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology*, 11th ed. New York, NY: McGraw-Hill; 2008.)

Refractive power is greatest when the curvature of a lens is greatest. The refractive power of a lens is measured in **diopters,** the number of diopters being the reciprocal of the principal focal distance in meters. For example, a lens with a principal focal distance of 0.25 m has a refractive power of 1/0.25, or 4

diopters. The human eye has a refractive power of 60 diopters at rest.

CLINICAL BOX 10–1

Glaucoma

IOP is not the only cause of **glaucoma,** a degenerative disease in which there is loss of retinal ganglia cells; however, it is a critical risk factor. In 20–50% of the patients with glaucoma, IOP is normal (10–20 mm Hg); however, increased IOP worsens glaucoma, and treatment is aimed at lowering the pressure. Indeed, elevations in IOP due to injury or surgery can cause glaucoma. Glaucoma is caused by poor drainage of the aqueous humor through the filtration angle formed between the iris and the cornea. **Open-angle glaucoma,** a chronic disease, is caused by decreased permeability through the trabeculae into the canal of Schlemm, which leads to an increase in IOP. In some cases, this type of glaucoma is due to a genetic defect. **Closed-angle glaucoma** results from a forward ballooning of the iris so that it reaches the back of the cornea and obliterates the filtration angle, thus reducing the outflow of aqueous humor. If left untreated, glaucoma can lead to blindness.

THERAPEUTIC HIGHLIGHTS

Glaucoma can be treated with agents that decrease the secretion or production of aqueous humor or with drugs that increase outflow of the aqueous humor. β-Adrenoceptor antagonists such as **timolol** decrease the secretion of aqueous fluid. Carbonic anhydrase inhibitors (eg, **dorzolamide, acetazolamide**) also exert their beneficial effects by decreasing the secretion of aqueous humor. Glaucoma can also be treated with cholinergic agonists (eg, **pilocarpine, carbachol, physostigmine**) that increase aqueous outflow by causing ciliary muscle contraction. Aqueous outflow is also increased by **prostaglandins.** Prolonged use of **corticosteroids** can lead to glaucoma and increase the risk of occurrence of ocular infections due to fungi or viruses. A common treatment is a combination of a β-blocker to reduce secretion and a prostaglandin to increase outflow.

In the eye, light is refracted at the anterior surface of the cornea and at the

anterior and posterior surfaces of the lens. The process of refraction can be represented diagrammatically, however, without introducing any appreciable error, by drawing the rays of light as if all refraction occurs at the anterior surface of the cornea (Figure 10–2). Note that the retinal image is inverted. The connections of the retinal receptors are such that from birth any inverted image on the retina is viewed right side up and projected to the visual field on the side opposite to the retinal area stimulated. This perception is present in infants and is innate. If retinal images are turned right side up by means of special lenses, the objects viewed look as if they are upside down.

COMMON DEFECTS OF THE IMAGE-FORMING MECHANISM

Table 10–1 shows some of the common tests included in a comprehensive ophthalmology exam in an adult. In some individuals, the eyeball is shorter than normal, and the parallel rays of light are brought to a focus behind the retina. This abnormality is called **hyperopia** or farsightedness **(Figure 10–3)**. Sustained accommodation, even when viewing distant objects, can partially compensate for the defect, but the prolonged muscular effort is tiring and may cause headaches and blurred vision. The prolonged convergence of the visual axes associated with the accommodation may lead eventually to **strabismus** **(Clinical Box 10–2)**. The defect can be corrected by using glasses with convex lenses, which aid the refractive power of the eye in shortening the focal distance.

TABLE 10–1 Components of a Comprehensive Ophthalmology Exam.

Assessment	Method	Result
Visual acuity	Snellen chart	Nearsightedness, farsightedness, astigmatism
Refraction	Phoropter; retinoscope	Lens power needed for vision correction
Color blindness	Ishihara chart	Protanopia, deuteranopia, or tritanopia
Pupillary light reflex	Penlight	Test integrity of sensory and motor innervation of the eye; aniscoria (uneven pupil size)
Intraocular pressure	Goldmann applanation tonometry	Risk factor for glaucoma
Optic nerve, macula, vessel integrity	Fundoscope, opthalmascope	Macular degeneration, retinal tear
Eyelids, cornea, conjunctiva, sclera, iris	Slit lamp	Cataract, corneal injury, dry eye syndrome, retinal detachment, macular degeneration, retinal vessel occlusion, diabetic retinopathy
Peripheral vision	Perimeter, penlight	Map of peripheral portions of the visual fields
Central vision	Tangent screen	Map of central visual field; identify objective scotoma (blind spots)
Extraocular muscle motility/alignment	Check six cardinal positions of gaze	Heterotropia

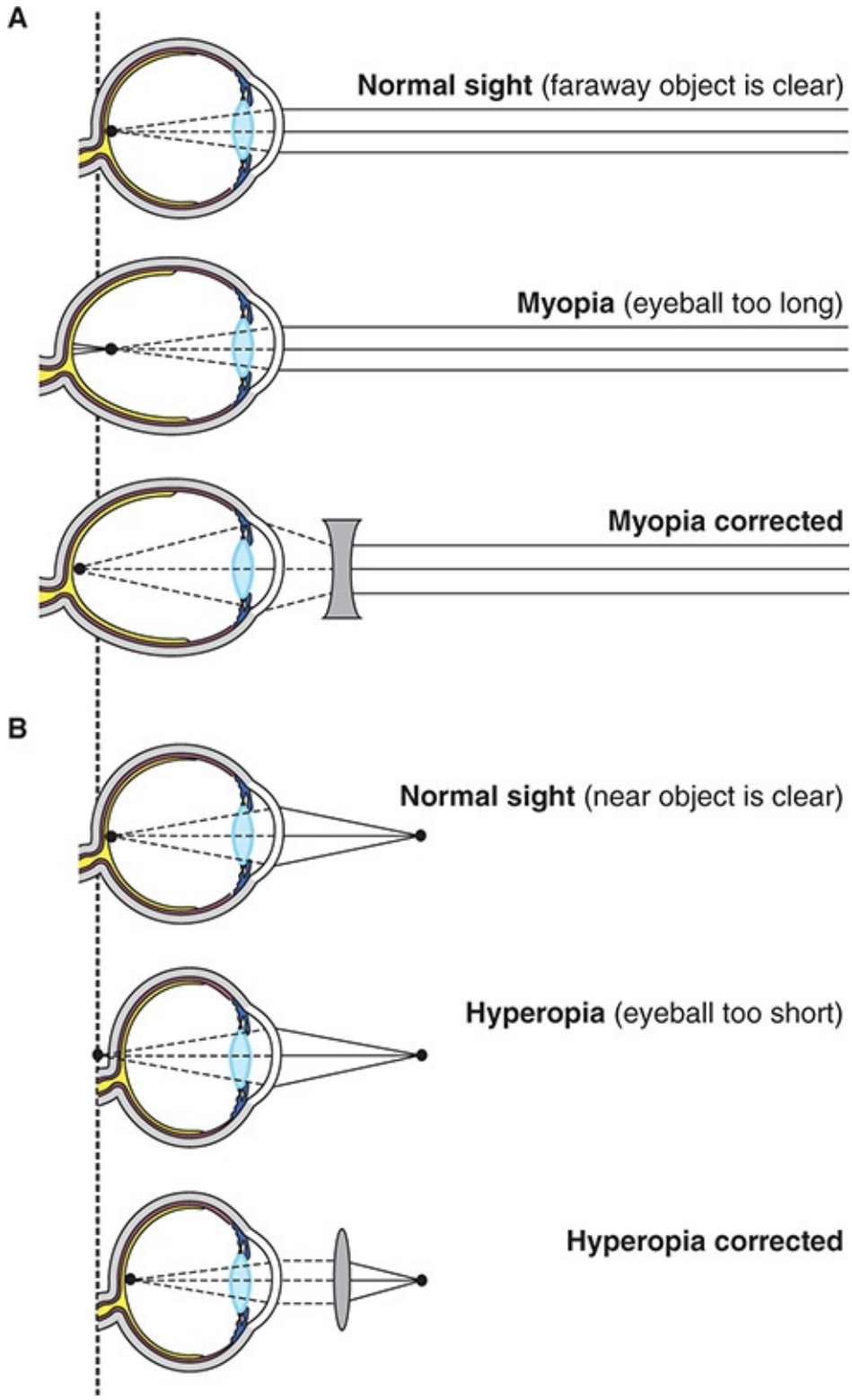


FIGURE 10–3 Common defects of the optical system of the eye. A) In myopia (nearsightedness), the eyeball is too long and light rays focus in front of the retina. Placing a biconcave lens in front of the eye causes the light rays to

diverge slightly before striking the eye, so that they are brought to a focus on the retina. **B)** In hyperopia (farsightedness), the eyeball is too short and light rays come to a focus behind the retina. A biconvex lens corrects this by adding to the refractive power of the lens of the eye. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology*, 11th ed. New York, NY: McGraw-Hill; 2008.)

In **myopia** (nearsightedness), the anteroposterior diameter of the eyeball is too long (Figure 10–3). Myopia may be genetic in origin; however, there is a positive correlation between sleeping in a lighted room before the age of 2 and the subsequent development of myopia. Thus, the shape of the eye appears to be determined in part by the refraction presented to it. In young adults, the extensive close work involved in activities such as studying accelerates the development of myopia. This defect can be corrected by glasses with biconcave lenses, which make parallel light rays diverge slightly before they strike the eye.

Astigmatism is a common condition in which the curvature of the cornea is not uniform. When the curvature in one meridian is different from that in others, light rays in that meridian are refracted to a different focus, so that part of the retinal image is blurred. A similar defect may be produced if the lens is pushed out of alignment or the curvature of the lens is not uniform, but these conditions are rare. Astigmatism can usually be corrected with cylindric lenses placed in such a way that they equalize the refraction in all meridians.

ACCOMMODATION

When the **ciliary muscle** is relaxed, parallel light rays striking the optically normal **(emmetropic)** eye are brought to a focus on the retina. As long as this relaxation is maintained, rays from objects closer than 6 m from the observer are brought to a focus behind the retina, and consequently the objects appear blurred. The problem of bringing diverging rays from close objects to a focus on the retina can be solved by increasing the distance between the lens and the retina or by increasing the curvature or refractive power of the lens.

The process by which the curvature of the lens is increased is called **accommodation.** At rest, the lens is held under tension by the lens ligaments. Because the lens substance is malleable and the lens capsule has considerable elasticity, the lens is pulled into a flattened shape. If the gaze is directed at a near object, the ciliary muscle contracts. This decreases the distance between the edges of the ciliary body and relaxes the lens ligaments, so that the lens springs

into a more convex shape **(Figure 10–4)**. The change is greatest in the anterior surface of the lens. In young individuals, the change in shape may add as many as 12 diopters to the refractive power of the eye. The relaxation of the lens ligaments produced by contraction of the ciliary muscle is due partly to the sphincter-like action of the circular muscle fibers in the ciliary body and partly to the contraction of longitudinal muscle fibers that attach anteriorly, near the corneoscleral junction. As these fibers contract, they pull the whole ciliary body forward and inward. This motion brings the edges of the ciliary body closer together. The nearest point to the eye at which an object can be brought into clear focus by accommodation is called the **near point of vision**. The near point recedes throughout life, slowly at first and then rapidly with advancing age, from approximately 9 cm at age 10 to approximately 83 cm at age 60. This recession is due principally to increasing hardness of the lens, with a resulting loss of accommodation due to the steady decrease in the degree to which the curvature of the lens can be increased. By the time a healthy individual reaches age 40–45 years, the loss of accommodation is usually sufficient to make reading and close work difficult. This condition, which is known as **presbyopia**, can be corrected by wearing glasses with convex lenses. In addition to accommodation, the visual axes converge, and the pupil constricts when an individual looks at a near object. This three-part response is called the **near response.**

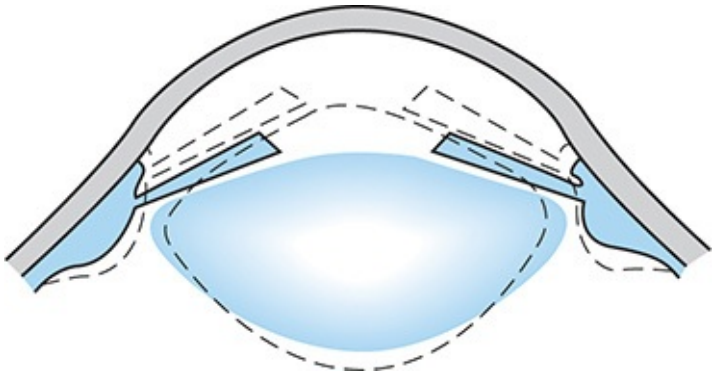


FIGURE 10–4 Accommodation. The solid lines represent the shape of the lens, iris, and ciliary body at rest, and the dashed lines represent the shape during accommodation. When gaze is directed at a near object, ciliary muscles contract. This decreases the distance between the edges of the ciliary body and relaxes the lens ligaments, and the lens becomes more convex.

CLINICAL BOX 10–2

Strabismus & Amblyopia

Strabismus is a misalignment of the eyes and one of the most common eye problems in children, affecting 4% of children under 6 years of age. It is characterized by one or both eyes turning inward (**esotropia**), outward (**exotropia**), upward, or downward. In some cases, more than one of these conditions is present. Strabismus is commonly called “wandering eye” or “crossed-eyes.” It results in visual images that do not fall on corresponding retinal points. When visual images chronically fall on noncorresponding points in the two retinas in young children, one is eventually suppressed **(suppression scotoma).** This suppression is a cortical phenomenon, and rarely develops in adults. It is important to institute treatment before age 6 in affected children, because if the suppression persists, the loss of visual acuity in the eye generating the suppressed image is permanent. A similar suppression with subsequent permanent loss of visual acuity can occur in children in whom vision in one eye is blurred or distorted due to a refractive error. The loss of vision in these cases is called **amblyopia ex anopsia,** a term that refers to uncorrectable loss of visual acuity that is not directly due to organic disease of the eye. Typically, an affected child has one weak eye with poor vision and one strong eye with normal vision. It affects about 3% of the general population. Amblyopia is also referred to as “lazy eye,” and it often coexists with strabismus.

THERAPEUTIC HIGHLIGHTS

Atropine (a cholinergic muscarinic receptor antagonist) and **miotics** such as **echothiophate iodide** can be administered in the eye to correct strabismus and ambylopia. Atropine will blur the vision in the good eye to force the individual to use the weaker eye. Eye muscle training through **optometric vision therapy** has also been proven to be useful, even in patients older than 17 years. Some types of strabismus can be corrected by surgical shortening of some of the eye muscles, by eye muscle training exercises or by using glasses with prisms that bend the light rays sufficiently to compensate for the abnormal position of the eyeball. However, subtle defects in **depth perception** persist. Congenital abnormalities of the visual tracking mechanisms may cause both strabismus and the defective depth perception. In infant monkeys, covering one eye with a patch for 3 months causes a loss of ocular dominance columns; input from the remaining eye spreads to take over all the cortical cells and the patched eye becomes functionally blind. Comparable changes may occur in children with strabismus.

PUPILLARY LIGHT REFLEXES

Light entering one eye constricts the pupil of that eye **(direct light reflex)** as well as the pupil of the other eye **(consensual light reflex).** The sensory fibers initiating this **pupillary light reflex** are in the **optic nerve** (second cranial nerve). The efferent pathway of the reflex is the ipsilateral and contralateral **Edinger-Westphal nuclei** that contain preganglionic parasympathetic neurons within the **oculomotor nerve** (third cranial nerve) that innervate postganglionic neurons in the ciliary ganglion. The reflex not only regulates the amount of light that enters the eye but it also affects the quality of the retinal image (a smaller pupil diameter gives a greater depth of focus).

PHOTO-TRANSDUCTION PROCESS

RETINA

The retina is organized into layers containing different types of cells and neural processes **(Figure 10–5)**. The **outer nuclear layer** contains the photoreceptors **(rods** and **cones).** The **inner nuclear layer** contains the cell bodies of the excitatory and inhibitory interneurons including **bipolar cells, horizontal cells,** and **amacrine cells.** The **ganglion cell layer** contains various types of **ganglion cells** that are the only output neurons of the retina; their axons form the optic nerve. The **outer plexiform layer** is interposed between the outer and inner nuclear layers; the **inner plexiform layer** is interposed between the inner nuclear and ganglion cell layers.

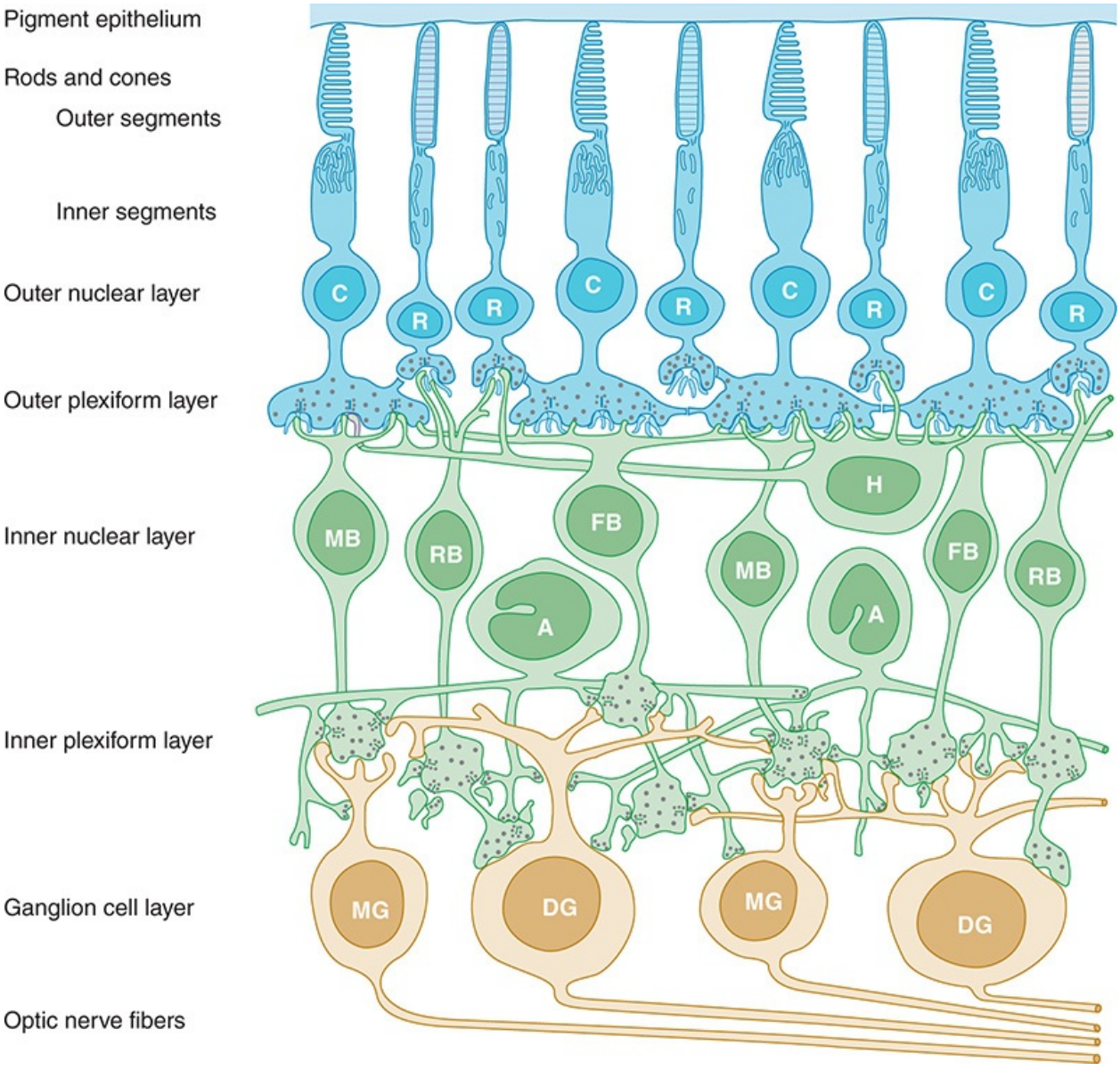


FIGURE 10–5 Neural components of the extrafoveal portion of the retina. C, cone; R, rod; MB, RB, and FB, midget, rod, and flat bipolar cells; DG and MG, diffuse and midget ganglion cells; H, horizontal cells; A, amacrine cells. (Modified with permission from Dowling JE, Boycott BB: Organization of the primate retina: Electron microscopy. Proc R Soc Lond Ser B [Biol] 1966 Nov 15;166(1002):80–111.)

The rods and cones, which are next to the choroid, synapse with bipolar cells; the bipolar cells synapse with ganglion cells. There are various types of bipolar cells that differ in terms of morphology and function. Horizontal cells connect photoreceptor cells to other photoreceptor cells in the outer plexiform layer.

Amacrine cells connect ganglion cells to one another in the inner plexiform layer via processes of varying length and patterns. Amacrine cells also connect with the terminals of bipolar cells. Retinal neurons are also connected via gap junctions.

Because the receptor layer of the retina rests on the **pigment epithelium** next to the choroid, light rays must pass through the ganglion cell and bipolar cell layers to reach the rods and cones. The pigment epithelium absorbs light rays, preventing the reflection of rays back through the retina that would otherwise produce blurring of the visual images.

PHOTORECEPTORS

Each rod and cone photoreceptor is divided into an outer segment, an inner segment that includes a nuclear region, and a synaptic terminal zone **(Figure 10–6)**. The outer segments are modified **cilia** composed of regular stacks of flattened **saccules** or membranous **disks.** The saccules and disks contain the photosensitive compounds that react to light, initiating action potentials in the visual pathways. The inner segments are rich in mitochondria; this is the region that synthesizes the photosensitive compounds. The inner and outer segments are connected by a ciliary stalk through which the photosensitive compounds travel from the inner segment to the outer segment of the rods and cones.

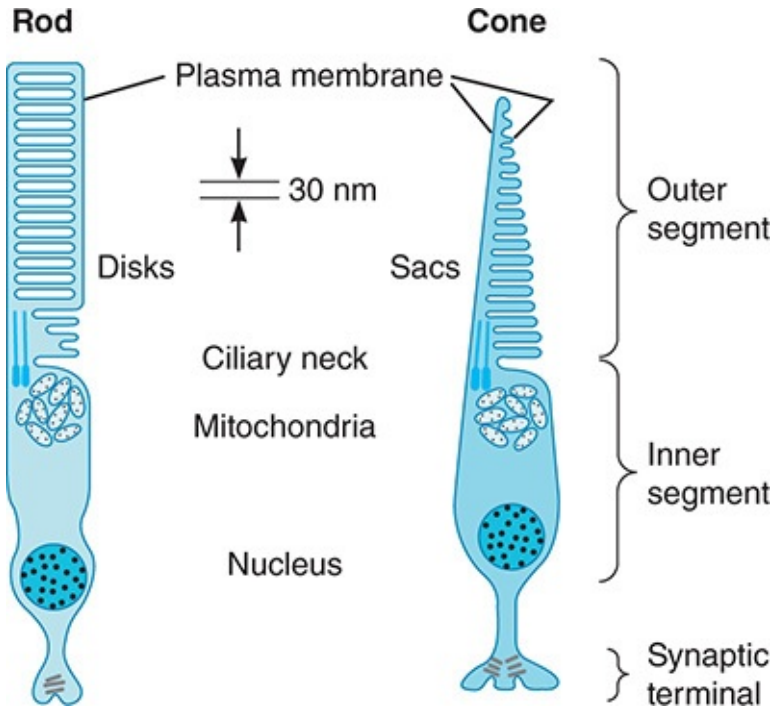


FIGURE 10–6 Schematic diagram of a rod and a cone. Each rod and cone is divided into an outer segment, an inner segment with a nuclear region, and a synaptic zone. The saccules and disks in the outer segment contain photosensitive compounds that react to light to initiate action potentials in the visual pathways. (Reproduced with permission from Lamb TD: Electrical responses of photoreceptors. In: *Recent Advances in Physiology*. No. 10. Baker PF [editor]. Churchill Livingstone, 1984.)

The rods are named for the thin, rodlike appearance of their outer segments. Each rod contains a stack of disk membranes that are flattened membrane-bound intracellular organelles that have detached from the outer membrane and are thus free floating. Cones generally have thick inner segments and conical outer segments, although their morphology varies throughout the retina. The saccules of the cones are formed by infolding of the membrane of the outer segment. Rod outer segments are being constantly renewed by the formation of new disks at the inner edge of the segment and phagocytosis of old disks from the outer tip by cells of the pigment epithelium. Cone renewal is a more diffused process and may occur at multiple sites in the outer segments.

In the extrafoveal portions of the retina, rods predominate **(Figure 10–7)**, and there is a good deal of convergence. *Flat* bipolar cells (Figure 10–5) make synaptic contact with several cones, and *rod* bipolar cells make synaptic contact with several rods. Because there are nearly 6 million cones and 120 million rods in each human eye but only 1.2 million nerve fibers in each optic nerve, the overall convergence of receptors through bipolar cells on ganglion cells is about 105:1. However, there is divergence from this point on. For example, in the visual cortex the number of neurons concerned with vision is 1000 times the number of fibers in the optic nerves.

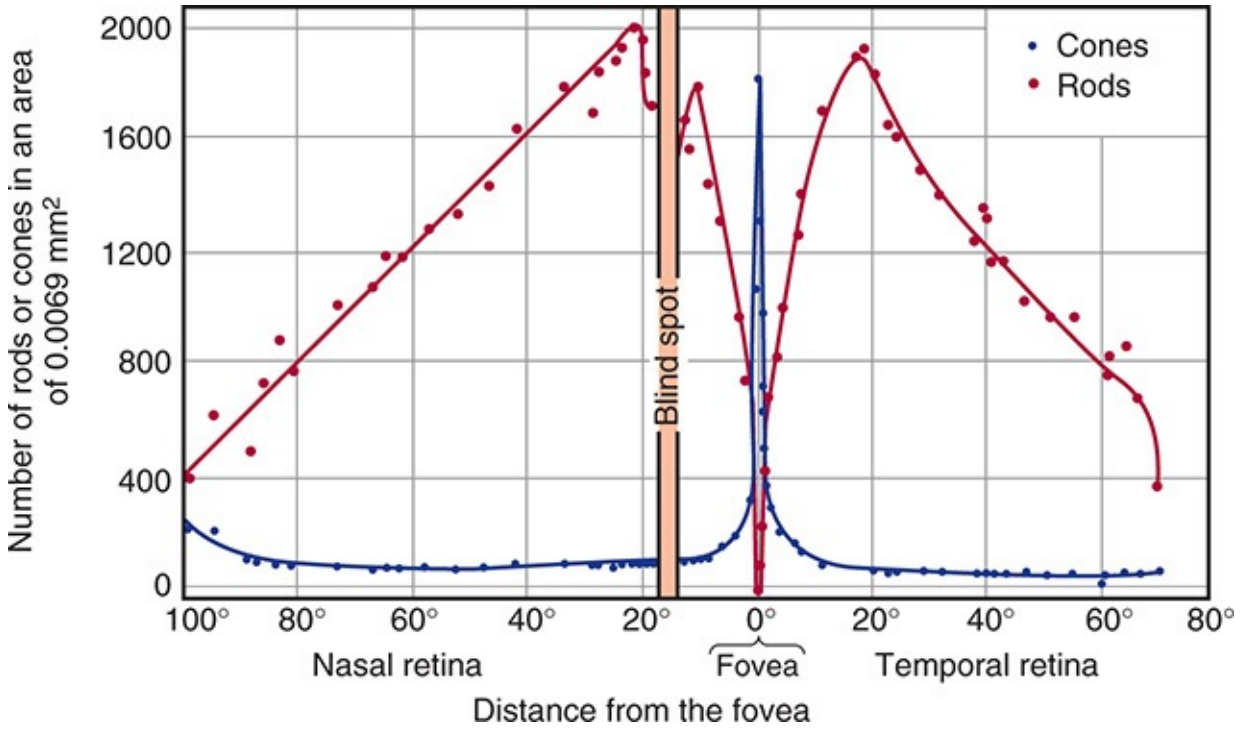


FIGURE 10–7 Rod and cone density along the horizontal meridian through the human retina. A plot of the relative acuity of vision in the various parts of the light-adapted eye would parallel the cone density curve; a similar plot of relative acuity of the dark-adapted eye would parallel the rod density curve.

One of the most important characteristics of the visual system is its ability to function over a wide range of light intensity. When one goes from near darkness to bright sunlight, light intensity increases by 10 log units or a factor of 10 billion. Adjustments in the diameter of the pupil can reduce the fluctuation in intensity; when the diameter is reduced from 8 to 2 mm, its area decreases by a factor of 16 and light intensity at the retina is reduced by more than 1 log unit.

The two types of photoreceptors contribute to our ability to react to fluctuations in intensity. The rods are extremely sensitive to light and are the receptors for night vision **(scotopic vision).** The scotopic visual apparatus resolves the details and boundaries of objects. The cones have a much higher threshold, but the cone system has a much greater acuity and is the system responsible for vision in bright light **(photopic vision)** and for color vision.

THE PHOTORECEPTOR MECHANISM

The potential changes that initiate action potentials in the retina are generated by

the action of light on photosensitive compounds in the rods and cones. When light is absorbed by these substances, their structure changes, and this triggers a sequence of events that initiates neural activity.

The eye is unique in that the receptor potentials of the photoreceptors and the electrical responses of most of the other neural elements in the retina are local, graded potentials, and it is only in the ganglion cells that all-or-none action potentials are generated. The responses of the rods, cones, and horizontal cells are hyperpolarizing, and the responses of the bipolar cells are either hyperpolarizing or depolarizing. Amacrine cells produce depolarizing potentials and spikes that may act as generator potentials for the propagated spikes produced in the ganglion cells.

The cone receptor potential has a sharp onset and offset, whereas the rod receptor potential has a sharp onset and slow offset. The curves relating the amplitude of receptor potentials to stimulus intensity have similar shapes in rods and cones, but the rods are much more sensitive. Therefore, rod responses are proportional to stimulus intensity at levels of illumination that are below the threshold for cones. On the other hand, cone responses are proportional to stimulus intensity at high levels of illumination when the rod responses are maximal and cannot change. Therefore, cones generate good responses to changes in light intensity above background but do not represent absolute illumination well, whereas rods detect absolute illumination.

IONIC BASIS OF PHOTORECEPTOR POTENTIALS

The cGMP-gated cation channels in the outer segments of the rods and cones are open in the dark, so current flows from the inner to the outer segment and to the synaptic ending of the photoreceptor **(Figure 10–8)**. The Na^+, K^+ ATPase in the inner segment maintains ionic equilibrium. Release of synaptic transmitter (glutamate) is steady in the dark. When light strikes the outer segment, the reactions that are initiated close some of the cGMP-gated cation channels to induce a hyperpolarizing receptor potential. The hyperpolarization reduces glutamate release and generates a signal in the bipolar cells that ultimately leads to action potentials in ganglion cells that are transmitted to the brain.

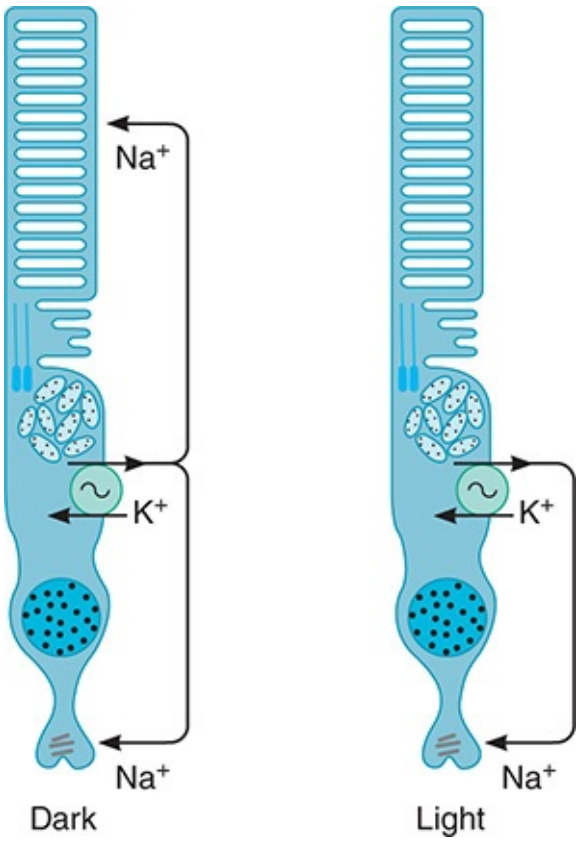


FIGURE 10–8 Effect of light on current flow in visual receptors. In the dark, cGMP-gated cation channels in the outer segment are held open by cGMP. Light leads to increased conversion of cGMP to 5′-GMP, and some of the channels close. This produces hyperpolarization of the synaptic terminal of the photoreceptor.

RHODOPSIN

Rhodopsin (visual purple) is the photosensitive pigment in the rods and is composed of **retinal,** an aldehyde of vitamin A, and the protein **opsin.** Vitamin A is needed for the synthesis of retinal, so a deficiency in this vitamin produces visual abnormalities **(Clinical Box 10–3)**.

Opsin has a molecular weight of 41 kDa and is found in the membranes of the rod disks; it makes up 90% of the total protein in these membranes. Opsin is part of the large family of G-protein-coupled receptors (GPCR). Retinal is parallel to the surface of the membrane **(Figure 10–9)** and is attached to a lysine residue at position 296 in the seventh transmembrane domain.

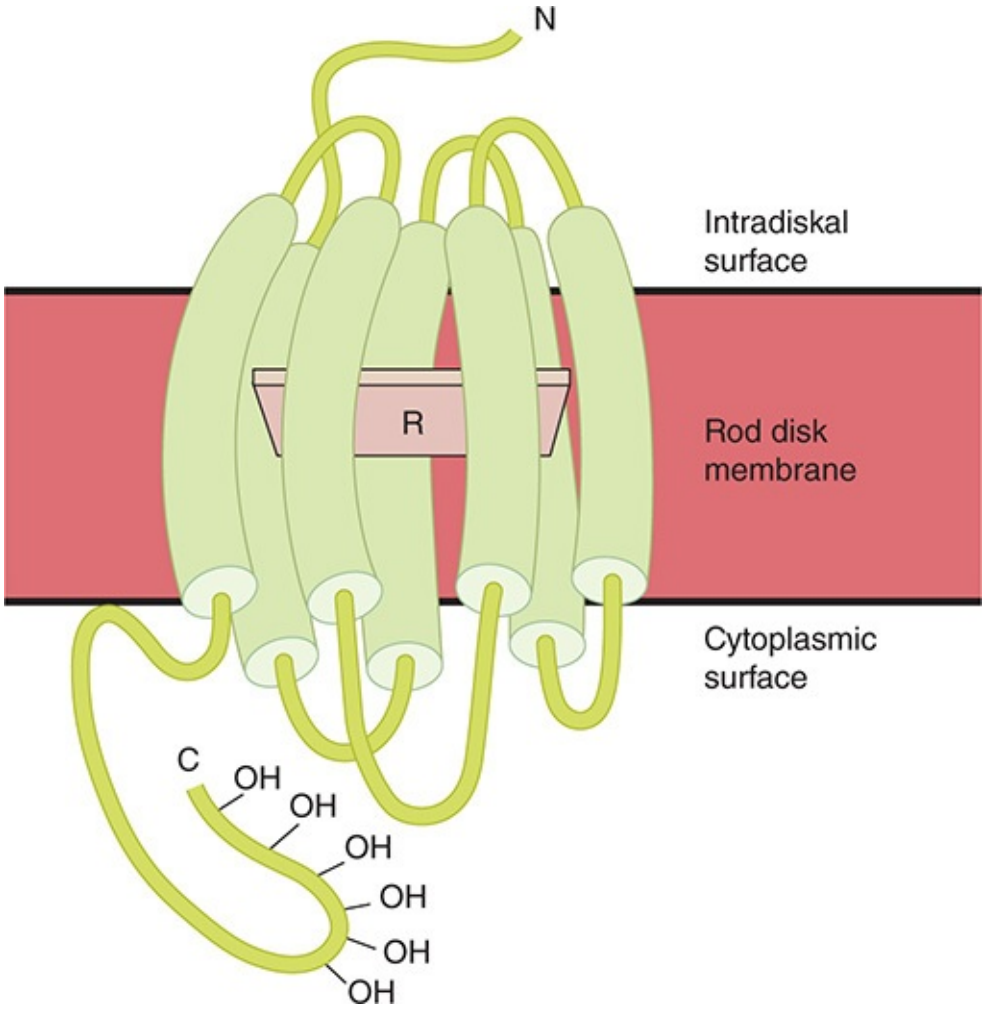


FIGURE 10–9 Diagrammatic representation of the structure of rhodopsin in the rod disk membrane. Rhodopsin is formed when retinal (R) is attached to opsin via a lysine residue at position 296 in the 7th transmembrane domain of the GPCR. GPCR, G-protein-coupled receptor.

CLINICAL BOX 10–3

Vitamin A Deficiency

Vitamin A was the first fat-soluble vitamin identified and is composed of a family of compounds called retinoids. Deficiency is rare in the United States, but it is still a major public health problem in the developing world. Annually, about 80,000 individuals worldwide (mostly children in underdeveloped countries) lose their sight from severe vitamin A deficiency. Vitamin A deficiency is due to inadequate intake of foods high in vitamin A (liver,

kidney, whole eggs, milk, cream, and cheese) or **β-carotene,** a precursor of vitamin A, found in dark green leafy vegetables and yellow or orange fruits and vegetables. One of the earliest visual defects to appear with vitamin A deficiency is night blindness **(nyctalopia).** Vitamin A deficiency also contributes to blindness by causing the eye to become very dry, damaging the cornea **(xerophthalmia)** and retina. Vitamin A first alters rod function, but concomitant cone degeneration occurs as vitamin A deficiency develops. Prolonged deficiency is associated with anatomic changes in the rods and cones followed by degeneration of the neural layers of the retina.

THERAPEUTIC HIGHLIGHTS

Treatment with vitamin A can restore retinal function if given before the receptors are destroyed. Vitamin A–rich foods include liver, chicken, beef, eggs, whole milk, yams, carrots, spinach, kale, and other green vegetables. Other vitamins, especially those of the B complex, are also necessary for the normal functioning of the retina and other neural tissues.

Figure 10–10 summarizes the sequence of events in photoreceptors by which incident light leads to production of a signal in the next succeeding neural unit in the retina. In the dark, the retinal in rhodopsin is in the 11-*cis* configuration. The only action of light is to change the shape of the retinal, converting it to the all-*trans* isomer. This, in turn, alters the configuration of the opsin, and the opsin change activates its associated heterotrimeric G-protein **(transducin)** that has several subunits (Tα, Gβ1, and Gγ1). After 11-*cis* retinal is converted to the all-*trans* configuration, it separates from the opsin in a process called bleaching. This changes the color from the rosy red of rhodopsin to the pale yellow of opsin.

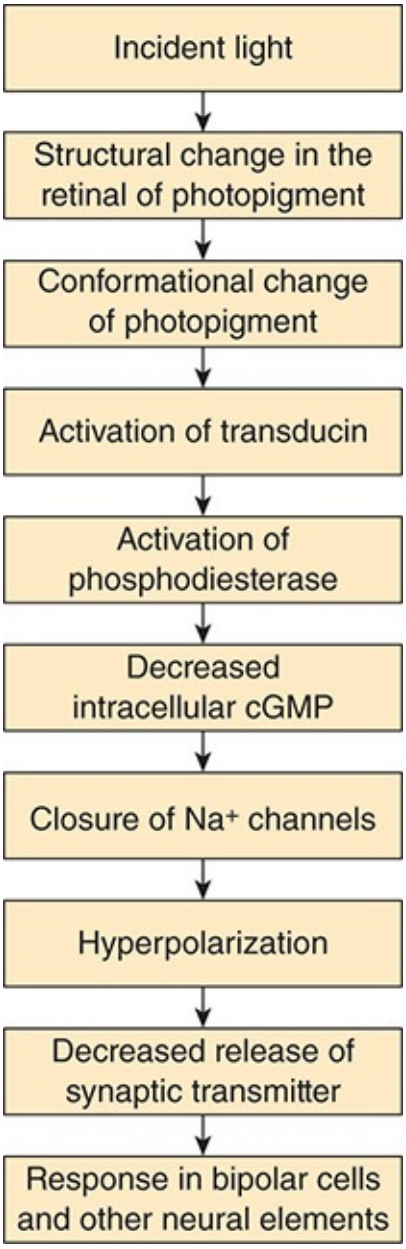


FIGURE 10–10 Sequence of events involved in phototransduction in rods and cones.

Some of the all-*trans* retinal is converted back to the 11-*cis* retinal by retinal isomerase, and then again associates with scotopsin to replenish the rhodopsin supply. Some 11-*cis* retinal is also synthesized from vitaminA. All of these reactions, except the formation of the all-*trans* isomer of retinal, are independent of the light intensity, proceeding equally well in light or darkness. The amount of rhodopsin in the receptors therefore varies inversely with the incident light level.

The G-protein transducin exchanges GDP for GTP, and the α subunit

separates. This subunit remains active until its intrinsic GTPase activity hydrolyzes the GTP. Termination of the activity of transducin is accelerated by its binding of β-arrestin. The α subunit activates cGMP phosphodiesterase to convert cGMP to 5′-GMP.

In darkness, when phosphodiesterase activity is low, cGMP-gated Na^+ channels are maintained in an open state so both Na^+ and Ca^{2+} enter the photoreceptor; the cell is depolarized, and glutamate is released. The light-induced decline in the cytoplasmic cGMP concentration causes some cGMP-gated Na^+ channels to close, reducing the entry of Na^+ and Ca^{2+} and producing the hyperpolarizing potential. This cascade of reactions occurs very rapidly and amplifies the light signal. The amplification helps explain the remarkable sensitivity of rod photoreceptors; these receptors can produce a detectable response to as little as one photon of light.

Calcium ions exert a negative feedback effect on the phototransduction process. In darkness, the relatively high intracellular Ca^{2+} concentration inhibits guanylyl cyclase activity, decreases the activity of the cGMP-gated Na^+ channels, and increases the activity of rhodopsin. The light-induced decrease in Ca^{2+} concentration influences these components of the phototransduction cascade.

PROCESSING OF VISUAL INFORMATION IN THE RETINA

A characteristic of the retinal bipolar and retinal ganglion cells (as well as the lateral geniculate neurons and the neurons in layer 4 of the visual cortex) is that they respond best to a small, circular stimulus and that, within their receptive field, an annulus of light around the center (surround illumination) antagonizes the response to the central spot **(Figure 10–11)**. The center can be excitatory with an inhibitory surround (an **on-center/off-surround cell**) or inhibitory with an excitatory surround (an **off-center/on-surround cell**). The inhibition of the center response by the surround is probably due to inhibitory feedback from one photoreceptor to another mediated via horizontal cells. Thus, activation of nearby photoreceptors by addition of the annulus triggers horizontal cell hyperpolarization, which in turn inhibits the response of the centrally activated photoreceptors. The inhibition of the response to central illumination by an increase in surrounding illumination is an example of **lateral inhibition**—that form of inhibition in which activation of a neural unit is associated with

inhibition of the activity of nearby units. It is a general phenomenon in mammalian sensory systems and helps sharpen the edges of a stimulus and improve discrimination.

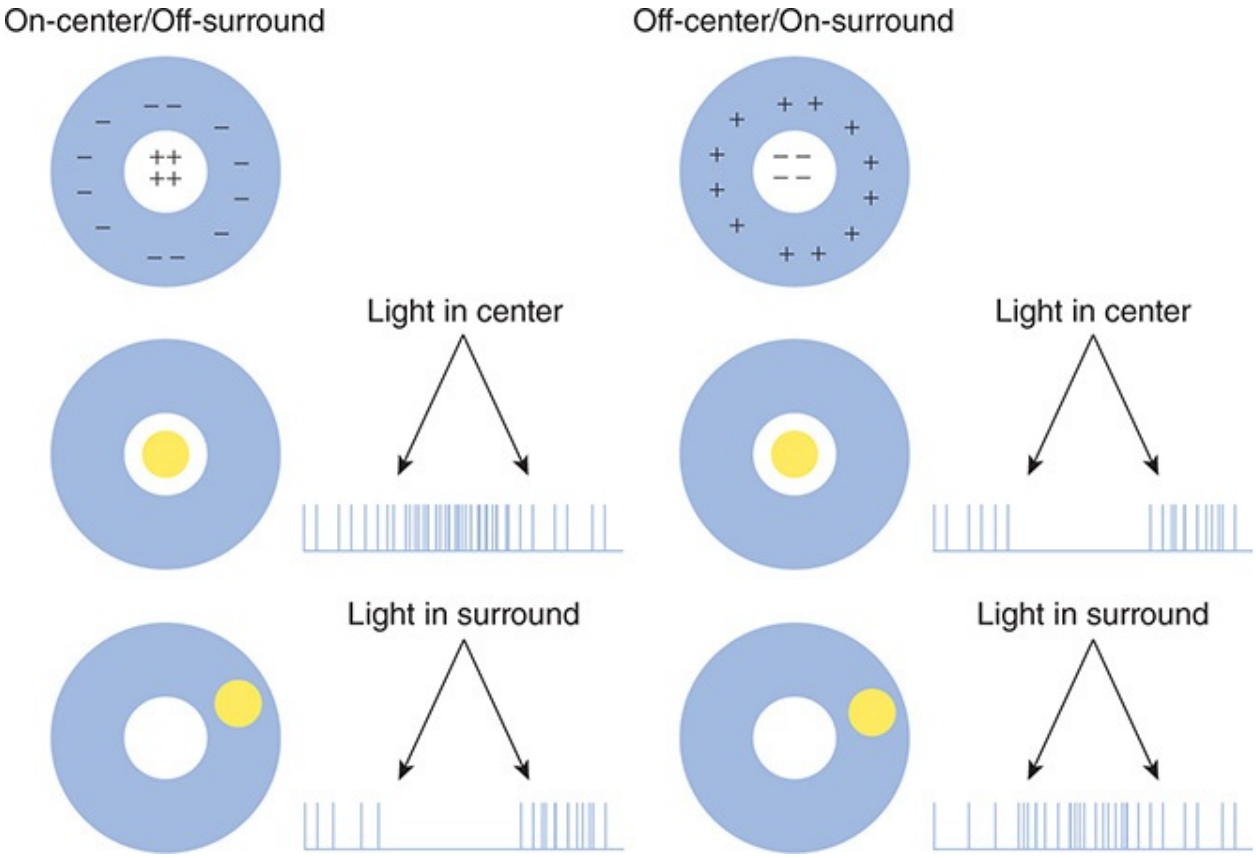


FIGURE 10–11 Responses of two types of retinal ganglion cells to light (yellow circle) focused on a portion of their receptive field. Left side: An *on-center/off-surround* cell responds with an increase in firing rate when the light is placed in the center of the receptive field and with a decrease in firing rate when the light is placed in the surround portion of the receptive field. **Right side:** An *off-center/on-surround* cell responds with a decrease in firing rate when the light is placed in the center and with an increase in firing rate when the light is placed in the surround.

DARK ADAPTATION

If a person spends a considerable length of time in brightly lighted surroundings and then moves to a dimly lighted environment, the retinas slowly become more sensitive to light as the individual becomes “accustomed to the dark.” This

decline in visual threshold is known as **dark adaptation.** It is nearly maximal in about 20 min, although some further decline occurs over longer periods. The time required for dark adaptation depends in part on the time needed to build up the rhodopsin stores that are continuously being broken down in bright light. When one passes suddenly from a dim to a brightly lighted environment, the light seems intensely and even uncomfortably bright until the eyes adapt to the increased illumination and the visual threshold rises. This adaptation occurs over a period of about 5 min and is called **light adaptation,** although it is merely the disappearance of dark adaptation.

The dark adaptation response has two components. The first drop in visual threshold, rapid but small in magnitude, is due to dark adaptation of the cones because when only the foveal, rod-free portion of the retina is tested, the decline proceeds no further. In the peripheral portions of the retina, a further drop occurs because of adaptation of the rods. The total change in threshold between the light-adapted and the fully dark-adapted eye is very great.

Radiologists, aircraft pilots, and others who need maximal visual sensitivity in dim light can avoid having to wait 20 min in the dark to become dark-adapted if they wear red goggles when in bright light. Light wavelengths in the red end of the spectrum stimulate the rods to only a slight degree while permitting the cones to function reasonably well. Therefore, a person wearing red glasses can see in bright light during the time it takes for the rods to become dark-adapted.

VISUAL ACUITY

The optic nerve leaves the eye at a point 3 mm medial to and slightly above the posterior pole of the globe. This region is visible through the ophthalmoscope as the **optic disk** **(Figure 10–12)**. Since there are no visual receptors over the disk, this area of the retina does not respond to light and is known as the **blind spot.** Near the posterior pole of the eye, there is a yellowish pigmented spot called the **macula.** The **fovea** is in the center of the macula; it is a thinned-out, rod-free portion of the retina in which cones are densely packed. Each cone synapses on a single bipolar cell, which, in turn, synapses on a single ganglion cell, providing a direct pathway to the brain. There are very few overlying cells and no blood vessels; thus, the fovea is the point where **visual acuity** is greatest. When attention is attracted to or fixed on an object, the eyes are normally moved so that light rays coming from the object fall on the fovea. **Age-related macular degeneration (AMD)** is a disease in which sharp, central vision is gradually destroyed **(Clinical Box 10–4)**.

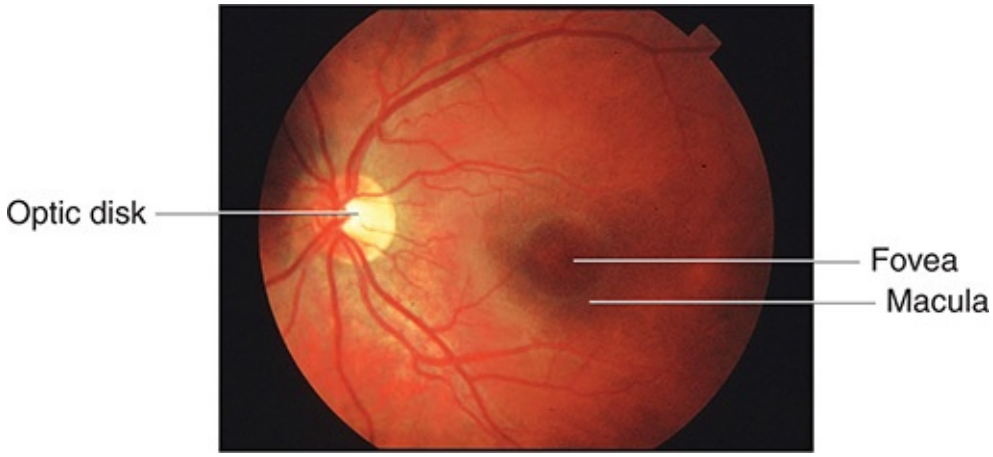


FIGURE 10–12 The fundus of the eye in a healthy human as seen through the ophthalmoscope. The fundus of the eye refers to the interior surface of the eye, opposite the lens, and includes the retina, optic disk, macula and fovea, and posterior pole. Optic nerve fibers leave the eyeball at the optic disk to form the optic nerve. The arteries, arterioles, and veins in the superficial layers of the retina near its vitreous surface can be seen through the ophthalmoscope. (Used with permission of Dr AJ Weber, Michigan State University.)

An ophthalmoscope is used to view the **fundus** which is the interior surface of the eye, opposite to the lens; it includes the retina, optic disk, macula and fovea, and posterior pole (Figure 10–12). The arteries, arterioles, and veins in the superficial layers of the retina near its vitreous surface can be examined. The retinal vessels supply the bipolar and ganglion cells, but the receptors are nourished primarily by the capillary plexus in the choroid. This is why **retinal detachment** is so damaging to the receptor cells. Because this is the one place in the body where arterioles are readily visible, an ophthalmoscope examination is of value in the diagnosis and evaluation of diabetes mellitus, hypertension, and other diseases that affect blood vessels **(Clinical Box 10–5)**.

Glaucoma (Clinical Box 10–1) causes changes in the appearance of the fundus of the eye as seen through an ophthalmoscope **(Figure 10–13)**. The photograph on the left is from a primate with a normal eye and shows an optic disk with a uniform “pinkish” color. The blood vessels appear relatively flat as they cross the margin of the disk. This is because there are a normal number of ganglion cell fibers, and the blood vessels have intact support tissue around them. The photograph on the right is from a primate with glaucoma that was experimentally induced by causing a chronic elevation in intraocular pressure. As is characteristic of glaucomatous optic neuropathy, the disk is pale, especially in the center. The retinal blood vessels are distorted, especially at the disk

margin, due to a lack of support tissue; and there is increased “cupping” of the disk.

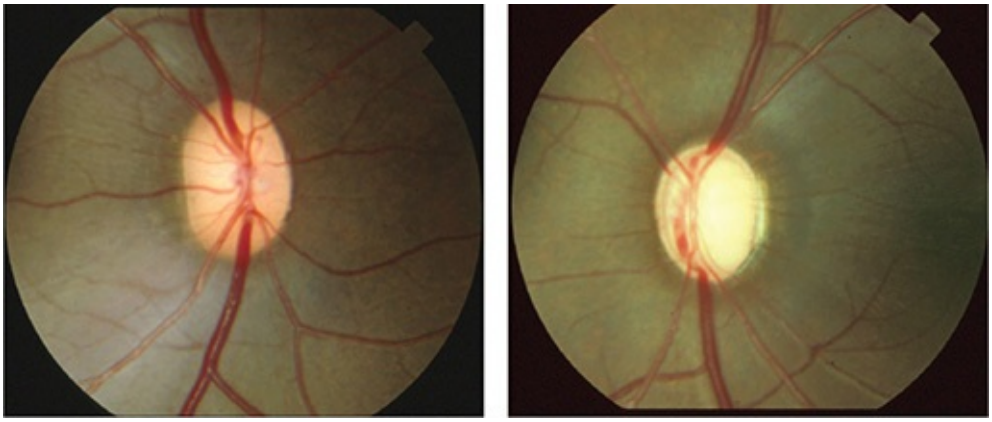


FIGURE 10–13 The fundus of the eye in a normal primate (left) and in a primate with experimentally induced glaucoma (right) as seen through an ophthalmoscope. Normal: uniform “pinkish” color vessels appear relatively flat crossing the margin of disk due to a normal number of ganglion cell fibers and since they have intact support tissue around them. Glaucomatous: disk is pale, especially in center, vessels are distorted, especially at the disk margin due to lack of support tissue and increased “cupping” of the disk. (Used with permission of Dr AJ Weber, Michigan State University.)

CLINICAL BOX 10–4

Visual Acuity and Age-Related Macular Degeneration

Visual acuity is the degree to which the details and contours of objects are perceived, and it is usually defined in terms of the shortest distance by which two lines can be separated and still be perceived as two lines. Visual acuity is often determined by using the **Snellen letter charts** viewed from a distance of 20 ft (6 m). The results are expressed as a fraction; the numerator is 20 (distance from which letters are read) and the denominator is the greatest distance from the chart at which an individual with normal visual acuity can read the line. Normal visual acuity is 20/20; a subject with 20/15 visual acuity has better than normal vision (not farsightedness); and one with 20/100 visual acuity has subnormal vision. Visual acuity is a complex phenomenon and is influenced by many factors, including optical factors (eg, the state of the image-forming mechanisms of the eye), retinal factors (eg, the state of the

cones), and stimulus factors (eg, illumination, brightness of the stimulus, contrast between the stimulus and the background, length of time the subject is exposed to the stimulus). Many drugs can also have adverse side effects on visual acuity. Patients treated with the antiarrhythmic drug **amiodarone** often report corneal changes (**kerotopathy**) including complaints of blurred vision, glare and halos around lights, or light sensitivity. **Aspirin** and other anticoagulants can cause conjunctival or retinal hemorrhaging that can impair vision. **Maculopathy** is a risk factor for those treated with **tamoxifen** for breast cancer. Antipsychotic therapies such as **thioridazine** can cause pigmentary changes that can affect visual acuity, color vision, and dark adaptation.

There are over 20 million individuals in the United States and Europe with age-related macular degeneration or **AMD**, a deterioration of central visual acuity. Nearly 30% of those aged 75 or older have this disorder, and it is the most common cause of visual loss in those over 50 years of age. Women are at greater risk than men for AMD; whites have a greater risk than blacks. There are two types: wet and dry. Wet AMD occurs when fragile blood vessels begin to form under the macula. Blood and fluid leak from these vessels and rapidly damage the macula. Vascular endothelial growth factors (VEGFs) may contribute to the growth of these blood vessels. Dry AMD occurs when the cones in the macula slowly break down, causing a gradual loss of central vision.

THERAPEUTIC HIGHLIGHTS

The US Food and Drug Administration has approved three VEGF inhibitors for the treatment of wet AMD: **ranibizumab** (Lucentis), **bevacizumab** (Avastin), and **aflibercept** (Eylea). **Photodynamic therapy** uses injection of **verteporfin** into the vein in an arm; this drug is activated by a laser light that produces a chemical reaction to destroy abnormal blood vessels. **Laser surgery** can be done to repair damaged blood vessels if they are at a distance from the fovea. However, new vessels may form after the surgery, and vision loss may progress.

CLINICAL BOX 10–5

Eyes Are a Window to Your Health

A thorough exam of the eye (Table 10–1) can reveal a lot about your overall health. In addition to determining your visual acuity and evidence for macular degeneration, a detached retina, or a cataract; an ophthalmologist is able to identify signs of diseases in other organs. **Exophthalmos** (bulging of the eye out of the orbit) can be indicative of **Graves disease** due to an overactive thyroid or an **orbital tumor**. The presence of a gray ring around the cornea **(arcus senilis)** often is linked to high cholesterol and hyperlipidemia. **Ptosis** (a droopy lid) can be evidence of **myasthenia gravis** (neuromuscular disease); a combination of ptosis, **aniscoria** (uneven pupil size), and **facial anhidrosis** (lack of sweating) is indicative of **Horner syndrome** (interruption of the sympathetic innervation of the eye). See Chapter 13 for more information on Horner's syndrome. **Hypertension** can be detected during a retinal exam because high blood pressure causes twisting and kinking of tiny retinal blood vessels. A **fundoscopic exam** can detect **diabetic retinopathy**, a leading cause of blindness, that is associated with **macular edema, microaneurysms** (microscopic bulges protruding from the arterial wall), retinal vessel dilation, **neovascularization** (development of new blood vessels), and **cotton wool spots** (fluffy white patches on the retina due to nerve damage). See Chapter 24 for more information on the adverse consequences of diabetes.

COLOR VISION

Colors have three attributes: **hue, intensity,** and **saturation** (degree of freedom from dilution with white). For any color there is a **complementary color** that, when properly mixed with it, produces a sensation of white. Black is the sensation produced by the absence of light, but it is probably a positive sensation because the blind eye does not "see black"; rather, it "sees nothing."

Another observation of basic importance is the demonstration that the sensation of white, any spectral color, and even the extraspectral color, purple, can be produced by mixing various proportions of red light (wavelength 723–647 nm), green light (575–492 nm), and blue light (492–450 nm). Red, green, and blue are therefore called the **primary colors.** A third important point is that the color perceived depends in part on the color of other objects in the visual field. Thus, for example, a red object is seen as red if the field is illuminated with green or blue light, but as pale pink or white if the field is illuminated with red light. **Clinical Box 10–6** describes color blindness.

RETINAL MECHANISMS

The **Young–Helmholtz theory** of color vision is based on the existence of three kinds of cones, each containing a different photopigment that is maximally sensitive to one of the three primary colors. The sensation of any given color is determined by the relative frequency of the impulses from each of these cone systems **(Figure 10–14)**. One pigment (the blue-sensitive, or short-wave, pigment) absorbs light maximally in the blue-violet portion of the spectrum. Another (the green-sensitive, or middle-wave, pigment) absorbs maximally in the green portion. The third (the red-sensitive, or long-wave, pigment) absorbs maximally in the yellow portion. Blue, green, and red are the primary colors, but the cones with their maximal sensitivity in the yellow portion of the spectrum are sensitive enough in the red portion to respond to red light at a lower threshold than green.

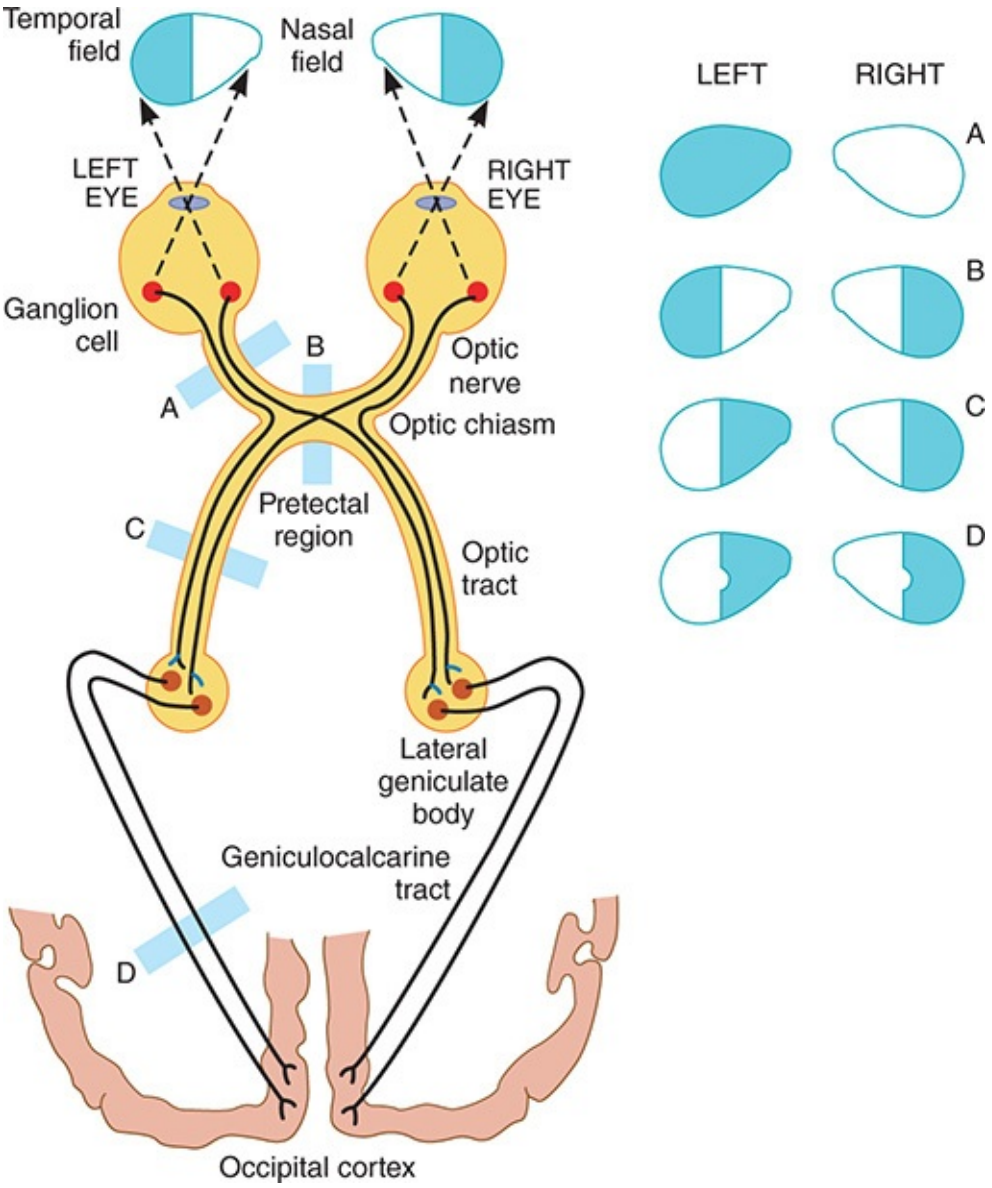


FIGURE 10–14 Visual pathways. Transection of the pathways at the locations indicated by the letters causes the visual field defects shown in the diagrams on the right. The fibers from the nasal half of each retina decussate in the optic chiasm, so that the fibers in the optic tracts are those from the temporal half of one retina and the nasal half of the other. A lesion that interrupts one optic nerve causes blindness in that eye (A). A lesion in one optic tract causes blindness in half of the visual field (C) and is called homonymous (same side of both visual fields) hemianopia (half-blindness). Lesions affecting the optic chiasm destroy fibers from both nasal hemiretinas and produce a heteronymous (opposite sides of the visual fields) hemianopia (B). Occipital lesions may spare the fibers from the macula (as in D) because of the separation in the brain of these fibers from

the others subserving vision.

CLINICAL BOX 10–6

Color Blindness

The most common test for **color blindness** uses the **Ishihara charts,** which are plates containing figures made up of colored spots on a background of similarly shaped colored spots. The figures are intentionally made up of colors that are liable to look the same as the background to an individual who is color blind. Some color-blind individuals are unable to distinguish certain colors, whereas others have only a color weakness. The prefixes “prot-,” “deuter-,” and “trit-” refer to defects of the red, green, and blue cone systems, respectively. Individuals with normal color vision are called **trichromats. Dichromats** are individuals with only two cone systems; they may have protanopia, deuteranopia, or tritanopia. **Monochromats** have only one cone system. Dichromats can match their color spectrum by mixing only two primary colors; monochromats match their color spectrum by varying the intensity of only one. Abnormal color vision is an inherited abnormality in 8% of white males and 0.4% of white females. Tritanopia is rare and shows no sexual selectivity. However, about 2% of color-blind males are dichromats who have protanopia or deuteranopia, and about 6% are anomalous trichromats in whom the red- sensitive or the green-sensitive pigment is shifted in its spectral sensitivity. These abnormalities are inherited as recessive and X-linked characteristics. Color blindness occurs in males if the X chromosome has the abnormal gene. Females show a defect only when both X chromosomes contain the abnormal gene. However, female children of a man with X-linked color blindness are carriers of color blindness and pass the defect on to half of their sons. Therefore, X-linked color blindness skips generations and appears in males of every second generation. Color blindness can also occur in individuals with lesions of area V8 of the visual cortex since this region is uniquely concerned with color vision in humans. This deficit is called **achromatopsia.** Transient blue-green color weakness occurs as a side effect in individuals taking sildenafil (Viagra) for the treatment of erectile dysfunction because the drug inhibits the retinal as well as the penile form of phosphodiesterase.

The gene for human rhodopsin is on chromosome 3, and the gene for the blue-sensitive S cone pigment is on chromosome 7. The other two cone pigments are encoded by genes arranged in tandem on the q arm of the X chromosome. The green-sensitive M and red-sensitive L pigments are very similar in structure. Their opsins show 96% homology of amino acid sequences, but each of these pigments has only 43% homology with the opsin of blue-sensitive pigment, and all three have about 41% homology with rhodopsin.

RESPONSES IN THE VISUAL PATHWAYS & CORTEX

NEURAL PATHWAYS

The axons of the ganglion cells pass caudally in the optic nerve and **optic tract** to end in the **lateral geniculate body** in the thalamus **(Figure 10–15)**. The fibers from each nasal hemiretina decussate in the **optic chiasm.** In the geniculate body, the fibers from the nasal half of one retina and the temporal half of the other synapse on the cells whose axons form the **geniculocalcarine tract.** This tract passes to the occipital lobe of the cerebral cortex. The effects of lesions in these pathways on visual function are discussed in the next section.

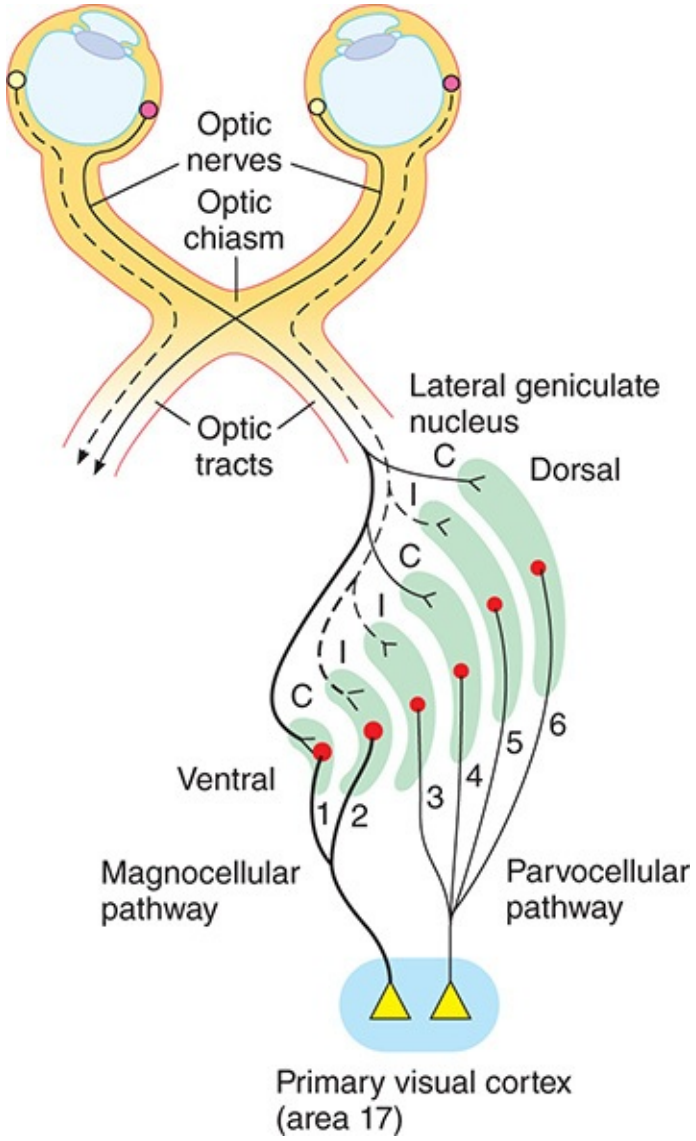


FIGURE 10–15 Ganglion cell projections from the right hemiretina of each eye to the right lateral geniculate body and from this nucleus to the right primary visual cortex. Note the six layers of the geniculate body. P ganglion cells project to layers 3–6, and M ganglion cells project to layers 1 and 2. The ipsilateral (I) and contralateral (C) eyes project to alternate layers. Not shown are the interlaminar area cells, which project via a separate component of the P pathway to blobs in the visual cortex. (Modified with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

The axons of retinal ganglion cells project a detailed spatial representation of the retina on the lateral geniculate body. Each geniculate body contains six well-

defined layers **(Figure 10–16)**. Layers 3–6 have small cells and are called parvocellular, whereas layers 1 and 2 have large cells and are called magnocellular. On each side, layers 1, 4, and 6 receive input from the contralateral eye, whereas layers 2, 3, and 5 receive input from the ipsilateral eye. In each layer, there is a precise point-for-point representation of the retina, and all six layers are in register so that along a line perpendicular to the layers, the receptive fields of the cells in each layer are almost identical. Only 10–20% of the input to the **lateral geniculate nucleus** (LGN) comes from the retina; inputs also occur from the visual cortex and other brain regions. The feedback pathway from the visual cortex is involved in visual processing related to the perception of orientation and motion.

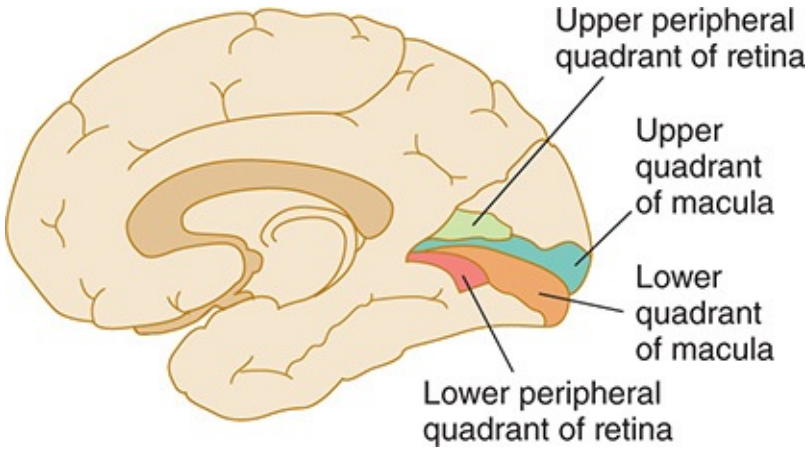


FIGURE 10–16 Medial view of the human right cerebral hemisphere showing projection of the retina on the primary visual cortex in the occipital cortex around the calcarine fissure. The geniculocalcarine fibers from the medial half of the lateral geniculate terminate on the superior lip of the calcarine fissure, and those from the lateral half terminate on the inferior lip. Also, the fibers from the lateral geniculate body that relay macular vision separate from those that relay peripheral vision and end more posteriorly on the lips of the calcarine fissure.

There are several types of retinal ganglion cells. These include large ganglion cells (magno, or M cells), which add responses from different kinds of cones and are concerned with movement and stereopsis. Another type is the small ganglion cells (parvo, or P cells), which subtract input from one type of cone from input from another and are concerned with color, texture, and shape. The M ganglion cells project to the magnocellular portion of the lateral geniculate, whereas the P ganglion cells project to the parvocellular portion. From the LGN, a

magnocellular pathway and a parvocellular pathway project to the visual cortex. The magnocellular pathway, from layers 1 and 2, carries signals for detection of movement, depth, and flicker. The parvocellular pathway, from layers 3–6, carries signals for color vision, texture, shape, and fine detail. The small-field bistratified ganglion cells may be involved in color vision and carry the short (blue) wavelength information to the intralaminar zones of the LGN.

EFFECT OF LESIONS IN THE OPTIC PATHWAYS

Lesions along the visual pathways can be localized with a high degree of accuracy by the effects they produce in the visual fields. The fibers from the nasal half of each retina decussate in the optic chiasm, so that the fibers in the optic tracts are those from the temporal half of one retina and the nasal half of the other. In other words, each optic tract subserves half of the field of vision. Therefore, a lesion that interrupts one optic nerve causes blindness in that eye, but a lesion in one optic tract causes blindness in half of the visual field (Figure 10–15). This defect is classified as a **homonymous** (same side of both visual fields) **hemianopia** (half-blindness).

Lesions affecting the optic chiasm, such as pituitary tumors expanding out of the sella turcica, cause destruction of the fibers from both nasal hemiretinas and produce a **heteronymous** (opposite sides of the visual fields) **hemianopia.** Because the fibers from the macula are located posteriorly in the optic chiasm, hemianopic scotomas developed before vision in the two hemiretinas are completely lost. Selective visual field defects are further classified as bitemporal, binasal, and right or left.

The optic nerve fibers from the upper retinal quadrants subserving vision in the lower half of the visual field terminate in the medial half of the lateral geniculate body, and the fibers from the lower retinal quadrants terminate in the lateral half. The geniculocalcarine fibers from the medial half of the lateral geniculate body terminate on the superior lip of the calcarine fissure, and those from the lateral half terminate on the inferior lip. Furthermore, the fibers from the lateral geniculate body that subserve macular vision separate from those that subserve peripheral vision and end more posteriorly on the lips of the calcarine fissure **(Figure 10–17)**. Because of this anatomic arrangement, occipital lobe lesions may produce discrete quadrantic visual field defects (upper and lower quadrants of each half visual field).

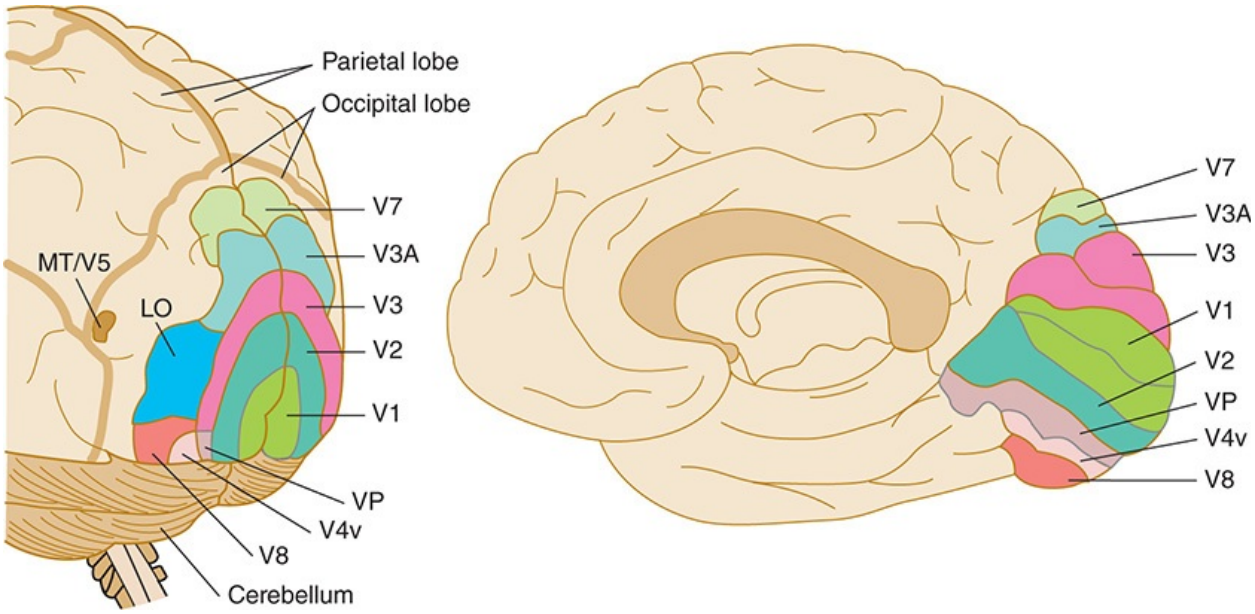


FIGURE 10–17 Some of the main areas to which the primary visual cortex (V1) projects in the human brain. Lateral and medial views. LO, lateral occipital; MT, medial temporal; VP, ventral parietal. See also Table 10–2. (Modified with permission from Logothetis N: Vision: A window on consciousness. Sci Am 1999 Nov;281(5):69–75.)

Macular sparing or loss of peripheral vision with intact macular vision is also common with occipital lesions (Figure 10–15) because the macular representation is separate from that of the peripheral fields and very large relative to that of the peripheral fields. Therefore, occipital lesions must extend considerable distances to destroy both macular and peripheral vision. The fibers to the pretectal region that subserves the pupillary reflex produced by shining a light into the eye leave the optic tracts near the geniculate bodies. Therefore, blindness with preservation of the pupillary light reflex is usually due to bilateral lesions caudal to the optic tract.

PRIMARY VISUAL CORTEX

The primary visual receiving area (**primary visual cortex;** also known as V1) is located on the sides of the calcarine fissure (Figure 10–17). Just as the ganglion cell axons project a detailed spatial representation of the retina on the lateral geniculate body, the lateral geniculate body projects a similar point-for-point representation on the primary visual cortex. In the visual cortex, many nerve cells are associated with each incoming fiber. Like the rest of the neocortex, the

visual cortex has six layers. The axons from the LGN that form the magnocellular pathway end in layer 4, specifically in its deepest part, layer 4C. Many of the axons that form the parvocellular pathway also end in layer 4C. However, the axons from the interlaminar region end in layers 2 and 3.

Layers 2 and 3 of the cortex contain clusters of cells about 0.2 mm in diameter that, unlike the neighboring cells, contain a high concentration of the mitochondrial enzyme cytochrome oxidase. The clusters have been named **blobs.** They are arranged in a mosaic in the visual cortex and are concerned with color vision. However, the parvocellular pathway also carries color opponent data to the deep part of layer 4.

Like the ganglion cells, the lateral geniculate neurons and the neurons in layer 4 of the visual cortex respond to stimuli in their receptive fields with on-centers and inhibitory surrounds or off-centers and excitatory surrounds. A bar of light covering the center is an effective stimulus for them because it stimulates the entire center and relatively little of the surround. However, the bar has no preferred orientation and, as a stimulus, is equally effective at any angle.

The responses of the neurons in other layers of the visual cortex are strikingly different. **Simple cells** respond to bars of light, lines, or edges, but only when they have a specific orientation. When a bar of light is rotated as little as 10° from the preferred orientation, the firing rate of the simple cell is usually decreased; and if the stimulus is rotated much more, the response disappears. **Complex cells** also have a preferred orientation of a linear stimulus but are less dependent on the location of a stimulus in the visual field than the simple cells and the cells in layer 4. They often respond maximally when a linear stimulus is moved laterally without a change in its orientation.

The visual cortex, like the somatosensory cortex, is arranged in vertical columns that are concerned with orientation **(orientation columns).** Each is about 1 mm in diameter. However, the orientation preferences of neighboring columns differ in a systematic way; as one moves from column to column across the cortex, sequential changes occur in orientation preference of 5–10°. Thus, it seems likely that for each ganglion cell receptive field in the visual field, there is a collection of columns in a small area of visual cortex, representing the possible preferred orientations at small intervals throughout the full 360°. The simple and complex cells are called **feature detectors** because they respond to and analyze certain features of the stimulus.

Another feature of the visual cortex is the presence of **ocular dominance columns.** The geniculate cells and the cells in layer 4 receive input from only one eye, and the layer 4 cells alternate with cells receiving input from the other

eye. If a large amount of a radioactive amino acid is injected into one eye, the amino acid is incorporated into protein and transported by axoplasmic flow to the ganglion cell terminals, across the geniculate synapses, and along the geniculocalcarine fibers to the visual cortex. In layer 4, labeled endings from the injected eye alternate with unlabeled endings from the uninjected eye. The result, when viewed from above, is a vivid pattern of stripes that covers much of the visual cortex and is separate from and independent of the grid of orientation columns.

About half of the simple and complex cells receive an input from both eyes. The inputs are identical or nearly so in terms of the portion of the visual field involved and the preferred orientation. However, they differ in strength, so that between the cells to which the input comes totally from the ipsilateral or the contralateral eye, there is a spectrum of cells influenced to different degrees by both eyes.

OTHER CORTICAL AREAS CONCERNED WITH VISION

The primary visual cortex (V1) projects to many other parts of the occipital lobes and other parts of the brain. These are often identified by number (V2, V3, etc) or by letters (LO, MT, etc). The distribution of some of these in the human brain is shown in **Figure 10–18**, and their putative functions are listed in **Table 10–2**. The visual projections from V1 can be divided roughly into a **dorsal** or **parietal pathway,** concerned primarily with motion, and a **ventral** or **temporal pathway,** concerned with shape and recognition of forms and faces. In addition, connections to the sensory areas are important. For example, in the occipital cortex, visual responses to an object are better if the object is felt at the same time. There are many other relevant connections to other systems.

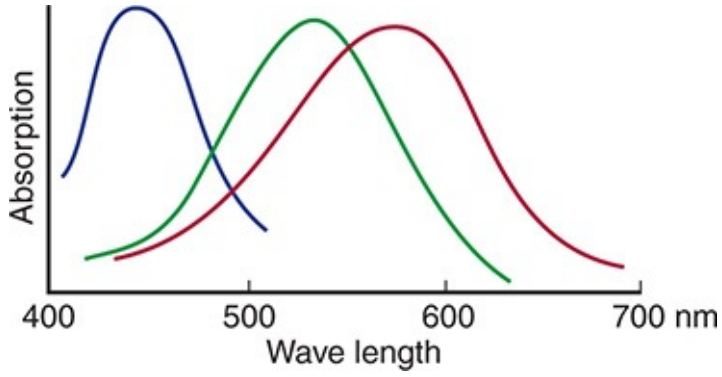


FIGURE 10–18 Absorption spectra of the three cone pigments in the human retina. The S pigment that peaks at 440 nm senses blue, and the M pigment that peaks at 535 nm senses green. The remaining L pigment peaks in the yellow portion of the spectrum, at 565 nm, but its spectrum extends far enough into the long wavelengths to sense red.

TABLE 10–2 Functions of visual projection areas in the human brain.

V1	Primary visual cortex; receives input from lateral geniculate nucleus, begins processing in terms of orientation, edges, etc
V2, V3, VP	Continued processing, larger visual fields
V3A	Motion
V4	Detection of shape, color, and texture of an object
MT/V5	Processes direction of movement
V6	May detect direction and speed of bodily movement
LO	Recognition of shape
V7	Representation of upper and lower quadrants of the visual field
V8	Color vision

LO, lateral occipital; MT, medial temporal; VP, ventral parietal.

NEURAL MECHANISMS OF COLOR VISION

Color is mediated by ganglion cells that subtract or add input from one type of cone to input from another type. Processing in the ganglion cells and the LGN produces impulses that pass along three types of neural pathways that project to V1: a red-green pathway that signals differences between L- and M-cone responses, a blue-yellow pathway that signals differences between S-cone and the sum of L- and M-cone responses, and a luminance pathway that signals the sum of L- and M-cone responses. These pathways project to the blobs and the deep portion of layer 4C of V1. From the blobs and layer 4, color information is projected to V8. However, it is not known how V8 converts color input into the sensation of color.

EYE MOVMENTS

The eye is moved within the orbit by six ocular muscles that are innervated by the oculomotor, trochlear, and abducens nerves. **Figure 10–19** shows the movements produced by the six pairs of muscles. Because the oblique muscles pull medially, their actions vary with the position of the eye. When the eye is turned nasally, the inferior oblique elevates it and the superior oblique depresses it. When it is turned laterally, the superior rectus elevates it and the inferior rectus depresses it. Because much of the visual field is binocular, a very high order of coordination of the movements of the two eyes is necessary if visual images are to fall at all times on corresponding points in the two retinas and diplopia is to be avoided.

There are four types of eye movements, each controlled by a different neural system but sharing the same final common path, the motor neurons that supply the external ocular muscles. **Saccades,** sudden jerky movements, occur as the gaze shifts from one object to another. They bring new objects of interest onto the fovea and reduce adaptation in the visual pathway that would occur if gaze were fixed on a single object for long periods. **Smooth pursuit movements** are tracking movements of the eyes as they follow moving objects. **Vestibular movements,** adjustments that occur in response to stimuli initiated in the semicircular canals, maintain visual fixation as the head moves. **Convergence movements** bring the visual axes toward each other as attention is focused on objects near the observer. Saccadic movements, pursuit movements, and vestibular movements depend on an intact visual cortex. Saccades are programmed in the frontal cortex and the superior colliculi and pursuit movements in the cerebellum.

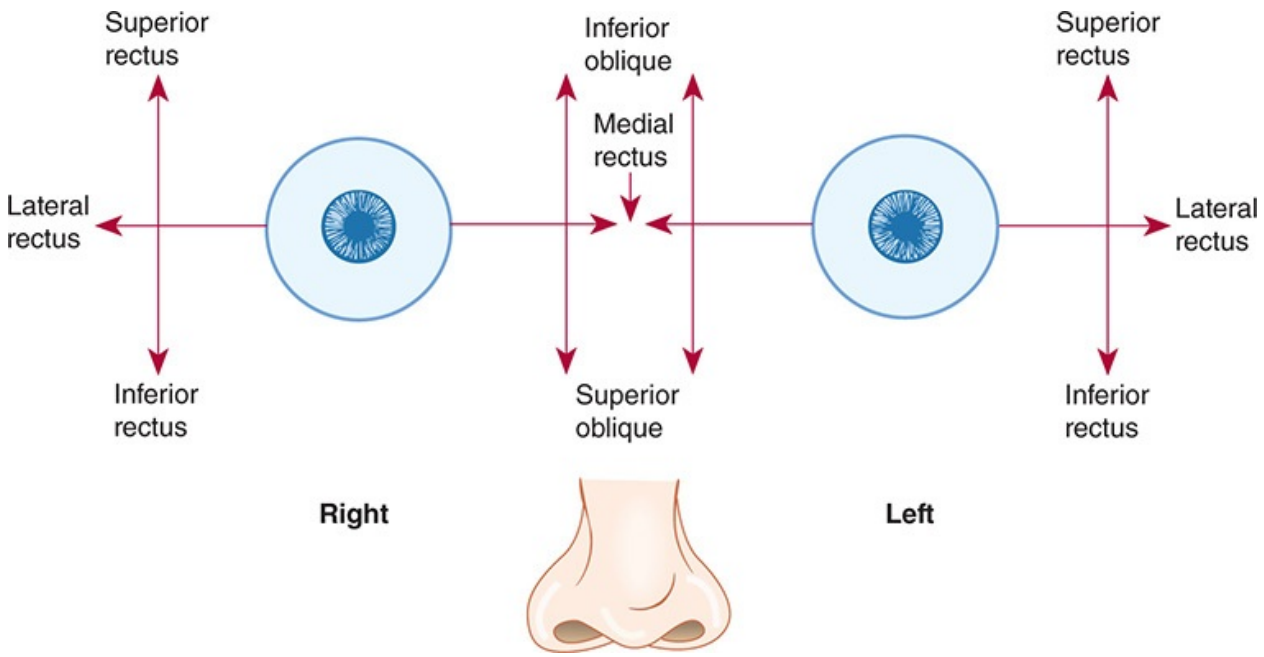


FIGURE 10–19 Diagram of eye muscle actions. The eye is adducted by the medial rectus and abducted by the lateral rectus. The adducted eye is elevated by the inferior oblique and depressed by the superior oblique; the abducted eye is elevated by the superior rectus and depressed by the inferior rectus. (Reproduced with permission from Waxman SG: *Clinical Neuroanatomy*, 26th ed. New York, NY: McGraw-Hill; 2010.)

CHAPTER SUMMARY

- The major parts of the eye are the sclera (protective covering), cornea (transfer light rays), choroid (nourishment), retina (photoreceptor cells), lens, and iris.
- The bending of light rays (refraction) allows one to focus an accurate image onto the retina. Light is refracted at the anterior surface of the cornea and at the anterior and posterior surfaces of the lens.
- To bring diverging rays from close objects to a focus on the retina, the curvature of the lens is increased, a process called accommodation. The pupillary light reflex regulates the amount of light that enters the eye and affects the quality of the retinal image (a smaller pupil diameter gives a greater depth of focus).
- In hyperopia (farsightedness), the eyeball is too short and light rays come to a focus behind the retina. In myopia (nearsightedness), the anteroposterior

diameter of the eyeball is too long. Astigmatism is a common condition in which the curvature of the cornea is not uniform. Presbyopia is the loss of accommodation for near vision.

- The retina is organized into several layers: the outer nuclear layer contains the photoreceptors (rods and cones); the inner nuclear layer contains bipolar cells, horizontal cells, and amacrine cells; and the ganglion cell layer contains the only output neuron of the retina.
- Rhodopsin is the photosensitive pigment in the rods and is composed of retinal and the protein opsin which is a GPCR. Exposure to light causes this sequence of events: structural change in retinal, a conformational change in opsin, activation of transducin, activation of phosphodiesterase, decreased cGMP, closure of cGMP-gated cation channels, hyperpolarization, decrease in neurotransmitter release, and responses in neural elements of the retina.
- In response to light, horizontal cells are hyperpolarized; bipolar cells are either hyperpolarized or depolarized; and amacrine cells are depolarized and develop spikes that may act as generator potentials for the propagated spikes produced in the ganglion cells.
- The decline in visual threshold after spending long periods of time in a dimly lit room is called dark adaptation. The fovea in the center of the retina is the point where visual acuity is greatest. Age-related macular degeneration is a disease in which sharp, central visual acuity is gradually destroyed.
- The visual pathway is from the rods and cones to bipolar cells to ganglion cells then via the optic tract to the thalamic lateral geniculate body to the occipital lobe of the cerebral cortex. The fibers from each nasal hemiretina decussate in the optic chiasm; the fibers from the nasal half of one retina and the temporal half of the other synapse on the cells whose axons form the geniculocalcarine tract.
- Neurons in layer 4 of the visual cortex respond to stimuli in their receptive fields with on-centers and inhibitory surrounds or off-centers and excitatory surrounds. Simple cells respond to bars of light, lines, or edges, but only when they have a particular orientation. Complex cells also have a preferred orientation of a linear stimulus but are less dependent on the location of a stimulus in the visual field. Projections from area V1 can be divided into a dorsal or parietal pathway (concerned primarily with motion) and a ventral or temporal pathway (concerned with shape and recognition of forms and faces).
- The Young–Helmholtz theory of color vision is based on the existence of

three kinds of cones, each containing a different photopigment that is maximally sensitive to one of the three primary colors, with the sensation of any given color being determined by the relative frequency of the impulses from each of these cone systems.

- The fibers from the nasal half of each retina decussate in the optic chiasm, so that the fibers in the optic tracts are those from the temporal half of one retina and the nasal half of the other. A lesion of one optic nerve causes blindness in that eye; a lesion in one optic tract causes blindness in half of the visual field (homonymous hemianopia). A lesion of the optic chiasm destroys fibers from both nasal hemiretinas (heteronymous hemianopia). Occipital lesions may spare the fibers from the macular region.
- Eye movement is controlled by six ocular muscles innervated by the oculomotor, trochlear, and abducens nerves. The inferior oblique muscle turns the eye upward and outward; the superior oblique turns it downward and outward. The superior rectus muscle turns the eye upward and inward; the inferior rectus turns it downward and inward. The medial rectus muscle turns the eye inward; the lateral rectus turns it outward.
- Saccades (sudden jerky movements) occur as the gaze shifts from one object to another, and they reduce adaptation in the visual pathway that would occur if gaze were fixed on a single object for long periods. Smooth pursuit movements are tracking movements of the eyes as they follow moving objects. Vestibular movements occur in response to stimuli in the semicircular canals to maintain visual fixation as the head moves. Convergence movements bring the visual axes toward each other as attention is focused on objects near the observer.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A visual exam in an 80-year-old man shows he has a reduced ability to see objects in the upper and lower quadrants of the left visual fields of both eyes but some vision remains in the central regions of the visual field. The diagnosis is

   A. central scotoma.

   B. heteronymous hemianopia with macular sparing.

   C. lesion of the optic chiasm.

D. homonymous hemianopia with macular sparing.

E. retinopathy.

2. A 45-year-old woman who had never needed to wear glasses had trouble reading a menu in a dimly-lit restaurant. She then recalled that as of late she needed to have the newspaper closer to her eyes to read it. Her ophthalmologist told her she was experiencing age-related loss of accommodation for near vision (presbyopis) that is due to

A. the inability to increase the tension on the lens ligaments.

B. the inability to increase the curvature of the lens.

C. relaxation of the sphincter muscle of the iris.

D. contraction of the ciliary muscle.

E. increased softness of the lens.

3. A 28-year-old man with severe myopia made an appointment to see his ophthalmologist when he began to notice flashing lights and floaters in his visual field. He was diagnosed with a retinal detachment. The inner nuclear layer of the retina is comprised of

A. the inner segments of the photoreceptors (rods and cones).

B. various types of ganglion cells.

C. bipolar cells, horizontal cells, and amacrine cells.

D. glial cells that generate new rods and cones.

E. cell bodies of the optic nerve.

4. A 65-year-old woman was diagnosed with dry age-related macular degeneration with a foveal-sparing scotoma. The fovea of the eye

A. has the lowest light threshold.

B. is the region of highest visual acuity.

C. contains only red and green cones.

D. contains only rods.

E. is situated over the head of the optic nerve.

5. A 62-year-old man went to his ophthalmologist for his routine eye exam. It included ophthalmoscopy to visualize the interior surface of his eye, opposite to the lens. This portion of the eye is called

A. the optic disk.

B. the macula.

C. the sclera.

D. the conjunctiva.

E. the fundus.

6. Which of the following parts of the eye has the greatest concentration of rods?

A. Ciliary body

B. Iris

C. Optic disk

D. Fovea

E. Parafoveal region

7. The correct sequence of events involved in phototransduction in rods and cones in response to light is:

A. activation of transducin, decreased release of glutamate, structural changes in rhodopsin, closure of cGMP-gated cation channels, and decrease in intracellular cGMP.

B. decreased release of glutamate, activation of transducin, closure of cGMP-gated cation channels, decrease in intracellular cGMP, and structural changes in rhodopsin.

C. structural changes in rhodopsin, decrease in intracellular cGMP, decreased release of glutamate, closure of cGMP-gated cation channels, and activation of transducin.

D. structural changes in rhodopsin, activation of transducin, decrease in intracellular cGMP, closure of cGMP-gated cation channels, and decreased release of glutamate.

E. activation of transducin, structural changes in rhodopsin, closure of cGMP-gated cation channels, decrease in intracellular cGMP, and decreased release of glutamate.

8. A 25-year-old medical student spent a summer volunteering in the sub-Saharan region of Africa. There he noted a high incidence of people reporting difficulty with night vision due to a lack of vitamin A in their diet. Vitamin A is a precursor for the synthesis of

A. rods and cones.

B. retinal.

C. rod transducin.

D. opsin.

E. cone transducin.

9. An 11-year-old boy was having difficulty reading the graphs that his teacher

was showing at the front of classroom. His teacher recommended he be seen by an ophthalmologist. Not only was he asked to look at a Snellen letter chart for visual acuity but he was also asked to identify numbers in an Ishihara chart. He responded that he merely saw a bunch of dots. Abnormal color vision is 20 times more common in males than females because most cases are caused by an abnormal

A. dominant gene on the Y chromosome.

B. recessive gene on the Y chromosome.

C. dominant gene on the X chromosome.

D. recessive gene on the X chromosome.

E. recessive gene on chromosome 22.

10. A 32-year-old man was brought to the emergency department after being found comatose by his wife. The resident in the emergency department assessed his pupillary light reflex as a useful gauge of his brainstem function. He found that when the light was shone into his left eye, neither pupil constricted; but when the light was shone in his right eye, both pupils constricted. The physician determined that damage was within

A. the left optic nerve.

B. the left oculomotor nerve.

C. the right optic nerve.

D. the right oculomotor nerve.

E. the sphincter muscle of the left eye.

11. A tumor was diagnosed near the base of the skull in a 56-year-old woman, impinging on her right optic tract. Which parts of the visual field of each eye are relayed through the right optic tract?

A. The temporal half of left retina and the nasal half of the right retina

B. The nasal half of left retina and the temporal half of the right retina

C. The temporal half of right retina and the nasal half of the left retina

D. The nasal half of right retina and the temporal half of the left retina

E. The temporal and nasal halves of the left retina

12. A 50-year-old man began having difficulty moving his left eye downward and laterally. His primary care physician did testing that revealed he had damage to a cranial nerve that innervated one of the muscles controlling eye movement. Which nerve and muscle allow the eye to move downward and laterally?

A. The oculomotor nerve and the inferior oblique muscle

B. The trochlear nerve and the superior oblique muscle

C. The abducens nerve and the lateral rectus muscle

D. The oculomotor nerve and the superior oblique muscle

E. The trochlear nerve and the inferior rectus muscle

13. A medical student was recording responses of retinal ganglion cells to light focused on a portion of their receptive field. He identified one cell as an off-center/on-surround cell. What might contribute to the reduced firing rate when the light was focused on the center of the receptive field of the ganglion cell?

A. The light caused the release of an inhibitory neurotransmitter released from the terminals of rod cells that synapse on the ganglion cell.

B. The activation of nearby photoreceptors leads to hyperpolarization of a horizontal cell that in turn reduces the activity of the ganglion cell.

C. The activation of nearby photoreceptors triggers an action potential in an amacrine cell that inhibits the ganglion cell.

D. The light caused an action potential in rod and cone receptors within the receptive field and they caused hyperpolarization in the ganglion cell.

CHAPTER 11

Hearing & Equilibrium

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the components and functions of the external, middle, and inner ear.
- Explain the roles of the tympanic membrane, the auditory ossicles (malleus, incus, and stapes), and scala vestibule in sound transmission.
- Describe the way that movements of molecules in the air are converted into impulses generated in hair cells in the cochlea.
- Explain how pitch, loudness, and timbre are coded in the auditory pathways.
- Describe the components of the auditory pathway from the cochlear hair cells to the cerebral cortex.
- Compare the causes of conductive and sensorineural hearing loss and the tests used to distinguish between them.
- Define the following terms: tinnitus, presbycusis, and syndromic and nonsyndromic deafness.
- Explain how cochlear implants and hearing aids function.
- Explain how the receptors in the semicircular canals detect rotational acceleration and how the receptors in the saccule and utricle detect linear acceleration.

- List the major sensory inputs that provide the information that is synthesized in the brain into the sense of position in space.
- Describe the neural mechanisms for vestibular nystagmus and how nystagmus can be used as a diagnostic indicator of the integrity of the vestibular system.
- Describe the cause and clinical signs of the following vestibular disturbances: vertigo, Ménière disease, and motion sickness.

INTRODUCTION

Our ears not only let us detect sounds, but they also help us maintain balance. Receptors for two sensory modalities (hearing and equilibrium) are housed in the ear. The external ear, the middle ear, and the cochlea of the inner ear are involved with hearing. The semicircular canals, the utricle, and the saccule of the inner ear are involved with equilibrium. Both hearing and equilibrium rely on a very specialized type of receptor called a hair cell. There are six groups of hair cells in each inner ear: one in each of the three semicircular canals, one in the utricle, one in the saccule, and one in the cochlea. Receptors in the semicircular canals detect rotational acceleration, those in the utricle detect linear acceleration in the horizontal direction, and the ones in the saccule detect linear acceleration in the vertical direction.

STRUCTURE & FUNCTION OF THE EAR

EXTERNAL & MIDDLE EAR

The external ear is composed of the **auricle** (pinna) that captures sound waves, the **external auditory meatus** (ear canal) through which sound waves travel, and the **tympanic membrane** (eardrum) that moves in and out in response to sound **(Figure 11–1)**. The tympanic membrane marks the beginning of the middle ear.

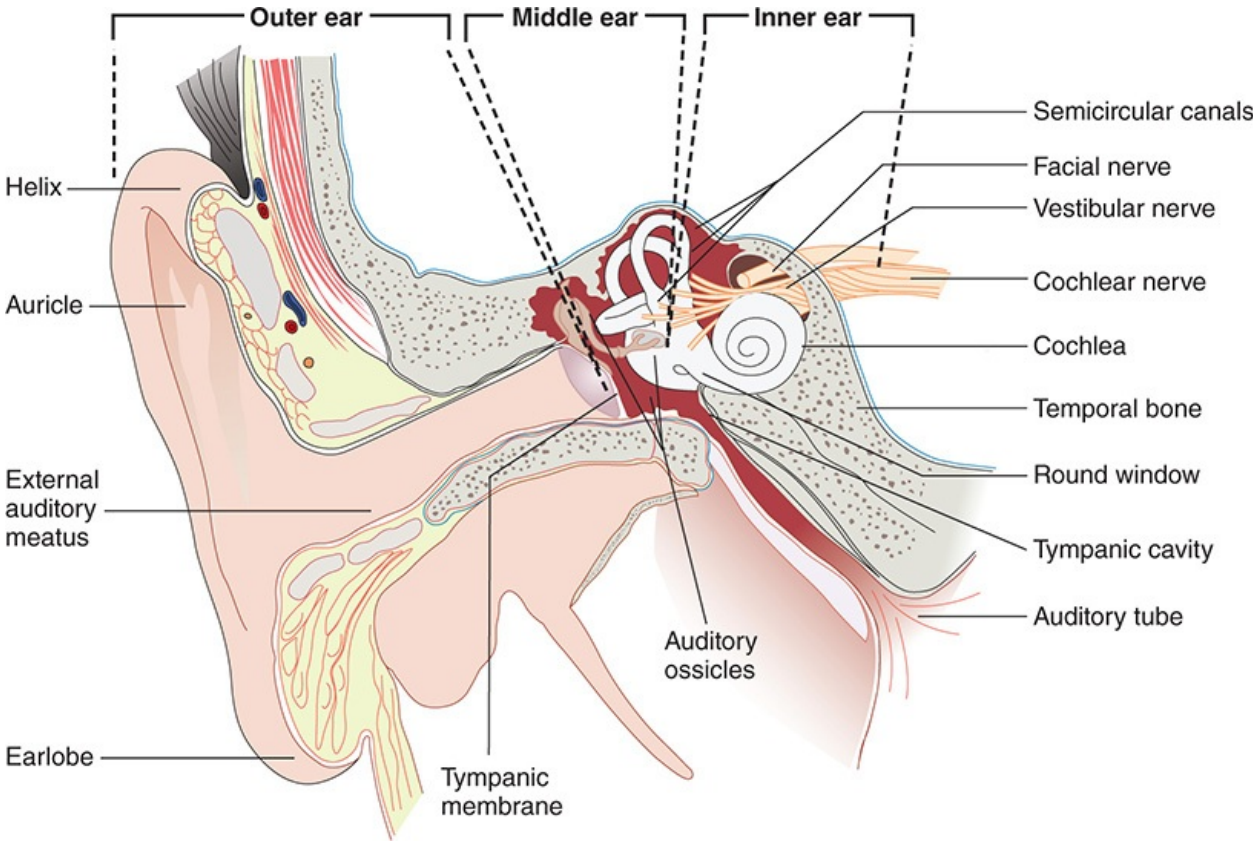


FIGURE 11–1 The structures of the external, middle, and inner portions of the human ear. Sound waves travel from the external ear to the tympanic membrane via the external auditory meatus. The middle ear is an air-filled cavity in the temporal bone; it contains the auditory ossicles. The inner ear is composed of the bony and membranous labyrinths. To make the relationships clear, the cochlea has been turned slightly and the middle ear muscles have been omitted. (Reproduced with permission from Fox SI: *Human Physiology*. New York, NY: McGraw-Hill; 2008.)

The middle ear is an air-filled cavity in the temporal bone that opens via the **eustachian (auditory) tube** into the nasopharynx and through the nasopharynx to the exterior. The tube is usually closed, but during swallowing, chewing, and yawning it opens, keeping the air pressure on the two sides of the eardrum equalized. Figure 11–2 shows the three tiny **auditory ossicles or bones** of the middle ear: **malleus** (hammer), **incus** (anvil), and **stapes** (stirrup). The **manubrium** (handle of the malleus) is attached to the back of the tympanic membrane. Its head is attached to the wall of the middle ear, and its short process is attached to the incus, which in turn articulates with the head of the stapes (stirrup). Its **footplate** is attached by an annular ligament to the walls of the **oval**

window that marks the beginning of the inner ear. Two small skeletal muscles **(tensor tympani** and **stapedius)** are also located in the middle ear. Contraction of the former pulls the manubrium of the malleus medially and decreases the vibrations of the tympanic membrane; contraction of the latter pulls the footplate of the stapes out of the oval window. The functions of the ossicles and muscles are detailed below.

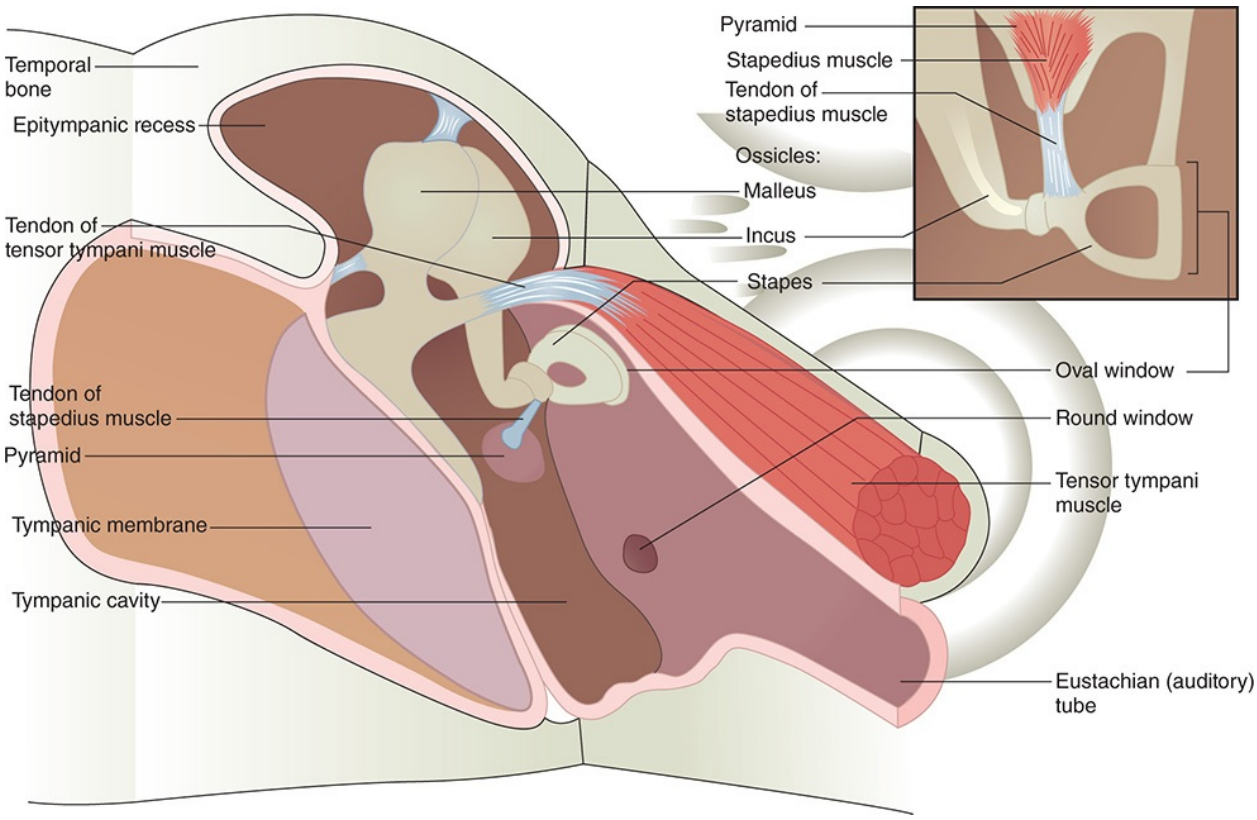


FIGURE 11–2 The medial view of the middle ear containing three auditory ossicles (malleus, incus, and stapes) and two small skeletal muscles (tensor tympani muscle and stapedius). The manubrium (handle of the malleus) is attached to the back of the tympanic membrane. Its head is attached to the wall of the middle ear, and its short process is attached to the incus, which in turn articulates with the head of the stapes. The footplate of the stapes is attached by an annular ligament to the walls of the oval window. Contraction of the tensor tympani muscle pulls the manubrium medially and decreases the vibrations of the tympanic membrane; contraction of the stapedius muscle pulls the footplate of the stapes out of the oval window. (Reproduced with permission from Fox SI: *Human Physiology*. New York, NY: McGraw-Hill; 2008.)

INNER EAR

The inner ear **(labyrinth)** is made up of two parts, one within the other. The **bony labyrinth** is a series of channels in the petrous portion of the **temporal bone** and is filled with a fluid called **perilymph** that has a relatively low concentration of K^+, similar to that of plasma or the cerebrospinal fluid. The **membranous labyrinth** is inside of these bony channels, surrounded by the perilymph; it more or less duplicates the shape of the bony channels and is filled with a K^+-rich fluid called **endolymph.** The labyrinth has three components: the **cochlea** that contains hair cells (receptors) for hearing, **semicircular canals** that contain hair cells that respond to head rotation, and the **otolith organs** that contain hair cells that respond to changes in gravity and head tilt **(Figure 11–3)**.

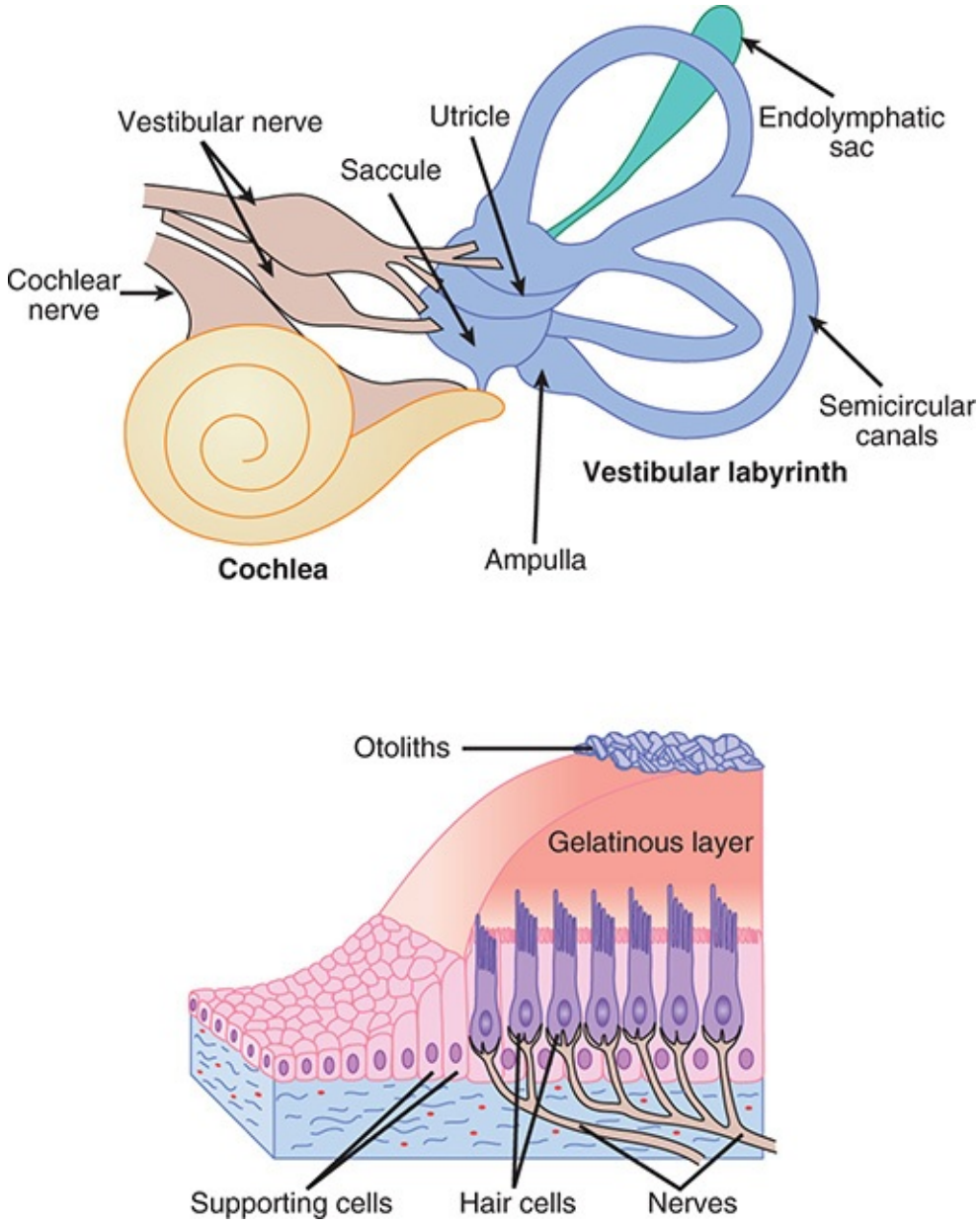


FIGURE 11–3 The membranous labyrinth of the inner ear has three components: semicircular canals, cochlea, and otolith organs. The semicircular canals are sensitive to angular accelerations that deflect the gelatinous cupula and associated hair cells. In the cochlea, hair cells spiral along the basilar membrane within the organ of Corti. Airborne sounds set the eardrum in motion, which is conveyed to the cochlea by bones of the middle ear. This flexes the membrane up and down. Hair cells in the organ of Corti are stimulated by shearing motion. The otolithic organs (saccule and utricle) are sensitive to linear acceleration in vertical and horizontal planes. Hair cells are attached to the otolithic membrane. Information from the cochlear hair cells is carried by the

cochlear division of the auditory (VIII cranial) nerve. Information from the hair cells in the semicircular canals and otolith organs is carried by the vestibular divisions of the auditory nerve.

The cochlea is a 35-mm-long coiled tube that makes two and three quarter turns. The basilar membrane and Reissner membrane divide it into three chambers or **scalae** **(Figure 11–4)**. The upper **scala vestibuli** and lower **scala tympani** contain perilymph and communicate with each other at the apex of the cochlea via a small opening **(helicotrema).** At the base of the cochlea, the scala vestibuli ends at the oval window that is closed by the footplate of the stapes. The scala tympani ends at the **round window,** a foramen on the medial wall of the middle ear that is closed by the flexible **secondary tympanic membrane.** The **scala media,** the middle cochlear chamber, is continuous with the membranous labyrinth and does not communicate with the other two scalae.

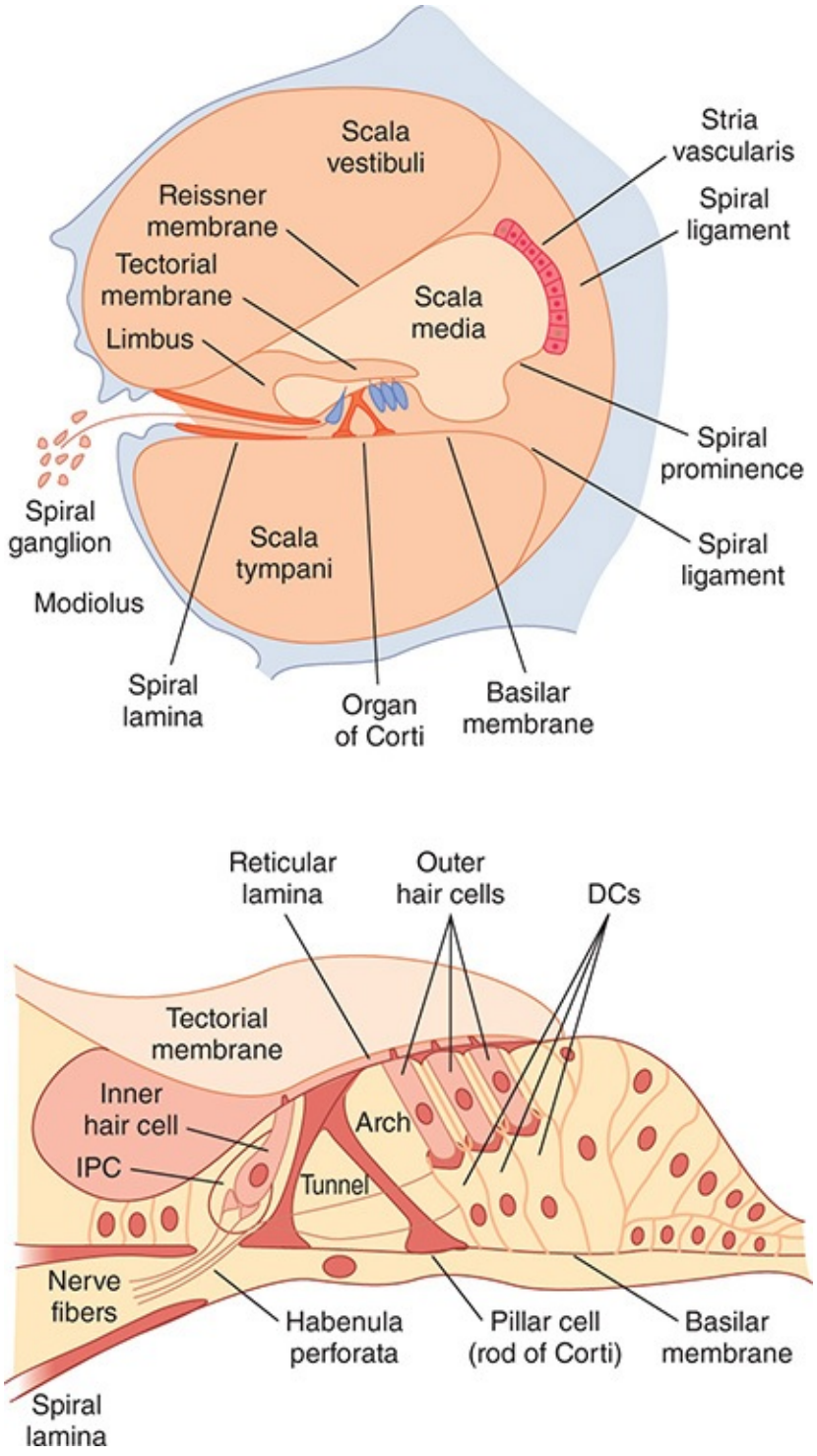


FIGURE 11–4 Schematic of the cochlea and organ of Corti in the membranous labyrinth of the inner ear. Top: The cross section of the cochlea shows the organ of Corti and the three scalae of the cochlea. **Bottom:** This shows the structure of the organ of Corti as it appears in the basal turn of the cochlea. DC, outer phalangeal cells (Deiters cells) supporting outer hair cells;

IPC, inner phalangeal cells supporting inner hair cell. (Reproduced with permission from Pickels JO: *An Introduction to the Physiology of Hearing*, 2nd ed. Academic Press; 1988.)

The spiral-shaped **organ of Corti** on the basilar membrane extends from the apex to the base of the cochlea. It contains the highly specialized auditory receptors (hair cells) whose processes pierce the tough, membrane-like **reticular lamina** that is supported by the **pillar cells** or **rods of Corti** (Figure 11–4). The hair cells are arranged in four rows: three rows of **outer hair cells** lateral to the tunnel formed by the rods of Corti and one row of **inner hair cells** medial to the tunnel. There are 20,000 outer hair cells and 3500 inner hair cells in each human cochlea. Covering the rows of hair cells is a thin, viscous, but elastic **tectorial membrane** in which the tips of the hairs of the outer but not the inner hair cells are embedded. The cell bodies of the sensory neurons that arborize around the bases of the hair cells are located in the **spiral ganglion** within the **modiolus,** the bony core around which the cochlea is wound. Most (90–95%) of these sensory neurons innervate the inner hair cells; only 5–10% innervate the more numerous outer hair cells. By contrast, most of the efferent fibers in the auditory nerve terminate on the outer rather than inner hair cells. The axons of the afferent neurons that innervate the hair cells form the **auditory (cochlear) division** of the eighth cranial nerve.

In the cochlea, gap junctions between the hair cells and the adjacent phalangeal cells prevent endolymph from reaching the base of the cell. However, the basilar membrane is relatively permeable to perilymph in the scala tympani and the tunnel of the organ of Corti. The bases of the hair cells are bathed in perilymph. Because of similar gap junctions, the arrangement is similar for the hair cells in other parts of the inner ear; that is, the processes of the hair cells are bathed in endolymph, whereas their bases are bathed in perilymph.

On each side of the head, the semicircular canals are perpendicular to each other, so that they are oriented in the three planes of space. The **crista ampullaris** (sensory organ of rotation) is located in the expanded end **(ampulla)** of each of the membranous canals (Figure 11–3). Each crista consists of hair cells and supporting cells surmounted by a gelatinous partition **(cupula)** that closes off the ampulla. The processes of the hair cells are embedded in the cupula, and the bases of the hair cells are in close contact with the afferent fibers of the **vestibular division** of the eighth cranial nerve.

A pair of otolith organs, the **saccule** and **utricle,** are located near the center of the membranous labyrinth. The **macula**, the sensory epithelium of these organs, are vertically oriented in the saccule and horizontally located in the utricle when

the head is upright. The maculae contain supporting cells and hair cells, surrounded by an otolithic membrane in which are embedded crystals of calcium carbonate, the **otoliths** (Figure 11–3). The otoliths or **otoconia** range from 3 to 19 µm in length. The processes of the hair cells are embedded in the membrane. The nerve fibers from the hair cells join those from the cristae in the vestibular division of the eighth cranial nerve.

SENSORY RECEPTORS IN THE EAR: HAIR CELLS

The specialized sensory mechanoreceptors in the ear consist of six patches of hair cells in the membranous labyrinth **(Figure 11–5)**. The hair cells in the organ of Corti signal hearing; the hair cells in the utricle signal horizontal acceleration; the hair cells in the saccule signal vertical acceleration; and a patch in each of the three semicircular canals signal rotational acceleration. Each hair cell is embedded in an epithelium made up of supporting cells, with the basal end in close contact with afferent neurons. A hair bundle projects from the apical end. It has one large **kinocilium,** a true but nonmotile cilium, with nine pairs of microtubules around a core and a central pair of microtubules. The kinocilium is lost from the cochlear hair cells in adults. The other 30–150 processes **(stereocilia)** are found in all hair cells; they have cores composed of parallel filaments of actin. There is an orderly structure within the clump of processes on each cell. Along an axis toward the kinocilium, the stereocilia increase progressively in height; along the perpendicular axis, all the stereocilia are of the same height.

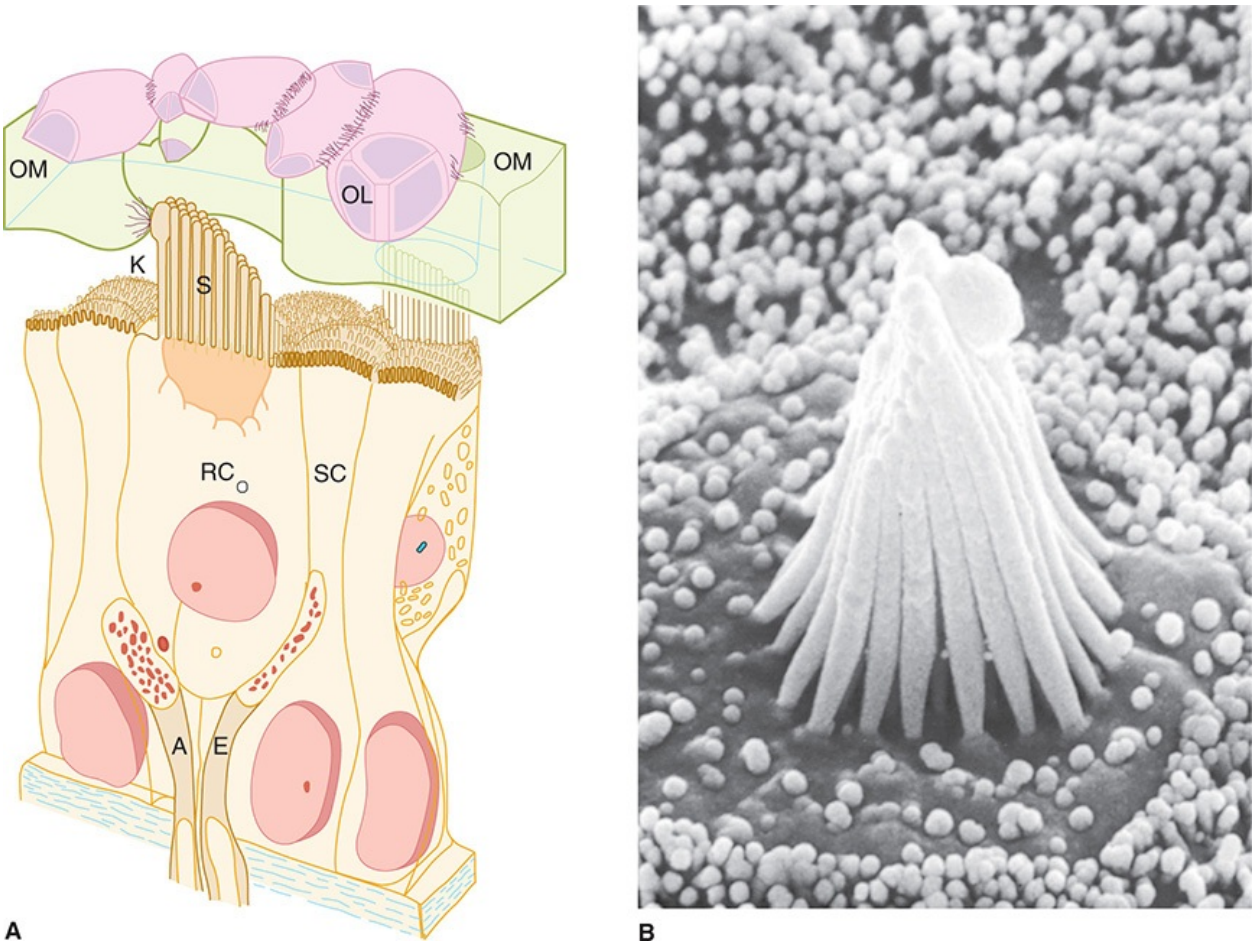


FIGURE 11–5 Structure of hair cell in the saccule. Left: Hair cells in the membranous labyrinth of the ear have a common structure, and each is within an epithelium of supporting cells (SC) surmounted by an otolithic membrane (OM) embedded with crystals of calcium carbonate, the otoliths (OL). Projecting from the apical end are rod-shaped processes, or hair cells (RC), in contact with afferent (A) and efferent (E) nerve fibers. Except in the cochlea, one of these, **kinocilium** (K), is a true but nonmotile cilium with nine pairs of microtubules around its circumference and a central pair of microtubules. The other processes, **stereocilia** (S), are found in all hair cells; they have cores of actin filaments coated with isoforms of myosin. Within the clump of processes on each cell there is an orderly structure. Along an axis toward the kinocilium, the stereocilia increase progressively in height; along the perpendicular axis, all the stereocilia are the same height. (Reproduced with permission from Llinas R, Precht W (eds): *Frog Neurobiology*. Springer; 1976.) **Right:** Scanning electron micrograph of processes on a hair cell in the saccule. The otolithic membrane has been removed. The small projections around the hair cell are microvilli on supporting cells. (Used with permission of AJ Hudspeth.)

ELECTRICAL RESPONSES IN HAIR CELLS

Very fine processes called **tip links** **(Figure 11–6)** tie the tip of each stereocilium to the side of its higher neighbor, and mechanically sensitive cation channels are at the junction in the taller process. When the shorter stereocilia are pushed toward the taller ones, the channel open time is increased. K^+, the most abundant cation in endolymph, and Ca^{2+} enter via the channel and induce depolarization. A myosin-based molecular motor in the taller neighbor then moves the channel toward the base, releasing tension in the tip link. This causes the channel to close and restores the resting state. Depolarization of hair cells causes them to release a neurotransmitter, probably glutamate, which initiates depolarization of neighboring afferent neurons.

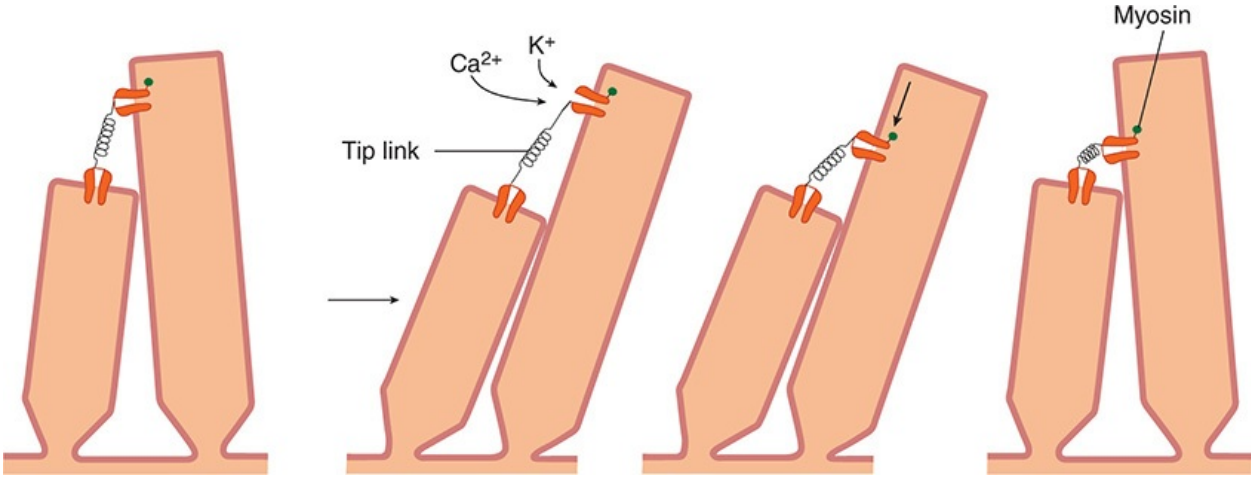


FIGURE 11–6 Schematic representation of the role of tip links in the responses of hair cells. When a stereocilium is pushed toward a taller stereocilium, the tip link is stretched and opens an ion channel in its taller neighbor. The channel next is moved down the taller stereocilium by a molecular motor, so the tension on the tip link is released. When the hairs return to the resting position, the motor moves back up the stereocilium.

The K^+ that enters hair cells via the mechanically sensitive cation channels is recycled **(Figure 11–7)**. It enters supporting cells and then passes on to other supporting cells by way of gap junctions. In the cochlea, it eventually reaches the stria vascularis and is secreted back into the endolymph, completing the cycle.

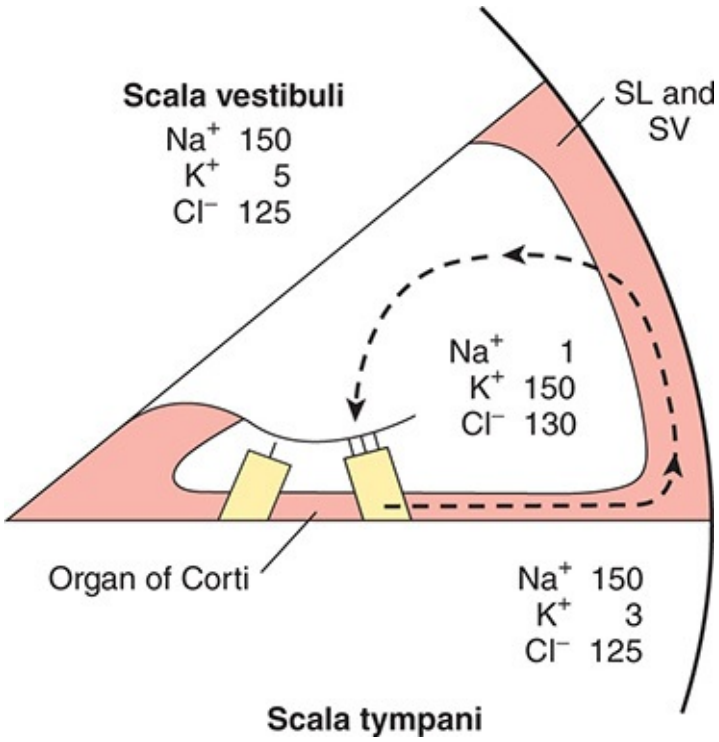


FIGURE 11–7 Ionic composition of perilymph in the scala vestibuli, endolymph in the scala media, and perilymph in the scala tympani. SL, spiral ligament. SV, stria vascularis. The dashed arrow indicates the path by which K^+ recycles from the hair cells to the supporting cells to the spiral ligament and is then secreted back into the endolymph by cells in the stria vascularis.

As described above, the processes of hair cells project into the endolymph and the bases are bathed in perilymph. This arrangement is necessary for the normal production of receptor potentials. The perilymph is formed mainly from plasma. On the other hand, endolymph is formed in the scala media by the stria vascularis and has a high concentration of K^+ and a low concentration of Na^+ (Figure 11–7). Cells in the stria vascularis have a high concentration of Na^+, K^+ ATPase. A unique electrogenic K^+ pump in the stria vascularis may account for the fact that the scala media is electrically positive by 85 mV relative to the scala vestibuli and scala tympani.

The resting membrane potential of hair cells is about −60 mV. When the stereocilia are pushed toward the kinocilium, the membrane potential is decreased to about −50 mV. When the hair bundle is pushed in the opposite direction, the cell is hyperpolarized. Displacing the processes in a direction perpendicular to this axis provides no change in membrane potential, and

displacing the processes in a direction that is intermediate between these two directions induces depolarization or hyperpolarization that is proportional to the degree to which the direction is toward or away from the kinocilium. Thus, the hair processes provide a mechanism for generating changes in membrane potential proportional to the direction and distance the hair moves.

HEARING

SOUND WAVES

Sound is the sensation produced when longitudinal vibrations of molecules in the external environment strike the tympanic membrane. **Figure 11–8** shows a plot of these movements as changes in pressure on the tympanic membrane per unit of time; such movements in the environment are called **sound waves.** The waves travel through air at a speed of 344 m/s (770 mph) at 20°C at sea level; the speed of sound increases with temperature or altitude. Other media can also conduct sound waves, but at a different speed. For example, the speed of sound is 1450 m/s at 20°C in fresh water and is even greater in salt water.

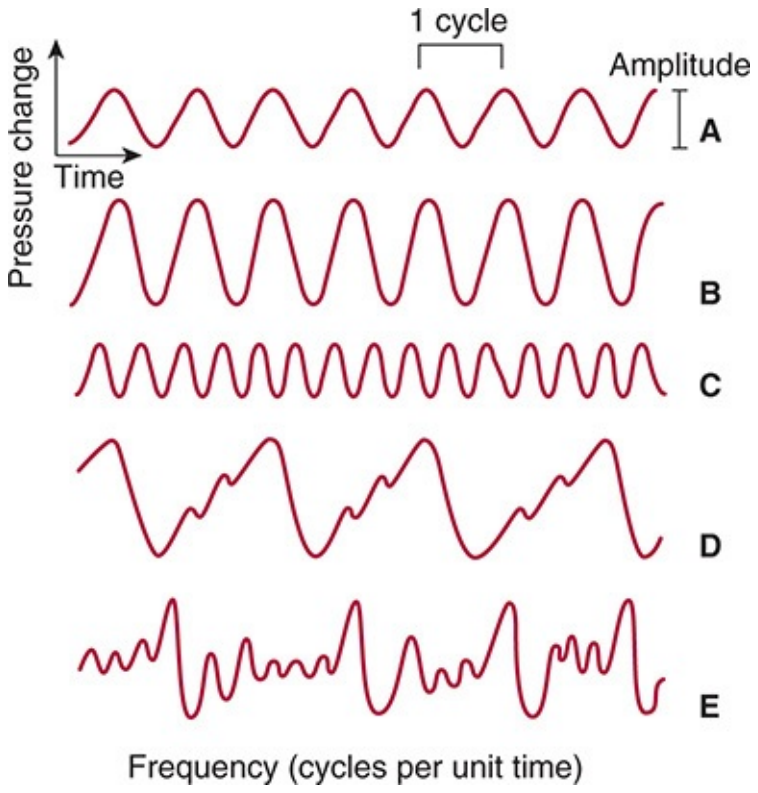


FIGURE 11–8 Characteristics of sound waves. A is the record of a pure tone.

B has a greater amplitude and is louder than **A. C** has the same amplitude as **A** but a greater frequency, and its pitch is higher. **D** is a complex wave form that is regularly repeated. Such patterns are perceived as musical sounds, whereas waves like that shown in **E,** which have no regular pattern, are perceived as noise.

In general, the **loudness** of a sound is directly correlated with the **amplitude** of a sound wave (Figure 11–8). The **pitch** of a sound is directly correlated with the **frequency** (number of waves per unit of time) of the sound wave. Sound waves that have repeating patterns, even though the individual waves are complex, are perceived as musical sounds; aperiodic nonrepeating vibrations cause a sensation of noise. Most musical sounds are made up of a wave with a primary frequency that determines the pitch of the sound plus a number of harmonic vibrations **(overtones)** that give the sound its characteristic **timbre** (quality). Variations in timbre allow us to distinguish the sounds of different musical instruments even though they are playing notes of the same pitch.

Although the pitch of a sound depends primarily on the frequency of the sound wave, loudness also plays a part; low tones (below 500 Hz) seem lower and high tones (above 4000 Hz) seem higher as their loudness increases. Duration also affects pitch to a minor degree. The pitch of a tone cannot be perceived unless it lasts for more than 0.01 s, and pitch rises as duration increases from 0.01 to 0.1 s. Finally, the pitch of complex sounds that include harmonics of a given frequency is still perceived even if the primary frequency is absent.

The amplitude of a sound wave is expressed on a **decibel scale**. The intensity of a sound in **bels** is the logarithm of the ratio of the intensity of that sound and a standard sound. A decibel (dB) is 0.1 bel. The standard sound reference level adopted by the Acoustical Society of America corresponds to 0 dB at a pressure level of 0.000204 × dyne/cm^2, a value that is just at the auditory threshold for the average human. A value of 0 dB does not mean the absence of sound but a sound level of an intensity equal to that of the standard. The 0- to 140-dB range from threshold pressure to a pressure that is potentially damaging to the organ of Corti actually represents a 10^7 (10 million)-fold variation in sound pressure.

A range of 120–160 dB (eg, firearms, jackhammer, and jet plane on takeoff) is classified as painful; 90–110 dB (eg, subway, bass drum, chain saw, and lawn mower) is extremely high; 60–80 dB (eg, alarm clock, busy traffic, dishwasher, and conversation) is very loud; 40–50 dB (eg, moderate rainfall and normal room noise) is moderate; and 30 dB (eg, whisper and library) is faint. Prolonged

or frequent exposure to sounds above 85 dB can cause hearing loss.

The sound frequencies audible to humans range from about 20 to a maximum of 20,000 cycles per second (Hz). The threshold of the human ear varies with the pitch of the sound **(Figure 11–9)**, the greatest sensitivity being in the 1000- to 4000-Hz range. The pitch of the average male voice in conversation is about 120 Hz and that of the average female voice about 250 Hz. The number of pitches that can be distinguished by an average individual is about 2000, but trained musicians can improve on this figure considerably. Pitch discrimination is best in the 1000- to 3000-Hz range and is poor at high and low pitches.

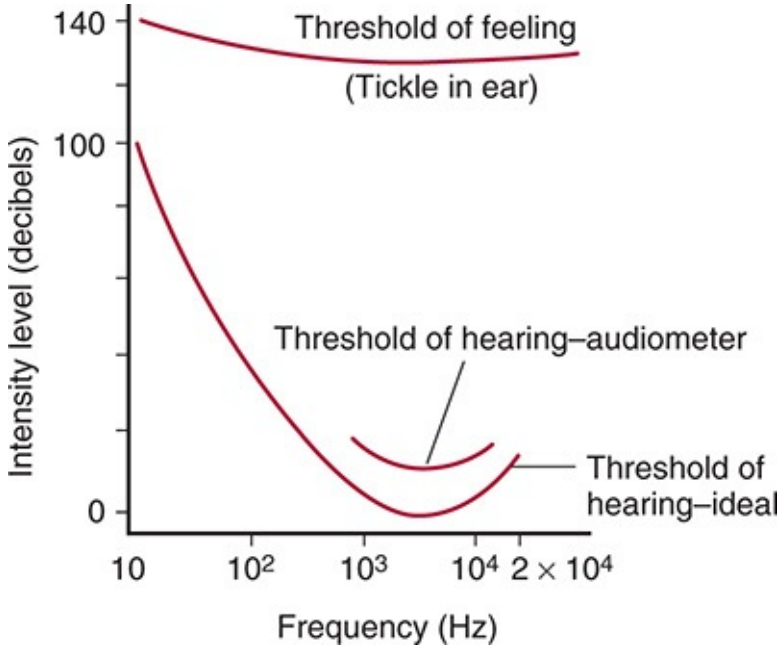


FIGURE 11–9 Human audibility curve. The middle curve is that obtained by audiometry under the usual conditions. The lower curve is that obtained under ideal conditions. At about 140 dB (top curve), sounds are felt as well as heard.

The presence of one sound decreases the ability to hear other sounds, a phenomenon known as **masking.** It is due to the relative or absolute refractoriness of previously stimulated auditory receptors and nerve fibers to other stimuli. The degree to which a given tone masks others is related to its pitch. The masking effect of the background noise in all but the most carefully soundproofed environments raises the auditory threshold by a definite and measurable amount.

SOUND TRANSMISSION

The ear converts sound waves in the external environment into action potentials in the auditory nerves. The waves are transformed by the eardrum and auditory ossicles into movements of the footplate of the stapes. These movements set up waves in the fluid of the inner ear **(Figure 11–10)**. The action of the waves on the organ of Corti generates action potentials in the nerve fibers.

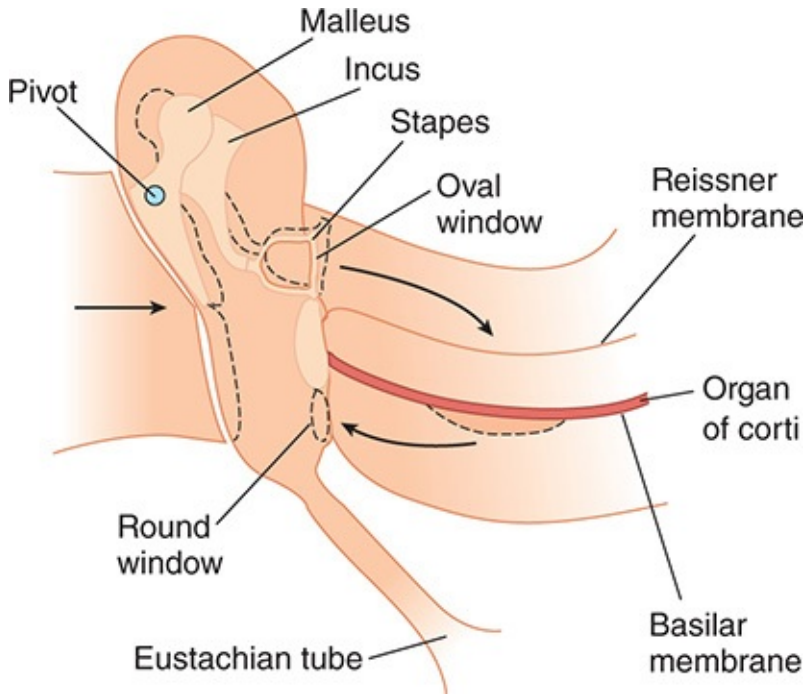


FIGURE 11–10 Schematic representation of the auditory ossicles and the way their movement translates movements of the tympanic membrane into a wave in the fluid of the inner ear. The wave is dissipated at the round window. The movements of the ossicles, the membranous labyrinth, and the round window are indicated by dashed lines. The waves are transformed by the eardrum and auditory ossicles into movements of the footplate of the stapes. These movements set up waves in the fluid of the inner ear. In response to the pressure changes produced by sound waves on its external surface, the tympanic membrane moves in and out to function as a resonator that reproduces the vibrations of the sound source. The motions of the tympanic membrane are imparted to the manubrium of the malleus, which rocks on an axis through the junction of its long and short processes, so that the short process transmits the vibrations of the manubrium to the incus. The incus moves so that the vibrations are transmitted to the head of the stapes. Movements of the head of the stapes swing its footplate.

When sound waves change the pressure on its external surface, the tympanic

membrane moves in and out. The motions of the tympanic membrane are imparted to the manubrium of the malleus. The malleus rocks on an axis through the junction of its long and short processes, so that the short process transmits the vibrations of the manubrium to the incus. The incus moves in such a way that the vibrations are transmitted to the head of the stapes. Movements of the head of the stapes swing its footplate to and fro like a door hinged at the posterior edge of the oval window. The auditory ossicles thus function as a lever system that converts the vibrations of the tympanic membrane into movements of the stapes against the perilymph-filled scala vestibuli of the cochlea (Figure 11–10). This system increases the sound pressure that arrives at the oval window, because the lever action of the malleus and incus multiplies the force 1.3 times and the area of the tympanic membrane is much greater than the area of the footplate of the stapes.

Contraction of the tensor tympani and stapedius muscles of the middle ear cause the manubrium of the malleus to be pulled inward and the footplate of the stapes to be pulled outward to reduce sound transmission (Figure 11–2). Loud sounds initiate the **tympanic reflex** that causes contraction of these muscles. This reflex prevents strong sound waves from causing excessive stimulation of the auditory receptors.

TRAVELING WAVES

The movements of the footplate of the stapes set up a series of traveling waves in the perilymph of the scala vestibuli. A diagram of such a wave is shown in **Figure 11–11**. As the wave moves up the cochlea, its height increases to a maximum and then drops off rapidly. The distance from the stapes to this point of maximum height varies with the frequency of the vibrations initiating the wave. High-pitched sounds generate waves that reach maximum height near the base of the cochlea; low-pitched sounds generate waves that peak near the apex. The bony walls of the scala vestibuli are rigid, but Reissner membrane is flexible. The basilar membrane is not under tension, and it is also readily depressed into the scala tympani by the peaks of waves in the scala vestibuli. Displacements of the fluid in the scala tympani are dissipated into air at the round window. Sound produces distortion of the basilar membrane, and the site at which this distortion is maximal is determined by the sound wave frequency. The tops of the hair cells in the organ of Corti are held rigid by the reticular lamina, and the hairs of the outer hair cells are embedded in the tectorial membrane (Figure 11–4). When the stapes moves, both membranes move in the

same direction, but they are hinged on different axes, so a shearing motion bends the hairs. The hairs of the inner hair cells are not attached to the tectorial membrane, but they are bent by fluid moving between the tectorial membrane and the underlying hair cells.

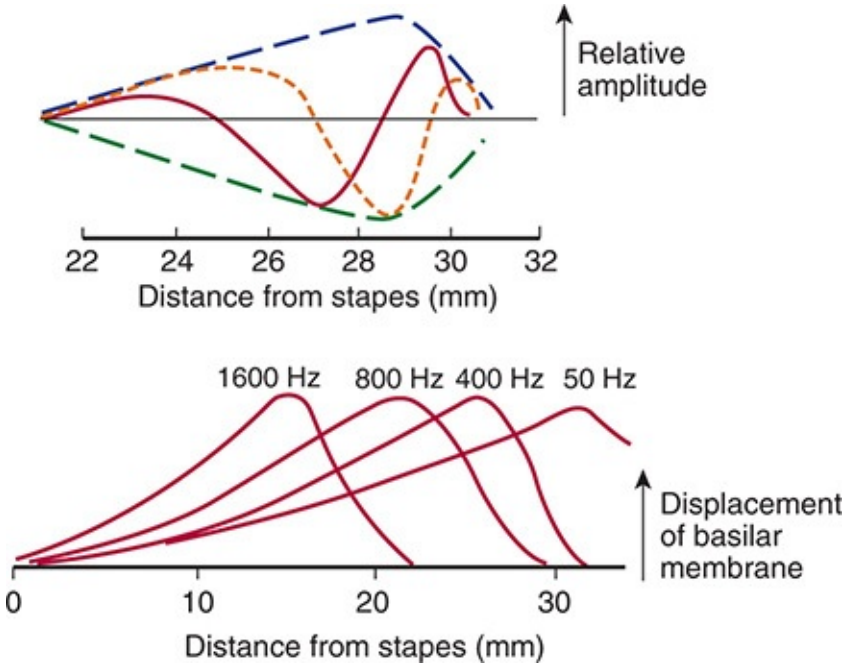


FIGURE 11–11 Traveling waves. Top: The solid and the short-dashed lines represent the wave at two instants of time. The long-dashed line shows the "envelope" of the wave formed by connecting the wave peaks at successive instants. **Bottom:** Displacement of the basilar membrane by the waves generated by stapes vibration of the frequencies shown at the top of each curve.

FUNCTIONS OF THE OUTER HAIR CELLS

The inner hair cells are the primary sensory receptors that generate action potentials in the auditory nerves and are stimulated by the fluid movements. On the other hand, the outer hair cells respond to sound like the inner hair cells, but depolarization makes them shorten and hyperpolarization makes them lengthen. They do this over a very flexible part of the basal membrane, and this action increases the amplitude and clarity of sounds. Thus, outer hair cells amplify sound vibrations entering the inner ear from the middle ear. These changes in outer hair cells occur in parallel with changes in a membrane protein **prestin,** the motor protein of outer hair cells.

The **olivocochlear bundle** is a prominent bundle of efferent fibers in each auditory nerve that arises from both ipsilateral and contralateral superior olivary complexes and ends primarily around the bases of the outer hair cells of the organ of Corti. The activity in this nerve bundle modulates the sensitivity of the hair cells via the release of acetylcholine. The effect is inhibitory, and it may function to block background noise while allowing other sounds to be heard.

ACTION POTENTIALS IN AUDITORY NERVE FIBERS

The frequency of the action potentials in a given auditory nerve fiber determines the loudness of a sound. At low sound intensities, each axon is activated by sounds of only one frequency that depends on the part of the cochlea from which the fiber originates. At higher sound intensities, the individual axons discharge to a wider spectrum of sound frequencies, particularly to frequencies lower than that at which threshold simulation occurs.

The place in the organ of Corti that is maximally stimulated determines the pitch perceived when a sound wave strikes the ear. The traveling wave set up by a tone produces peak depression of the basilar membrane, and consequently maximal receptor stimulation, at one point. As noted above, the distance between this point and the stapes is inversely related to the pitch of the sound, with low tones producing maximal stimulation at the apex of the cochlea and high tones producing maximal stimulation at the base. The pathways from the various parts of the cochlea to the brain are distinct.

CENTRAL AUDITORY PATHWAY

The afferent fibers in the auditory division of the eighth cranial nerve end in **dorsal** and **ventral cochlear nuclei** **(Figure 11–12)**. From there, auditory impulses pass by various routes to the **inferior colliculi,** the centers for auditory reflexes, and via the **medial geniculate body** in the thalamus to the **auditory cortex** located on the superior temporal gyrus of the temporal lobe. Information from both ears converges on each superior olive, and beyond this, most of the neurons respond to inputs from both sides. In humans, low tones are represented anterolaterally and high tones posteromedially in the auditory cortex.

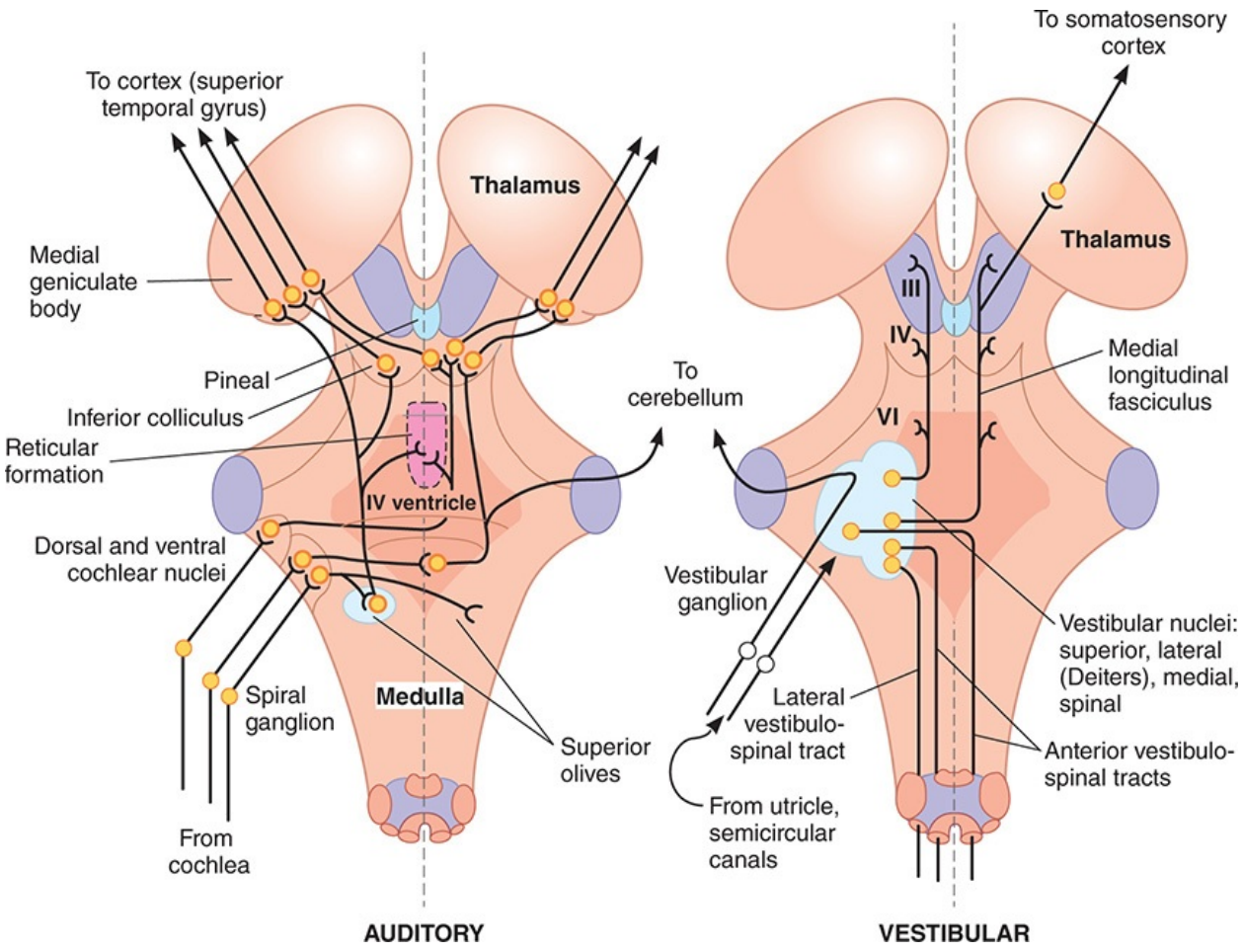


FIGURE 11–12 Simplified diagram of main auditory (left) and vestibular (right) pathways superimposed on a dorsal view of the brainstem. Cerebellum and cerebral cortex have been removed. For the auditory pathway, eighth cranial nerve afferent fibers form the cochlea end in dorsal and ventral cochlear nuclei. From there, most fibers cross the midline and terminate in the contralateral inferior colliculus. From there, fibers project to the medial geniculate body in the thalamus and then to the auditory cortex located on the superior temporal gyrus of the temporal lobe. For the vestibular pathway, the vestibular nerve terminates in the ipsilateral vestibular nucleus. Most fibers from the semicircular canals terminate in the superior and medial divisions of the vestibular nucleus and project to nuclei controlling eye movement. Most fibers from the utricle and saccule terminate in the lateral division, which then projects to the spinal cord. They also terminate on neurons that project to the cerebellum and the reticular formation. The vestibular nuclei also project to the thalamus and from there to the primary somatosensory cortex. The ascending connections to cranial nerve nuclei are concerned with eye movements.

The responses of individual second-order neurons in the cochlear nuclei to sound stimuli are like those of the individual auditory nerve fibers. The frequency at which sounds of the lowest intensity evoke a response varies from unit to unit; with increased sound intensities, the band of frequencies to which a response occurs becomes wider. The major difference between the responses of the first- and second-order neurons is the presence of a sharper "cutoff" on the low-frequency side in the medullary neurons. This greater specificity of the second-order neurons is probably due to an inhibitory process in the brainstem. In the primary auditory cortex, most neurons respond to inputs from both ears, but strips of cells are stimulated by input from the contralateral ear and inhibited by input from the ipsilateral ear.

The increasing availability of positron emission tomography (PET) scanning and functional magnetic resonance imaging (fMRI) has greatly improved the level of knowledge about auditory association areas in humans. The auditory pathways in the cortex resemble the visual pathways in that increasingly complex processing of auditory information takes place along them. An interesting observation is that although the auditory areas look very much the same on the two sides of the brain, there is marked hemispheric specialization. For example, **Wernicke area** (see Figure 8–7) is concerned with the processing of auditory signals related to speech. During language processing, this area is much more active on the left side than on the right side. Wernicke area on the right side is more concerned with melody, pitch, and sound intensity. The auditory pathways are also very plastic, and, like the visual and somatosensory pathways, they are modified by experience. Examples of auditory plasticity in humans include the observation that in individuals who become deaf before language skills are fully developed, sign language activates auditory association areas. Conversely, individuals who become blind early in life are demonstrably better at localizing sound than individuals with normal eyesight.

Musicians provide additional examples of cortical plasticity. In these individuals, the size of the auditory areas activated by musical tones is increased. In addition, violinists have altered somatosensory representation of the areas to which the fingers they use in playing their instruments project. Musicians also have larger cerebellums than nonmusicians, presumably because of learned precise finger movements.

SOUND LOCALIZATION

Determination of the direction from which a sound emanates in the horizontal

plane depends on detecting the difference in time between the arrival of the stimulus in the two ears and the consequent difference in phase of the sound waves on the two sides; it also depends on the fact that the sound is louder on the side closest to the source. The detectable time difference, which can be as little as 20 µs, is the most important factor at frequencies below 3000 Hz; the loudness difference is the most important at frequencies above 3000 Hz. Neurons in the auditory cortex that receive input from both ears respond maximally or minimally when the time of arrival of a stimulus at one ear is delayed by a fixed period relative to the time of arrival at the other ear. This fixed period varies from neuron to neuron.

Sounds coming from directly in front of the individual differ in quality from those coming from behind because each pinna (the visible portion of the exterior ear) is turned slightly forward. Reflections of sound waves from the pinnal surface change as sounds move up or down, and the change in the sound waves is the primary factor in locating sounds in the vertical plane. Sound localization is markedly disrupted by lesions of the auditory cortex.

HEARING LOSS

There are two major categories of hearing loss or deafness: sensorineural and conductive **(Clinical Box 11–1)**. **Sensorineural hearing loss** (deafness) is the most common type of hearing loss; it is usually due to the loss of cochlear hair cells but can be result from damage to the eighth cranial nerve or within central auditory pathways. It often impairs the ability to hear certain pitches while others are unaffected. Aminoglycoside antibiotics such as streptomycin and gentamicin obstruct the mechanosensitive channels in the stereocilia of hair cells (especially outer hair cells) and can cause the cells to degenerate, producing sensorineural hearing loss and abnormal vestibular function. Damage to the hair cells by prolonged exposure to noise is also associated with hearing loss. Other causes include autoimmune disorders, traumatic injuries, acoustic neuromas, tumors of the eighth cranial nerve and cerebellopontine angle, and vascular damage in the medulla.

Conductive hearing loss refers to impaired sound transmission in the external or middle ear and impacts all sound frequencies. Among the causes of conduction hearing loss are plugging of the external auditory canals with wax (cerumen) or foreign bodies, otitis externa (inflammation of the outer ear, “swimmer’s ear”) and otitis media (inflammation of the middle ear) causing fluid accumulation or scarring, or perforation of the eardrum. Severe conductive

deafness can result from **otosclerosis** in which bone is resorbed and replaced with sclerotic bone that grows over the oval window.

Auditory acuity is commonly measured with an **audiometer.** This device presents the subject with pure tones of various frequencies through earphones. At each frequency, the threshold intensity is determined and plotted on a graph as a percentage of normal hearing. This provides an objective measurement of the degree of deafness and a picture of the tonal range most affected.

Conduction and sensorineural deafness can be differentiated by simple tests with a tuning fork. Three of these tests are outlined in Table 11–1. The **Rinne test** compares sound conduction through air and bone. **Bone conduction** is the transmission of vibrations of the bones of the skull to the fluid of the inner ear. The **Weber and Schwabach tests** demonstrate the important masking effect of environmental noise on the auditory threshold.

TABLE 11–1 Common tests with a tuning fork to distinguish between sensorineural and conduction hearing loss.

	Weber	Rinne	Schwabach
Method	Base of vibrating tuning fork placed on vertex of skull	Base of vibrating tuning fork placed on mastoid process until subject no longer hears it, then held in air next to ear	Bone conduction of patient compared with that of healthy subject
Normal	Hears equally on both sides	Hears vibration in air after bone conduction is over	
Conduction deafness (one ear)	Sound louder in diseased ear because masking effect of environmental noise is absent on diseased side	Vibrations in air not heard after bone conduction is over	Bone conduction better than normal (conduction defect excludes masking noise)
Sensorineural deafness (one ear)	Sound louder in normal ear	Vibration heard in air after bone conduction is over, as long as nerve deafness is partial	Bone conduction worse than normal

CLINICAL BOX 11–1

Hearing Loss

Hearing loss is the most common sensory defect in humans. According to the World Health Organization, over 270 million people worldwide have moderate to profound hearing loss, with one-fourth of these cases beginning in childhood. According to the National Institutes of Health, ~15% of Americans between 20 and 69 years of age have high-frequency hearing loss due to exposure to loud sounds or noise at work or in leisure activities (**noise-induced hearing loss**). Both inner and outer hair cells are damaged by excessive noise, but outer hair cells appear to be more vulnerable. The use of

various chemicals (**ototoxins)** also causes hearing loss. These include some antibiotics (streptomycin), loop diuretics (furosemide), and platinum-based chemotherapy agents (cisplatin). These ototoxic agents damage the outer hair cells or the stria vascularis. **Presbycusis,** the gradual hearing loss associated with aging, affects more than one-third of those over age 75 and is probably due to gradual cumulative loss of hair cells and neurons. In most cases, hearing loss is a multifactorial disorder caused by both genetic and environmental factors. Single-gene mutations can cause hearing loss. This type of hearing loss is a monogenic disorder with an autosomal dominant, autosomal recessive, X-linked, or mitochondrial mode of inheritance. Monogenic forms of deafness can be defined as **syndromic** (hearing loss associated with other abnormalities) or **nonsyndromic** (only hearing loss). About 0.1% of newborns have genetic mutations leading to deafness. Nonsyndromic deafness due to genetic mutations can first appear in adults rather than in children and may account for many of the 16% of all adults who have significant hearing impairment. It is estimated that the products of 100 or more genes are essential for normal hearing, and deafness loci have been identified in all but 5 of the 24 human chromosomes. The most common mutation leading to congenital hearing loss is that of the protein connexin 26. This defect prevents the normal recycling of K^+ through the sustentacular cells. Mutations in three nonmuscle myosins also cause deafness. These are myosin-VIIa, associated with the actin in the hair cell processes; myosin-Ib, which is probably part of the "adaptation motor" that adjusts tension on the tip links; and myosin-VI, which is essential in some way for the formation of normal cilia. Deafness is also associated with mutant forms of α-tectin, one of the major proteins in the tectorial membrane. An example of syndromic deafness is **Pendred syndrome,** in which a mutant multifunctional anion exchanger causes deafness and goiter. Another example is one form of the **long QT syndrome** in which one of the K^+ channel proteins, **KVLQT1,** is mutated. In the stria vascularis, the normal form of this protein is essential for maintaining the high K^+ concentration in endolymph, and in the heart it helps maintain a normal QT interval. Individuals who are homozygous for mutant KVLQT1 are deaf and predisposed to the ventricular arrhythmias and sudden death that characterize the long QT syndrome. Mutations of the membrane protein **barttin** can cause deafness as well as the renal manifestations of Bartter syndrome.

THERAPEUTIC HIGHLIGHTS

Cochlear implants are used to treat both children and adults with severe hearing loss. The US Food and Drug Administration reported that, as of December 2012, approximately 324,000 cochlear devices have been implanted worldwide and at least 50,000 are implanted every year. They may be used in children as young as 12 months old. These devices consist of a microphone (picks up environmental sounds), a speech processor (selects and arranges these sounds), a transmitter and receiver/stimulator (converts these sounds into electrical impulses), and an electrode array (sends the impulses to the auditory nerve). Although the implant cannot restore normal hearing, it provides a useful representation of environmental sounds to a deaf person. Those with adult-onset deafness who receive cochlear implants can learn to associate the signals it provides with sounds they remember. Children who receive cochlear implants in conjunction with intensive therapy have been able to acquire speech and language skills. Research is also underway to develop cells that can replace the hair cells in the inner ear. For example, researchers at Stanford University were able to generate cells resembling mechanosensitive hair cells from mouse **embryonic** and **pluripotent stem cells.**

Hearing aids can also be used to treat sensorineural hearing loss in individuals who have significant residual hearing. The microphone component of an analog hearing aid receives sound that converts sound waves (vibrations) to electrical signals; an amplifier then increases the power of the signal and sends it to the ear via a speaker. Digital hearing aids convert sound waves into numerical codes akin to a computer binary code before amplifying them. The code also includes information about pitch or loudness, so it can be programmed to amplify selectively those sound wave frequencies to which the wearer is least sensitive.

Injury to inner ear hair cells can induce random electrical impulses that are then relayed to the auditory cortex, leading to an intermittent or steady, high-pitched ringing in the ear **(tinnitus)**. Tinnitus affects about 50 million Americans and can be a symptom of age-related hearing loss, excessive exposure to loud noises, ear infections, or otosclerosis. In most cases, its cause is unknown. Hypertension and atherosclerosis are risk factors for the development of tinnitus. Also, it can be triggered or worsened by the use of antimalarial drugs, antibiotics, chemotherapeutic drugs, diuretics, or high doses of aspirin.

VESTIBULAR SYSTEM

The vestibular system can be divided into the **vestibular apparatus** and central **vestibular nuclei.** The vestibular apparatus within the inner ear detects head motion and position and transduces this information to a neural signal (Figure 11–3). The vestibular nuclei are primarily concerned with maintaining the position of the head in space. The tracts that descend from these nuclei mediate head-on-neck and head-on-body adjustments.

CENTRAL VESTIBULAR PATHWAY

The cell bodies of the 19,000 neurons supplying the cristae and maculae on each side are located in the vestibular ganglion. Each vestibular nerve terminates in the ipsilateral four-part vestibular nucleus (Figure 11–12) and in the flocculonodular lobe of the cerebellum (not shown in the figure). Fibers from the semicircular canals terminate primarily in the superior and medial divisions of the vestibular nucleus; neurons in this region project mainly to nuclei controlling eye movement (see Chapter 10). Fibers from the utricle and saccule project predominantly to the lateral division (Deiters nucleus) of the vestibular nucleus which then projects to the contralateral spinal cord. The descending vestibular nucleus receives input from the otolith and projects to the cerebellum, reticular formation, and the spinal cord. The ascending connections to cranial nerve nuclei are control eye movements; the vestibular nuclei also ascend to project to thalamocortical neurons.

RESPONSES TO ROTATIONAL ACCELERATION

Rotational acceleration in the plane of a given semicircular canal stimulates its crista. The endolymph, because of its inertia, is displaced in a direction opposite to the direction of rotation. The fluid pushes on the cupula, deforming it. This bends the processes of the hair cells (Figure 11–3). When a constant speed of rotation is reached, the fluid spins at the same rate as the body and the cupula swings back into the upright position. When rotation is stopped, deceleration produces displacement of the endolymph in the direction of the rotation, and the cupula is deformed in a direction opposite to that during acceleration. It returns to mid position in 25–30 s. Movement of the cupula in one direction increases the firing rate of single nerve fibers from the crista; movement in the opposite direction inhibits neural activity **(Figure 11–13)**.

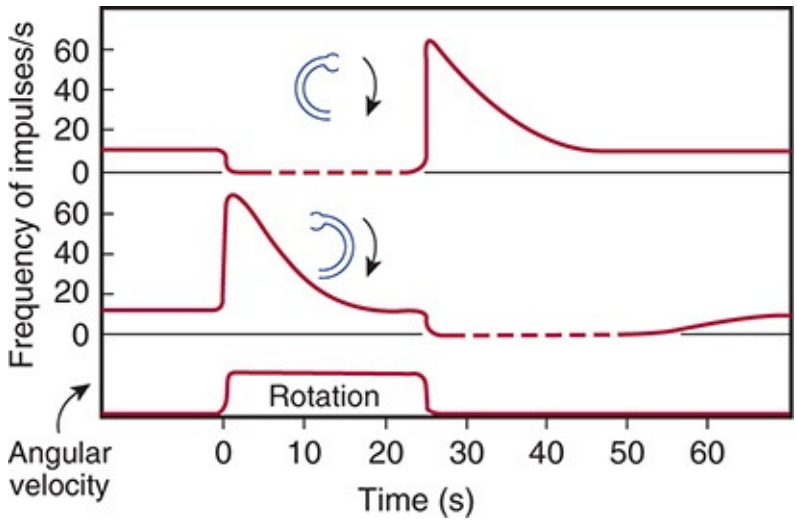


FIGURE 11–13 Ampullary responses to rotation. Average time course of impulse discharge from the ampulla of two semicircular canals during rotational acceleration, steady rotation, and deceleration. Movement of the cupula in one direction increases the firing rate of single nerve fibers from the crista, and movement in the opposite direction inhibits neural activity. (Reproduced with permission from Adrian ED: Discharges from vestibular receptors in the cat. J Physiol 1943; Mar 25; 101(4):389–407.)

Rotation causes maximal stimulation of the semicircular canals most nearly in the plane of rotation. Because the canals on one side of the head are a mirror image of those on the other side, the endolymph is displaced toward the ampulla on one side and away from it on the other. The pattern of stimulation reaching the brain varies with the direction as well as the plane of rotation. Linear acceleration probably fails to displace the cupula and therefore does not stimulate the cristae. However, when one part of the labyrinth is destroyed, other parts take over its functions. **Clinical Box 11–2** describes the characteristic eye movements that occur during a period of rotation.

RESPONSES TO LINEAR ACCELERATION

The utricular and saccular maculae respond to horizontal and vertical acceleration, respectively. The otoliths in the surrounding membrane are denser than the endolymph, and acceleration in any direction causes them to be displaced in the opposite direction, distorting the hair cell processes and generating activity in the vestibular nerve. The maculae also discharge in the absence of head movement due to the pull of gravity on the otoliths.

The impulses generated from these receptors are partly responsible for **labyrinth righting reflexes.** The reflex is initiated by tilting of the head that stimulates the otolithic organs; the response is a compensatory contraction of the neck muscles to keep the head level. A **vestibulo-ocular reflex** stabilizes images on the retina during head movements. Vestibular stimulation during the rotation leads to inhibition of extraocular muscles on one side and activation on the extraocular muscles on the other side.

CLINICAL BOX 11–2

Nystagmus

Nystagmus is the characteristic jerky movement of the eye observed at the start and end of a period of rotation. It is actually a reflex that maintains visual fixation on stationary points while the body rotates. When rotation starts, the eyes move slowly in a direction opposite to the direction of rotation, maintaining visual fixation (**vestibulo-ocular reflex**). When the limit of this movement is reached, the eyes quickly snap back to a new fixation point and then again move slowly in the other direction. The slow component is initiated by impulses from the vestibular labyrinths; the quick component is triggered by a center in the brainstem. Nystagmus is frequently horizontal (ie, the eyes move in the horizontal plane), but it can be vertical (when the head is tipped sideways during rotation) or rotatory (when the head is tipped forward). By convention, the direction of eye movement in nystagmus is identified by the direction of the quick component. The direction of the quick component during rotation is the same as that of the rotation, but the postrotatory nystagmus that occurs due to displacement of the cupula when rotation is stopped is in the opposite direction. When nystagmus is seen at rest, it is a sign of a pathology. Two examples of this are **congenital nystagmus** that is seen at birth and **acquired nystagmus** that occurs later in life. In these clinical cases, nystagmus can persist for hours at rest. Acquired nystagmus can be seen in patients with acute **temporal bone fracture** affecting **semicircular canals** or after damage to the **flocculonodular lobe** or the **fastigial nucleus.** It can also occur as a result of stroke, multiple sclerosis, head injury, and brain tumors. Some drugs (especially antiseizure drugs), alcohol, and sedatives can cause nystagmus.

Nystagmus can be used as a diagnostic indicator of the integrity of the vestibular system. Caloric stimulation can be used to test the function of the

vestibular labyrinth. The semicircular canals are stimulated by instilling warm (40°C) or cold (30°C) water into the external auditory meatus. The temperature difference sets up convection currents in the **endolymph,** with consequent motion of the cupula. In healthy persons, warm water causes nystagmus that bears toward the stimulus, whereas cold water induces nystagmus that bears toward the opposite ear. This test is given the mnemonic **COWS (C**old water nystagmus is **O**pposite side; **W**arm water nystagmus is **S**ame side). In the case of a unilateral lesion in the vestibular pathway, nystagmus is reduced or absent on the side of the lesion. To avoid nystagmus, vertigo, and nausea when irrigating the ear canals in the treatment of ear infections, the fluid used should be at body temperature.

THERAPEUTIC HIGHLIGHTS

There is no cure for acquired nystagmus, and treatment depends on the cause. Correcting the underlying cause (stopping drug usage, surgical removal of a tumor) is often the treatment of choice. Also, rectus muscle surgery has been used successfully to treat some cases of acquired nystagmus. Short-term correction of nystagmus can result from injections of **botulinum toxin** (Botox) to paralyze the ocular muscles.

Although most of the responses to stimulation of the maculae are reflex in nature, vestibular impulses also reach the cerebral cortex. These impulses may mediate conscious perception of motion and supply part of the information necessary for orientation in space. **Vertigo** is the sensation of rotation in the absence of actual rotation and is a prominent symptom when one labyrinth is inflamed.

SPATIAL ORIENTATION

Orientation in space depends in part on input from the vestibular receptors, but visual cues are also important. Spatial orientation also uses information from proprioceptors in joint capsules and from cutaneous touch and pressure receptors. These four inputs are synthesized at a cortical level into a continuous picture of the individual's orientation in space. **Clinical Box 11–3** describes some common vestibular disorders.

CLINICAL BOX 11–3

Vestibular Disorders

Vestibular balance disorders are the ninth most common reason for visits to a primary care clinician. It is one of the most common reasons elderly people seek medical advice. Patients often describe balance problems in terms of vertigo, dizziness, lightheadedness, and motion sickness. Neither lightheadedness nor dizziness is necessarily a symptom of vestibular problems, but **vertigo** is a prominent symptom of a disorder of the inner ear or vestibular system, especially when one labyrinth is inflamed. **Benign paroxysmal positional vertigo** is the most common vestibular disorder and is characterized by episodes of vertigo that occur with particular changes in body position (eg, turning over in bed and bending over). One possible cause is that **otoconia** from the utricle separate from the otolith membrane and become lodged in the canal or cupula of the semicircular canal. This causes abnormal deflections when the head changes position relative to gravity.

Ménière disease is an abnormality of the inner ear causing vertigo or severe dizziness, tinnitus, fluctuating hearing loss, and the sensation of pressure or pain in the affected ear lasting several hours. Symptoms can occur suddenly and recur daily or very rarely. The hearing loss is initially transient but can become permanent. The pathophysiology may involve an immune reaction. An inflammatory response can increase fluid volume within the membranous labyrinth, causing it to rupture and allowing the endolymph and perilymph to mix together. The worldwide prevalence for Ménière disease is ∼12 per 1000 individuals. It is diagnosed most often between the ages of 30 and 60, and it affects both sexes similarly.

The nausea, blood pressure changes, sweating, pallor, and vomiting that are the well-known symptoms of **motion sickness** are produced by excessive vestibular stimulation and occur when conflicting information is fed into the vestibular and other sensory systems. **Space motion sickness** (ie, the nausea, vomiting, and vertigo experienced by astronauts) develops when they are first exposed to microgravity and often wears off after a few days of space flight. It can then recur with reentry, as the force of gravity increases again. It is due to mismatches in neural input created by changes in the input from some parts of the vestibular apparatus and other gravity sensors without corresponding changes in the other spatial orientation inputs.

THERAPEUTIC HIGHLIGHTS

Symptoms of benign paroxysmal positional vertigo often subside over weeks or months, but if treatment is needed, one option is a procedure called **canalith repositioning.** This consists of simple and slow maneuvers to position your head to move the otoconia from the semicircular canals back into the vestibule that houses the utricle. There is no cure for Ménière disease, but the symptoms can be controlled by reducing the fluid retention through dietary changes (low-salt or salt-free diet, no caffeine, no alcohol) or medications such as diuretics (eg, **hydrochlorothiazide**). Individuals with Ménière disease often respond to drugs used to alleviate the symptoms of vertigo. **Vestibulosuppressants** such as the **antihistamine meclizine** decrease the excitability of the middle ear labyrinth and block conduction in middle ear vestibular-cerebellar pathway. Motion sickness commonly can be prevented with the use of antihistamines or **scopolamine,** a **cholinergic muscarinic receptor antagonist.**

CHAPTER SUMMARY

- The external ear includes the auricle, external auditory meatus, and tympanic membrane; the external ear captures sound waves and directs them toward the middle ear. The middle ear contains three bones (malleus, incus, and stapes), the Eustachian tube, and the tensor tympani and stapedius muscles. The inner ear contains the cochlea with the receptors (hair cells) for hearing and the semicircular canals with hair cell receptors for balance.
- The pressure changes produced by sound waves cause the tympanic membrane to move in and out; thus, it functions as a resonator to reproduce the vibrations of the sound source. The auditory ossicles serve as a lever system to convert the vibrations of the tympanic membrane into movements of the stapes against the perilymph-filled scala vestibuli of the cochlea.
- Sound is the sensation produced when longitudinal vibrations of air molecules strike the tympanic membrane. The hair cells in the organ of Corti signal hearing. The stereocilia provide a mechanism for generating changes in membrane potential proportional to the direction and distance the hair moves.
- Loudness is correlated with the amplitude of a sound wave, pitch with the frequency, and timbre with harmonic vibrations.

- The activity within the auditory pathway passes from the eighth cranial nerve afferent fibers to the dorsal and ventral cochlear nuclei to the inferior colliculi to the thalamic medial geniculate body and then to the auditory cortex.
- Tinnitus, a high-pitched ringing in the ear, can result from injury to inner ear hair cells. Presbycusis is age-related hearing loss due to the gradual cumulative loss of hair cells. Single-gene mutations can cause hearing loss; these monogenic forms of deafness are defined as either syndromic (hearing loss associated with other abnormalities) or nonsyndromic (only hearing loss).
- Sensorineural hearing loss is usually due to loss of cochlear hair cells but can result from damage to the eighth cranial nerve or central auditory pathway. Conductive hearing loss is due to impaired sound transmission in the external or middle ear and impacts all sound frequencies. Conductive and sensorineural deafness can be differentiated by simple tests (eg, Rinne test) with a tuning fork.
- Cochlear implants consist of a microphone, a speech processor, a transmitter and receiver/stimulator that converts sounds into electrical impulses, and an electrode array that sends the impulses to the auditory nerve. Hearing aids consist of a microphone, an amplifier and a speaker. Analog hearing aids convert sound waves to electrical signals; digital hearing aids convert sound waves into numerical codes that provide information about pitch or loudness.
- Rotational acceleration stimulates the crista in the semicircular canals, displacing the endolymph in a direction opposite to the direction of rotation, deforming the cupula and bending the hair cell. The utricle responds to horizontal acceleration and the saccule to vertical acceleration. Acceleration in any direction displaces the otoliths, distorting the hair cell processes and generating neural activity.
- Spatial orientation is dependent on input from vestibular receptors, visual cues, proprioceptors in joint capsules, and cutaneous touch and pressure receptors.
- Nystagmus is the characteristic jerky movement of the eye at the start and end of a period of rotation; it is actually a reflex that maintains visual fixation on stationary points while the body rotates (vestibulo-ocular reflex). A test that induces nystagmus (COWS) by instilling warm or cold water into the external auditory meatus can be used to evaluate the integrity of the

vestibular system.

- Benign paroxysmal positional vertigo is the most common vestibular disorder and is characterized by episodes of vertigo that occur with particular changes in body. Ménière disease is an abnormality of the inner ear that causes vertigo, tinnitus, hearing loss, and sensation of pressure or pain in the affected ear. Symptoms of motion sickness (eg, nausea, sweating, pallor, and vomiting) occur when conflicting information is fed into the vestibular and other sensory systems.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A 9-year-old girl complained of ear pain due to an inflammation and build-up of fluids in the middle ear. She was diagnosed with a middle ear infection, acute otitis media of bacterial origin, and she was treated with an antibiotic. The middle ear contains

   A. hair cells that mediate linear acceleration.

   B. the membranous labyrinth containing endolymph.

   C. the bony labyrinth containing perilymph fluid.

   D. cochlear hair cells that mediate hearing.

   E. the auditory ossicles and the tensor tympani and stapedius muscles.

2. A 2-year-old girl was diagnosed with a type of sensorineural deafness. After being evaluated by an audiologist, she was determined to be a good candidate for a cochlear implant. A cochlear implant is comprised of

   A. a microphone, transmitter, and receiver that converts sound into vibrating waves.

   B. artificial hair cells that are able to replace damaged hair cells in the inner ear.

   C. a microphone, amplifier, and speaker that can convert sound waves into electrical signals.

   D. a microphone, speech processor, transmitter, and receiver that converts sound into electrical impulses.

   E. a microphone, transmitter, and receiver that converts sound into numerical codes that provide information about pitch or loudness.

3. After playing the violin for the Boston Symphony Orchestra for 18 years, a

40-year-old man was given the opportunity to follow the dream he had as a child to be a physician like his father. Compared to other students in his medical school class, what distinctive features might be expressed in his auditory system?

A. The pitch of his conversational voice will be about 120 Hz compared to his younger male classmates whose voices are likely to have a pitch of about 250 Hz.

B. When presented with musical tones, a larger area of his auditory cortex will be activated compared to the area activated in the cortex of his classmates who are not musically inclined.

C. As a musician, he will be better able to localize sound than most of his classmates.

D. The Wernicke area on the left side of his brain will be more concerned with melody, pitch, and sound.

E. He will be able to distinguish about 2000 pitches; in contrast, his younger classmates will be able to distinguish only about 1000 pitches.

4. A 40-year-old man, employed as a road construction worker for nearly 20 years, went to his clinician to report that he recently began to notice difficulty hearing during normal conversations. A Weber test showed that sound from a vibrating tuning fork was localized to the right ear. A Schwabach test showed that bone conduction was below normal. A Rinne test showed that both air and bone conductions were abnormal, but air conduction lasted longer than bone conduction. The diagnosis was

A. sensorial hearing loss in both ears.

B. conduction deafness in the right ear.

C. sensorial deafness in the right ear.

D. conduction deafness in the left ear.

E. sensorineural deafness in the left ear.

5. A medical resident was asked to give a lecture to medical students on how sound is transmitted from the environment through the ear. He would have said the following about the roles of the auditory ossicles in this process.

A. The movement of the malleus transmits the vibrations of the manubrium to the incus.

B. The movement of the incus allows the sound wave to go through the oval window.

C. The movement of the stapes transmits the vibrations of the manubrium to

the incus.

D. The movement of the stapes allows the sound wave to go through the round window.

E. The movement of the malleus transmits the vibrations of the tympanic membrane to the stapes.

6. A medical resident was asked to give a lecture to medical students on how the brain synthesizes information to provide the sense of the position of the body in space. He would have said that the following sensory inputs play an important role in spatial orientation.

A. vestibular receptors, cochlear receptors, retinal receptors, and cutaneous pressure receptors

B. cochlear receptors, retinal receptors, proprioceptors, and cutaneous touch and pressure receptors

C. vestibular receptors, visual cues, proprioceptors in joints, and cutaneous touch and pressure receptors

D. middle ear receptors, visual cues, IA and IB sensory fibers, and touch receptors

E. organ of Corti hair cells, membranous labyrinth hair cells, visual clues, and touch receptors

7. The components of the auditory pathway are

A. sensory fibers in the acoustic branch of the eighth cranial nerve, the lateral cochlear nucleus, the superior colliculus, and the auditory cortex.

B. sensory fibers in the auditory branch of the eighth cranial nerve, the lateral cochlear nucleus, the inferior colliculus, lateral geniculate body, and the superior temporal gyrus of the cortex.

C. afferent fibers of the eighth cranial nerve, the dorsal and ventral cochlear nuclei, the inferior colliculi, the lateral geniculate body, and the auditory cortex.

D. sensory fibers in the cochlear branch of the eighth cranial nerve, the dorsal and ventral cochlear nuclei, the inferior colliculus, the medial geniculate body, and the auditory cortex.

E. afferent fibers of the cochlear branch of the eighth cranial nerve, the ventral cochlear nuclei, the superior colliculi, the lateral geniculate body, and the superior temporal gyrus of the cortex.

8. A healthy male medical student volunteered to undergo evaluation of the function of his vestibular system for a class demonstration. The direction of

his nystagmus is expected to be vertical when he is rotated

A. after warm water is put in one of his ears.

B. with his head tipped backward.

C. after cold water is put in both of his ears.

D. with his head tipped sideways.

E. with his head tipped forward.

9. A 65-year old man had an acute injury to the utricle of the inner ear, causing problems with balance. In the utricle, tip links in hair cells are involved in

A. formation of perilymph.

B. depolarization of the stria vascularis.

C. movements of the basement membrane.

D. perception of sound.

E. regulation of distortion-activated ion channels.

10. A 40-year old woman made an appointment with an otolaryngologist due to ringing in her ear that has been interfering with her ability to concentrate. What is a likely diagnosis and a potential cause of this symptom?

A. Pendred syndrome due to inflammation of the middle ear hair cells that induce random electrical impulses in the auditory nerve.

B. Syndromic hearing due to a single gene mutation.

C. Presbycusis that can result from injury to inner ear hair cells that induce random electrical impulses in the eighth cranial nerve.

D. Tinnitus that can be caused by injury to inner ear hair cells that induce random electrical impulses in the auditory nerve.

E. Tinnitus due to an accumulation of perilymph moving over inner ear hair cells.

11. An MD/PhD candidate was doing research on the generation of changes in the membrane potential of cochlear inner hair cells. What steps are involved in this process?

A. When the shorter stereocilia are pushed toward the taller ones, the channel open time is increased, and K^+ and Ca^{2+} enter via the channel and induce hyperpolarization.

B. When the shorter stereocilia are pushed toward the taller ones, the channel open time is increased, and K^+ and Ca^{2+} enter via the channel and induce depolarization.

C. When the taller stereocilia are pushed toward the shorter ones, the channel

open time is increased, and Na^+ and Ca^{2+} exit via the channel and induce hyperpolarization.

D. When the stereocilia move toward the sound source, the channel open time is decreased, and K^+ and Cl^- exit via the channel and induce depolarization.

E. When the stereocilia move toward the sound source, the channel open time is increased, and K^+ and Cl^- enter via the channel and induce depolarization.

12. A 45-year-old woman sought medical advice when she experienced sudden onset of vertigo, tinnitus and hearing loss in her left ear, nausea, and vomiting. She was referred to an otolaryngologist to rule out Ménière disease. Which of the following are possible causes of Ménière disease?

A. Ménière disease is autosomal dominant genetic disorder that weakens the membranous labyrinth of the inner ear.

B. The hair cells of the cochlea are altered to give the sensation of motion even at rest.

C. The otoliths dislodge, enter the semicircular canal, and stimulate the hair cells.

D. An inflammatory response increases fluid volume within the membranous labyrinth, causing it to rupture and allowing the endolymph and perilymph to intermix.

E. The membranous labyrinth on one side has become inflamed.

CHAPTER 12

Reflex & Voluntary Control of Posture & Movement

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the basic elements of a reflex pathway.
- Identify the components, function, and afferent nerve fibers of the muscle spindle.
- Explain the neuronal response initiated by striking the patellar tendon (patellar tendon or knee jerk reflex) that leads to contraction of a muscle.
- Explain how the activity of γ-motor neurons alters the response to muscle stretch.
- Describe the role of Golgi tendon organs in the control of skeletal muscle.
- Define physiologic tremor, clonus, and muscle tone.
- Identify the components and function of the withdrawal reflex pathway.
- Define spinal shock and describe the initial and long-term changes in spinal reflexes that follow spinal cord injury.
- Describe how skilled movements are planned and carried out.
- Contrast the organization of the central pathways involved in the control of axial (posture) and distal (skilled movement, fine motor movements)

muscles.

- Describe the clinical tests and findings that distinguish between upper and lower motor neuron disorders, including the Babinski sign and clonus.
- Identify the pathophysiology and characteristics of cerebral palsy, decerebrate rigidity, and decorticate rigidity.
- Identify the components of the basal ganglia and the pathways that interconnect them, along with the neurotransmitters in each pathway.
- Explain the pathophysiology and symptoms of Parkinson disease, Huntington disease, and other pathologies of the basal ganglia pathways.
- Discuss the functions of the cerebellum and the neurologic abnormalities produced by diseases of this part of the brain.

INTRODUCTION

Somatic motor function depends ultimately on the activity of the spinal motor neurons and homologous neurons in the motor nuclei of the cranial nerves. These neurons, the final common pathways to skeletal muscle, are bombarded by impulses from an immense array of descending pathways, other spinal neurons, and peripheral afferents. Some of these inputs end directly on α-motor neurons, but many exert their effects via interneurons or via γ-motor neurons to the muscle spindles and back through the Ia afferent fibers to the spinal cord. It is the integrated activity of these multiple inputs from spinal, brainstem, midbrain, and cortical levels that regulates the posture of the body and makes coordinated movement possible.

The inputs converging on motor neurons have three major functions: to induce voluntary activity, to adjust body posture, and to make movements smooth and precise. The patterns of voluntary activity are planned within the brain, and the commands are sent to the muscles primarily via the corticospinal and corticobulbar systems. Posture is continually adjusted not only before but also during movement by information carried in descending brainstem pathways and peripheral afferents. Movement is smoothed and coordinated by the medial and intermediate portions of the cerebellum (spinocerebellum) and its connections. The basal ganglia and the lateral portions of the cerebellum (cerebrocerebellum) are part of a feedback circuit to the premotor and motor cortex that is concerned with planning and organizing voluntary movement.

Chapter 11 introduced somatomotor control by describing the role of the

vestibular system in control of balance and equilibrium. This chapter considers two types of motor output: reflex (involuntary) and voluntary. A subdivision of reflex responses includes some rhythmic movements such as swallowing, chewing, scratching, and walking, which are largely involuntary but are subject to voluntary adjustment and control.

GENERAL PROPERTIES OF REFLEXES

The basic unit of integrated reflex activity is the **reflex arc** that consists of a sense organ, an afferent neuron, one or more synapses within a central integrating station, an efferent neuron, and an effector. The afferent neurons enter via the dorsal roots or cranial nerves and have their cell bodies in dorsal root ganglia or homologous cranial nerve ganglia. The efferent fibers leave via the ventral roots or motor cranial nerves.

Activity in the reflex arc starts in a sensory receptor with a **receptor potential** (see Chapter 8) whose magnitude is proportional to the strength of the stimulus **(Figure 12–1)**. This generates all-or-none action potentials in the afferent nerve, the number of action potentials being proportional to the size of the receptor potential. In the central nervous system (CNS), the responses are again graded in terms of excitatory postsynaptic potentials (EPSPs) and inhibitory postsynaptic potentials (IPSPs) at the synaptic junctions (see Chapter 6). When the action potentials in the efferent motor nerve reach the effector, a graded response is induced. If the graded response is adequate to produce action potentials in the muscle, muscle contraction will ensue. The activity in the reflex arc is modified by the multiple inputs converging on the efferent neurons or at any synaptic station within the reflex arc.

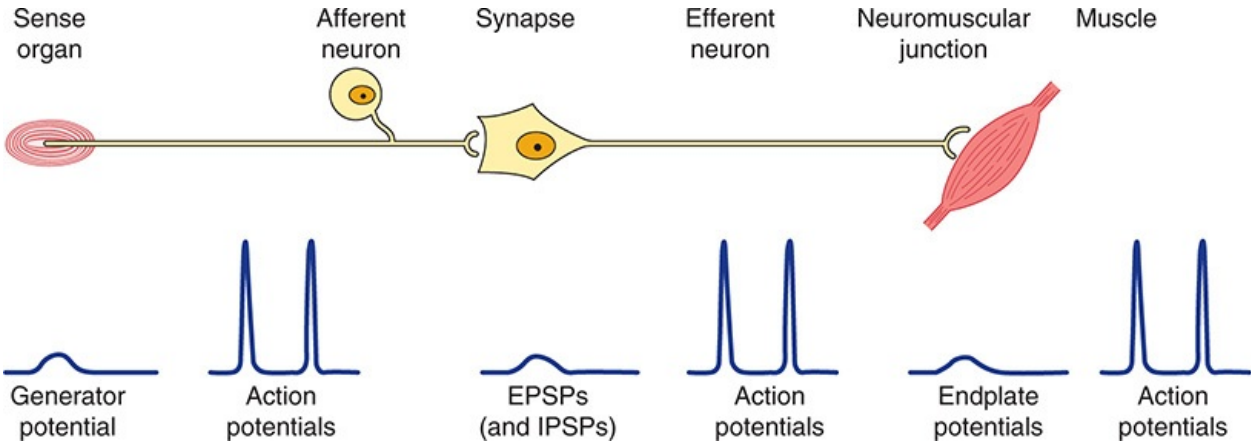


FIGURE 12–1 The reflex arc. Note that at the receptor and in the CNS a

nonpropagated graded response occurs that is proportional to the magnitude of the stimulus. The response at the neuromuscular junction is also graded, though under normal conditions it is always large enough to produce a response in skeletal muscle. On the other hand, in the portions of the arc specialized for transmission (afferent and efferent nerve fibers, muscle membrane), the responses are all-or-none action potentials.

Reflex activity is stereotyped and specific in that a particular stimulus elicits a particular response. The fact that reflex responses are stereotyped does not exclude the possibility of their being modified by experience. Reflexes are adaptable and can be modified to perform motor tasks and maintain balance. Descending inputs from higher brain regions play an important role in modulating and adapting spinal reflexes.

The **α-motor neurons** that supply the extrafusal fibers in skeletal muscles are the efferent side of many reflex arcs. All neural influences affecting muscular contraction ultimately funnel through them to the muscles, thus they are called the **final common pathway.** Numerous inputs converge on α-motor neurons; the surface of an average motor neuron and its dendrites accommodates about 10,000 synaptic knobs. At least five inputs go from the same spinal segment to a typical spinal motor neuron. In addition to these, there are excitatory and inhibitory inputs, generally relayed via interneurons, from other levels of the spinal cord and multiple long-descending tracts from the brain. All of these pathways converge on and determine the activity in the final common pathway.

MONOSYNAPTIC REFLEXES: THE STRETCH REFLEX

The simplest reflex arc is the one with a single synapse between the afferent and efferent neurons (i.e., **monosynaptic reflexes**). Reflex arcs in which interneurons are interposed between the afferent and efferent neurons are called **polysynaptic reflexes.** There can be anywhere from two to hundreds of synapses in a polysynaptic reflex arc.

When a skeletal muscle with an intact nerve supply is stretched, it contracts. This response is called the **stretch reflex** or **myotatic reflex.** The stimulus that initiates this reflex is stretch of the muscle, and the response is contraction of the muscle being stretched. The sense organ is a small encapsulated spindle-like or fusiform-shaped structure called the **muscle spindle,** located within the fleshy part of the muscle. The impulses originating from the spindle are transmitted to

the CNS by fast sensory fibers that pass directly to the motor neurons that supply the same muscle. The neurotransmitter at the central synapse is glutamate. The stretch reflex is the best known and studied monosynaptic reflex and is typified by the **knee jerk reflex** (Clinical Box 12–1).

CLINICAL BOX 12–1

Knee Jerk Reflex

Tapping the patellar tendon elicits the **knee jerk,** a stretch reflex of the quadriceps femoris muscle, because the tap on the tendon stretches the muscle. A similar contraction is observed if the quadriceps is stretched manually. The knee jerk reflex is an example of a **deep tendon reflex (DTR)** in a neurologic exam and is graded on the following scale: 0 (absent), 1+ (hypoactive), 2+ (brisk, normal), 3+ (hyperactive without clonus), 4+ (hyperactive with mild clonus), and 5+ (hyperactive with sustained clonus). Absence of the knee jerk can signify an abnormality anywhere within the reflex arc, including the muscle spindle, the Ia afferent nerve fibers, or the motor neurons to the quadriceps muscle. The most common cause is a peripheral neuropathy from such things as diabetes, alcoholism, and toxins. A hyperactive reflex can signify an interruption of corticospinal and other descending pathways that suppress the activity in the reflex arc. DTR is not specific to an assessment of the knee jerk reflex; stretch reflexes can be elicited from most of the large muscles of the body. Tapping on the tendon of the triceps brachii, for example, causes an extensor response at the elbow as a result of reflex contraction of the triceps. Tapping on the Achilles tendon causes an ankle jerk due to reflex contraction of the gastrocnemius, and tapping on the side of the face causes a stretch reflex in the masseter. The spinal nerve involved in testing various DTRs is as follows: biceps (C5, C6 spinal nerve); triceps (C7 spinal nerve); patellar (L4 spinal nerve); and Achilles tendon (S1 spinal nerve).

STRUCTURE OF MUSCLE SPINDLES

Each muscle spindle has three essential elements: (1) a group of specialized **intrafusal muscle fibers** with contractile polar ends and a noncontractile center, (2) large diameter myelinated afferent nerves (types Ia and II) originating in the

central portion of the intrafusal fibers, and (3) small diameter myelinated efferent nerves supplying the polar contractile regions of the intrafusal fibers (Figure 12–2A). It is important to understand the relationship of these elements to each other and to the muscle itself to appreciate the role of this sense organ in signaling changes in the length of the muscle in which it is located. Changes in muscle length are associated with changes in joint angle; thus, muscle spindles provide information on position (ie, **proprioception**).

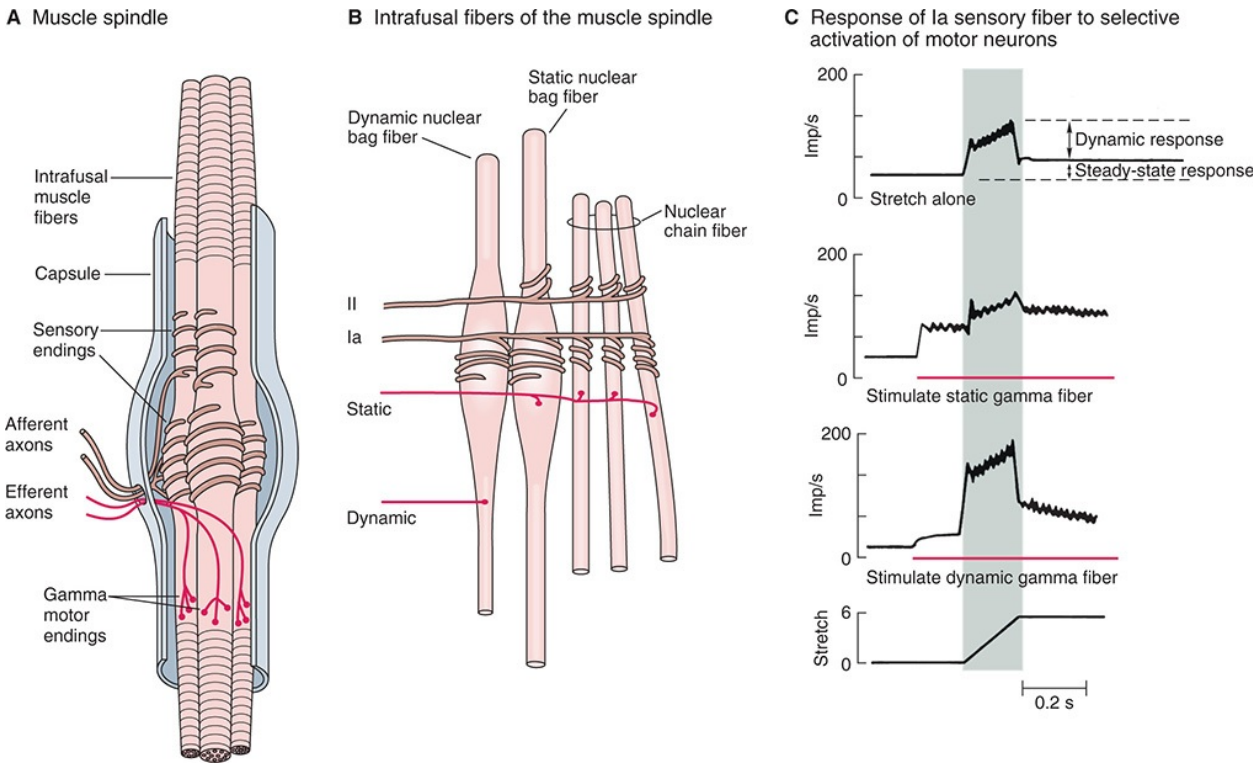


FIGURE 12–2 Mammalian muscle spindle. A) Diagrammatic representation of the main components of mammalian muscle spindle including intrafusal muscle fibers, afferent sensory fiber endings, and efferent motor fibers (γ-motor neurons). **B)** Three types of intrafusal muscle fibers: dynamic nuclear bag, static nuclear bag, and nuclear chain fibers. A single Ia afferent fiber innervates all three types of fibers to form a primary sensory ending. A group II sensory fiber innervates nuclear chain and static bag fibers to form a secondary sensory ending. Dynamic γ-motor neurons innervate dynamic bag fibers; static γ-motor neurons innervate combinations of chain and static bag fibers. **C)** Comparison of discharge pattern of Ia afferent activity during stretch alone and during stimulation of static or dynamic γ-motor neurons. Without γ-stimulation, Ia fibers show a small dynamic response to muscle stretch and a modest increase in steady-state firing. When static γ-motor neurons are activated, the steady-state

response increases and the dynamic response decreases. When dynamic γ-motor neurons are activated, the dynamic response is markedly increased but the steady-state response gradually returns to its original level. (Reproduced with permission from Gray H: *Gray's Anatomy: The Anatomical Basis of Clinical Practice*, 40th ed. St. Louis, MO: Churchill Livingstone/Elsevier; 2009.)

The **intrafusal fibers** are positioned in parallel to the **extrafusal fibers** (the regular contractile units of the muscle) with the ends of the spindle capsule attached to the tendons at either end of the muscle. Intrafusal fibers do not contribute to the overall contractile force of the muscle, but rather serve a pure sensory function. There are two types of intrafusal fibers in mammalian muscle spindles. The first type contains many nuclei in a dilated central area and is called a **nuclear bag fiber** (**Figure 12–2B**). There are two subtypes of nuclear bag fibers: **dynamic** and **static.** The second intrafusal fiber type, the **nuclear chain fiber,** is thinner and shorter and lacks a definite bag. Typically, each muscle spindle contains two or three nuclear bag fibers and about five nuclear chain fibers.

There are two kinds of sensory endings in each spindle, a single **primary (group Ia) ending** and up to eight **secondary (group II) endings** (Figure 12–2B). The Ia afferent fiber wraps around the center of the dynamic and static nuclear bag fibers and nuclear chain fibers. Group II sensory fibers are located adjacent to the centers of the static nuclear bag and nuclear chain fibers; these fibers do not innervate the dynamic nuclear bag fibers. Ia afferents are very sensitive to the velocity of the change in muscle length during a stretch **(dynamic response);** thus, they provide information about the speed of movements and allow for quick corrective movements. The steady-state (tonic) activity of group Ia and II afferents provide information on steady-state length of the muscle **(static response).** The top trace in **Figure 12–2C** shows the dynamic and static components of activity in a Ia afferent during muscle stretch. Note that they discharge most rapidly while the muscle is being stretched (shaded area of graphs) and less rapidly during sustained stretch.

The spindles have a motor nerve supply of their own called **γ-motor neurons;** they are 3–6 µm in diameter and constitute about 30% of the fibers in the ventral roots. There are two types of γ-motor neurons: **dynamic** that supply dynamic nuclear bag fibers and **static** that supply static nuclear bag fibers and nuclear chain fibers. Activation of dynamic γ-motor neurons increases the dynamic sensitivity of the group Ia endings. Activation of the static γ-motor neurons increases the tonic level of activity in both group Ia and II endings, decreases the dynamic sensitivity of group Ia afferents, and can prevent silencing

of Ia afferents during muscle stretch (Figure 12–2C).

CENTRAL CONNECTIONS OF AFFERENT FIBERS

Ia fibers end directly on motor neurons supplying the extrafusal fibers of the same muscle **(Figure 12–3)**. The **reaction time** is the interval between the application of a stimulus and the response. In humans, the reaction time for a stretch reflex is 19–24 ms. Weak stimulation of the sensory nerve from the muscle that stimulates only Ia fibers causes a contractile response with a similar latency. Because the conduction velocities of the afferent and efferent fiber types are known and the distance from the muscle to the spinal cord can be measured, it is possible to calculate how much of the reaction time was taken up by conduction to and from the spinal cord. When this value is subtracted from the reaction time, the remainder (**central delay)** is the time taken for the reflex activity to traverse the spinal cord. The central delay for the knee jerk reflex is 0.6–0.9 ms. Because the minimum synaptic delay is 0.5 ms, only one synapse could have been traversed.

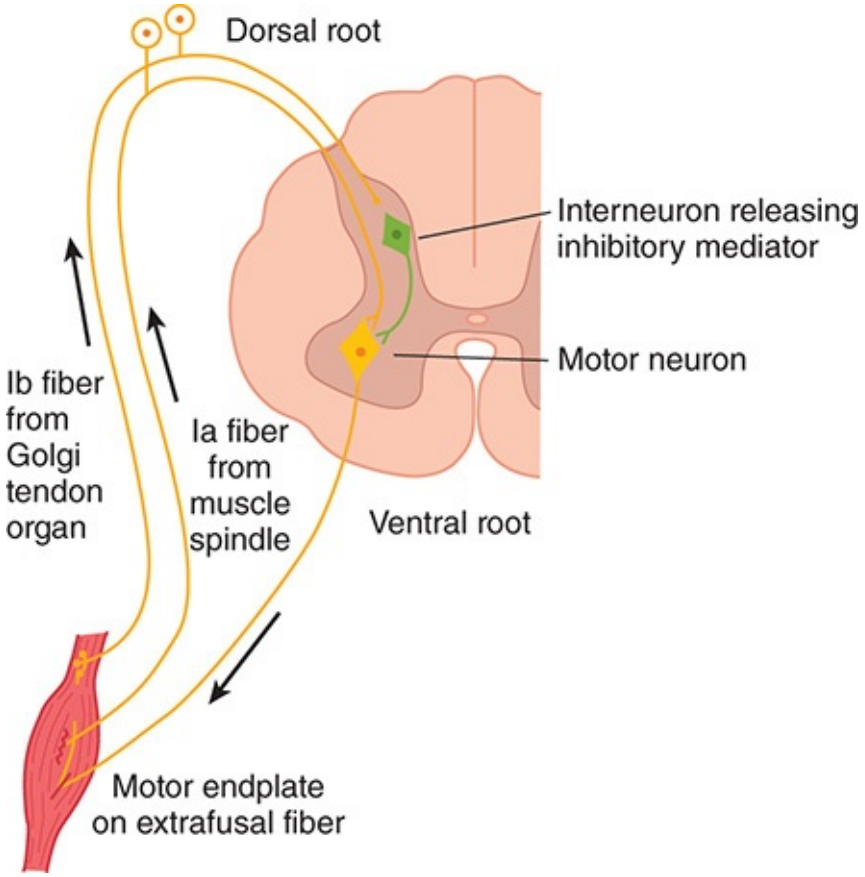


FIGURE 12–3 Diagram illustrating the pathways responsible for the stretch reflex and the inverse stretch reflex. Stretch stimulates the muscle spindle, which activates Ia fibers that excite the motor neuron. Stretch also stimulates the Golgi tendon organ, which activates Ib fibers that excite an interneuron that releases the inhibitory mediator glycine. With strong stretch, the resulting hyperpolarization of the motor neuron is so great that it stops discharging.

FUNCTION OF MUSCLE SPINDLES

When the muscle spindle is stretched, its sensory endings are distorted and receptor potentials are generated. These in turn set up action potentials in the sensory fibers at a frequency proportional to the degree of stretching. Because the spindle is in parallel with the extrafusal fibers, when the muscle is passively stretched, the spindles are also stretched, referred to as **“loading the spindle.”** This initiates reflex contraction of the extrafusal fibers in the muscle. On the other hand, the spindle afferents characteristically stop firing when the muscle is made to contract by electrical stimulation of the α-motor neurons to the extrafusal fibers because the muscle shortens while the spindle is unloaded

(Figure 12–4).

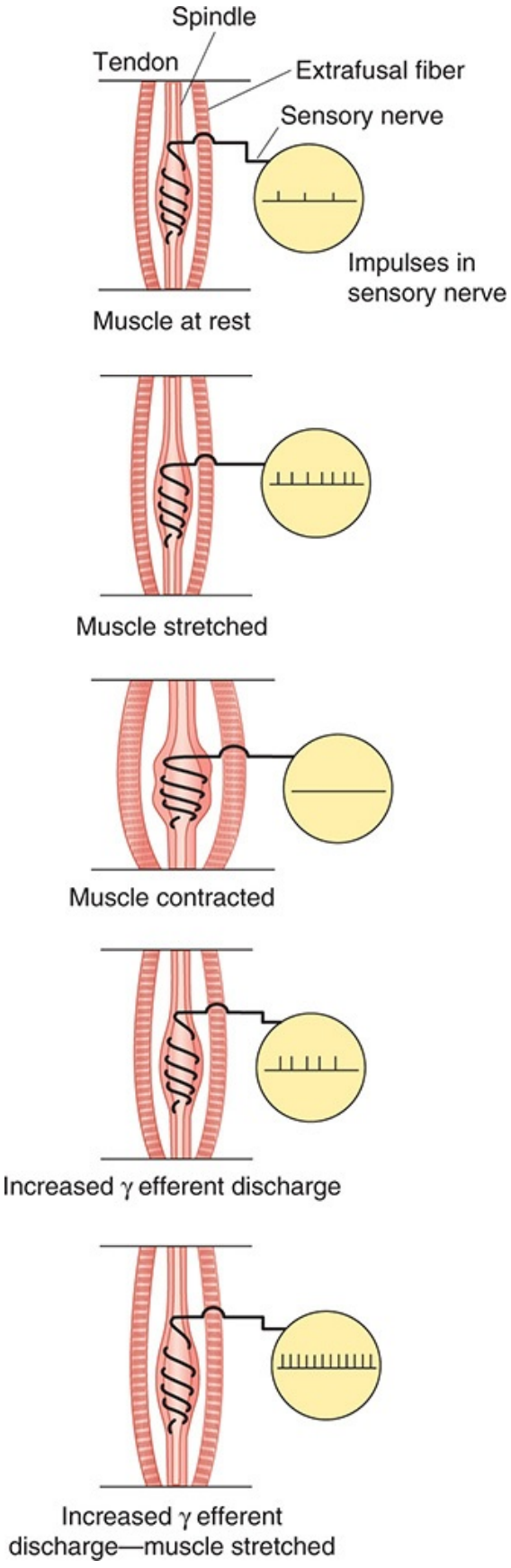

Spindle
Tendon
Extrafusal fiber
Sensory nerve
Impulses in sensory nerve
Muscle at rest
Muscle stretched
Muscle contracted
Increased γ efferent discharge
Increased γ efferent discharge—muscle stretched

FIGURE 12–4 Effect of various conditions on muscle spindle discharge. When the whole muscle is stretched, the muscle spindle is also stretched and its sensory endings are activated at a frequency proportional to the degree of stretching ("loading the spindle"). Spindle afferents stop firing when the muscle contracts ("unloading the spindle"). Stimulation of γ-motor neurons causes the contractile ends of the intrafusal fibers to shorten. This stretches the nuclear bag region, initiating impulses in sensory fibers. If the whole muscle is stretched during stimulation of the γ-motor neurons, the rate of discharge in sensory fibers is further increased.

The muscle spindle and its reflex connections constitute a feedback device that operates to maintain muscle length. If the muscle is stretched, spindle discharge increases and reflex shortening is produced. If the muscle is shortened without a change in γ-motor neuron discharge, spindle afferent activity decreases and the muscle relaxes.

Dynamic and static responses of muscle spindle afferents influence **physiologic tremor.** The response of the Ia sensory fiber endings to the dynamic (phasic) as well as the static events in the muscle is important because the prompt, marked phasic response helps dampen oscillations caused by conduction delays in the feedback loop regulating muscle length. Normally a small oscillation occurs in this feedback loop. This physiologic tremor has low amplitude (barely visible to the naked eye) and a frequency of approximately 10 Hz. Physiologic tremor is a normal phenomenon that affects everyone while maintaining posture or during movements. However, the tremor would be more prominent if it were not for the sensitivity of the spindle to velocity of stretch. It can become exaggerated in some situations such as when we are anxious or tired or because of drug toxicity. Numerous factors contribute to the genesis of physiologic tremor. It is dependent on both central **(inferior olive)** sources and peripheral factors including motor unit firing rates, reflexes, and mechanical resonance.

RECIPROCAL INNERVATION

When a stretch reflex occurs, the opposing muscles relax due to **reciprocal innervation.** Impulses in the Ia fibers from the muscle spindles of the protagonist muscle cause postsynaptic inhibition of the motor neurons to the antagonists. A collateral from each Ia fiber passes in the spinal cord to an

inhibitory interneuron that synapses on a motor neuron supplying the antagonist muscles. This example of postsynaptic inhibition is discussed in Chapter 6, and the pathway is illustrated in Figure 6–5.

EFFECTS OF γ-MOTOR NEURON DISCHARGE

Activation of γ-motor neurons produces a very different picture from that produced by activation of the α-motor neurons. Activation of γ-motor neurons does not lead directly to detectable contraction of the muscles because intrafusal fibers are not strong enough or plentiful enough to cause shortening. However, their activation does cause the contractile ends of the intrafusal fibers to shorten and therefore stretches the nuclear bag portion of the spindles, deforming the endings, and initiating impulses in Ia fibers (Figure 12–4). This in turn can lead to reflex contraction of the muscle. Thus, muscles can be made to contract via activation of the α-motor neurons that innervate the extrafusal fibers or the γ-motor neurons that initiate contraction indirectly via the stretch reflex.

If the whole muscle is stretched during stimulation of the γ-motor neurons, the rate of discharge in the Ia fibers is further increased (Figure 12–4). Increased γ-motor neuron activity thus increases **spindle sensitivity** during stretch.

In response to descending excitatory input to spinal motor circuits, both α- and γ-motor neurons are activated. Because of this **"α-γ coactivation,"** intrafusal and extrafusal fibers shorten together, and spindle afferent activity can occur throughout the period of muscle contraction. In this way, the spindle remains capable of responding to stretch and reflexively adjusting α-motor neuron discharge.

CONTROL OF γ-MOTOR NEURON DISCHARGE

The γ-motor neurons are regulated by descending tracts originating in areas of the brain that also control α-motor neurons (described below). Via these pathways, the sensitivity of the muscle spindles and hence the threshold of the stretch reflexes in various parts of the body can be adjusted and shifted to meet the needs of postural control.

Other factors also influence γ-motor neuron activity. Anxiety causes an increased discharge, a fact that probably explains the hyperactive tendon reflexes sometimes seen in anxious patients. In addition, unexpected movement is associated with a greater efferent discharge. Stimulation of the skin, especially

by noxious agents, increases γ-motor neuron activity to ipsilateral flexor muscle spindles while decreasing that to extensors and produces the opposite pattern in the opposite limb. Trying to pull the hands apart when the flexed fingers are hooked together facilitates the knee jerk reflex **(Jendrassik maneuver),** and this may also be due to increase γ-motor neuron discharge initiated by afferent impulses from the hands.

INVERSE STRETCH REFLEX

Up to a point, the harder a muscle is stretched, the stronger is the reflex contraction. However, when the tension becomes great enough, contraction suddenly ceases and the muscle relaxes. This relaxation in response to strong stretch is called the **inverse stretch reflex.** The receptor for the inverse stretch reflex is in the **Golgi tendon organ** (Figure 12–5) that consists of a netlike collection of knobby nerve endings among the fascicles of a tendon. There are 3–25 muscle fibers per tendon organ. The fibers from the Golgi tendon organs are the Ib group of myelinated, rapidly conducting sensory nerve fibers. Activation of these Ib fibers leads to IPSPs in the motor neurons that supply the muscle from which the fibers arise. The Ib fibers end in the spinal cord on inhibitory interneurons that terminate directly on the motor neurons (Figure 12–3). They also make excitatory connections with motor neurons supplying antagonists to the muscle.

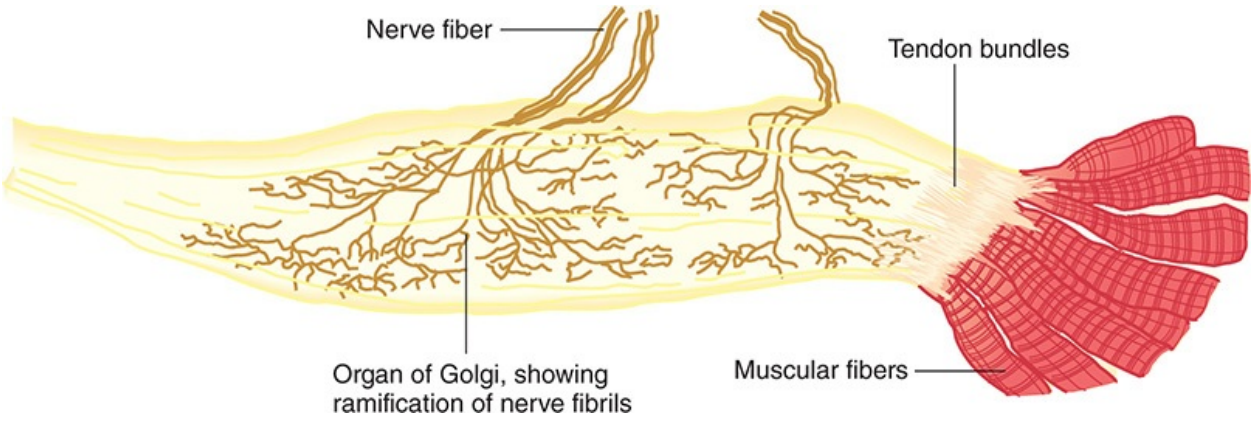


FIGURE 12–5 Golgi tendon organ. This organ is the receptor for the inverse stretch reflex and consists of a netlike collection of knobby nerve endings among the fascicles of a tendon. The innervation is the Ib group of myelinated, rapidly conducting sensory nerve fibers. (Reproduced with permission from Gray H: *Gray's Anatomy: The Anatomical Basis of Clinical Practice,* 40th ed. St. Louis,

MO: Churchill Livingstone/Elsevier; 2009.)

Because the Golgi tendon organs, unlike the spindles, are in series with the muscle fibers, they are stimulated by both passive stretch and active contraction of the muscle. The threshold of the Golgi tendon organs is low. The degree of stimulation by passive stretch is not great because the more elastic muscle fibers take up much of the stretch; therefore, it takes a strong stretch to produce relaxation. However, discharge is regularly produced by contraction of the muscle, and the Golgi tendon organ thus functions as a transducer in a feedback circuit that regulates muscle force in a manner analogous to the spindle feedback circuit that regulates muscle length.

CLINICAL BOX 12–2

Clonus

Clonus is characteristic of states in which there is increased γ-motor neuron activity. This neurologic sign is the occurrence of regular, repetitive, rhythmic contractions of a muscle subjected to sudden, maintained stretch. Only sustained clonus with five or more beats is considered abnormal. Ankle clonus is initiated by brisk, maintained dorsiflexion of the foot; the response is rhythmic plantar flexion at the ankle. The **stretch reflex-inverse stretch reflex sequence** may contribute to this response. However, it can occur on the basis of synchronized motor neuron discharge without Golgi tendon organ activation. The spindles of the tested muscle are hyperactive, and the burst of impulses from them activates all the motor neurons supplying the muscle at once. The consequent muscle contraction stops spindle discharge. However, the stretch has been maintained, and as soon as the muscle relaxes it is again stretched and the spindles stimulated. There are numerous causes of abnormal clonus including traumatic brain injury, brain tumors, strokes, and multiple sclerosis. Clonus may also occur in spinal cord injury that disrupts the descending cortical input to a spinal glycinergic inhibitory interneuron called the **Renshaw cell.** This cell receives excitatory input from α-motor neurons via axon collaterals (and in turn it inhibits the same α-motor neuron). In addition, cortical fibers activating ankle flexors synapse on Renshaw cells (as well as type Ia inhibitory interneurons) that inhibit the antagonistic ankle extensors. This circuitry prevents reflex stimulation of the extensors when flexors are active. Therefore, when the descending cortical fibers are

damaged **(upper motor neuron lesion),** the inhibition of antagonists is absent. The result is repetitive, sequential contraction of ankle flexors and extensors (clonus). Clonus may be seen in patients with amyotrophic lateral sclerosis (ALS), stroke, multiple sclerosis, spinal cord damage, epilepsy, liver or kidney failure, and hepatic encephalopathy.

THERAPEUTIC HIGHLIGHTS

Treatment of clonus often centers on its underlying cause. For some individuals, physical therapy that includes stretching exercises can reduce episodes of clonus. **Immunosuppressants** (eg, **azathioprine** and **corticosteroids**), **anticonvulsants** (eg, **primidone** and **levetiracetam**), and **tranquilizers** (eg, **clonazepam**) are beneficial in the treatment of clonus. **Botulinum toxin** has also been used to block the release of acetylcholine in the muscle, which triggers the rhythmic muscle contractions that are characteristic of clonus.

The primary endings in the spindles and the Golgi tendon organs together regulate the velocity of the muscle contraction, muscle length, and muscle force. The interaction of spindle discharge, tendon organ discharge, and reciprocal innervation determines the firing rate of α-motor neurons **(Clinical Box 12–2)**.

MUSCLE TONE

The resistance of a muscle to stretch is often referred to as its **tone** or **tonus.** If the motor nerve to a muscle is severed, the muscle offers very little resistance and is said to be **flaccid.** A **hypertonic (spastic)** muscle is one in which the resistance to stretch is high because of hyperactive stretch reflexes. Somewhere between the states of flaccidity and spasticity is the ill-defined area of normal tone. The muscles are generally hypotonic when the rate of γ-motor neuron discharge is low and hypertonic when it is high.

When the muscles are hypertonic, the sequence of moderate stretch → muscle contraction, strong stretch → muscle relaxation is clearly seen. Passive flexion of the elbow, for example, meets immediate resistance as a result of the stretch reflex in the triceps muscle. Further stretch activates the inverse stretch reflex. The resistance to flexion suddenly collapses, and the arm flexes. Continued passive flexion stretches the muscle again, and the sequence may be repeated. This sequence of resistance followed by a sudden decrease in resistance when a

limb is moved passively is known as the **clasp-knife effect** because of its resemblance to the closing of a pocket knife.

WITHDRAWAL REFLEX

The **withdrawal reflex** is a typical polysynaptic reflex that occurs in response to a noxious stimulus to the skin or subcutaneous tissues and muscle. The response is flexor muscle contraction and inhibition of extensor muscles, so that the body part stimulated is flexed and withdrawn from the stimulus. When a strong stimulus is applied to a limb, the response includes not only flexion and withdrawal of that limb but also extension of the opposite limb. This **crossed extensor response** is properly part of the withdrawal reflex. Strong stimuli can generate activity in the interneuron pool that spreads to all four extremities.

IMPORTANCE OF THE WITHDRAWAL REFLEX

Flexor responses can be produced by innocuous stimulation of the skin or by stretch of the muscle, but strong flexor responses with withdrawal are initiated only by stimuli that are noxious or at least potentially harmful (ie, **nociceptive stimuli**). The withdrawal reflex serves a protective function as flexion of the stimulated limb gets it away from the source of irritation, and extension of the other limb supports the body.

A weak noxious stimulus to one foot evokes a minimal flexion response; stronger stimuli produce greater and greater flexion as the stimulus irradiates to more and more of the motor neuron pool supplying the muscles of the limb. Stronger stimuli also cause a more prolonged response. A weak stimulus causes one quick flexion movement; a strong stimulus causes prolonged flexion and sometimes a series of flexion movements. This prolonged response is due to prolonged, repeated firing of the motor neurons **(after-discharge)** that is due to continued bombardment of motor neurons by impulses arriving by complicated and circuitous polysynaptic paths.

As the strength of a noxious stimulus is increased, the reaction time is shortened. Spatial and temporal facilitation occur at synapses in the polysynaptic pathway. Stronger stimuli produce more action potentials per second in the active branches and cause more branches to become active; summation of the EPSPs to the threshold level for action potential generation occurs more rapidly.

SPINAL INTEGRATION OF REFLEXES

The responses to spinal cord injury (**SCI**) illustrate the integration of reflexes at the spinal level. The deficits seen after SCI vary, of course, depending on the level of the injury. **Clinical Box 12–3** provides information on long-term problems related to SCI and recent advancements in treatment options.

CLINICAL BOX 12–3

Spinal Cord Injury

It has been estimated that the worldwide annual incidence of sustaining **SCI** is between 10 and 83 per million of the population. Leading causes are vehicular accidents, violence, and sports injuries. The mean age of patients who sustain SCI is 33 years old, and men outnumber women with a nearly 4:1 ratio. Approximately 52% of SCI cases result in quadriplegia and about 42% lead to paraplegia. With quadriplegia, the threshold for the withdrawal reflex is very low; even minor noxious stimuli may cause not only prolonged withdrawal of one extremity but marked flexion-extension patterns in the other three limbs. Stretch reflexes are also hyperactive. Afferent stimuli irradiate from one reflex center to another after SCI. When even a relatively minor noxious stimulus is applied to the skin, it may activate autonomic neurons and produce evacuation of the bladder and rectum, sweating, pallor, and blood pressure swings in addition to the withdrawal response. This distressing **mass reflex** can sometimes be used to give patients with paraplegia a degree of bladder and bowel control. They can be trained to initiate urination and defecation by stroking or pinching their thighs, thus producing an intentional mass reflex. If the cord section is incomplete, the flexor spasms initiated by noxious stimuli can be associated with bursts of pain that are particularly bothersome. They can be treated with considerable success with baclofen, a $GABA_B$ receptor agonist that crosses the blood-brain barrier and facilitates inhibition.

THERAPEUTIC HIGHLIGHTS

Treatment of SCI patients presents complex problems. Administration of corticosteroids such as **methylprednisolone** may have beneficial effects by fostering recovery and minimizing loss of function after SCI. They need to be

given soon after the injury and then discontinued because of the well-established deleterious effects of long-term corticosteroid treatment. Their immediate value is likely due to reduction of the inflammatory response in the damaged tissue. Because SCI patients are immobile, a negative nitrogen balance develops and large amounts of body protein are catabolized. Their body weight compresses the circulation to the skin over bony prominences, causing formation of **pressure ulcers.** The ulcers heal poorly and are prone to infection because of body protein depletion. The tissues that are broken down include the protein matrix of bone and this, plus the immobilization, cause Ca^{2+} to be released in large amounts, leading to **hypercalcemia, hypercalciuria,** and formation of **calcium stones** in the urinary tract. The combination of stones and bladder paralysis cause urinary stasis, which predisposes to **urinary tract infection,** the most common complication of SCI. The search continues for ways to get axons of neurons in the spinal cord to regenerate. Administration of **neurotrophins** shows some promise in experimental animals, and so does implantation of **embryonic stem cells** at the site of injury. Another possibility being explored is bypassing the site of SCI with **brain–computer interface devices.** However, these novel approaches are a long way from routine clinical use.

In all vertebrates, transection of the spinal cord is followed by a period of **spinal shock** during which all spinal reflex responses are profoundly depressed. Subsequently, reflex responses return and become hyperactive. The duration of spinal shock is proportional to the degree of encephalization of motor function in the various species. In frogs and rats it lasts for minutes; in dogs and cats it lasts for 1–2 h; in monkeys it lasts for days; and in humans it usually lasts for a minimum of 2 weeks.

Cessation of tonic bombardment of spinal neurons by excitatory impulses in descending pathways (see below) plays a role in development of spinal shock. In addition, spinal inhibitory interneurons that normally are themselves inhibited may be released from this descending inhibition to become disinhibited. This, in turn, would inhibit motor neurons. The recovery of reflex excitability may be due to the development of denervation hypersensitivity to the mediators released by the remaining spinal excitatory endings. Another contributing factor is sprouting of collaterals from existing neurons, with the formation of additional excitatory endings on interneurons and motor neurons.

The first reflex response to appear as spinal shock wears off in humans is

often a slight contraction of the leg flexors and adductors in response to a noxious stimulus (ie, the withdrawal reflex). In some patients, the knee jerk reflex recovers first. The interval between cord transection and the return of reflex activity is about 2 weeks in the absence of any complications, but if complications are present it is much longer. Once the spinal reflexes begin to reappear after spinal shock, their threshold steadily drops.

Circuits intrinsic to the spinal cord can produce walking movements when stimulated in a suitable manner even after spinal cord transection in cats and dogs. There are two **locomotor pattern generators** in the spinal cord: one in the cervical region and one in the lumbar region. However, this does not mean that spinal animals or humans can walk without stimulation; the pattern generator has to be turned on by tonic discharge of a discrete area in the midbrain, the **mesencephalic locomotor region,** and, of course, this is only possible in patients with incomplete spinal cord transection. Progress is being made in teaching humans with SCI to take a few steps by placing them, with support, on a treadmill.

GENERAL PRINCIPLES OF CENTRAL ORGANIZATION OF MOTOR PATHWAYS

To voluntarily move a limb, the brain must plan a movement, arrange appropriate motion at many different joints at the same time, and adjust the motion by comparing plan with performance. The motor system "learns by doing" and performance improves with repetition. This involves synaptic plasticity. Damage to the cerebral cortex before or during childbirth or during the first 2–3 years of development can lead to cerebral palsy, a disorder that affects muscle tone, movement, and coordination **(Clinical Box 12–4)**.

CLINICAL BOX 12–4

Cerebral Palsy

Cerebral palsy (CP) is a term used to describe any one of several nonprogressive neurologic disorders that occur before or during childbirth or during early childhood. Prenatal factors, including exposure of the developing brain to hypoxia, infections, or toxins, may account for 70–80% of cases of CP. Typical symptoms of the disorder include spasticity, ataxia, deficits in

fine motor control, and abnormal gait (crouched or "scissored gait"). Sensory deficits including loss of vision and hearing as well as learning difficulties and seizures often occur in children with CP. In developed countries, the prevalence of CP is 2–2.5 cases per 1000 live births; however, the incidence of CP in children who are born prematurely is much higher compared with children born at term. Based on differences in the resting tone in muscles and the limbs involved, CP is classified into different groups. The most prevalent type is **spastic CP** that is characterized by **spasticity, hyperreflexia,** clonus, and a **positive Babinski sign.** These are all signs of damage to the corticospinal tract (Clinical Box 12–5). **Dyskinetic CP** is characterized by abnormal involuntary movements (**chorea** and **athetosis**) and may reflect damage to extrapyramidal motor areas. It is not uncommon to have signs of both types of CP. The rarest type is **hypotonic CP** that presents with truncal and extremity hypotonia, hyperreflexia, and persistent primitive reflexes.

THERAPEUTIC HIGHLIGHTS

There is no cure for CP. Treatment often includes physical and occupational therapy. Botulinum toxin injections into affected muscles have been used to reduce muscle spasticity, especially in the gastrocnemius muscle. Other drugs used to treat muscle spasticity in patients with CP include **diazepam** (a benzodiazepine that binds to the $GABA_A$ receptor), **baclofen** (an agonist at presynaptic $GABA_B$ receptors in the spinal cord), and **dantrolene** (a direct muscle relaxant). Various surgeries have been used to treat CP, including **selective dorsal rhizotomy** (section of the dorsal roots) and **tenotomy** (severing the tendon in the gastrocnemius muscles).

Figure 12–6 shows the general motor control scheme with the commands for voluntary movement originating in cortical association areas. The movements are planned in the cortex, the basal ganglia, and the lateral portions of the cerebellar hemispheres, as indicated by increased electrical activity before the movement. The basal ganglia and cerebellum funnel information to the premotor and motor cortex by way of the thalamus. Motor commands from the motor cortex are relayed predominantly via the corticospinal tracts to the spinal cord and the corresponding corticobulbar tracts to motor neurons in the brainstem. However, collaterals from these pathways and a few direct connections from the motor cortex end on brainstem nuclei that project to motor neurons in the

brainstem and spinal cord. These pathways can also mediate voluntary movement. Movement sets up alterations in sensory input from the special senses and from muscles, tendons, joints, and the skin. This feedback information that adjusts and smooths movement is relayed directly to the motor cortex and spinocerebellum. The spinocerebellum then projects to the brainstem. The main brainstem pathways that are concerned with posture and coordination are the rubrospinal, reticulospinal, tectospinal, and vestibulospinal tracts.

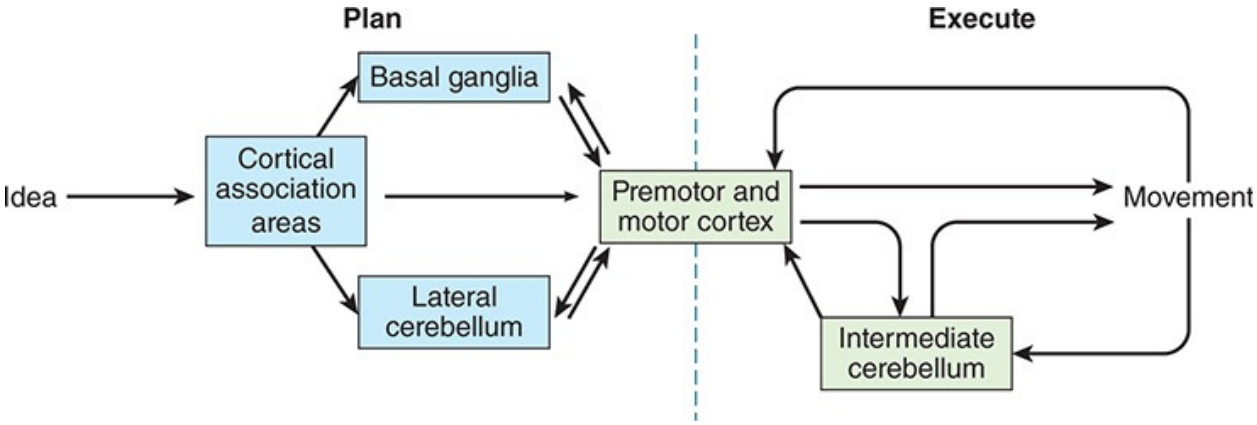


FIGURE 12–6 Control of voluntary movement. Commands for voluntary movement originate in cortical association areas. The cortex, basal ganglia, and cerebellum work cooperatively to plan movements. Movement executed by the cortex is relayed via the corticospinal tracts and corticobulbar tracts to motor neurons. The cerebellum provides feedback to adjust and smoothen movement.

MOTOR CORTEX & VOLUNTARY MOVEMENT

PRIMARY MOTOR CORTEX

The reader can refer to Figure 8–8 for the locations of the major cortical regions involved in motor control. The **primary motor cortex (M1)** is in the precentral gyrus of the frontal lobe, extending into the central sulcus. By means of stimulation experiments in patients undergoing craniotomy under local anesthesia, this region was mapped to show where various parts of the body are represented in the precentral gyrus. **Figure 12–7** shows the **motor homunculus** with the feet at the top of the gyrus and the face at the bottom. The cortical representation of each body part is proportional in size to the skill with which the part is used in fine, voluntary movement. The areas involved in speech and hand movements are especially large in the cortex; use of the pharynx, lips, and

tongue to form words and of the fingers and opposable thumbs to manipulate the environment are activities in which humans are especially skilled.

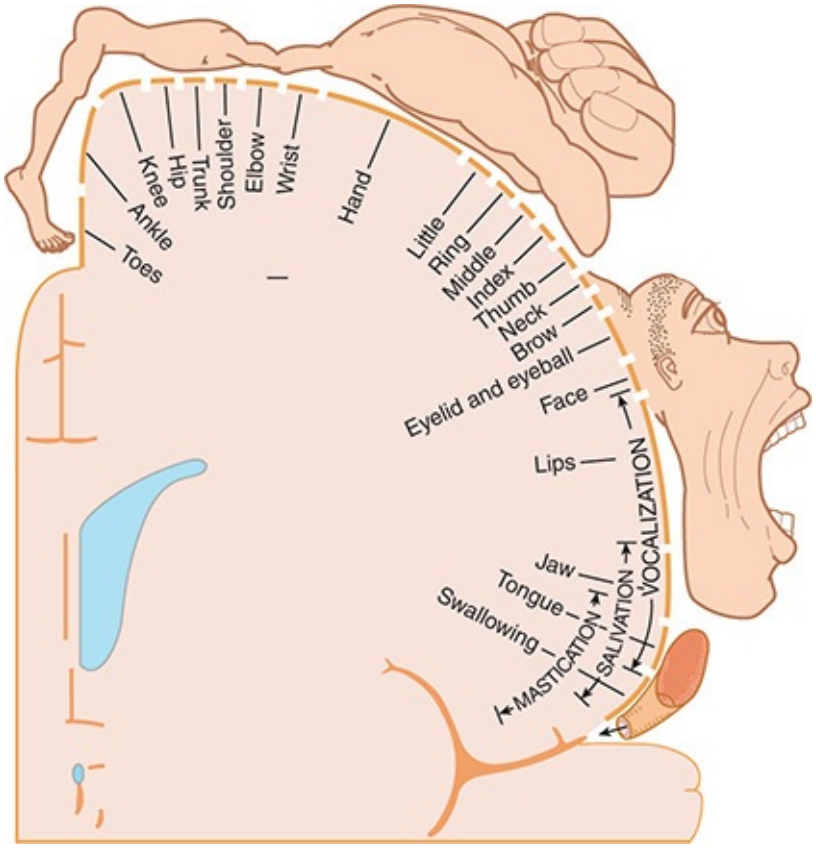


FIGURE 12–7 Motor homunculus. The figure represents, on a coronal section of the precentral gyrus, the location of the cortical representation of the various parts. The size of the various parts is proportional to the cortical area devoted to them. Compare with Figure 8–9 which shows the sensory homunculus. (Reproduced with permission from Penfield W, Rasmussen G: *The Cerebral Cortex of Man*. Macmillan, 1950.)

Modern brain imaging techniques such as **positron emission tomography (PET)** and **functional magnetic resonance imaging (fMRI)** can be used to map the cortex to identify motor areas. **Figure 12–8** shows activation of the hand area of the motor cortex while repetitively squeezing a ball with either the right or left hand.

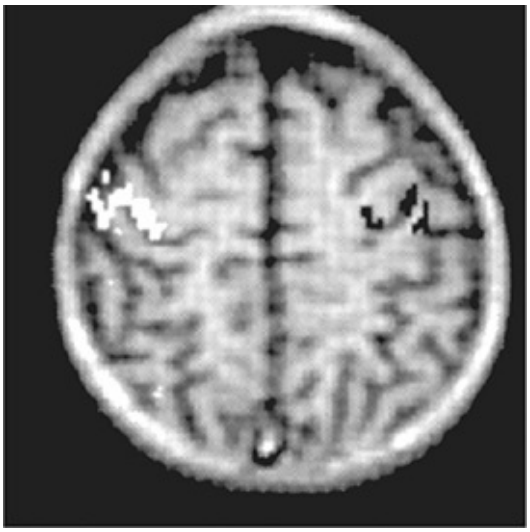


FIGURE 12–8 Hand area of motor cortex demonstrated by functional magnetic resonance imaging (fMRI) in a 7-year-old boy. Changes in signal intensity, measured using a method called echoplanar MRI, result from changes in the flow, volume, and oxygenation of the blood. The child was instructed to repetitively squeeze a foam-rubber ball at the rate of two to four squeezes per second with the right or left hand. Changes in cortical activity with the ball in the right hand are shown in black. Changes in cortical activity with the ball in the left hand are shown in white. (Data from Novotny EJ, et al: Functional magnetic resonance imaging [fMRI] in pediatric epilepsy. Epilepsia 1994;35(Supp 8):36.)

The cells in the cortical motor areas are arranged in a **columnar organization.** M1 neurons represent movements of groups of muscles for different tasks. Neurons in several cortical columns project to the same muscle; moreover, the cells in each column receive sensory input from the peripheral area in which they produce movement, providing the basis for feedback control of movement. Some of this input may be direct and some is relayed from other parts of the cortex.

SUPPLEMENTARY MOTOR AREA

The supplementary motor area is on and above the superior bank of the cingulate sulcus on the medial side of the hemisphere. It projects to M1 and contains a map of the body; but it is less precise than in M1. The supplementary motor area is involved in organizing or planning motor sequences, while M1 executes the movements.

When human subjects count to themselves without speaking, the motor cortex is quiescent, but when they speak the numbers aloud as they count, blood flow increases in M1 and the supplementary motor area. Thus, both regions are involved in voluntary movement when the movements being performed are complex and involve planning.

PREMOTOR CORTEX

The premotor cortex is located anterior to the precentral gyrus, on the lateral and medial cortical surface; it contains a somatotopic map. This region receives input from sensory regions of the parietal cortex and projects to M1, the spinal cord, and the brainstem reticular formation. This region is concerned with setting posture at the start of a planned movement and with getting the individual prepared to move. It is most involved in control of proximal limb muscles needed to orient the body for movement.

POSTERIOR PARIETAL CORTEX

The somatic sensory area and related portions of the posterior parietal lobe project to the premotor cortex. Lesions of the somatic sensory area cause defects in motor performance that are characterized by inability to execute learned sequences of movements such as eating with a knife and fork. Some of the neurons are concerned with aiming the hands toward an object and manipulating it, whereas other neurons are concerned with hand-eye coordination. As described below, neurons in this posterior parietal cortex contribute to the descending pathways involved in motor control.

PLASTICITY

A striking discovery made possible by PET and fMRI is that the motor cortex shows the same kind of plasticity as already described for the sensory cortex in Chapter 8. For example, the finger areas of the contralateral motor cortex enlarge as a pattern of rapid finger movement is learned with the fingers of one hand; this change is detectable at 1 week and maximal at 4 weeks. Cortical areas of output to other muscles also increase in size when motor learning involves these muscles. When a small focal ischemic lesion is produced in the hand area of the motor cortex of monkeys, the hand area may reappear, with return of motor

function, in an adjacent undamaged part of the cortex. Thus, the maps of the motor cortex are not immutable; they change with experience.

CONTROL OF AXIAL & DISTAL MUSCLES

Within the brainstem and spinal cord, pathways and neurons that are concerned with the control of skeletal muscles of the trunk (axial) and proximal portions of the limbs are located medially or ventrally, whereas pathways and neurons that are concerned with the control of skeletal muscles in the distal portions of the limbs are located laterally. The axial muscles are concerned with postural adjustments and gross movements, whereas the distal limb muscles mediate fine, skilled movements. Neurons in the medial portion of the ventral horn innervate proximal limb muscles, particularly the flexors, and lateral ventral horn neurons innervate distal limb muscles. Similarly, the ventral corticospinal tract and medial descending brainstem pathways (tectospinal, reticulospinal, and vestibulospinal tracts) are concerned with adjustments of proximal muscles and posture, and the lateral corticospinal and rubrospinal tracts are concerned with distal limb muscles and, particularly in the case of the lateral corticospinal tract, with skilled voluntary movements.

CORTICOSPINAL & CORTICOBULBAR TRACTS

The somatotopic organization just described for the motor cortex continues throughout the pathways from the cortex to the motor neurons. The axons of neurons from the motor cortex that project to spinal motor neurons form the **corticospinal tracts,** a large bundle of about 1 million fibers. About 80% of these fibers cross the midline in the medullary pyramids to form the **lateral corticospinal tract** **(Figure 12–9)**. The remaining 20% make up the **ventral corticospinal tract,** which does not cross the midline until it reaches the level of the spinal cord at which it terminates. Lateral corticospinal tract neurons make monosynaptic connections to motor neurons, especially those concerned with skilled movements. Many corticospinal tract neurons also synapse on spinal interneurons antecedent to motor neurons; this indirect pathway is important in coordinating groups of muscles.

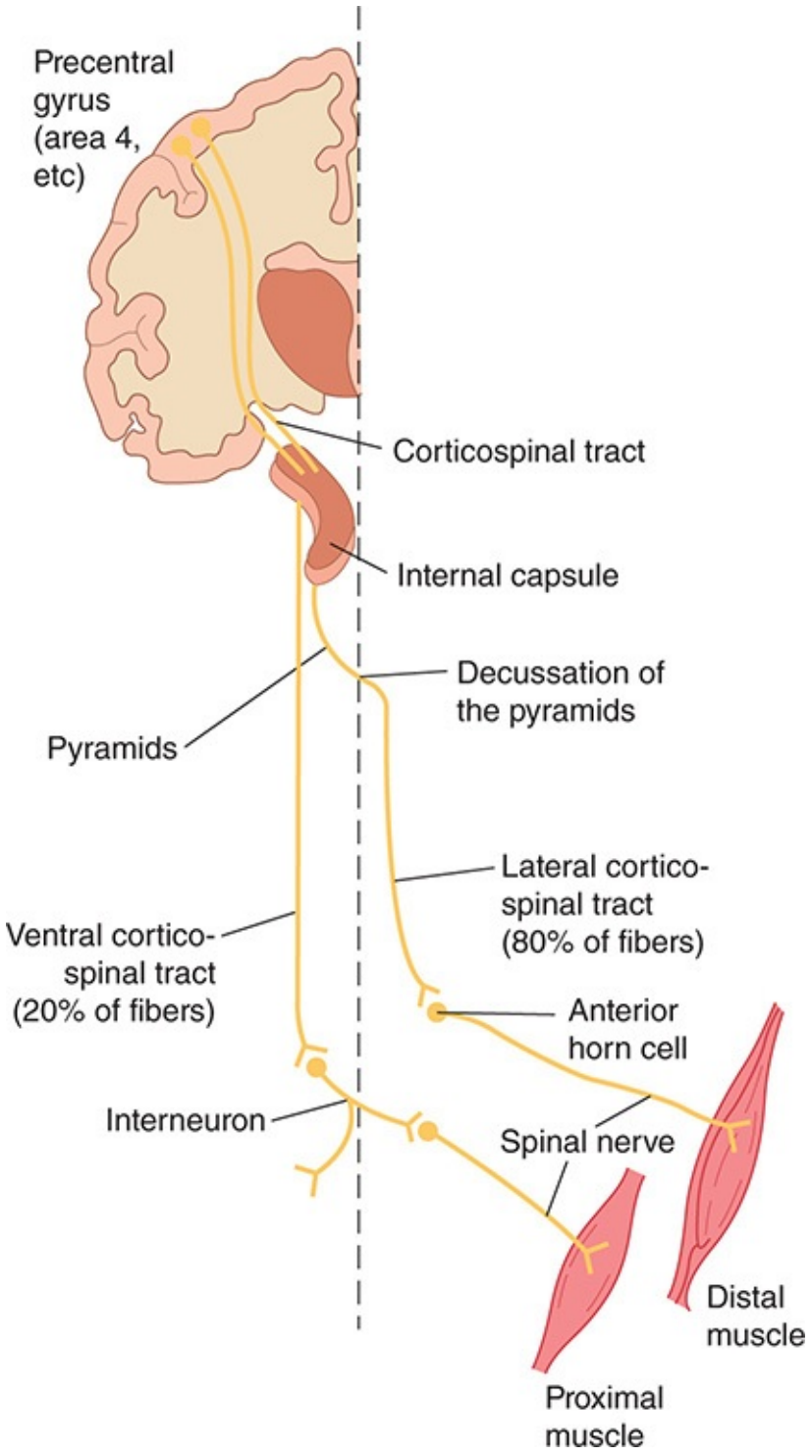


FIGURE 12–9 The corticospinal tracts. This tract originates in the precentral gyrus and passes through the internal capsule. Most fibers decussate in the pyramids and descend in the lateral white matter of the spinal cord to form the lateral division of the tract which can make monosynaptic connections with spinal motor neurons. The ventral division of the tract remains uncrossed until

reaching the spinal cord where axons terminate on spinal interneurons antecedent to motor neurons.

The trajectory from the cortex to the spinal cord passes through the **corona radiata** to the posterior limb of the **internal capsule.** Within the midbrain they traverse the **cerebral peduncle** and the basilar pons until they reach the **medullary pyramids** on their way to the spinal cord.

The **corticobulbar tract** is composed of the fibers that pass from the motor cortex to motor neurons in the trigeminal, facial, and hypoglossal nuclei. Corticobulbar neurons end either directly on the cranial nerve nuclei or on their antecedent interneurons within the brainstem. Their axons traverse through the genu of the internal capsule, the cerebral peduncle (medial to corticospinal tract neurons), to descend with corticospinal tract fibers in the pons and medulla.

The motor system can be divided into lower and upper motor neurons. **Lower motor neurons** refer to the spinal and cranial motor neurons that directly innervate skeletal muscles. **Upper motor neurons** are those in the cortex and brainstem that activate the lower motor neurons. The pathophysiologic responses to damage to lower and upper motor neurons are very distinctive **(Clinical Box 12–5)**.

ORIGINS OF CORTICOSPINAL & CORTICOBULBAR TRACTS

Corticospinal and corticobulbar tract neurons are pyramidal shaped and located in layer V of the cerebral cortex (see Chapter 14). The cortical areas from which these tracts originate were identified on the basis of electrical stimulation that produced prompt discrete movement. About 31% of the corticospinal tract neurons are from M1. The premotor cortex and supplementary motor cortex account for 29% of the corticospinal tract neurons. The other 40% of corticospinal tract neurons originate in the parietal lobe and primary somatosensory area in the postcentral gyrus. The corticospinal and corticobulbar system is the primary pathway for the initiation of skilled voluntary movement.

BRAINSTEM PATHWAYS INVOLVED IN POSTURE & VOLUNTARY MOVEMENT

As mentioned above, spinal motor neurons are organized so that those

innervating the most proximal muscles are located most medially and those innervating the more distal muscles are located more laterally. This organization is also reflected in descending brainstem pathways **(Figure 12–10)**.

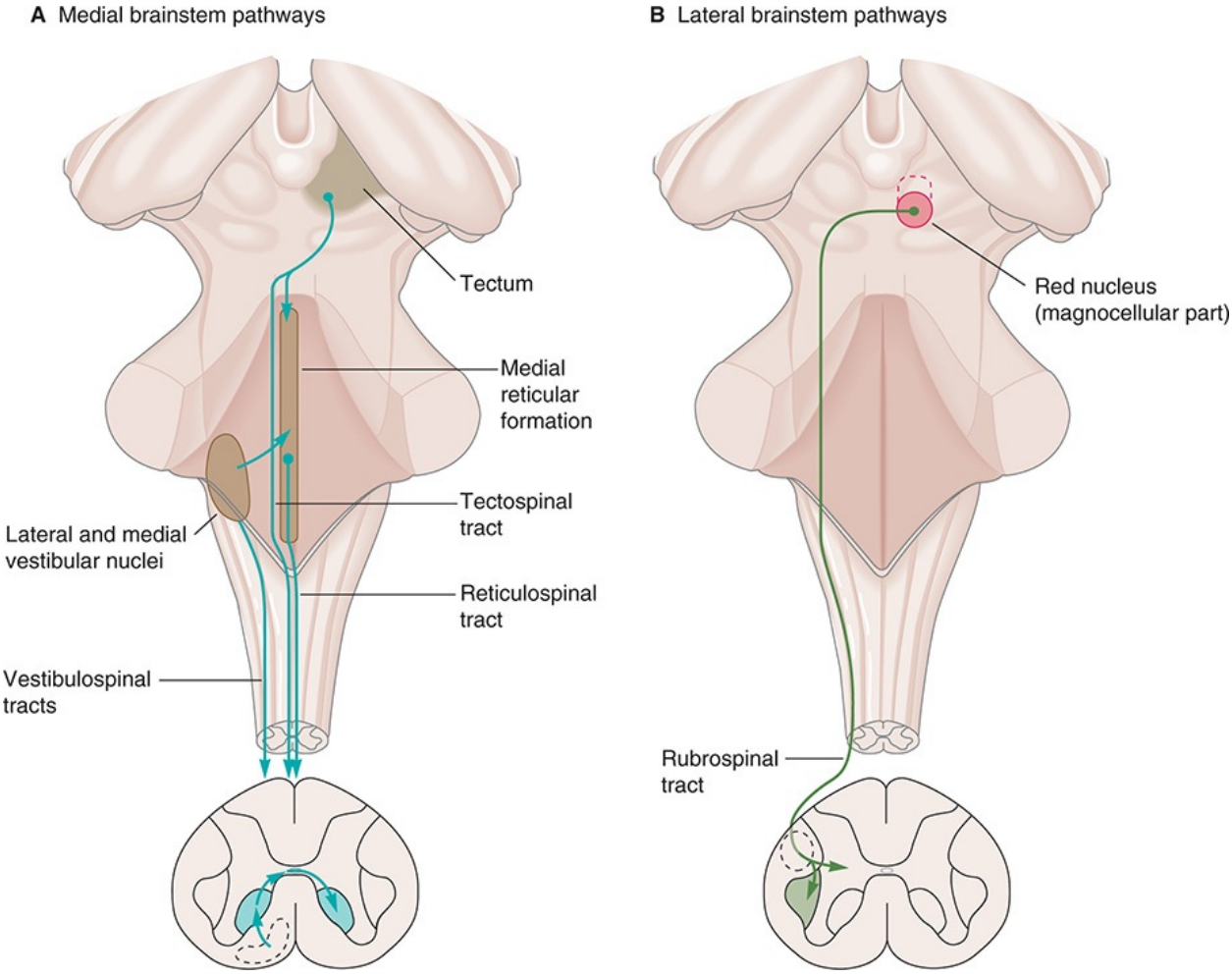


FIGURE 12–10 Medial and lateral descending brainstem pathways involved in motor control. A) Medial pathways (reticulospinal, vestibulospinal, and tectospinal) terminate in ventromedial area of spinal gray matter and control axial and proximal muscles. **B)** Lateral pathway (rubrospinal) terminates in dorsolateral area of spinal gray matter and controls distal muscles. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

CLINICAL BOX 12–5

Lower versus Upper Motor Neuron Damage

Lower motor neurons are those whose axons terminate on skeletal muscles. Damage to these neurons is associated with **flaccid paralysis, muscular atrophy, fasciculations** (visible muscle twitches that appear as flickers under the skin), **hypotonia** (decreased muscle tone), and **hyporeflexia** or **areflexia.** An example of a disease that leads to lower motor neuron damage is **ALS.** "Amyotrophic" means "no muscle nourishment" and describes the atrophy that muscles undergo because of disuse. "Sclerosis" refers to the hardness felt when a pathologist examines the spinal cord on autopsy; the hardness is due to proliferation of astrocytes and scarring of the lateral columns of the spinal cord. ALS is a selective, progressive degeneration of α-motor neurons. This fatal disease is also known as **Lou Gehrig disease** in remembrance of a famous American baseball player who died of ALS. The worldwide annual incidence of ALS has been estimated to be 0.5–3 cases per 100,000 people. The disease has no racial, socioeconomic, or ethnic boundaries. The life expectancy of ALS patients is usually 3–5 years after diagnosis. ALS is most commonly diagnosed in middle age and affects men more often than women. Most cases of ALS are sporadic in origin; but 5–10% of the cases have a familial link. Possible causes include viruses, neurotoxins, heavy metals, DNA defects (especially in familial ALS), immune system abnormalities, and enzyme abnormalities. About 40% of the familial cases have a mutation in the gene for Cu/Zn superoxide dismutase *(SOD-1)* on chromosome 21. SOD is a free radical scavenger that reduces oxidative stress. A defective *SOD-1* gene permits free radicals to accumulate and kill neurons. An increase in the excitability of deep cerebellar nuclei due to the inhibition of **small-conductance calcium-activated potassium (SK) channels** may contribute to the development of cerebellar ataxia.

Upper motor neurons typically refer to corticospinal tract neurons that innervate spinal motor neurons, but they can also include brainstem neurons that control spinal motor neurons. Damage to these neurons initially causes muscles to become weak and flaccid but eventually leads to spasticity, **hypertonia** (increased resistance to passive movement), hyperactive stretch reflexes, and abnormal plantar extensor reflex (positive Babinski sign). The Babinski sign is dorsiflexion of the great toe and fanning of the other toes when the lateral aspect of the sole of the foot is scratched. In adults, the normal response to this stimulation is plantar flexion in all the toes. The Babinski sign is of value in the localization of disease processes, but its physiologic significance is unknown. In infants whose corticospinal tracts are

not well developed, dorsiflexion of the great toe and fanning of the other toes is the natural response to stimuli applied to the sole of the foot.

THERAPEUTIC HIGHLIGHTS

One of the few drugs shown to modestly slow the progression of ALS is **riluzole,** a drug that opens the SK channels and may be effective in preventing nerve damage caused by excessive release of the excitatory amino acid, glutamate. Spasticity associated with motor neuron disease can be reduced by the muscle relaxant **baclofen** (a derivative of GABA); in some cases, a subarachnoid infusion of baclofen is given via an implanted lumbar pump. Spasticity can also be treated with **tizanidine,** a centrally acting α_2-adrenoceptor agonist; its effectiveness is due to increasing presynaptic inhibition of spinal motor neurons. **Botulinum toxin** is also approved for the treatment of spasticity; this toxin acts by binding to receptors on the cholinergic nerve terminals to decrease the release of acetylcholine, causing neuromuscular blockade.

MEDIAL BRAINSTEM PATHWAYS

The medial brainstem pathways, which work in concert with the ventral corticospinal tract, are the **pontine and medullary reticulospinal, vestibulospinal,** and **tectospinal tracts.** These pathways descend in the ipsilateral ventral columns of the spinal cord and terminate predominantly on interneurons and long propriospinal neurons in the ventromedial part of the ventral horn to control axial and proximal muscles. A few medial pathway neurons synapse directly on motor neurons controlling axial muscles.

The medial and lateral vestibulospinal tracts are involved in vestibular function and are described in Chapter 11. The medial tract originates in the medial and inferior vestibular nuclei and projects bilaterally to cervical spinal motor neurons that control neck musculature. The lateral tract originates in the lateral vestibular nuclei and projects ipsilaterally to neurons at all spinal levels. It activates motor neurons to antigravity muscles (eg, proximal limb extensors) to control posture and balance.

The pontine and medullary reticulospinal tracts project to all spinal levels. They are involved in the maintenance of posture and in modulating muscle tone, especially via an input to γ-motor neurons. Pontine reticulospinal neurons are

primarily excitatory and medullary reticulospinal neurons are primarily inhibitory. The tectospinal tract originates in the superior colliculus of the midbrain and projects to the contralateral cervical spinal cord to control head and eye movements.

LATERAL BRAINSTEM PATHWAY

The main control of distal muscles arises from the lateral corticospinal tract, but neurons within the red nucleus of the midbrain cross the midline and project to interneurons in the dorsolateral part of the spinal ventral horn to also influence motor neurons that control distal limb muscles. This **rubrospinal tract** excites flexor motor neurons and inhibits extensor motor neurons. This pathway is not very prominent in healthy humans, but it may play a role in the posture typical of decorticate rigidity (see below).

POSTURE-REGULATING SYSTEMS

When damage occurs somewhere along the neural axis, the activities integrated below the injury are released from the control of higher brain centers and often are accentuated. Release of this type, long a cardinal principle in neurology, may be due in some situations to disruption of an inhibitory control by higher neural regions.

DECEREBRATION

A complete transection of the brainstem between the superior and inferior colliculi permits the brainstem pathways to function independent of their input from higher brain structures. This **midcollicular decerebration** is diagramed in **Figure 12–11** by the dashed line labeled A. This lesion interrupts all inputs from the cortex (corticospinal and corticobulbar tracts) and red nucleus (rubrospinal tract), primarily to distal muscles of the extremities. The excitatory and inhibitory reticulospinal pathways (primarily to postural extensor muscles) remain intact. The dominance of drive from ascending sensory pathways to the excitatory reticulospinal pathway leads to hyperactivity in extensor muscles in all four extremities **(decerebrate rigidity).** This resembles what ensues after **uncal herniation** due to a supratentorial lesion. Uncal herniation can occur in patients with large tumors or a hemorrhage in the cerebral hemisphere. **Figure**

12–12A shows the posture typical of such a patient. Clinical Box 12–6 describes complications related to uncal herniation.

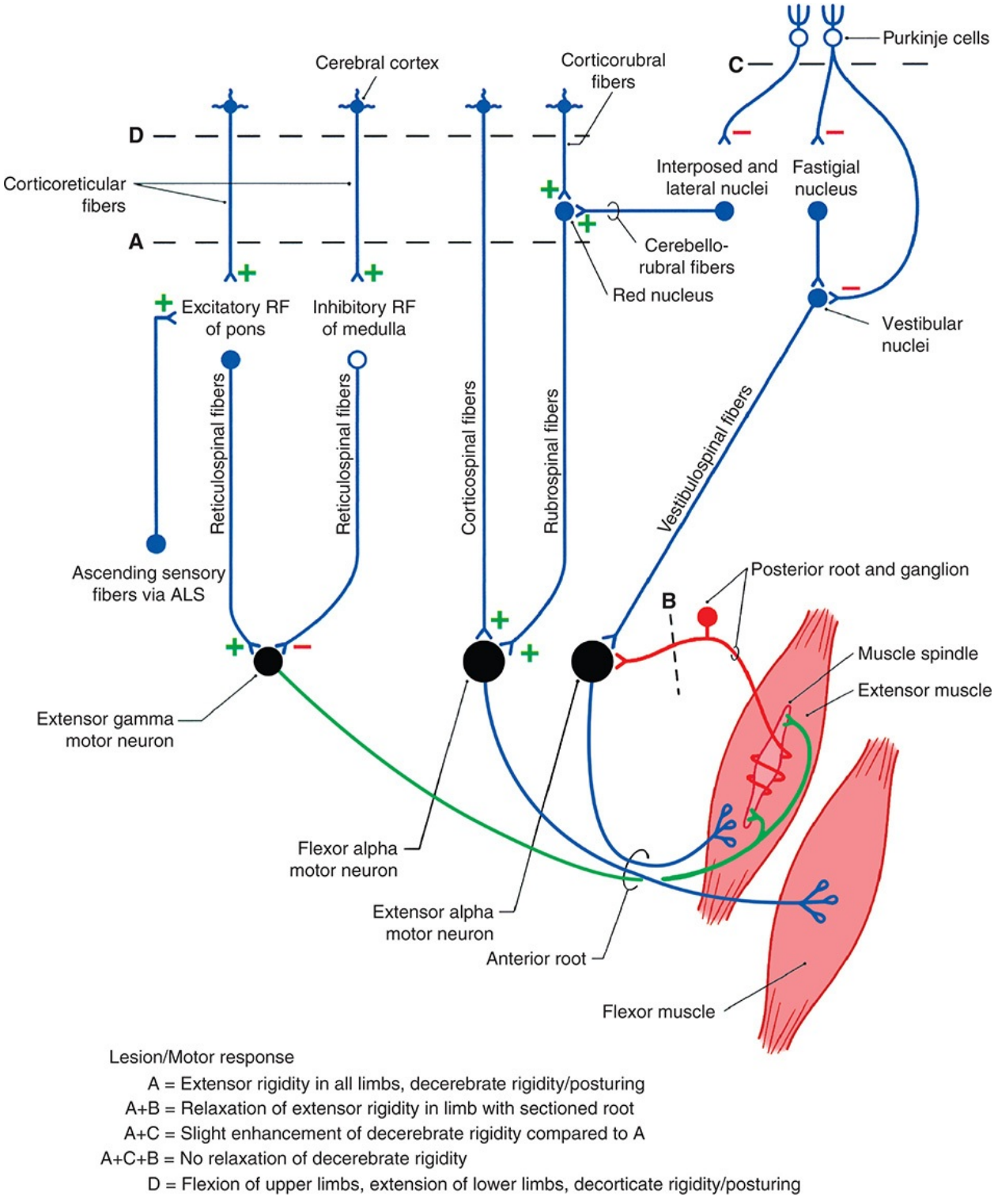


FIGURE 12–11 A circuit drawing representing lesions produced in

experimental animals to replicate decerebrate and decorticate deficits seen in humans. Bilateral transections are indicated by dashed lines A, B, C, and D. Decerebration is at a midcollicular level (A), decortication is rostral to the superior colliculus, dorsal roots sectioned for one extremity (B), and damage to the anterior lobe of cerebellum (C). The objective was to identify anatomic substrates responsible for decerebrate or decorticate rigidity/posturing seen in humans with lesions that either isolate the forebrain from the brainstem or separate rostral from caudal brainstem and spinal cord. (Reproduced with permission from Haines DE (ed): *Fundamental Neuroscience for Basic and Clinical Applications*, 3rd ed. St. Louis, MO: Elsevier; 2006.)

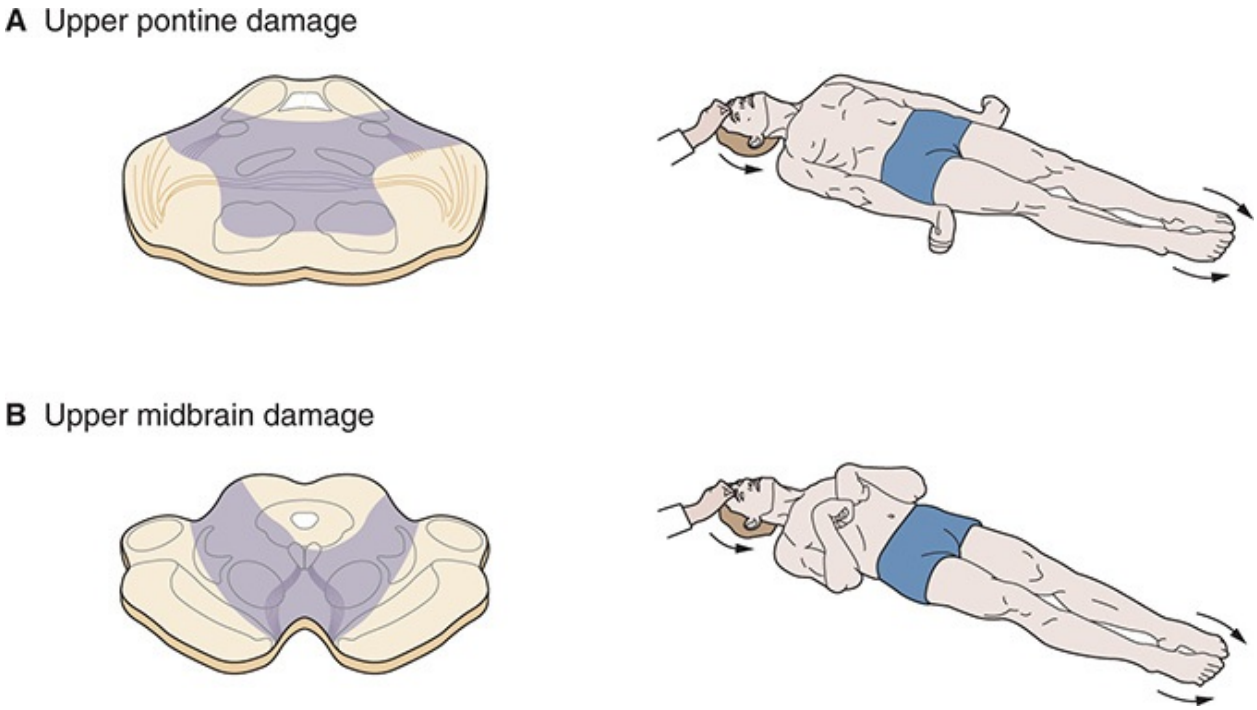


FIGURE 12–12 Decerebrate and decorticate postures. A) Damage to lower midbrain and upper pons causes decerebrate posturing in which lower extremities are extended with toes pointed inward and upper extremities extended with fingers flexed and forearms pronate. Neck and head are extended. **B)** Damage to upper midbrain may cause decorticate posturing in which upper limbs are flexed, lower limbs are extended with toes pointed slightly inward, and head is extended. (Modified with permission from Kandel ER, Schwartz JH, Jessell TM (eds): *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

Decerebrate rigidity is spasticity due to facilitation of the myotatic stretch

reflex. That is, the excitatory input from the reticulospinal pathway activates γ-motor neurons which indirectly activate α-motor neurons (via Ia spindle afferent activity). This is called the **gamma loop.** This was demonstrated by cutting dorsal roots to a limb in midcollicular decerebrate cats (dashed line labeled B in Figure 12–11) that immediately eliminated the hyperactivity of extensor muscles.

Decerebrate rigidity can also lead to direct activation of α-motor neurons. If the anterior lobe of the cerebellum is removed in a decerebrate animal (dashed line labeled C in Figure 12–11), extensor muscle hyperactivity is exaggerated **(decerebellate rigidity).** This lesion eliminates cortical inhibition of the cerebellar fastigial nucleus and secondarily increases excitation to vestibular nuclei. Subsequent dorsal root section does not reverse the rigidity due to activation of α-motor neurons independent of the gamma loop.

DECORTICATION

Injury to the cerebral cortex **(decortication;** dashed line labeled D in Figure 12–11) produces **decorticate rigidity** that is characterized by flexion of the upper extremities at the elbow and extensor hyperactivity in the lower extremities **(Figure 12–12B)**. The flexion is explained by rubrospinal excitation of flexor muscles in the upper extremities; the hyperextension of lower extremities is due to the same changes that occur after midcollicular decerebration.

CLINICAL BOX 12–6

Uncal Herniation

Space-occupying lesions from large tumors, hemorrhages, strokes, or abscesses in the cerebral hemisphere can drive the uncus of the temporal lobe over the edge of the cerebellar tentorium, compressing the ipsilateral cranial nerve III **(uncal herniation).** Before the herniation these patients experience a decreased level of consciousness, lethargy, poorly reactive pupils, deviation of the eye to a “down and out” position, hyperactive reflexes, and a bilateral Babinski sign (due to compression of the ipsilateral corticospinal tract). After the brain herniates, the patients are decerebrate and comatose, have fixed and dilated pupils, and eye movements are absent. Once damage extends to the midbrain, a **Cheyne-Stokes respiratory pattern** develops, characterized by a pattern of waxing-and-waning depth of respiration with interposed periods of

apnea. Eventually, medullary function is lost, breathing ceases, and recovery is unlikely. Hemispheric masses closer to the midline compress the thalamic reticular formation and can cause coma before eye findings develop **(central herniation).** As the mass enlarges, midbrain function is affected, the pupils dilate, and a decerebrate posture ensues. With progressive herniation, pontine vestibular and then medullary respiratory functions are lost.

Decorticate rigidity is seen on the hemiplegic side after hemorrhages or thromboses in the internal capsule. The small arteries in the internal capsule are especially prone to rupture or thrombotic obstruction, so this type of decorticate rigidity is fairly common. Sixty percent of intracerebral hemorrhages occur in the internal capsule, 10% in the cerebral cortex, 10% in the pons, 10% in the thalamus, and 10% in the cerebellum.

BASAL GANGLIA

ORGANIZATION OF THE BASAL GANGLIA

The term **basal ganglia** (or **basal nuclei**) is applied to five interactive structures on each side of the brain **(Figure 12–13)**. These are the **caudate nucleus, putamen,** and **globus pallidus** (three large nuclear masses underlying the cortical mantle), the **subthalamic nucleus,** and **substantia nigra.** The caudate nucleus and putamen collectively form the **striatum;** the putamen and globus pallidus collectively form the **lenticular nucleus.**

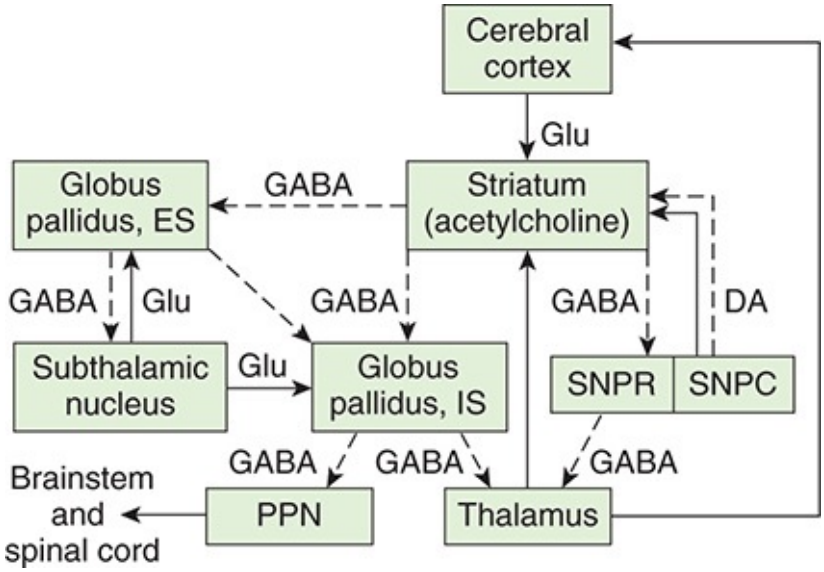


FIGURE 12–13 The basal ganglia. The basal ganglia are composed of the caudate nucleus, putamen, and globus pallidus and the functionally related subthalamic nucleus and substantia nigra. The frontal (coronal) section shows the location of the basal ganglia in relation to surrounding structures.

The globus pallidus is divided into external and internal segments (GPe and GPi); both regions contain inhibitory GABAergic neurons. The substantia nigra is divided into **pars compacta** that uses dopamine as a neurotransmitter and **pars reticulata** that uses GABA as a neurotransmitter. About 95% of striatal neurons are medium spiny neurons that use GABA as a neurotransmitter. The remaining striatal neurons are aspiny interneurons that differ in terms of size and neurotransmitters: large (acetylcholine), medium (somatostatin), and small (GABA).

Figure 12–14 shows the major connections to and from and within the basal ganglia along with the neurotransmitters within these pathways. There are two main inputs to the basal ganglia; they are both excitatory (glutamate), and they both terminate in the striatum. They are from a wide region of the cerebral cortex **(corticostriate pathway)** and from intralaminar nuclei of the thalamus **(thalamostriatal pathway).** The two major outputs of the basal ganglia are from GPi and substantia nigra pars reticulata. Both are inhibitory (GABAergic) and both project to the thalamus. From the thalamus, there is an excitatory (glutamatergic) projection to the prefrontal and premotor cortex. This completes the cortical-basal ganglia-thalamic-cortical loop.

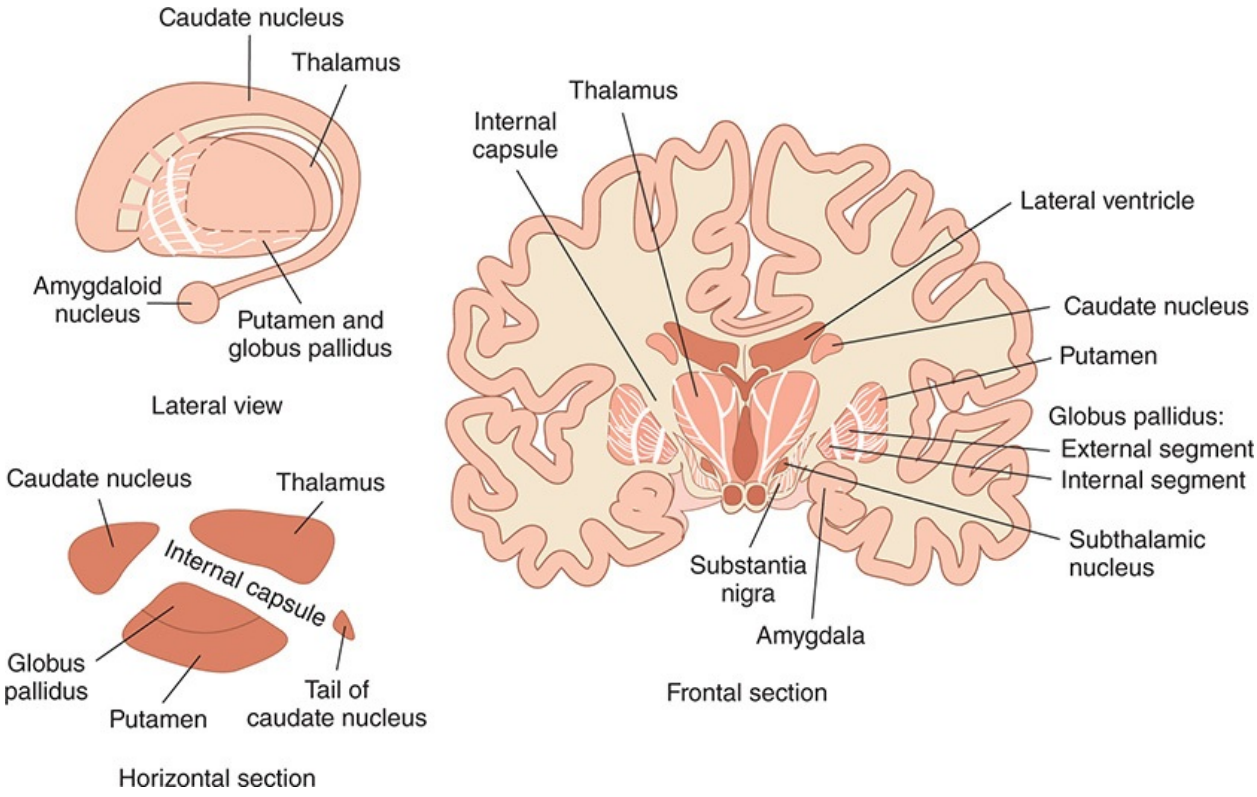


FIGURE 12–14 Diagrammatic representation of the principal connections of the basal ganglia. Solid lines indicate excitatory pathways, dashed lines indicate inhibitory pathways. The transmitters are indicated in the pathways, where they are known. DA, dopamine; Glu, glutamate. Acetylcholine is the transmitter produced by interneurons in the striatum. ES, external segment; IS, internal segment; PPN, pedunculopontine nuclei; SNPC, substantia nigra, pars compacta; SNPR, substantia nigra, pars reticulata. The subthalamic nucleus also projects to the pars compacta of the substantia nigra; this pathway has been omitted for clarity.

The connections within the basal ganglia include a dopaminergic **nigrostriatal projection** from the substantia nigra pars compacta to the striatum and a GABAergic projection from the striatum to substantia nigra pars reticulata. There is an inhibitory projection from the striatum to both GPe and GPi. The subthalamic nucleus receives an inhibitory input from GPe, and in turn the subthalamic nucleus has an excitatory (glutamatergic) projection to both GPe and GPi.

FUNCTION

Neurons in the basal ganglia, like those in the lateral portions of the cerebellar hemispheres, discharge before movements begin. The basal ganglia are involved in the planning and programming of movement or, more broadly, in the processes by which an abstract thought is converted into voluntary action (Figure 12–6). They influence the motor cortex via the thalamus, and the corticospinal pathways provide the final common pathway to motor neurons. In addition, GPi projects to nuclei in the brainstem, and from there to motor neurons in the brainstem and spinal cord. The basal ganglia, particularly the caudate nuclei, also play a role in some cognitive processes. Possibly because of the interconnections of this nucleus with the frontal portions of the neocortex, lesions of the caudate nuclei disrupt performance on tests involving object reversal and delayed alternation. In addition, lesions of the head of the left but not the right caudate nucleus and nearby white matter in humans are associated with a dysarthric form of aphasia that resembles Wernicke aphasia (see Chapter 15).

DISEASES OF THE BASAL GANGLIA IN HUMANS

Three distinct biochemical pathways in the basal ganglia normally operate in a balanced manner: (1) the nigrostriatal dopaminergic system, (2) the intrastriatal cholinergic system, and (3) the GABAergic system that projects from the striatum to the globus pallidus and substantia nigra. When one or more of these pathways become dysfunctional, characteristic motor abnormalities occur. Diseases of the basal ganglia lead to two general types of disorders: **hyperkinetic** and **hypokinetic.** The hyperkinetic conditions are those in which movement is excessive and abnormal, including chorea, athetosis, and ballism. Hypokinetic abnormalities include akinesia and bradykinesia.

Chorea is characterized by rapid, involuntary “dancing” movements. **Athetosis** is characterized by continuous, slow writhing movements. Choreiform and athetotic movements have been likened to the start of voluntary movements occurring in an involuntary, disorganized way. In **ballism,** involuntary flailing, intense, and violent movements occur. **Akinesia** is difficulty in initiating movement and decreased spontaneous movement. **Bradykinesia** is slowness of movement.

In addition to Parkinson disease (described below), there are several other disorders that involve a malfunction within the basal ganglia. A few of these are

described in Clinical Box 12–7. **Huntington disease** is one of an increasing number of human genetic diseases affecting the nervous system that are characterized by **trinucleotide repeat** expansion. Most of these involve cytosine-adenine-guanine (CAG) repeats (Table 12–1), but one involves cytosine-guanine-guanine (CGG) repeats and another involves CTG repeats (T refers to thymine). Increased numbers of a 12-nucleotide repeat are also associated with a rare form of epilepsy.

TABLE 12–1 Examples of trinucleotide repeat diseases.

Disease	Expanded Trinucleotide Repeat	Affected Protein
Huntington disease	CAG	Huntingtin
Spinocerebellar ataxia, types 1, 2, 3, 7	CAG	Ataxin 1, 2, 3, 7
Spinocerebellar ataxia, type 6	CAG	α_{1A} subunit of Ca^{2+} channel
Dentatorubral-pallidoluysian atrophy	CAG	Atrophin
Spinobulbar muscular atrophy	CAG	Androgen receptor
Fragile X syndrome	CGG	FMR-1
Myotonic dystrophy	CTG	DM protein kinase
Friedreich ataxia	GAA	Frataxin

PARKINSON DISEASE

Parkinson disease has both hypokinetic and hyperkinetic features. It was originally described in 1817 by James Parkinson and is named for him. Parkinson disease is the first disease identified as being due to a deficiency in a specific neurotransmitter (Clinical Box 12–8). It results from the degeneration of dopaminergic neurons in the substantia nigra pars compacta. The fibers to the putamen (part of the striatum) are most severely affected.

The hypokinetic features of Parkinson disease are akinesia and bradykinesia, and the hyperkinetic features are **cogwheel rigidity** and **tremor at rest.** The absence of motor activity and the difficulty in initiating voluntary movements are striking. There is a decrease in the normal, unconscious movements such as swinging of the arms during walking, the panorama of facial expressions related to the emotional content of thought and speech, and the multiple "fidgety" actions and gestures that occur in all of us. The rigidity is different from spasticity because motor neuron discharge increases to both the agonist and antagonist muscles. Passive motion of an extremity meets with a plastic, dead-feeling resistance that has been likened to bending a lead pipe and is therefore called **lead pipe rigidity.** Sometimes a series of "catches" takes place during passive motion (cogwheel rigidity), but the sudden loss of resistance seen in a spastic extremity is absent. The tremor, which is present at rest and disappears with activity, is due to regular, alternating 8-Hz contractions of antagonistic muscles.

Figure 12–15 shows the current view of the pathogenesis of the movement disorders in Parkinson disease. In healthy individuals, basal ganglia output is inhibitory via GABAergic nerve fibers. The dopaminergic neurons that project from the substantia nigra to the putamen normally have two effects. They stimulate the D_1 dopamine receptors, which inhibit GPi via direct GABAergic receptors; and they inhibit D_2 receptors, which also inhibit the GPe. In addition, the inhibition reduces the excitatory discharge from the subthalamic nucleus to the GPi. This balance between inhibition and excitation somehow maintains normal motor function. In Parkinson disease, the dopaminergic input to the putamen is lost. This results in decreased inhibition and increased excitation from the subthalamic nucleus to the GPi. The overall increase in inhibitory output to the thalamus and brainstem disorganizes movement.

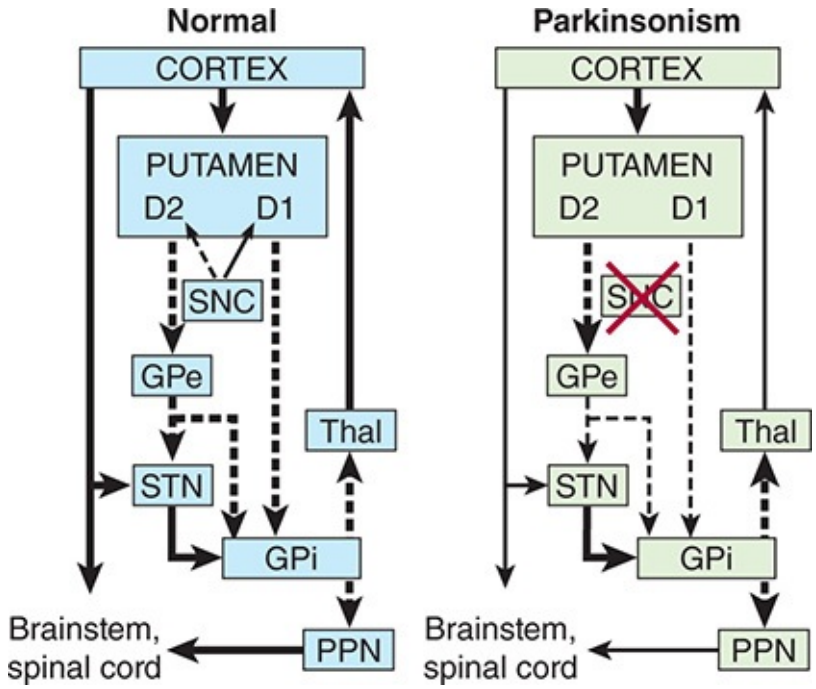


FIGURE 12–15 Probable basal ganglia-thalamocortical circuitry in Parkinson disease. Solid arrows indicate excitatory outputs and dashed arrows inhibitory outputs. The strength of each output is indicated by the width of the arrow. GPe, external segment of the globus pallidus; GPi, internal segment of the globus pallidus; PPN, pedunculopontine nuclei; SNC, pars compacta of the substantia nigra; STN, subthalamic nucleus; Thal, thalamus. See text for details. (Modified with permission from Grafton SC, DeLong M: Tracing the brain circuitry with functional imaging, Nat Med 1997 Jun; 3(6):602–603.)

CLINICAL BOX 12–7

Basal Ganglia Diseases

The initial detectable damage in **Huntington disease** is to medium spiny neurons in the striatum. The loss of this GABAergic pathway to the GPe releases inhibition, permitting the hyperkinetic features of the disease to develop. An early sign is a jerky trajectory of the hand when reaching to touch a spot, especially toward the end of the reach. Later, hyperkinetic **choreiform movements** appear and gradually increase until they incapacitate the patient. Speech becomes slurred and then incomprehensible, and a progressive dementia is followed by death, usually within 10–15 years after the onset of symptoms. Huntington disease affects 5 out of 100,000 people

worldwide. It is inherited as an autosomal dominant disorder, and its onset is usually between the ages of 30 and 50. The abnormal gene responsible for the disease is located near the end of the short arm of chromosome 4. It normally contains 11–34 CAG repeats, each coding for glutamine. In patients with Huntington disease, this number is increased to 42–86 or more copies; and the greater the number of repeats, the earlier the age of onset and the more rapid the progression of the disease. The gene codes for **huntingtin,** a protein that plays a role in axonal trafficking, regulation of gene transcription, and cell survival. Poorly soluble protein aggregates, which are toxic, form in cell nuclei and elsewhere. However, the correlation between aggregates and symptoms is less than perfect. The loss of the function of huntingtin that occurs is proportional to the size of the CAG insert. In animal models of the disease, intrastriatal grafting of fetal striatal tissue improves cognitive performance. In addition, tissue caspase-1 activity is increased in the brains of humans with the disease. Moreover, in mice in which the gene for this apoptosis-regulating enzyme has been knocked out, progression of the disease is slowed.

Wilson disease (or **hepatolenticular degeneration**) is a rare disorder of copper metabolism that has an onset between 6 and 25 years of age, affecting about four times as many females as males. Wilson disease affects about 30,000 people worldwide. It is a genetic autosomal recessive disorder due to a mutation on the long arm of chromosome 13q. It affects the copper-transporting ATPase gene (*ATP7B*) in the liver, leading to an accumulation of copper in the liver and resultant progressive liver damage. About 1% of the population carries a single abnormal copy of this gene but does not develop any symptoms. The disease may develop in a child who inherits the gene from both parents. In affected individuals, copper accumulates in the periphery of the cornea in the eye accounting for the characteristic yellow **Kayser-Fleischer rings.** The dominant neuronal pathology is degeneration of the putamen, a part of the **lenticular nucleus.** Motor disturbances include "wing-beating" tremor or **asterixis, dysarthria,** unsteady gait, and rigidity. **Tardive dyskinesia** is a disease that involves the basal ganglia, but it is caused by medical treatment of another disorder with **neuroleptic drugs** such as phenothiazines or haloperidol. Therefore, tardive dyskinesia is iatrogenic in origin. Long-term use of these drugs may produce biochemical abnormalities in the striatum. The motor disturbances include either temporary or permanent uncontrolled involuntary movements of the face and tongue and cogwheel rigidity. The neuroleptic drugs act via blockade of dopaminergic transmission. Prolonged drug use leads to hypersensitivity of

D_3 dopaminergic receptors and an imbalance in nigrostriatal influences on motor control.

THERAPEUTIC HIGHLIGHTS

Treatment for Huntington disease is directed at treating the symptoms and maintaining quality of life because there is no cure. In general, drugs used to treat the symptoms of this disease have side effects such as fatigue, nausea, and restlessness. In August 2008, the US Food and Drug Administration approved the use of **tetrabenazine** to reduce choreiform movements that characterize the disease. This drug binds reversibly to vesicular monoamine transporters (VMATs) and thus inhibits the uptake of monoamines into synaptic vesicles. It also acts as a dopamine receptor antagonist. Tetrabenazine is the first drug to receive approval for individuals with Huntington disease. It is also used to treat other hyperkinetic movement disorders such as tardive dyskinesia. **Chelating agents** (eg, **penicillamine and trienthine**) are used to reduce the copper in the body in individuals with Wilson disease. Tardive dyskinesia has proven to be difficult to treat. Treatment in patients with psychiatric disorders is often directed at prescribing a neuroleptic with less likelihood of causing the disorder. **Clozapine** is an atypical neuroleptic drug that has been an effective substitute for traditional neuroleptic drugs but with less risk for development of tardive dyskinesia.

Familial cases of Parkinson disease occur, but these are uncommon. The genes for at least five proteins can be mutated. These proteins appear to be involved in ubiquitination. Two of the proteins, **α-synuclein** and **parkin,** interact and are found in **Lewy bodies.** The Lewy bodies are inclusion bodies in neurons that occur in all forms of Parkinson disease.

An important consideration in Parkinson disease is the balance between the excitatory discharge of cholinergic interneurons and the inhibitory dopaminergic input in the striatum. Some improvement is produced by decreasing the cholinergic influence with anticholinergic drugs. More dramatic improvement is produced by administration of L-dopa. Unlike dopamine, this dopamine precursor crosses the blood-brain barrier and helps repair the dopamine deficiency. However, the degeneration of these neurons continues and in 5–7 years the beneficial effects of L-dopa disappear.

CEREBELLUM

The cerebellum sits astride the main sensory and motor systems in the brainstem **(Figure 12–16)**. The medial **vermis** and lateral **cerebellar hemispheres** are more extensively folded and fissured than the cerebral cortex. The cerebellum weighs only 10% as much as the cerebral cortex, but its surface area is about 75% of that of the cerebral cortex. Anatomically, the cerebellum is divided into three parts by two transverse fissures. The posterolateral fissure separates the medial nodulus and the lateral flocculus on either side from the rest of the cerebellum, and the primary fissure divides the remainder into an anterior and a posterior lobe. Lesser fissures divide the vermis into smaller sections, so that it contains 10 primary lobules numbered I–X from superior to inferior.

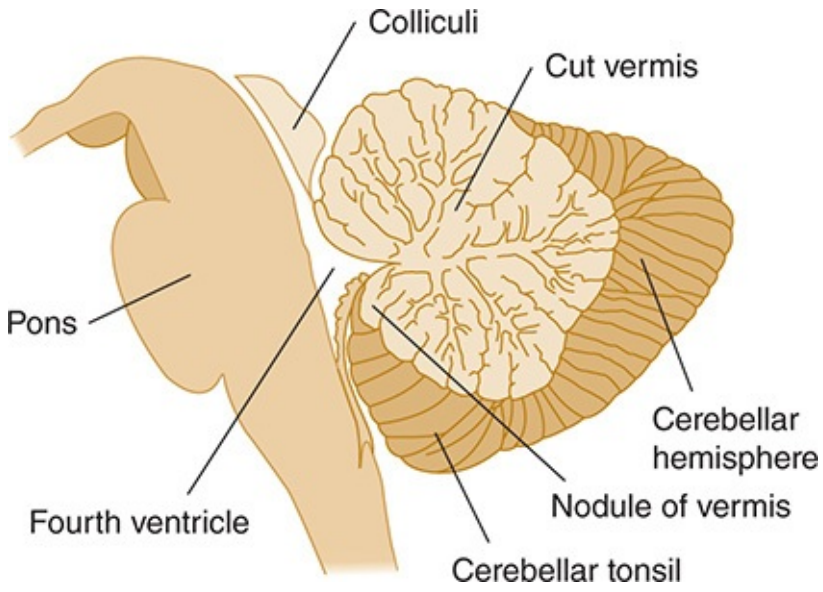


FIGURE 12–16 A midsagittal section through the cerebellum. The medial vermis and lateral cerebellar hemispheres have many narrow, ridge-like folds called folia. Although not shown, the cerebellum is connected to the brainstem by three pairs of peduncles (superior, middle, and inferior). (Reproduced with permission from Waxman SG: *Clinical Neuroanatomy*, 26th ed. New York, NY: McGraw-Hill; 2010.)

The cerebellum is connected to the brainstem by three pairs of peduncles that are located above and around the fourth ventricle. The **superior cerebellar peduncle** includes fibers from deep cerebellar nuclei that project to the brainstem, red nucleus, and thalamus. The **middle cerebellar peduncle** contains only afferent fibers from the contralateral pontine nuclei, and the **inferior**

cerebellar peduncle a mixture of afferent fibers from the brainstem and spinal cord and efferent fibers to the vestibular nuclei.

The cerebellum has an external **cerebellar cortex** separated by white matter from the **deep cerebellar nuclei.** The middle and inferior cerebellar peduncles carry afferent fibers into the cerebellum where they are called **mossy** and **climbing fibers.** These fibers emit collaterals to the deep nuclei and pass to the cortex. There are four deep nuclei: the **dentate,** the **globose,** the **emboliform,** and the **fastigial** nuclei. The globose and the emboliform nuclei are sometimes lumped together as the **interpositus nucleus.**

ORGANIZATION OF THE CEREBELLUM

The cerebellar cortex has three layers: an external molecular layer, a Purkinje cell layer that is only one cell thick, and an internal granular layer. There are five types of neurons in the cortex: Purkinje, granule, basket, stellate, and Golgi cells **(Figure 12–17)**. The **Purkinje cells** are among the largest neurons in the CNS. They have extensive dendritic arbors that extend throughout the molecular layer. Their axons, which are the only output from the cerebellar cortex, project to the deep cerebellar nuclei, especially the dentate nucleus, where they form inhibitory synapses. They also make inhibitory connections with neurons in the vestibular nuclei.

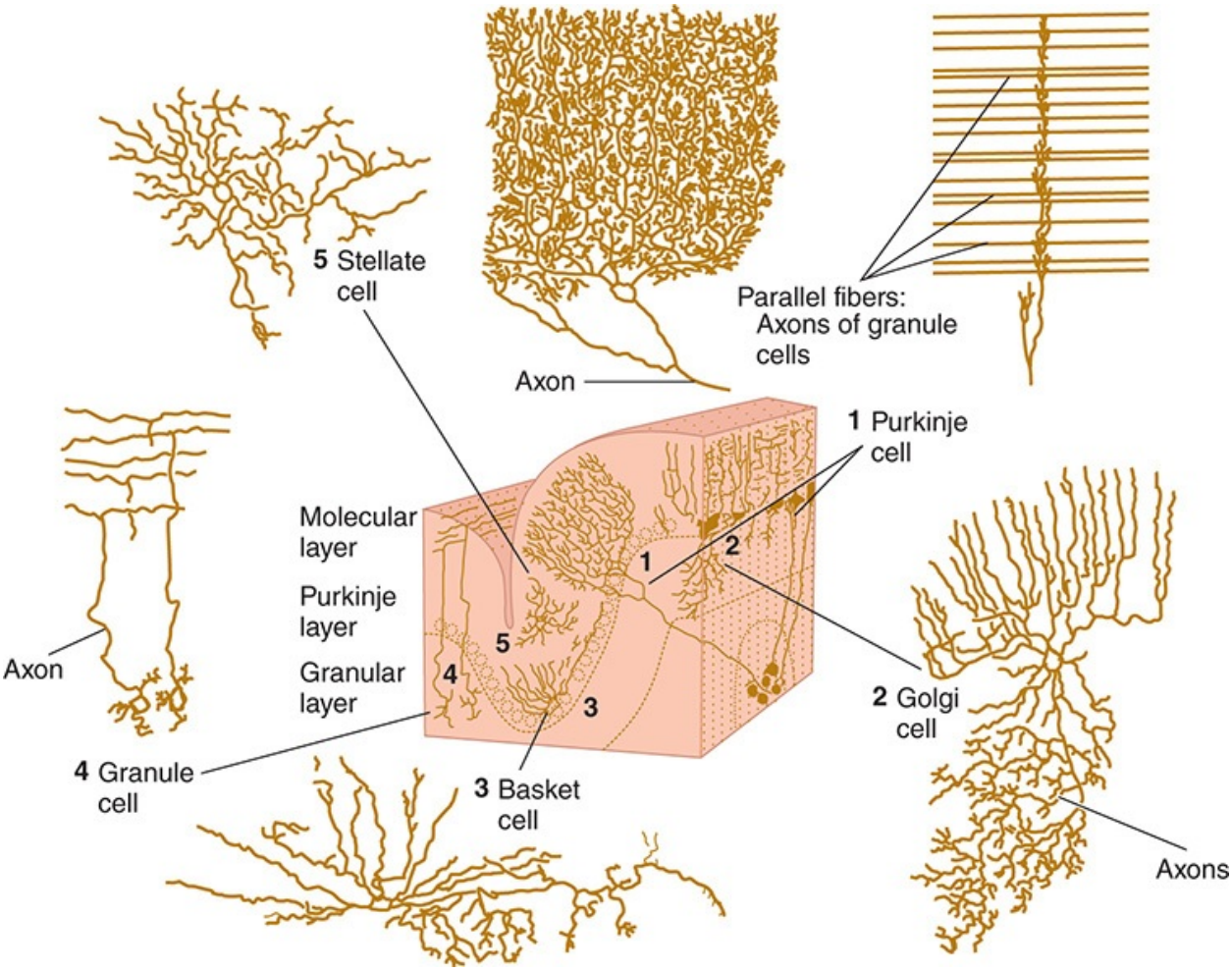


FIGURE 12–17 Location and structure of five neuronal types in the cerebellar cortex. Drawings are based on Golgi-stained preparations. Purkinje cells (1) have processes aligned in one plane; their axons are the only output from the cerebellum. Axons of granule cells (4) traverse and make connections with Purkinje cell processes in molecular layer. Golgi (2), basket (3), and stellate (5) cells have characteristic positions, shapes, branching patterns, and synaptic connections. (For 1 and 2, Reproduced with permission of Ramon y Cajal S: Histologie du Systeme Nerveux II., C.S.I.C. Madrid; For 3–5, Reproduced with permission of Palay SL, Chan-Palay V: *Cerebellar Cortex*. Berlin: Springer-Verlag, 1975.)

The cerebellar **granule cells,** whose cell bodies are in the granular layer, receive excitatory input from the mossy fibers and innervate the Purkinje cells **(Figure 12–18)**. Each sends an axon to the molecular layer, where the axon bifurcates to form a T. The branches of the T are straight and run long distances; thus, they are called **parallel fibers.** The dendritic trees of the Purkinje cells are

markedly flattened and oriented at right angles to the parallel fibers. The parallel fibers form excitatory synapses on the dendrites of many Purkinje cells, and the parallel fibers and Purkinje dendritic trees form a grid of remarkably regular proportions.

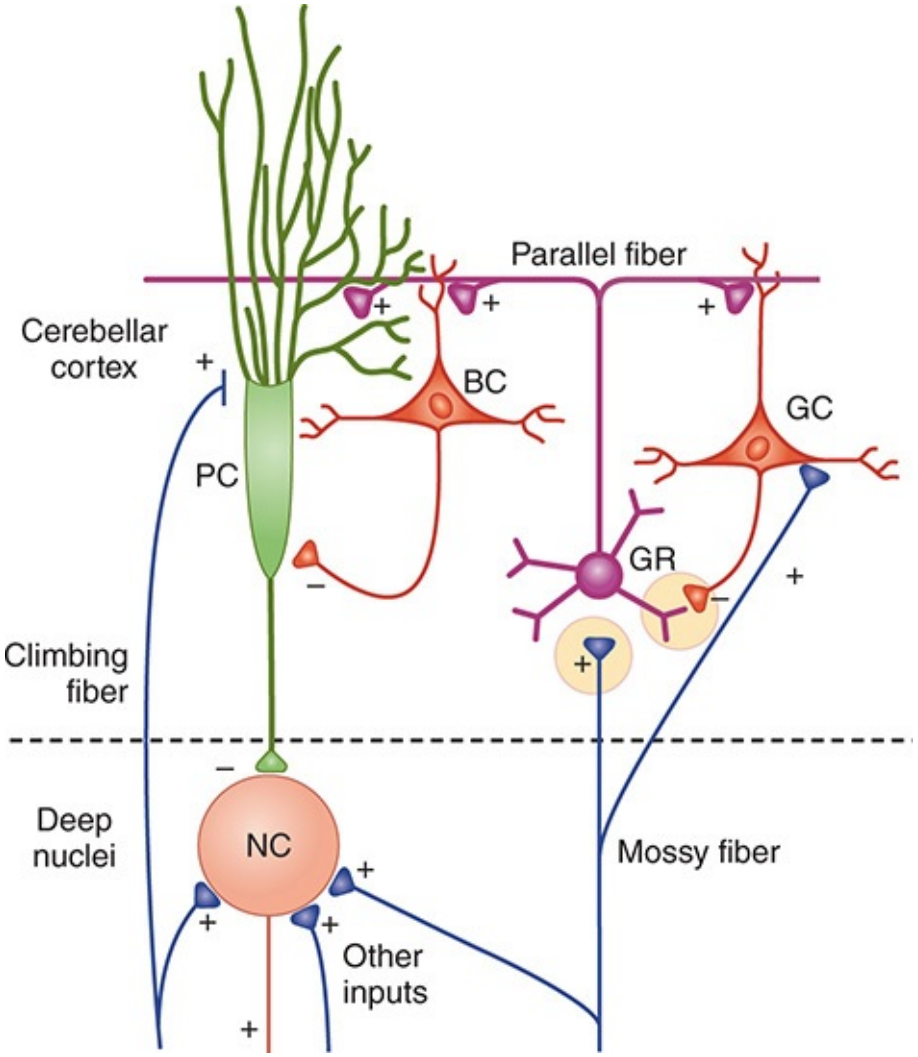


FIGURE 12–18 Diagram of neural connections in the cerebellum. Plus (+) and minus (–) signs indicate whether endings are excitatory or inhibitory. BC, basket cell; GC, Golgi cell; GR, granule cell; NC, cell in deep nucleus; PC, Purkinje cell. Note that PCs and BCs are inhibitory. The connections of the stellate cells, which are not shown, are similar to those of the basket cells, except that they end for the most part on Purkinje cell dendrites.

CLINICAL BOX 12–8

Parkinson Disease

There are 7–10 million people worldwide in whom Parkinson disease has been diagnosed. The disease is 1.5 times more prevalent in men than in women. Each year in the United States, nearly 60,000 new cases are diagnosed. Parkinsonism occurs in sporadic idiopathic form in many middle-aged and elderly individuals and is one of the most common neurodegenerative diseases. It is estimated to occur in 1–2% of individuals over age 65. Dopaminergic neurons and dopamine receptors are steadily lost with age in the basal ganglia in healthy individuals, and an acceleration of these losses apparently precipitates parkinsonism. Symptoms appear when 60–80% of the nigrostriatal dopaminergic neurons degenerate.

Parkinsonism is also seen as a complication of treatment with the phenothiazine group of antipsychotic drugs and other drugs that block dopaminergic D_2 receptors. It can be produced in rapid and dramatic form by injection of 1-methyl-4-phenyl-1,2,5,6-tetrahydropyridine (MPTP). This effect was discovered by chance when a drug dealer in northern California supplied some of his clients with a homemade preparation of synthetic heroin that contained MPTP. MPTP is a prodrug that is metabolized in astrocytes by the enzyme monoamine oxidase (MOA-B) to produce a potent oxidant, 1-methyl-4-phenylpyridinium (MPP+). MPP+ is taken up by the dopamine transporter into dopaminergic neurons in the substantia nigra, which it destroys without affecting other dopaminergic neurons to any appreciable degree.

THERAPEUTIC HIGHLIGHTS

There is no cure for Parkinson disease, and drug therapies are designed to treat the symptoms. **Sinemet,** a combination of **levodopa** (L-dopa) and **carbidopa,** is the most commonly used drug for the treatment of Parkinson disease. The addition of carbidopa to L-dopa increases its effectiveness and prevents the conversion of L-dopa to dopamine in the periphery and thus reduces some of the adverse side effects of L-dopa (nausea, vomiting, and cardiac rhythm disturbances). Dopamine agonists, including **apomorphine, bromocriptine, pramipexole,** and **ropinirole,** have also proven effective in some patients with Parkinson disease. Taken in combination with L-dopa, cathechol-O-methyltransferase (COMT) inhibitors (eg, **entacapone**) are another class of drugs used to treat this disease. They act by blocking the breakdown of L-dopa, allowing more of it to reach the brain to increase the level of dopamine. MAO-

B inhibitors (eg, **selegiline**) also prevent the breakdown of dopamine. They can be given soon after diagnosis and delay the need for L-dopa.

The US Food and Drug Administration has approved the use of deep brain stimulation (DBS) for the treatment of Parkinson disease. DBS reduces the amount of L-dopa patients need and thus reduces its adverse side effects (eg, involuntary movements called **dyskinesias**). DBS has been associated with reduction of tremors, slowness of movements, and gait problems in some patients. Surgical treatments in general are reserved for those who have exhausted drug therapies or who have not responded favorably to them. Lesions in GPi **(pallidotomy)** or in the subthalamic nucleus **(thalamotomy)** have been performed to help restore the output balance of the basal ganglia toward normal (see Figure 12–15). Another surgical approach is to implant dopamine-secreting tissue in or near the basal ganglia. Transplants of the patient's own adrenal medullary tissue or carotid body works for a while, apparently by functioning as a sort of dopamine minipump, but long-term results have been disappointing. Results with transplantation of fetal striatal tissue have been better, and transplanted cells have been shown to not only survive but also make appropriate connections in the host's basal ganglia. However, some patients with transplants develop dyskinesias due to excessive levels of dopamine.

The other three types of neurons in the cerebellar cortex are inhibitory interneurons. **Basket cells** (Figure 12–17) are located in the molecular layer. They receive excitatory input from the parallel fibers and each projects to many Purkinje cells (Figure 12–18). Their axons form a basket around the cell body and axon hillock of each Purkinje cell they innervate. **Stellate cells** are similar to the basket cells but are located in the more superficial molecular layer. **Golgi cells** are located in the granular layer. Their dendrites project into the molecular layer and receive excitatory input from the parallel fibers (Figure 12–18). Their cell bodies receive excitatory input via collaterals from the incoming mossy fibers. Their axons project to the dendrites of the granule cells where they form an inhibitory synapse.

As already mentioned, the two excitatory main inputs to the cerebellar cortex are climbing fibers and mossy fibers (Figure 12–18). The climbing fibers come from a single source, the inferior olivary nuclei. Each projects to the primary dendrites of a Purkinje cell, around which it entwines like a climbing plant. Proprioceptive input to the inferior olivary nuclei comes from all over the body.

On the other hand, the mossy fibers provide direct proprioceptive input from all parts of the body plus input from the cerebral cortex via the pontine nuclei to the cerebellar cortex. They end on the dendrites of granule cells in complex synaptic groupings called **glomeruli.** The glomeruli also contain the inhibitory endings of the Golgi cells mentioned above.

The fundamental circuits of the cerebellar cortex are relatively simple (Figure 12–18). Climbing fiber inputs exert a strong excitatory effect on single Purkinje cells, whereas mossy fiber inputs exert a weak excitatory effect on many Purkinje cells via the granule cells. The basket and stellate cells are also excited by granule cells via their parallel fibers; and the basket and stellate cells, in turn, inhibit the Purkinje cells **(feed-forward inhibition).** Golgi cells are excited by the mossy fiber collaterals and parallel fibers, and they inhibit transmission from mossy fibers to granule cells. The neurotransmitter released by the stellate, basket, Golgi, and Purkinje cells is GABA, whereas the granule cells release glutamate. GABA acts via $GABA_A$ receptors, but the combinations of subunits in these receptors vary from one cell type to the next. The granule cell is unique in that it appears to be the only type of neuron in the CNS that has a $GABA_A$ receptor containing the α6 subunit.

The output of the Purkinje cells is in turn inhibitory to the deep cerebellar nuclei. As noted above, these nuclei also receive excitatory inputs via collaterals from the mossy and climbing fibers. It is interesting, in view of their inhibitory Purkinje cell input, that the output of the deep cerebellar nuclei to the brainstem and thalamus is always excitatory. Thus, almost all the cerebellar circuitry seems to be concerned solely with modulating or timing the excitatory output of the deep cerebellar nuclei to the brainstem and thalamus. The primary afferent systems that converge to form the mossy fiber or climbing fiber input to the cerebellum are summarized in **Table 12–2**.

TABLE 12–2 Function of principal afferent systems to the cerebellum.[a]

Afferent Tracts	Transmits
Vestibulocerebellar	Vestibular impulses from labyrinths, direct and via vestibular nuclei
Dorsal spinocerebellar	Proprioceptive and exteroceptive impulses from muscle spindles, Golgi tendon organ, and joint receptors of lower limbs and trunk
Ventral spinocerebellar	Proprioceptive and exteroceptive impulses from muscle spindles, Golgi tendon organ, and joint receptors of upper and lower limbs
Cuneocerebellar	Proprioceptive impulses from muscle spindles, Golgi tendon organ, and joint receptors of upper limb and upper thorax
Tectocerebellar	Auditory and visual impulses via inferior and superior colliculi, respectively
Pontocerebellar	Impulses from motor and other parts of cerebral cortex via pontine nuclei
Olivocerebellar	Proprioceptive input from whole body via relay in inferior olive

[a]The olivocerebellar pathway projects to the cerebellar cortex via climbing fibers; the rest of the listed paths project via mossy fibers. Several other pathways transmit impulses from nuclei in the brainstem to the cerebellar cortex and to the deep nuclei, including a serotonergic input from the raphe nuclei to the granular and molecular layers and a noradrenergic input from the locus coeruleus to all three layers.

FUNCTIONAL DIVISIONS

From a functional point of view, the cerebellum is divided into three parts **(Figure 12–19)**. The nodulus in the vermis and the flanking flocculus in the hemisphere on each side form the **vestibulocerebellum** (or **flocculonodular lobe**). This lobe is phylogenetically the oldest part of the cerebellum and has

vestibular connections concerned with equilibrium and eye movements. The rest of the vermis and the adjacent medial portions of the hemispheres form the **spinocerebellum,** the region that receives proprioceptive input from the body as well as a copy of the "motor plan" from the motor cortex. By comparing plan with performance, it smooths and coordinates movements that are ongoing. The vermis projects to the brainstem area concerned with control of axial and proximal limb muscles (medial brainstem pathways, and the hemispheres of the spinocerebellum project to the brainstem areas concerned with control of distal limb muscles (lateral brainstem pathways). The lateral portions of the cerebellar hemispheres **(cerebrocerebellum)** are the newest from a phylogenetic point of view, reaching their greatest development in humans. They interact with the motor cortex in planning and programming movements.

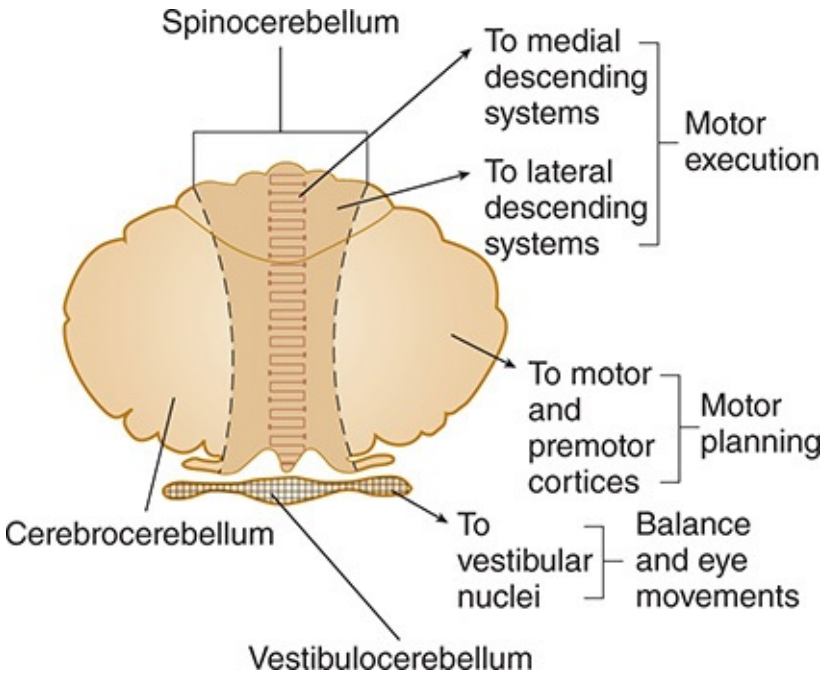


FIGURE 12–19 Three functional divisions of the cerebellum. The nodulus in the vermis and the flanking flocculus in the hemisphere on each side form the vestibulocerebellum which has vestibular connections and is concerned with equilibrium and eye movements. The rest of the vermis and the adjacent medial portions of the hemispheres form the spinocerebellum, the region that receives proprioceptive input from the body as well as a copy of the "motor plan" from the motor cortex. The lateral portions of the cerebellar hemispheres are called the cerebrocerebellum which interacts with the motor cortex in planning and programming movements. (Modified with permission from Kandel ER, Schwartz JH, Jessell TM (eds): *Principles of Neural Science*, 4th ed. New York,

NY: McGraw-Hill; 2000.)

Most of the vestibulocerebellar output passes directly to the brainstem; but the rest of the cerebellar cortex projects to the deep nuclei that then project to the brainstem. The deep nuclei provide the only output for the spinocerebellum and the cerebrocerebellum. The medial portion of the spinocerebellum projects to the fastigial nuclei and from there to the brainstem. The adjacent hemispheric portions of the spinocerebellum project to the emboliform and globose nuclei and from there to the brainstem. The cerebrocerebellum projects to the dentate nucleus and from there either directly or indirectly to the ventrolateral nucleus of the thalamus.

CEREBELLAR DISEASE

Damage to the cerebellum leads to several characteristic abnormalities, including **hypotonia, ataxia,** and **intention tremor.** Motor abnormalities associated with cerebellar damage vary depending on the region involved. **Figure 12–20** illustrates some of these abnormalities. Additional information is provided in **Clinical Box 12–9**.

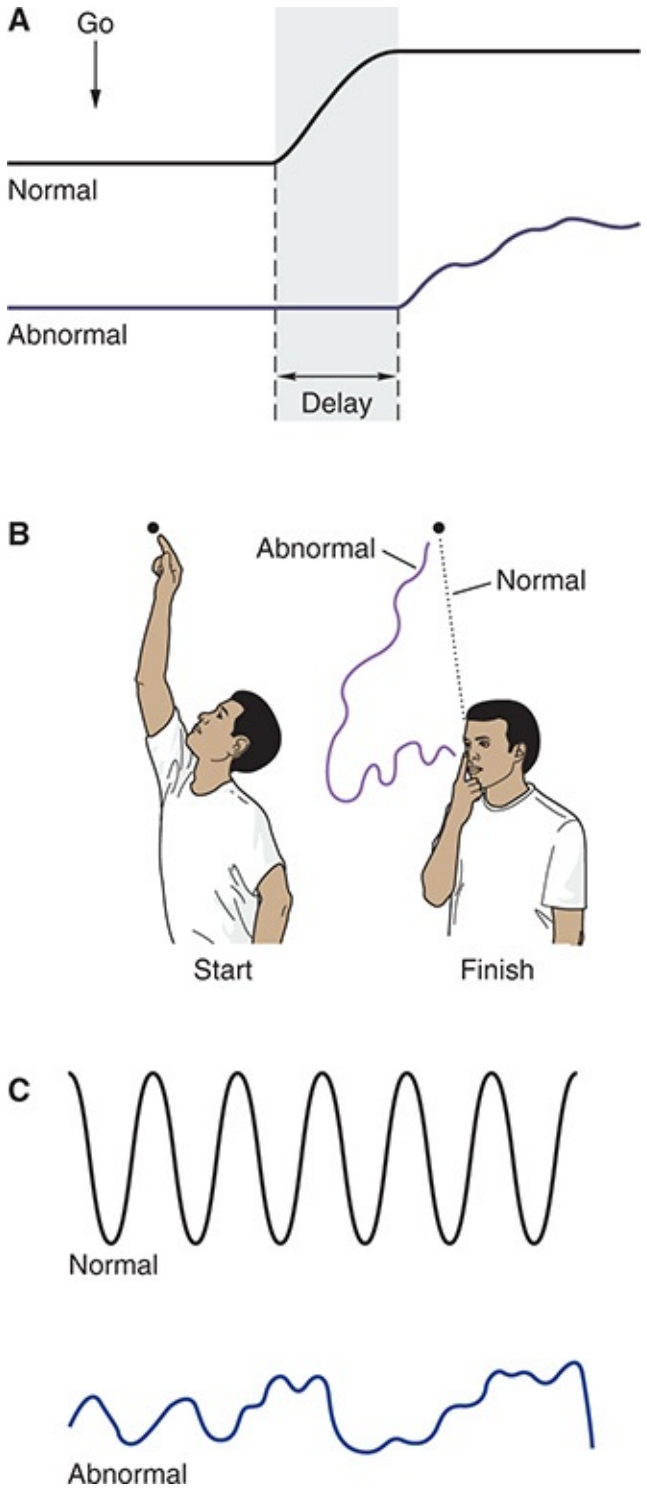


FIGURE 12–20 Typical defects associated with cerebellar disease. A) Lesion of the right cerebellar hemisphere delays initiation of movement. The patient is told to clench both hands simultaneously; right hand clenches later than left (shown by recordings from a pressure bulb transducer squeezed by the patient).

B) Dysmetria and decomposition of movement shown by patient moving his arm from a raised position to his nose. Tremor increases on approaching the nose. **C)** Dysdiadochokinesia occurs in the abnormal position trace of hand and forearm as a cerebellar subject tries alternately to pronate and supinate forearm while flexing and extending elbow as rapidly as possible. (Used with permission from Kandel ER, Schwartz JH, Jessell TM (eds): *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

THE CEREBELLUM & LEARNING

The cerebellum is concerned with learned adjustments that make coordination easier when a given task is performed over and over. As a motor task is learned, activity in the brain shifts from the prefrontal areas to the parietal and motor cortex and the cerebellum. The basis of the learning in the cerebellum is probably the input via the olivary nuclei. The mossy fiber-granule cell-Purkinje cell pathway is highly divergent, allowing an individual Purkinje cell to receive input from many mossy fibers arising from different regions. In contrast, a Purkinje cell receives input from a single climbing fiber but it makes 2000–3000 synapses on it. Climbing fiber activation produces a large, complex spike in the Purkinje cell, and this spike produces long-term modification of the pattern of mossy fiber input to that particular Purkinje cell. Climbing fiber activity is increased when a new movement is being learned, and selective lesions of the olivary complex abolish the ability to produce long-term adjustments in certain motor responses.

CLINICAL BOX 12–9

Cerebellar Disease

Most abnormalities associated with damage to the cerebellum are apparent during movement. The marked **ataxia** is characterized as incoordination due to errors in the rate, range, force, and direction of movement. Ataxia is manifest not only in the wide-based, unsteady, “drunken” gait of patients but also in defects of the skilled movements involved in the production of speech, so that slurred, **scanning speech** results. Many types of ataxia are hereditary, including **Friedreich ataxia** and **Machado-Joseph disease.** There is no cure for hereditary ataxias. Voluntary movements are also highly abnormal when the cerebellum is damaged. For example, attempting to touch an object with a

finger results in overshooting. This **dysmetria** promptly initiates a gross corrective action, but the correction overshoots to the other side, and the finger oscillates back and forth. This oscillation is called an **intention tremor.** Another characteristic of cerebellar disease is the inability to stop movement promptly. Normally, for example, flexion of the forearm against resistance is quickly checked when the resistance force is suddenly broken off. A patient with cerebellar disease cannot stop the movement of the limb, and the forearm flies backward in a wide arc. This abnormal response is known as the **rebound phenomenon** and is one of the reasons these patients show **dysdiadochokinesia,** the inability to perform rapidly alternating opposite movements such as repeated pronation and supination of the hands. Finally, patients with cerebellar disease have difficulty performing actions that involve simultaneous motion at more than one joint. They dissect such movements and carry them out one joint at a time, a phenomenon known as **decomposition of movement.** Other signs of cerebellar deficit in humans point to the importance of the cerebellum in the control of movement. Motor abnormalities associated with cerebellar damage vary depending on the region involved. The major dysfunction seen after damage to the vestibulocerebellum is ataxia, **disequilibrium,** and **nystagmus.** Damage to the vermis and fastigial nucleus (part of the spinocerebellum) leads to scanning speech and disturbances in the control of axial and trunk muscles during attempted antigravity postures. Degeneration of this portion of the cerebellum can result from thiamine deficiency in alcoholics or malnourished individuals. The major dysfunction seen after damage to the cerebrocerebellum is a delay in initiating movements and decomposition of movement.

THERAPEUTIC HIGHLIGHTS

Management of ataxia is primarily supportive and often includes physical, occupational, and speech therapy. Attempts to identify effective drug therapies have met with little success. **DBS** of the ventral intermediate nucleus of the thalamus may reduce cerebellar tremor, but it is less effective in reducing ataxia. A deficiency in **coenzyme Q10** (CoQ10) may contribute to the abnormalities seen in some forms of familial ataxia. If low levels of CoQ10 are detected, treatment to replace the missing CoQ10 has been beneficial.

CHAPTER SUMMARY

- The basic unit of integrated reflex activity is the reflex arc that consists of a sense organ, an afferent neuron, one or more synapses within a central integrating station, an efferent neuron, and an effector.
- A muscle spindle is a group of specialized intrafusal muscle fibers with contractile polar ends and a noncontractile center that is parallel to the extrafusal muscle fibers and innervated by types Ia and II afferent fibers and efferent γ-motor neurons.
- Tapping the patellar tendon causes muscle stretch that activates type Ia fibers that synapse directly on the motor neurons to the extrafusal muscle fibers in the same muscle to cause it to contract (knee jerk reflex).
- Activation of γ-motor neurons causes the contractile ends of the intrafusal fibers to shorten and thus stretches the nuclear bag portion of the spindles, deforming the endings, and initiating impulses in Ia fibers. This can lead to reflex contraction of the muscle. Thus, the γ-motor neurons can initiate contraction indirectly via the stretch reflex.
- A Golgi tendon organ is a netlike collection of knobby nerve endings among the fascicles of a tendon that is in series with extrafusal muscle fibers and innervated by type Ib afferents. They are stimulated by both passive stretch and active contraction of the muscle to relax the muscle (inverse stretch reflex) and function as a transducer to regulate muscle force.
- Physiologic tremor is a rapid (10 Hz), low-amplitude transient tremor in the limbs of healthy individuals; it can become accentuated with anxiety. Clonus is the occurrence of regular, repetitive, rhythmic contractions of a muscle subjected to sudden, maintained stretch; clonus with five or more beats is considered abnormal. Muscle tone is the resistance to muscle stretch. If the muscle offers very little resistance and is said to be flaccid; if resistance to stretch is high, the muscle is spastic. Damage to the cerebral cortex before or during childbirth or during the first 2–3 years of development can lead to cerebral palsy, a disorder that affects muscle tone, movement, and coordination.
- The flexor withdrawal reflex is a polysynaptic reflex that is initiated by nociceptive stimuli; it can serve as a protective mechanism to present further injury.
- Spinal cord injury is followed by a period of spinal shock during which all spinal reflex responses are depressed. In humans, recovery begins about 2

weeks after the injury.

- The supplementary cortex, basal ganglia, and cerebellum participate in the planning of skilled movements. Commands from the primary motor cortex and other cortical regions are relayed via the corticospinal and corticobulbar tracts to spinal and brainstem motor neurons. Cortical areas and motor pathways descending from the cortex are somatotopically organized.
- The ventral corticospinal tract and medial descending brainstem pathways (tectospinal, reticulospinal, and vestibulospinal tracts) regulate proximal muscles and posture. The lateral corticospinal and rubrospinal tracts control distal limb muscles for fine motor control and skilled voluntary movements.
- Lower motor neurons are those whose axons terminate on skeletal muscles; damage to these neurons (eg, ALS) is associated with flaccid paralysis, muscular atrophy, fasciculations, hypotonia, and hyporeflexia or areflexia. Upper motor neurons include corticospinal tract neurons that innervate spinal motor neurons; damage to these neurons initially causes muscles to become weak and flaccid but eventually leads to spasticity, hypertonia, hyperactive stretch reflexes, and a positive Babinski sign.
- Damage to the cerebral cortex before or during childbirth can lead to cerebral palsy, a disorder that affects muscle tone, movement, and coordination. Decerebrate rigidity leads to hyperactivity in extensor muscles in all four extremities; the spasticity is due to facilitation of the stretch reflex. It resembles what occurs with uncal herniation due to a supratentorial lesion. Decorticate rigidity is flexion of the upper extremities at the elbow and extensor hyperactivity in the lower extremities. It occurs on the hemiplegic side after hemorrhage or thrombosis in the internal capsule.
- The basal ganglia include the caudate nucleus, putamen, globus pallidus, subthalamic nucleus, and substantia nigra. The connections between the parts of the basal ganglia include a dopaminergic nigrostriatal projection from the substantia nigra to the striatum and a GABAergic projection from the striatum to substantia nigra.
- Parkinson disease is due to degeneration of the nigrostriatal dopaminergic neurons and is characterized by akinesia, bradykinesia, cogwheel rigidity, and tremor at rest. Huntington disease is characterized by choreiform movements due to the loss of the GABAergic inhibitory pathway to the globus pallidus.
- The cerebellar cortex contains five types of neurons: Purkinje, granule, basket, stellate, and Golgi cells. The two main inputs to the cerebellar cortex

are climbing fibers and mossy fibers. Purkinje cells are the only output from the cerebellar cortex, and they generally project to the deep nuclei. Damage to the cerebellum leads to several characteristic abnormalities, including hypotonia, ataxia, and intention tremor.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A 48-year-old man was undergoing a thorough neurological exam after falling from a construction platform. The test included an evaluation of his deep tendon reflex. What are the elements of the deep tendon reflex?

   A. Golgi tendon organ, group Ib afferent fibers, spinal inhibitory interneuron, contralateral α-motor neuron, and extrafusal fibers of the skeletal muscle

   B. Golgi tendon organ, group II afferent fibers, spinal inhibitory interneuron, ipsilateral α-motor neuron, and intrafusal fibers of the skeletal muscle

   C. Group Ia sensory fibers originating in the contractile portion of the intrafusal fibers of the muscle spindle, spinal dorsal horn neuron, ipsilateral α-motor neuron, and extrafusal fibers of the skeletal muscle

   D. Group Ia sensory fibers originating in the central portion of the intrafusal fibers of the muscle spindle, ipsilateral α-motor neuron, and extrafusal fibers of the skeletal muscle

   E. Group II sensory fibers originating in the contractile portion of the intrafusal fibers of the muscle spindle, spinal inhibitory interneuron, ipsilateral α-motor neuron, and extrafusal fibers of the skeletal muscle

2. When dynamic γ-motor neurons are activated at the same time as α-motor neurons to muscle,

   A. prompt inhibition of discharge in spindle Ia afferents takes place.

   B. clonus is likely to occur.

   C. the muscle will not contract.

   D. the number of impulses in spindle Ia afferents is smaller than when α discharge alone is increased.

   E. the number of impulses in spindle Ia afferents is greater than when α discharge alone is increased.

3. The inverse stretch reflex

   A. occurs when Ia spindle afferents are inhibited.

B. is a monosynaptic reflex initiated by activation of the Golgi tendon organ.

C. is a disynaptic reflex with a single interneuron inserted between the afferent and efferent limbs.

D. is a polysynaptic reflex with many interneurons inserted between the afferent and efferent limbs.

E. uses type II afferent fibers from the Golgi tendon organ.

4. Withdrawal reflexes are

A. initiated by innocuous stimulation of the skin.

B. can lead to the appearance of clonus.

C. prolonged if the stimulus is strong.

D. an example of a stretch reflex.

E. accompanied by the same response on both sides of the body.

5. While exercising, a 42-year-old woman developed sudden onset of tingling in her right leg and an inability to control movement in that limb. A neurologic exam showed a hyperactive knee jerk reflex and a positive Babinski sign. What is a possible basis for these findings and does it reflect damage to an upper or lower motor neuron?

A. She had a lower thoracic disk rupture that damaged the right side of her spinal cord (upper motor neuron damage).

B. She had a mid-cervical disk rupture that damaged the left side of her spinal cord (upper motor neuron damage).

C. She had a lower lumbar disk rupture that compressed the spinal nerve at that segmental level (lower motor neuron damage).

D. She had a sacral disk rupture that put pressure on the ventral root at that segmental level (lower motor neuron damage).

E. She was experiencing dysfunction of both a lower motor neuron and an upper motor neuron.

6. Increased neural activity before a skilled voluntary movement is *first* seen in the

A. spinal motor neurons.

B. precentral motor cortex.

C. midbrain.

D. cerebellum.

E. cortical association areas.

7. A 58-year-old woman was brought to the emergency department of her local

hospital because of a sudden change of consciousness. All four limbs were extended, suggestive of decerebrate rigidity. A brain CT showed a rostral pontine hemorrhage. What are the underlying neurophysiological changes that lead to the appearance of decerebrate rigidity?

A. Destruction of the rubrospinal tract eliminates inhibition of the cerebellar fastigial nucleus and secondarily increases excitation to vestibular nuclei to activate extensor muscles in the limbs.

B. Loss of the corticospinal pathway disrupts excitatory input to motor neurons controlling flexor muscles, leaving extensor muscles to undergo sustained contraction.

C. Sensory input activates the medullary reticulospinal pathway, which then directly activates motor neurons to extensor muscles in all four extremities.

D. Sensory input activates neurons in the rubrospinal tract that inhibit flexor α-motor neurons and excite extensor α-motor neurons in all four limbs.

E. Sensory input activates the pontine reticulospinal pathway, which then activates primarily γ-motor neurons to extensor muscles in all four extremities.

8. Which of the following correctly describes components of the central pathway responsible for control of posture?

A. The tectospinal pathway originates in the superior colliculus and terminates on neurons in the dorsolateral area of the spinal ventral horn that innervate limb muscles.

B. The medullary reticulospinal pathway terminates on neurons in the ventromedial area of the spinal ventral horn that innervate axial and proximal muscles.

C. The pontine reticulospinal pathway terminates on neurons in the dorsomedial area of the spinal ventral horn that innervate limb muscles.

D. The medial vestibular pathway terminates on neurons in the dorsomedial area of the spinal ventral horn that innervate axial and proximal muscles.

E. The lateral vestibular pathway terminates on neurons in the dorsolateral area of the spinal ventral horn that innervate axial and proximal muscles.

9. Which of the following describes a connection between components of the basal ganglia?

A. The subthalamic nucleus releases glutamate to excite the globus pallidus, internal segment.

B. The substantia nigra pars reticulata releases dopamine to inhibit the

striatum.

C. The substantia nigra pars compacta releases dopamine to excite the globus pallidus, external segment.

D. The striatum releases acetylcholine to excite the substantia nigra pars reticulata.

E. The globus pallidus, external segment releases glutamate to excite the striatum.

10. A 60-year-old man with Parkinson disease, which was diagnosed 15 years ago, has been taking carbidopa and levodopa (Sinemet); until recently, he has been able to continue to work and help with routine jobs around the house. Now his tremor and rigidity interfere with these activities. His clinician has suggested that he undergo deep brain stimulation therapy. The therapeutic effect of L-dopa in patients with Parkinson disease eventually wears off because

A. antibodies to dopamine receptors develop.

B. inhibitory pathways grow into the basal ganglia from the frontal lobe.

C. there is an increase in circulating α-synuclein.

D. the normal action of nerve growth factor (NGF) is disrupted.

E. the dopaminergic neurons in the substantia nigra continue to degenerate.

11. An 8-year-old girl was brought to her pediatrician because her parents noted frequent episodes of gait unsteadiness and speech difficulties. Her mother was concerned because of a family history of Friedreich ataxia. Which of the following is a correct description of connections involving cerebellar neurons?

A. Basket cells release glutamate to activate Purkinje cells.

B. Climbing fiber inputs exert a strong excitatory effect on Purkinje cells, and mossy fiber inputs exert a strong inhibitory effect on Purkinje cells.

C. Granule cells release glutamate to excite basket cells and stellate cells.

D. The axons of Purkinje cells are the sole output of the cerebellar cortex, and they release glutamate to excite the deep cerebellar nuclei.

E. Golgi cells are inhibited by mossy fiber collaterals.

12. After falling down a flight of stairs, a young woman is found to have partial loss of voluntary movement on the right side of her body and loss of pain and temperature sensation on the left side below the midthoracic region. It is probable that she has a lesion

A. damaging the left half of the spinal cord in the lumbar region.

B. damaging the left half of the spinal cord in the upper thoracic region.

C. damaging sensory and motor pathways on the right side of the pons.

D. damaging the right half of the spinal cord in the upper thoracic region.

E. damaging the dorsal half of the spinal cord in the upper thoracic region.

13. At the age of 30, a male postal worker reported weakness in his right leg. Within a year the weakness had spread to his entire right side. A neurologic examination revealed flaccid paralysis, muscular atrophy, fasciculations, hypotonia, and hyporeflexia of muscles in the right arm and leg. Sensory and cognitive function tests were normal. Which of the following diagnosis is likely?

A. A large tumor in the left primary motor cortex

B. A cerebral infarct in the region of the corona radiate

C. A vestibulocerebellar tumor

D. Damage to the basal ganglia

E. Amyotrophic lateral sclerosis

CHAPTER 13

Autonomic Nervous System

OBJECTIVES

After studying this chapter, you should be able to:

- Identify the location of the cell bodies and axonal trajectories of preganglionic and postganglionic sympathetic and parasympathetic neurons.
- Identify the neurotransmitters and receptor types involved in neurotransmission within the autonomic nervous system and its target organs.
- Describe how various drugs alter neurotransmitter synthesis, storage, release, and reuptake and receptor activation and blockade within the autonomic nervous system.
- Describe the ways that the autonomic nervous system contributes to homeostasis.
- Compare the overall functions of the parasympathetic and sympathetic nervous systems.
- Compare and contrast the functions of sympathetic and parasympathetic nerves at targets where they act as functional antagonists, synergistically, and independently.
- Identify the location of forebrain and brainstem neurons and sensory

afferents that are involved in the control of the autonomic nervous system.

- Identify examples of autonomic dysfunction due to primary damage within the autonomic nervous system or as a consequence of other pathologies.
- Describe the composition and functions of the enteric nervous system.

INTRODUCTION

The autonomic nervous system (ANS) is comprised of the **sympathetic nervous system**, the **parasympathetic nervous system**, and the **enteric nervous system**. The ANS is sometimes called the involuntary nervous system because it carries out its functions without requiring a conscious effort. It influences a wide array of physiological processes via its innervation of smooth muscle, cardiac muscle, and pacemaker cells, exocrine and endocrine glands, adipose tissue, liver cells, and lymphatic tissue. In fact, skeletal muscle is the only innervated part of the body that is not under the control of the ANS. The ultimate responsibility of the ANS is to maintain **homeostasis** despite perturbations exerted by the external and internal environments. Although survival may be possible without the ANS, the ability to adapt to environmental stressors and other challenges is severely compromised. The ANS also plays a role in the body's response to an emotional experience. The importance of understanding the functions of the ANS is underscored by the fact that so many prescription and over-the-counter drugs exert their actions on elements of the ANS or its effector targets. Changes in autonomic activity contribute to many diseases (eg, hypertension, heart failure). Also, many neurologic disorders are associated with autonomic dysfunction **(Clinical Box 13–1)**.

CLINICAL BOX 13–1

Multiple System Atrophy & Shy–Drager Syndrome

Multiple system atrophy (MSA) is a neurodegenerative disorder associated with autonomic failure due to loss of preganglionic autonomic neurons in the spinal cord and brainstem. In the absence of an autonomic nervous system, it is difficult to regulate body temperature, fluid and electrolyte balance, and blood pressure. In addition to these autonomic abnormalities, MSA presents with cerebellar, basal ganglia, locus coeruleus, inferior olivary nucleus, and pyramidal tract deficits. MSA is defined as "a sporadic, progressive, adult-

onset disorder characterized by autonomic dysfunction, parkinsonism, and cerebellar ataxia in any combination." **Shy–Drager syndrome** is a subtype of MSA in which autonomic failure dominates. The pathologic hallmark of MSA is cytoplasmic and nuclear inclusions in oligodendrocytes and neurons in central motor and autonomic areas. There is also depletion of monoaminergic, cholinergic, and peptidergic markers in several brain regions and in the cerebrospinal fluid. The cause of MSA remains elusive, but there is some evidence that a neuroinflammatory mechanism causing activation of microglia and production of toxic cytokines may occur in brains of MSA patients. Basal levels of sympathetic activity and plasma norepinephrine levels are normal in MSA patients, but they fail to increase in response to standing or other stimuli and leads to severe **orthostatic hypotension.** In addition to the fall in blood pressure, orthostatic hypotension leads to dizziness, dimness of vision, and even fainting. MSA is also accompanied by parasympathetic dysfunction, including urinary and sexual dysfunction. MSA is most often diagnosed in individuals between 50 and 70 years of age; it affects more men than women. Erectile dysfunction is often the first symptom of the disease. There are also abnormalities in baroreceptor reflex and respiratory control mechanisms. Although autonomic abnormalities are often the first symptoms, 75% of patients with MSA also experience motor disturbances.

THERAPEUTIC HIGHLIGHTS

There is no cure for MSA but various therapies are used to treat specific signs and symptoms of the disease. **Corticosteroids** are often prescribed to retain salt and water to increase blood pressure. In some individuals, parkinsonian-like signs can be alleviated by administration of **levodopa** and **carbidopa (Sinemet).** Various clinical trials are underway to test the effectiveness of using intravenous **immunoglobulins** to counteract the neuroinflammatory process that occurs in MSA; **fluoxetine** (a **serotonin uptake inhibitor**) to prevent orthostatic hypotension, improve mood, and alleviate sleep, pain, and fatigue in MSA patients; and **rasagiline** (a **monoamine oxidase inhibitor**) in MSA patients with parkinsonism.

ANATOMIC ORGANIZATION OF AUTONOMIC

OUTFLOW

GENERAL FEATURES

Figure 13–1 compares some fundamental characteristics of the innervation to skeletal muscles with innervation to smooth muscle, cardiac muscle, and glands. As discussed in Chapter 12, the final common pathway linking the central nervous system (CNS) to skeletal muscles is the α-motor neuron. Similarly, sympathetic and parasympathetic neurons serve as the final common pathway from the CNS to visceral targets. However, in marked contrast to the somatomotor nervous system, the peripheral motor portions of the ANS are made up of two neurons: **preganglionic** and **postganglionic neurons.** The cell bodies of the preganglionic neurons are located in the intermediolateral (IML) column of the spinal cord and in motor nuclei of some cranial nerves. In contrast to the large diameter and rapidly conducting α-motor neurons, preganglionic axons are small-diameter, myelinated, relatively slowly conducting B fibers. The axons of the postganglionic neurons are mostly unmyelinated C fibers and terminate on the visceral effectors.

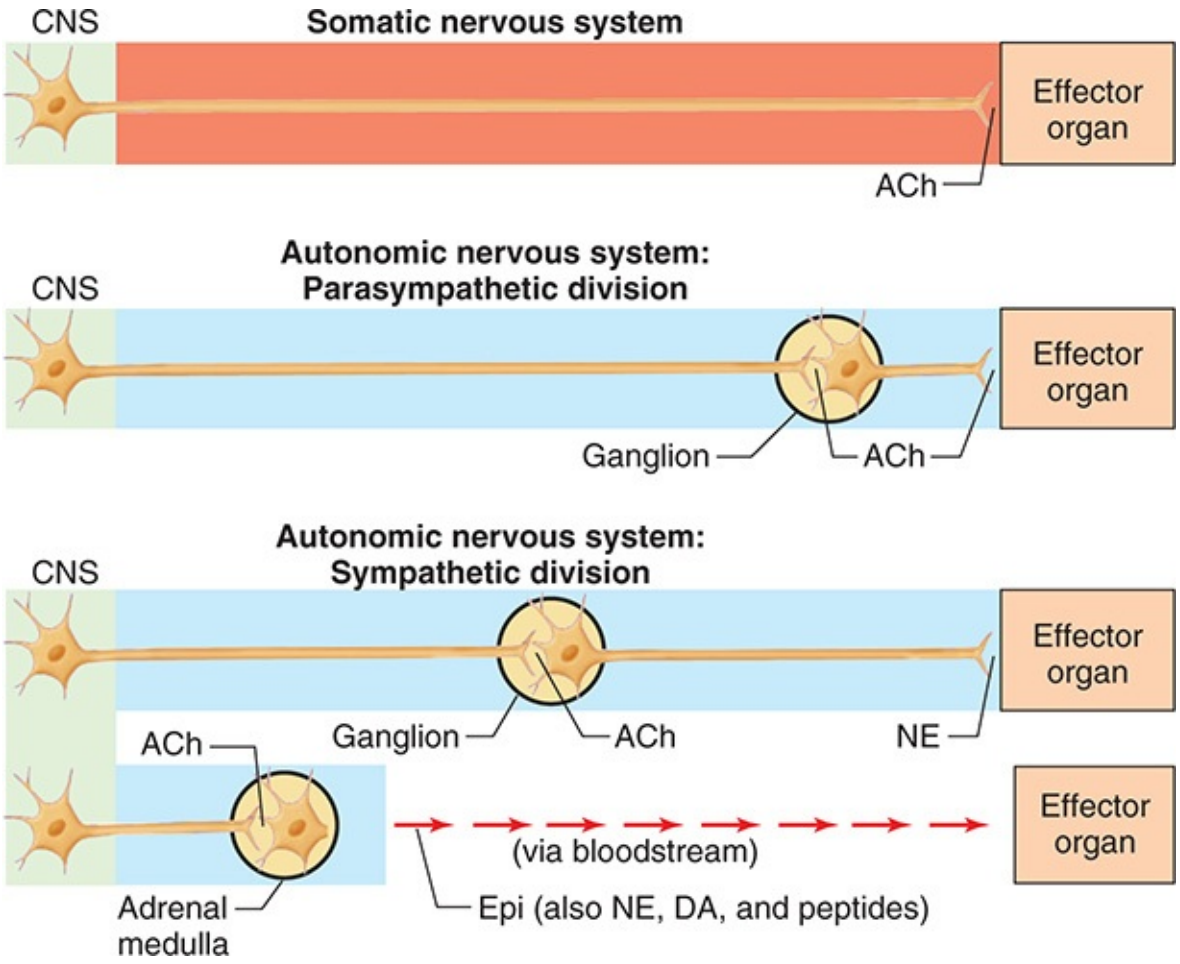


FIGURE 13–1 Comparison of peripheral organization and transmitters released by somatomotor and autonomic nervous systems. In the case of the somatomotor nervous system, the neuron that leaves the spinal cord projects directly to the effector organ. In the case of the autonomic nervous system, there is a synapse between the neuron that leaves the spinal cord and the effector organ (except for neurons that innervate the adrenal medulla). Note that all neurons that leave the central nervous system release acetylcholine (ACh). CNS, central nervous system; DA, dopamine; Epi, epinephrine; NE, norepinephrine. (Used with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology*. New York, NY: McGraw-Hill; 2008.)

One similar feature of autonomic preganglionic neurons and α-motor neurons is that acetylcholine is released at their nerve terminals (Figure 13–1). This is the neurotransmitter released by all neurons whose axons exit the CNS, including cranial motor neurons, α-motor neurons, γ-motor neurons, preganglionic sympathetic neurons, and preganglionic parasympathetic neurons.

Postganglionic parasympathetic neurons also release acetylcholine; postganglionic sympathetic neurons release either norepinephrine or acetylcholine.

SYMPATHETIC DIVISION OF THE ANS

In contrast to α-motor neurons, which are located at all spinal segments, sympathetic preganglionic neurons are located in the IML of only the first thoracic to the third or fourth lumbar segments. Thus, the sympathetic nervous system is called the thoracolumbar division of the ANS. The axons of the sympathetic preganglionic neurons leave the spinal cord at the level at which their cell bodies are located and exit via the ventral root along with axons of α- and γ-motor neurons **(Figure 13–2)**. They then separate from the ventral root via the **white rami communicans** and project to the adjacent **sympathetic paravertebral ganglion,** where some of them end on the cell bodies of the postganglionic neurons. Paravertebral ganglia are located adjacent to each thoracic and upper lumbar spinal segment; in addition, there are a few ganglia adjacent to the cervical and sacral spinal segments. The ganglia are connected to each other via the axons of preganglionic neurons that travel rostrally or caudally to terminate on postganglionic neurons located at some distance. Together these ganglia and axons form the **sympathetic chain** bilaterally. This arrangement is seen in Figure 13–2 and **Figure 13–3**. Note that, despite the fact that there are no cervical or sacral sympathetic preganglionic neurons, the sympathetic chain includes cervical ganglia that provide innervation to structures in the head (eg, eye and salivary glands) and sacral ganglia that provide innervation of pelvic organs.

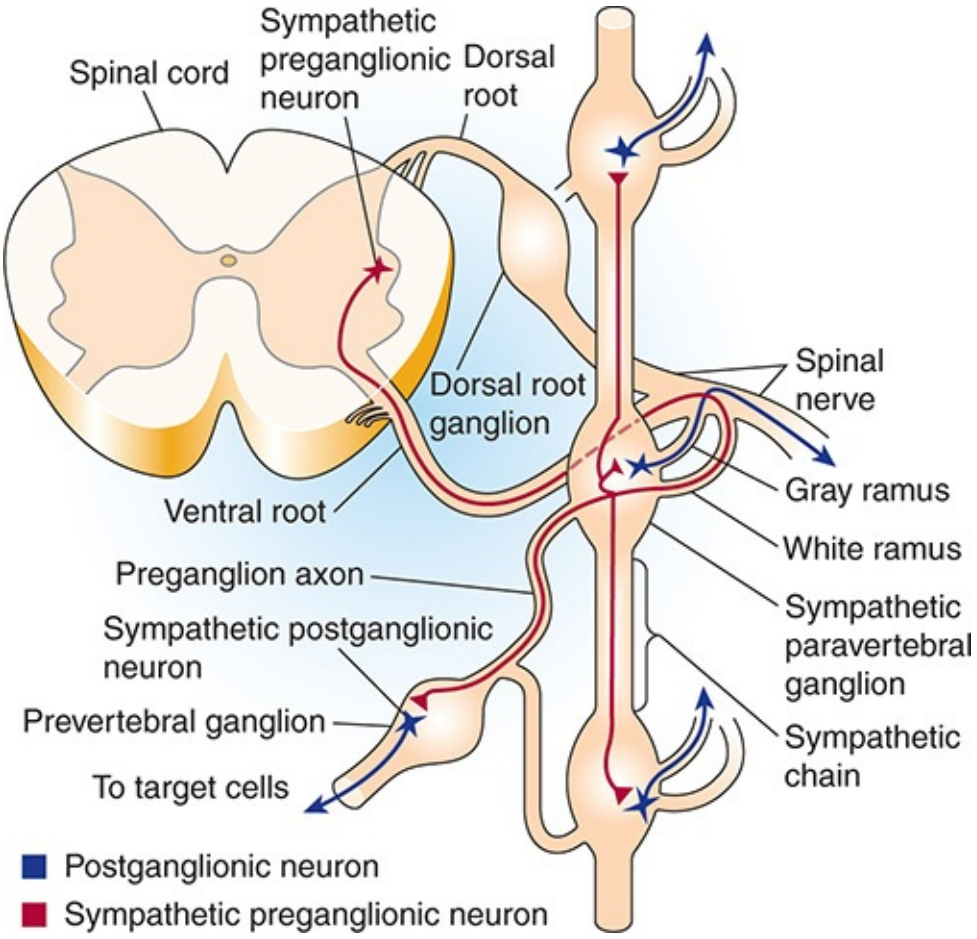


FIGURE 13–2 Projection of sympathetic preganglionic and postganglionic fibers. The drawing shows the thoracic spinal cord, paravertebral, and prevertebral ganglia. Preganglionic neurons are shown in red, and postganglionic neurons in dark blue.

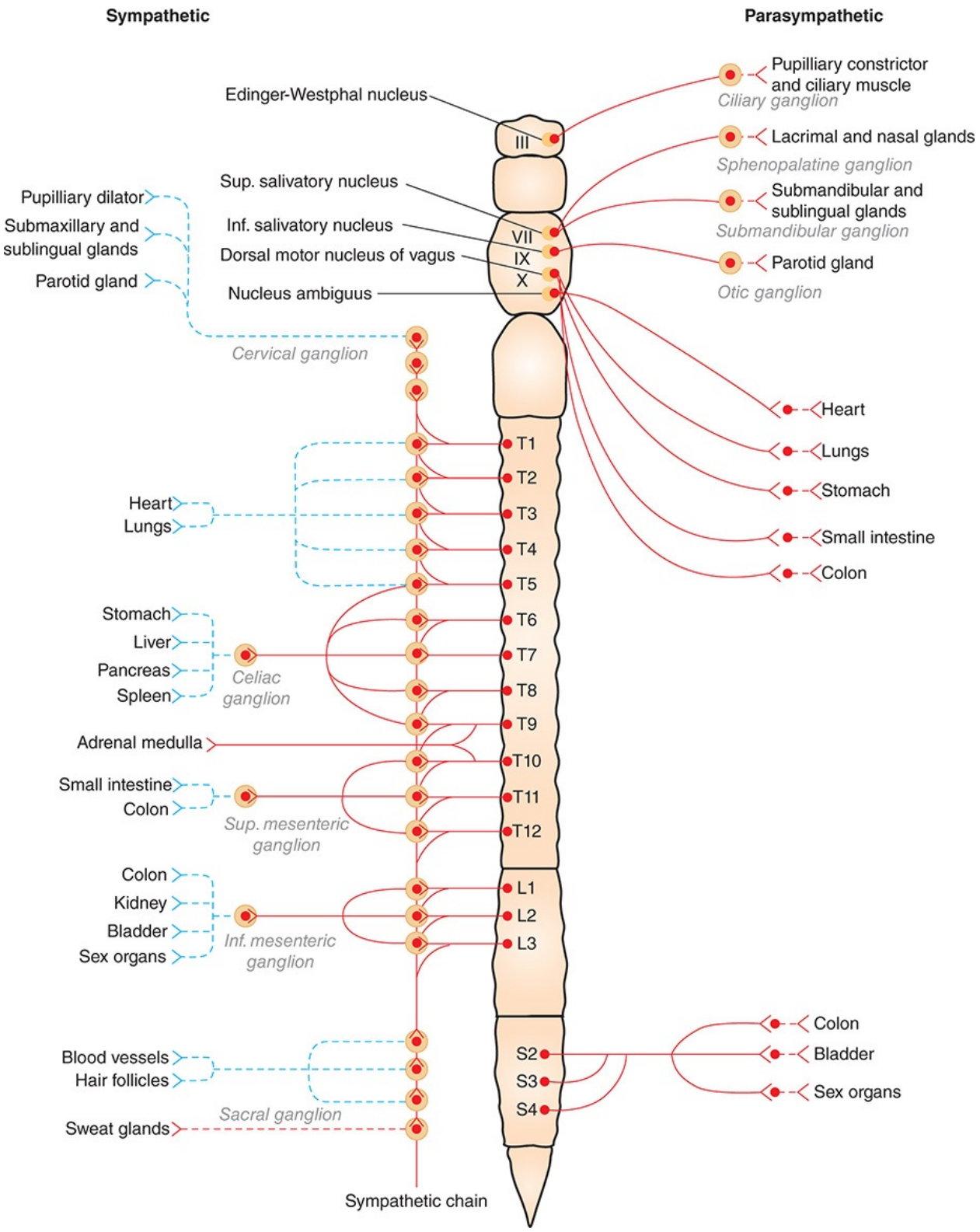


FIGURE 13–3 Organization of sympathetic (left) and parasympathetic (right) nervous systems. Cholinergic nerves are shown in red and noradrenergic

nerves are shown in blue. Preganglionic nerves are solid lines; postganglionic nerves are dashed lines.

Some preganglionic neurons pass through the paravertebral ganglion chain and end on postganglionic neurons located in **prevertebral** (or **collateral**) **ganglia** close to the viscera, including the celiac, superior mesenteric, and inferior mesenteric ganglia (Figure 13–3). There are also preganglionic neurons whose axons terminate directly on the effector organ, the adrenal gland.

The axons of some of the postganglionic neurons leave the chain ganglia and reenter the spinal nerves via the **gray rami communicans** and are distributed to autonomic effectors in the areas supplied by these spinal nerves (Figure 13–2). These postganglionic sympathetic nerves terminate mainly on smooth muscle (eg, blood vessels and hair follicles) and on sweat glands in the limbs. Other postganglionic fibers leave the chain ganglia to enter the thoracic cavity to terminate in visceral organs. Postganglionic fibers from prevertebral ganglia also terminate in visceral targets.

PARASYMPATHETIC DIVISION OF THE ANS

The parasympathetic nervous system is called the **craniosacral division** of the ANS because of the location of its preganglionic neurons within several cranial nerve nuclei (III, VII, IX, and X) and the IML of the sacral spinal cord (Figure 13–3). The cell bodies in the **Edinger-Westphal nucleus** of the oculomotor nerve project to the **ciliary ganglia** to innervate the sphincter (constrictor) muscle of the iris and the ciliary muscle. Neurons in the **superior salivatory nucleus** of the facial nerve project to the **sphenopalatine ganglia** to innervate the lacrimal glands and nasal and palatine mucous membranes and to the **submandibular ganglia** to innervate the submandibular (also called submaxillary) and sublingual glands. The cell bodies in the **inferior salivatory nucleus** of the glossopharyngeal nerve project to the **otic ganglion** to innervate the parotid salivary gland. Vagal preganglionic fibers synapse on ganglia cells clustered within the walls of visceral organs; thus, these parasympathetic postganglionic fibers are very short. Neurons in the **nucleus ambiguus** innervate the sinoatrial (SA) and atrioventricular (AV) nodes in the heart; and neurons in the **dorsal motor vagal nucleus** innervate the esophagus, trachea, lungs, and gastrointestinal tract. The parasympathetic sacral outflow **(pelvic nerve)** supplies the pelvic viscera via branches of the second to fourth sacral spinal nerves.

CHEMICAL TRANSMISSION AT AUTONOMIC JUNCTIONS

ACETYLCHOLINE & NOREPINEPHRINE

Acetylcholine and **norepinephrine** are the principal neurotransmitters synthesized and released by autonomic neurons. The cholinergic autonomic neurons (ie, release acetylcholine) are (1) all preganglionic neurons, (2) all parasympathetic postganglionic neurons, (3) sympathetic postganglionic neurons that innervate sweat glands, and (4) sympathetic postganglionic neurons that end on blood vessels in some skeletal muscles and produce vasodilation when stimulated (sympathetic vasodilator nerves). The remaining sympathetic postganglionic neurons are **noradrenergic** (ie, release norepinephrine). The adrenal medulla is essentially a sympathetic ganglion in which the postganglionic cells have lost their axons and secrete both norepinephrine and epinephrine directly into the bloodstream.

Table 13–1 shows the types of cholinergic and adrenergic receptors at various junctions within the ANS. The junctions in the peripheral autonomic motor pathways are logical sites for pharmacologic manipulation of visceral function. The neurotransmitters are synthesized, stored in the nerve endings, and released near the neurons, muscle cells, or gland cells where they bind to various ion channels or G-protein-coupled receptors (GPCR) to initiate their characteristic actions. The neurotransmitters are then removed from the area by reuptake or metabolism. Each of these steps can be stimulated or inhibited, with predictable consequences. **Table 13–2** lists how various drugs can affect neurotransmission in autonomic neurons and effector sites.

TABLE 13–1 Responses of some effector organs to autonomic nerve activity.

Effector Organs	Parasympathetic Nervous System	Sympathetic Nervous System	
		Receptor Type	Response
Eyes			
Radial muscle of iris	—	α_1	Contraction (mydriasis)
Sphincter muscle of iris	Contraction (miosis)		—
Ciliary muscle	Contraction for near vision		—
Heart			
SA node	Decreased heart rate	β_1	Increased heart rate
Atria and ventricle	Decreased atrial contractility	β_1, β_2	Increased contractility
AV node and Purkinje fibers	Decreased conduction velocity	β_1	Increased conduction velocity
Arterioles			
Skin, splanchnic vessels	—	α_1	Constriction
Skeletal muscle	—	α_1/ β_2, M	Constriction/Dilation
Systemic veins	—	α_1, α_2/ β_2	Constriction/Dilation
Bronchial smooth muscle	Contraction	β_2	Relaxation
Stomach and Intestine			
Motility and tone	Increased	α_1, α_2, β_2	Decreased
Sphincters	Relaxation	α_1	Contraction
Secretion	Stimulation	—	
Gallbladder	Contraction	β_2	Relaxation
Urinary bladder			
Detrusor	Contraction	β_2	Relaxation
Sphincter	Relaxation	α_1	Contraction
Uterus (pregnant)	—	α_1/ β_2	Contraction/Relaxation
Male sex organs	Erection	α_1	Ejaculation
Skin			
Pilomotor muscles	—	α_1	Contraction
Sweat glands	—	M	Secretion
Liver	—	α_1, β_2	Glycogenolysis
Pancreas			
Acini	Increased secretion	α	Decreased secretion
Islet cells	—	α_2/β_2	Decreased/Increased secretion
Salivary glands	Profuse, watery secretion	α_1/β	Thick, viscous secretion/Amylase secretion
Lacrimal glands	Secretion		—
Adipose tissue	—	β_3	Lipolysis

A dash means the target tissue is not innervated by this division of the autonomic nervous system. AV, atrioventricular; SA, sinoatrial. Modified with permission from Brunton LL, Chabner BA, Knollmann BC (eds): *Goodman and Gilman's The Pharmacological Basis of Therapeutics*, 12th ed. New York, NY: McGraw-Hill; 2011.

TABLE 13–2 Examples of drugs that affect processes involved in autonomic neurotransmission.

Transmission Process	Drug	Site of Drug Action	Drug Action
Neurotransmitter synthesis	Hemicholinium	Membrane of cholinergic nerve terminals	Blocks choline uptake; slows synthesis
	Metyrosine	Cytoplasm of noradrenergic nerve terminals	Inhibits tyrosine hydroxylase; blocks synthesis
Neurotransmitter storage mechanism	Vesamicol	Vesicles in cholinergic nerve terminals	Prevents storage of acetylcholine
	Reserpine	Vesicles in noradrenergic nerve terminals	Prevents storage of norepinephrine
Neurotransmitter release mechanism	Norepinephrine, dopamine, acetylcholine, prostaglandins	Receptors on cholinergic and adrenergic nerve terminals	Modulates transmitter release
Neurotransmitter reuptake mechanism	Cocaine, tricyclic antidepressants	Adrenergic nerve terminals	Inhibits uptake; prolongs transmitter's action on postsynaptic receptors
Inactivation of neurotransmitter	Edrophonium, neostigmine, physostigmine	Acetylcholinesterase in cholinergic synapses	Inhibits enzyme; prolongs and intensifies actions of acetylcholine
Adrenoceptor activation	α_1: Phenylephrine α_2: Clonidine	Sympathetic postganglionic nerve-effector organ junctions (eg, blood vessels, hair follicles, and radial muscle)	Binds to and activates α-adrenoceptors; ↑ IP_3/DAG cascade (α_1) or ↓ cAMP (α_2)
	β_1: Dobutamine β_2: Albuterol, ritodrine, salmeterol, terbutaline	Sympathetic postganglionic nerve-effector organ junctions (eg, heart, bronchial smooth muscle, and uterine smooth muscle)	Binds to and activates β-adrenoceptors; ↑ cAMP
Adrenoceptor blockade	Nonselective: Phenoxybenzamine α_1: Prazosin, terazosin α_2: Yohimbine	Sympathetic postganglionic nerve-effector organ junctions (eg, blood vessels)	Binds to and blocks α-adrenoceptors
	β_1, β_2: Propranolol $\beta_1 > \beta_2$: Atenolol, esmolol	Sympathetic postganglionic nerve-effector organ junctions (eg, heart and bronchial smooth muscle)	Binds to and blocks β-adrenoceptors
Nicotinic receptor activation	Nicotine	Receptors on autonomic ganglia	Binds to nicotinic receptors; opens Na^+, K^+ channels in postsynaptic membrane
Nicotinic receptor blockade	Hexamethonium, trimethaphan	Receptors on autonomic ganglia	Binds to and blocks nicotinic receptors
Muscarinic receptor activation	Bethanecol	Cholinergic receptors on smooth muscle, cardiac muscle, and glands	Binds to and activates muscarinic receptors; ↑ IP_3/DAG cascade or ↓ cAMP
Muscarinic receptor blockade	Atropine, ipratropium, scopolamine, tropicamide	Cholinergic receptors on smooth muscle, cardiac muscle, and glands	Binds to and blocks muscarinic receptors

Many of these drugs can also act on cholinergic and adrenergic receptors in the central nervous system. For example, the main actions of clonidine and yohimbine to alter blood pressure are by their actions in the brain. cAMP, cyclic adenosine monophosphate; IP_3/DAG, inositol 1,4,5-triphosphate, and diacylglycerol.

CHOLINERGIC NEUROTRANSMISSION

The processes involved in the synthesis and breakdown of acetylcholine were described in Chapter 7. Acetylcholine does not usually circulate in the blood, and the effects of localized cholinergic discharge are generally discrete and of short duration because of the high concentration of acetylcholinesterase at cholinergic nerve endings. This enzyme rapidly breaks down the acetylcholine, terminating its actions.

Transmission in autonomic ganglia is mediated primarily by the actions of acetylcholine on nicotinic cholinergic receptors that are blocked by

hexamethonium **(Figure 13–4)**. These are called N_N receptors to distinguish them from the nicotinic cholinergic receptors (N_M) that are located at the neuromuscular junction and are blocked by D-tubocurarine (curare). Nicotinic receptors are ligand-gated ion channels; binding of an agonist to these receptors opens N^+ and K^+ channels to cause depolarization.

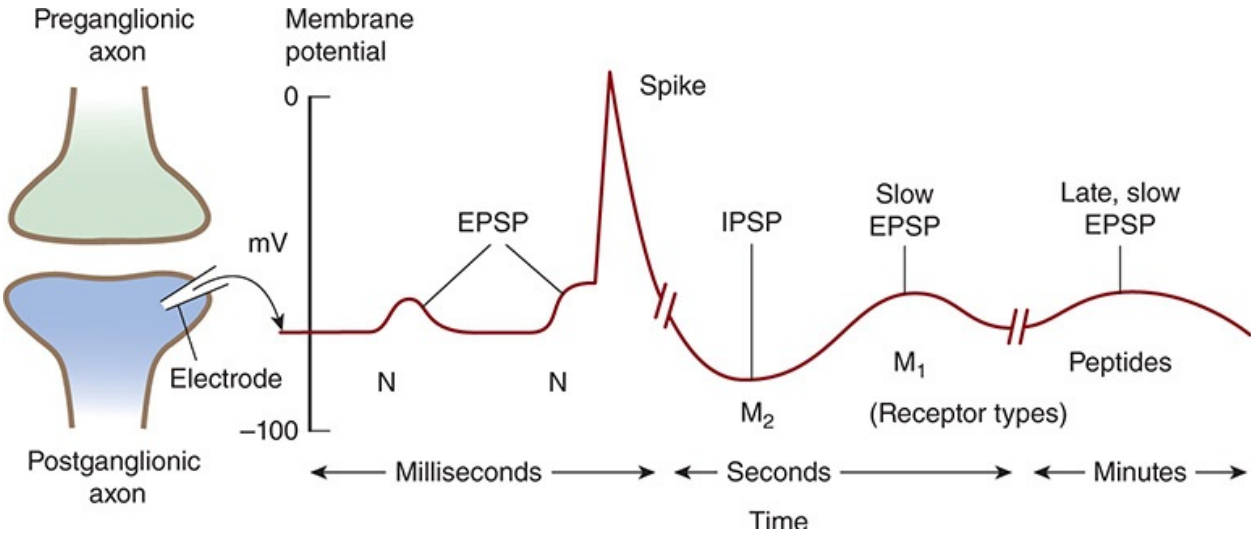


FIGURE 13–4 Schematic of excitatory and inhibitory postsynaptic potentials (EPSP and IPSP) recorded via an electrode in an autonomic ganglion cell. In response to acetylcholine release from the preganglionic neuron, two EPSPs were generated in the postganglionic neuron due to nicotinic (N) receptor activation. The first EPSP was below the threshold for eliciting an action potential, but the second EPSP was suprathreshold and evoked an action potential. This was followed by an IPSP, probably evoked by muscarinic (M_2) receptor activation. The IPSP is then followed by a slower, M_1-dependent EPSP, and this can be followed by an even slower peptide-induced EPSP. (Used with permission from Katzung BG, Maters SB, Trevor AJ: *Basic & Clinical Pharmacology*, 11th ed. New York, NY: McGraw-Hill; 2009.)

The responses produced in postganglionic neurons by stimulation of their preganglionic innervation include both a **fast excitatory postsynaptic potential (EPSP)** that generates action potentials and a prolonged excitatory postsynaptic potential **(slow EPSP).** The slow response may modulate and regulate transmission through the sympathetic ganglia. The initial depolarization is produced by acetylcholine acting on the N_N receptor. The slow EPSP is produced by acetylcholine acting on a muscarinic receptor on the membrane of

the postganglionic neuron.

The release of acetylcholine from postganglionic fibers acts on muscarinic cholinergic receptors that are blocked by atropine. Muscarinic receptors are GPCR and are divided into subtypes M_1–M_5; M_2 and M_3 are the main subtypes found in autonomic target organs. M_2 receptors are found in the heart; binding of an agonist to these receptors opens K^+ channels and inhibits **adenylyl cyclase.** M_3 receptors are located on smooth muscle and glands; binding of an agonist to these receptors leads to the formation of **inositol 1,4,5-triphosphate (IP_3)** and **diacylglycerol (DAG)** and an increase in intracellular Ca^{2+}.

Compounds with muscarinic actions include congeners of acetylcholine and drugs that inhibit acetylcholinesterase. **Clinical Box 13–2** describes some of the signs and therapeutic strategies for the treatment of acute intoxication from **organophosphate cholinesterase inhibitors.** **Clinical Box 13–3** describes an example of **cholinergic poisoning** due to digestion of toxic mushrooms.

NORADRENERGIC NEUROTRANSMISSION

The processes involved in the synthesis, reuptake, and breakdown of norepinephrine were described in Chapter 7. Norepinephrine spreads farther and has a more prolonged action than acetylcholine. Norepinephrine, epinephrine, and dopamine are all found in plasma. The epinephrine and some of the dopamine come from the adrenal medulla, but most of the norepinephrine diffuses into the bloodstream from sympathetic nerve endings. Metabolites of norepinephrine and dopamine also enter the circulation.

The norepinephrine released from sympathetic postganglionic fibers binds to adrenoceptors. These are GPCR and are divided into several subtypes: α_1, α_2, β_1, β_2, and β_3. Table 13–1 shows the locations of these receptor subtypes on smooth muscles, cardiac muscle, and glands. Binding of an agonist to α_1-adrenoceptors activates the G_q-coupling protein that leads to the formation of IP_3 and DAG and an increase in intracellular Ca^{2+}. Binding of an agonist to α_2-adrenoceptors causes dissociation of the inhibitory G-protein G_i to inhibit adenylyl cyclase and decrease **cyclic adenosine monophosphate (cAMP).** Binding of an agonist to β-adrenoceptors activates the G_s-coupling protein to activate adenylyl cyclase and increase cAMP.

There are several diseases or syndromes that result from dysfunction of sympathetic innervation of specific body regions. **Clinical Box 13–4** describes

Horner syndrome that is due to interruption of sympathetic nerves to the face. Clinical Box 13–5 describes a vasospastic condition **(Raynaud phenomenon)** in which blood flow to the fingers and toes is transiently reduced, typically when a sensitive individual is exposed to stress or cold.

CLINICAL BOX 13–2

Organophosphates: Pesticides and Nerve Gases

The World Health Organization estimates that 1–3% of agricultural workers worldwide suffer from **acute pesticide poisoning;** it accounts for significant morbidity and mortality, especially in developing countries. Like **organophosphate** pesticides (eg, **parathion and malathion**), **nerve gases** (eg, **soman** and **sarin**) used in chemical warfare and terrorism inhibit acetylcholinesterase at peripheral and central cholinergic synapses, prolonging the actions of acetylcholine at these synapses. The organophosphate **cholinesterase inhibitors** are readily absorbed by the skin, lung, gut, and conjunctiva, making them very dangerous. They bind to the enzyme and undergo hydrolysis, resulting in a phosphorylated active site on the enzyme. The covalent phosphorous-enzyme bond is very stable and hydrolyzes at a very slow rate. The phosphorylated enzyme complex undergoes a process called **aging** in which one of the oxygen-phosphorous bonds breaks down, which strengthens the phosphorous-enzyme bond. This process occurs within 10 min of exposure to soman. The earliest signs of organophosphate toxicity are indicative of excessive activation of autonomic muscarinic receptors; these include miosis, salivation, sweating, bronchial constriction, vomiting, and diarrhea. CNS signs of toxicity include cognitive disturbances, convulsions, seizures, and even coma; these signs are often accompanied by nicotinic effects such as depolarizing neuromuscular blockade.

THERAPEUTIC HIGHLIGHTS

The muscarinic cholinergic receptor antagonist **atropine** can be given parenterally to control signs of excessive activation of muscarinic cholinergic receptors. Nucleophiles (eg, **pralidoxime)** can break the bond between the organophosphate and the acetylcholinesterase if given soon after exposure to the organophosphate and before aging has occurred. Thus, this drug is called a “cholinesterase regenerator.” If **pyridostigmine** is administered in advance of

exposure to a cholinesterase inhibitor, it binds to the enzyme and prevents binding by the toxic organophosphate agent. The protective effects of pyridostigmine dissipate within 3–6 h, but this provides enough time for clearance of the organophosphate from the body. Since the drug cannot cross the blood-brain barrier, protection is limited to peripheral cholinergic synapses. A mixture of pyridostigmine, **carbamate,** and atropine can be administered prophylactically to soldiers and civilians who are at risk for exposure to nerve gases. **Benzodiazepines** can be used to abort the seizures caused by exposure to organophosphates.

CLINICAL BOX 13–3

Mushroom Poisoning

Of more than 5000 species of mushrooms found in the United States, approximately 100 are poisonous and ingestion of about 12 of these can result in death. Estimates are an annual incidence of five cases per 100,000 individuals in the United States. Mushroom poisoning or **mycetism** is divided into rapid-onset (15–30 min after ingestion) and delayed-onset (6–12 h after ingestion) types. In rapid-onset cases caused by mushrooms of the *Inocybe* genus, the symptoms are due to excessive activation of muscarinic cholinergic synapses. The major signs of **muscarinic poisoning** include nausea, vomiting, diarrhea, urinary urgency, vasodilation, sweating, and salivation. Ingestion of mushrooms such as the *Amanita muscaria* exhibit signs of the **antimuscarinic syndrome** rather than muscarinic poisoning because they also contain alkaloids that block muscarinic cholinergic receptors. The classic symptoms of this syndrome are being “red as a beet” (flushed skin), “hot as a hare” (hyperthermia), “dry as a bone” (dry mucous membranes, no sweating), “blind as a bat” (blurred vision and cycloplegia), and “mad as a hatter” (confusion and delirium). The delayed-onset type of mushroom poisoning occurs after ingestion of *Amanita phalloides, Amanita virosa, Galerina autumnalis,* and *Galerina marginata*. These mushrooms cause abdominal cramping, nausea, vomiting, and profuse diarrhea; but the major toxic effects are due to hepatic injury (jaundice and bruising) and associated central effects (confusion, lethargy, and coma). These mushrooms contain **amatoxins** that inhibit RNA polymerase. There is a 60% mortality rate associated with ingestion of these mushrooms.

THERAPEUTIC HIGHLIGHTS

The rapid-onset type muscarinic poisoning can be treated effectively with atropine. Individuals who exhibit the antimuscarinic syndrome can be treated with **physostigmine,** a cholinesterase inhibitor with a 2–4 h duration of action that acts centrally and peripherally. If agitated, these individuals may require sedation with a benzodiazepine or an **antipsychotic agent.** The delayed-onset of toxicity due to ingestion of mushrooms containing amatoxins does not respond to cholinergic drugs. Treatment of amatoxin ingestion includes intravenous administration of fluids and electrolytes to maintain adequate hydration. Administering a combination of a high-dose of **penicillin G** and **silibinin** (a **flavonolignan** found in certain herbs with antioxidant and hepatoprotective properties) can improve survival. If necessary, vomiting can be induced by using activated charcoal to reduce the absorption of the toxin.

CLINICAL BOX 13–4

Horner Syndrome

Horner syndrome is a rare disorder resulting from interruption of preganglionic or postganglionic sympathetic innervation to the face. The problem can result from injury to the nerves, injury to the carotid artery, a stroke or lesion in the brainstem, or a tumor in the lung. In most cases the problem is unilateral, with symptoms occurring only on the side of the damage. The hallmark of Horner syndrome is the triad of **anhidrosis** (reduced sweating), **ptosis** (drooping eyelid), and **miosis** (constricted pupil). Symptoms also include **enophthalmos** (sunken eyeball) and vasodilation.

THERAPEUTIC HIGHLIGHTS

There is no specific pharmacologic treatment for Horner syndrome, but drugs affecting noradrenergic neurotransmission can be used to determine whether the source of the problem is interruption of the preganglionic or postganglionic innervation to the face. Since the iris of the eye responds to topical **sympathomimetic drugs** (ie, direct agonists on adrenoceptors or drugs that increase the release or prevent reuptake of norepinephrine from the nerve terminal), one can easily test the viability of the noradrenergic nerves to the eye.

If the postganglionic sympathetic fibers are damaged, their terminals would degenerate and there would be a loss of stored catecholamines. If the preganglionic fibers are damaged, the postganglionic noradrenergic nerve would remain intact (but be inactive) and would still have stored catecholamines in its terminal. If a drug that causes release of catecholamine stores (eg, **hydroxyamphetamine**) is administered and the constricted pupil does not dilate, one would conclude that the noradrenergic nerve is damaged. If the eye dilates in response to this drug, the catecholamine stores are still able to be released, so the damage must be preganglionic. Administration of **phenylephrine** (α-adrenoceptor agonist) would dilate the pupil regardless of the site of injury as the drug binds to the receptor on the radial muscle of the iris.

CLINICAL BOX 13–5

Raynaud Phenomenon

Approximately 5% of men and 8% of women experience an episodic reduction in blood flow primarily to the fingers, often during exposure to cold or during a stressful situation. Vasospasms in the toes, tip of nose, ears, and penis can also occur. Smoking is associated with an increase in the incidence and severity of the symptoms of Raynaud phenomenon. The symptoms begin to occur between the age of 15 and 25; it is most common in cold climates. The symptoms often include a triphasic change in color of the skin of the digits. Initially, the skin becomes pale or white **(pallor),** cold, and numb. This can be followed by a **cyanotic** period in which the skin turns blue or even purple, during which time the reduced blood flow can cause intense pain. Once the blood flow recovers, the digits often turn deep red **(rubor)** and there can be swelling and tingling. Primary Raynaud phenomenon or **Raynaud disease** refers to the idiopathic appearance of the symptoms in individuals who do not have another underlying disease to account for the symptoms. In such cases, the vasospastic attacks may merely be an exaggeration of a normal response to cold temperature or stress. Secondary Raynaud phenomenon or **Raynaud syndrome** refers to the presence of these symptoms due to another disorder such as **scleroderma, lupus, rheumatoid arthritis, Sjögren syndrome, carpel tunnel syndrome,** and **anorexia.** Although initially thought to reflect an increase in sympathetic activity to the

vasculature of the digits, this is no longer regarded as the mechanism underlying the episodic vasospasms.

THERAPEUTIC HIGHLIGHTS

The first treatment strategy for Raynaud phenomenon is to avoid exposure to the cold, reduce stress, quit smoking, and avoid the use of medications that are vasoconstrictors (eg, β-adrenoceptor antagonists, cold medications, caffeine, and opioids). If the symptoms are severe, drugs may be needed to prevent tissue damage. These include **calcium channel blockers** (eg, **nifedipine**) and **α-adrenoceptor antagonists** (eg, **prazosin**). In individuals who do not respond to pharmacologic treatments, surgical sympathectomy has been done.

NONADRENERGIC & NONCHOLINERGIC TRANSMITTERS

In addition to the "classical neurotransmitters," some autonomic fibers also release neuropeptides. The small granulated vesicles in postganglionic sympathetic neurons contain **adenosine triphosphate (ATP)** and norepinephrine, and the large granulated vesicles contain **neuropeptide Y (NPY).** Low-frequency activation of these nerves promotes release of ATP, and high-frequency stimulation causes release of NPY. Some visceral organs contain purinergic receptors, and ATP is a mediator along with norepinephrine in some autonomic targets.

Many sympathetic fibers innervating the vasculature of viscera, skin, and skeletal muscles release NPY and **galanin** in addition to norepinephrine. **Vasoactive intestinal polypeptide (VIP), calcitonin gene-related peptide (CGRP),** or **substance P** are co-released with acetylcholine by sympathetic nerves to sweat glands **(sudomotor fibers).** VIP is co-localized with acetylcholine in many cranial parasympathetic postganglionic neurons supplying glands. Vagal parasympathetic postganglionic neurons in the gastrointestinal tract contain VIP and the enzymatic machinery to synthesize **nitric oxide (NO).**

RESPONSES OF EFFECTOR ORGANS TO AUTONOMIC NERVE IMPULSES

HOMEOSTASIS

Homeostasis refers to the ability to maintain a stable internal environment despite challenges imposed by shifts in things such as air temperature, atmospheric and blood oxygen and carbon dioxide levels, physical activity, exposure to toxins, disease, drug therapies, fever, and diet. With the wide distribution of autonomic nerves throughout the body, it is not surprising that the ANS works in concert with the endocrine system (see Chapter 16) to maintain homeostatic variables within a physiological range. The ANS plays a fundamental role in the regulation and coordination of many physiologic functions that are critical for homeostasis. These include airflow through the bronchial tree, blood flow, blood gas composition, blood glucose, blood pressure, body temperature, digestion, electrolyte balance, glandular secretions, heart rate, and urination. The failure to maintain homeostasis **(homeostatic imbalance)** can lead to death or a disease. Diseases such as type I diabetes, dehydration, hyperthermia, or hypothermia, heart failure, and hypertension are examples of the consequence of homeostatic imbalance and loss of negative feedback mechanisms to return physiological parameters to normal levels. **Clinical Box 13–6** describes examples of the deleterious consequences of homeostatic imbalance.

CLINICAL BOX 13–6

Homeostatic Imbalance

A failure to maintain homeostasis can cause major dysfunction of organs under the control of the ANS. If we lose our ability to regulate body temperature, one can experience hyperthermia or hypothermia. Failure to maintain energy balance can lead to the development of obesity or diabetes. If your kidney is no longer able to regulate salt and water balance, the body will retain water, salts and metabolic wastes that has many dire consequences (eg, cardiac arrhythmias, hypertension, anemia, and neurological complications). For many physiological systems, the maintenance of homeostasis involves having a sensor (eg, thermoreceptor) to detect the deviation from normal value and a negative feedback mechanism to return the physiological measure (eg, body temperature) back to a normal level (eg, 98.6°F/37°C, normothermia). The body has a complex mechanism to maintain body temperature within a thermal neutral zone of 36–37°C; this is the temperature

range between the low point at which shivering begins and the high point at which sweating begins. When body temperature is elevated by an exposure to high environmental temperatures or physical exertion, thermoreceptors in the skin (see Chapter 8) and the hypothalamus are activated. These receptors send signals to a central control area in the **medial preoptic and anterior hypothalamic nuclei.** Activation of these neurons engages the effector autonomic pathways that promote sweating (activation of sudomotor sympathetic nerves) and cutaneous vasodilation (activation of skin sympathetic nerves). The combination of sweating and cutaneous vasodilation allow for heat dissipation and a return to normothermia. This, in turn, would silence the thermoreceptors and reduce activity emanating from the hypothalamus. If one loses the ability to dissipate heat, hyperthermia or **heat stroke** can ensue. Heat stroke is considered a medical emergency and leads to many indices of homeostatic imbalance: tachycardia, rapid shallow breathing, dizziness, anhidrosis, dry and hot skin, muscle cramping, confusion, and seizures. See Chapter 17 for more details on the mechanisms involved in temperature regulation. Other conditions that can cause anhidrosis and thus the loss of a key homeostatic mechanism to control body temperature include diabetic neuropathy, Parkinson disease, MSA, Sjögren syndrome, Horner syndrome, and alcoholism.

PARASYMPATHETIC AND SYMPATHETIC NERVOUS SYSTEM: PHYSIOLOGICAL ANTAGONISTS, SYNERGISTIC FUNCTIONS, OR INDEPENDENT ACTIONS

The effects of stimulation of the noradrenergic and cholinergic postganglionic nerve fibers are listed in Table 13–1, and they point to another difference between the ANS and the somatomotor nervous system. The release of acetylcholine by α-motor neurons only leads to contraction of skeletal muscles. In contrast, release of acetylcholine onto smooth muscle of some organs leads to contraction (eg, walls of the gastrointestinal tract) while release onto other organs promotes relaxation (eg, sphincters in the gastrointestinal tract). The only way to relax a skeletal muscle is to inhibit the discharges of the α-motor neurons; but for some targets innervated by the ANS, contraction can be shifted to relaxation by switching from activation of the parasympathetic nervous system

to activation of the sympathetic nervous system. This is the case for the many organs that receive dual innervation with antagonistic effects, including the heart, airways, digestive tract, and urinary bladder. For example, stimulation of sympathetic nerves increases heart rate, and stimulation of parasympathetic nerves decreases heart rate.

In other cases, the effects of sympathetic and parasympathetic activation can be considered complementary. An example is the innervation of salivary glands. Parasympathetic activation causes release of watery saliva, while sympathetic activation causes the production of thick, viscous saliva.

The two divisions of the ANS can also act in a synergistic or cooperative manner in the control of some functions. One example is the control of pupil diameter in the eye. Both sympathetic and parasympathetic innervations are excitatory, but the former contracts the radial (or dilator) muscle to cause mydriasis (widening of the pupil) and the latter contracts the sphincter (or constrictor) muscle to cause miosis (narrowing of the pupil). Another example is the synergistic actions of these nerves on sexual function. Activation of parasympathetic nerves to the penis increases penile blood flow and leads to erection while activation of sympathetic nerves to the male reproductive tract causes ejaculation.

There are also several organs that are innervated by only one division of the ANS. In addition to the adrenal gland, most blood vessels, the pilomotor muscles in the skin (hair follicles), and sweat glands are innervated exclusively by sympathetic nerves (sudomotor fibers). The lacrimal muscle (tear gland), ciliary muscle (for accommodation for near vision; see Chapter 10), and the nasopharyngeal gland are innervated exclusively by parasympathetic nerves.

The functions promoted by activity in the parasympathetic nervous system are those concerned with the vegetative aspects of day-to-day living. Parasympathetic action favors digestion and absorption of food by increasing the activity of the intestinal musculature, increasing gastric secretion, and relaxing the pyloric sphincter. For this reason, it is sometimes called the “rest and digest” division of the ANS.

The sympathetic division discharges as a unit in emergency situations and can be called the catabolic nervous system or the “flight or fight” division of the ANS. The effect of this discharge prepares the individual to cope with an emergency. Sympathetic activity dilates the pupils (letting more light into the eyes), accelerates the heartbeat and raises the blood pressure (providing better perfusion of the vital organs and muscles), and constricts the blood vessels of the skin (which limits bleeding from wounds). Noradrenergic discharge also leads to

elevated plasma glucose and free fatty acid levels (supplying more energy). On the basis of effects like these, Walter Cannon called the emergency-induced discharge of the sympathetic nervous system the “preparation for flight or fight.”

The emphasis on mass discharge in stressful situations should not obscure the fact that the sympathetic fibers also subserve other functions and, in fact, there is activity in sympathetic nerves even under resting conditions. For example, tonic sympathetic discharge to the arterioles maintains arterial pressure, and variations in this tonic discharge are the mechanism by which carotid sinus feedback regulation of blood pressure occurs (see Chapter 32). In addition, sympathetic discharge is decreased in fasting animals and increased when fasted animals are again fed. These changes may explain the decrease in blood pressure and metabolic rate produced by fasting and the opposite changes produced by feeding.

DESCENDING INPUTS TO AUTONOMIC PREGANGLIONIC NEURONS

As is the case for α-motor neurons, the activity of autonomic nerves is dependent on both reflexes (eg, baroreceptor and chemoreceptor reflexes) and a balance between descending excitatory and inhibitory inputs from several brain regions. Chapter 32 describes the roles of baroreceptor and chemoreceptor reflexes and medullary neurons in maintaining homeostasis within the cardiovascular system. **Figure 13–5** shows the source of some forebrain and brainstem inputs to sympathetic preganglionic neurons. There are parallel pathways from the hypothalamic paraventricular nucleus, pontine catecholaminergic A5 cell group, rostral ventrolateral medulla, and medullary raphe nuclei. This is analogous to projections from the brainstem and cortex converging on somatomotor neurons in the spinal cord. The rostral ventrolateral medulla is the major source of excitatory input to sympathetic neurons. In addition to these direct pathways to preganglionic neurons, there are many brain regions that feed into these pathways, including the amygdala, mesencephalic periaqueductal gray, caudal ventrolateral medulla, nucleus of the tractus solitarius, and medullary lateral tegmental field. This is analogous to the control of somatomotor function by areas such as the basal ganglia and cerebellum.

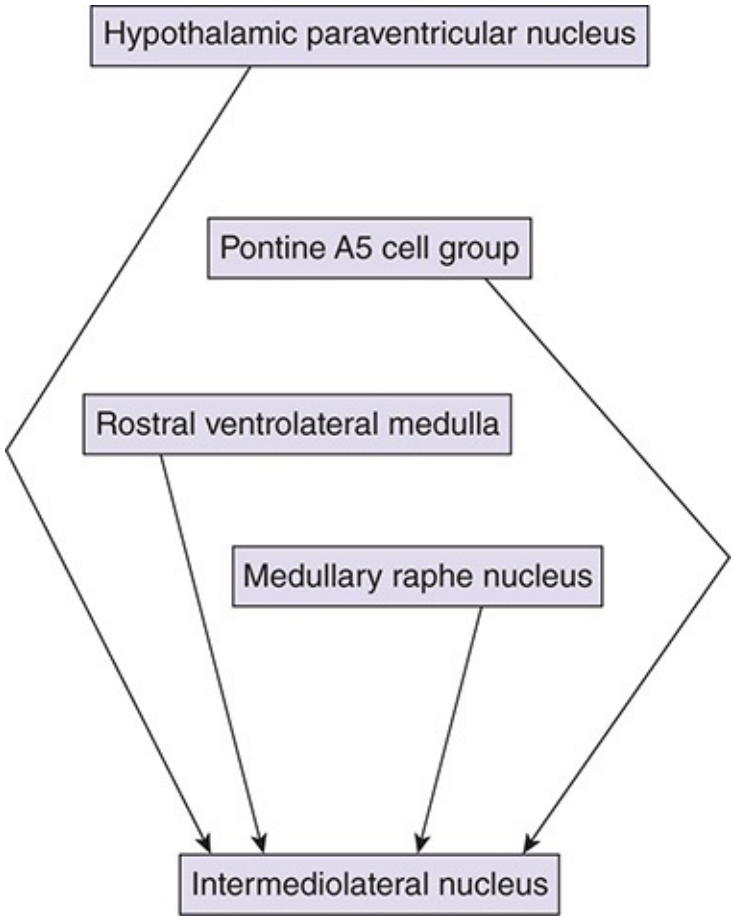


FIGURE 13–5 Pathways that control autonomic responses. Direct projections (solid lines) to autonomic preganglionic neurons include the hypothalamic paraventricular nucleus, pontine A5 cell group, rostral ventrolateral medulla, and medullary raphe.

AUTONOMIC DYSFUNCTION

Drugs, neurodegenerative diseases, trauma, inflammatory processes, and neoplasia are a few examples of factors that can lead to dysfunction of the ANS (Clinical Boxes 13–1 through 13–4). The types of dysfunction can range from complete autonomic failure to autonomic hyperactivity. Among disorders associated with autonomic failure are orthostatic hypotension, neurogenic syncope (vasovagal response), erectile dysfunction, neurogenic bladder, gastrointestinal dysmotility, sudomotor failure, and Horner syndrome. Autonomic hyperactivity can be the basis for neurogenic hypertension, cardiac arrhythmias, neurogenic pulmonary edema, myocardial injury, hyperhidrosis, hyperthermia, and hypothermia.

ENTERIC NERVOUS SYSTEM

The enteric nervous system or third division of the ANS is located within the wall of the digestive tract from the esophagus to the anus. It is composed of two well-organized neural plexuses. The **myenteric plexus** is located between longitudinal and circular layers of muscle and is involved in control of digestive tract motility. The **submucosal plexus** is located between the circular muscle and the luminal mucosa; it senses the environment of the lumen and regulates gastrointestinal blood flow and epithelial cell function.

The enteric nervous system contains all the elements of a nervous system (sensory neurons, interneurons, and motor neurons) leading to it being referred to as a "mini-brain." Sensory neurons innervate receptors in the mucosa that respond to mechanical, thermal, osmotic, and chemical stimuli. Motor neurons control motility, secretion, and absorption by acting on smooth muscle and secretory cells. Interneurons integrate information from sensory neurons and feedback to the enteric motor neurons.

Parasympathetic and sympathetic nerves connect the CNS to the enteric nervous system or directly to the digestive tract. Although the enteric nervous system can function autonomously, normal digestive function often requires communication between the CNS and the enteric nervous system (see Chapter 25).

CHAPTER SUMMARY

- Preganglionic sympathetic neurons are located in the spinal thoracolumbar IML and project to postganglionic neurons in the paravertebral or prevertebral ganglia or the adrenal medulla. Preganglionic parasympathetic neurons are located in motor nuclei of cranial nerves III, VII, IX, and X and the sacral IML and project to ganglia located in or near the effector target. The targets of the ANS include smooth muscle, cardiac muscle and pacemaker cells, exocrine and endocrine glands, adipose tissue, liver cells, and lymphatic tissue.
- Acetylcholine is released at nerve terminals of all preganglionic neurons, postganglionic parasympathetic neurons, and a few postganglionic sympathetic neurons (sweat glands and sympathetic vasodilator fibers). The remaining sympathetic postganglionic neurons release norepinephrine.
- Ganglionic transmission is mediated by activation of nicotinic receptors.

Postganglionic cholinergic transmission is mediated by activation of muscarinic receptors. Postganglionic adrenergic transmission is mediated by activation of α_1-, β-, or β_2-adrenoceptors, depending on the target organ.

- Many commonly used drugs exert their therapeutic actions by serving as agonists (eg, bethanecol, phenylephrine, albuterol) or antagonists (eg, atropine, phenoxybenzamine, atenolol) at autonomic synapses, by blocking neurotransmitter synthesis (eg, metyrosine), or by blocking neurotransmitter release (eg, tricyclic antidepressants).
- The ANS works in concert with the endocrine system to maintain homeostasis or a stable internal environment despite challenges imposed by shifts in things such as air temperature, oxygen and carbon dioxide levels, physical activity, exposure to toxins, disease, drug therapies, fever, and diet.
- Sympathetic activity prepares the individual to cope with an emergency by accelerating the heartbeat, raising blood pressure (perfusion of the vital organs), and constricting the blood vessels of the skin (limits bleeding from wounds). Parasympathetic activity is concerned with the vegetative aspects of day-to-day living and favors digestion and absorption of food by increasing the activity of the intestinal musculature, increasing gastric secretion, and relaxing the pyloric sphincter.
- At many organs that receive dual innervation (eg, heart, airways, digestive tract, and urinary bladder), the two divisions of the ANS act as physiological antagonists. In other cases, the two divisions of the ANS can act in a synergistic manner in the control of some functions (eg, control of pupil diameter). Some organs are innervated by only the sympathetic (eg, blood vessels) or only parasympathetic (eg, ciliary muscle) nervous system.
- Direct projections to sympathetic preganglionic neurons in the IML originate in the hypothalamic paraventricular nucleus, pontine catecholaminergic A5 cell group, rostral ventrolateral medulla, and medullary raphe nuclei.
- The enteric nervous system is located within the wall of the digestive tract and is composed of the myenteric plexus (control of digestive tract motility) and the submucosal plexus (regulates gastrointestinal blood flow and epithelial cell function).

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Hypertension developed in a 26-year-old man after he began taking amphetamine to boost his energy and to suppress his appetite. Which of the following drugs would be expected to mimic the effects of increased sympathetic discharge on blood vessels?

   A. Phenylephrine

   B. Trimethaphan

   C. Atropine

   D. Reserpine

   E. Albuterol

2. A 35-year-old woman in whom multiple system atrophy was diagnosed had symptoms indicative of failure of sympathetic nerve activity. List expected findings resulting from failure of sympathetic nerve activity to the ventricle of the heart, bronchial smooth muscle, sweat glands, and blood vessels.

   A. Bradycardia, bronchial dilation, reduced sweating, and vasodilation.

   B. Decreased ventricular contractility, bronchial constriction, profuse sweating, and vasodilation.

   C. Tachycardia, bronchial constriction, reduced sweating, and vasoconstriction.

   D. Decreased ventricular contractility, bronchial constriction, reduced sweating, and vasodilation.

   E. Bradycardia, bronchial dilation, profuse sweating, and vasodilation.

3. A 45-year-old man had a meal containing wild mushrooms that he picked in a field earlier in the day. Within a few hours after eating the mushrooms, he developed signs of muscarinic poisoning. List an expected finding resulting from activation of muscarinic receptors on the lacrimal gland, SA node of the heart, sphincter muscle in the urinary tract, and sweat glands.

   A. Increased salivation, bradycardia, relaxation of the urinary tract sphincter, and no change in sweating.

   B. Decreased tear production, tachycardia, contraction of the urinary tract sphincter, and decreased sweating.

   C. Increased tear production, bradycardia, relaxation of the urinary tract sphincter, and increased sweating.

   D. Increased salivation, tachycardia, contraction of the urinary tract sphincter, and no change in sweating.

   E. Increased tear production, bradycardia, contraction of the urinary tract

sphincter, and increased sweating.

4. An MD/PhD candidate was studying control of pupillary diameter by stimulation of the ANS. What is the location of the cell bodies of preganglionic parasympathetic, postganglionic parasympathetic, preganglionic sympathetic, and postganglionic sympathetic nerves controlling pupillary diameter, respectively?

   A. Pupillary nucleus, ciliary ganglion, cervical IML, and cervical paravertebral ganglion.

   B. Second cranial nerve nucleus, pupillary ganglion, cervical IML, and cervical prevertebral ganglion.

   C. Pupillary nucleus, otic ganglion, thoracic IML, and thoracic prevertebral ganglion.

   D. Edinger-Westphal nucleus, pupillary ganglion, thoracic IML, and cervical prevertebral ganglion.

   E. Edinger-Westphal nucleus, ciliary ganglion, thoracic IML, and cervical paravertebral ganglion.

5. An MD/PhD candidate was studying autonomic control of pupillary diameter and accommodation for near vision. How does stimulation of the parasympathetic and sympathetic nerve activity affect these responses?

   A. Parasympathetic nerve activity causes mydriasis by contraction of the sphincter muscle and makes the lens more concave by contraction of the ciliary muscle; sympathetic nerve activity causes miosis by relaxation of the sphincter muscle and makes the lens more convex by contraction of the ciliary muscle.

   B. Parasympathetic nerve activity causes miosis by contraction of the constrictor muscle and causes contraction of the ciliary muscle to make the lens more convex; sympathetic nerve activity causes mydriasis by contraction of the radial muscle and causes relaxation of the ciliary muscle to make the lens more concave.

   C. Parasympathetic nerve activity causes mydriasis by contraction of the sphincter muscle and makes the lens more convex by contraction of the ciliary muscle; sympathetic nerve activity causes miosis by contraction of the dilator muscle and makes the lens more concave by contraction of the ciliary muscle.

   D. Parasympathetic nerve activity causes miosis by contraction of the sphincter muscle and makes the lens more convex by contraction of the ciliary muscle; sympathetic nerve activity causes mydriasis by contraction

of the radial muscle and does not alter the shape of the lens.

E. Parasympathetic nerve activity causes miosis by contraction of the sphincter muscle and makes the lens more convex by contraction of the ciliary muscle; sympathetic nerve activity causes mydriasis by relaxation of the sphincter muscle and does not alter the shape of the lens.

6. A 57-year-old man had severe hypertension that was found to result from a tumor compressing on the surface of the medulla. Which one of the following statements about pathways involved in the control of sympathetic nerve activity is correct?

A. Preganglionic sympathetic nerves receive inhibitory input from the rostral ventrolateral medulla.

B. The major source of excitatory input to preganglionic sympathetic nerves is the paraventricular nucleus of the hypothalamus.

C. The activity of sympathetic preganglionic neurons can be affected by the activity of neurons in the amygdala.

D. Unlike the activity in δ-motor neurons, sympathetic preganglionic neurons are not under any significant reflex control.

E. Under resting conditions, the sympathetic nervous system is not active; it is active only during stress giving rise to the term “flight or fight” response.

7. Diabetic autonomic neuropathy was diagnosed a few years ago in a 53-year-old woman with diabetes. She recently noted abdominal distension and a feeling of being full after eating only a small portion of food, suggesting the neuropathy had extended to her enteric nervous system to cause gastroparesis. What are the components of the enteric nervous system?

A. The enteric nervous system is a specialized subdivision of the parasympathetic nervous system for control of gastrointestinal function and includes specialized preganglionic and postganglionic cholinergic neurons.

B. The enteric nervous system contains the myenteric plexus that regulates gastrointestinal motility and the submucosal plexus that regulates gastrointestinal blood flow and epithelial cell function and includes motor neurons, sensory neurons, and interneurons.

C. The enteric nervous system contains the submucosal plexus that contains motor neurons that control gastric secretions and motility and a myenteric plexus that contains sensory neurons that signal information about the environment, and mucosal interneurons that relay sensory information to the central nervous system.

D. The enteric nervous system contains motor neurons within the circular muscle, sensory neurons within the longitudinal muscle, and interneurons within the mucosa that relay sensory information to the central nervous system.

E. The enteric nervous system the submucosal plexus comprised exclusively of sensory neurons transmit information about the contents of the gastrointestinal tract via interneurons to the myenteric plexus that is comprised exclusively of motor neurons.

8. A respiratory therapist was giving a lecture on how the body responds to changes in arterial blood gases. As part of her lecture, she explained homeostasis as follows.

A. Homeostasis prevents blood gases from deviating from normal values for even brief periods of time.

B. Homeostasis maintains blood gases in a normal range by activation of chemoreceptors that sense the deviation from normal and then engages a positive feedback system to return blood gases to a normal level.

C. Homeostasis maintains blood gases in a normal range by activation of sympathetic and parasympathetic chemoreceptors that then stimulate increased respiratory activity.

D. Homeostasis maintains blood gases in a normal range by activation of chemoreceptors that sense the deviation from normal and then engages a negative feedback system to return blood gases to a normal level.

CHAPTER 14

Electrical Activity of the Brain, Sleep–Wake States, & Circadian Rhythms

OBJECTIVES

After studying this chapter, you should be able to:

- Explain the function of the thalamocortical pathway and ascending arousal system in the control of arousal and consciousness.
- Explain the interplay between brainstem neurons that contain norepinephrine, serotonin, and acetylcholine and diencephalic histaminergic and GABAergic neurons in mediating transitions between sleep and wakefulness.
- Explain the physiological basis and the main clinical uses of the electroencephalogram (EEG).
- Describe possible causes of seizure activity and explain the differences between generalized and partial seizures.
- Identify the primary types of cortical rhythms recorded in an EEG that reflect different states of wakefulness and sleep.
- Summarize the behavioral and EEG characteristics of rapid eye movement (REM) sleep and the four stages of non-REM sleep.
- Describe the pattern of normal nighttime sleep in adults and the variations in

this pattern from birth to old age.

- Describe the symptoms of narcolepsy, sleep apnea, and other sleep disorders.
- Describe the roles of the suprachiasmatic nuclei (SCN) and melatonin in regulation of the circadian rhythm.

INTRODUCTION

Most of the sensory systems introduced in Chapters 8–12 relay impulses from receptors via multiunit pathways to specific sites in the cerebral cortex. The impulses are responsible for perception and localization of individual sensations; however, they must be processed in the awake brain to be perceived. There is a spectrum of behavioral states ranging from deep sleep through alertness with focused attention. Each distinct state is correlated with a discrete pattern of brain electrical activity. Feedback oscillations within the cerebral cortex and between the thalamus and the cortex produce this activity and are determinants of the behavioral state. Arousal can be produced by sensory stimulation and by impulses ascending from the brainstem to the thalamus and then to the cortex. Some of these activities have rhythmic fluctuations that are approximately 24 h in length **(circadian rhythm).**

THALAMOCORTICAL PATHWAYS & ASCENDING AROUSAL SYSTEM

THALAMIC NUCLEI

The **thalamus** within the diencephalon is comprised of groups of nuclei that participate in sensory, motor, and limbic functions. The thalamus is the “gateway to the cerebral cortex” because it processes virtually all information that reaches the cortex. The thalamus also receives input from the cortex.

The thalamus has two major groups of nuclei: those that project diffusely to wide areas of the neocortex **(midline and intralaminar nuclei)** and those that project to discrete regions of the neocortex and limbic system **(specific sensory relay nuclei).** The latter group includes the medial and lateral geniculate bodies that relay auditory and visual impulses to the auditory and visual cortices, respectively, and the ventral posterior lateral (VPL) and ventral posteromedial

nuclei that relay somatosensory information to the postcentral gyrus. The ventral anterior and ventral lateral nuclei receive input from the basal ganglia and the cerebellum and project to the motor cortex. The anterior nuclei receive input from the mammillary bodies and project to the limbic cortex to influence memory and emotion. Most thalamic neurons are excitatory and release glutamate. The thalamus also contains inhibitory neurons in the **thalamic reticular nucleus.** These neurons release **GABA,** and unlike many thalamic neurons, their axons do not project to the cortex. Rather, they are thalamic interneurons and modulate the responses of other thalamic neurons to input coming from the cortex.

CORTICAL ORGANIZATION

The neocortex is arranged in six layers **(Figure 14–1)**. Afferents from the specific nuclei of the thalamus terminate primarily in layer IV; the nonspecific nuclei project to layers I–IV. **Pyramidal neurons,** the most common cell type in the cortex, have extensive vertical dendritic trees that reach toward the cortical surface **(Figure 14–2)**. Their cell bodies are found in all cortical layers except layer I. The axons of pyramidal cells have recurrent collaterals that turn back and synapse on the superficial portions of the dendritic trees. Pyramidal neurons are excitatory neurons that release **glutamate** at their terminals, and they are the only projection neurons of the cortex.

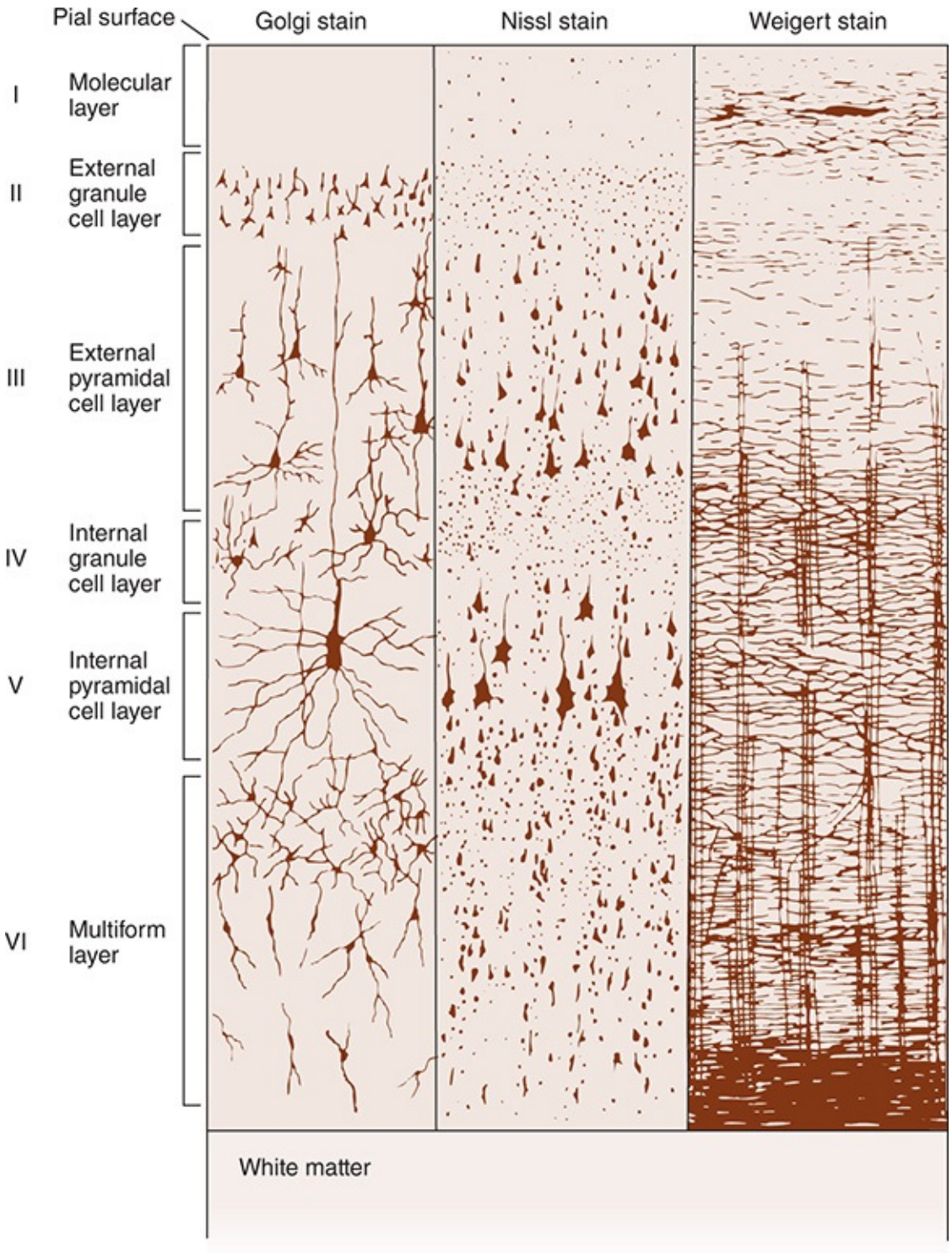


FIGURE 14–1 Structure of the cerebral cortex. The cortical layers are indicated by the numbers. Golgi stain shows neuronal cell bodies and dendrites, Nissl stain shows cell bodies, and Weigert myelin sheath stain shows myelinated nerve fibers. (Modified with permission from Ranson SW, Clark SL: *The*

Anatomy of the Nervous System, 10th ed. St. Louis, MO: Saunders; 1959.)

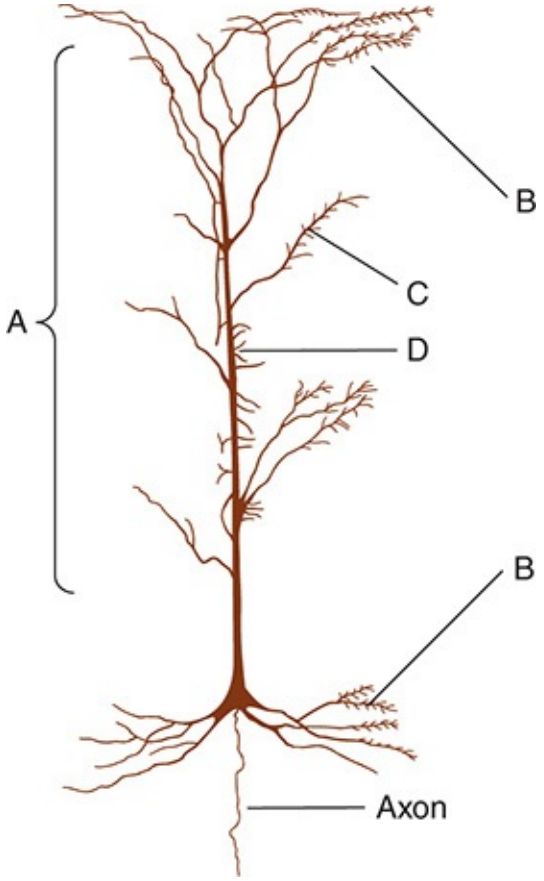


FIGURE 14–2 Neocortical pyramidal cell, showing the distribution of neurons that terminate on it. A denotes nonspecific afferents from the brainstem and the thalamus; **B** denotes recurrent collaterals of pyramidal cell axons; **C** denotes commissural fibers from mirror image sites in the contralateral hemisphere; **D** denotes specific afferents from thalamic sensory relay nuclei. (Based on Scheibel ME, Scheibel AB: Structural organization of nonspecific thalamic nuclei and their projection toward cortex. Brain Res 1967 Sep; 6(1):60–94.)

The other cortical cell types are local circuit interneurons and are classified based on their shape, pattern of projection, and neurotransmitter. Inhibitory interneurons **(basket cells** and **chandelier cells**) release GABA as their neurotransmitter. Basket cells have long axonal endings that surround the soma of pyramidal neurons; they account for most inhibitory synapses on the pyramidal soma and dendrites. Chandelier cells are a powerful source of inhibition of pyramidal neurons because their axonal endings terminate

exclusively on the initial segment of the pyramidal cell axon. Their terminal boutons form short vertical rows that resemble candlesticks, thus accounting for their name. **Spiny stellate cells** are excitatory neurons that release glutamate; these multipolar interneurons are located primarily in layer IV and are a major recipient of sensory information arising from the thalamus.

In addition to being organized into layers, the cerebral cortex is also organized into columns. Neurons within a column have similar response properties, suggesting they comprise a local processing network (eg, orientation and ocular dominance columns in the visual cortex).

Evoked Cortical Potentials

The electrical events that occur in the cortex after stimulation of a sensory receptor can be monitored with a recording electrode. If the electrode is over the primary receiving area for a particular sense, a surface-positive wave appears with a latency of 5–12 ms. This is followed by a small negative wave, and then a larger, more prolonged positive deflection frequently occurs with a latency of 20–80 ms. The first positive-negative wave sequence is the **primary evoked potential;** the second is the **diffuse secondary response.**

The primary evoked potential is highly specific in its location and occurs only where the pathways from a particular sensory system project The positive-negative wave sequence recorded from the surface of the cortex occurs because the superficial cortical layers are positive relative to the initial negativity, then negative relative to the deep hyperpolarization. The surface-positive diffuse secondary response, unlike the primary response, is not highly localized. It appears at the same time over most of the cortex and is due to activity in projections from the midline and intralaminar thalamic nuclei.

ASCENDING AROUSAL SYSTEM

The **ascending arousal system** is a complex polysynaptic pathway comprised of monoaminergic, cholinergic, and histaminergic neurons that project to the intralaminar and reticular nuclei of the thalamus which, in turn, project diffusely to wide regions of the cortex including the frontal, parietal, temporal, and occipital cortices **(Figure 14–3)**. Collaterals funnel into it not only from the long ascending sensory tracts but also from the trigeminal, auditory, visual, and olfactory systems. The complexity of the ascending arousal system and the degree of convergence in it abolish modality specificity, and most neurons are

activated with equal facility by different sensory stimuli. Components of the arousal system include norepinephrine-containing neurons in the pontine locus coeruleus, serotoninergic neurons in the brainstem raphé nuclei, cholinergic neurons in the pontine and midbrain pedunculopontine and laterodorsal tegmental nuclei, and histaminergic neurons in the hypothalamic tuberomammilary nucleus.

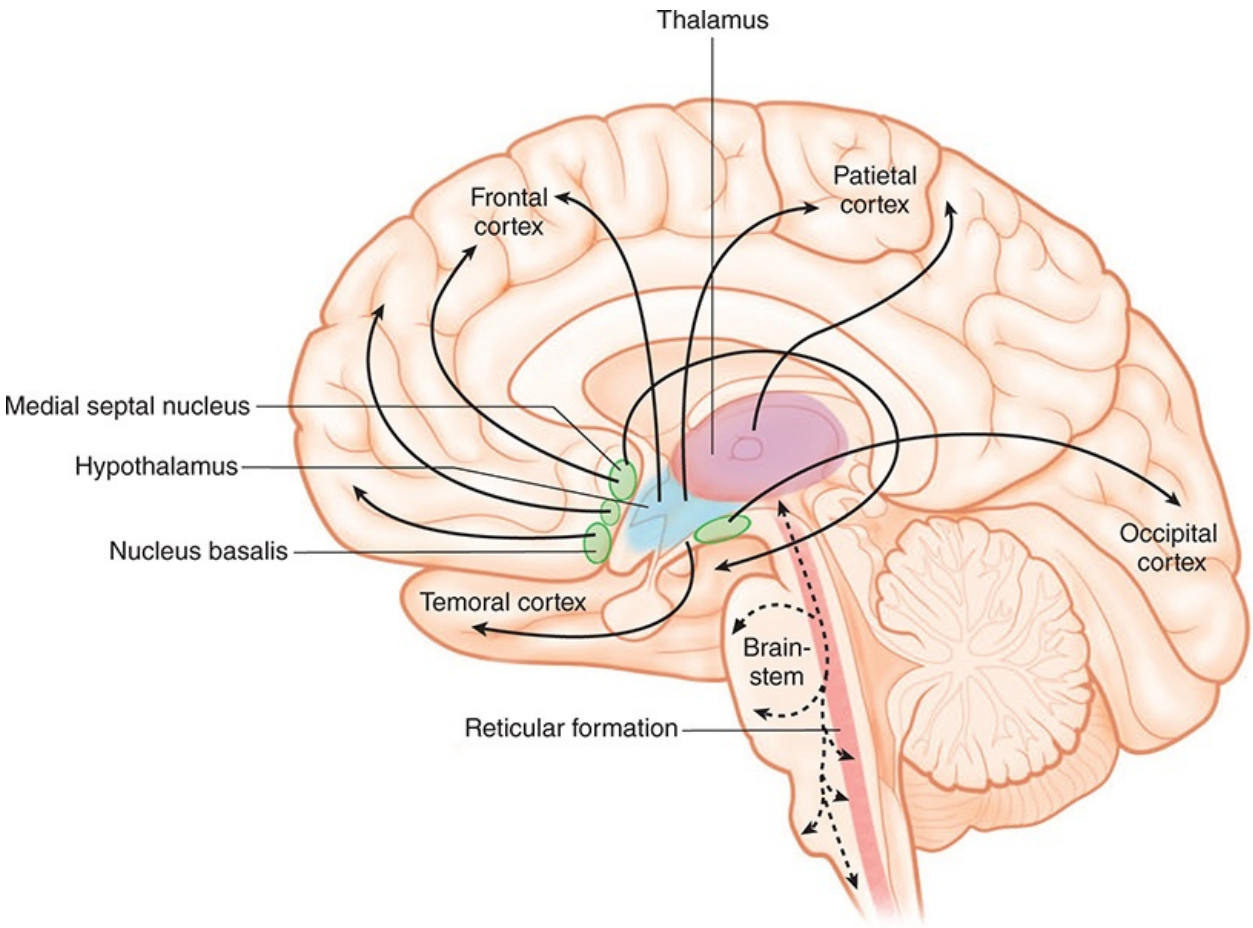


FIGURE 14–3 Cross section through the midline of the human brain showing the ascending arousal system in the brainstem with projections to the intralaminar nuclei of the thalamus and the output from the intralaminar nuclei to many parts of the cerebral cortex. Activation of these areas can be shown by positive emission tomography scans when subjects shift from a relaxed awake state to an attention-demanding task.

NEUROCHEMICAL MECHANISMS PROMOTING SLEEP & AROUSAL

Transitions between sleep and wakefulness manifest a circadian rhythm consisting of an average of 6–8 h of sleep and 16–18 h of wakefulness. Nuclei in both the brainstem and hypothalamus are critical for the transitions between these states of consciousness. As described above, the brainstem ascending arousal system is comprised of several groups of neurons that release norepinephrine, serotonin, acetylcholine, or histamine. The locations and wide projections of these neuronal populations are shown in Figure 7–2. The forebrain is also involved in the control of the sleep–wake cycles via hypothalamic **preoptic neurons** that release **GABA** and **tuberomamilary neurons** that release **histamine.** Also, hypothalamic neurons release **orexin** to play a role in switching between sleep and wakefulness.

One theory regarding the basis for transitions from sleep to wakefulness involves alternating reciprocal activity of different groups of neurons in the ascending arousal system. In this model **(Figure 14–4)**, wakefulness and **rapid eye movement (REM) sleep** are at opposite extremes. When the activity of norepinephrine- and serotonin-containing neurons (locus coeruleus and raphé nuclei) is dominant, activity in acetylcholine-containing pontine neurons is reduced. This pattern of activity contributes to the appearance of the awake state. The reverse of this pattern leads to REM sleep. When there is a more even balance between the activity of the aminergic and cholinergic neurons, **non-REM sleep** occurs. The orexin released from hypothalamic neurons may regulate the changes in activity in these brainstem neurons. An increased release of GABA and reduced release of histamine increase the likelihood of non-REM sleep via deactivation of the thalamus and cortex. Wakefulness occurs when GABA release is reduced and histamine release is increased.

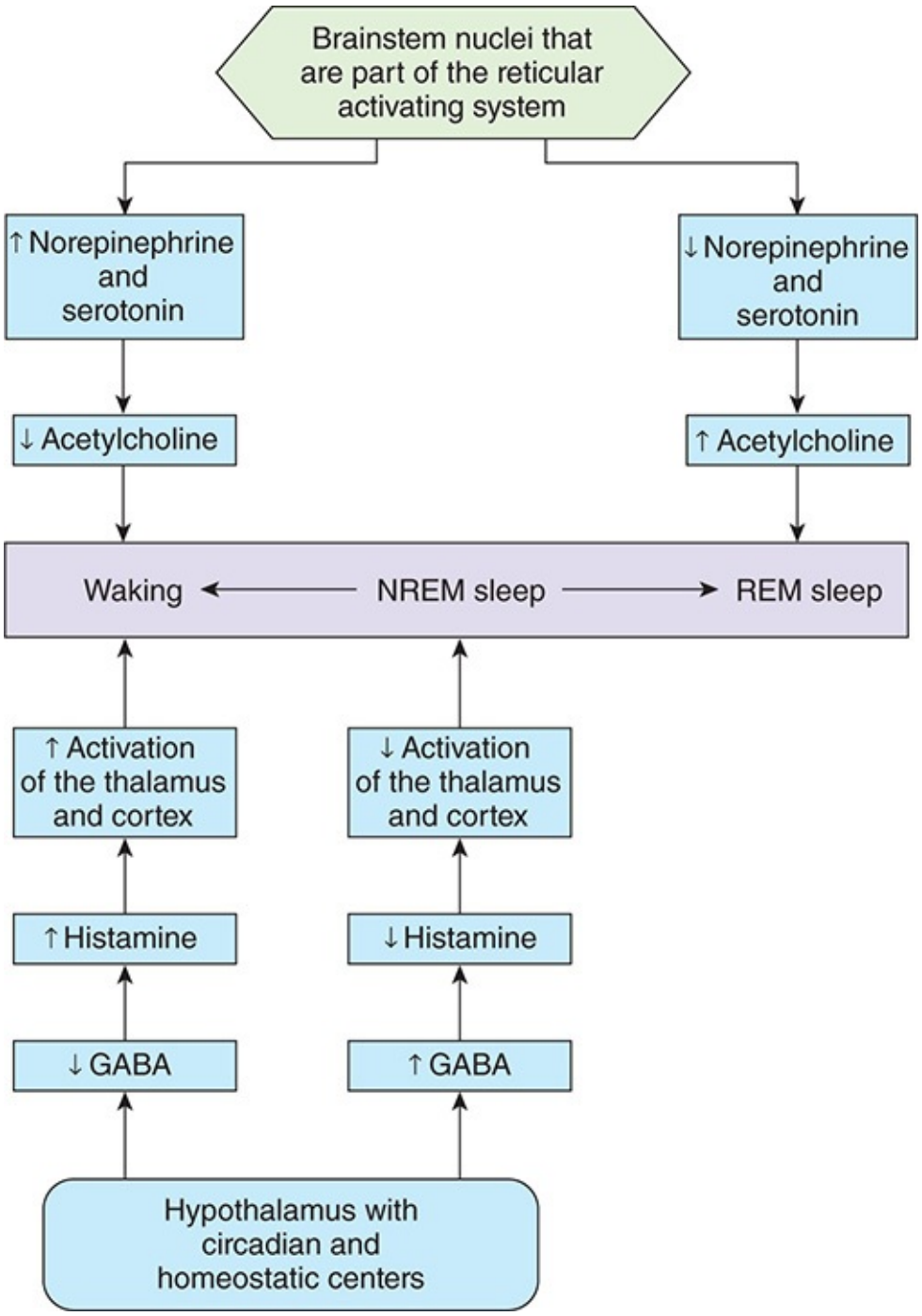


FIGURE 14–4 A model of how alternating activity of brainstem and hypothalamic neurons may influence the different states of consciousness. In this model, wakefulness and REM sleep are at opposite extremes. When the activity of norepinephrine- and serotonin-containing neurons (locus coeruleus and raphe nuclei) is dominant, there is a reduced level of activity in acetylcholine-containing pontine neurons leading to wakefulness. The reverse of

this pattern leads to REM sleep. A more even balance in the activity of these groups of neurons is associated with non-REM (NREM) sleep. Increases in GABA and decreases in histamine promote non-REM sleep via deactivation of the thalamus and cortex. Wakefulness occurs when GABA is reduced and histamine is released. (Used with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology*, 11th ed. New York, NY: McGraw-Hill; 2008.)

THE ELECTROENCEPHALOGRAM

The term **electroencephalogram (EEG)** refers to a recording that represents the electrical activity of the brain. The EEG can be recorded with scalp electrodes through the unopened skull. The term **electrocorticogram** is used for the recording obtained with electrodes on the pial surface of the cortex.

The EEG recorded from the scalp is a measure of the summation of dendritic postsynaptic potentials rather than action potentials **(Figure 14–5)**. The dendrites of the cortical neurons are a forest of similarly oriented, densely packed units in the superficial layers of the cerebral cortex (Figure 14–1). Propagated potentials can be generated in dendrites. In addition, recurrent axon collaterals end on dendrites in the superficial layers. As excitatory and inhibitory endings on the dendrites of each cell become active, current flows into and out of these current sinks and sources from the rest of the dendritic processes and the cell body. The cell body–dendrite relationship is that of a constantly shifting dipole. Current flow in the dipole produces wavelike potential fluctuations in a volume conductor (Figure 14–5). When the sum of the dendritic activity is negative relative to the cell body, the neuron is depolarized and hyperexcitable; when it is positive, the neuron is hyperpolarized and less excitable.

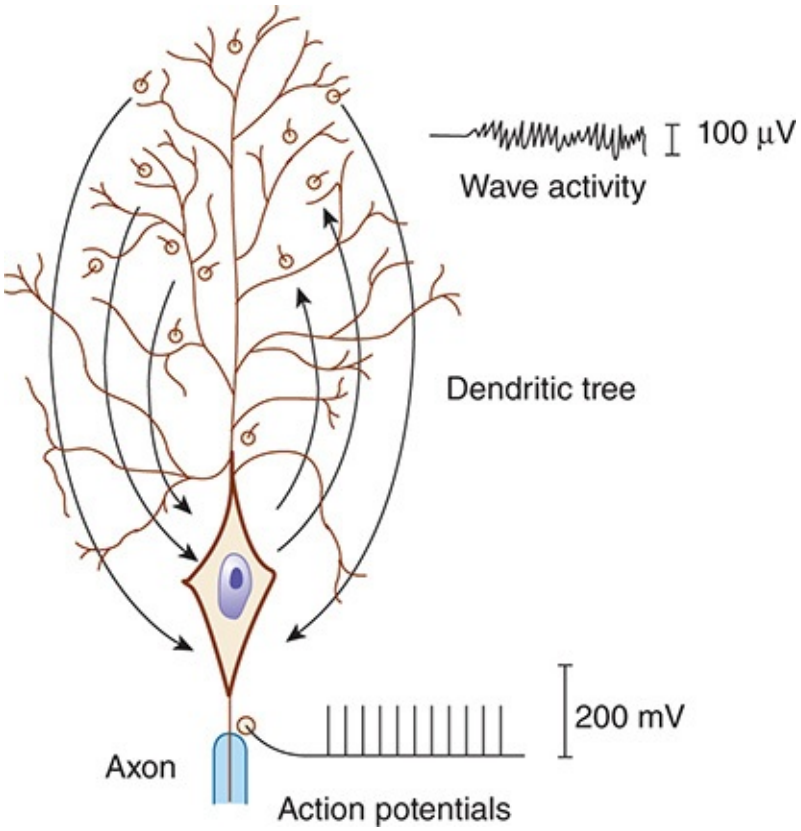


FIGURE 14–5 Diagrammatic comparison of the electrical responses of the axon and the dendrites of a large cortical neuron. Current flow to and from active synaptic knobs on the dendrites produces wave activity, while all-or-none action potentials are transmitted along the axon. When the sum of the dendritic activity is negative relative to the cell body, the neuron is depolarized; when it is positive, the neuron is hyperpolarized. The electroencephalogram recorded from the scalp is a measure of the summation of dendritic postsynaptic potentials rather than action potentials.

CLINICAL USES OF THE EEG

The EEG can be of value in localizing neuropathological processes. When fluid collection overlies a portion of the cortex, activity over this area may be damped. This fact may aid in diagnosing and localizing conditions such as subdural hematomas. Lesions in the cerebral cortex cause local formation of transient disturbances in brain activity, marked by high-voltage abnormal waves that can be recorded with an EEG. Seizure activity can occur because of increased firing of excitatory neurons (eg, release of glutamate) or decreased firing of inhibitory neurons (eg, release GABA).

TYPES OF SEIZURES

Epilepsy is a condition in which there are recurring, unprovoked seizures that may result from damage to the brain. The seizures represent abnormal, highly synchronous neuronal activity. Epilepsy is a syndrome with multiple causes. In some forms, characteristic EEG patterns occur during seizures or between attacks; however, abnormalities are often difficult to demonstrate. Seizures are divided into **partial (focal) seizures** and **generalized seizures.**

Partial seizures originate in a small group of neurons and can result from head injury, brain infection, stroke, or tumor; but often the cause is unknown. Symptoms depend on the seizure focus. They are further subdivided into **simple partial seizures** (without loss of consciousness) and **complex partial seizures** (with altered consciousness). An example of a simple partial seizure is localized jerking movements in one hand progressing to clonic movements of the entire arm lasting about 60–90 s. **Auras** typically precede the onset of a partial seizure and include abnormal sensations. The time after the seizure until normal neurologic function returns is called the **postictal period.**

Generalized seizures are associated with widespread electrical activity and involve both hemispheres simultaneously. They are further subdivided into **convulsive** and **nonconvulsive** categories depending on whether tonic or clonic movements occur. **Absence seizures** (formerly called petit mal seizures) are a form of nonconvulsive generalized seizures characterized by a momentary loss of consciousness. They are associated with 3/s doublets, each consisting of a typical **spike-and-wave** pattern of activity that lasts for about 10 s **(Figure 14–6)**. They are not accompanied by auras or postictal periods. These spike and waves are likely generated by low threshold T-type Ca^{2+} channels in thalamic neurons.



FIGURE 14–6 Absence seizures. This is a recording of four cortical EEG leads from a 6-year-old boy who, during the recording, had one of his “blank spells” in

which he was transiently unaware of his surroundings and blinked his eyelids. Absence seizures are associated with 3/s doublets, each consisting of a typical spike-and-wave pattern of activity that lasts for about 10 s. Time is indicated by the horizontal calibration line. EEG, electroencephalogram. (Reproduced with permission from Waxman SG: *Neuroanatomy with Clinical Correlations*, 25th ed. New York, NY: McGraw-Hill; 2003.)

Tonic-clonic seizures (formerly called grand mal seizure) are the most common convulsive generalized seizure. It is associated with sudden onset of contraction of limb muscles **(tonic phase)** lasting about 30 s, followed by a clonic phase with symmetric jerking of the limbs as a result of alternating contraction and relaxation **(clonic phase)** lasting 1–2 min. There is fast EEG activity during the tonic phase. Slow waves, each preceded by a spike, occur at the time of each clonic jerk; slow waves persist for a while after the attack.

The release of glutamate from astrocytes may play a role in in the pathophysiology of epilepsy. Also, there is evidence that reorganization of astrocytes along with dendritic sprouting and new synapse formation is the structural basis for recurrent excitation in the epileptic brain. **Clinical Box 14–1** describes information on the role of genetic mutations in some forms of epilepsy.

CLINICAL BOX 14–1

Genetic Mutations & Epilepsy

Epilepsy has no geographic, racial, sex, or social bias. It can occur at any age, but is most often diagnosed in infancy, childhood, adolescence, and old age. It is the second most common neurologic disorder after stroke. According to the World Health Organization, an estimated 50 million people worldwide (8.2 per 1000 individuals) experience epileptic seizures. The prevalence in developing countries (such as Colombia, Ecuador, India, Liberia, Nigeria, Panama, United Republic of Tanzania, and Venezuela) is more than 10 per 1000. Many affected individuals experience unprovoked seizures, for no apparent reason, and without any other neurologic abnormalities. These are called **idiopathic epilepsies** and are assumed to be genetic in origin. Mutations in voltage-gated potassium, sodium, and chloride channels have been linked to some forms of idiopathic epilepsy. Mutated ion channels can lead to neuronal hyperexcitability via various pathogenic mechanisms. Scientists have recently identified the mutated gene responsible for

development of **childhood absence epilepsy (CAE);** several patients with CAE have mutations in a subunit gene of the GABA receptor called **GABRB3.** Also, **SCN1A** and **SCN1B mutations** have been identified in an inherited form of epilepsy called **generalized epilepsy with febrile seizures.** SCN1A and SCN1B are sodium channel subunit genes that are widely expressed within the central nervous system. SCN1A mutations are suspected in several other forms of epilepsy.

THERAPEUTIC HIGHLIGHTS

Only about two-thirds of those who experience seizure activity respond to drug therapies. Some respond to surgical interventions (eg, those with temporal lobe seizures), and others respond to vagal nerve stimulation (eg, those with partial seizures). Prior to the 1990s, the most common drugs used to treat seizures **(anticonvulsants)** included phenytoin, valproate, and barbiturates. Newer drugs have become available but, as is the case with the older drugs, they are palliative rather than curative. There are three broad mechanisms of action of anticonvulsant drugs: enhancing inhibitory neurotransmission (increased GABA release), reducing excitatory neurotransmission (decreased glutamate release), or altering ionic conductance. **Gabapentin** is a GABA analog that acts by decreasing Ca^{2+} entry into cells and reducing glutamate release; it is used to treat generalized seizures. **Topiramate** blocks voltage-gated Na^{+} channels associated with glutamate receptors and potentiates the inhibitory effect of GABA; it is also used to treat generalized seizures. **Ethosuximide** reduces the low threshold T-type Ca^{2+} currents in thalamic neurons and is particularly effective in treatment of absence seizures. **Valproate** and **phenytoin** block high-frequency firing of neurons by acting on voltage-gated Na^{+} channels to reduce glutamate release.

SLEEP–WAKE CYCLE: VARIATIONS IN EEG RHYTHMS

ALPHA AND BETA RHYTHMS

In adult humans who are awake but at rest with the mind wandering and the eyes closed, the most prominent component of the EEG is a fairly regular pattern of

waves at a frequency of 8–13 Hz and amplitude of 50–100 µV when recorded from the scalp. This pattern is the **alpha rhythm** (Figure 14–7). It is most marked in the parietal and occipital lobes and is associated with decreased levels of attention. The frequency and magnitude of the EEG rhythm can vary with age, with the use of some drugs, and in some pathological conditions (Clinical Box 14–2).

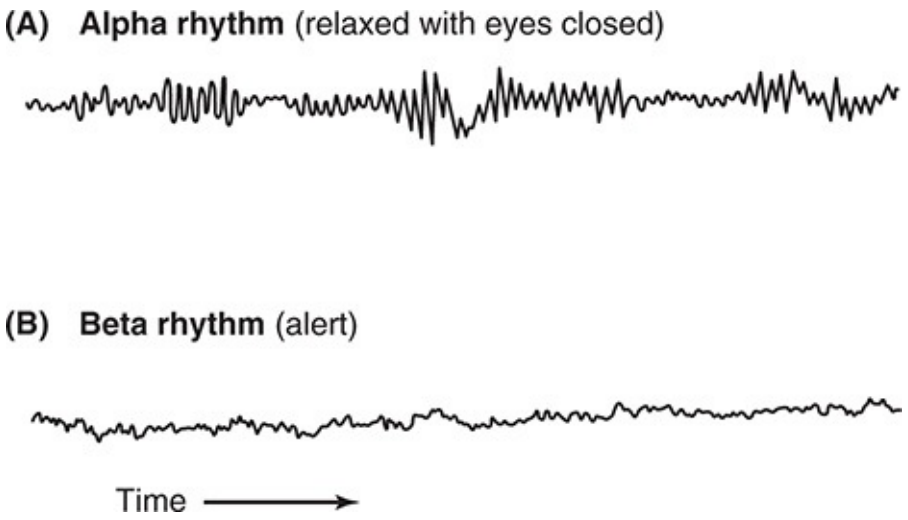


FIGURE 14–7 EEG records showing the alpha and beta rhythms. When attention is focused on something, the 8–13 Hz alpha rhythm is replaced by an irregular 13–30 Hz low-voltage activity, the beta rhythm. This phenomenon is referred to as alpha block, arousal, or the alerting response. (Used with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology*, 11th ed. New York, NY: McGraw-Hill; 2008.)

When attention is focused on something, the alpha rhythm is replaced by an irregular 13–30 Hz low-voltage activity, the **beta rhythm** (Figure 14–7). This phenomenon is called alpha block and can be produced by any form of sensory stimulation or mental concentration, such as solving arithmetic problems. Another term for this phenomenon is the **arousal** or **alerting response** because it is correlated with the aroused, alert state. It has also been called **desynchronization** because it represents breaking up of the obviously synchronized neural activity necessary to produce regular waves. However, the rapid EEG activity seen in the alert state is also synchronized but at a higher rate. Therefore, the term "desynchronization" is misleading.

SLEEP STAGES: NON-REM & REM SLEEP

Non-REM sleep is divided into four stages **(Figure 14–8)**. Stage 1 non-REM sleep is the transition from wakefulness to sleep, the EEG shows a low-voltage, mixed frequency pattern. A **theta rhythm** (4–7 Hz) can be seen at this stage of sleep. Stage 2 of non-REM sleep is marked by the appearance of sinusoidal waves called **sleep spindles** (7–15 Hz) and occasional high voltage biphasic waves called **K complexes.** Muscle tone is reduced during this time. Stage 3 of non-REM sleep is characterized by the appearance of a high-amplitude **delta rhythm** (0.5–4 Hz) in the EEG, reflecting a further reduction is arousal, and a further reduction in muscle tone. Maximum slowing with large waves is seen in stage 4 of non-REM sleep. Thus, the characteristic of deep sleep is a pattern of rhythmic slow waves, indicative of marked synchronization of cortical and thalamic activity; it is sometimes referred to as slow-wave sleep. While the occurrence of theta and delta rhythms is normal during sleep, their appearance during wakefulness is a sign of brain dysfunction.

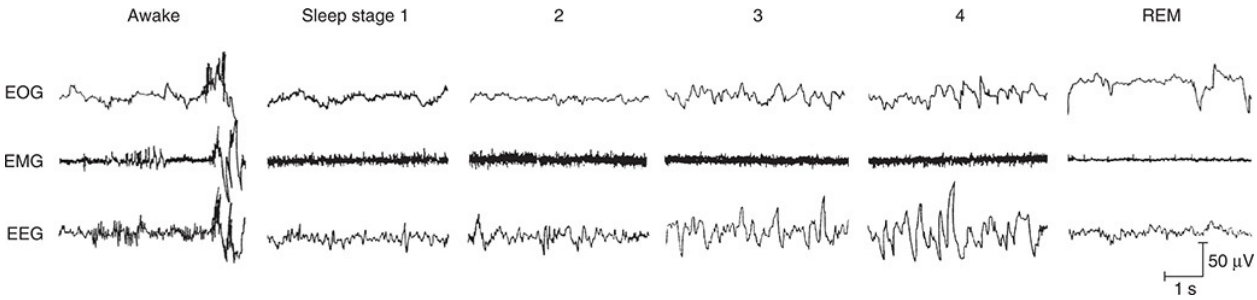


FIGURE 14–8 EEG and muscle activity during various stages of the sleep–wake cycle. Non-REM sleep has four stages. Stage 1 is characterized by a slight slowing of the EEG. Stage 2 has high-amplitude K complexes and spindles. Stages 3 and 4 have slow, high-amplitude delta waves. REM sleep is characterized by eye movements, loss of muscle tone, and a low-amplitude, high-frequency activity pattern. The higher voltage activity in the EOG tracings during stages 2 and 3 reflect high amplitude EEG activity in the prefrontal areas rather than eye movements. EOG, electrooculogram registering eye movements; EMG, electromyogram registering skeletal muscle activity. (Reproduced with permission from Rechtschaffen A, Kales A: *A Manual of Standardized Terminology, Techniques and Scoring System and Sleep Stages of Human Subjects*. Los Angeles: University of California Brain Information Service; 1968.)

CLINICAL BOX 14–2

Variations in the Alpha Rhythm

In humans, the frequency of the dominant EEG rhythm at rest varies with age. In infants, there is fast, beta-like activity, but the occipital rhythm is a slow 0.5–2-Hz pattern. During childhood this latter rhythm speeds up, and the adult alpha pattern gradually appears during adolescence. The frequency of the alpha rhythm is decreased by low blood glucose levels, low body temperature, low levels of adrenal glucocorticoid hormones, and high arterial partial pressure of CO_2 ($PaCO_2$). It is increased by the reverse conditions. Forced over-breathing to lower the $PaCO_2$ is sometimes used clinically to bring out latent EEG abnormalities. The frequency and magnitude of the alpha rhythm is also decreased by metabolic and toxic encephalopathies including those due to hyponatremia and vitamin B_{12} deficiency. The frequency of the alpha rhythm is reduced during acute intoxication with **alcohol, amphetamines, barbiturates, phenytoin,** and **antipsychotics. Propofol,** a hypnotic/sedative drug, can induce a rhythm in the EEG that is analogous to the classic alpha rhythm.

The high-amplitude slow waves seen in the EEG during non-REM sleep are periodically replaced by rapid, low-voltage EEG activity in REM sleep (Figure 14–8). REM sleep gets its name from the characteristic rapid, roving eye movements that occur during this stage of sleep and are recorded as an **electrooculogram (EOG)**. Except for eye movement, there is almost a complete loss of skeletal muscle tone in REM sleep. The threshold for arousal from sleep by sensory stimuli is elevated during this time. Another characteristic of REM sleep is the occurrence of large phasic potentials that originate in the cholinergic neurons in the pons and pass rapidly to the lateral geniculate body and from there to the occipital cortex. They are called **pontogeniculo-occipital (PGO) spikes.**

Positron emission tomography (PET) scans in REM sleep show increased activity in the pontine area, amygdala, and anterior cingulate gyrus, but decreased activity in the prefrontal and parietal cortex. Activity in visual association areas is increased, but activity is decreased in the primary visual cortex. This is consistent with increased emotion and operation of a closed neural system cut off from the areas that relate brain activity to the external world.

DISTRIBUTION OF SLEEP STAGES

In a typical night of sleep, a young adult first enters non-REM sleep, passes through stages 1 and 2, and spends 70–100 min in stages 3 and 4. Sleep then lightens, and a REM period follows. This cycle is repeated at intervals of about 90 min throughout the night **(Figure 14–9)**. The cycles are similar, though there is less stage 3 and 4 sleep and more REM sleep toward morning; thus, four to six REM periods occur per night. REM sleep occupies 80% of total sleep time in premature infants and 50% in full-term neonates. Thereafter, the proportion of REM sleep falls rapidly and plateaus at about 25% until it falls to about 20% in the elderly. Children have more total sleep time (8–10 h) compared to most adults (about 6 h).

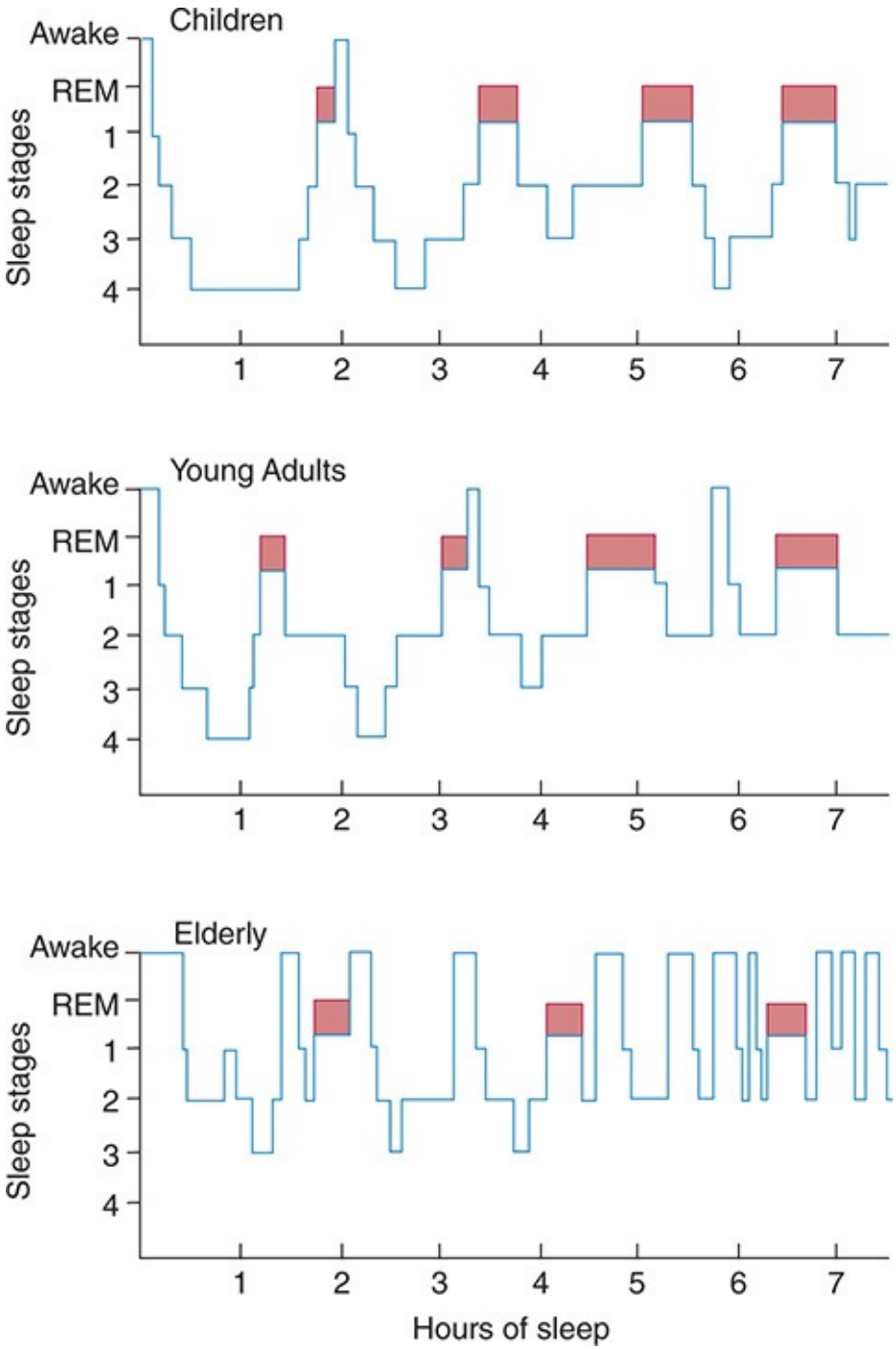


FIGURE 14–9 Normal sleep cycles at various ages. REM sleep is indicated by the darker colored areas. In a typical night of sleep, a young adult first enters non-REM sleep, passes through stages 1 and 2, and spends 70–100 min in stages 3 and 4. Sleep then lightens, and a REM period follows. This cycle is repeated at intervals of about 90 min throughout the night. The cycles are similar, though there is less stage 3 and 4 sleep and more REM sleep toward morning. REM

sleep occupies 50% of total sleep time in neonates; this proportion declines rapidly and plateaus at ~25% until it falls further in the elderly. (Reproduced with permission from Kales AM, Kales JD: Sleep disorders. N Engl J Med 1974; Feb 28; 290(9):487–499.)

Dreaming occurs in both REM and non-REM sleep stages, but their characteristics differ. Dreams that occur during REM sleep tend to be longer and more visual and emotional than those that occur during non-REM sleep.

IMPORTANCE OF SLEEP

Various studies imply that sleep is needed to maintain metabolic-caloric balance, thermal equilibrium, and immune competence. **Clinical Box 14–3** describes several common sleep disorders. If humans are awakened every time they show REM sleep and then permitted to sleep without interruption, they show a great deal more than the normal amount of REM sleep for a few nights. Relatively prolonged REM deprivation does not seem to have adverse psychological effects.

CLINICAL BOX 14–3

Sleep Disorders

Narcolepsy is a chronic neurologic disorder caused by the brain's inability to regulate sleep–wake cycles normally. The affected individual experiences a sudden loss of voluntary muscle tone **(cataplexy),** an eventual irresistible urge to sleep during daytime, and possibly brief episodes of total paralysis at the beginning or end of sleep. Narcolepsy is also characterized by a sudden onset of REM sleep, unlike normal sleep that begins with non-REM, slow-wave sleep. The prevalence of narcolepsy ranges from 1 in 600 in Japan to 1 in 500,000 in Israel, with 1 in 1000 Americans being affected. Narcolepsy has a familial incidence strongly associated with a class II antigen of the major histocompatibility complex on chromosome 6 at the HLA-DR2 or HLA-DQW1 locus, implying a genetic susceptibility to narcolepsy. The HLA complexes are interrelated genes that regulate the immune system (see Chapter 3). Compared to brains from healthy persons, the brains of persons with narcolepsy often contain fewer **hypocretin (orexin)**-producing neurons in the hypothalamus. The HLA complex may increase susceptibility to an

immune attack on these neurons, leading to their degeneration.

Obstructive sleep apnea (OSA) is the most common cause of daytime sleepiness due to fragmented sleep at night; it affects about 24% of middle-aged men and 9% of women in the United States. Breathing ceases for more than 10 s during frequent episodes of obstruction of the upper airway (especially the pharynx) due to a reduction in muscle tone. The apnea causes brief arousals from sleep in order to reestablish upper airway tone. An individual with OSA typically begins to snore soon after falling asleep. The snoring gets progressively louder until it is interrupted by an episode of apnea, which is followed by a loud snort and gasp as the individual tries to breathe. OSA is not associated with a reduction in total sleep time, but individuals with OSA experience a much greater time in stage 1 non-REM sleep (from an average of 10% of total sleep to 30–50%) and a marked reduction in slow-wave sleep (stages 3 and 4 non-REM sleep). The pathophysiology of OSA includes both a reduction in neuromuscular tone at the onset of sleep and a change in the central respiratory drive.

Periodic limb movement disorder (PLMD) is a stereotypical rhythmic extension of the big toe and dorsiflexion of the ankle and knee during sleep lasting for about 0.5–10 s and recurring at intervals of 20–90 s. Movements can actually range from shallow, continual movement of the ankle or toes, to wild and strenuous kicking and flailing of the legs and arms. Electromyograph (EMG) recordings show bursts of activity during the first hours of non-REM sleep associated with brief EEG signs of arousal. The duration of stage 1 non-REM sleep may be increased and that of stages 3 and 4 may be decreased compared to age-matched controls. PLMD is reported to occur in 5% of individuals between the ages of 30 and 50 and increases to 44% of those over the age of 65. PLMD is similar to **restless leg syndrome** or Willis–Ekbom disease in which individuals have an irresistible urge to move their legs while at rest all day long.

Sleepwalking **(somnambulism),** bed-wetting **(nocturnal enuresis),** and **night terrors** are referred to as **parasomnias,** which are sleep disorders associated with arousal from non-REM and REM sleep. Episodes of sleepwalking are more common in children than in adults and occur predominantly in males. They may last several minutes. Somnambulists walk with their eyes open and avoid obstacles, but when awakened they cannot recall the episodes.

THERAPEUTIC HIGHLIGHTS

Excessive daytime sleepiness in patients with narcolepsy can be treated with amphetamine-like stimulants, including **modafinil, methylphenidate (Ritalin),** and **methamphetamine. Gamma hydroxybutyrate (GHB)** is used to reduce the frequency of cataplexy attacks and the incidences of daytime sleepiness. Cataplexy is often treated with antidepressants such as **imipramine** and **desipramine,** but these drugs are not officially approved by the US Federal Drug Administration for such use. The most common treatment for OSA is **continuous positive airflow pressure (CPAP),** a machine that increases airway pressure to prevent airway collapse. Drugs have generally proven to have little or no benefit in treating OSA. **Dopamine agonists,** which are used to treat Parkinson disease, can be used to treat PLMD and restless leg syndrome.

CIRCADIAN RHYTHMS

ROLE OF SUPRACHIASMATIC NUCLEI

Most, if not all, living cells in plants and animals have rhythmic fluctuations in their function on a circadian cycle. Normally they become entrained or synchronized to the day–night light cycle in the environment. If they are not entrained, they become progressively more out of phase with the light–dark cycle because they are longer or shorter than 24 h. In most cases, the **suprachiasmatic nuclei (SCN)** play a major role in the entrainment process **(Figure 14–10)**. The SCN receive information about the light–dark cycle via a special neural pathway, the **retinohypothalamic fibers.** Efferent fibers from the SCN initiate neural and humoral signals that entrain a wide variety of well-known circadian rhythms including the sleep–wake cycle and **melatonin** release from the richly vascularized pineal gland.

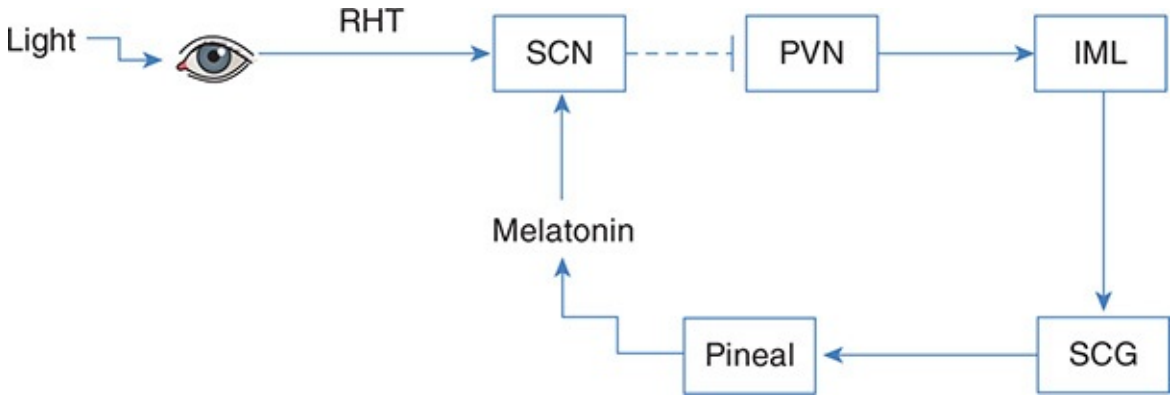


FIGURE 14–10 Secretion of melatonin. The cyclic activity of the suprachiasmatic nucleus (SCN) sets up a circadian rhythm for melatonin release. This rhythm is entrained to light/dark cycles by neurons in the retina. Light signals are relayed via the retinohypothalamic (RHT) fibers to the SCN. GABAergic neurons in the SCN inhibit neurons in the hypothalamic paraventricular nucleus (PVN) which then reduces the activity of sympathetic preganglionic neurons in the spinal intermediolateral nucleus (IML). These sympathetic preganglionic neurons innervate postganglionic neurons in the superior cervical ganglion (SCG) that regulate release of melatonin from the pineal gland.

GABAergic neurons in the SCN inhibit neurons in the hypothalamic paraventricular nucleus. From the hypothalamus, descending pathways converge onto preganglionic sympathetic neurons that innervate the superior cervical ganglion, the site of origin of the postganglionic neurons to the pineal gland.

The SCN have two peaks of circadian activity that may correlate with the observation that exposure to bright light can either advance, delay, or have no effect on the sleep–wake cycle depending on the time of day when it is experienced. During the usual daytime it has no effect, just after dark it delays the onset of the sleep period, and just before dawn it accelerates the onset of the next sleep period. Injections of melatonin have similar effects. **Clinical Box 14–4** describes circadian rhythm disorders that impact the sleep–wake state.

MELATONIN & CIRCADIAN RHYTHMS

Pineal pinealocytes contain melatonin and the enzymes responsible for its synthesis from serotonin by N-acetylation and O-methylation, and they secrete the hormone into the blood and the cerebrospinal fluid **(Figure 14–11)**. Two melatonin G-protein-coupled receptors (MT_1 and MT_2) are found on neurons in the SCN. Activation of MT_1 receptors inhibits adenylyl cyclase and results in sleepiness. Activation of MT_2 receptors stimulates phosphoinositide hydrolysis and may function to synchronize the light–dark cycle.

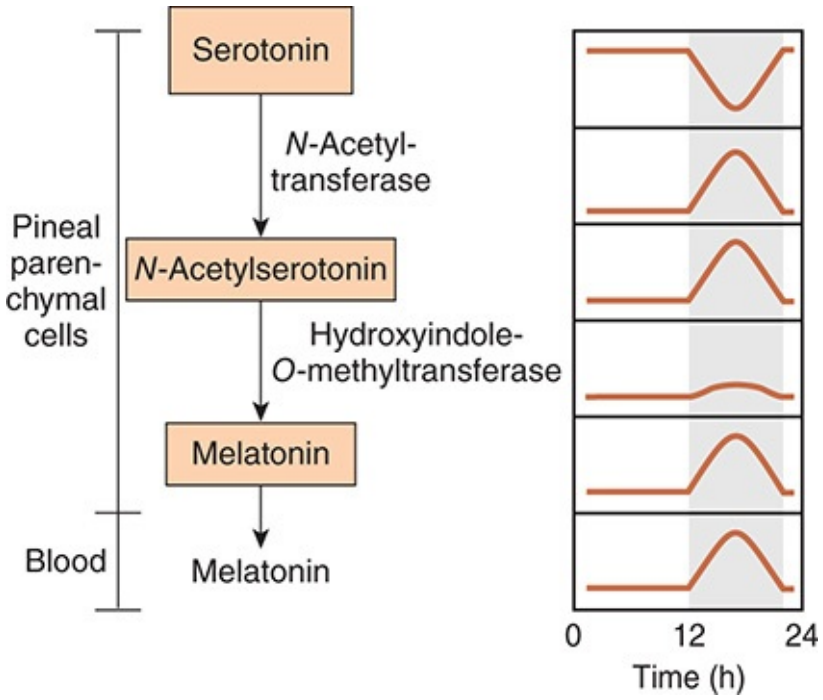


FIGURE 14–11 Diurnal rhythms of compounds involved in melatonin synthesis in the pineal. Melatonin and the enzymes responsible for its synthesis from serotonin are found in pineal pinealocytes; melatonin is secreted into the bloodstream. Melatonin synthesis and secretion are increased during the dark period (shaded area) and maintained at a low level during the light period.

The diurnal change in melatonin secretion may function as a timing signal to coordinate events with the light–dark cycle in the environment. Melatonin synthesis and secretion are increased during the dark period of the day and maintained at a low level during daylight hours (Figure 14–11). This diurnal variation in secretion is due to norepinephrine released from postganglionic sympathetic nerves that innervate the pineal gland (Figure 14–10). Norepinephrine acts via β-adrenoceptors to increase intracellular cAMP, and the cAMP in turn produces a marked increase in *N*-acetyltransferase activity. This results in increased melatonin synthesis and secretion.

CLINICAL BOX 14–4

Insomnia & Circadian Rhythm Disturbances of the Sleep–Wake State

Insomnia is defined as difficulty in initiating and/or maintaining sleep several times a week. Nearly 30% of the adult population experience episodes

of insomnia, and more than 50% of those aged 65 or older have sleep problems. Individuals with persistent episodes of insomnia are more likely to experience accidents, a diminished work experience, and a poorer overall quality of life. Insomnia is often comorbid with **depression,** and both disorders show abnormal regulation of **corticotropin-releasing factor.**

There are two major types of sleep disorders associated with disruption of the circadian rhythm. These are **transient sleep disorders** (jet lag, altered sleep cycle because of shift work, and illness) and **chronic sleep disorders (delayed or advanced sleep phase syndrome).** Those with delayed sleep phase syndrome have the inability to fall asleep in the evenings and awaken in the mornings. However, they have a normal total sleep time. Those with advanced sleep phase syndrome consistently fall asleep in early evening and awaken in early morning. This is seen most often in the elderly and the depressed.

THERAPEUTIC HIGHLIGHTS

Light therapy is an effective treatment in individuals who experience disturbances in their circadian cycle. **Melatonin** can be used to treat jet lag and insomnia in elderly individuals. **Ramelteon** is a MT_1 and MT_2 melatonin receptor agonist that is more effective than melatonin in treating insomnia. **Zolpidem** (Ambien) is an example of a sedative-hypnotic that slows brain activity to promote sleep onset. In addition to treating daytime sleepiness in narcolepsy, **modafinil** has also been used successfully in the treatment of daytime sleepiness due to shift work and to treat delayed sleep disorder syndrome.

CHAPTER SUMMARY

- The thalamus is the gateway to the cortex and includes neurons that project diffusely to wide areas of the neocortex (midline and intralaminar nuclei) and neurons that project to discrete regions of the neocortex (specific sensory relay nuclei).
- The ascending arousal system is comprised of monoaminergic, cholinergic, and histaminergic neurons that project to the intralaminar and reticular nuclei of the thalamus that project diffusely to wide regions of the cortex including the frontal, parietal, temporal, and occipital cortices.

- Wakefulness and REM sleep are at opposite extremes of consciousness. When norepinephrine- and serotonin-containing neurons (locus coeruleus and raphé nuclei) are most active, pontine cholinergic neurons are less active and wakefulness ensues. The reverse of this pattern leads to REM sleep. A more even balance of the activity in these groups of neurons is associated with non-REM sleep. Increases in GABA and decreases in histamine also promote non-REM sleep via deactivation of the thalamus and cortex.
- The EEG reflects the electrical activity (summation of dendritic postsynaptic potentials) of the brain and can be of value in localizing pathologic processes and in characterizing different types of seizures.
- Partial (focal) seizures originate in a small group of neurons and are subdivided into simple (without loss of consciousness) and complex (with altered consciousness). Generalized seizures are associated with widespread electrical activity and are subdivided into convulsive and non-convulsive categories depending on whether tonic or clonic movements occur.
- The major rhythms in the EEG are alpha (8–13 Hz) when awake with eyes closed, beta (13–30 Hz) when alert; theta (4–7 Hz) and delta (0.5–4 Hz) oscillations appear during deep sleep.
- Throughout non-REM sleep, there is some activity of skeletal muscle. A theta rhythm can be seen during stage 1 of sleep. Stage 2 is marked by the appearance of sleep spindles and occasional K complexes. In stage 3, a delta rhythm is dominant. Maximum slowing with large slow waves is seen in stage 4. REM sleep is characterized by low-voltage, high-frequency EEG activity and rapid, roving movements of the eyes.
- A young adult typically passes through stages 1 and 2, and spends 70–100 min in stages 3 and 4. Sleep then lightens, and a REM period follows. This cycle repeats at 90-min intervals throughout the night. REM sleep occupies 50% of total sleep time in full-term neonates; this proportion declines rapidly and plateaus at about 25% until it falls further in old age.
- Narcolepsy is a sudden loss of voluntary muscle tone (cataplexy), an irresistible urge to sleep during daytime, and a sudden onset of REM sleep. OSA is the most common cause of daytime sleepiness; breathing ceases for more than 10 s during frequent episodes of obstruction of the upper airway. Compared to normal sleep patterns, individuals spend more time in stage 1 non-REM sleep and less time in stages 3 and 4 non-REM sleep. PLMD is a stereotypical rhythmic extension of the big toe and dorsiflexion of the ankle

and knee during sleep lasting for about 0.5–10 s and recurring at intervals of 20–90 s.

- The entrainment of biologic processes to the light–dark cycle is regulated by the SCN. The diurnal change in melatonin release from the pineal gland may coordinate events with the light–dark cycle.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. An MD/PhD student was comparing sleep patterns in different age groups. What is expected to be the different about sleep as a function of age?

   A. Non-REM sleep occupies about 50% of total sleep in young adults and falls to about 20% in the elderly.

   B. Young adults experience about 4–6 episodes of REM sleep each night, and infants experience about 5–10 episodes of REM each night.

   C. REM sleep occupies about 80% of total sleep in full-term infants and falls to about 20% in the elderly.

   D. REM sleep occupies about 80% of total sleep in premature infants and falls until it plateaus at about 25% in adults.

   E. In a typical night, a young adult spends 70–100 min in stages 3 and 4 of non-REM sleep, whereas a child spends only about 30 min in these stages of deep sleep.

2. An MD/PhD student was studying the role of thalamocortical pathways in control of arousal. He was preparing his thesis proposal and included the following information on the two major types of thalamic projections to the cortex.

   A. Most thalamic neurons release glutamate throughout the cortex, but thalamic reticular neurons release GABA in the cortex.

   B. Afferents from the specific nuclei of the thalamus terminate in cortical layer I–IV, whereas the nonspecific afferents terminate primarily on spiny stellate cells of layer IV.

   C. Afferents from the specific nuclei of the thalamus terminate primarily on pyramidal neurons in layer IV, whereas the nonspecific afferents terminate primarily on inhibitory basket calls of layer IV.

   D. Thalamic projections to wide regions of the neocortex originate from the midline and intralaminar nuclei; thalamic projections to discrete regions of

the neocortex originate in specific sensory relay nuclei.

3. In a healthy, alert adult sitting with their eyes closed, the dominant EEG rhythm observed with electrodes over the occipital lobes is

   A. delta (0.5–4 Hz).

   B. theta (4–7 Hz).

   C. alpha (8–13 Hz).

   D. beta (18–30 Hz).

   E. fast, irregular low-voltage activity.

4. A 35-year-old man spent the evening in a sleep clinic to determine if he had obstructive sleep apnea. The tests showed that non-REM sleep accounted for over 30% of his total sleep time. Which of the following pattern of changes in central neurotransmitters or neuromodulators are associated with the transition from non-REM to wakefulness?

   A. Decrease in norepinephrine, increase in serotonin, increase in acetylcholine, decrease in histamine, and decrease in GABA.

   B. Decrease in norepinephrine, increase in serotonin, increase in acetylcholine, decrease in histamine, and increase in GABA.

   C. Decrease in norepinephrine, decrease in serotonin, increase in acetylcholine, increase in histamine, and increase in GABA.

   D. Increase in norepinephrine, increase in serotonin, decrease in acetylcholine, increase in histamine, and decrease in GABA.

   E. Increase in norepinephrine, decrease in serotonin, decrease in acetylcholine, increase in histamine, and decrease in GABA.

5. A healthy medical student participated in a research project on the effect of sleep deprivation on their EEG. During the baseline testing phase, no abnormalities were found. The following behavioral and EEG characteristics were likely recorded during stage 2 of non-REM sleep.

   A. Skeletal muscle movements were observed and the EEG showed a mixture of sleep spindles and theta waves.

   B. Skeletal muscle tone was reduced and sleep spindles and K complexes appeared in the EEG.

   C. Skeletal muscle movements were detected and the dominant rhythm in the EEG was delta waves.

   D. There was a complete absence of eye movements and the dominant rhythm in the EEG was theta waves.

E. There was an absence of skeletal muscle and eye movements and the EEG was desynchronized.

6. For the past several months, a 67-year-old woman experienced difficulty initiating and/or maintaining sleep several times a week. A friend suggested that she take melatonin to regulate her sleep–wake cycle. Endogenous melatonin secretion would be increased by

A. reducing the synthesis of serotonin.

B. inhibition of the paraventricular nucleus.

C. stimulation of the superior cervical ganglion.

D. stimulation of the optic nerve.

E. by blockade of hydroxyindole-*O*-methyltransferase.

7. Childhood absence epilepsy was diagnosed in a 10-year-old boy. His EEG showed a bilateral synchronous, symmetric 3-Hz spike-and-wave discharge. Absence seizures are a form of

A. nonconvulsive generalized seizures accompanied by momentary loss of consciousness.

B. complex partial seizures accompanied by momentary loss of consciousness.

C. nonconvulsive generalized seizures without a loss of consciousness.

D. simple partial seizures without a loss of consciousness.

E. convulsive generalized seizures accompanied by momentary loss of consciousness.

8. A child diagnosed with absence epilepsy began treatment with ethosuximide. What is the mechanism of action by which ethosuximide is an effective antiseizure drug?

A. It is a GABA analog that decreases Ca^{2+} entry into cells.

B. It blocks voltage-gated Na^{+} channels associated with glutamate receptors.

C. It potentiates GABA transmission.

D. It is a dopamine receptor agonist.

E. It inhibits T-type Ca^{2+} channels.

9. A 57-year-old professor at a medical school experienced numerous episodes of a sudden loss of muscle tone and an irresistible urge to sleep in the middle of the afternoon. The diagnosis was narcolepsy, which

A. is characterized by a sudden onset of non-REM sleep.

B. has a familial incidence associated with a class II antigen of the major histocompatibility complex.

C. may be due to the presence of an excessive number of orexin-producing neurons in the hypothalamus.

D. is often effectively treated with dopamine receptor agonists.

E. is the most common cause of daytime sleepiness.

CHAPTER 15

Learning, Memory, Language, & Speech

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the role of brain imaging techniques in identifying normal brain function and changes caused by brain damage.
- List the common causes, symptoms, and methods to assess traumatic brain injury (TBI).
- Describe the various forms of memory and identify the parts of the brain involved in memory processing and storage.
- Define synaptic plasticity, long-term potentiation (LTP), long-term depression (LTD), habituation, and sensitization, and explain their roles in learning and memory.
- Identify the abnormalities of brain structure and function that are characteristic of Alzheimer disease.
- Define the terms categorical hemisphere and representational hemisphere and summarize the differences between them.
- Identify the cortical areas important for language and their interconnections.
- Summarize the differences between fluent and nonfluent aphasia and explain each type on the basis of its pathophysiology.

INTRODUCTION

The understanding of brain function in humans has been revolutionized by the development and widespread availability of **positron emission tomographic (PET), functional magnetic resonance imaging (fMRI), computed tomography (CT) scanning,** and other imaging and diagnostic techniques. CT scans provide a high-resolution 3-dimensional image of the brain; it is useful for examining damage to the skull and detecting acute subarachnoid hemorrhage. PET imaging can measure local glucose metabolism, blood flow, and oxygen; fMRI can measure local amounts of oxygenated blood. PET and fMRI provide an index of the level of the activity in various parts of the brain in healthy humans and in those with pathologies or brain injuries **(see Clinical Box 15–1)**. They are used to study not only simple responses but also complex aspects of learning, memory, and perception. Different portions of the cortex are activated when a person is hearing, seeing, speaking, or generating words. **Figure 15–1** shows examples of the use of imaging to compare the functions of the cerebral cortex in processing words in a male versus a female subject.

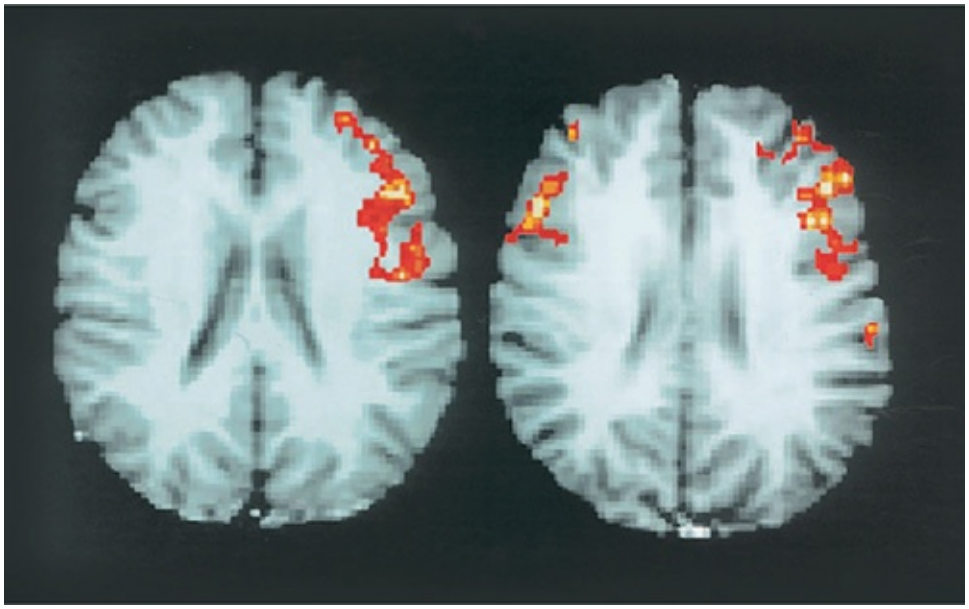


FIGURE 15–1 Comparison of the images of the active areas of the brain in a man (left) and a woman (right) during a language-based activity. Women use both sides of their brain whereas men use only a single side. This difference may reflect different strategies used for language processing. (Used with permission of Shaywitz et al, 1995. NMR Research/Yale Medical School.)

CLINICAL BOX 15–1

Traumatic Brain Injury

Traumatic brain injury (TBI) is defined as a nondegenerative, noncongenital insult to the brain due to an excessive mechanical force or penetrating injury to the head. It can lead to a permanent or temporary impairment of cognitive, physical, emotional, and behavioral functions, and it can be associated with a diminished or altered state of consciousness. TBI is one of the leading causes of death or disability worldwide. According to the Centers for Disease Control and Prevention, each year at least 1.5 million individuals in the United States sustain a TBI. It is most common in children under age 4, in adolescents aged 15–19 years of age, and in adults over the age of 65. In all age groups, TBI occurs twice as often in males compared to females. In about 75% of the cases, the TBI is considered mild and manifests as a concussion. Adults with severe TBI who are treated have a mortality rate of about 30%, but about 50% regain most if not all of their functions with therapy. The leading causes of TBI include falls, motor vehicle accidents, being struck by an object, and assaults. In some cases, areas remote from the actual injury also begin to malfunction, a process called **diaschisis.** TBI is often divided into primary and secondary stages. Primary injury is that caused by the mechanical force (eg, skull fracture and surface contusions) or acceleration–deceleration due to unrestricted movement of the head leading to shear, tensile, and compressive strains. These injuries can cause **intracranial hematoma** (epidural, subdural, or subarachnoid) and **diffuse axonal injury.** Secondary injury is often a delayed response and may be due to impaired cerebral blood flow that can eventually lead to cell death. A **Glasgow Coma Scale** is the most common system used to define the severity of TBI and evaluates motor responses, verbal responses, and eye opening to assess the levels of consciousness and neurologic functioning after an injury. Symptoms of mild TBI include headache, confusion, dizziness, blurred vision, ringing in the ears, a bad taste in the mouth, fatigue, disturbances in sleep, mood changes, and problems with memory, concentration, or thinking. Individuals with moderate or severe TBI show these symptoms as well as vomiting or nausea, convulsions or seizures, an inability to be roused, fixed and dilated pupils, slurred speech, limb weakness, loss of coordination, and increased confusion, restlessness, or agitation. In the most severe cases of TBI, the affected individual may go into a **permanent vegetative state**.

THERAPEUTIC HIGHLIGHTS

The advancements in brain imaging technology have improved the ability of medical personnel to diagnose and evaluate the extent of brain damage. Since little can be done to reverse the brain damage, therapy is initially directed at stabilizing the patient and trying to prevent further (secondary) injury. Medications that can be administered include diuretics (to reduce pressure in the brain), anticonvulsant drugs during the first week post injury (to avoid additional brain damage resulting from a seizure), and coma-inducing drugs (to reduce oxygen demands). This is followed by rehabilitation that includes physical, occupational, and speech/language therapies. Recovery of brain function can be due to several factors: brain regions that were suppressed but not damaged can regain their function, axonal sprouting and redundancy allows other areas of the brain to take over the functions that were lost due to the injury, and behavioral substitution, by learning new strategies to compensate for the deficits.

LEARNING & MEMORY

A characteristic of humans is their ability to alter behavior based on experience. **Learning** is acquisition of the information that makes this possible and **memory** is the retention and storage of that information. The two are obviously closely related and are considered together in this chapter.

FORMS OF MEMORY

From a physiological point of view, memory is divided into explicit and implicit forms **(Figure 15–2)**. **Explicit** or **declarative memory** is associated with consciousness, or at least awareness, and is dependent on the **hippocampus** and other parts of the **medial temporal lobes** of the brain for its retention. **Clinical Box 15–2** describes how tracking a patient with brain damage has led to an awareness of the role of the temporal lobe in declarative memory. **Implicit** or **nondeclarative memory** does not involve awareness, and its retention does not usually involve processing in the hippocampus.

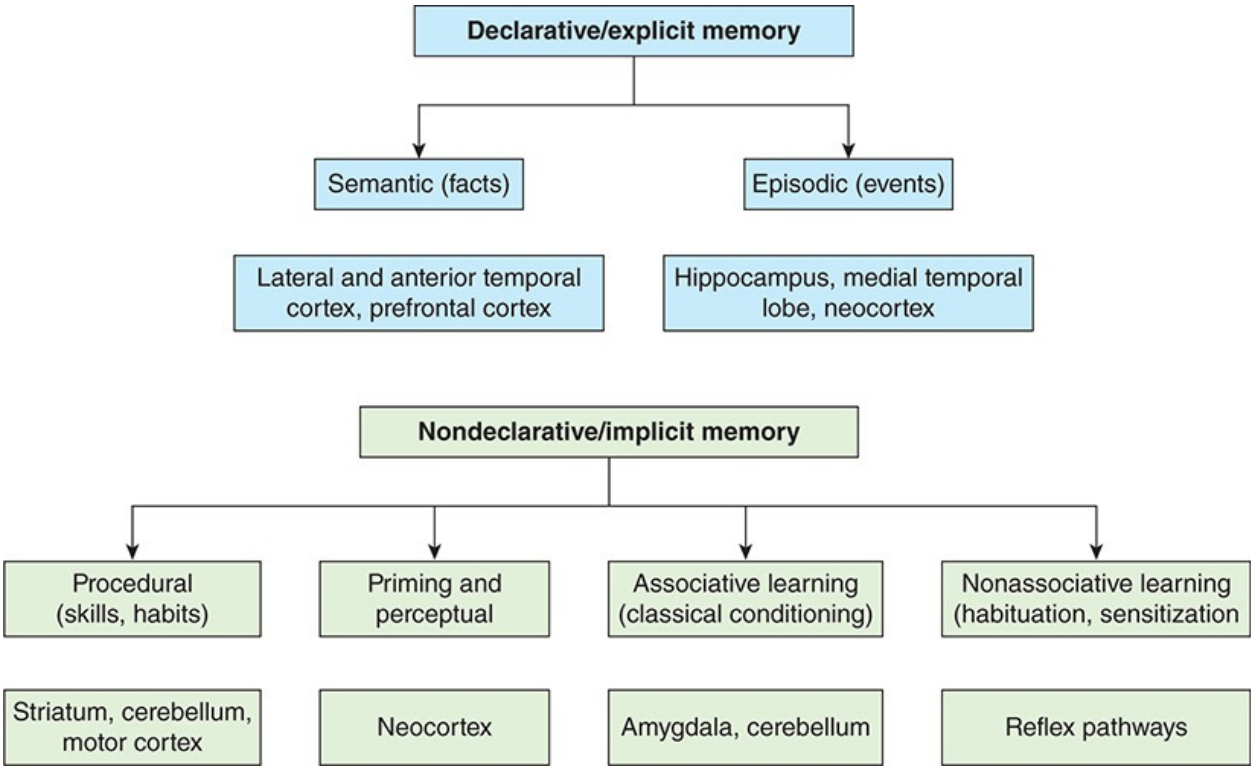


FIGURE 15–2 Forms of memory. Explicit (declarative) memory is associated with consciousness and is dependent on the integrity of the hippocampus, temporal lobes, neocortex, and prefrontal cortex for its retention. Explicit memory is for factual knowledge about people, places, things, and events. Implicit (nondeclarative) memory does not involve awareness, and it does not involve processing in the hippocampus. It requires the integrity of the amygdala, cerebellum, striatum, and neocortex as well as reflex pathways. Implicit memory is important for training reflexive motor or perceptual skills, classical conditioning, and habituation and sensitization. (Modified with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

Explicit memory is for factual knowledge about people, places, and things. It is divided into **semantic memory** for facts (eg, words, rules, and language) and **episodic memory** for events. Explicit memories that are initially required for activities such as riding a bicycle can become implicit once the task is thoroughly learned.

Implicit memory is important for training reflexive motor or perceptual skills and is subdivided into four types. **Priming** is the facilitation of the recognition of words or objects by prior exposure to them and is dependent on the **neocortex.** An example of priming is the improved recall of a word when presented with the

first few letters of it. **Procedural memory** includes skills and habits, which, once acquired, become unconscious and automatic. This type of memory is processed in the **striatum. Associative learning** relates to **classical** and **operant conditioning** in which one learns about the relationship between one stimulus and another. This type of memory is dependent on the **amygdala** for its emotional responses and the **cerebellum** for the motor responses. **Nonassociative** learning includes **habituation** and **sensitization** and is dependent on various reflex pathways.

CLINICAL BOX 15–2

The Case of HM: Defining a Link between Brain Function & Memory

HM was a patient who suffered from bilateral temporal lobe seizures that began following a bicycle accident at age 9. His case has been studied by many scientists and has led to a greater understanding of the link between the **temporal lobe** and **declarative memory.** HM had partial seizures for many years, and then several tonic-clonic seizures by age 16. In 1953, at the age of 27, HM underwent bilateral surgical removal of the amygdala, large portions of the hippocampal formation, and portions of the association area of the temporal cortex. HM's seizures were better controlled after surgery, but removal of the temporal lobes led to devastating memory deficits. He maintained **long-term memory** for events that occurred prior to surgery, but he suffered from **anterograde amnesia.** His **short-term memory** was intact, but he could not commit new events to long-term memory. He had normal procedural memory, and he could learn new puzzles and motor tasks. His case was the first to bring attention to the critical role of temporal lobes in formation of long-term declarative memories and to implicate this region in the conversion of short-term to long-term memories. Later work showed that the **hippocampus** is the primary structure within the temporal lobe involved in this conversion. Because HM retained memories from before surgery, his case also shows that the hippocampus is not involved in the storage of declarative memory. HM died in 2008 and only at that time was his identity released; his full name was Henry Gustav Molaison. An audio-recording and a transcript of the dialog by National Public Radio from the 1990s of HM talking to scientists was released in 2007 and is available at http://www.npr.org/templates/story/story.php?storyId=7584970.

Explicit memory and many forms of implicit memory involve (1) **short-term memory,** which lasts seconds to hours, during which processing in the hippocampus and elsewhere lays down long-term changes in synaptic strength and (2) **long-term memory,** which stores memories for years and sometimes for life. During short-term memory, the memory traces are subject to disruption by trauma and various drugs, whereas long-term memory traces are remarkably resistant to disruption. **Working memory** is a form of short-term memory that keeps information available, usually for very short periods of time, while the individual plans action based on it.

NEURAL BASIS OF MEMORY

The key to memory is alteration in the strength of selected synaptic connections. Second messenger systems contribute to the changes in neural circuitry required for learning and memory. Alterations in cellular membrane channels are often correlated to learning and memory. In all but the simplest of cases, the alteration involves the synthesis of proteins and the activation of genes. This occurs during the change from short-term working memory to long-term memory.

Acquisition of long-term learned responses can be prevented in some cases. For example, there is a loss of memory for the events immediately preceding a brain concussion or electroshock therapy **(retrograde amnesia).** This amnesia can actually encompass many days preceding the event that triggered it; remote memories remain intact.

SYNAPTIC PLASTICITY & LEARNING

Short- and long-term changes in synaptic function can occur because of the history of discharge at a synapse; that is, synaptic conduction can be strengthened or weakened on the basis of past experience. These changes are of great interest because they represent forms of learning and memory. They can be presynaptic or postsynaptic in location.

One form of plastic change is **posttetanic potentiation,** the production of enhanced postsynaptic potentials in response to stimulation. This enhancement lasts up to 60 s and occurs after a brief tetanizing train of stimuli in the presynaptic neuron. The tetanizing stimulation causes Ca^{2+} to accumulate in the presynaptic neuron to such a degree that the intracellular binding sites that keep cytoplasmic Ca^{2+} low are overwhelmed.

Habituation is a simple form of learning in which a neutral stimulus is repeated many times. The first time it is applied it is novel and evokes a reaction (the orienting reflex or "what is it?" response). However, it evokes less and less electrical response as it is repeated. Eventually, the subject becomes habituated to the stimulus and ignores it. This is associated with decreased release of neurotransmitter from the presynaptic terminal because of decreased intracellular Ca^{2+}. The decrease in intracellular Ca^{2+} is due to a gradual inactivation of Ca^{2+} channels. It can be short term, or it can be prolonged if exposure to the benign stimulus is repeated many times. Habituation is a classic example of nonassociative learning.

Sensitization is in a sense the opposite of habituation. Sensitization is the prolonged occurrence of augmented postsynaptic responses after a stimulus to which one has become habituated is paired once or several times with a noxious stimulus. The sensitization is due to presynaptic facilitation. Sensitization may occur as a transient response, or if it is reinforced by additional pairings of the noxious stimulus and the initial stimulus, it can exhibit features of short-term or long-term memory. The short-term prolongation of sensitization is due to a Ca^{2+}-mediated change in adenylyl cyclase that leads to a greater production of cAMP. The **long-term potentiation (LTP)** also involves protein synthesis and growth of the presynaptic and postsynaptic neurons and their connections.

LTP is a rapidly developing persistent enhancement of the postsynaptic potential response to presynaptic stimulation after a brief period of rapidly repeated stimulation of the presynaptic neuron. It resembles posttetanic potentiation but is much more prolonged and can last for days. There are multiple mechanisms by which LTP can occur, some are dependent on changes in the **N-methyl-D-aspartate (NMDA) receptor** and some are independent of the NMDA receptor. LTP is initiated by an increase in intracellular Ca^{2+} in either the presynaptic or postsynaptic neuron.

LTP occurs in many parts of the nervous system but has been studied in greatest detail in a synapse within the hippocampus, specifically the connection of a pyramidal cell in the CA3 region and a pyramidal cell in the CA1 region via the **Schaffer collateral.** This is an example of an NMDA receptor-dependent form of LTP involving an increase in Ca^{2+} in the postsynaptic neuron. Recall that NMDA receptors are permeable to Ca^{2+} as well as to Na^{+} and K^{+}. **Figure 15–3** summarizes the hypothetical basis of the Schaffer collateral LTP. At the resting membrane potential, glutamate release from a presynaptic neuron binds to both NMDA and non-NMDA receptors on the postsynaptic neuron. In the case of the Schaffer collateral, the non-NMDA receptor of interest is the **α-amino-3-**

hydroxy-5-methylisoxazole-4 propionic acid (AMPA) receptor. Na^+ and K^+ can flow only through the AMPA receptor because the presence of Mg^{2+} on the NMDA receptor blocks it. However, the membrane depolarization that occurs in response to high frequency tetanic stimulation of the presynaptic neuron is sufficient to expel the Mg^{2+} from the NMDA receptor, allowing the influx of Ca^{2+} into the postsynaptic neuron. This leads to activation of Ca^{2+}/calmodulin kinase, protein kinase C, and tyrosine kinase which together induce LTP. The Ca^{2+}/calmodulin kinase phosphorylates the AMPA receptors, increasing their conductance, and moves more of these receptors into the synaptic cell membrane from cytoplasmic storage sites. Once LTP is induced, a chemical signal (possibly nitric oxide, NO) is released by the postsynaptic neuron and passes retrogradely to the presynaptic neuron, producing a long-term increase in the quantal release of glutamate.

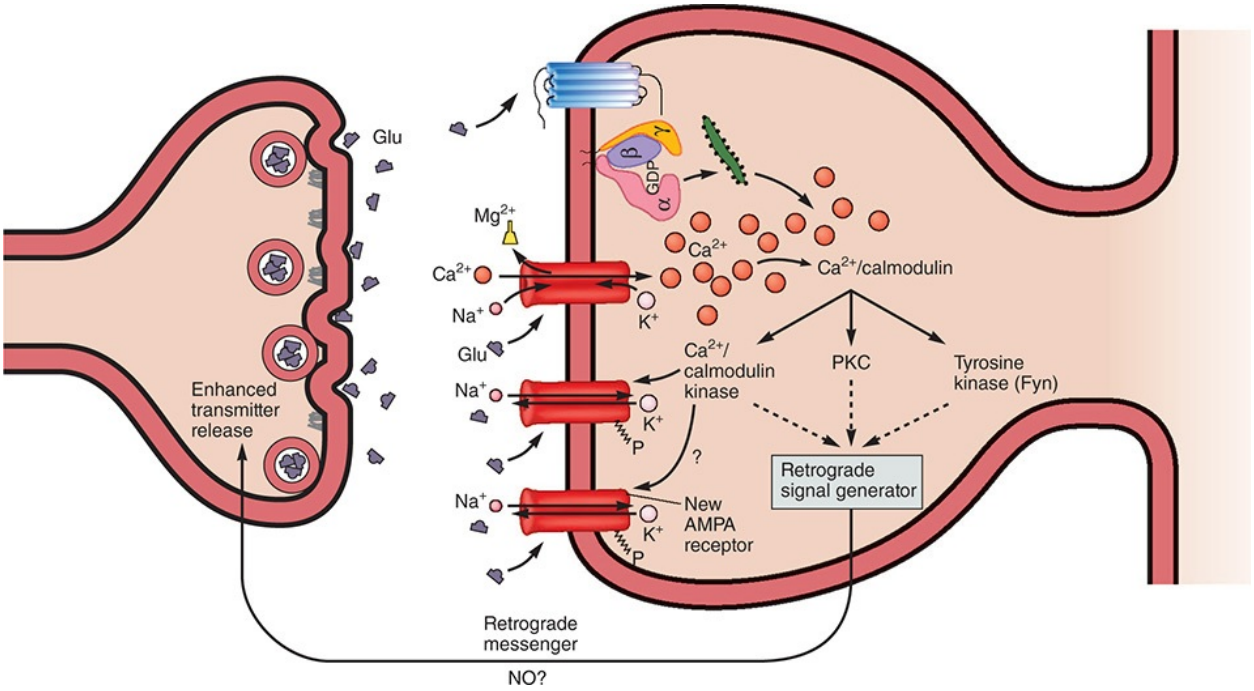


FIGURE 15–3 Production of LTP in Schaffer collaterals in the hippocampus. Glutamate (Glu) released from the presynaptic neuron binds to AMPA and NMDA receptors in the membrane of the postsynaptic neuron. The depolarization triggered by activation of the AMPA receptors relieves the Mg^{2+} block in the NMDA receptor channel, and Ca^{2+} enters the neuron with Na^+. The increase in cytoplasmic Ca^{2+} activates Ca^{2+}/calmodulin kinase, protein kinase C, and tyrosine kinase which together induce LTP. The Ca^{2+}/calmodulin kinase II phosphorylates the AMPA receptors, increasing their conductance, and moves

more AMPA receptors into the synaptic cell membrane from cytoplasmic storage sites. In addition, once LTF is induced, a chemical signal (possibly nitric oxide, NO) is released by the postsynaptic neuron and passes retrogradely to the presynaptic neuron, producing a long-term increase in the quantal release of glutamate. AMPA, α-amino-3-hydroxy-5-methylisoxazole-4 propionic acid; LTP, long-term potentiation; NMDA, N-methyl-D-aspartate.

LTP identified in the mossy fibers of the hippocampus (connecting granule cells in the dentate cortex) is due to an increase in Ca^{2+} in the presynaptic rather than the postsynaptic neuron in response to tetanic stimulation and is independent of NMDA receptors. The influx of Ca^{2+} in the presynaptic neuron is thought to activate Ca^{2+}/calmodulin-dependent adenylyl cyclase to increase cAMP.

Long-term depression (LTD) was first noted in the hippocampus but occurs throughout the brain in the same fibers as LTP. LTD is the opposite of LTP. It resembles LTP in many ways, but it is characterized by a decrease in synaptic strength. It is produced by slower stimulation of presynaptic neurons and is associated with a smaller rise in intracellular Ca^{2+} than occurs in LTP. In the cerebellum, its occurrence requires the phosphorylation of the GluR2 subunit of the AMPA receptors. It may be part of the mechanism by which learning occurs in the cerebellum.

NEUROGENESIS

The traditional view that brain cells are not added after birth is wrong; new neurons form from stem cells throughout life in at least two areas: the olfactory bulb and the hippocampus. This is a process called **neurogenesis.** Experience-dependent growth of new granule cells in the dentate gyrus of the hippocampus may contribute to learning and memory. A reduction in the number of new neurons formed reduces at least one form of hippocampal memory production.

ASSOCIATIVE LEARNING: CONDITIONED REFLEXES

A classic example of associative learning is a **conditioned reflex.** A conditioned reflex is a reflex response to a stimulus that previously elicited little or no response, acquired by repeatedly pairing the stimulus with another stimulus that

normally does produce the response. In Pavlov's classic experiments, the salivation normally induced by placing meat in the mouth of a dog was studied. A bell was rung just before the meat was placed in the dog's mouth, and this was repeated several times until the animal would salivate when the bell was rung even though no meat was placed in its mouth. In this experiment, the meat placed in the mouth was the **unconditioned stimulus (US),** the stimulus that normally produces an innate response. The **conditioned stimulus (CS)** was the bell ringing. After the CS and US had been paired a sufficient number of times, the CS produced the response originally evoked only by the US. The CS had to precede the US. Many somatic, visceral, and other neural changes can be made to occur as conditioned reflex responses. Conditioning of visceral responses is called **biofeedback.**

WORKING MEMORY

As noted above, working memory keeps incoming information available for a short time while deciding what to do with it. It is that form of memory which permits us, for example, to look up a telephone number, and then remember the number while we pick up the telephone and dial the number. It consists of a **central executive** located in the prefrontal cortex, and two "rehearsal systems": a **verbal system** for retaining verbal memories and a parallel **visuospatial system** for retaining visual and spatial aspects of objects. The executive steers information into these rehearsal systems.

HIPPOCAMPUS & MEDIAL TEMPORAL LOBE

Working memory areas are connected to the hippocampus and the adjacent parahippocampal portions of the medial temporal cortex. Output from the hippocampus leaves via the subiculum and the entorhinal cortex and binds together and strengthens circuits in different neocortical areas, forming over time the stable remote memories that can now be triggered by many different cues.

Bilateral destruction of the ventral hippocampus, or Alzheimer disease and similar disease processes that destroy its CA1 neurons, can cause striking defects in short-term memory. Humans with such destruction have intact working memory and remote memory. Their implicit memory processes are generally intact. They perform adequately in terms of conscious memory if they concentrate on what they are doing. However, if they are distracted even briefly,

all memory of what they were doing and what they proposed to do is lost. They are thus capable of new learning and retain old prelesion memories, but they cannot form new long-term memories.

The hippocampus is closely associated with the overlying parahippocampal cortex in the medial frontal lobe. Memory processes can be studied with fMRI and with measurement of evoked potentials (event-related potentials; ERPs) in epileptic patients with implanted electrodes. When subjects recall words, activity in their left frontal lobe and their left parahippocampal cortex increases. In contrast, when they recall pictures or scenes, activity takes place in their right frontal lobe and the parahippocampal cortex on both sides.

The connections of the hippocampus to the diencephalon are also involved in memory. Some people with alcoholism-related brain damage develop impairment of recent memory, and the memory loss correlates well with the presence of pathologic changes in the mamillary bodies that have extensive efferent connections to the hippocampus via the fornix. The mamillary bodies project to the anterior thalamus via the mamillothalamic tract, and lesions of the thalamus cause loss of recent memory. From the thalamus, the fibers concerned with memory project to the prefrontal cortex and from there to the basal forebrain. From the **nucleus basalis of Meynert** in the basal forebrain, a diffuse cholinergic projection goes to the entire neocortex, the amygdala, and the hippocampus. Severe loss of these fibers occurs in Alzheimer disease.

The amygdala is closely associated with the hippocampus and is concerned with encoding and recalling emotionally charged memories. During retrieval of fearful memories, the theta rhythms of the amygdala and the hippocampus become synchronized. In healthy subjects, events associated with strong emotions are remembered better than events without an emotional charge, but in patients with bilateral lesions of the amygdala, this difference is absent.

Individuals with lesions of the ventromedial prefrontal cortex perform poorly on memory tests, but they spontaneously describe events that never occurred. This phenomenon is called confabulation or false memories.

LONG-TERM MEMORY

While the encoding process for short-term explicit memory involves the hippocampus, long-term memories are stored in various parts of the neocortex. Apparently, the various parts of the memories (visual, olfactory, auditory, etc) are localized to the cortical regions concerned with these functions. These pieces are tied together by long-term changes in the strength of transmission at relevant

synaptic junctions so that all the components are brought to consciousness when the memory is recalled.

Once long-term memories have been established, they can be recalled or accessed by many different associations. For example, the memory of a vivid scene can be evoked not only by a similar scene but also by a sound or smell associated with the scene and by words such as “scene,” “vivid,” and “view.” Thus, each stored memory must have multiple routes or keys. Furthermore, many memories have an emotional component or “color,” that is, in simplest terms, memories can be pleasant or unpleasant.

STRANGENESS & FAMILIARITY

Stimulation of some parts of the temporal lobes causes a change in interpretation of one’s surroundings. For example, when the stimulus is applied, the subject may feel strange in a familiar place or may feel that what is happening now has happened before. The occurrence of a sense of familiarity or a sense of strangeness in appropriate situations may help the healthy individual adjust to the environment. In strange surroundings, one is alert and on guard, whereas in familiar surroundings, vigilance is relaxed. An inappropriate feeling of familiarity with new events or in new surroundings is known as the **déjà vu phenomenon** from the French words meaning “already seen.” This occurs occasionally in healthy persons, and it may also occur as an aura (a sensation immediately preceding a seizure) in patients with temporal lobe epilepsy.

ALZHEIMER DISEASE & SENILE DEMENTIA

Alzheimer disease is the most common age-related neurodegenerative disorder. Memory decline initially manifests as a loss of episodic memory, which impedes recollection of recent events. Loss of short-term memory is followed by general loss of cognitive and other brain functions, agitation, depression, the need for constant care, and, eventually, death. **Clinical Box 15–3** describes the etiology and therapeutic strategies for the treatment of Alzheimer disease.

The cytopathologic hallmarks of Alzheimer disease are intracellular **neurofibrillary tangles,** made up in part of hyperphosphorylated forms of the **tau protein** that normally binds to microtubules, and extracellular **amyloid plaques** that have a core of **β-amyloid peptides** surrounded by altered nerve fibers and reactive glial cells **(Figure 15–4)**. The β-amyloid peptides are

products of **amyloid precursor protein (APP),** a transmembrane protein that projects into the extracellular fluid from all nerve cells. This protein is hydrolyzed at three different sites by α-secretase, β-secretase, and γ-secretase, respectively. When APP is hydrolyzed by α-secretase, nontoxic peptide products are produced. However, when it is hydrolyzed by β-secretase and γ-secretase, polypeptides with 40–42 amino acids are produced; the actual length varies because of variation in the site at which γ-secretase cuts the protein chain. These polypeptides are toxic, the most toxic being $A\beta\sigma^{1-42}$. The polypeptides form extracellular aggregates that can stick to AMPA receptors and Ca^{2+} ion channels, increasing Ca^{2+} influx. The polypeptides also initiate an inflammatory response with production of intracellular tangles. The damaged cells eventually die, leading to a third characterization of the brain pathology in individuals with this neurodegenerative disease—atrophy associated with narrowing of the gyri, widening of the sulci, enlargement of the ventricles, and reduction in brain weight.

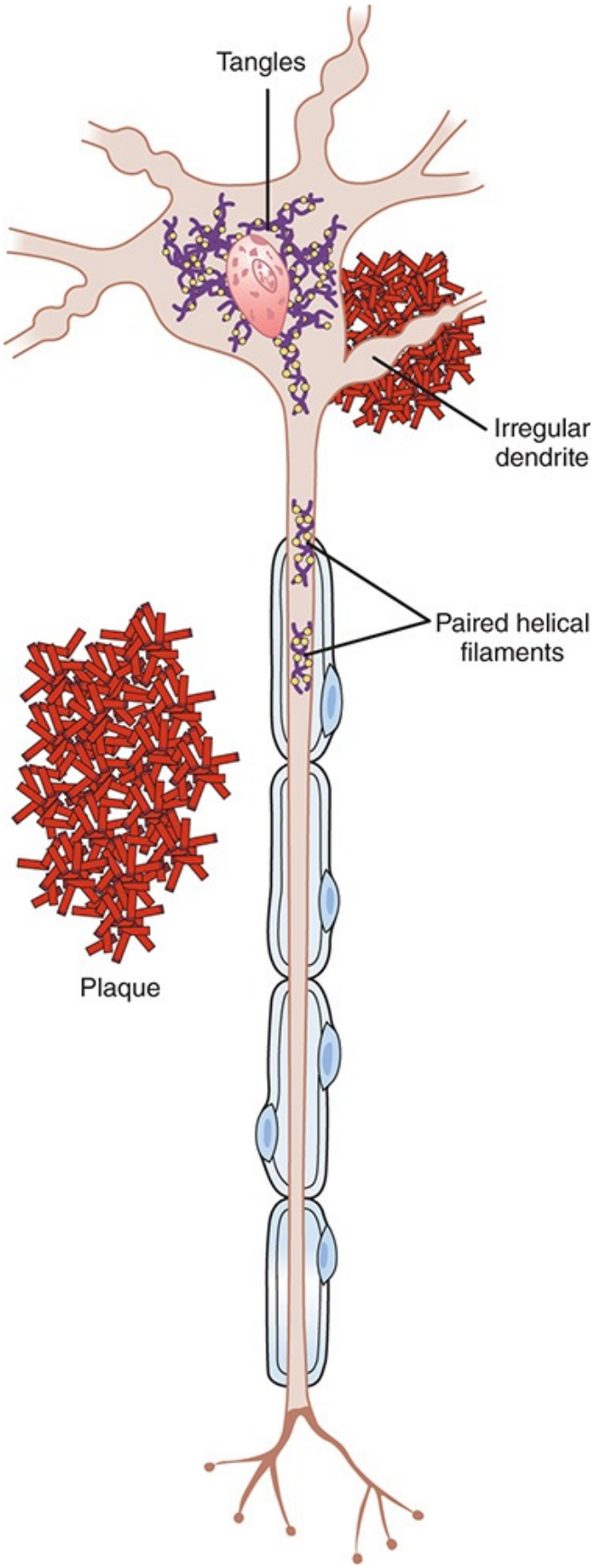

Tangles
Irregular dendrite
Paired helical filaments
Plaque

FIGURE 15–4 Abnormalities in a neuron are associated with Alzheimer disease. The cytopathologic hallmarks are intracellular neurofibrillary tangles and extracellular amyloid plaques that have a core of β-amyloid peptides surrounded by altered nerve fibers and reactive glial cells. A third characterization is brain atrophy. (Used with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

CLINICAL BOX 15–3

Alzheimer Disease

Alzheimer disease was originally characterized in middle-aged people, and similar deterioration in elderly individuals is technically **senile dementia** of the Alzheimer type, though it is frequently just called Alzheimer disease. Both genetic and environmental factors can contribute to the etiology of the disease **(Table 15–1)**. Most cases are sporadic, but a familial form of the disease (accounting for about 5% of the cases) is seen in an early-onset form of the disease. In these cases, the disease is caused by mutations in genes for the amyloid precursor protein on chromosome 21, presenilin 1 on chromosome 14, or presenilin 2 on chromosome 1. It is transmitted in an autosomal dominant mode, so offspring in the same generation have a 50/50 chance of developing familial Alzheimer disease if one of their parents is affected. Each mutation leads to an overproduction of the β-amyloid protein found in neuritic plaques. Senile dementia can be caused by vascular disease and other disorders, but Alzheimer disease is the most common cause, accounting for 50–60% of the cases. The most common risk factor for developing Alzheimer disease is age. This neurodegenerative disease is present in 8–17% of the population over the age of 65, with the incidence nearly doubling every 5 years after reaching the age of 60. In those who are 95 years of age and older, the incidence is 40–50%. It is estimated that by the year 2050, up to 16 million people age 65 and older in the United States will have Alzheimer disease. Although the prevalence of the disease appears to be higher in women, this may be due to their longer life span as the incidence rates are similar for men and women. Alzheimer disease plus the other forms of senile dementia are a major medical problem.

THERAPEUTIC HIGHLIGHTS

Research is aimed at identifying strategies to prevent the occurrence, delay the onset, slow the progression, or alleviate the symptoms of Alzheimer disease. The use of **acetylcholinesterase inhibitors** (eg, **donepezil, galantamine, rivastigmine,** or **tacrine**) in early stages of the disease increases the availability of acetylcholine in the synaptic cleft. This class of drugs has shown some promise in ameliorating global cognitive dysfunction, but not learning and memory impairments in these patients. These drugs also delay the worsening of symptoms for up to 12 months in about 50% of the cases studied. **Memantine** (an NMDA receptor antagonist) prevents glutamate-induced excitotoxicity in the brain and is used to treat moderate to severe Alzheimer disease. It delays but does not prevent worsening of symptoms (eg, loss of memory and confusion) in some patients. Drugs used to block the production of β-amyloid proteins are under development. Also attempts are underway to develop vaccines that would allow the body's immune system to produce antibodies to attack these proteins.

TABLE 15–1 Risk factors associated with the development and progression of Alzheimer disease.

Common	Genes	Environment
Age	ApoE4	Low education level
Family members with dementia	APP (chromosome 21) PS-1 (chromosome 14) PS-2 (chromosome 1) Down syndrome (trisomy 21)	Head trauma Prions Toxins Viruses

APP, amyloid precursor protein; ApoE4, apolipoprotein E4 allele; PS-1, presenilin-1; PS-2, presenilin-2.

An interesting finding that may well have broad physiologic implications is the observation that frequent effortful mental activities, such as doing difficult crossword puzzles and playing board games, slow the onset of cognitive dementia due to Alzheimer disease and vascular disease. The explanation for this "use it or lose it" phenomenon is unknown, but it suggests that the hippocampus and its connections have plasticity like other parts of the brain and skeletal and cardiac muscles.

LANGUAGE & SPEECH

Memory and learning are functions of large parts of the brain, but the centers controlling some of the other "higher functions of the nervous system," particularly the mechanisms related to language, are more or less localized to the neocortex. Speech and other intellectual functions are especially well developed in humans, the animal species in which the neocortical mantle is most highly developed.

COMPLEMENTARY SPECIALIZATION OF THE HEMISPHERES VERSUS "CEREBRAL DOMINANCE"

The term language includes the understanding of the spoken and printed word and expressing ideas in speech and writing. Human language functions depend more on one cerebral hemisphere than on the other. This hemisphere is concerned with categorization and symbolization and is called the **dominant hemisphere.** The **nondominant hemisphere** is specialized in the area of spatiotemporal relations; for example, it is involved in the identification of objects by their form and the recognition of musical themes and in facial recognition. Consequently, the concept of "cerebral dominance" and a dominant and nondominant hemisphere has been replaced by a concept of complementary specialization of the hemispheres, one for sequential-analytic processes (the **categorical hemisphere**) and one for visuospatial relations (the **representational hemisphere**). The categorical hemisphere is concerned with language functions. **Clinical Box 15–4** describes deficits that occur in subjects with representational or categorical hemisphere lesions.

In 96% of right-handed individuals (91% of the human population), the left hemisphere is the dominant or categorical hemisphere; and in the remaining 4%, the right hemisphere is dominant. In 70% of left-handed individuals, the left hemisphere is the dominant hemisphere; in 15% of left-handed persons, the right hemisphere is the categorical hemisphere and in remaining 15%, there is no clear lateralization. Learning disabilities such as **dyslexia** **(see Clinical Box 15–5),** an impaired ability to learn to read, are 12 times as common in left-handers compared to right-handers. The spatial talents of left-handers may be above average; a disproportionately large number of artists, musicians, and mathematicians are left-handed.

CLINICAL BOX 15–4

Lesions of Representational & Categorical Hemispheres

Lesions in the categorical hemisphere produce language disorders, whereas extensive lesions in the representational hemisphere do not. Instead, lesions in the representational hemisphere produce **astereognosis**—the inability to identify objects by feeling them—and other agnosias. **Agnosia** is the general term used for the inability to recognize objects by a particular sensory modality even though the sensory modality itself is intact. Lesions producing these defects are generally in the parietal lobe. Especially when they are in the representational hemisphere, lesions of the inferior parietal lobule, a region in the posterior part of the parietal lobe that is close to the occipital lobe, cause **unilateral inattention** and **neglect.** Individuals with such lesions do not have any apparent primary visual, auditory, or somatesthetic defects, but they ignore stimuli from one side of their bodies or the surrounding space. This leads to failure to care for half their bodies and, in extreme cases, to situations in which individuals shave half their faces, dress half their bodies, or read half of each page. This inability to put together a picture of visual space on one side is due to a shift in visual attention to the side of the brain lesion and can be improved, if not totally corrected, by wearing eyeglasses that contain prisms. Hemispheric specialization extends to other parts of the cortex as well. Patients with lesions in the categorical hemisphere are disturbed about their disability and often depressed, whereas patients with lesions in the representational hemisphere are sometimes unconcerned and even euphoric. Lesions of different parts of the categorical hemisphere produce **fluent, nonfluent,** and **anomic aphasias.** Although aphasias are produced by lesions of the categorical hemisphere, lesions in the representational hemisphere also have effects. For example, they may impair the ability to tell a story or make a joke. They may also impair a subject's ability to get the point of a joke and, more broadly, to comprehend the meaning of differences in inflection and the "color" of speech. This is an example of the way the hemispheres are specialized rather than simply being dominant and nondominant.

THERAPEUTIC HIGHLIGHTS

Treatments for agnosia and aphasia are symptomatic and supportive. Individuals with agnosia can be taught exercises to help them identify objects

that are a necessity for independence. Therapy for individuals with aphasia helps them use remaining language abilities, compensate for language problems, and learn other methods of communicating. Some individuals with aphasia experience recovery but often some disabilities remain. Factors that influence the degree of improvement include the cause and extent of the brain damage, the area of the brain that was damaged, and the age and health of the individual. Computer-assisted therapies can improve retrieval of certain parts of speech as well as allowing an alternative way to communicate.

CLINICAL BOX 15–5

Dyslexia

Dyslexia is characterized by difficulties with learning how to decode at the word level, to spell, and to read accurately and fluently despite having a normal or even higher than normal level of intelligence. It is often due to an inherited abnormality that affects 5% of the population with a similar incidence in boys and girls. Dyslexia is the most common and prevalent of all known learning disabilities. It often coexists with attention deficit disorder. Many individuals with dyslexic symptoms also have problems with short-term memory skills and problems processing spoken language. Acquired dyslexias can occur with brain damage in the left hemisphere's key language areas. Also, in many cases, there is a decreased blood flow in the angular gyrus in the categorical hemisphere. There are several theories to explain the cause of dyslexia. The **phonologic hypothesis** is that dyslexics have a specific impairment in the representation, storage, and/or retrieval of speech sounds. The **rapid auditory processing theory** proposes that the primary deficit is the perception of short or rapidly varying sounds. The **visual theory** is that a defect in the magnocellular portion of the visual system slows processing and also leads to phonemic deficit. More selective speech defects have also been described. For example, lesions limited to the left temporal pole cause inability to retrieve names of places and persons but preserves the ability to retrieve common nouns, verbs, and adjectives.

THERAPEUTIC HIGHLIGHTS

Treatments for children with dyslexia frequently rely on modified teaching

strategies that include the involvement of various senses (hearing, vision, and touch) to improve reading skills. The sooner the diagnosis is made and interventions are applied, the better the prognosis.

PHYSIOLOGY OF LANGUAGE

Language is one of the fundamental bases of human intelligence and a key part of human culture. The primary brain areas concerned with language are arrayed along and near the sylvian fissure (lateral cerebral sulcus) of the categorical hemisphere. A region at the posterior end of the superior temporal gyrus called the **Wernicke area** (Figure 15–5) is concerned with comprehension of auditory and visual information. It projects via the **arcuate fasciculus** to the **Broca area** in the frontal lobe immediately in front of the inferior end of the motor cortex. Broca area processes the information received from Wernicke area into a detailed and coordinated pattern for vocalization and then projects the pattern via a speech articulation area in the insula to the motor cortex. This then initiates the appropriate movements of the lips, tongue, and larynx to produce speech. The probable sequence of events that occurs when a subject names a visual object is shown in Figure 15–6. The angular gyrus behind the Wernicke area appears to process information from words that are read in such a way that they can be converted into the auditory forms of the words in Wernicke area.

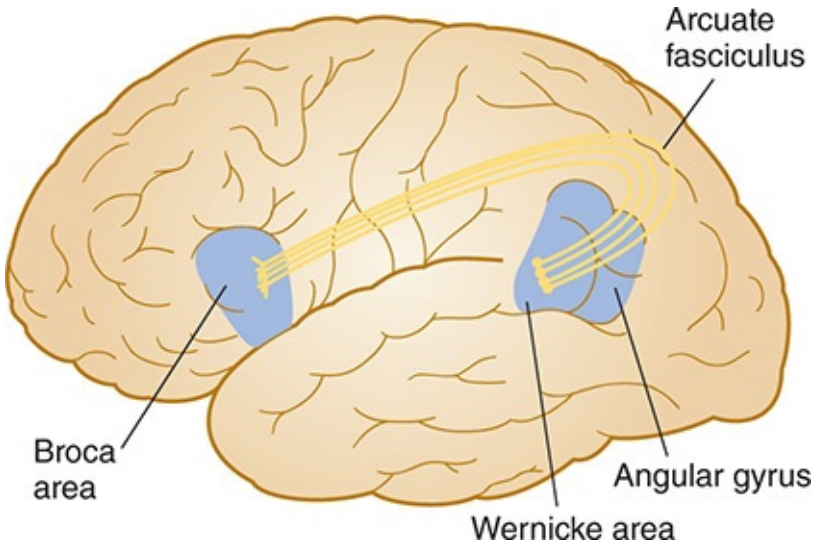


FIGURE 15–5 Location of some of the areas in the categorical hemisphere that are concerned with language functions. Wernicke area is in the posterior

end of the superior temporal gyrus and is concerned with comprehension of auditory and visual information. It projects via the arcuate fasciculus to Broca area in the frontal lobe. Broca area processes information received from Wernicke area into a detailed and coordinated pattern for vocalization and then projects the pattern via a speech articulation area in the insula to the motor cortex, which initiates the appropriate movements of the lips, tongue, and larynx to produce speech.

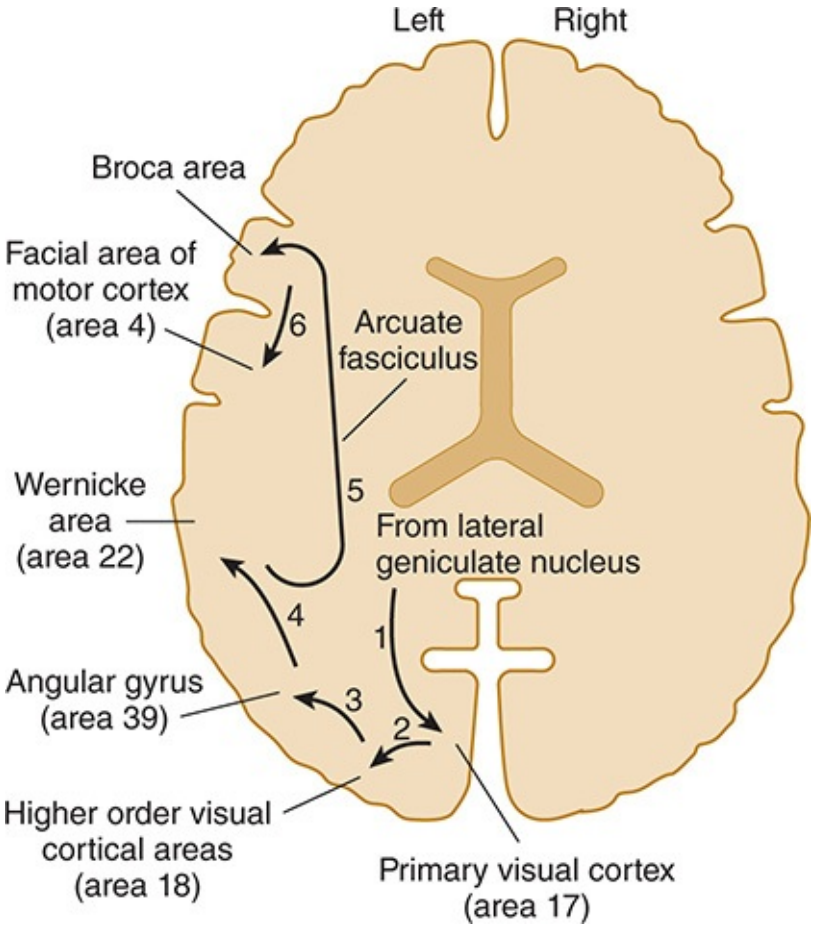


FIGURE 15–6 Path taken by impulses when a subject identifies a visual object, projected on a horizontal section of the human brain. Information travels from the lateral geniculate nucleus in the thalamus to the primary visual cortex, to higher order visual critical areas, and to the angular gyrus. Information then travels from Wernicke area to Broca area via the arcuate fasciculus. Broca area processes the information into a detailed and coordinated pattern for vocalization and then projects the pattern via a speech articulation area in the insula to the motor cortex, which initiates the appropriate movements of the lips, tongue, and larynx to produce speech.

In individuals who learn a second language in adulthood, fMRI reveals that the portion of Broca area concerned with it is adjacent to but separate from the area concerned with the native language. However, in children who learn two languages early in life, only a single area is involved with both. Children acquire fluency in a second language more easily than adults.

LANGUAGE DISORDERS

Aphasias are abnormalities of language functions that are not due to defects of vision or hearing or to motor paralysis. They are caused by lesions in the categorical hemisphere (see Clinical Box 15–4). The most common cause is embolism or thrombosis of a cerebral blood vessel. Aphasias can be classified as **nonfluent, fluent,** or **anomic aphasias.** A lesion of Broca area causes a nonfluent aphasia denoted as an expressive or motor aphasia. Affected individuals have slow speech and difficulty in generating verbal or written words. Patients with severe damage to this area are limited to two or three words to express the whole range of meaning and emotion. Sometimes the words retained are those that were being spoken at the time of the injury or vascular accident that caused the aphasia.

A lesion in Wernicke area produces a type of fluent aphasia in which speech itself is normal but it is full of jargon and neologisms that make little sense. The patient also fails to comprehend the meaning of spoken or written words, so other aspects of the use of language are compromised. Another form of fluent aphasia is a condition in which patients can speak relatively well and have good auditory comprehension but cannot put parts of words together or conjure up words.

When a lesion damages the angular gyrus in the categorical hemisphere without affecting Wernicke or Broca areas, there is no difficulty with speech or the understanding of auditory information; instead there is trouble understanding written language or pictures, because visual information is not processed and transmitted to Wernicke area. The result is a condition called **anomic aphasia.**

The isolated lesions that cause the selective defects described above occur in some patients, but brain destruction is often more general. Consequently, more than one form of aphasia is often present. Frequently, the aphasia is general **(global),** involving both receptive and expressive functions. In this situation, speech is scant as well as nonfluent. Writing is abnormal in all aphasias in which speech is abnormal, but the neural circuits involved are unknown. In addition, deaf persons in whom a lesion develops in the categorical hemisphere lose their

ability to communicate using sign language.

Stuttering is associated with right cerebral dominance and widespread elevated activity in the cerebral cortex and cerebellum, including increased activity in the supplementary motor area. Stimulation of this area can produce **laughter,** with the duration and intensity of the laughter proportional to the intensity of the stimulus.

RECOGNITION OF FACES

An important part of the visual input goes to the inferior temporal lobe, where representations of objects, particularly faces, are stored **(Figure 15–7)**. Faces are particularly important in distinguishing friends from foes and the emotional state of those seen. Storage and recognition of faces is more strongly represented in the right inferior temporal lobe in right-handed individuals, though the left lobe is also active. Damage to this area can cause **prosopagnosia,** the inability to recognize faces. Patients with this abnormality can recognize forms and reproduce them. They can recognize people by their voices, and many of them show autonomic responses when they see familiar as opposed to unfamiliar faces. However, they cannot identify the familiar faces they see. The presence of an autonomic response to a familiar face in the absence of recognition implicates the existence of a separate dorsal pathway for processing information about faces that leads to recognition at only a subconscious level.

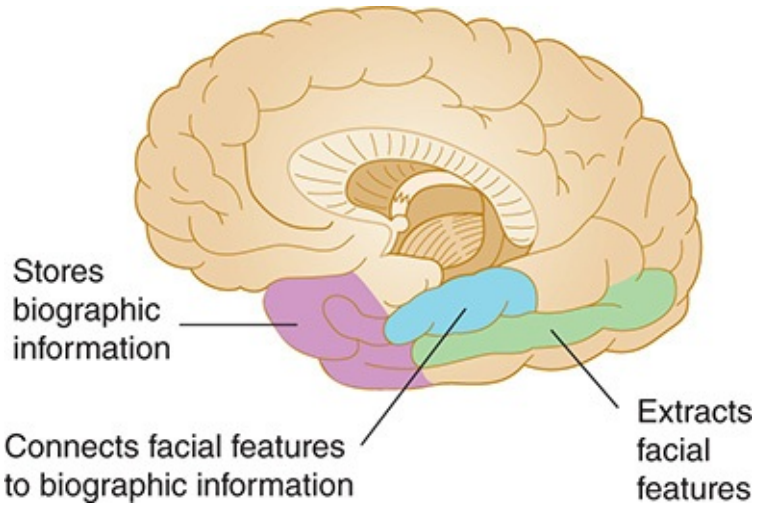


FIGURE 15–7 Areas in the right cerebral hemisphere, in right-handed individuals, that are concerned with recognition of faces. An important part of the visual input goes to the inferior temporal lobe, where representations of

objects, particularly faces, are stored. In humans, storage and recognition of faces is more strongly represented in the right inferior temporal lobe in right-handed individuals, though the left lobe is also active. (Modified with permission from Szpir M: Accustomed to your face. Am Sci 1992; 80:539.)

LOCALIZATION OF OTHER FUNCTIONS

Use of fMRI and PET scanning combined with study of patients with strokes and head injuries has provided insight into the ways serial processing of sensory information produce cognition, reasoning, comprehension, and language. Analysis of the brain regions involved in arithmetic calculations has highlighted two areas. In the inferior portion of the left frontal lobe is an area concerned with number facts and calculations. Frontal lobe lesions can cause **acalculia,** a selective impairment of mathematical ability. There are areas around the intraparietal sulci of both parietal lobes that are concerned with visuospatial representations of numbers.

Two right-sided subcortical structures play a role in accurate navigation. One is the right hippocampus that is concerned with learning the location of places, and the other is the right caudate nucleus that facilitates movement to the places. Men have larger brains than women and are said to have superior spatial skills and ability to navigate.

Other defects seen in patients with localized cortical lesions include, for example, the inability to name animals, though the ability to name other living things is intact. One patient with a left parietal lesion had difficulty with the second half but not the first half of words. Some patients with parietooccipital lesions write only with consonants, omitting vowels. The pattern that emerges from such observations is one of precise sequential processing of information in localized brain areas.

CHAPTER SUMMARY

- CT scans provide a high-resolution 3-dimensional image of the brain or other organ. Both PET imaging and fMRI provide an index of the level of the activity in various parts of the brain in health or disease.
- TBI results from an excessive mechanical force or penetrating injury to the head (eg, falls, motor vehicle accidents, and assaults). It can lead to impaired cognitive, physical, emotional, and behavioral functions and can

be associated with as altered state of consciousness. A Glasgow Coma Scale is used to define the severity of TBI and imaging can identify the extent of the brain damage.

- Memory is divided into explicit (declarative) and implicit (nondeclarative). Explicit memory is further subdivided into semantic and episodic. Implicit memory is further subdivided into priming, procedural, associative learning, and nonassociative learning.
- Declarative memory involves the hippocampus and the medial temporal lobe for retention. Priming is dependent on the neocortex. Procedural memory is processed in the striatum. Associative learning is dependent on the amygdala for its emotional responses and the cerebellum for the motor responses. Nonassociative learning is dependent on various reflex pathways.
- Synaptic plasticity is the ability of neural tissue to change as reflected by LTP (an increased effectiveness of synaptic activity) or LTD (a reduced effectiveness of synaptic activity) after continued use. Habituation is a simple form of learning in which a neutral stimulus is repeated many times. Sensitization is the prolonged occurrence of augmented postsynaptic responses after a stimulus to which one has become habituated is paired once or several times with a noxious stimulus.
- Alzheimer disease is characterized by progressive loss of short-term memory followed by general loss of cognitive function. The cytopathologic hallmarks of Alzheimer disease are intracellular neurofibrillary tangles and extracellular senile plaques.
- Categorical and representational hemispheres are for sequential-analytic processes and visuospatial relations, respectively. Lesions in the categorical hemisphere produce language disorders, whereas lesions in the representational hemisphere produce astereognosis.
- The main cortical regions involved in language are Wernicke area in the upper temporal lobe that projects via the arcuate fasciculus to Broca area in the frontal lobe. Wernicke area is important for comprehension of auditory and visual information; Broca area processes the information from the Wernicke area produces a coordinated pattern for vocalization.
- Aphasias are abnormalities of language functions and are caused by lesions in the categorical hemisphere. They are classified as fluent (Wernicke area), nonfluent (Broca area), and anomic (angular gyrus) based on the location of brain lesions.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A 23-year-old medical student suffered a traumatic brain injury as a result of a motorcycle accident. He was unconscious and was rushed to the emergency department of the local hospital. A CT scan was performed and appropriate interventions were taken. What kind of information is obtained from a CT scan?

   A. Blood flow in both superficial and deep parts of the brain

   B. Changes in regional brain activity

   C. Functional rather than detailed anatomic information

   D. A high-resolution 3-dimensional image of the brain

   E. Local glucose metabolism, blood flow, and oxygen

2. A 15-year-old boy who was not wearing a helmet was in a bicycle accident. He was rushed to the emergency department of the local hospital where it was diagnosed with mild traumatic brain injury based on the Glasgow Coma Scale. Which of the following symptoms are consistent with this level of traumatic brain injury?

   A. Nausea and vomiting, slurred speech, and blurred vision

   B. Fatigue, sleep disturbances, and mood changes

   C. Limb weakness, confusion, and agitation

   D. Mood changes, restlessness, and nausea

   E. Slurred speech, limb weakness, and fatigue

3. A 52-year-old man had a closed-head injury that caused damage to the prefrontal cortex. As a result, this individual might be expected to experience the following memory disturbance.

   A. Disappearance of remote memories.

   B. Loss of working memory.

   C. Loss of the ability to encode events of the recent past into long-term memory.

   D. Loss of the ability to recall faces and forms.

   E. Production of inappropriate emotional responses when recalling events of the recent past.

4. A 70-year-old woman fell down a flight of stairs, hitting her head on the concrete sidewalk. The trauma caused a severe intracranial hemorrhage

resulting in brain damage. Following this injury, she seemed to ignore her left side; for example, she would only wash the right side of her body or only put a shoe on her right foot. What area of the brain was most likely damaged as a result of this fall down the stairs?

A. Inferior parietal lobe of the representational hemisphere

B. Nucleus basalis of Meynert and related areas of the forebrain

C. Mammillary bodies

D. Angular gyrus in the categorical hemisphere

E. Parietal lobe of the categorical hemisphere

5. An MD/PhD candidate was studying the role of the neocortex in language development. What information is known about the interconnection of cortical areas important for language?

A. Broca area projects to Wernicke area via the arcuate fasciculus.

B. Temporal lobe projects to the frontal lobe via the corpus callosum

C. Broca area projects to Wernicke area via the planum temporale.

D. Temporal lobe projects to Wernicke area via the corpus callosum.

E. Wernicke area projects to Broca area via the arcuate fasciculus.

6. A 67-year-old woman suffered a stroke that damaged the posterior end of the superior temporal gyrus. A lesion of Wernicke area in the categorical hemisphere causes her to

A. lose her short-term memory.

B. experience nonfluent aphasia in which she speaks in a slow, halting voice.

C. experience déjà vu.

D. talk rapidly but make little sense, which is characteristic of fluent aphasia.

E. lose the ability to recognize faces, which is called prosopagnosia.

7. Which of the following is most likely involved in production of LTP?

A. NO release, activation of NMDA receptors, and membrane hyperpolarization

B. Decreased Ca^{2+} in presynaptic or postsynaptic neurons, activation of NMDA receptors, and membrane depolarization

C. Activation of NMDA receptors, NO-induced reduction in glutamate release in a presynaptic neuron, and membrane depolarization

D. Increased Ca^{2+} in presynaptic or postsynaptic neurons, activation of NMDA receptors, and membrane depolarization

E. NO-induced increase in glutamate release in a presynaptic neuron, activation of non-NMDA receptors, membrane hyperpolarization

8. A 79-year-old woman has been experiencing difficulty finding her way back home after her morning walks. Her husband has also noted that she takes much longer to do routine chores around the home and often appears to be confused. He is hoping that this is just due to “old age” but fears it may be a sign of Alzheimer disease. Which of the following is the definitive sign of this disease?

A. Loss of short-term memory

B. The presence of intracellular neurofibrillary tangles and extracellular neuritic plaques with a core of β-amyloid peptides

C. A mutation in genes for amyloid precursor protein (APP) on chromosome 21

D. Rapid reversal of symptoms with the use of acetylcholinesterase inhibitors

E. A loss of cholinergic neurons in the nucleus basalis of Meynert

SECTION III

Endocrine & Reproductive Physiology

The key role of the endocrine system is to maintain whole body homeostasis. This is accomplished via the coordination of hormonal signaling pathways that regulate cellular activity in target organs throughout the body. Endocrine mechanisms are also concerned with the ability of humans to reproduce, and the sexual maturation required for this function. Classic **endocrine glands** are scattered throughout the body and secrete **hormones** into the circulatory system, usually via ductless secretion into the interstitial fluid. **Target organs** express receptors that bind the specific hormone to initiate a cellular response. The endocrine system can be contrasted with the neural regulation of physiologic function that was the focus of the previous section. Endocrine effectors typically provide “broadcast” regulation of multiple tissues and organs simultaneously, with specificity provided for by the expression of relevant receptors. A change in environmental conditions, for example, often calls for an integrated response across many organ systems. Neural regulation, on the other hand, is often exquisitely spatially delimited, such as the ability to contract just a single muscle. Nevertheless, both systems must work collaboratively to allow for minute-to-minute as well as longer term stability of the body’s interior milieu.

Hormones are the soluble messengers of the endocrine system and are classified into steroids, peptides, and amines (see Chapters 1 and 2). Steroid hormones can cross the lipid-containing plasma membrane of cells and usually bind to intracellular receptors. Peptide and amine hormones bind to cell surface receptors. Steroid hormones are produced by the adrenal cortex (Chapter 19), the ovaries (Chapter 22) and the gonads, testes (Chapter 23), in addition to steroid hormones that are made by the placenta during pregnancy (Chapter 22). Amine hormones are derivatives of the amino acid tyrosine and are made by the thyroid

(Chapter 20) and the adrenal medulla (Chapter 19). Interestingly, the tyrosine-derived thyroid hormone behaves more like a steroid than a peptide hormone by binding to an intracellular receptor. The majority of hormones, however, are peptides and they are usually synthesized as preprohormones before being cleaved first to prohormones in the endoplasmic reticulum and then to the active hormone in secretory vesicles.

Diseases of the endocrine system are numerous. Indeed, endocrine and metabolic disorders are among the most common afflictions in developed countries, particularly when nutrition and access to health care provisions are generous and high-risk individuals are identified by regular screening. At least 11 endocrine and metabolic disorders are present in 5% or more of the adult US population, including diabetes mellitus, osteopenia, dyslipidemia, metabolic syndrome, and thyroiditis. For example, type 2 diabetes mellitus is one of the most prevalent endocrine disorders of the 21st century and involves an inability of the body to respond to insulin. The resulting high blood glucose damages many tissues leading to secondary complications (see Chapter 24). In large part, the high and increasing prevalence of diabetes and other metabolic disorders rests on the substantial prevalence of obesity in developed countries, with as many as one-third of the US adult population now considered to be obese, and two-thirds overweight. Indeed, based on a 2009 report, obesity also affects 28% of US children aged 12–17, and while the current prevalence of type 2 diabetes in children is quite low, this prevalence is accordingly expected to rise. Further, a number of endocrine disorders are more prevalent in specific ethnic groups, or in a particular gender. Overall, the burden of endocrine and metabolic disorders, with their protean manifestations and complications, is a serious public health crisis and even highlights an apparent national shortage of trained endocrinologists. Many endocrine disorders must be managed by primary care clinicians as a result.

CHAPTER 16

Basic Concepts of Endocrine Regulation

OBJECTIVES

After studying this chapter, you should be able to:

- Describe hormones and their contribution to whole body homeostatic mechanisms.
- Understand the chemical nature of different classes of hormones and how this determines their mechanism of action on target cells. Contrast membrane bound and intracellular hormone receptors and understand the principles of desensitization, down-regulation, and inactivation.
- Define how hormones are synthesized and secreted by cells of endocrine glands, including how peptide hormones are cleaved from longer precursors.
- Understand the effects of plasma hormone binding proteins (thyroid and steroid hormones) on the availability of hormones to their sites of action, and on the mechanisms that regulate hormone secretion.
- Explain the principles of feedback control (negative and positive) of hormone secretion.
- Understand the principles governing disease states that result from over- or under-production of key hormones.

INTRODUCTION

This section of the text deals with the various endocrine glands that control the function of multiple organ systems of the body. In general, endocrine physiology is concerned with the maintenance of various aspects of **homeostasis.** The mediators of such control mechanisms are soluble factors known as **hormones.** The word hormone was derived from the Greek *horman*, meaning to set in motion. In preparation for specific discussions of the various endocrine systems and their hormones, this chapter will address some concepts of endocrine regulation that are common among all systems.

Another feature of endocrine physiology to keep in mind is that, unlike other physiologic systems that are considered in this text, the endocrine system cannot be cleanly defined along anatomic lines. Rather, the endocrine system is a distributed system of glands and circulating messengers that is often stimulated by the central nervous system or the autonomic nervous system, or both.

HORMONAL SPECIFICITY OF ACTION ON TARGET CELLS

As noted in the introduction to this section, hormones comprise steroids, amines, and peptides. Peptide hormones are by far the most numerous. Many hormones can be grouped into families reflecting their structural similarities as well as the similarities of the receptors they activate.

Steroids and thyroid hormones are distinguished by their predominantly intracellular sites of action, since they can diffuse freely through the cell membrane. They bind to a family of largely cytoplasmic proteins known as nuclear receptors. Upon ligand binding, the receptor–ligand complex translocates to the nucleus where it either homodimerizes, or associates with a distinct liganded nuclear receptor to form a heterodimer. In either case, the dimer binds to DNA to either increase or decrease gene transcription in the target tissue. Individual members of the nuclear receptor family have a considerable degree of homology, and share many functional domains, such as the zinc fingers that permit DNA binding. However, sequence variations allow for ligand specificity as well as binding to specific DNA motifs. In this way, the transcription of distinct genes is regulated by individual hormones.

HORMONE SECRETION

SYNTHESIS & PROCESSING

The regulation of hormone synthesis, of course, depends on their chemical nature. For peptide hormones as well as hormone receptors, synthesis is controlled predominantly at the level of transcription. For amine and steroid hormones, synthesis is controlled indirectly by regulating the production of key synthetic enzymes as well as by substrate availability.

Interestingly, the majority of peptide hormones are synthesized initially as much larger polypeptide chains, and then processed intracellularly by specific proteases to yield the final hormone molecule. In some cases, multiple hormones may be derived from the same initial precursor, depending on the specific processing steps present in a given cell type. Presumably this provides for a level of genetic "economy." It is also notable that the hormone precursors themselves are typically inactive. This may be a mechanism that provides for an additional measure of regulatory control, or, in the case of thyroid hormones, may dictate the site of highest hormone availability.

The synthesis of all of the proteins/peptides discussed above is subject to the normal mechanisms of transcriptional control in the cell (see Chapter 2). In addition, there is provision for exquisitely specific regulation by other hormones, since the regulatory regions of many peptide hormone genes contain binding motifs for the nuclear receptors discussed above. For example, thyroid hormone directly suppresses TSH expression via the thyroid hormone receptor. These specific mechanisms to regulate hormone transcription are essential to the function of feedback loops, as will be addressed in greater detail below. In some cases, the abundance of selected hormones may also be regulated via effects on translation. For example, elevated levels of circulating glucose stimulate the translation of insulin mRNA. These effects are mediated by the ability of glucose to increase the interaction of the insulin mRNA with specific RNA-binding proteins, which increase its stability and enhance its translation. The net effect is a more precise and timely regulation of insulin levels, and thus energy metabolism, than could be accomplished with transcriptional regulation alone.

The precursors for peptide hormones are processed through the cellular machinery that handles proteins destined for export, including trafficking through specific vesicles where the propeptide form can be cleaved to the final active hormones. Mature hormones are also subjected to a variety of posttranslational processing steps, such as glycosylation, which can influence their ultimate biologic activity and/or stability in the circulation. Ultimately, all hormones enter either the constitutive or regulated secretory pathway (see

Chapter 2).

SECRETION

The secretion of many hormones is via a process of exocytosis of stored granules, as discussed in Chapter 2. The exocytotic machinery is activated when the cell type that synthesizes and stores the hormone in question is activated by a specific signal, such as a neurotransmitter or peptide releasing factor. One should, however, contrast the secretion of stored hormones with that of those that are continually released by diffusion (eg, steroids). Control of the secretion of the latter molecules occurs via kinetic influences on the synthetic enzymes or carrier proteins involved in hormone production. For example, the steroidogenic acute regulatory protein (StAR) is a labile protein whose expression, activation, and deactivation are regulated by intracellular signaling cascades and their effectors, including a variety of protein kinases and phosphatases. StAR traffics cholesterol from the outer to the inner membrane leaflet of the mitochondrion. Because this is a rate-limiting first step in the synthesis of the steroid precursor, pregnenolone, this arrangement permits changes in the rate of steroid synthesis, and thus secretion, in response to homeostatic cues such as trophic hormones, cytokines, and stress **(Figure 16–1)**.

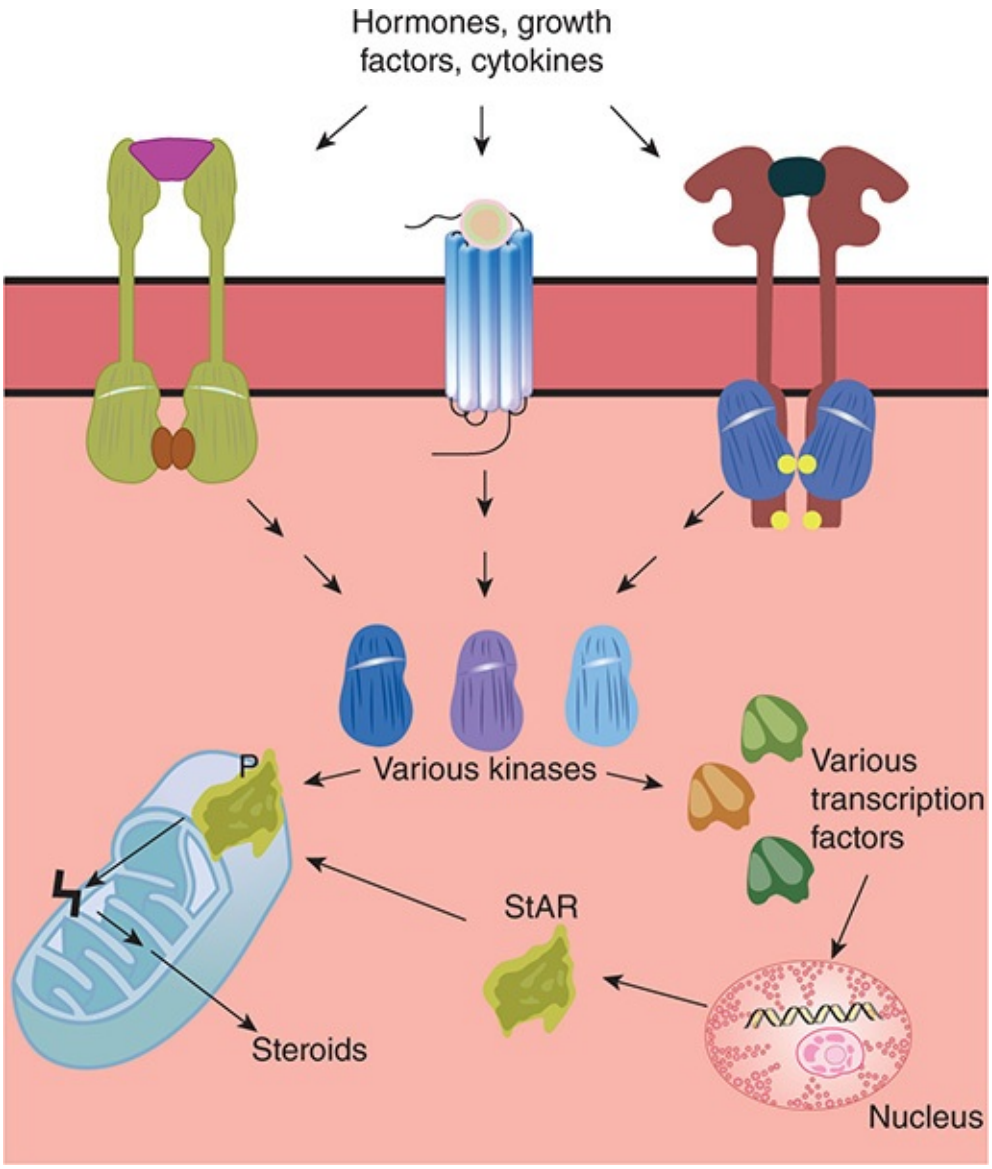


FIGURE 16–1 Regulation of steroid biosynthesis by the steroidogenic acute regulatory protein (StAR). Extracellular signals activate intracellular kinases that, in turn, phosphorylate transcription factors that upregulate StAR expression. StAR is activated by phosphorylation, and facilitates transfer of cholesterol from the outer to inner mitochondrial membrane leaflet. This then allows conversion of cholesterol into pregnenolone, which is the first intermediate in the steroid biosynthetic pathway.

An additional complexity related to hormone secretion relates to the fact that some hormones are secreted in a pulsatile manner. Secretion rates may peak and ebb relative to circadian rhythms, in response to the timing of meals, or as regulated by other pattern generators whose periodicity may range from

milliseconds to years. Pulsatile secretion is often related to the activity of oscillators in the hypothalamus that regulate the membrane potential of neurons, in turn secreting bursts of hormone releasing factors into the hypophysial blood flow that then cause the release of pituitary and other downstream hormones in a similar pulsatile manner (see Chapters 17 and 18). There is evidence that these hormone pulses convey different information to the target tissues that they act upon compared to a steady exposure to a single concentration of the hormone. Therapeutically, pulsatile secretion may pose challenges if, due to deficiency, it proves necessary to replace a particular hormone that is normally secreted in this way.

HORMONE TRANSPORT IN THE BLOOD

In addition to the rate of secretion and its nature (steady vs. pulsatile), a number of factors influence the circulating levels of hormones. These include the rates of hormone degradation and/or uptake, receptor binding and availability of receptors, and the affinity of a given hormone for plasma carriers **(Figure 16–2)**. Stability influences the circulating half-life of a given hormone and has therapeutic implications for hormone replacement therapy, in addition to those posed by pulsatile secretion as discussed above.

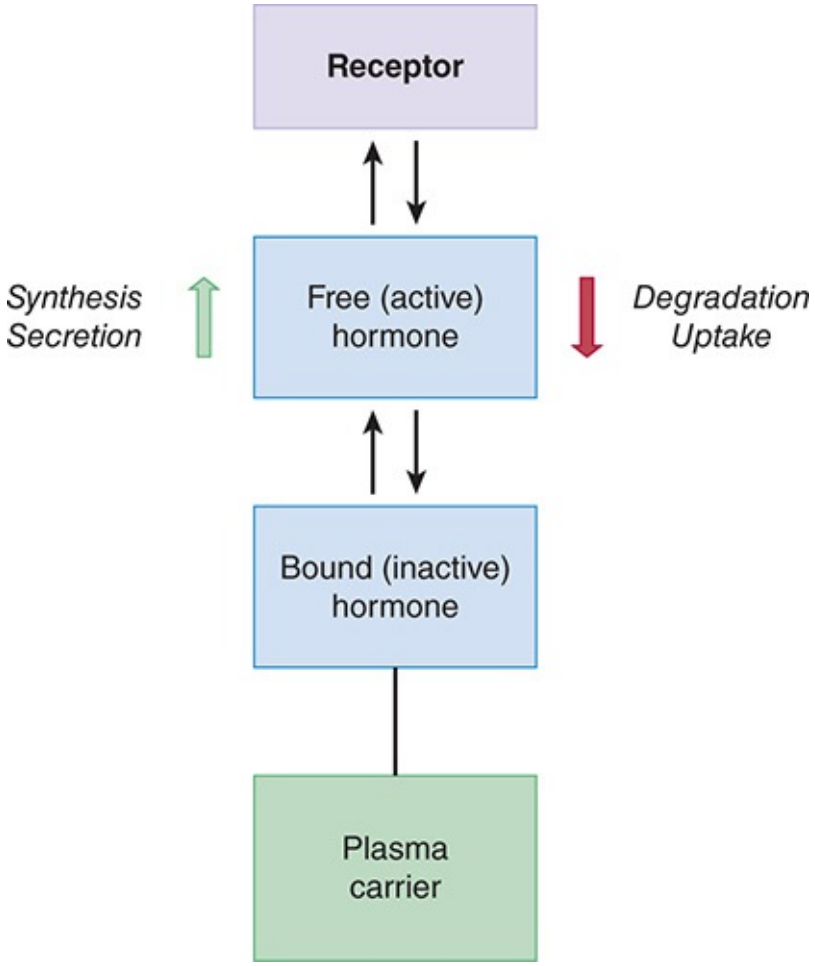


FIGURE 16–2 Summary of factors that determine the level of free hormones circulating in the bloodstream. Factors that increase (green upward arrow) or decrease (red downward arrow) hormone levels are shown. Free hormones also equilibrate with the forms bound to either receptors or plasma carrier proteins.

Plasma carriers for specific hormones have a number of important physiologic functions. First, they serve as a reservoir of inactive hormone and thus provide a hormonal reserve. Bound hormones are typically prevented from degradation or uptake. Thus, the bound hormone reservoir can allow fluctuations in hormonal levels to be smoothed over time. Plasma carriers also restrict the access of the hormone to some sites. Ultimately, plasma carriers may be vital in modulating levels of the free hormone in question. Typically, it is only the free hormone that is biologically active in target tissues or can mediate feedback regulation (see below) since it is the only form able to access the extravascular compartment.

Catecholamine and most peptide hormones are soluble in plasma and are transported as such. In contrast, steroid hormones are hydrophobic and are mostly bound to large proteins called **steroid binding proteins** (SBP), which are synthesized in the liver. As a result, only small amounts of the free hormone are dissolved in the plasma. Specifically, **sex hormone–binding globulin** (SHBG) is a glycoprotein that binds to the sex hormones, testosterone and 17β-estradiol. Progesterone, cortisol, and other corticosteroids are bound by transcortin.

The SBP–hormone complex and the free hormone are in equilibrium in the plasma, and only the free hormone is able to diffuse across cell membranes. SBP have three main functions: they increase the solubility of lipid-based hormones in the blood; they reduce the rate of hormone loss in the urine by preventing the hormones from being filtered in the kidney; and as mentioned above, they provide a source of hormone in the bloodstream that can release free hormone as the equilibrium changes. It follows that an additional way to regulate the availability of hormones that bind to carrier proteins, such as steroids, is to regulate the expression and secretion of the carrier proteins themselves. This is a critical mechanism that regulates the bioavailability of thyroid hormones, for example (see Chapter 20).

In a pathophysiologic setting, some medications can alter levels of binding proteins or displace hormones that are bound to them. In addition, some binding proteins are promiscuous and bind multiple hormones (eg, SHBG). These observations may have clinical implications for endocrine homeostasis, since free hormones are needed to feedback and control their rates of synthesis and secretion (see below).

Finally, the anatomic relationship of sites of release and action of hormones may play a key role in their regulation. For example, a number of hormones are destroyed by passage through the pulmonary circulation or the liver. This may markedly curtail the temporal window within which a given hormone can act.

HORMONE ACTION

Hormones exert a wide range of distinctive actions on a huge number of target cells to effect changes in metabolism, release of other hormones and regulatory substances, changes in ion channel activity, and cell growth, among others **(Clinical Box 16–1)**. Ultimately, the concerted action of the hormones of the body ensures the maintenance of homeostasis. Indeed, all hormones affect homeostasis to some degree. However, a subset of the hormones, including thyroid hormone, cortisol, parathyroid hormone, vasopressin, the

mineralocorticoids, and insulin, are the key contributors to homeostasis **(Table 16–1)**. Detailed information on the precise biologic effects of these molecules can be found in subsequent chapters.

TABLE 16–1 Major hormonal contributors to homeostasis.

Hormone	Source	Action
Thyroid hormone	Thyroid	Controls basal metabolism in most tissues
Cortisol	Adrenal cortex	Energy metabolism; permissive action for other hormones
Mineralocorticoids	Adrenal cortex	Regulate plasma volume via effects on serum electrolytes
Vasopressin	Posterior pituitary	Regulates plasma osmolality via effects on water excretion
Parathyroid hormone	Parathyroids	Regulates calcium and phosphorus levels
Insulin	Pancreas	Regulates plasma glucose concentration

Hydrophilic hormones, including peptides and catecholamines, exert their acute effects by binding to cell surface receptors. Most of these are from the GPCR family. Hydrophobic hormones, on the other hand, predominantly exert their actions via nuclear receptors. There are two classes of nuclear receptors that are important in endocrine physiology. The first class provides direct stimulation of transcription via induction of the binding of a transcriptional co-activator when the hormonal ligand is bound. In the second class, hormone binding triggers simultaneous dislodging of a transcriptional co-repressor and recruitment of a co-activator. The latter class of receptor allows for a wider dynamic range of regulation of the genes targeted by the hormone in question.

In recent years, it has become apparent that a number of receptors for steroid and other hydrophobic hormones are extranuclear, and some may even be present on the cell surface. The characterization of such receptors at a molecular

level, their associated signaling pathways, and indeed proof of their existence has been complicated by the ability of hydrophobic hormones to diffuse relatively freely into all cellular compartments. These **extranuclear receptors,** some of which may be structurally related or even identical to the more classic nuclear receptors, are proposed to mediate rapid responses to steroids and other hormones that do not require alterations in gene transcription. The physiologic effects at these receptors may therefore be distinct from those classically associated with a given hormone. Evidence is accumulating, for example, that plasma membrane receptors for estrogen can mediate acute arterial vasodilation as well as reducing cardiac hypertrophy in pathophysiologic settings. Functions such as these may account for differences in the prevalence of cardiovascular disease in premenopausal and postmenopausal women. In any event, this active area of biomedical investigation is likely to broaden our horizons of the full spectrum of action of steroid hormones.

CLINICAL BOX 16–1

Breast Cancer

Breast cancer is the most common malignancy of women, with about 1 million new cases diagnosed each year worldwide. The proliferation of more than two-thirds of breast tumors are driven by the ovarian hormone, estrogen, by virtue of the fact that the tumor cells express high levels of posttranslationally modified estrogen receptors (ERs). The clinical significance of these molecular findings has been known for more than 100 years, since the Scottish surgeon, Sir Thomas Beatson, reported delayed disease progression in patients with advanced breast cancer following removal of their ovaries. In modern times, determination of whether a given breast cancer is, or is not, **ER-positive** is a critical diagnostic test that guides treatment decisions, as well as an important prognosticator. ER-positive tumors are typically of lower grade, and patients with such tumors have improved survival (although the latter is likely due, at least in part, to the availability of excellent treatment options for ER-positive tumors compared with those that are ER- negative—see below).

THERAPEUTIC HIGHLIGHTS

Estrogen-responsive breast tumors are dependent on the presence of the hormone for growth. In modern times, cells can be deprived of the effects of

estrogen pharmacologically, rather than resorting to oophorectomy. **Tamoxifen** and related agents specifically inhibit the receptor and may also hasten its degradation. In postmenopausal women, where estrogen is derived from the metabolism of testosterone in extragonadal tissues rather than from the ovaries, **aromatase inhibitors** inhibit the conversion of androgens to estrogen, and thereby deprive tumor cells of their critical signal for continued proliferation.

PRINCIPLES OF FEEDBACK CONTROL

A principle that is critical for endocrine physiology is that of **feedback regulation.** This holds that the responsiveness of target cells to hormonal action subsequently "feeds back" to control the inciting endocrine organ. Feedback can regulate the further release of the hormone in either a negative feedback or (more rarely) a positive feedback loop. Positive feedback relates to the enhancement or continued stimulation of the original release mechanism/stimulus. Such mechanisms are really only seen in settings that need to gather momentum to an eventual outcome, such as parturition. Negative feedback is a far more common control mechanism and involves the inhibition or dampening of the initial hormone release mechanism/stimulus. A general scheme for feedback inhibition of endocrine axes is depicted in **Figure 16–3**.

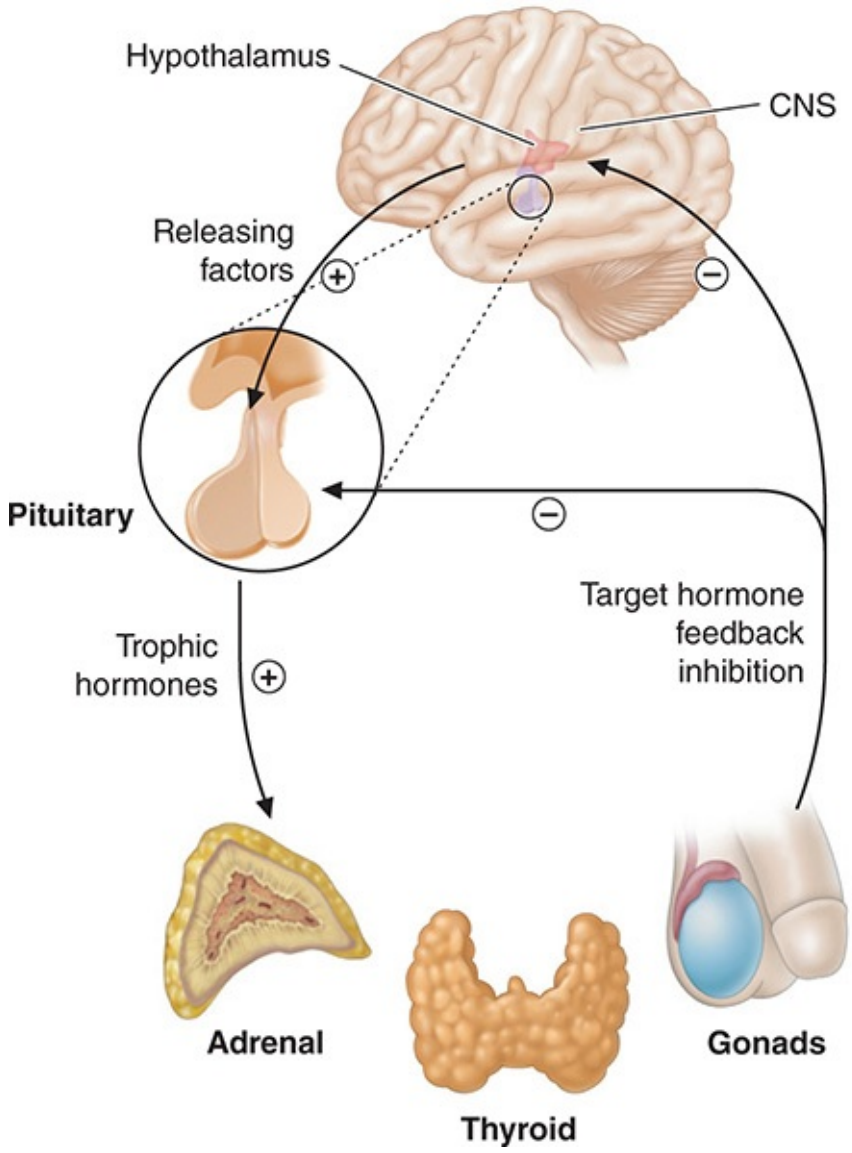


FIGURE 16–3 Summary of feedback loops regulating endocrine axes. CNS, central nervous system. (Reproduced with permission from Jameson JL (ed): *Harrison's Endocrinology*, 2nd ed. New York, NY: McGraw Hill; 2010.)

In general, the endocrine system uses a network of feedback responses to maintain a steady state. Steady state can be explained using blood osmolality as an example **(Figure 16–4)**. Blood osmolality in humans must be maintained within a physiologic range of 275–299 mOsm, and to maintain homeostasis this variable should not exceed that range. To ensure that osmolality does not change in the context of an open system, processes are in place that will add or remove water from the system to ensure a constant osmolality. The osmolality of blood will increase with dehydration and decrease with overhydration. If blood

osmolality increases outside the ideal range (by 10 mOsm or more), osmoreceptors are activated. These signal release of the peptide hormone, vasopressin, into the circulation (from the pituitary). Vasopressin acts on the renal collecting duct, and increases the permeability of the apical (lumenal) plasma membrane to water via the insertion of a protein called an aquaporin. Water is then moved from the urine into the circulation via transcellular transport. The reabsorption of water from the urine to the blood resets the osmolality of the blood to within the physiologic range. The decrease in blood osmolality then exerts a negative feedback on the cells of the hypothalamus and the pituitary and vasopressin release is inhibited, meaning that water reabsorption from the urine is reduced. Further details of this collaboration between the kidneys, hypothalamus, and pituitary are found in Chapter 38.

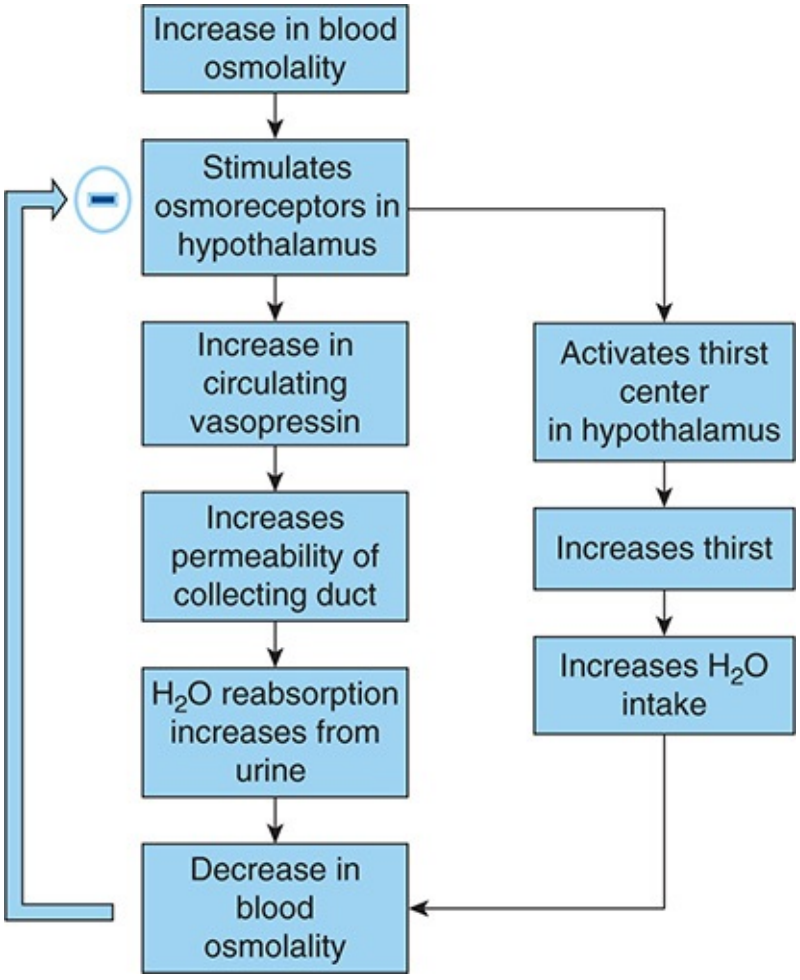


FIGURE 16–4 Feedback loop that ensures homeostasis of blood osmolality. An increase in blood osmolality triggers the thirst mechanism as well as renal conservation of water via the release of vasopressin from the hypothalamus.

Both outcomes decrease blood osmolality back toward the normal range, which feeds back to terminate hypothalamic signaling.

Negative feedback control systems such as those described are the most common feedback/homeostatic systems in the body. Other examples include temperature regulation (see Chapter 17) and the regulation of blood glucose concentrations (see Chapter 24). Feedback control loops also provide for diagnostic strategies in evaluating patients with suspected endocrine disorders. For example, in a patient being evaluated for hypothyroidism, normal levels of TSH (see Chapter 20) tend to rule out a primary defect at the level of the thyroid gland itself, and rather suggest that a defect at the level of the anterior pituitary should be sought. Conversely, if TSH is elevated, it suggests that the normal ability of circulating thyroid hormone to suppress TSH synthesis has been lost, likely due to a reduction in the ability of the thyroid gland to synthesize the hormone **(Clinical Box 16–2)**.

TYPES OF ENDOCRINE DISORDERS

It is pertinent also to discuss briefly the types of disease states where endocrine physiology can become deranged. Additional details of these disease states can be found in ensuing chapters.

HORMONE DEFICIENCY

Deficiencies of particular hormones are most commonly seen in the setting where there is destruction of the glandular structure responsible for their production, often as a result of inappropriate autoimmune attack. For example, in type 1 diabetes mellitus, pancreatic β cells are destroyed leading to an inability to synthesize insulin, often from a very young age. Similarly, hormonal deficiencies arise when there are inherited mutations in the factors responsible for their release or in the receptors for these releasing factors. Defects in the enzymatic machinery needed for hormone production, or a lack of appropriate precursors (eg, iodine deficiency leads to hypothyroidism), will also reduce the amount of the relevant hormone available for bodily requirements.

HORMONE RESISTANCE

Many of the consequences of hormone deficiency can be reproduced in disease states where adequate levels of a given hormone are synthesized and released, but the target tissues become resistant to the hormone's effects. Indeed, there is often overproduction of the implicated hormone in these conditions because the feedback loops that normally serve to shut off hormone synthesis and/or secretion are similarly desensitized. Mutations in hormone receptors (especially nuclear receptors) may result in heritable syndromes of hormone resistance. These syndromes, while relatively rare, usually exhibit severe outcomes, and have provided insights into the basic cell biology of hormone signaling. Functional hormone resistance that develops over time is also seen. Resistance arises from a relative failure of receptor signaling to couple efficiently to downstream intracellular effector pathways that normally mediate the effects of the hormone. The most common example of this is seen in type 2 diabetes mellitus. Target tissues for insulin gradually become more and more resistant to its actions, secondary to reduced activation of phosphatidylinositol 3-kinase and other intracellular signaling pathways. A key factor precipitating this outcome is obesity. In addition, because of excessive insulin secretion, pancreatic β cells become "exhausted" and may eventually fail, necessitating treatment with exogenous insulin. An important therapeutic goal, therefore, is to minimize progression to β cell exhaustion before irreversible insulin resistance sets in, with diet, exercise, and treatment with so-called **insulin sensitizers** (such as metformin and rosiglitazone).

CLINICAL BOX 16–2

Approach to the Patient with Suspected Endocrine Disease

Unlike many of the disorders of individual organ systems considered elsewhere in this volume, the symptoms of endocrine disease may be protean because of the number of body systems that are impacted by hormonal action. Further, many endocrine glands are relatively inaccessible to direct physical examination. Endocrine disorders must therefore be diagnosed on the basis of the symptoms they produce in concert with appropriate biochemical testing. Radioimmunoassays for specific hormones remain the mainstay of diagnostic endocrinolology and can be used to establish steady state concentrations as well as dynamic changes of the hormone in question (the latter requiring repeated blood sampling over time). In addition, the principles of feedback regulation of hormone synthesis and release may allow the clinician to

pinpoint the likely locus of any defect by comparing the levels of hormones in the same axis. For example, if testosterone levels are low but those of luteinizing hormone (LH) are high, this suggests that the testes are unable to respond to LH. Conversely, if both testosterone and LH are low, the problem is more likely to be at the level of the pituitary. Synthetic hormones can also be administered exogenously to test whether increased basal levels of a given hormone can be suppressed, or abnormally low levels can be stimulated by a relevant upstream agent. An example of applying this type of reasoning to the evaluation of suspected hypothyroidism is provided in **Figure 16–5**.

THERAPEUTIC HIGHLIGHTS

The appropriate treatment of endocrine disorders depends on their underlying basis. For example, if a particular hormone or its releasing factor is deficient, hormone replacement therapy is often indicated to ameliorate symptoms as well as long-term negative outcomes (Figure 16–5).

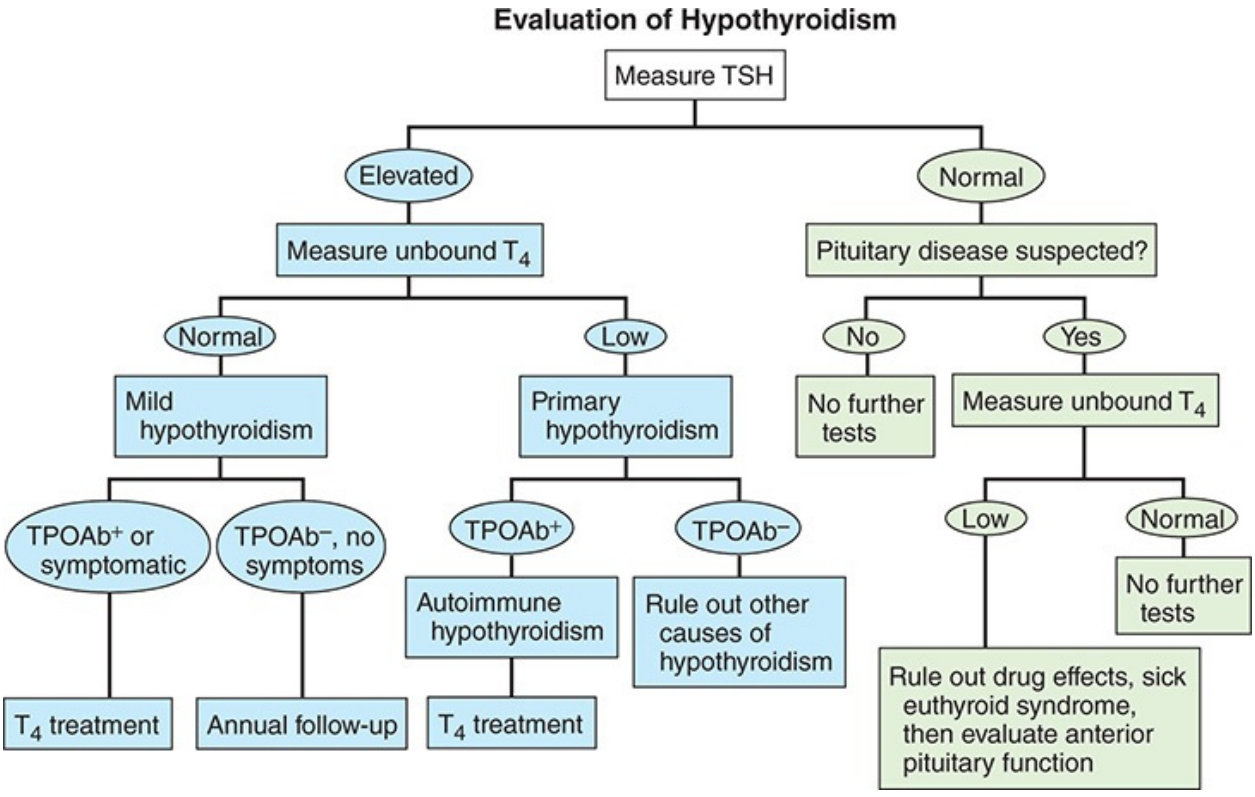


FIGURE 16–5 Summary of a strategy for the laboratory evaluation of hypothyroidism. TSH, thyroid-stimulating hormone; T_4, thyroid hormone;

TPOAb^{+}, positive for autoantibodies to thyroid peroxidase; TPOAb^{-}, antiperoxidase antibodies not present. (Reproduced with permission from Jameson JL [editor]: *Harrison's Endocrinology*, 2nd ed. New York, NY: McGraw Hill; 2010.)

HORMONE EXCESS

The converse of disorders of hormone deficiency or resistance is seen in diseases associated with hormone excess or overstimulation of hormone receptors (or both). A wide variety of endocrine tumors may produce hormones in an excessive and uncontrolled manner. Note that the secretion of hormones from tumor cells may not be subject to the same types of feedback regulation that are seen for the normal source of that hormone. In the setting of an endocrine tumor, exaggerated effects of the hormone are seen. For example, acromegaly, or gigantism, occurs in patients afflicted with an adenoma derived from pituitary somatotropes that secretes excessive quantities of growth hormone (see Chapter 18). In addition, other endocrine tumors may secrete hormones other than those characteristic of the cell type or tissue from which they are originally derived. When hormone production is increased in all of these cases, there usually will also be downregulation of upstream releasing factors due to the triggering of negative feedback loops.

Disorders of hormone excess can also be mimicked by antibodies that bind to, and activate, the receptor for the hormone. A classic example of such a condition is Graves disease, where susceptible individuals generate thyroid-stimulating immunoglobulins (TSIs) that bind to the receptor for TSH. This causes a conformational change that elicits receptor activation, and thus secretion of thyroid hormone in the absence of a physiologic trigger for this event. Diseases associated with hormone excess can also occur in a heritable manner secondary to activating mutations of hormone releasing factor receptors or their downstream targets. As seen for endocrine tumors, these pathophysiologic triggers of excessive hormone release are of course not subject to dampening by negative feedback loops.

CHAPTER SUMMARY

- The endocrine system consists of a distributed set of glands and the chemical messengers that they produce, referred to as hormones. Hormones play a

critical role in ensuring homeostasis (ie, the relative stability of body systems).

- Hormones can be grouped into peptide/protein, amine, and steroid categories. Water-soluble hormones (peptides and catecholamines) bind to cell surface receptors; hydrophobic hormones diffuse into the cell and activate nuclear receptors to regulate gene transcription.
- Hormone availability is dictated by the rate of synthesis, the presence of releasing factors, and rates of degradation or uptake. Free hydrophobic hormones are also in equilibrium with a form bound to plasma protein carriers, the latter representing a hormone reservoir as well as an additional mechanism to regulate hormone availability.
- The synthesis and release of many hormones is subject to regulation by negative feedback loops.
- Disease states can arise in the setting of both hormone deficiency and excess. Hormone deficiencies may be mimicked by inherited defects in their receptors or downstream signaling pathways; hormone excess may be mimicked by autoantibodies that bind to and activate hormone receptors, or by activating mutations of these receptors.

CHAPTER 17

Hypothalamic Regulation of Hormonal Functions

OBJECTIVES

After reading this chapter, you should be able to:

- Describe the anatomic connections between the hypothalamus and the pituitary gland and the functional significance of each connection.
- Define the role of the hypothalamus in producing and secreting hormones of the posterior pituitary.
- Discuss the effects of vasopressin, the receptors on which it acts, and how its secretion is regulated.
- Discuss the effects of oxytocin, the receptors on which it acts, and how its secretion is regulated.
- Outline the hypophysiotropic hormones, and the effect each has on anterior pituitary function.
- List the temperature-regulating mechanisms, and describe the way in which they are integrated under hypothalamic control to maintain normal body temperature in both cold and heat stress.
- Explain how the hypothalamus regulates water intake, and outline how thirst is regulated

- Discuss the pathophysiology of fever.

INTRODUCTION

The **hypothalamus** regulates complex autonomic mechanisms that maintain the chemical constancy of the internal environment, and regulates metabolic endocrine processes to control body temperature and satiety. It synthesizes and secretes **hypothalamic** hormones, and these in turn stimulate or inhibit the secretion of pituitary hormones. The hypothalamus also functions with the limbic system as a unit that regulates emotional and instinctual behavior.

HYPOTHALAMUS: ANATOMIC CONSIDERATIONS

The hypothalamus **(Figure 17–1)** is located in the lower part of the brain above the pituitary gland (Chapter 16, Figure 16–3) and releases hormones directly into the hypophysial portal system, which carries them directly to the pituitary gland. The hypothalamus is the portion of the anterior end of the diencephalon that lies below the hypothalamic sulcus and in front of the interpeduncular nuclei. It is divided into a variety of nuclei and nuclear areas.

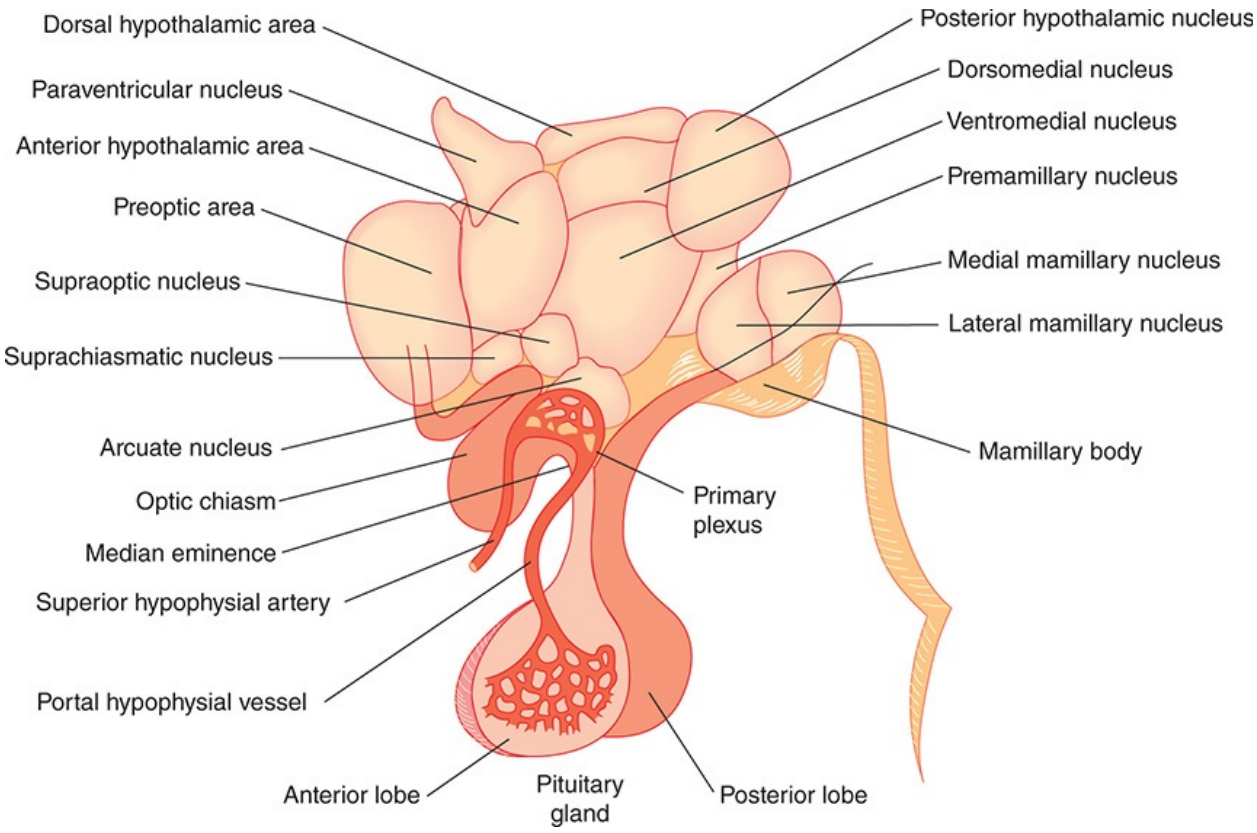


FIGURE 17–1 Human hypothalamus, with a superimposed diagrammatic representation of the portal hypophysial vessels.

AFFERENT & EFFERENT CONNECTIONS OF THE HYPOTHALAMUS

The principal afferent and efferent neural pathways to and from the hypothalamus are mostly unmyelinated. Many connect the hypothalamus to the limbic system. Important connections also exist between the hypothalamus and nuclei in the midbrain tegmentum, pons, and hindbrain.

Norepinephrine-secreting neurons with their cell bodies in the hindbrain end in many different parts of the hypothalamus (see Figure 7–2). Paraventricular neurons that secrete oxytocin and vasopressin project in turn to the hindbrain and the spinal cord. Neurons that secrete epinephrine have their cell bodies in the hindbrain and end in the ventral hypothalamus.

An intrahypothalamic system is composed of dopamine-secreting neurons that have their cell bodies in the arcuate nucleus and end on or near the capillaries that form the portal vessels in the median eminence. Serotonin-

secreting neurons project to the hypothalamus from the raphe nuclei.

RELATION TO THE PITUITARY GLAND

There are neural connections between the hypothalamus and the posterior lobe of the pituitary gland and vascular connections between the hypothalamus and the anterior lobe. Embryologically, the posterior pituitary arises as an evagination of the floor of the third ventricle. It is made up in large part of the endings of axons that arise from cell bodies in the supraoptic and paraventricular nuclei and pass to the posterior pituitary **(Figure 17–2)** via the **hypothalamohypophysial tract**. Most of the supraoptic fibers end in the posterior lobe itself, whereas some of the paraventricular fibers end in the median eminence. The anterior and intermediate lobes of the pituitary arise in the embryo from the rathke pouch, an evagination from the roof of the pharynx (see Figure 18–1). **Sympathetic** nerve fibers reach the anterior lobe from its capsule, and parasympathetic fibers reach it from the petrosal nerves, but few if any nerve fibers pass to it from the hypothalamus. **However**, the **portal hypophysial vessels** form a direct vascular link between the hypothalamus and the anterior pituitary. Arterial twigs from the carotid arteries and circle of willis form a network of fenestrated capillaries called the **primary plexus** on the ventral surface of the hypothalamus (Figure 17–1). Capillary loops also penetrate the median eminence. The capillaries drain into the sinusoidal portal hypophysial vessels that carry blood down the pituitary stalk to the capillaries of the anterior pituitary. This system begins and ends in capillaries without going through the heart and is therefore a true portal system. In birds and some mammals, including humans, there is no other anterior hypophysial arterial supply other than capsular vessels and anastomotic connections from the capillaries of the posterior pituitary. The **median eminence** is generally defined as the portion of the ventral hypothalamus from which the portal vessels arise. This region is outside the blood-brain barrier (see Chapter 33).

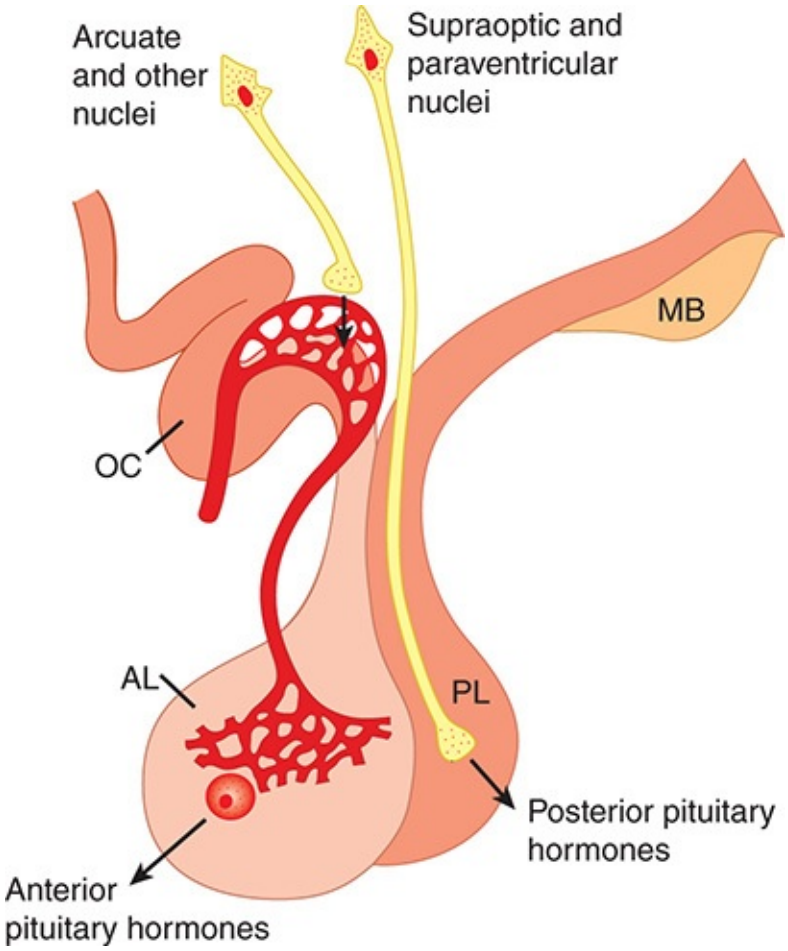


FIGURE 17–2 Secretion of hypothalamic hormones. The hormones of the posterior lobe (PL) are released into the general circulation from the endings of supraoptic and paraventricular neurons, whereas hypophysiotropic hormones are secreted into the portal hypophysial circulation from the endings of arcuate and other hypothalamic neurons. AL, anterior lobe; MB, mamillary bodies; OC, optic chiasm.

HYPOTHALAMIC FUNCTION

The major functions of the hypothalamus are summarized in **Table 17–1**. Some are fairly clear-cut visceral reflexes, and others include complex behavioral and emotional reactions; however, all involve a specific response to a specific stimulus. It is important to keep this in mind in considering hypothalamic function.

TABLE 17–1 Summary of principal hypothalamic regulatory mechanisms.

Function	Afferents from	Integrating Areas
Temperature regulation	Temperature receptors in the skin, deep tissues, spinal cord, hypothalamus, and other parts of the brain	Anterior hypothalamus, response to heat; posterior hypothalamus, response to cold
Neuroendocrine control of:		
Catecholamines	Limbic areas concerned with emotion	Dorsal and posterior hypothalamus
Vasopressin	Osmoreceptors, "volume receptors," others	Supraoptic and paraventricular nuclei
Oxytocin	Touch receptors in breast, uterus, genitalia	Supraoptic and paraventricular nuclei
Thyroid-stimulating hormone (thyrotropin, TSH) via TRH	Temperature receptors in infants, perhaps others	Paraventricular nuclei and neighboring areas
Adrenocorticotropic hormone (ACTH) and β-lipotropin (β-LPH) via CRH	Limbic system (emotional stimuli); reticular formation ("systemic" stimuli); hypothalamic and anterior pituitary cells sensitive to circulating blood cortisol level; suprachiasmatic nuclei (diurnal rhythm)	Paraventricular nuclei
Follicle-stimulating hormone (FSH) and luteinizing hormone (LH) via GnRH	Hypothalamic cells sensitive to estrogens, eyes, touch receptors in skin and genitalia of reflex ovulating species	Preoptic area; other areas
Prolactin via PIH and PRH	Touch receptors in breasts, other unknown receptors	Arcuate nucleus; other areas (hypothalamus inhibits secretion)
Growth hormone via somatostatin and GRH	Unknown receptors	Periventricular nucleus, arcuate nucleus
"Appetitive" behavior:		
Thirst	Osmoreceptors, probably located in the organum vasculosum of the lamina terminalis; angiotensin II uptake in the subfornical organ	Lateral superior hypothalamus
Hunger	Glucostat cells sensitive to rate of glucose utilization; leptin receptors; receptors for other polypeptides	Ventromedial, arcuate, and paraventricular nuclei; lateral hypothalamus
Sexual behavior	Cells sensitive to circulating estrogen and androgen, others	Anterior ventral hypothalamus plus, in the male, piriform cortex
Defensive reactions (fear, rage)	Sense organs and neocortex, paths unknown	Diffuse, in limbic system and hypothalamus
Control of body rhythms	Retina via retinohypothalamic fibers	Suprachiasmatic nuclei

RELATION TO AUTONOMIC FUNCTION

Many years ago, Sherrington called the hypothalamus "the head ganglion of the autonomic system." Stimulation of the hypothalamus produces autonomic responses, but the hypothalamus does not seem to be concerned with the regulation of visceral function per se. Rather, the autonomic responses triggered in the hypothalamus are part of more complex phenomena such as eating, and emotions such as rage. For example, stimulation of various parts of the hypothalamus, especially the lateral areas, produces diffuse sympathetic discharge and increased adrenal medullary secretion—the mass sympathetic discharge seen in animals exposed to stress (the flight or fight reaction; see Chapter 13).

It has been claimed that separate hypothalamic areas control epinephrine and norepinephrine secretion. Differential secretion of one or the other of these adrenal medullary catecholamines does occur in certain situations (see Chapter 20), but the selective increases are small.

Body weight depends on the balance between caloric intake and utilization of calories. Obesity results when the former exceeds the latter. The hypothalamus and related parts of the brain play a key role in the regulation of food intake. Obesity is considered in detail in Chapter 26, and the relation of obesity to diabetes mellitus is discussed in Chapter 24.

Hypothalamic regulation of sleep and circadian rhythms are discussed in Chapter 14.

THIRST

Another appetitive mechanism under hypothalamic control is thirst. Drinking is regulated by plasma osmolality and extracellular fluid (ECF) volume in much the same fashion as vasopressin secretion (see Chapter 38). Water intake is increased by increased effective osmotic pressure of the plasma **(Figure 17–3)**, by decreases in ECF volume, and by psychological and other factors. Osmolality acts via **osmoreceptors,** receptors that sense the osmolality of the body fluids. These osmoreceptors are located in the anterior hypothalamus.

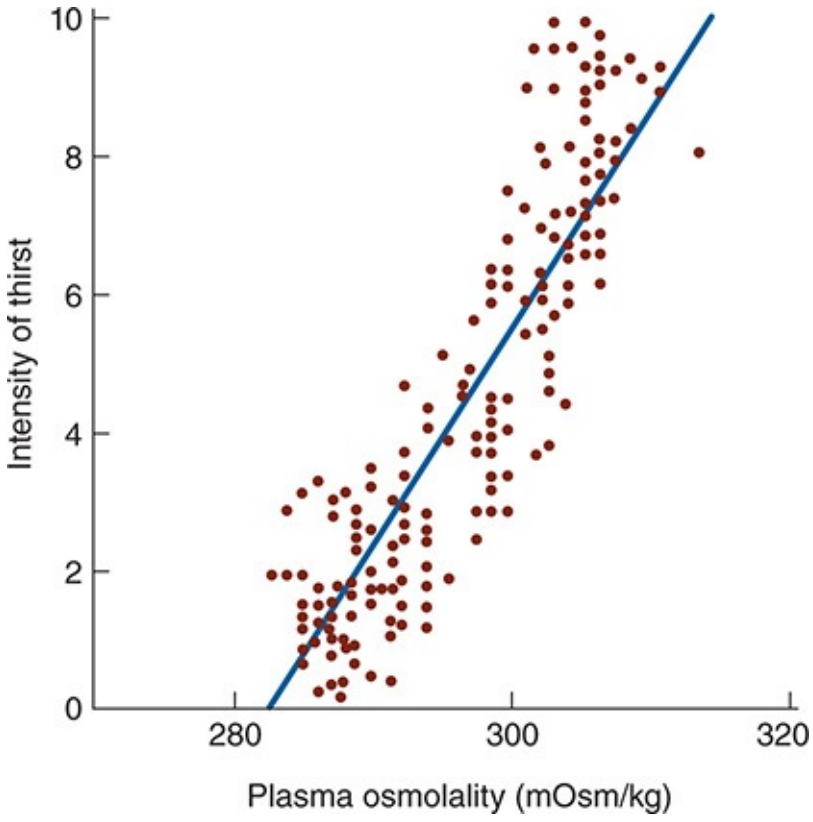


FIGURE 17–3 Relation of plasma osmolality to thirst in healthy adult

humans during infusion of hypertonic saline. The intensity of thirst is measured on a special analog scale. (Reproduced with permission from Thompson CJ et al: The osmotic thresholds for thirst and vasopressin release are similar in healthy humans. Clin Sci Lond 1986; Dec; 71(6): 651–656.)

Decreases in ECF volume also stimulate thirst by a pathway independent of that mediating thirst in response to increased plasma osmolality **(Figure 17–4)**. Thus, hemorrhage causes increased drinking even if there is no change in the osmolality of the plasma. The effect of ECF volume depletion on thirst is mediated in part via the renin–angiotensin system (see Chapter 38). Renin secretion is increased by hypovolemia and results in an increase in circulating angiotensin II. The angiotensin II acts on the **subfornical organ,** a specialized receptor area in the diencephalon (see Figure 33–7), to stimulate the neural areas concerned with thirst. Some evidence suggests that it acts on the **organum vasculosum of the lamina terminalis (OVLT)** as well. These areas are highly permeable and are two of the circumventricular organs located outside the blood-brain barrier (see Chapter 33). However, drugs that block the action of angiotensin II do not completely block the thirst response to hypovolemia, and it appears that the baroreceptors in the heart and blood vessels are also involved.

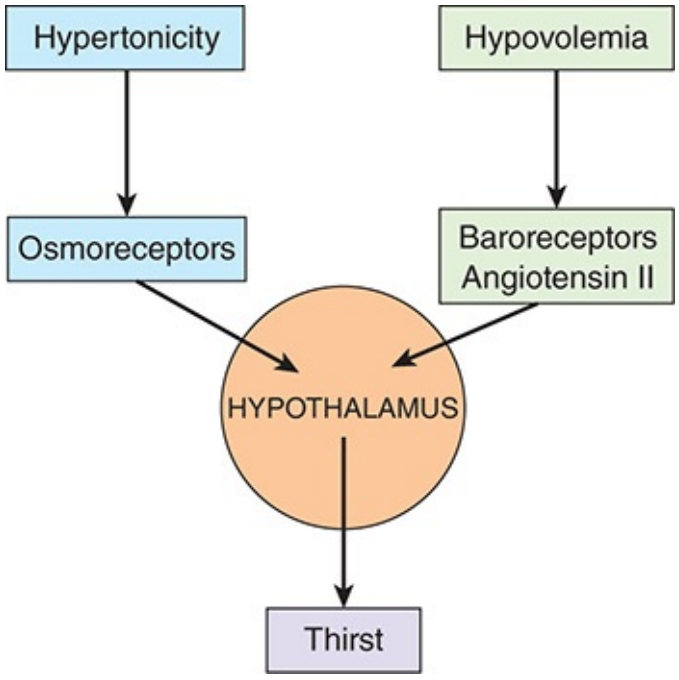


FIGURE 17–4 Diagrammatic representation of the way in which changes in plasma osmolality and changes in ECF volume affect thirst by separate pathways. ECF, extracellular fluid.

The intake of liquids is increased during eating **(prandial drinking).** The increase has been called a learned or habit response, but it has not been investigated in detail. One factor is an increase in plasma osmolality that occurs as food is absorbed. Another may be an action of one or more gastrointestinal hormones on the hypothalamus.

When the sensation of thirst is obtunded, either by direct damage to the diencephalon or by depressed or altered states of consciousness, patients stop drinking adequate amounts of fluid. Dehydration results if appropriate measures are not instituted to maintain water balance. If the protein intake is high, the products of protein metabolism cause an osmotic diuresis (see Chapter 38), and the amounts of water required to maintain hydration are large. Most cases of **hypernatremia** are actually due to simple dehydration in patients with psychoses or hypothalamic disease who do not or cannot increase their water intake when their thirst mechanism is stimulated. Lesions of the anterior communicating artery can also obtund thirst because branches of this artery supply the hypothalamic areas concerned with thirst.

OTHER FACTORS REGULATING WATER INTAKE

A number of other well-established factors contribute to the regulation of water intake. Psychological and social factors are important. Dryness of the pharyngeal mucous membrane causes a sensation of thirst. Patients in whom fluid intake must be restricted sometimes get appreciable relief of thirst by sucking ice chips or a wet cloth.

Dehydrated dogs, cats, camels, and some other animals rapidly drink just enough water to make up their water deficit. They stop drinking before the water is absorbed (while their plasma is still hypertonic), so some kind of pharyngeal gastrointestinal “metering” must be involved. Some evidence suggests that humans have a similar metering ability, though it is not well developed.

CONTROL OF POSTERIOR PITUITARY SECRETION

VASOPRESSIN & OXYTOCIN

In most mammals, the hormones secreted by the posterior pituitary gland are

arginine vasopressin (AVP) and **oxytocin.** In hippopotami and most pigs, arginine in the vasopressin molecule is replaced by lysine to form **lysine vasopressin.** The posterior pituitaries of some species of pigs and marsupials contain a mixture of arginine and lysine vasopressin. The posterior lobe hormones are nanopeptides with a disulfide ring at one end **(Figure 17–5)**.

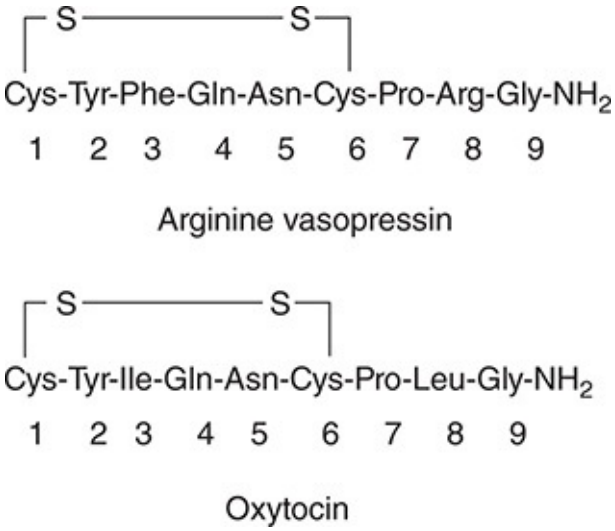


FIGURE 17–5 Arginine vasopressin and oxytocin.

BIOSYNTHESIS, INTRANEURONAL TRANSPORT, & SECRETION

The hormones of the posterior pituitary gland are synthesized in the cell bodies of the magnocellular neurons in the supraoptic and paraventricular nuclei and transported down the axons of these neurons to their endings in the posterior lobe, where they are secreted in response to electrical activity in the endings. Some of the neurons make oxytocin and others make vasopressin, and oxytocin-containing and vasopressin-containing cells are found in both nuclei.

Oxytocin and vasopressin are typical **neural hormones,** that is, hormones secreted into the circulation by nerve cells. This type of neural regulation is compared with other types in **Figure 17–6**. The term **neurosecretion** was originally coined to describe the secretion of hormones by neurons, but the term is somewhat misleading because it appears that all neurons secrete chemical messengers (see Chapter 7).

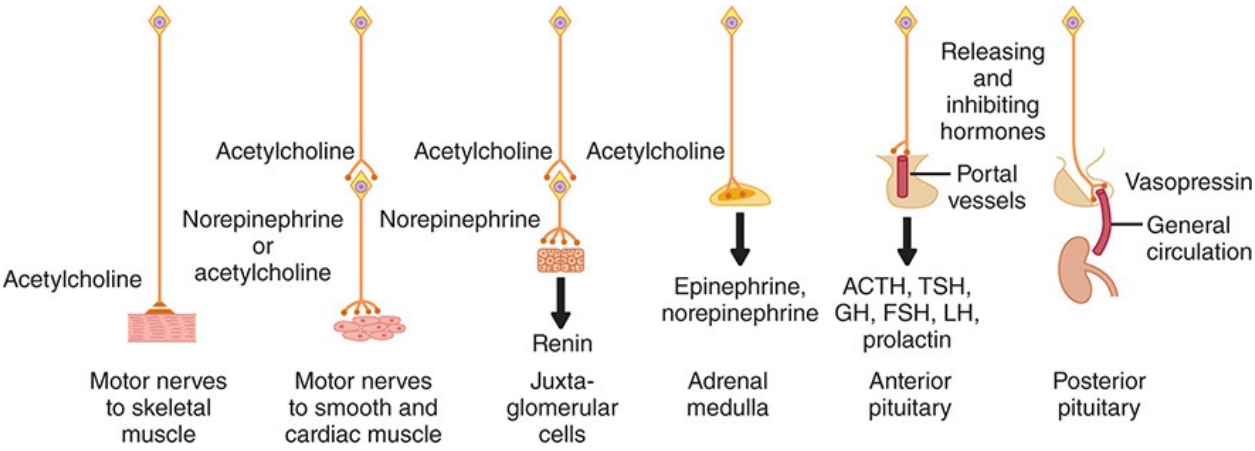


FIGURE 17–6 Neural control mechanisms. In the two situations on the left, neurotransmitters act at nerve endings on muscle; in the two in the middle, neurotransmitters regulate the secretion of endocrine glands; and in the two on the right, neurons secrete hormones into the hypophysial portal or general circulation. ACTH, adrenocorticotropic hormone; FSH, follicle-stimulating hormone; GH, growth hormone; LH, luteinizing hormone; TSH, thyroid-stimulating hormone.

Like other peptide hormones, the posterior lobe hormones are synthesized as part of larger precursor molecules. Vasopressin and oxytocin each have a characteristic **neurophysin** associated with them in the granules in the neurons that secrete them—neurophysin I in the case of oxytocin and neurophysin II in the case of vasopressin. The neurophysins were originally thought to be binding polypeptides, but it now appears that they are simply parts of the precursor molecules. The precursor for AVP, **prepropressophysin,** contains a 19-amino-acid residue leader sequence followed by AVP, neurophysin II, and a glycopeptide **(Figure 17–7)**. **Prepro-oxyphysin,** the precursor for oxytocin, is a similar but smaller molecule that lacks the glycopeptide.

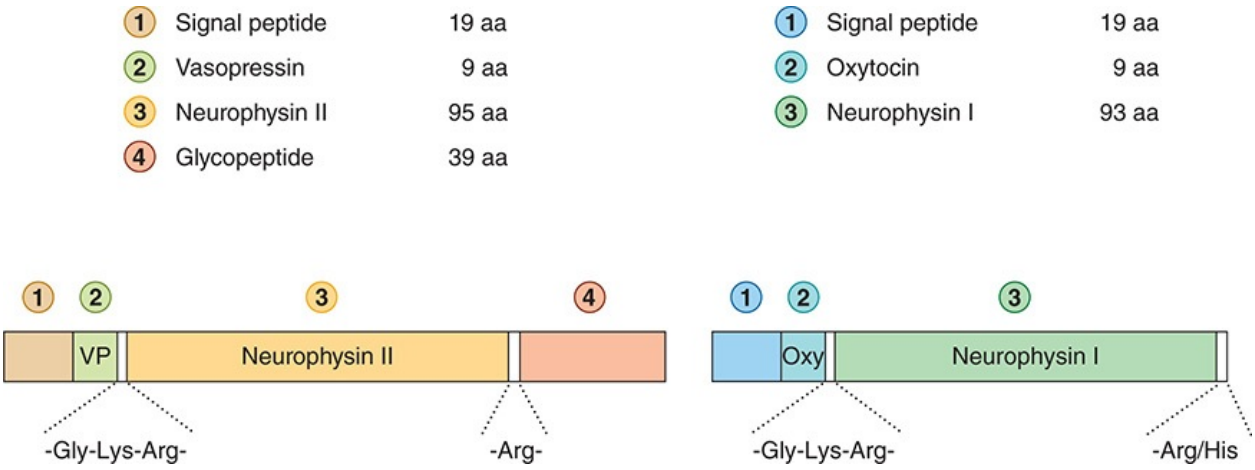


FIGURE 17–7 Structure of bovine prepropressophysin (left) and prepro-oxyphysin (right). Gly in the 10 position of both peptides is necessary for amidation of the Gly residue in position 9. aa, amino acid residues. (Reproduced with permission from Richter D: Molecular events in expression of vasopressin and oxytocin and their cognate receptors. Am J Physiol 1988; Aug; 255(2 Pt 2):F207–F219.)

The precursor molecules are synthesized in the ribosomes of the cell bodies of the neurons. They have their leader sequences removed in the endoplasmic reticulum, are packaged into secretory granules in the Golgi apparatus, and are transported down the axons by axoplasmic flow to the endings in the posterior pituitary. The secretory granules, called **Herring bodies,** are easy to stain in tissue sections, and they have been extensively studied. Cleavage of the precursor molecules occurs as they are being transported, and the storage granules in the endings contain free vasopressin or oxytocin and the corresponding neurophysin. In the case of vasopressin, the glycopeptide is also present. All these products are secreted, but the functions of the components other than the established posterior pituitary hormones are unknown. Physiologic control of vasopressin secretion is described in detail in Chapter 38.

ELECTRICAL ACTIVITY OF MAGNOCELLULAR NEURONS

The oxytocin-secreting and vasopressin-secreting neurons also generate and conduct action potentials, and action potentials reaching their endings trigger the release of hormones by Ca^{2+}-dependent exocytosis. At least in anesthetized rats,

these neurons are silent at rest or discharge at low, irregular rates (0.1–3 spikes/s). However, their response to stimulation varies **(Figure 17–8)**. Stimulation of the nipples causes a synchronous, high-frequency discharge of the oxytocin neurons after an appreciable latency. This discharge causes release of a pulse of oxytocin and consequent milk ejection in postpartum females. On the other hand, stimulation of vasopressin-secreting neurons by a stimulus such as an increase in blood osmolality during dehydration, or loss of blood volume due to hemorrhage, causes an initial steady increase in firing rate followed by a prolonged pattern of phasic discharge in which periods of high-frequency discharge alternate with periods of electrical quiescence **(phasic bursting).** These phasic bursts are generally not synchronous in different vasopressin-secreting neurons. They are well-suited to maintain a prolonged increase in the output of vasopressin, as opposed to the synchronous, relatively short, high-frequency discharge of oxytocin-secreting neurons in response to stimulation of the nipples.

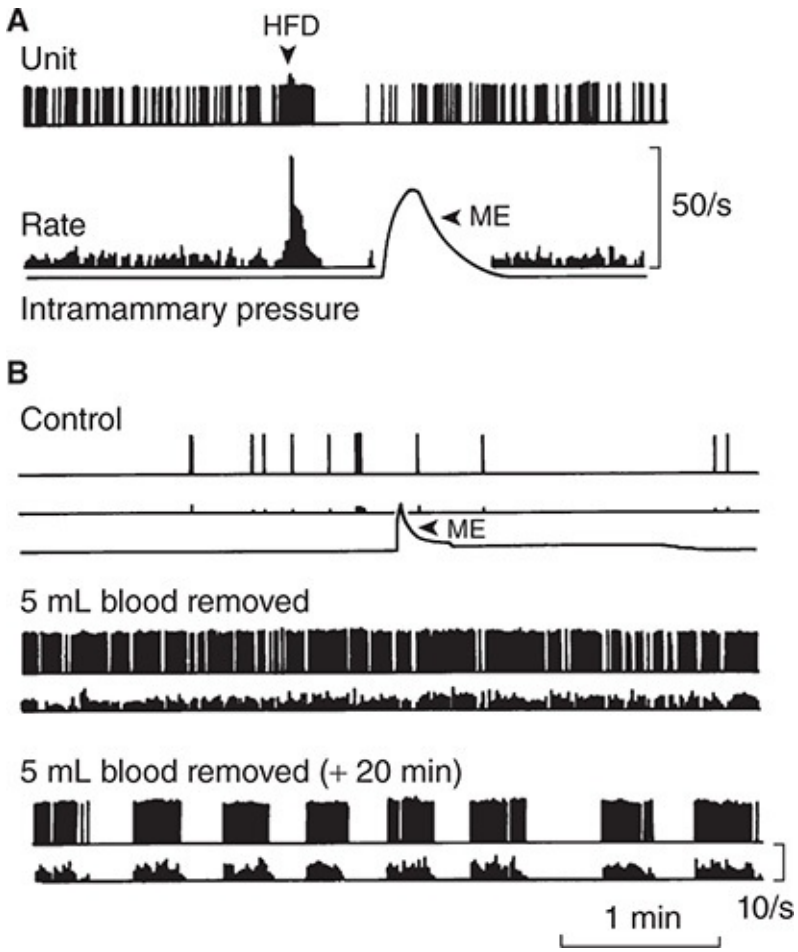


FIGURE 17–8 Responses of magnocellular neurons to stimulation. The

tracings show individual extracellularly recorded action potentials, discharge rates, and intramammary duct pressure. **A)** Response of an oxytocin-secreting neuron. HFD, high-frequency discharge; ME, milk ejection. Stimulation of nipples started before the onset of recording. **B)** Responses of a vasopressin-secreting neuron, showing no change in the slow firing rate in response to stimulation of nipples and a prompt increase in the firing rate when 5 mL of blood was drawn, followed by typical phasic discharge. (Modified with permission from Wakerly JB: Hypothalamic neurosecretory function: Insights from electrophysiological studies of the magno-cellular nuclei. IBRO News 1985; 4:15.)

VASOPRESSIN & OXYTOCIN IN OTHER LOCATIONS

Vasopressin-secreting neurons are found in the suprachiasmatic nuclei, and vasopressin and oxytocin are also found in the endings of neurons that project from the paraventricular nuclei to the brainstem and spinal cord. These neurons appear to be involved in cardiovascular control. In addition, vasopressin and oxytocin are synthesized in the gonads and the adrenal cortex, and oxytocin is present in the thymus. The functions of the peptides in these organs are unsettled.

Vasopressin Receptors

There are at least three kinds of vasopressin receptors: V_{1A}, V_{1B}, and V_2. All are G-protein-coupled. The V_{1A} and V_{1B} receptors act through phosphatidylinositol hydrolysis to increase intracellular Ca^{2+} concentrations. The V_2 receptors act through G_s to increase cyclic adenosine monophosphate levels.

Effects of Vasopressin

Because one of its principal physiologic effects is the retention of water by the kidney, vasopressin is often called the **antidiuretic hormone (ADH).** It increases the permeability of the collecting ducts of the kidney so that water enters the hypertonic interstitium of the renal pyramids (see Chapter 37). The urine becomes concentrated and its volume decreases. The overall effect is

therefore retention of water in excess of solute; consequently, the effective osmotic pressure of the body fluids is decreased. In the absence of vasopressin, the urine is hypotonic to plasma, urine volume is increased, and there is a net water loss. Consequently, the osmolality of the body fluid rises.

Effects of Oxytocin

In humans, oxytocin acts primarily on the breasts and uterus, though it appears to be involved in luteolysis as well (see Chapter 22). A G-protein-coupled oxytocin receptor has been identified in human myometrium, and a similar or identical receptor is found in mammary tissue and the ovary. It triggers increases in intracellular Ca^{2+} levels.

The Milk Ejection Reflex

Oxytocin causes contraction of the **myoepithelial cells** that line the ducts of the breast. This squeezes the milk out of the alveoli of the lactating breast into the large ducts (sinuses) and thence out of the nipple **(milk ejection).** Many hormones acting in concert are responsible for breast growth and the secretion of milk into the ducts (see Chapter 22), but milk ejection in most species requires oxytocin.

Milk ejection is normally initiated by a neuroendocrine reflex. The receptors involved are touch receptors, which are plentiful in the breast—especially around the nipple. Impulses generated in these receptors are relayed from the somatic touch pathways to the supraoptic and paraventricular nuclei. Discharge of the oxytocin-containing neurons causes secretion of oxytocin from the posterior pituitary (Figure 17–8). The suckling of an infant at the breast stimulates the touch receptors, the nuclei are stimulated, oxytocin is released, and the milk is expressed into the sinuses, ready to flow into the mouth of the waiting infant. In lactating women, genital stimulation and emotional stimuli also produce oxytocin secretion, sometimes causing milk to spurt from the breasts.

Other Actions of Oxytocin

Oxytocin causes contraction of the smooth muscle of the uterus. The sensitivity of the uterine musculature to oxytocin is enhanced by estrogen and inhibited by progesterone. The inhibitory effect of progesterone is due to a direct action of

the steroid on uterine oxytocin receptors. In late pregnancy, the uterus becomes very sensitive to oxytocin coincident with a marked increase in the number of oxytocin receptors and oxytocin receptor mRNA (see Chapter 22). Oxytocin secretion is then increased during labor. After dilation of the cervix, descent of the fetus down the birth canal initiates impulses in the afferent nerves that are relayed to the supraoptic and paraventricular nuclei, causing secretion of sufficient oxytocin to enhance labor (Figure 22–22). The amount of oxytocin in plasma is normal at the onset of labor. It is possible that the marked increase in oxytocin receptors at this time allows normal oxytocin levels to initiate contractions, setting up a positive feedback. However, the amount of oxytocin in the uterus is also increased, and locally produced oxytocin may also play a role.

Oxytocin may also act on the nonpregnant uterus to facilitate sperm transport. The passage of sperm up the female genital tract to the uterine tubes, where fertilization normally takes place, depends not only on the motile powers of the sperm but also, at least in some species, on uterine contractions. The genital stimulation involved in coitus releases oxytocin, but whether oxytocin initiates the rather specialized uterine contractions that transport the sperm is as yet unproven. The secretion of oxytocin is also increased by stressful stimuli and, like that of vasopressin, is inhibited by alcohol.

Circulating oxytocin increases at the time of ejaculation in males, and it is possible that this causes increased contraction of the smooth muscle of the vas deferens, propelling sperm toward the urethra.

CONTROL OF ANTERIOR PITUITARY SECRETION

ANTERIOR PITUITARY HORMONES

The anterior pituitary secretes six hormones: **adrenocorticotropic hormone (corticotropin, ACTH), thyroid-stimulating hormone (thyrotropin, TSH), growth hormone, follicle-stimulating hormone (FSH), luteinizing hormone (LH),** and **prolactin (PRL).** An additional polypeptide, β-lipotropin (β-LPH), is secreted with ACTH, but its physiologic role is unknown. The actions of the anterior pituitary hormones are summarized in **Figure 17–9**. The hormones are discussed in detail in subsequent chapters. The hypothalamus plays an important stimulatory role in regulating the secretion of ACTH, β-LPH, TSH, growth hormone, FSH, and LH. It also regulates prolactin secretion, but its effect is predominantly inhibitory rather than stimulatory.

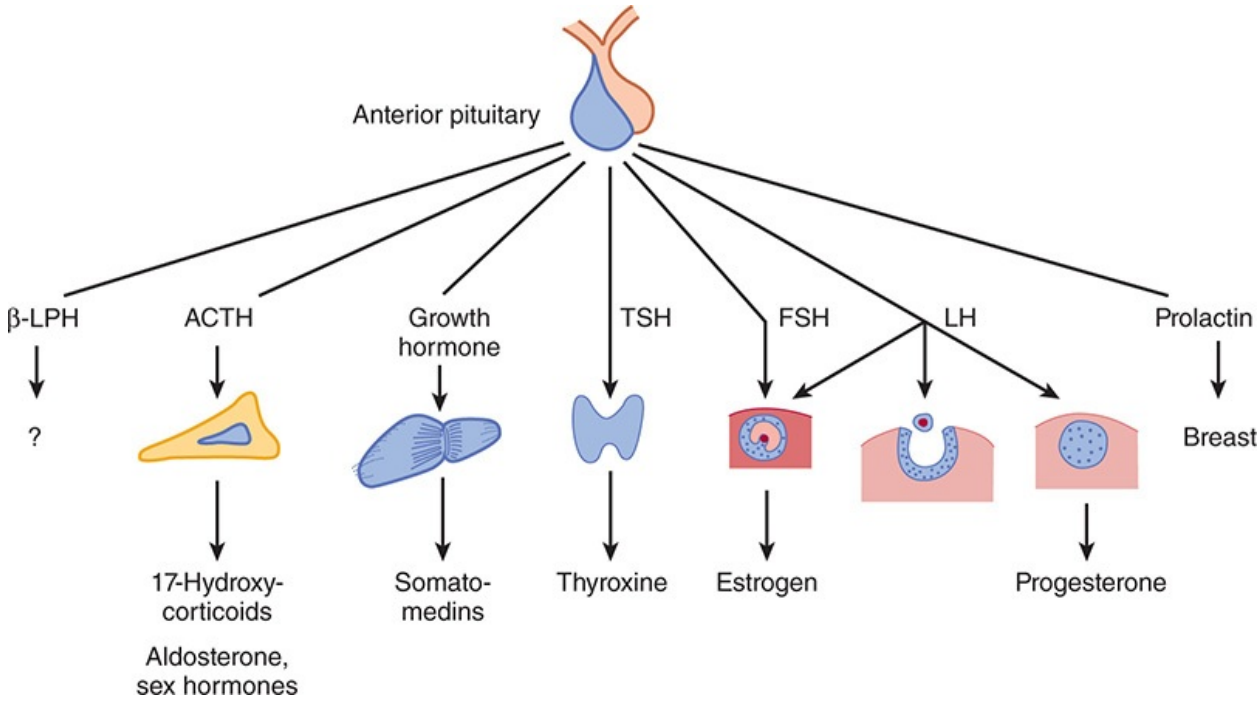


FIGURE 17–9 Anterior pituitary hormones. In women, FSH and LH act in sequence on the ovary to produce growth of the ovarian follicle, ovulation, and formation and maintenance of the corpus luteum. Prolactin stimulates lactation. In men, FSH and LH control the functions of the testes. ACTH, adrenocorticotropic hormone; β-LPH, β-lipotropin; FSH, follicle-stimulating hormone; LH, luteinizing hormone; TSH, thyroid-stimulating hormone.

NATURE OF HYPOTHALAMIC CONTROL

Anterior pituitary secretion is controlled by chemical agents carried in the portal hypophysial vessels from the hypothalamus to the pituitary. These substances used to be called releasing and inhibiting factors, but now they are commonly called **hypophysiotropic hormones.** The latter term seems appropriate since they are secreted into the bloodstream and act at a distance from their site of origin. Small amounts escape into the general circulation, but they are at their highest concentration in portal hypophysial blood.

HYPOPHYSIOTROPIC HORMONES

There are six established hypothalamic releasing and inhibiting hormones **(Figure 17–10)**: **corticotropin-releasing hormone (CRH); thyrotropin-**

releasing hormone (TRH); growth hormone–releasing hormone (GRH); growth hormone–inhibiting hormone (GIH, now generally called **somatostatin); luteinizing hormone–releasing hormone (LHRH,** now generally known as **gonadotropin-releasing hormone [GnRH]);** and **prolactin-inhibiting hormone (PIH).** In addition, hypothalamic extracts contain prolactin-releasing activity, and a **prolactin-releasing hormone (PRH)** has been postulated to exist. TRH, VIP, and several other polypeptides found in the hypothalamus stimulate prolactin secretion, but it is uncertain whether one or more of these peptides is the physiologic PRH. Recently, an orphan receptor was isolated from the anterior pituitary, and the search for its ligand led to the isolation of a 31-amino-acid polypeptide from the human hypothalamus. This polypeptide stimulated prolactin secretion by an action on the anterior pituitary receptor, but additional research is needed to determine whether it is the physiologic PRH. GnRH stimulates the secretion of FSH as well as that of LH, and it seems unlikely that a separate FSH-releasing hormone exists.

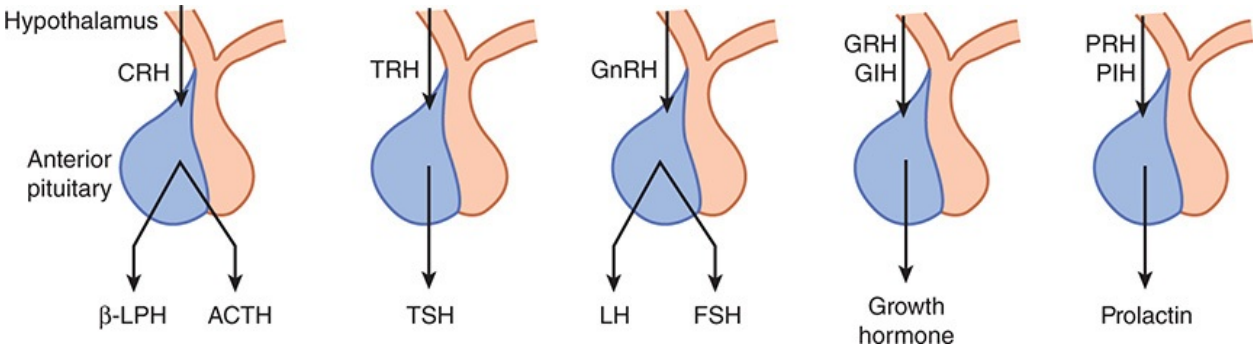


FIGURE 17–10 Effects of hypophysiotropic hormones on the secretion of anterior pituitary hormones. ACTH, adrenocorticotropic hormone; β-LPH, β-lipotropin; CRH, corticotropin-releasing hormone; FSH, follicle-stimulating hormone; GIH, growth hormone–inhibiting hormone; GnRH, gonadotropin-releasing hormone; GRH, growth hormone–releasing hormone; LH, luteinizing hormone; PIH, prolactin-inhibiting hormone; PRH, prolactin-releasing hormone; TRH, thyrotropin-releasing hormone; TSH, thyroid-stimulating hormone.

The structures of the genes and preprohormones for TRH, GnRH, somatostatin, CRH, and GRH are known. PreproTRH contains six copies of TRH. Several other preprohormones may contain other hormonally active peptides in addition to the hypophysiotropic hormones.

The area from which the hypothalamic releasing and inhibiting hormones are secreted is the median eminence of the hypothalamus. This region contains few

nerve cell bodies, but many nerve endings are in close proximity to the capillary loops from which the portal vessels originate.

The locations of the cell bodies of the neurons that project to the external layer of the median eminence and secrete the hypophysiotropic hormones are shown in **Figure 17–11**, which also shows the location of the neurons secreting oxytocin and vasopressin. The GnRH-secreting neurons are primarily in the medial preoptic area, the somatostatin-secreting neurons are in the periventricular nuclei, the TRH-secreting and CRH-secreting neurons are in the medial parts of the paraventricular nuclei, and the GRH-secreting (and dopamine-secreting) neurons are in the arcuate nuclei.

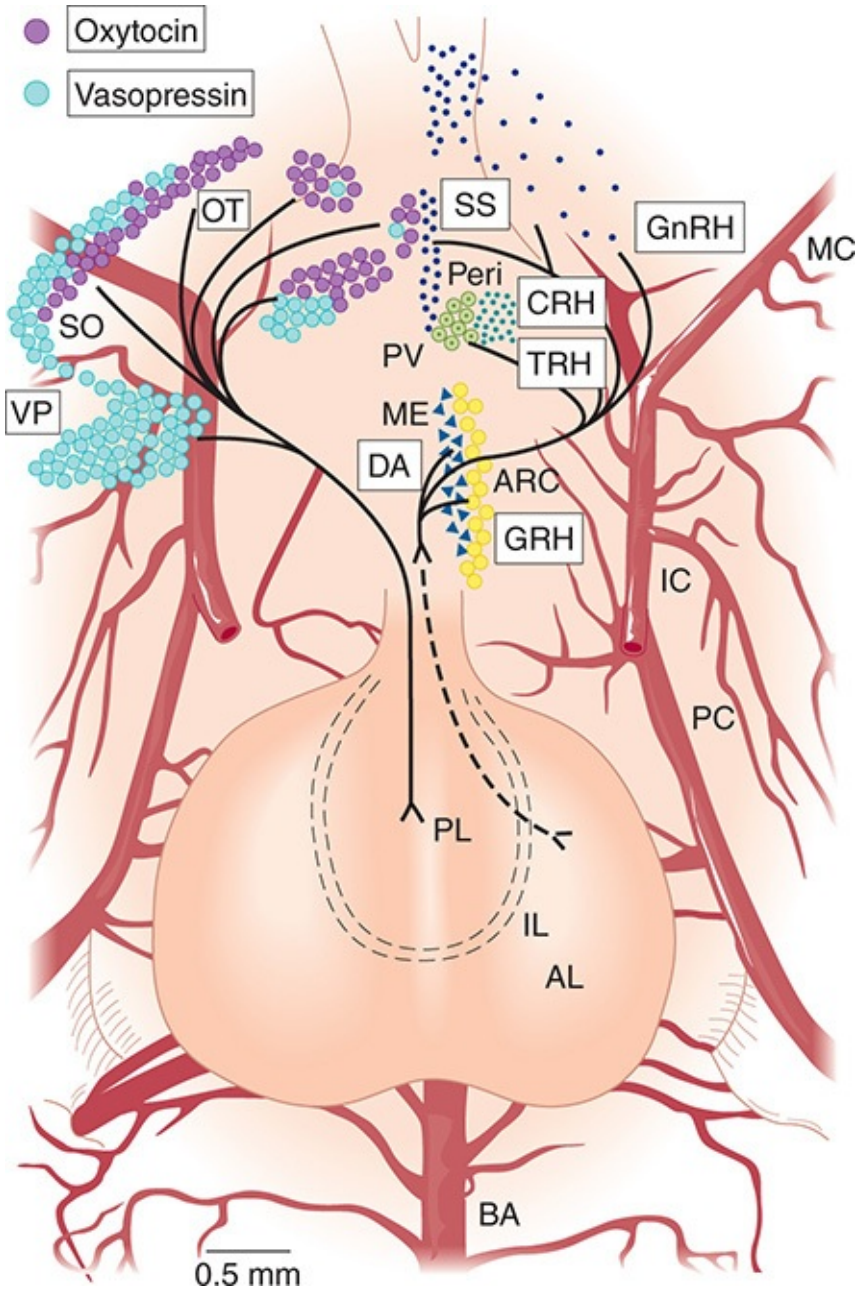


FIGURE 17–11 Location of cell bodies of hypophysiotropic hormone-secreting neurons projected on a ventral view of the hypothalamus and pituitary of the rat. AL, anterior lobe; ARC, arcuate nucleus; BA, basilar artery; CRH, corticotropin-releasing hormone; DA, dopamine; GRH, growth hormone-releasing hormone; GnRH, gonodotropin-releasing hormone; IC, internal carotid artery; IL, intermediate lobe; MC, middle cerebral artery; ME, median eminence; PC, posterior cerebral artery; Peri, periventricular nucleus; PL, posterior lobe; PV, paraventricular nucleus; SO, supraoptic nucleus; TRH, thyroid-releasing hormone. The names of the hormones are enclosed in boxes.

(Used with permission of LW Swanson and ET Cunningham Jr.)

Most, if not all, of the hypophysiotropic hormones affect the secretion of more than one anterior pituitary hormone (Figure 17–10). The FSH-stimulating activity of GnRH has been mentioned previously. TRH stimulates the secretion of prolactin as well as TSH. Somatostatin inhibits the secretion of TSH as well as growth hormone. It does not normally inhibit the secretion of the other anterior pituitary hormones, but it inhibits the abnormally elevated secretion of ACTH in patients with Nelson syndrome. CRH stimulates the secretion of ACTH and β-LPH.

Hypophysiotropic hormones function as neurotransmitters in other parts of the brain, the retina, and the autonomic nervous system (see Chapter 7). In addition, somatostatin is found in the pancreatic islets (see Chapter 24), GRH is secreted by pancreatic tumors, and somatostatin and TRH are found in the gastrointestinal tract (see Chapter 25).

Receptors for most of the hypophysiotropic hormones are coupled to G-proteins. There are two human CRH receptors: hCRH-RI and hCRH-RII. The physiologic role of hCRH-RII is unsettled, though it is found in many parts of the brain. In addition, a **CRH-binding protein** in the peripheral circulation inactivates CRH. It is also found in the cytoplasm of corticotropes in the anterior pituitary, and in this location it might play a role in receptor internalization. However, the exact physiologic role of this protein is unknown. Other hypophysiotropic hormones do not have known binding proteins.

SIGNIFICANCE & CLINICAL IMPLICATIONS

Research delineating the multiple neuroendocrine regulatory functions of the hypothalamus is important because it helps explain how endocrine secretion is matched to the demands of a changing environment. The nervous system receives information about changes in the internal and external environment from the sense organs. It brings about adjustments to these changes through effector mechanisms that include not only somatic movement but also changes in the rate at which hormones are secreted.

The manifestations of hypothalamic disease are neurologic defects, endocrine changes, and metabolic abnormalities such as hyperphagia and hyperthermia. The relative frequencies of the signs and symptoms of hypothalamic disease in one large series of cases are shown in **Table 17–2**. The possibility of hypothalamic pathology should be kept in mind in evaluating all patients with

pituitary dysfunction, especially those with isolated deficiencies of single pituitary tropic hormones.

TABLE 17–2 Symptoms and signs in 60 patients with hypothalamic disease.

Symptoms and Signs	Percentage of Cases
Endocrine and metabolic findings	
Precocious puberty	40
Hypogonadism	32
Diabetes insipidus	35
Obesity	25
Abnormalities of temperature regulation	22
Emaciation	18
Bulimia	8
Anorexia	7
Neurologic findings	
Eye signs	78
Pyramidal and sensory deficits	75
Headache	65
Extrapyramidal signs	62
Vomiting	40
Psychic disturbances, rage attacks, etc	35
Somnolence	30
Convulsions	15

Data from Bauer HG: Endocrine and other clinical manifestations of hypothalamic disease. J Clin Endocrinol 1954;14:13. See also Kahana L, et al: Endocrine manifestations of intracranial extrasellar lesions. J Clin Endocrinol 1962;22:304.

A condition of considerable interest in this context is **Kallmann syndrome,** the combination of hypogonadism due to low levels of circulating gonadotropins **(hypogonadotropic hypogonadism)** with partial or complete loss of the sense of smell (**hyposmia** or **anosmia**). Embryologically, GnRH neurons develop in the nose and migrate up the olfactory nerves and then through the brain to the hypothalamus. If this migration is prevented by congenital abnormalities in the olfactory pathways, the GnRH neurons do not reach the hypothalamus and pubertal maturation of the gonads does not occur. The syndrome is most common in men, and the cause in many cases is mutation of the *KALIG1* gene, a gene on the X chromosome that codes for an adhesion molecule necessary for

the normal development of the olfactory nerve. However, the condition also occurs in women and can be due to other genetic abnormalities.

TEMPERATURE REGULATION

In the body, heat is produced by muscular exercise, assimilation of food, and all the vital processes that contribute to the basal metabolic rate. It is lost from the body by radiation, conduction, and vaporization of water in the respiratory passages and on the skin. Small amounts of heat are also removed in the urine and feces. The balance between heat production and heat loss determines the body temperature. Because the speed of chemical reactions varies with temperature and because the enzyme systems of the body have narrow temperature ranges in which their function is optimal, normal body function depends on a relatively constant body temperature.

Invertebrates generally cannot adjust their body temperatures and so are at the mercy of the environment. Vertebrates have evolved mechanisms for maintaining body temperature by adjusting heat production and heat loss. In reptiles, amphibians, and fish, the adjusting mechanisms are relatively rudimentary, and these species are called “cold-blooded” **(poikilothermic)** because their body temperature fluctuates over a considerable range. In birds and mammals, the “warm-blooded” **(homeothermic)** animals, a group of reflex responses that are primarily integrated in the hypothalamus operate to maintain body temperature within a narrow range in spite of wide fluctuations in environmental temperature. The hibernating mammals are a partial exception. While awake they are homeothermic, but during hibernation their body temperature falls.

NORMAL BODY TEMPERATURE

In homeothermic animals, the actual temperature at which the body is maintained varies from species to species and, to a lesser degree, from individual to individual. In humans, the traditional normal value for the oral temperature is 37°C (98.6°F), but in one large series of normal young adults, the morning oral temperature averaged 36.7°C, with a standard deviation of 0.2°C. Therefore, 95% of all young adults would be expected to have a morning oral temperature of 36.3–37.1°C (97.3–98.8°F; mean ± 1.96 standard deviations). Various parts of the body are at different temperatures, and the magnitude of the temperature difference between the parts varies with the environmental temperature. The

extremities are generally cooler than the rest of the body. The temperature of the scrotum is carefully regulated at 32°C. The rectal temperature is representative of the temperature at the core of the body and varies least with changes in environmental temperature. The oral temperature is normally 0.5°C lower than the rectal temperature, but it is affected by many factors, including ingestion of hot or cold fluids, gum chewing, smoking, and mouth breathing.

The normal human core temperature undergoes a regular circadian fluctuation of 0.5–0.7°C. In individuals who sleep at night and are awake during the day (even when hospitalized at bed rest), it is lowest at about 6:00 AM and highest in the evenings **(Figure 17–12)**. It is lowest during sleep, is slightly higher in the awake but relaxed state, and rises with activity. In women, an additional monthly cycle of temperature variation is characterized by a rise in basal temperature at the time of ovulation (Figure 22–14). Temperature regulation is less precise in young children and they may normally have a temperature that is 0.5°C or so above the established norm for adults.

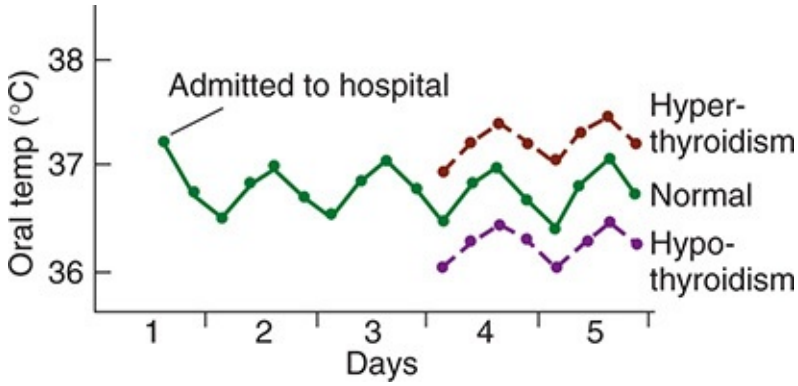


FIGURE 17–12 Typical temperature chart of a hospitalized patient who does not have a febrile disease. Note the slight rise in temperature, due to excitement and apprehension, at the time of admission to the hospital, and the regular circadian temperature cycle.

During exercise, the heat produced by muscular contraction accumulates in the body and the rectal temperature normally rises as high as 40°C (104°F). This rise is due in part to the inability of the heat-dissipating mechanisms to handle the greatly increased amount of heat produced, but evidence suggests that during exercise in addition there is an elevation of the body temperature at which the heat-dissipating mechanisms are activated. Body temperature also rises slightly during emotional excitement, probably owing to unconscious tensing of the muscles. It is chronically elevated by as much as 0.5°C when the metabolic rate

is high, as in hyperthyroidism, and lowered when the metabolic rate is low, as in hypothyroidism (Figure 17–12). Some apparently normal adults chronically have a temperature above the normal range (constitutional hyperthermia).

HEAT PRODUCTION

A variety of basic chemical reactions contribute to body heat production at all times. Ingestion of food increases heat production, but the major source of heat is the contraction of skeletal muscle **(Table 17–3)**. Heat production can be varied by endocrine mechanisms in the absence of food intake or muscular exertion. Epinephrine and norepinephrine produce a rapid but short-lived increase in heat production; thyroid hormones produce a slowly developing but prolonged increase. Furthermore, sympathetic discharge decreases during fasting and is increased by feeding.

TABLE 17–3 Body heat production and heat loss.

Body heat is produced by:	
Basic metabolic processes	
Food intake (specific dynamic action)	
Muscular activity	
Body heat is lost by:	**Percentage of heat lost at 21°C**
Radiation and conduction	70
Vaporization of sweat	27
Respiration	2
Urination and defecation	1

A source of considerable heat, particularly in infants, is **brown adipose tissue (BAT).** This fat has a high rate of metabolism (cells contain mitochondria), and its thermogenic function has been likened to that of an electric blanket. Cold-induced BAT activity is associated with enhanced thermogenesis and energy expenditure in humans, leading to beneficial effects on glucose metabolism and fat mass.

HEAT LOSS

The processes by which heat is lost from the body when the environmental

temperature is below body temperature are listed in Table 17–3. **Conduction** is heat exchange between objects or substances at different temperatures that are in contact with one another. A basic characteristic of matter is that its molecules are in motion, with the amount of motion proportional to the temperature. These molecules collide with the molecules in cooler objects, transferring thermal energy to them. The amount of heat transferred is proportional to the temperature difference between the objects in contact **(thermal gradient).** Conduction is aided by **convection,** the movement of molecules away from the area of contact. Thus, for example, an object in contact with air at a different temperature changes the specific gravity of the air, and because warm air rises and cool air falls, a new supply of air is brought into contact with the object. Of course, convection is greatly aided if the object moves about in the medium or the medium moves past the object, for example, if a subject swims through water or a fan blows air through a room. **Radiation** is the transfer of heat by infrared electromagnetic radiation from one object to another at a different temperature with which it is not in contact. When an individual is in a cold environment, heat is lost by conduction to the surrounding air and by radiation to cool objects in the vicinity. Conversely, of course, heat is transferred to an individual and the heat load is increased by these processes when the environmental temperature is above body temperature. Note that because of radiation, an individual can feel chilly in a room with cold walls even though the room is relatively warm. On a cold but sunny day, the heat of the sun reflected off bright objects exerts an appreciable warming effect. It is the heat reflected from the snow, for example, that in part makes it possible to ski in fairly light clothes even though the air temperature is below freezing.

Because conduction occurs from the surface of one object to the surface of another, the temperature of the skin determines to a large extent the degree to which body heat is lost or gained. The amount of heat reaching the skin from the deep tissues can be varied by changing the blood flow to the skin. When the cutaneous vessels are dilated, warm blood wells into the skin, whereas in the maximally vasoconstricted state, heat is held centrally in the body. The rate at which heat is transferred from the deep tissues to the skin is called the **tissue conductance.** Further, birds have a layer of feathers next to the skin, and most mammals have a significant layer of hair or fur. Heat is conducted from the skin to the air trapped in this layer and from the trapped air to the exterior. When the thickness of the trapped layer is increased by fluffing the feathers or erection of the hairs **(horripilation),** heat transfer across the layer is reduced and heat losses (or, in a hot environment, heat gains) are decreased. “Goose pimples” are the result of horripilation in humans; they are the visible manifestation of cold-

induced contraction of the piloerector muscles attached to the rather meager hair supply. Humans usually supplement this layer of hair with one or more layers of clothes. Heat is conducted from the skin to the layer of air trapped by the clothes, from the inside of the clothes to the outside, and from the outside of the clothes to the exterior. The magnitude of the heat transfer across the clothing, a function of its texture and thickness, is the most important determinant of how warm or cool the clothes feel, but other factors, especially the size of the trapped layer of warm air, are also important. Dark clothes absorb radiated heat and light-colored clothes reflect it back to the exterior.

The other major process transferring heat from the body in humans and other animals that sweat is vaporization of water on the skin and mucous membranes of the mouth and respiratory passages. Vaporization of 1 g of water removes about 0.6 kcal of heat. A certain amount of water is vaporized at all times. This **insensible water loss** amounts to 50 mL/h in humans. When sweat secretion is increased, the degree to which the sweat vaporizes depends on the humidity of the environment. It is common knowledge that one feels hotter on a humid day. This is due in part to the decreased vaporization of sweat, but even under conditions in which vaporization of sweat is complete, an individual in a humid environment feels warmer than an individual in a dry environment. The reason for this difference is unknown, but it seems related to the fact that in the humid environment sweat spreads over a greater area of skin before it evaporates. During muscular exertion in a hot environment, sweat secretion reaches values as high as 1600 mL/h, and in a dry atmosphere, most of this sweat is vaporized. Heat loss by vaporization of water therefore varies from 30 to over 900 kcal/h.

Some mammals lose heat by **panting.** This rapid, shallow breathing greatly increases the amount of water vaporization in the mouth and respiratory passages and therefore the amount of heat lost. Because the breathing is shallow, it produces relatively little change in the composition of alveolar air (see Chapter 34).

The relative contribution of each of the processes that transfer heat away from the body (Table 17–3) varies with the environmental temperature. At 21°C, vaporization is a minor component in humans at rest. As the environmental temperature approaches body temperature, radiation losses decline and vaporization losses increase.

TEMPERATURE-REGULATING MECHANISMS

The reflex and semireflex thermoregulatory responses in humans are listed in

Table 17–4. They include autonomic, somatic, endocrine, and behavioral changes. One group of responses increases heat loss and decreases heat production; the other decreases heat loss and increases heat production. In general, exposure to heat stimulates the former group of responses and inhibits the latter, whereas exposure to cold does the opposite.

TABLE 17–4 Temperature-regulating mechanisms.

Mechanisms activated by cold
Shivering
Hunger
Increased voluntary activity
Increased secretion of norepinephrine and epinephrine
Decreased heat loss
Cutaneous vasoconstriction
Curling up
Horripilation
Mechanisms activated by heat
Increased heat loss
Cutaneous vasodilation
Sweating
Increased respiration
Decreased heat production
Anorexia
Apathy and inertia

Curling up “in a ball” is a common reaction to cold in animals and has a counterpart in the position some people assume on climbing into a cold bed. Curling up decreases the body surface exposed to the environment. Shivering is an involuntary response of the skeletal muscles, but cold also causes a semiconscious general increase in motor activity. Examples include foot stamping and dancing up and down on a cold day. Increased catecholamine secretion is an important endocrine response to cold. Mice unable to make norepinephrine and epinephrine because their dopamine β-hydroxylase gene is

knocked out do not tolerate cold; they have deficient vasoconstriction and are unable to increase thermogenesis in BAT through UCP 1. TSH secretion is increased by cold and decreased by heat in laboratory animals, but the change in TSH secretion produced by cold in adult humans is small and of questionable significance. It is common knowledge that activity is decreased in hot weather—the "it's too hot to move" reaction.

Thermoregulatory adjustments involve local responses as well as more general reflex responses. When cutaneous blood vessels are cooled they become more sensitive to catecholamines and the arterioles and venules constrict. This local effect of cold directs blood away from the skin. Another heat-conserving mechanism that is important in animals living in cold water is heat transfer from arterial to venous blood in the limbs. The deep veins **(venae comitantes)** run alongside the arteries supplying the limbs and heat is transferred from the warm arterial blood going to the limbs to the cold venous blood coming from the extremities (**countercurrent exchange;** see Chapter 37). This limits the ability to maintain heat in the tips of the extremities but conserves body heat.

The reflex responses activated by cold are controlled from the posterior hypothalamus. Those activated by warmth are controlled primarily from the anterior hypothalamus, although some thermoregulation against heat still occurs after decerebration at the level of the rostral midbrain. Stimulation of the anterior hypothalamus causes cutaneous vasodilation and sweating, and lesions in this region cause hyperthermia, with rectal temperatures sometimes reaching 43°C (109.4°F). Posterior hypothalamic stimulation causes shivering, and the body temperature of animals with posterior hypothalamic lesions falls toward that of the environment.

AFFERENTS

The hypothalamus is said to integrate body temperature information from sensory receptors (primarily cold receptors) in the skin, deep tissues, spinal cord, extrahypothalamic portions of the brain, and the hypothalamus itself. Each of these five inputs contributes about 20% of the information that is integrated. There are threshold core temperatures for each of the main temperature-regulating responses and when the threshold is reached the response begins. The threshold is 37°C for sweating and vasodilation, 36.8°C for vasoconstriction, 36°C for nonshivering thermogenesis, and 35.5°C for shivering.

FEVER

Fever is perhaps the oldest and most universally known hallmark of disease. It occurs not only in mammals but also in birds, reptiles, amphibia, and fish. When it occurs in homeothermic animals, the thermoregulatory mechanisms behave as if they were adjusted to maintain body temperature at a higher than normal level, that is, “as if the thermostat had been reset” to a new point above 37°C. The temperature receptors then signal that the actual temperature is below the new set point, and the temperature-raising mechanisms are activated. This usually produces chilly sensations due to cutaneous vasoconstriction and occasionally enough shivering to produce a shaking chill. However, the nature of the response depends on the ambient temperature. The temperature rise in experimental animals injected with a pyrogen is due mostly to increased heat production if they are in a cold environment and mostly to decreased heat loss if they are in a warm environment.

The pathogenesis of fever is summarized in **Figure 17–13**. Toxins from bacteria, such as endotoxin, act on monocytes, macrophages, and Kupffer cells to produce cytokines that act as **endogenous pyrogens (EPs).** There is good evidence that IL-1β, IL-6, IFN-β, IFN-γ, and TNF-α (see Chapter 3) can act independently to produce fever. These circulating cytokines are polypeptides and it is unlikely that they penetrate the brain. Instead, evidence suggests that they act on the OVLT, one of the circumventricular organs (see Chapter 33). This in turn activates the preoptic area of the hypothalamus. Cytokines are also produced by cells in the central nervous system (CNS) when these are stimulated by infection, and these may act directly on the thermoregulatory centers.

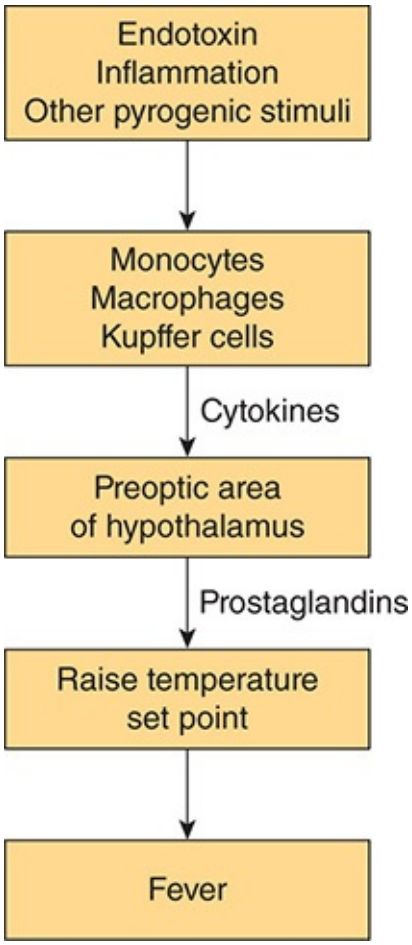


FIGURE 17–13 Pathogenesis of fever.

The fever produced by cytokines is probably due to local release of prostaglandins in the hypothalamus. Intrahypothalamic injection of prostaglandins produces fever. In addition, the antipyretic effect of aspirin is exerted directly on the hypothalamus, and aspirin inhibits prostaglandin synthesis. PGE_2 is one of the prostaglandins that causes fever. It acts on four subtypes of prostaglandin receptors—EP_1, EP_2, EP_3, and EP_4—and knockout of the EP_3 receptor impairs the febrile response to PGE_2, IL-1β, and endotoxin, or bacterial lipopolysaccharide (LPS).

The benefit of fever to the organism is uncertain. A beneficial effect is assumed because fever has evolved and persisted as a response to infections and other diseases. Many microorganisms grow best within a relatively narrow temperature range and a rise in temperature inhibits their growth. In addition, antibody production is increased when body temperature is elevated. Before the advent of antibiotics, fevers were artificially induced for the treatment of

neurosyphilis and proved to be beneficial. Hyperthermia also benefits individuals infected with anthrax; pneumococcal pneumonia; leprosy; and various fungal, rickettsial, and viral diseases. Hyperthermia also slows the growth of some tumors. However, very high temperatures are harmful. A rectal temperature over 41°C (106°F) for prolonged periods results in some permanent brain damage. When the temperature is over 43°C, heat stroke develops and death is common.

In **malignant hyperthermia,** various mutations of the gene coding for the ryanodine receptor (see Chapter 5) lead to excess Ca^{2+} release during muscle contraction triggered by stress. This in turn leads to contractures of the muscles, increased muscle metabolism, and a great increase in heat production in muscle. The increased heat production causes a marked rise in body temperature that is fatal if not treated.

Periodic fevers also occur in humans with mutations in the gene for **pyrin,** a protein found in neutrophils; the gene for mevalonate kinase, an enzyme involved in cholesterol synthesis; and the gene for the type 1 TNF receptor, which is involved in inflammatory responses. However, how any of these three mutant gene products cause fever is unknown.

HYPOTHERMIA

In hibernating mammals, body temperature drops to low levels without causing any demonstrable ill effects on subsequent arousal. This observation led to experiments on induced hypothermia. When the skin or the blood is cooled enough to lower the body temperature in nonhibernating animals or in humans, metabolic and physiologic processes slow down. Respiration and heart rate are very slow, blood pressure is low, and consciousness is lost. At rectal temperatures of about 28°C, the ability to spontaneously return the temperature to normal is lost, but the individual continues to survive and, if rewarmed with external heat, returns to a normal state. If care is taken to prevent the formation of ice crystals in the tissues, the body temperature of experimental animals can be lowered to subfreezing levels without producing any detectable damage after subsequent rewarming.

Humans tolerate body temperatures of 21–24°C (70–75°F) without permanent ill effects, and induced hypothermia has been used in surgery. On the other hand, accidental hypothermia due to prolonged exposure to cold air or cold water is a serious condition and requires careful monitoring and prompt rewarming.

CHAPTER SUMMARY

- Neural connections run between the hypothalamus and the posterior lobe of the pituitary gland, and vascular connections between the hypothalamus and the anterior lobe of the pituitary.
- In most mammals, the hormones secreted by the posterior pituitary gland are vasopressin and oxytocin. Vasopressin increases the permeability of the collecting ducts of the kidney to water, thus concentrating the urine. Oxytocin acts on the breasts (lactation) and the uterus (contraction).
- The anterior pituitary secretes six hormones: adrenocorticotropic hormone (corticotropin, ACTH), thyroid-stimulating hormone (thyrotropin, TSH), growth hormone, follicle-stimulating hormone (FSH), luteinizing hormone (LH), and prolactin (PRL).
- Other complex autonomic mechanisms that maintain the chemical constancy and temperature of the internal environment, such as thirst and temperature regulation, are integrated in the hypothalamus.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Thirst is stimulated by

   A. increases in plasma osmolality and volume.

   B. an increase in plasma osmolality and a decrease in plasma volume.

   C. a decrease in plasma osmolality and an increase in plasma volume.

   D. decreases in plasma osmolality and volume.

   E. injection of vasopressin into the hypothalamus.

2. When an individual is naked in a room in which the air temperature is 21°C (69.8°F) and the humidity 80%, the greatest amount of heat is lost from the body by

   A. elevated metabolism.

   B. respiration.

   C. urination.

   D. vaporization of sweat.

   E. radiation and conduction.

In questions 3–8, select A if the item is associated with (a), B if the item is associated with (b), C if the item is associated with both (a) and (b), and D if the item is associated with neither (a) nor (b).

(a) V_{1A} vasopressin receptors

(b) V_2 vasopressin receptors

3. Activation of G_s
4. Vasoconstriction
5. Increase in intracellular inositol triphosphate
6. Movement of aquaporin
7. Proteinuria
8. Milk ejection

CHAPTER 18

The Pituitary Gland

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the development and structure of the pituitary gland and its relationship to the hypothalamus.
- Identify the hormones secreted by the anterior and posterior lobes of the pituitary and their target organs, and how the numbers of the various cell types in the anterior pituitary are controlled in response to physiologic demands.
- Understand the function of hormones derived from proopiomelanocortin (POMC) and how they are involved in regulating skin coloration.
- Describe how growth hormone is secreted from the anterior pituitary and circulates and activates its receptors, and the stimuli that regulate growth hormone secretion with their underlying mechanisms.
- Understand the role of growth hormone in growth and metabolic function, and how somatomedins (such as insulin-like growth factors) may mediate some of its actions in the periphery.
- Define the normal timeline of growth in humans and identify factors in addition to growth hormone that contribute to its regulation.
- Understand pituitary secretion of gonadotropins and prolactin, how these are

regulated, and the actions of these hormones on reproductive tissues.

- Understand the basis of conditions where pituitary function is abnormal, and how they can be treated.

INTRODUCTION

The pituitary gland, or hypophysis, lies in a pocket of the sphenoid bone at the base of the brain and is closely related to the hypothalamus (see Figure 17–2). It is a coordinating center for control of many downstream endocrine glands, some of which are discussed in subsequent chapters. In many ways, it can be considered to consist of two separate endocrine organs that contain a plethora of hormonally active substances. The anterior lobe of the pituitary secretes **thyroid-stimulating hormone (TSH, thyrotropin), adrenocorticotropic hormone (ACTH), luteinizing hormone (LH), follicle-stimulating hormone (FSH), prolactin,** and **growth hormone** (see Figure 17–9), and receives almost all of its blood supply from the portal hypophysial vessels that pass initially through the median eminence, a structure immediately below the hypothalamus. This vascular arrangement positions the cells of the anterior pituitary to respond efficiently to regulatory factors released from the hypothalamus. Of the listed hormones, prolactin acts on the breast. The remaining five are, at least in part, **tropic hormones;** that is, they stimulate secretion of hormonally active substances by other endocrine glands or, in the case of growth hormone, the liver and other tissues (see below). The tropic hormones for some endocrine glands are discussed in the chapter on that gland: TSH in Chapter 19 and ACTH in Chapter 20. However, the gonadotropins FSH and LH, along with prolactin, are covered here. In some species, there is a well-developed intermediate lobe of the pituitary, which contains hormonally active derivatives of the proopiomelanocortin (POMC) molecule that regulate skin pigmentation, among other functions (see below). In humans, the intermediate lobe is rudimentary, and the cells that secrete derivatives of POMC are present in the anterior pituitary.

The posterior lobe of the pituitary consists predominantly of nerves that have their cell bodies in the hypothalamus, and stores **oxytocin** and **vasopressin** in the termini of these neurons, to be released into the bloodstream. The secretion of these hormones, as well as a discussion of the overall role of the hypothalamus and median eminence in regulating both the anterior and posterior pituitary, was covered in Chapter 17.

To avoid redundancy, this chapter will focus predominantly on growth

hormone and its role in growth and facilitating the activity of other hormones, along with a number of general considerations about the pituitary. The melanocyte-stimulating hormones (MSHs) of the intermediate lobe of the pituitary, α-MSH and β-MSH, will also be touched upon.

DEVELOPMENT, STRUCTURE, & CELL TYPES OF THE PITUITARY

GROSS ANATOMY

The anatomy of the pituitary gland is summarized in **Figure 18–1** and discussed in detail in Chapter 17. The posterior pituitary is made up largely of the endings of axons from the supraoptic and paraventricular nuclei of the hypothalamus and arises initially as an extension of this structure. The anterior pituitary, on the other hand, contains endocrine cells that store its characteristic hormones and arises embryologically as an invagination of the pharynx **(Rathke pouch).** In species where it is well developed, the intermediate lobe is formed in the embryo from the dorsal half of Rathke pouch, but is closely adherent to the posterior lobe in the adult. It is separated from the anterior lobe by the remains of the cavity in Rathke pouch, the **residual cleft.**

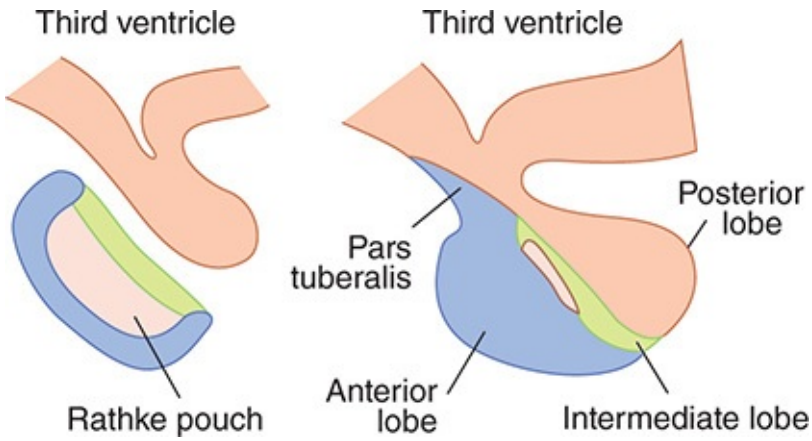


FIGURE 18–1 Diagrammatic outline of the formation of the pituitary (left) and the various parts of the organ in the adult (right).

HISTOLOGY

In the posterior lobe, the endings of the supraoptic and paraventricular axons can

be observed in close relation to blood vessels. **Pituicytes,** stellate cells that are modified astrocytes, are also present.

As noted above, the intermediate lobe is rudimentary in humans—most of its cells are incorporated in the anterior lobe. The anterior pituitary is made up of interlacing cell cords and an extensive network of sinusoidal capillaries. The endothelium of the capillaries is fenestrated, like that in other endocrine organs. The cells contain granules of stored hormone that are extruded from the cells by exocytosis. Their constituents then enter the capillaries to be conveyed to target tissues.

CELL TYPES IN THE ANTERIOR PITUITARY

Five types of secretory cells have been identified in the anterior pituitary by immunocytochemistry and electron microscopy. The cell types are the somatotropes, which secrete growth hormone; the lactotropes (also called mammotropes), which secrete prolactin; the corticotropes, which secrete ACTH; the thyrotropes, which secrete TSH; and the gonadotropes, which secrete FSH and LH. The characteristics of these cells are summarized in **Table 18–1**. Some cells may contain two or more hormones. It is also notable that the three pituitary glycoprotein hormones, FSH, LH, and TSH, while being made up of two subunits, all share a common α subunit that is the product of a single gene and has the same amino acid composition in each hormone, although their carbohydrate residues vary. The α subunit must be combined with a β subunit characteristic of each hormone for maximal physiologic activity. The β subunits, which are produced by separate genes and differ in structure, confer hormonal specificity (see Chapter 16). The α subunits are remarkably interchangeable and hybrid molecules can be created.

TABLE 18–1 Hormone-secreting cells of the human anterior pituitary gland.

Cell Type	Hormones Secreted	Percentage of Total Secretory Cells
Somatotrope	Growth hormone	50
Lactotrope	Prolactin	10–30
Corticotrope	ACTH	10
Thyrotrope	TSH	5
Gonadotrope	FSH, LH	20

ACTH, adrenocorticotropic hormone; FSH, follicle-stimulating hormone; LH, luteinizing hormone; TSH, thyroid-stimulating hormone.

The anterior pituitary also contains folliculostellate cells that send processes between the granulated secretory cells. These cells produce paracrine factors that regulate the growth and function of the secretory cells discussed above. Indeed, the anterior pituitary can adjust the relative proportion of secretory cell types to meet varying requirements for different hormones at different life stages. This plasticity has recently been ascribed to the presence of a small number of pluripotent stem cells that persist in the adult gland.

FORMATION AND FUNCTION OF PROOPIOMELANOCORTIN & DERIVATIVES

BIOSYNTHESIS

Corticotropes of the anterior lobe (or intermediate lobe cells, when present) synthesize a large precursor protein that is cleaved to form a family of hormones. Removal of the signal peptide results in the formation of the prohormone POMC. This molecule is also synthesized in the hypothalamus, the lungs, the gastrointestinal tract, and the placenta. The structure of POMC, as well as its derivatives, is shown in **Figure 18–2**. In corticotropes, it is hydrolyzed to ACTH and β-lipotropin (β-LPH), plus a small amount of β-endorphin, and these substances are secreted. Predominantly in intermediate lobe cells, POMC is further hydrolyzed to corticotropin-like intermediate-lobe peptide (CLIP), γ-LPH, and appreciable quantities of β-endorphin. The functions, if any, of CLIP and γ-LPH are unknown, whereas β-endorphin is an opioid peptide (see Chapter

7) that has the five amino acid residues of met-enkephalin at its amino terminal end. The **melanotropins** α- and β-MSH are also formed but apparently are not secreted in adult humans. In some species, however, these melanotropins have important physiologic functions, as discussed below.

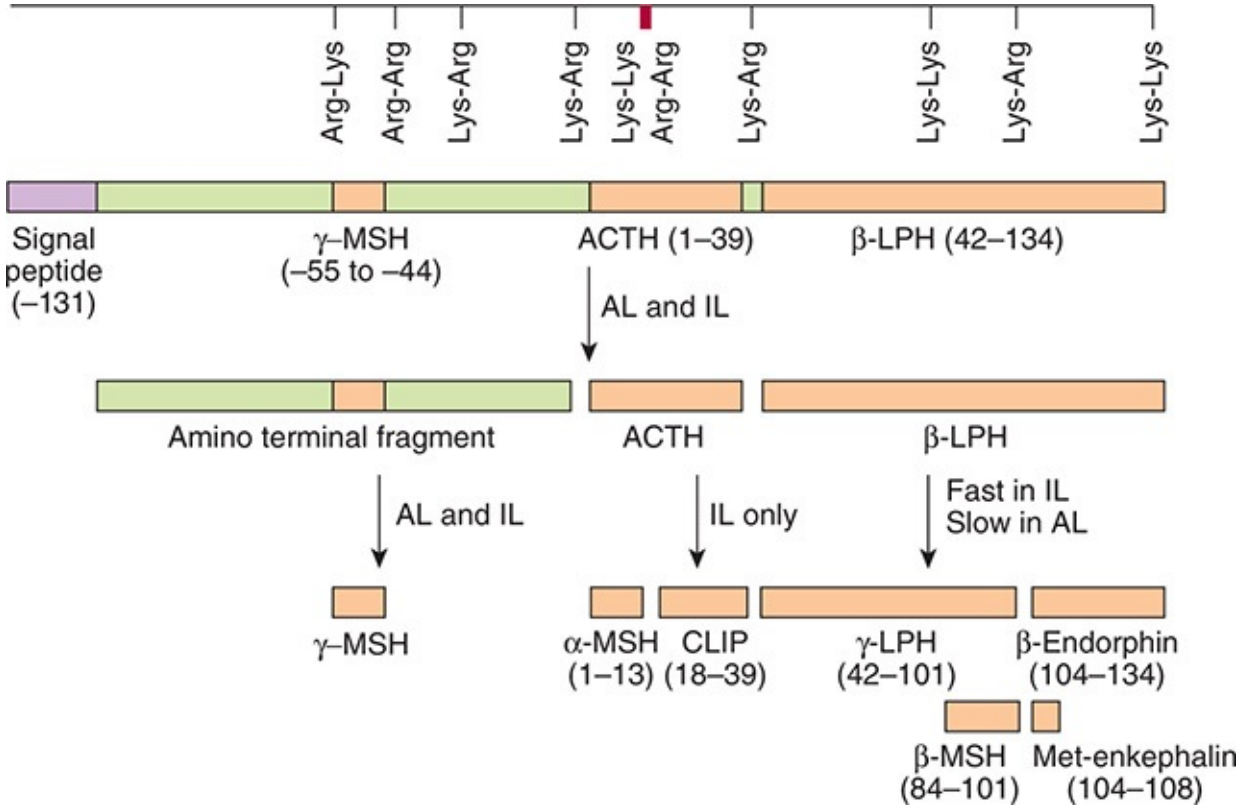


FIGURE 18–2 Schematic representation of the preproopiomelanocortin molecule formed in pituitary cells, neurons, and other tissues. The numbers in parentheses identify the amino acid sequences in each of the polypeptide fragments. The locations of Lys–Arg and other pairs of basic amino acids residues are also indicated; these are the sites of proteolytic cleavage in the formation of the smaller fragments of the parent molecule. ACTH, adrenocorticotropic hormone; AL, anterior lobe; CLIP, corticotropin-like intermediate-lobe peptide; IL, intermediate lobe; LPH, lipotropin; MSH, melanocyte-stimulating hormone.

CONTROL OF SKIN COLORATION & PIGMENT ABNORMALITIES

Fish, reptiles, and amphibia change the color of their skin for thermoregulation,

camouflage, and behavioral displays. They do this in part by moving black or brown granules into or out of the periphery of pigment cells called **melanophores.** The granules are made up of **melanins,** which are synthesized from dopamine (see Chapter 7) and dopaquinone. The movement of these granules is controlled by a variety of hormones and neurotransmitters, including α- and β-MSH, melanin-concentrating hormone, melatonin, and catecholamines.

Mammals have no melanophores containing pigment granules that disperse and aggregate, but they do have **melanocytes,** which have multiple processes containing melanin granules. Melanocytes express **melanotropin-1** receptors. Treatment with MSHs accelerates melanin synthesis, causing readily detectable darkening of the skin in humans in 24 h. As noted above, α- and β-MSH do not circulate in adult humans, and their function is unknown. However, ACTH binds to melanotropin-1 receptors. Indeed, pigmentary changes in several human endocrine diseases are due to changes in circulating ACTH. For example, abnormal pallor is a hallmark of hypopituitarism. Hyperpigmentation occurs in patients with adrenal insufficiency due to primary adrenal disease. Indeed, the presence of hyperpigmentation in association with adrenal insufficiency rules out the possibility that the insufficiency is secondary to pituitary or hypothalamic disease because in these conditions, plasma ACTH is not increased (see Chapter 20). Other disorders of pigmentation result from peripheral mechanisms. Thus, **albinos** have a congenital inability to synthesize melanin resulting from a variety of different genetic defects in the pathways for melanin synthesis. **Piebaldism** is characterized by patches of skin that lack melanin as a result of congenital defects in the migration of pigment cell precursors from the neural crest during embryonic development. Not only the condition but also the precise pattern of the loss is passed from one generation to the next. **Vitiligo** involves a similar patchy loss of melanin, but the loss develops progressively after birth secondary to an autoimmune process that targets melanocytes.

GROWTH HORMONE SECRETION

BIOSYNTHESIS & CHEMISTRY

The long arm of human chromosome 17 contains the growth hormone-hCS cluster that contains five genes: one, *hGH-N*, codes for the most abundant ("normal") form of growth hormone; a second, *hGH-V*, codes for the variant form of growth hormone; two code for human chorionic somatomammotropin (hCS) (see Chapter 22); and the fifth is probably an hCS pseudogene. Only

hGH-N is secreted by the pituitary; hGH-V and hCS are primarily products of the placenta, and as a consequence are only found in appreciable quantities in the circulation during pregnancy (see Chapter 22).

The structure of growth hormone varies considerably from one species to another. Porcine and simian growth hormones have only a transient effect in the guinea pig. In monkeys and humans, bovine and porcine growth hormones do not even have a transient effect on growth, although monkey and human growth hormones are fully active in both monkeys and humans. These facts are relevant to public health discussions surrounding the presence of bovine growth hormones (used to increase milk production) in dairy products, as well as the popularity of growth hormone supplements, marketed via the Internet, with body builders. Controversially, recombinant human growth hormone has also been given to children who are short in stature, but otherwise healthy (ie, without growth hormone deficiency), with apparently limited results.

PLASMA LEVELS, BINDING, & METABOLISM

A portion of circulating growth hormone is bound to a plasma protein that is a large fragment of the extracellular domain of the growth hormone receptor (see below). It appears to be produced by cleavage of receptors in humans, and its concentration is an index of the number of growth hormone receptors in the tissues. Approximately 50% of the circulating pool of growth hormone activity is in the bound form, providing a reservoir of the hormone to compensate for the wide fluctuations that occur in secretion (see below).

The basal plasma growth hormone level measured by radioimmunoassay in adult humans is normally less than 3 ng/mL. This represents both the protein-bound and free forms. Growth hormone is metabolized rapidly, at least in part in the liver. The half-life of circulating growth hormone in humans is 6–20 min, and the daily growth hormone output has been calculated to be 0.2–1.0 mg/d in adults.

HYPOTHALAMIC & PERIPHERAL CONTROL OF GROWTH HORMONE SECRETION

The secretion of growth hormone is not stable over time. Adolescents have the highest circulating levels of growth hormone, followed by children and finally adults. Levels decline in old age, and there has been considerable interest in

injecting growth hormone to counterbalance the effects of aging. There are also diurnal variations in growth hormone secretion superimposed on these developmental stages. Growth hormone is found at relatively low levels during the day, unless specific triggers for its release are present (see below). During sleep, on the other hand, large pulsatile bursts of growth hormone secretion occur. Therefore, it is not surprising that the secretion of growth hormone is under hypothalamic control. The hypothalamus controls growth hormone production by secreting **growth hormone–releasing hormone** (GHRH) as well as somatostatin, which inhibits growth hormone release (see Chapter 17). Thus, the balance between the effects of these hypothalamic factors on the pituitary will determine the level of growth hormone release. The stimuli of growth hormone secretion can therefore act by increasing hypothalamic secretion of GHRH, decreasing secretion of somatostatin, or both. A third regulator of growth hormone secretion is **ghrelin**. The main site of ghrelin synthesis and secretion is the stomach, but it is also produced in the hypothalamus and has marked growth hormone–stimulating activity. In addition, it appears to be involved in the regulation of food intake (see Chapter 26).

Growth hormone secretion is under feedback control (see Chapter 16), like the secretion of other anterior pituitary hormones. It acts on the hypothalamus to antagonize GHRH release. Growth hormone also increases circulating IGF-I, and IGF-I in turn exerts a direct inhibitory action on growth hormone secretion from the pituitary. It also stimulates somatostatin secretion **(Figure 18–3)**.

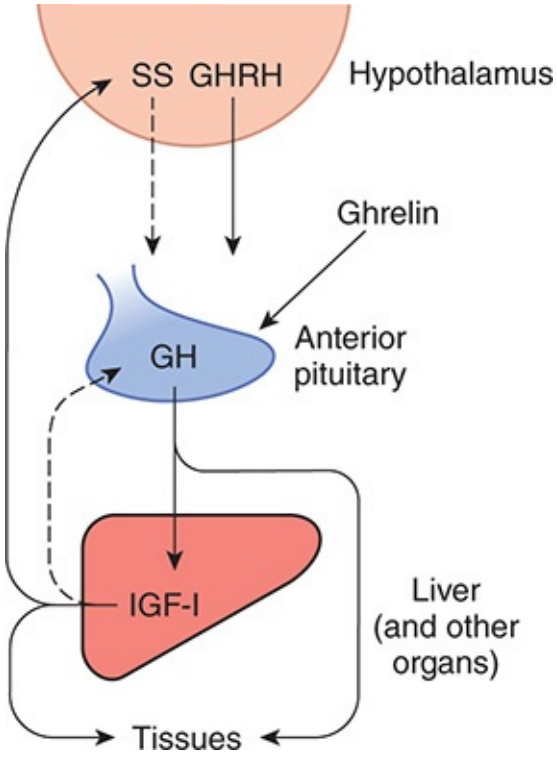


FIGURE 18–3 Feedback control of growth hormone secretion. Solid arrows represent positive effects and dashed arrows represent inhibition. GH, growth hormone; GHRH, growth hormone–releasing hormone; IGF-I, insulin-like growth factor-I; SS, somatostatin.

Other Stimuli Affecting Growth Hormone Secretion

The basal plasma growth hormone concentration ranges from 0 to 3 ng/mL in normal adults. However, secretory rates cannot be estimated from single values because of their irregular nature. Thus, average values over 24 h (see below) and peak values may be more meaningful, albeit difficult to assess. The stimuli that increase and decrease growth hormone secretion are summarized in **Table 18–2**. The stimuli that increase secretion fall into three general categories: (1) conditions such as hypoglycemia and/or fasting in which there is an actual or threatened decrease in the substrate for energy production in cells, (2) conditions in which the amounts of certain amino acids are increased in the plasma, and (3) stressful stimuli. Growth hormone secretion is also increased in persons deprived of rapid eye movement (REM) sleep (see Chapter 14) and inhibited during normal REM sleep.

TABLE 18–2 Stimuli that affect growth hormone secretion in humans.

Stimuli that increase secretion
Hypoglycemia
2-Deoxyglucose
Exercise
Fasting
Increase in circulating levels of certain amino acids
Protein meal
Infusion of arginine and some other amino acids
Glucagon
Lysine vasopressin
Going to sleep
L-dopa and α-adrenergic agonists that penetrate the brain
Apomorphine and other dopamine receptor agonists
Estrogens and androgens
Stressful stimuli (including various psychological stresses)
Pyrogen
Stimuli that decrease secretion
REM sleep
Glucose
Cortisol
FFA
Medroxyprogesterone
Growth hormone and IGF-I

FFA, free fatty acid; IGF, insulin-like growth factor; REM, rapid eye movement.

Glucose infusions lower plasma growth hormone levels and inhibit the response to exercise. The increase produced by 2-deoxyglucose is presumably due to intracellular glucose deficiency, since this compound blocks the catabolism of glucose-6-phosphate. Sex hormones induce growth hormone secretion, increase growth hormone responses to provocative stimuli such as arginine and insulin, and also serve as permissive factors for the action of growth

hormone in the periphery. This likely contributes to the relatively high levels of circulating growth hormone and associated growth spurt in puberty. Growth hormone secretion is also induced by thyroid hormones. Growth hormone secretion is inhibited, on the other hand, by cortisol, free fatty acids (FFA), and medroxyprogesterone.

Growth hormone secretion is increased by L-dopa, which increases the release of dopamine and norepinephrine in the brain, and by the dopamine receptor agonist apomorphine.

GROWTH HORMONE RECEPTORS

The receptor that mediates the cellular effects of growth hormone has a large extracellular portion, a transmembrane domain, and a large cytoplasmic portion. It is a member of the cytokine receptor superfamily, which is discussed in Chapter 3. Growth hormone has two domains that can bind to its receptor, and when it binds to one receptor, the second binding site attracts another, producing a homodimer **(Figure 18–4)**. Dimerization is essential for receptor activation.

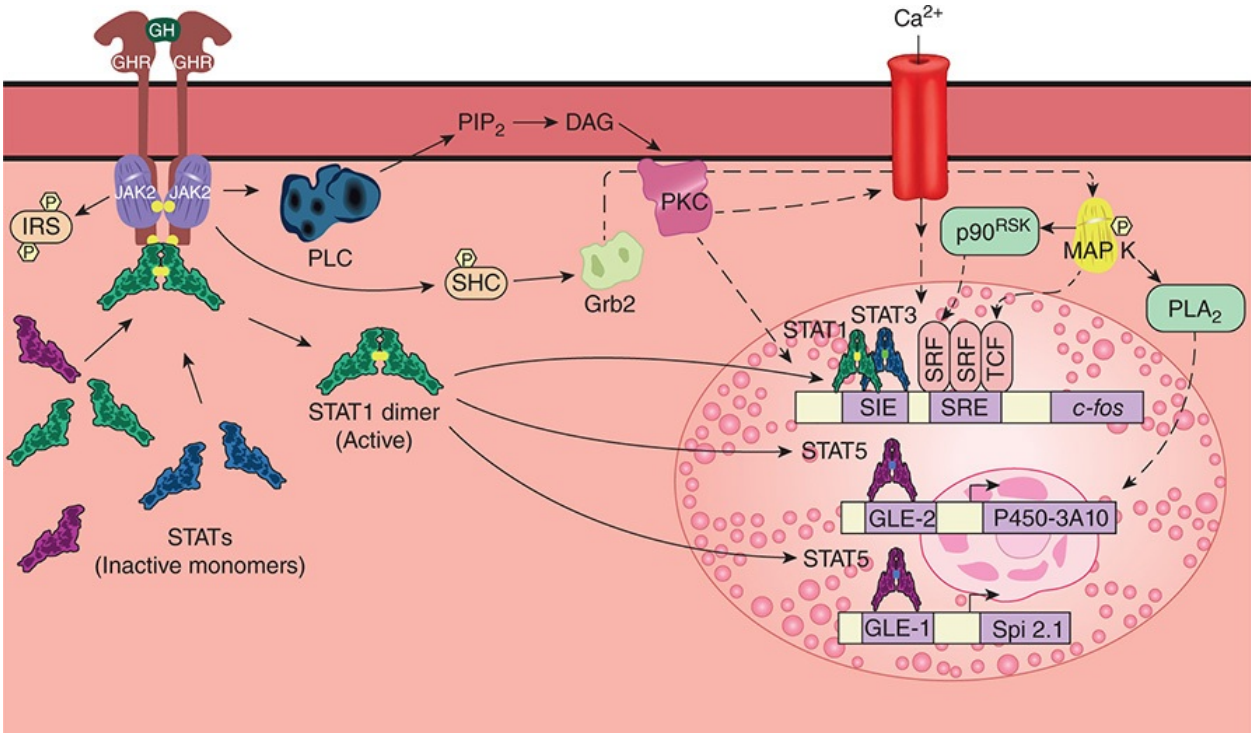


FIGURE 18–4 Some of the principal signaling pathways activated by the dimerized growth hormone receptor (GHR). Solid arrows indicate established pathways; dashed arrows indicate probable pathways. The details of the PLC

pathway and the pathway from Grb2 to MAP K are discussed in Chapter 2. The small uppercase letter Ps in yellow hexagons represent phosphorylation of the factor indicated. GLE-1 and GLE-2, interferon γ-activated response elements; IRS, insulin receptor substrate; $p90^{RSK}$, an S6 kinase; PLA_2, phospholipase A_2; SIE, Sis-induced element; SRE, serum response element; SRF, serum response factor; TCF, ternary complex factor.

Growth hormone has widespread effects in the body (see below), so even though it is not yet possible precisely to correlate intracellular and whole body effects, it is not surprising that, like insulin, growth hormone activates many different intracellular signaling cascades (Figure 18–4). Of particular note is its activation of the JAK2–STAT pathway. JAK2 is a member of the Janus family of cytoplasmic tyrosine kinases. STATs (for signal transducers and activators of transcription) are a family of cytoplasmic transcription factors that, upon phosphorylation by JAK kinases, migrate to the nucleus where they activate various genes. JAK–STAT pathways are known also to mediate the effects of prolactin and various other growth factors.

EFFECTS OF GROWTH HORMONE ON GROWTH

In young animals in which the epiphyses have not yet fused to the long bones (see Chapter 21), growth is inhibited by hypophysectomy and stimulated by growth hormone. Chondrogenesis is accelerated, and as the cartilaginous epiphysial plates widen, they lay down more bone matrix at the ends of long bones. In this way, stature is increased. Prolonged treatment of animals with growth hormone leads to gigantism.

When the epiphyses are closed, linear growth is no longer possible. In this case, an overabundance of growth hormone produces the pattern of bone and soft tissue deformities known in humans as **acromegaly.** The sizes of most of the viscera are increased. The protein content of the body is increased, and the fat content is decreased **(Clinical Box 18–1)**.

EFFECTS FOR GROWTH HORMONE ON PROTEIN & ELECTROLYTE HOMEOSTASIS

Growth hormone is a protein anabolic hormone and produces a positive nitrogen and phosphorus balance, a rise in plasma phosphorus, and a fall in blood urea nitrogen and amino acid levels. In adults with growth hormone deficiency, recombinant human growth hormone produces an increase in lean body mass and a decrease in body fat, along with an increase in metabolic rate and a fall in plasma cholesterol. Gastrointestinal absorption of Ca^{2+} is increased. Na^{+} and K^{+} excretion is reduced by an action independent of the adrenal glands, probably because these electrolytes are diverted from the kidneys to the growing tissues. On the other hand, excretion of the amino acid 4-hydroxyproline is increased during this growth, reflective of the ability of growth hormone to stimulate the synthesis of soluble collagen.

CLINICAL BOX 18–1

Gigantism & Acromegaly

Tumors of the somatotropes of the anterior pituitary (pituitary adenomas) secrete large amounts of growth hormone, leading to **gigantism** in children and to **acromegaly** in adults. If the tumor arises before puberty, the individual may grow to an extraordinary height. After linear growth is no longer possible, on the other hand, the characteristic features of acromegaly arise, including greatly enlarged hands and feet, vertebral changes attributable to osteoarthritis, soft tissue swelling, hirsutism, and protrusion of the brow and jaw. Abnormal growth of internal organs may eventually impair their function such that the condition, which has an insidious onset, can prove fatal if left untreated. Hypersecretion of growth hormone is accompanied by hypersecretion of prolactin in 20–40% of patients with acromegaly. About 25% of patients have abnormal glucose tolerance tests, and 4% develop lactation in the absence of pregnancy. Acromegaly can be caused by extra-pituitary as well as intrapituitary growth hormone–secreting tumors and by hypothalamic tumors that secrete GHRH, but the latter are rare.

THERAPEUTIC HIGHLIGHTS

The mainstay of therapy for acromegaly remains the use of somatostatin analogues that inhibit the secretion of growth hormone. A growth hormone receptor antagonist has become available and has been found to produce clinical improvement in cases of acromegaly that do not respond to other treatments. Surgical removal of the pituitary tumor is also helpful in both

acromegaly and gigantism, but sometimes challenging to perform due to the tumor's often invasive nature. In any case, adjuvant pharmacologic therapy must often be continued after surgery to control ongoing symptoms.

EFFECTS OF GROWTH HORMONE ON CARBOHYDRATE & FAT METABOLISM

The actions of growth hormone on carbohydrate metabolism are discussed in Chapter 24. At least some forms of growth hormone are diabetogenic because they increase hepatic glucose output and exert an anti-insulin effect in muscle. Growth hormone is also ketogenic and increases circulating FFA levels. The increase in plasma FFA, which takes several hours to develop, provides a ready source of energy for the tissues during hypoglycemia, fasting, and stressful stimuli. Growth hormone does not stimulate β cells of the pancreas directly, but it increases the ability of the pancreas to respond to insulinogenic stimuli such as arginine and glucose. This is an additional way growth hormone promotes growth, since insulin has a protein anabolic effect (see Chapter 24).

ROLE OF SOMATOMEDINS IN RESPONSE TO GROWTH HORMONE

The effects of growth hormone on growth, cartilage, and protein metabolism depend on an interaction between growth hormone and **somatomedins,** which are polypeptide growth factors secreted by the liver and other tissues. The first of these factors isolated was called sulfation factor because it stimulated the incorporation of sulfate into cartilage. However, it also stimulated collagen formation, and its name was changed to somatomedin. It then became clear that there are a variety of different somatomedins and that they are members of an increasingly large family of growth factors that affect many different tissues and organs.

In humans probably the only circulating somatomedins are **insulin-like growth factor I (IGF-I, somatomedin C)** and **IGF-II.** These factors are closely related to insulin, except that their C chains are not separated **(Figure 18–5)** and they have an extension of the A chain called the D domain. The hormone relaxin (see Chapter 22) is also a member of this family. Humans have two related

relaxin isoforms, and both resemble IGF-II.



FIGURE 18–5 Structure of human IGF-I, IGF-II, and insulin (ins) (top). The lower panel shows the structure of human IGF-II with its disulfide bonds, as well as three variant structures: a 21-aa extension of the C-terminus, a tetrapeptide substitution at Ser-29, and a tripeptide substitution of Ser-33.

The properties of insulin, IGF-I, and IGF-II are compared in **Table 18–3**. Both IGF-I and IGF-II are tightly bound to proteins in the plasma, prolonging their half-life in the circulation. The contribution of the IGFs to the insulin-like activity in blood is discussed in Chapter 24. The IGF-I receptor is very similar to the insulin receptor and probably uses similar or identical intracellular signaling pathways. The IGF-II receptor has a distinct structure (see Figure 24–5) and is involved in the targeting of acid hydrolases and other proteins to intracellular organelles. Secretion of IGF-I is independent of growth hormone before birth but is stimulated by growth hormone after birth, and it has pronounced growth-stimulating activity. Its concentration in plasma rises during childhood and peaks at the time of puberty, then declines to low levels in old age. IGF-II is largely independent of growth hormone and plays a role in the growth of the fetus before birth. In human fetuses in which it is overexpressed, several organs,

especially the tongue, other muscles, kidneys, heart, and liver, develop out of proportion to the rest of the body. In adults, the gene for IGF-II is expressed only in the choroid plexus and meninges.

TABLE 18–3 Comparison of insulin and the insulin-like growth factors (IGFs).

	Insulin	IGF-I	IGF-II
Other names	—	Somatomedin C	Multiplication-stimulating activity (MSA)
Number of amino acids	51	70	67
Source	Pancreatic β cells	Liver and other tissues	Diverse tissues
Level regulated by	Glucose	Growth hormone after birth, nutritional status	Unknown
Plasma levels	0.3–2 ng/mL	10–700 ng/mL; peaks at puberty	300–800 ng/mL
Plasma-binding proteins	No	Yes	Yes
Major physiologic role	Control of metabolism	Skeletal and cartilage growth	Growth during fetal development

DIRECT & INDIRECT ACTIONS OF GROWTH HORMONE

The understanding of the mechanism of action of growth hormone has evolved. It was originally thought to produce growth by a direct action on tissues, and then later was believed to act solely through its ability to induce somatomedins. However, if growth hormone is injected into one proximal tibial epiphysis, a

unilateral increase in cartilage width is produced, and cartilage, like other tissues, makes IGF-I. A current hypothesis to explain these results holds that growth hormone acts on cartilage to convert stem cells into cells that respond to IGF-I. Locally produced as well as circulating IGF-I then makes the cartilage grow. However, the independent role of circulating IGF-I remains important, since infusion of IGF-I in hypophysectomized rats restores bone and body growth. Overall, it seems that growth hormone and somatomedins can act both in cooperation and independently to stimulate pathways that lead to growth.

Figure 18–6 is a summary of current views of the actions of growth hormone and IGF-I. However, growth hormone probably combines with circulating and locally produced IGF-I in various proportions to produce at least some of the latter effects.

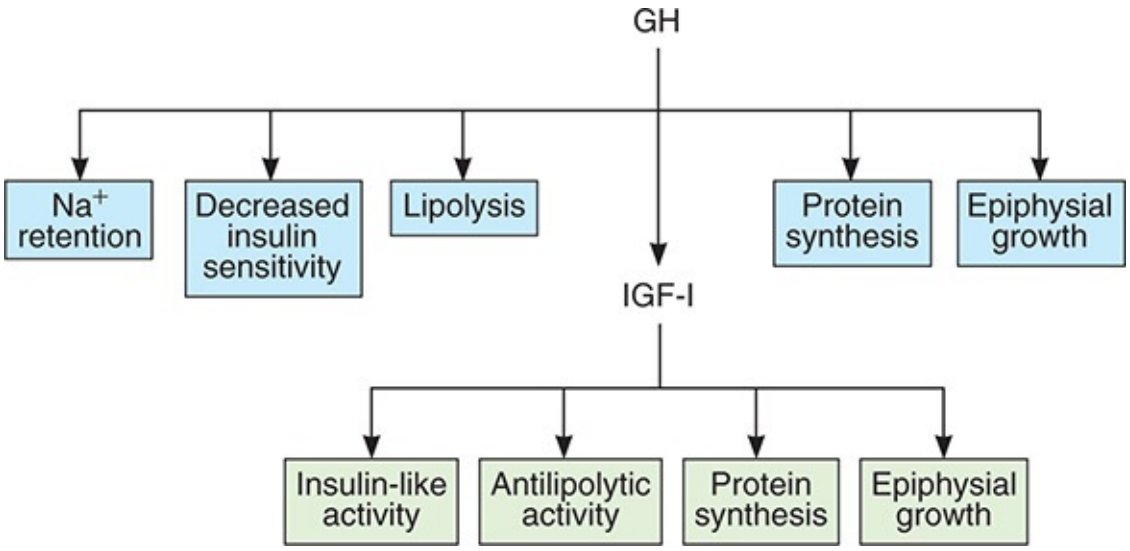


FIGURE 18–6 Direct and indirect actions of growth hormone (GH). The latter are mediated by the ability of GH to induce production of insulin-like growth factor-I (IGF-I). (Used with permission of R. Clark and N. Gesundheit.)

TIMELINE OF GROWTH & OTHER REGULATORS

Growth hormone, while being essentially unimportant for fetal development, is the most important hormone for postnatal growth. However, growth overall is a complex phenomenon that is affected not only by growth hormone and somatomedins but also by thyroid hormones, androgens, estrogens, glucocorticoids, and insulin. It is also affected, of course, by genetic factors, and

it depends on adequate nutrition. It is normally accompanied by an orderly sequence of maturational changes, and it involves accretion of protein and an increase in length and size, not just an increase in weight (which could reflect the formation of fat or retention of salt and water rather than growth per se).

ROLE OF NUTRITION

The food supply is the most important extrinsic factor affecting growth. The diet must be adequate not only in protein content but also in essential vitamins and minerals (see Chapter 26) and in calories, so that ingested protein is not burned for energy. However, the age at which a dietary deficiency occurs appears to be an important consideration. For example, once the pubertal growth spurt has commenced, considerable linear growth continues even if caloric intake is reduced. Injury and disease, on the other hand, stunt growth because they increase protein catabolism.

GROWTH PERIODS

In humans, two periods of rapid growth occur **(Figure 18–7)**: the first in infancy and the second in late puberty just before growth stops. The first period of accelerated growth is partly a continuation of the fetal growth period. The second growth spurt, at the time of puberty, is due to growth hormone, androgens, and estrogens. Because girls mature earlier than boys, this growth spurt appears earlier in girls. Of course, in both sexes the rate of growth of individual tissues varies **(Figure 18–8)**. The eventual cessation of growth is due in large part to closure of the epiphyses in the long bones by estrogens (see Chapter 21). After this time, further increases in height are not possible.

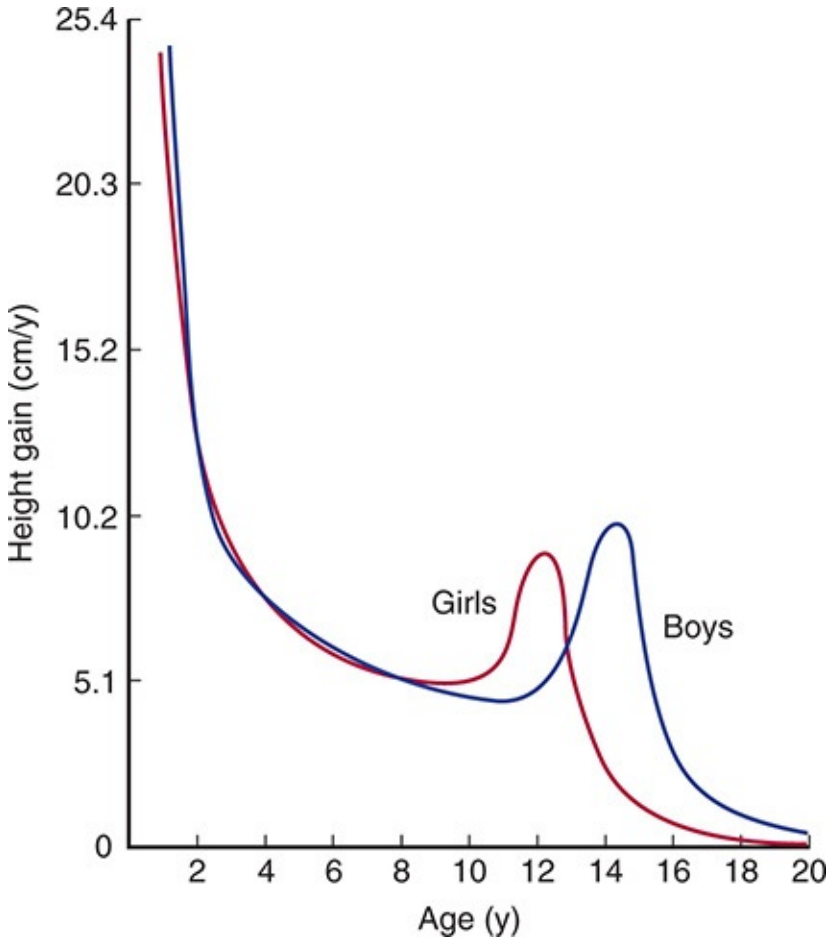


FIGURE 18–7 Rate of growth in boys and girls from birth to age 20.

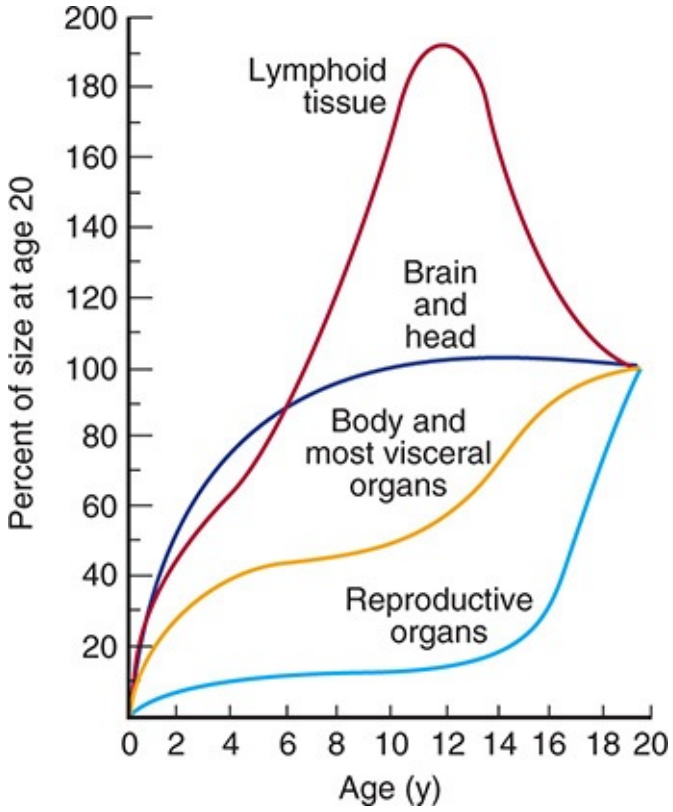


FIGURE 18–8 Growth of different tissues at various ages as a percentage of size at age 20. The curves are composites that include data for both boys and girls.

It is interesting that at least during infancy, growth is not a continuous process but is episodic or saltatory. Increases in the length of human infants of 0.5–2.5 cm in a few days are separated by periods of 2–63 days during which no measurable growth can be detected. The saltatory frequency varies between individuals and summates to final height. This episodic growth apparently reflects discrete times at which chondrocytes are susceptible to differentiation into a hypertrophic phase.

HORMONAL EFFECTS

The contributions of hormones to growth after birth are shown diagrammatically in **Figure 18–9**. Plasma growth hormone is elevated in newborns. Subsequently, average resting levels fall but the spikes of growth hormone secretion are larger, especially during puberty, so the mean plasma level over 24 h is increased; it is 2–4 ng/mL in normal adults, but 5–8 ng/mL in children. Plasma IGF-I levels also

rise during childhood, reaching a peak at 13–17 years of age. In contrast, IGF-II levels are constant throughout postnatal growth.

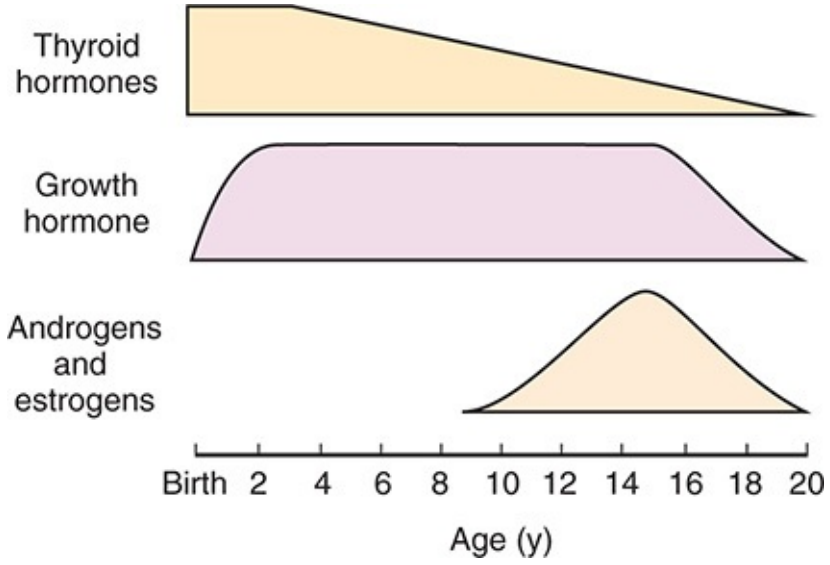


FIGURE 18–9 Relative importance of hormones in human growth at various ages. Thyroid hormones and growth hormones drive the rapid growth rate in the neonatal period and for the first few years of life, with the effect of the thyroid hormones tapering off thereafter. The growth spurt of puberty, on the other hand, involves interactions between the effects of growth hormone and the sex steroids. (Used with permission of D. A. Fisher.)

The growth spurt that occurs at the time of puberty (Figure 18–7) is due in part to the secretion of adrenal androgens at this time in both sexes and the protein anabolic effect of these androgens; however, it is also due to an interaction among sex steroids, growth hormone, and IGF-I. Treatment with estrogens and androgens increases the secretion of growth hormone in response to various stimuli and increases plasma IGF-I secondary to this increase in circulating growth hormone. This, in turn, causes growth.

Although androgens and estrogens initially stimulate growth, estrogens ultimately terminate growth by causing the epiphyses to fuse to the long bones (epiphysial closure). Once the epiphyses have closed, linear growth ceases (see Chapter 21). This is why patients with sexual precocity are apt to be dwarfed. On the other hand, men who were castrated before puberty tend to be tall because their estrogen production is decreased and their epiphyses remain open, allowing some growth to continue past the normal age of puberty.

In hypophysectomized animals, growth hormone increases growth, but this

effect is potentiated by thyroid hormones, which by themselves have no effect on growth. The action of thyroid hormones in this situation is therefore permissive to that of growth hormone, possibly via potentiation of the actions of somatomedins. Thyroid hormones also often appear to be necessary for stimulated growth hormone secretion; basal growth hormone levels are normal in hypothyroidism, but the response to hypoglycemia is frequently blunted. Thyroid hormones have widespread effects on the ossification of cartilage, the growth of teeth, the contours of the face, and the proportions of the body. Hypothyroid dwarfs (also known as **cretins**) therefore have infantile features **(Figure 18–10)**. Patients who are dwarfed because of panhypopituitarism have features consistent with their chronologic age until puberty, but since they do not mature sexually, they have juvenile features in adulthood **(Clinical Box 18–2)**.

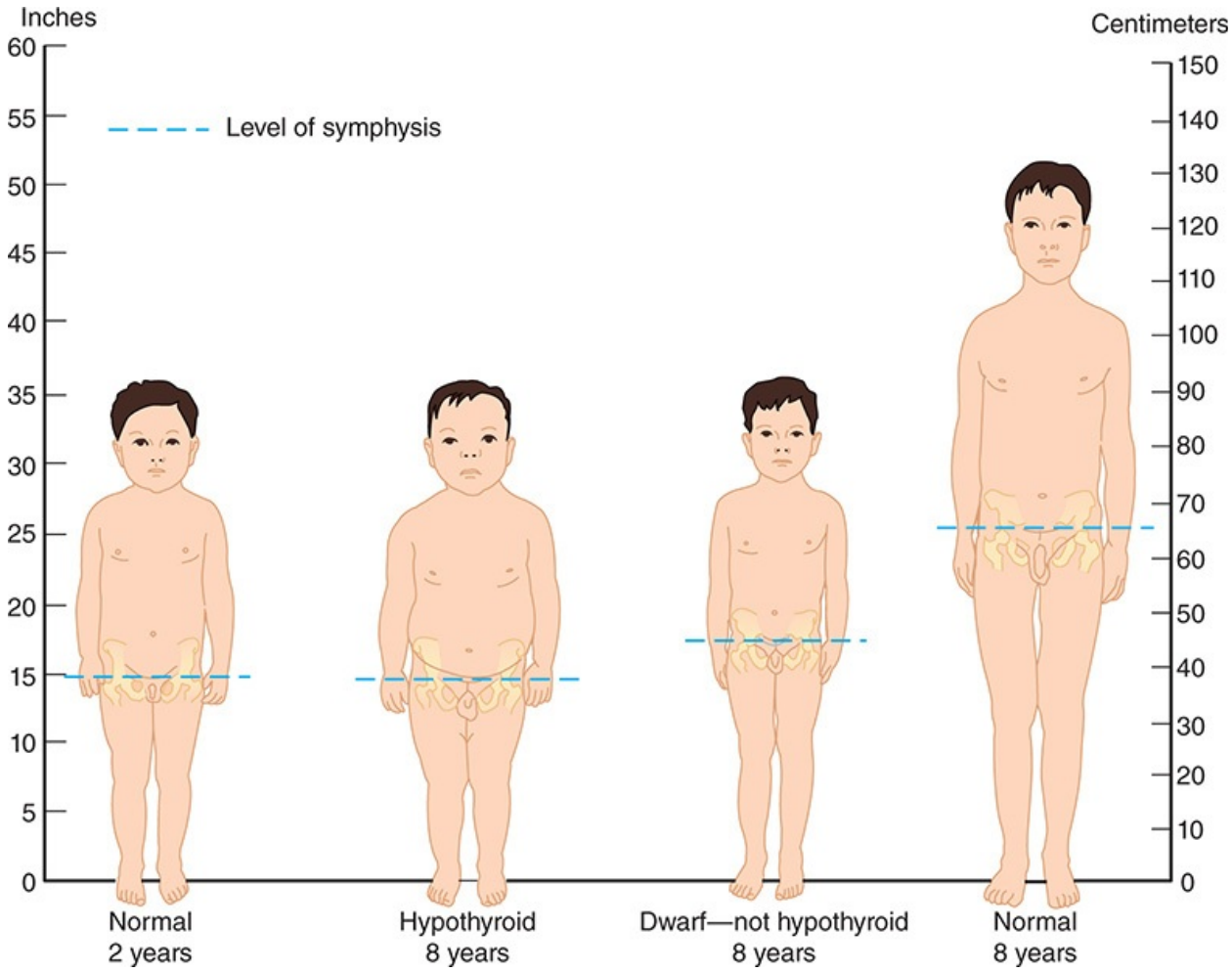


FIGURE 18–10 Normal and abnormal growth. Hypothyroid dwarfs (cretins) retain their infantile proportions, whereas dwarfs of the constitutional type and, to a lesser extent, of the hypopituitary type have proportions characteristic of

their chronologic age. See also Clinical Box 18–2. (Reproduced with permission from Wilkins L: *The Diagnosis and Treatment of Endocrine Disorders in Childhood and Adolescence*, 3rd ed. Thomas; 1966.)

The effect of insulin on growth is discussed in Chapter 24. Diabetic animals fail to grow, and insulin causes growth in hypophysectomized animals. However, the growth is appreciable only when large amounts of carbohydrate and protein are supplied with the insulin.

Adrenocortical hormones other than androgens exert a permissive action on growth in the sense that adrenalectomized animals fail to grow unless their blood pressure and circulations are maintained by replacement therapy. On the other hand, glucocorticoids are potent inhibitors of growth because of their direct action on cells, and treatment of children with pharmacologic doses of corticosteroids slows or stops growth for as long as the treatment is continued.

CATCH-UP GROWTH

Following illness or starvation in children, a period of **catch-up growth** **(Figure 18–11)** takes place during which the growth rate is greater than normal. The accelerated growth usually continues until the previous growth curve is reached, then slows to normal. The mechanisms that bring about and control catch-up growth are poorly understood.

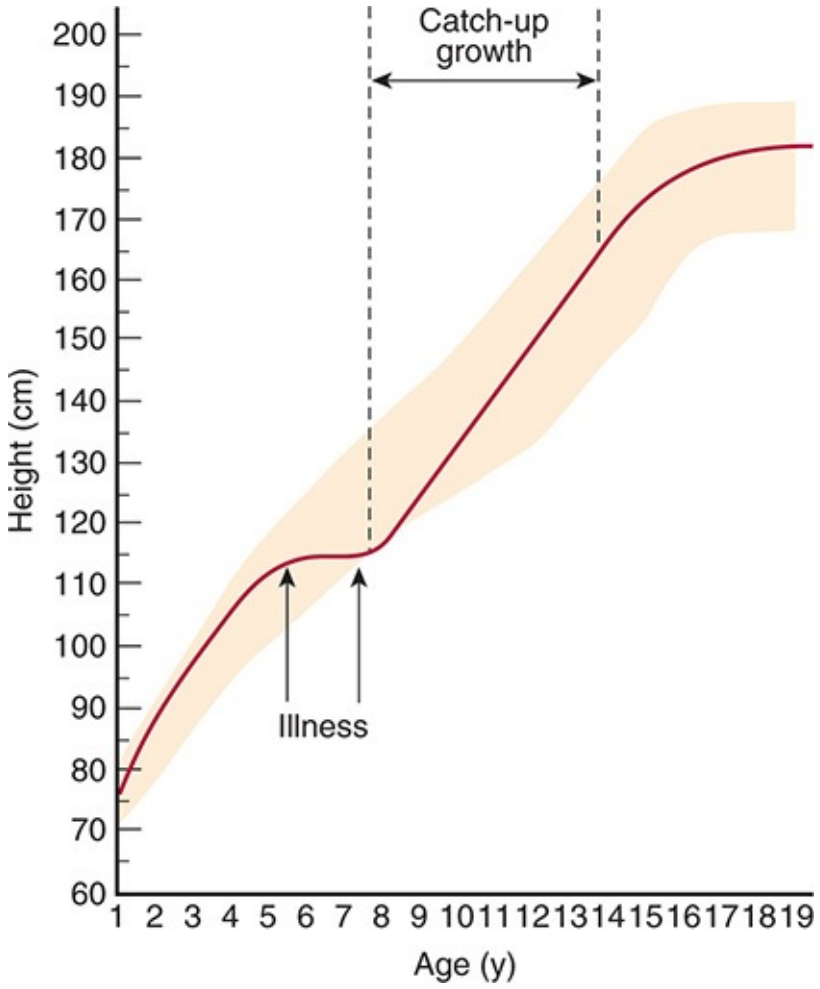


FIGURE 18–11 Growth curve for a normal boy who had an illness beginning at age 5 and ending at age 7. The shaded area shows the range of normal heights for a given age. The red line shows actual growth of the boy studied. Catch-up growth eventually returned his height to his previous normal growth curve. (Modified with permission from Boersma B, Wit JM: Catch-up growth. Endocr Rev 1997 Oct;18(5):646–661.)

PITUITARY GONADOTROPINS & PROLACTIN

CHEMISTRY

FSH and LH are each made up of an α and a β subunit. They are glycoproteins, and their carbohydrate residues increase their potency by markedly slowing their metabolism. The half-life of human FSH is about 170 min; the half-life of LH is about 60 min. Loss-of-function mutations in the FSH receptor cause

hypogonadism. Gain-of-function mutations cause a spontaneous form of **ovarian hyperstimulation syndrome,** a condition in which many follicles are stimulated and cytokines are released from the ovary, causing increased vascular permeability and shock.

CLINICAL BOX 18–2

Dwarfism

The accompanying discussion of growth control should suggest several possible etiologies of short stature. It can be due to GHRH deficiency, growth hormone deficiency, or deficient secretion of IGF-I. Isolated growth hormone deficiency is often due to GHRH deficiency, and in these instances, the growth hormone response to GHRH is normal. However, some patients with isolated growth hormone deficiency have abnormalities of their growth hormone secreting cells. In another group of dwarfed children, the plasma growth hormone concentration is normal or elevated, but their growth hormone receptors are unresponsive as a result of loss-of-function mutations. The resulting condition is known as **growth hormone insensitivity** or **Laron dwarfism.** Plasma IGF-I is markedly reduced, along with its binding protein. African pygmies have normal plasma growth hormone levels and a modest reduction in the plasma level of growth hormone–binding protein. However, their plasma IGF-I concentration fails to increase at the time of puberty and they experience less growth than nonpygmy controls throughout the prepubertal period.

Short stature may also be caused by mechanisms independent of specific defects in the growth hormone axis. It is characteristic of childhood hypothyroidism (cretinism) and occurs in patients with precocious puberty. It is also part of the syndrome of **gonadal dysgenesis** seen in patients who have an XO chromosomal pattern instead of an XX or XY pattern (see Chapter 22). Various bone and metabolic diseases also cause stunted growth, and in many cases there is no known cause (“constitutional delayed growth”). Chronic abuse and neglect can also cause dwarfism in children, independent of malnutrition. This condition is known as **psychosocial dwarfism** or the **Kaspar Hauser syndrome,** named for the patient with the first reported case. Finally, **achondroplasia,** the most common form of dwarfism in humans, is characterized by short limbs with a normal trunk. It is an autosomal dominant condition caused by a mutation in the gene that codes for **fibroblast growth**

factor receptor 3 (FGFR3). This member of the fibroblast growth receptor family is normally expressed in cartilage and the brain.

THERAPEUTIC HIGHLIGHTS

The treatment of dwarfism is dictated by its underlying cause. If treatment to replace the relevant hormone is commenced promptly in appropriate childhood cases, almost normal stature can often be attained. Thus, the availability of recombinant forms of growth hormone and IGF-I has greatly improved treatment in cases where these hormones are deficient.

Human pituitary prolactin has considerable structural similarity to human growth hormone and human chorionic somatomammotropin (hCS). The half-life of prolactin, like that of growth hormone, is about 20 min. Structurally similar prolactins are secreted by the endometrium and by the placenta.

REGULATION OF PROLACTIN SECRETION

The regulatory factors for prolactin secretion by the pituitary overlap, in part, with those causing secretion of growth hormone, but there are important differences, and some stimuli increase prolactin secretion while decreasing that of growth hormone (and vice versa) **(Table 18–4)**. The normal plasma prolactin concentration is approximately 5 ng/mL in men and 8 ng/mL in women. Secretion is tonically inhibited by the hypothalamus, and section of the pituitary stalk leads to an increase in circulating prolactin. Thus, the effect of the hypothalamic prolactin-inhibiting hormone, dopamine, must normally be greater than the effects of the various hypothalamic peptides with prolactin-releasing activity. In humans, prolactin secretion is increased by exercise, surgical and psychological stresses, and stimulation of the nipple (Table 18–4). The plasma prolactin level rises during sleep, the rise starting after the onset of sleep and persisting throughout the sleep period. Secretion is increased during pregnancy, reaching a peak at the time of parturition. After delivery, the plasma concentration falls to nonpregnant levels in about 8 days. Suckling produces a prompt increase in secretion, but the magnitude of this rise gradually declines after a woman has been nursing for more than 3 months. With prolonged lactation, milk secretion occurs with prolactin levels that are in the normal range.

TABLE 18–4 Comparison of factors affecting the secretion of human prolactin and growth hormone.

Factor	Prolactin	Growth Hormone
Sleep	I+	I+
Nursing	I++	N
Breast stimulation in nonlactating women	I	N
Stress	I+	I+
Hypoglycemia	I	I+
Strenuous exercise	I	I
Sexual intercourse in women	I	N
Pregnancy	I++	N
Estrogens	I	I
Hypothyroidism	I	N
TRH	I+	N
Phenothiazines, butyrophenones	I+	N
Opioids	I	I
Glucose	N	D
Somatostatin	N	D+
L-dopa	D+	I+
Apomorphine	D+	I+
Bromocriptine and related ergot derivatives	D+	I

D, moderate decrease; D+, marked decrease; I, moderate increase; I+, marked increase; I++, very marked increase; N, no change; TRH, thyrotropin-releasing hormone.

L-dopa decreases prolactin secretion by increasing the formation of dopamine; bromocriptine and other dopamine agonists inhibit secretion because they stimulate dopamine receptors. Chlorpromazine and related drugs that block dopamine receptors increase prolactin secretion. Thyrotropin-releasing hormone (TRH) stimulates the secretion of prolactin in addition to TSH, and additional

polypeptides with prolactin-releasing activity are present in hypothalamic tissue. Estrogens produce a slowly developing increase in prolactin secretion as a result of a direct action on the lactotropes.

It has now been established that prolactin facilitates the secretion of dopamine in the median eminence. Thus, prolactin acts in the hypothalamus in a negative feedback manner to inhibit its own secretion.

RECEPTORS

The receptors for FSH and LH are G-protein-coupled receptors coupled to adenylyl cyclase through a stimulatory G-protein (G_s; see Chapter 2). In addition, each has an extended, glycosylated extracellular domain.

The human prolactin receptor resembles the growth hormone receptor and is one of the superfamily of receptors that includes the growth hormone receptor and receptors for many cytokines and hematopoietic growth factors (see Chapters 2 and 3). It dimerizes and activates the Janus kinase/signal transducers and activators of transcription (JAK–STAT) pathway and other intracellular enzyme cascades (Figure 18–4).

ACTIONS OF FSH, LH, & PROLACTIN

The testes and ovaries become atrophic when the pituitary is removed or destroyed. The related actions of prolactin and the gonadotropins FSH and LH, as well as those of the gonadotropin secreted by the placenta, are described in detail in Chapters 22 and 23. In brief, FSH helps maintain the spermatogenic epithelium by stimulating Sertoli cells in the male and is responsible for the early growth of ovarian follicles in the female. LH is tropic for the Leydig cells and, in females, is responsible for the final maturation of the ovarian follicles and estrogen secretion from them. It is also responsible for ovulation, the initial formation of the corpus luteum, and secretion of progesterone.

Prolactin causes milk secretion from the breast after estrogen and progesterone priming. Its effect on the breast involves increasing messenger RNA (mRNA) levels and subsequent production of casein and lactalbumin. However, the action of the hormone is not exerted on the cell nucleus and is prevented by inhibitors of microtubules. Prolactin also inhibits the effects of gonadotropins, possibly by an action at the level of the ovary. It thereby prevents ovulation in lactating women. The function of prolactin in normal males is

unsettled, but excess prolactin secreted by tumors causes erectile dysfunction.

EFFECTS OF PITUITARY INSUFFICIENCY

CHANGES IN OTHER ENDOCRINE GLANDS

The widespread changes that develop when the pituitary is removed surgically or destroyed by disease in humans or animals are predictable in terms of the known hormonal functions of the gland. In hypopituitarism, the adrenal cortex atrophies and the secretion of adrenal glucocorticoids and sex hormones falls to low levels. Stress-induced increases in aldosterone secretion are absent, but basal aldosterone secretion and increases induced by salt depletion are normal, at least for some time. Since no mineralocorticoid deficiency is present, salt loss and hypovolemic shock do not develop, but the inability to increase glucocorticoid secretion makes patients with pituitary insufficiency sensitive to stress. The development of salt loss in long-standing hypopituitarism is discussed in Chapter 20. Growth is inhibited (see Clinical Box 18–2). Thyroid function is depressed to low levels, and cold is tolerated poorly. The gonads atrophy, sexual cycles stop, and some of the secondary sex characteristics disappear.

INSULIN SENSITIVITY

Hypophysectomized animals have a tendency to become hypoglycemic, especially when fasted. Hypophysectomy ameliorates diabetes mellitus (see Chapter 24) and markedly increases the hypoglycemic effect of insulin. This is due in part to the deficiency of adrenocortical hormones, but hypophysectomized animals are more sensitive to insulin than adrenalectomized animals because they also lack the anti-insulin effect of growth hormone.

WATER METABOLISM

Although selective destruction of the supraoptic–posterior pituitary causes diabetes insipidus (see Chapter 17), removal of both the anterior and posterior pituitary usually causes no more than a transient polyuria. In the past, there was speculation that the anterior pituitary secreted a “diuretic hormone,” but the amelioration of the diabetes insipidus is actually explained by a decrease in the osmotic load presented for excretion. Osmotically active particles hold water in

the renal tubules (see Chapter 38). Because of the ACTH deficiency, the rate of protein catabolism is decreased in hypophysectomized animals. Because of the TSH deficiency, the metabolic rate is low. Consequently, fewer osmotically active products of catabolism are filtered and urine volume declines, even in the absence of vasopressin. Growth hormone deficiency contributes to the depression of the glomerular filtration rate in hypophysectomized animals, and growth hormone increases the glomerular filtration rate and renal plasma flow in humans. Finally, because of the glucocorticoid deficiency, there is the same defective excretion of a water load that is seen in adrenalectomized animals. The "diuretic" activity of the anterior pituitary can thus be explained in terms of the actions of ACTH, TSH, and growth hormone.

OTHER DEFECTS

When growth hormone deficiency develops in adulthood, it is usually accompanied by deficiencies in other anterior pituitary hormones. The deficiency of ACTH and other pituitary hormones with MSH activity may be responsible for the pallor of the skin in patients with hypopituitarism. There may be some loss of protein in adults, but wasting is not a feature of hypopituitarism in humans, and most patients with pituitary insufficiency are well nourished.

CAUSES OF PITUITARY INSUFFICIENCY IN HUMANS

Tumors of the anterior pituitary cause pituitary insufficiency. Suprasellar cysts, remnants of Rathke pouch that enlarge and compress the pituitary, are another cause of hypopituitarism. In women who have an episode of shock due to postpartum hemorrhage, the pituitary may become infarcted, with the subsequent development of postpartum necrosis **(Sheehan syndrome).** The blood supply to the anterior lobe is vulnerable because it descends on the pituitary stalk through the rigid diaphragma sellae, and during pregnancy the pituitary is enlarged. Pituitary infarction is usually extremely rare in men.

CHAPTER SUMMARY

- The pituitary gland consists of two functional sections in humans: the anterior lobe, which secretes mainly tropic hormones; and the posterior

lobe, which contains nerve endings that project from the hypothalamus and release oxytocin and vasopressin. The anterior lobe receives almost all of its blood supply from the portal hypophysial vessels that carry regulatory factors released by the hypothalamus.

- The pituitary plays a critical role in regulating the function of downstream glands, and also exerts independent endocrine actions on a wide variety of peripheral organs and tissues. Cell types in the anterior lobe include somatotropes (growth hormone), lactotropes (prolactin), corticotropes (ACTH), thyrotropes (TSH), and gonadotropes (FSH, LH), which release the hormones listed in parentheses. The relative proportions of each cell type vary depending on needs at different life stages.
- Corticotropes of the anterior lobe synthesize proopiomelanocortin, which is the precursor of ACTH, endorphins, and melanotropins. ACTH is a primary regulator of skin pigmentation in mammals.
- Growth hormone is synthesized by somatotropes. It is secreted in an episodic manner in response to hypothalamic factors, and secretion is subject to feedback inhibition. A portion of the circulating pool is protein-bound.
- Growth hormone activates growth and influences protein, carbohydrate, and fat metabolism to react to stressful conditions. Many, but not all, of the peripheral actions of growth hormone can be attributed to its ability to stimulate production of IGF-I.
- Growth reflects a complex interplay of growth hormone, IGF-I, and many other hormones as well as extrinsic influences and genetic factors. The consequences of overproduction or underproduction of such influences depends on whether this occurs before or after puberty. Deficiencies in components of the growth hormone pathway in childhood lead to dwarfism; overproduction results in gigantism, acromegaly, or both.
- The pituitary also supplies hormones that regulate reproductive tissues and lactation—FSH, LH, and prolactin. Prolactin, in particular, is regulated by many of the factors that also regulate growth hormone secretion, although specific regulators may have opposing effects.
- Pituitary insufficiency is accompanied by atrophy of the adrenal cortex, sensitivity to stress, growth inhibition, depressed thyroid function, hypoglycemia, pallor, and atrophy of the gonads. Pituitary insufficiency may be caused by tumors or, in women, by infarction following shock due to postpartum hemorrhage.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A neuroscientist is studying communication between the hypothalamus and pituitary in a rat model. She interrupts blood flow through the median eminence and then measures circulating levels of pituitary hormones following appropriate physiologic stimulation. Secretion of which of the following hormones will be unaffected by the experimental manipulation?

   A. Growth hormone

   B. Prolactin

   C. TSH

   D. FSH

   E. Vasopressin

2. Loss of which of the following pituitary hormones might be expected to increase responses to painful stimuli?

   A. α-Melanocyte–stimulating hormone (α-MSH)

   B. β-MSH

   C. ACTH

   D. Growth hormone

   E. β-Endorphin

3. A 20-year-old African American woman is evaluated for evaluation of patches of skin on her face and hands that have lost pigmentation over a period of weeks. She is otherwise healthy. Blood tests reveal the presence of autoantibodies to melanocytes. The most likely diagnosis is

   A. albinism.

   B. piebaldism.

   C. primary adrenal insufficiency.

   D. vitiligo.

   E. hypopituitarism.

4. A scientist finds that infusion of growth hormone into the median eminence of the hypothalamus in experimental animals inhibits the secretion of growth hormone and concludes that this proves that growth hormone feeds back to inhibit GHRH secretion. Do you accept this conclusion?

   A. No, because growth hormone does not cross the blood-brain barrier.

   B. No, because the infused growth hormone could be stimulating dopamine

secretion.

C. No, because substances placed in the median eminence could be transported to the anterior pituitary.

D. Yes, because systemically administered growth hormone inhibits growth hormone secretion.

E. Yes, because growth hormone binds GHRH, inactivating it.

5. The growth hormone receptor

A. activates G_s.

B. requires dimerization to exert its effects.

C. must be internalized to exert its effects.

D. resembles the IGF-I receptor.

E. resembles the ACTH receptor.

6. A 7-year-old boy is evaluated for short stature. His average circulating growth hormone level is within the normal range for his age, but levels of IGF-I are reduced. His growth failure is most likely due to a defect in

A. GHRH release from the hypothalamus.

B. GHRH receptors.

C. androgen synthesis.

D. estrogen synthesis.

E. growth hormone receptors.

7. A young woman in the first trimester of her first pregnancy comes to the emergency room complaining of lower abdominal pain, nausea, and vomiting that have been resistant to treatment. An ultrasound scan reveals bilateral enlarged ovaries containing multiple cysts and free fluid in the pelvis and abdomen. Her symptoms worsen over time and progress to hemoconcentration and ascites before spontaneously resolving at 18 weeks of gestation. She reports that her mother had developed similar symptoms with each of her own pregnancies. DNA sequencing would be expected to reveal which of the following:

A. A loss of function mutation in FSH

B. A loss of function mutation in LH

C. A gain of function mutation in the FSH receptor

D. A gain of function mutation in the LH receptor

E. A loss of function mutation in gonadotropin-releasing hormone

8. During childbirth, a woman suffers a serious hemorrhage and goes into shock. After she recovers, she displays symptoms of hypopituitarism. Which of the following will not be expected in this patient?

A. Cachexia

B. Infertility

C. Pallor

D. Low basal metabolic rate

E. Intolerance to stress

CHAPTER 19

The Adrenal Medulla & Adrenal Cortex

OBJECTIVES

After studying this chapter, you should be able to:

- Name the three catecholamines secreted by the adrenal medulla and summarize their biosynthesis, metabolism, and function.
- List the stimuli that increase adrenal medullary secretion.
- Differentiate between C_{18}, C_{19}, and C_{21} steroids and give examples of each.
- Outline the steps involved in steroid biosynthesis in the adrenal cortex.
- Name the plasma proteins that bind adrenocortical steroids and discuss their physiologic role.
- Name the major site of adrenocortical hormone metabolism and the principal metabolites produced from glucocorticoids, adrenal androgens, and aldosterone.
- Describe the mechanisms by which glucocorticoids and aldosterone produce changes in cellular function.
- Define the physiological and pharmacological effects of glucocorticoids.
- Contrast the physiological and pathological effects of adrenal androgens.
- Describe the mechanisms that regulate secretion of glucocorticoids and adrenal sex hormones.

- Explain the actions of aldosterone and describe the mechanisms that regulate aldosterone secretion.
- Describe the main features of the diseases caused by an excess or deficiency of each of the hormones of the adrenal gland.

INTRODUCTION

The **adrenal glands** are endocrine organs that produce several hormones including **catecholamines** and **steroid hormones**. There are two adrenal glands, one sitting on top of each kidney **(Figure 19–1)**. Each adrenal gland has an outer cortex that secretes the steroid hormones, **mineralocorticoids, glucocorticoids and androgens**, and an inner **medulla** that secretes the catecholamines, **epinephrine, norepinephrine,** and **dopamine**.

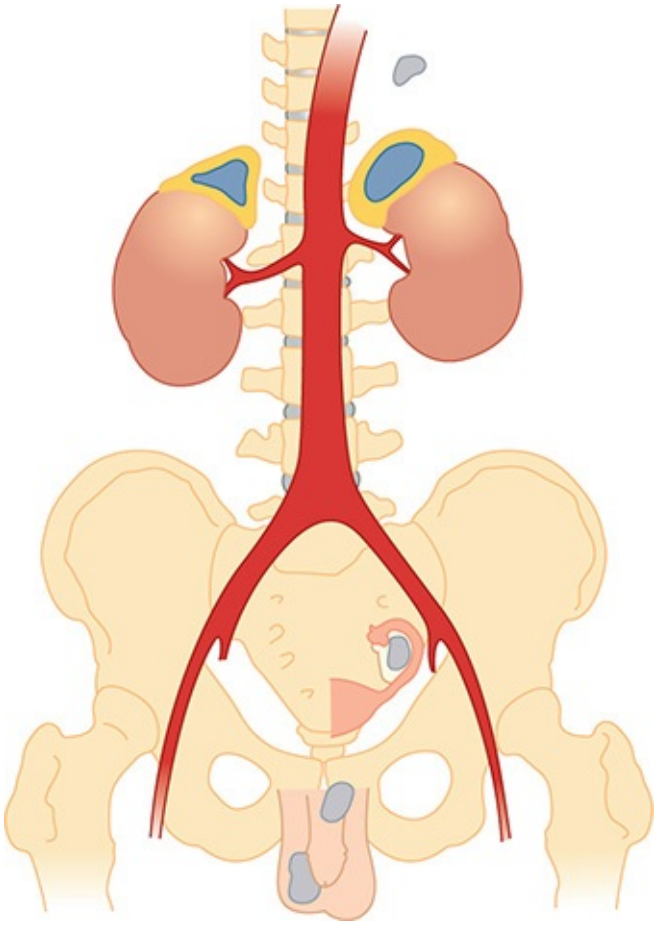


FIGURE 19–1 Human adrenal glands. Adrenocortical tissue is yellow; adrenal medullary tissue is blue. Note the location of the adrenals at the superior

pole of each kidney. Also shown are extra-adrenal sites (gray) at which cortical and medullary tissue is sometimes found. (Reproduced with permission from Williams RH: *Textbook of Endocrinology*, 4th ed. St. Louis, MO: Saunders; 1968.)

The adrenal cortex secretes **glucocorticoids (eg, cortisol)** which are steroids with widespread effects on the metabolism of carbohydrate and protein; and a **mineralocorticoid (aldosterone**) essential to the maintenance of Na^+ balance and extracellular fluid (ECF) volume.Mineralocorticoids and the glucocorticoids are necessary for survival. It is also a secondary site of **androgen** synthesis, secreting sex hormones such as testosterone, which can exert effects on reproductive function. Adrenocortical secretion is controlled primarily by adrenocorticotropic hormone (ACTH), from the anterior pituitary (Chapter 20), but mineralocorticoid secretion is also subject to independent control by circulating factors, of which the most important is **angiotensin II,** a peptide formed in the bloodstream by the action of **renin.**

The adrenal medulla is in effect a sympathetic ganglion in which the postganglionic neurons have lost their axons and become secretory cells. The cells secrete when stimulated by the preganglionic nerve fibers that reach the gland via the splanchnic nerves. Adrenal medullary hormones (epinephrine, norepinephrine, and dopamine) work mostly to prepare the body for emergencies, the so-called “fight-or-flight” responses.

ADRENAL MORPHOLOGY

The adrenal medulla, which constitutes 28% of the mass of the adrenal gland, is made up of interlacing cords of densely innervated granule-containing cells that abut on venous sinuses. Two cell types can be distinguished morphologically: an epinephrine-secreting type that has larger, less dense granules; and a norepinephrine-secreting type in which smaller, very dense granules fail to fill the vesicles in which they are contained. In humans, 90% of the cells are the epinephrine-secreting type and 10% are the norepinephrine-secreting type. The type of cell that secretes dopamine is unknown. **Paraganglia,** small groups of cells resembling those in the adrenal medulla, are found near the thoracic and abdominal sympathetic ganglia (Figure 19–1).

In adult mammals, the adrenal cortex is divided into three zones **(Figure 19–2)**. The outer **zona glomerulosa** is made up of whorls of cells that are continuous with the columns of cells that form the **zona fasciculata.** These

columns are separated by venous sinuses. The inner portion of the zona fasciculata merges into the **zona reticularis,** where the cell columns become interlaced in a network. The zona glomerulosa makes up 15% of the mass of the adrenal gland; the zona fasciculata, 50%; and the zona reticularis, 7%. The adrenocortical cells contain abundant lipid, especially in the outer portion of the zona fasciculata. All three cortical zones secrete **corticosterone,** but the active enzymatic mechanism for aldosterone biosynthesis is limited to the zona glomerulosa, whereas the enzymatic mechanisms for forming cortisol and sex hormones are found in the two inner zones. Furthermore, subspecialization occurs within the inner two zones, with the zona fasciculata secreting mostly glucocorticoids and the zona reticularis secreting mainly sex hormones.

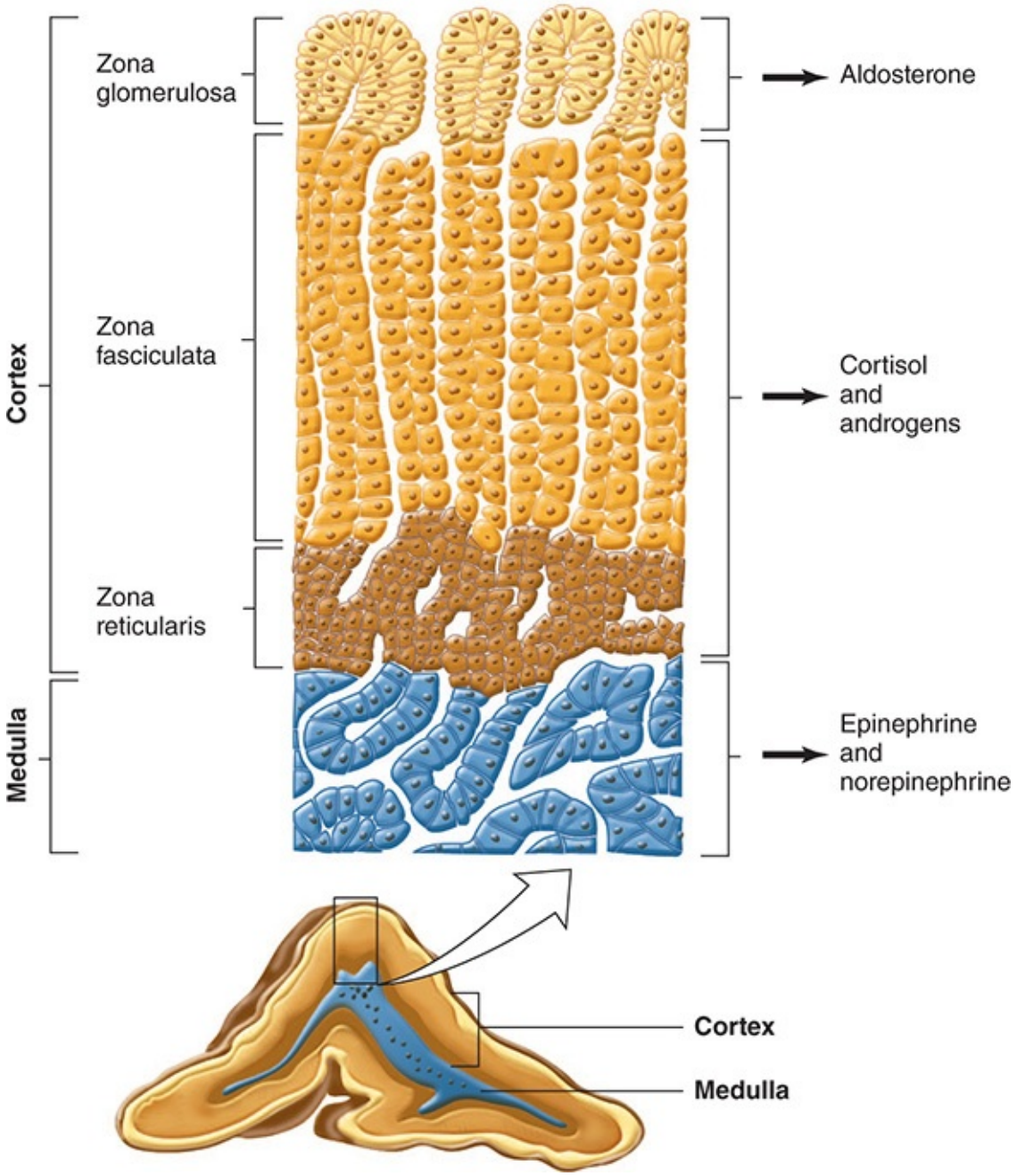


FIGURE 19–2 Section through an adrenal gland showing both the medulla and the zones of the cortex, as well as the hormones they secrete. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology: The Mechanisms of Body Function*, 11th ed. New York, NY: McGraw-Hill; 2008.)

Arterial blood reaches the adrenal from many small branches of the phrenic and renal arteries and the aorta. From a plexus in the capsule, blood flows

through the cortex to the sinusoids of the medulla. The medulla is also supplied by a few arterioles that pass directly to it from the capsule. In most species, including humans, blood from the medulla flows into a central adrenal vein. The blood flow through the adrenal is large, as it is in most endocrine glands.

During fetal life, the human adrenal is large and under pituitary control, but the three zones of the permanent cortex represent only 20% of the gland. The remaining 80% is the large **fetal adrenal cortex,** which undergoes rapid degeneration at the time of birth. A major function of this fetal adrenal is synthesis and secretion of sulfate conjugates of androgens that are converted in the placenta to estrogens (see Chapter 22). No structure is comparable to the human fetal adrenal in laboratory animals.

An important function of the zona glomerulosa, in addition to aldosterone synthesis, is the formation of new cortical cells. The adrenal medulla does not regenerate, but when the inner two zones of the cortex are removed, a new zona fasciculata and zona reticularis regenerate from glomerular cells attached to the capsule. Small capsular remnants regrow large pieces of adrenocortical tissue. Immediately after hypophysectomy, the zona fasciculata and zona reticularis begin to atrophy, whereas the zona glomerulosa is unchanged because of the action of angiotensin II on this zone. The ability to secrete aldosterone and conserve Na^+ is normal for some time after hypophysectomy, but in long-standing hypopituitarism, aldosterone deficiency may develop, apparently because of the absence of a pituitary factor that maintains the responsiveness of the zona glomerulosa. Injections of ACTH and stimuli that cause endogenous ACTH secretion produce hypertrophy of the zona fasciculata and zona reticularis but actually decrease, rather than increase, the size of the zona glomerulosa.

The cells of the adrenal cortex contain large amounts of smooth endoplasmic reticulum, which is involved in the steroid-forming process. Other steps in steroid biosynthesis occur in the mitochondria. The structure of steroid-secreting cells is very similar throughout the body. The typical features of such cells are shown in **Figure 19–3**.

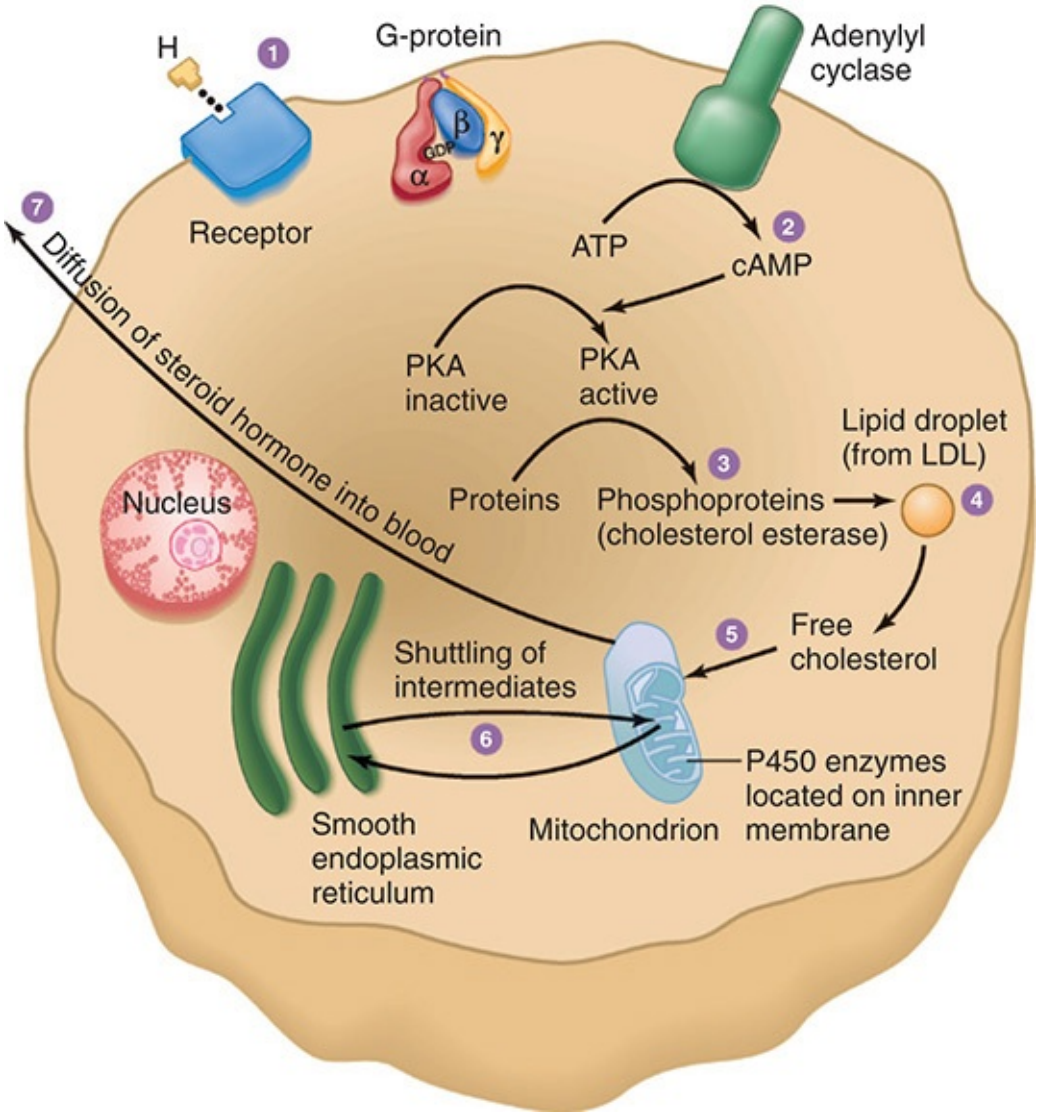


FIGURE 19–3 Schematic overview of the structures of steroid-secreting cells and the intracellular pathway of steroid synthesis. LDL, low-density lipoprotein; PKA, protein kinase A. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology: The Mechanisms of Body Function*, 11th ed. New York, NY: McGraw-Hill; 2008.)

ADRENAL MEDULLA: STRUCTURE & FUNCTION OF MEDULLARY HORMONES

CATECHOLAMINES

Norepinephrine, epinephrine, and small amounts of dopamine are synthesized by

the adrenal medulla. Cats and some other species secrete mainly norepinephrine, but in dogs and humans, most of the catecholamine output in the adrenal vein is epinephrine. Norepinephrine also enters the circulation from noradrenergic nerve endings.

The structures of norepinephrine, epinephrine, and dopamine and the pathways for their biosynthesis and metabolism are discussed in Chapter 7. Norepinephrine is formed by hydroxylation and decarboxylation of tyrosine, and epinephrine by methylation of norepinephrine. Phenylethanolamine- N-methyltransferase (PNMT), the enzyme that catalyzes the formation of epinephrine from norepinephrine, is found in appreciable quantities only in the brain and the adrenal medulla. Adrenal medullary PNMT is induced by glucocorticoids. Although relatively large amounts are required, the glucocorticoid concentration is high in the blood draining from the cortex to the medulla. After hypophysectomy, the glucocorticoid concentration of this blood falls and epinephrine synthesis is decreased. In addition, glucocorticoids are apparently necessary for the normal development of the adrenal medulla; in 21β-hydroxylase deficiency, glucocorticoid secretion is reduced during fetal life and the adrenal medulla is dysplastic. In untreated 21β-hydroxylase deficiency, circulating catecholamines are low after birth.

In plasma, about 95% of the dopamine and 70% of the norepinephrine and epinephrine are conjugated to sulfate. Sulfate conjugates are inactive and their function is unsettled. In recumbent humans, the normal plasma level of free norepinephrine is about 300 pg/mL (1.8 nmol/L). On standing, the level increases 50–100% **(Figure 19–4)**. The plasma norepinephrine level is generally unchanged after adrenalectomy, but the free epinephrine level, which is normally about 30 pg/mL (0.16 nmol/L), falls to essentially zero. The epinephrine found in tissues other than the adrenal medulla and the brain is for the most part absorbed from the bloodstream rather than synthesized in situ. Interestingly, low levels of epinephrine reappear in the blood some time after bilateral adrenalectomy, and these levels are regulated like those secreted by the adrenal medulla. They may come from cells such as the intrinsic cardiac adrenergic (ICA) cells (see Chapter 13), but their exact source is unknown.

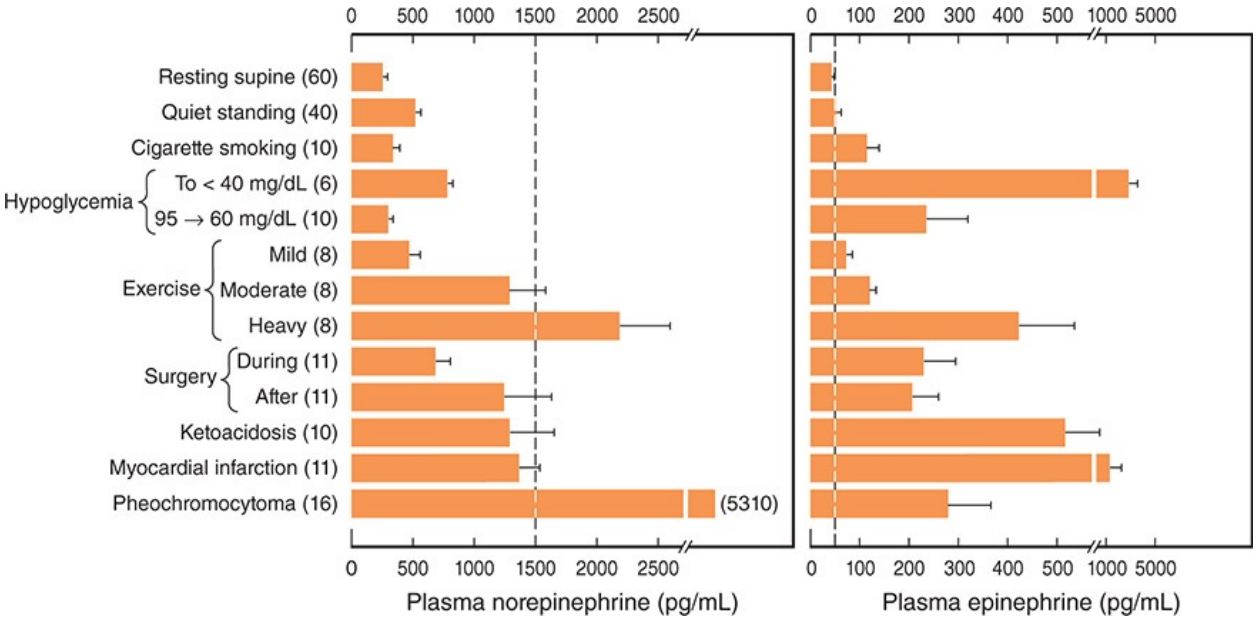


FIGURE 19–4 Norepinephrine and epinephrine levels in human venous blood in various physiologic and pathologic states. Note that the horizontal scales are different. The numbers to the left in parentheses are the numbers of subjects tested. In each case, the vertical dashed line identifies the threshold plasma concentration at which detectable physiologic changes are observed. (Modified and reproduced with permission from Cryer PE: Physiology and pathophysiology of the human sympathoadrenal neuroendocrine system. N Engl J Med 1980; Aug 21; 303(8):436–444.)

Plasma dopamine levels are normally very low, about 0.13 nmol/L. Most plasma dopamine is thought to be derived from sympathetic noradrenergic ganglia.

The catecholamines have a half-life of about 2 min in the circulation. For the most part, they are methoxylated and then oxidized to 3-methoxy-4-hydroxymandelic acid (vanillylmandelic acid [VMA]; see Chapter 7). About 50% of the secreted catecholamines appear in the urine as free or conjugated metanephrine and normetanephrine, and 35% as VMA. Only small amounts of free norepinephrine and epinephrine are excreted. In normal humans, about 30 µg of norepinephrine, 6 µg of epinephrine, and 700 µg of VMA are excreted per day.

OTHER SUBSTANCES SECRETED BY THE ADRENAL MEDULLA

In the medulla, norepinephrine and epinephrine are stored in granules with ATP. The granules also contain chromogranin A (see Chapter 7). Secretion is initiated by acetylcholine released from the preganglionic neurons that innervate the secretory cells. Acetylcholine activates cation channels allowing Ca^{2+} to enter the cells from the ECF and trigger the exocytosis of the granules. In this fashion, catecholamines, ATP, and proteins from the granules are all released into the blood together.

Epinephrine-containing cells of the medulla also contain and secrete opioid peptides (see Chapter 7). The precursor molecule is preproenkephalin. Most of the circulating metenkephalin comes from the adrenal medulla. The circulating opioid peptides do not cross the blood-brain barrier.

Adrenomedullin, a vasodepressor polypeptide found in the adrenal medulla, is discussed in Chapter 32.

EFFECTS OF EPINEPHRINE & NOREPINEPHRINE

In addition to mimicking the effects of noradrenergic nervous discharge, norepinephrine and epinephrine exert metabolic effects that include glycogenolysis in liver and skeletal muscle, mobilization of free fatty acids (FFA), increased plasma lactate, and stimulation of the metabolic rate. The effects of norepinephrine and epinephrine are brought about by actions on two classes of receptors: α- and β-adrenergic receptors. α-Receptors are subdivided into two groups, α_1 and α_2 receptors, and β-receptors into three groups, β_1, β_2, and β_3 receptors (see Chapter 7). There are three subtypes of α_1-receptors and three subtypes of α_2-receptors (see Table 7–2).

Norepinephrine and epinephrine both increase the force and rate of contraction of the isolated heart. These responses are mediated by β_1-receptors. The catecholamines also increase myocardial excitability, causing extrasystoles and, occasionally, more serious cardiac arrhythmias. Norepinephrine produces vasoconstriction in most if not all organs via α_1-receptors, but epinephrine dilates the blood vessels in skeletal muscle and the liver via β_2-receptors. This usually overbalances the vasoconstriction produced by epinephrine elsewhere, and the total peripheral resistance drops. When norepinephrine is infused slowly in normal animals or humans, the systolic and diastolic blood pressures rise. The **hypertension** stimulates the carotid and aortic baroreceptors, producing reflex

bradycardia that overrides the direct cardioacceleratory effect of norepinephrine. Consequently, cardiac output per minute falls. Epinephrine causes a widening of the pulse pressure but because baroreceptor stimulation is insufficient to obscure the direct effect of the hormone on the heart, cardiac rate and output increase. These changes are summarized in **Figure 19–5**.

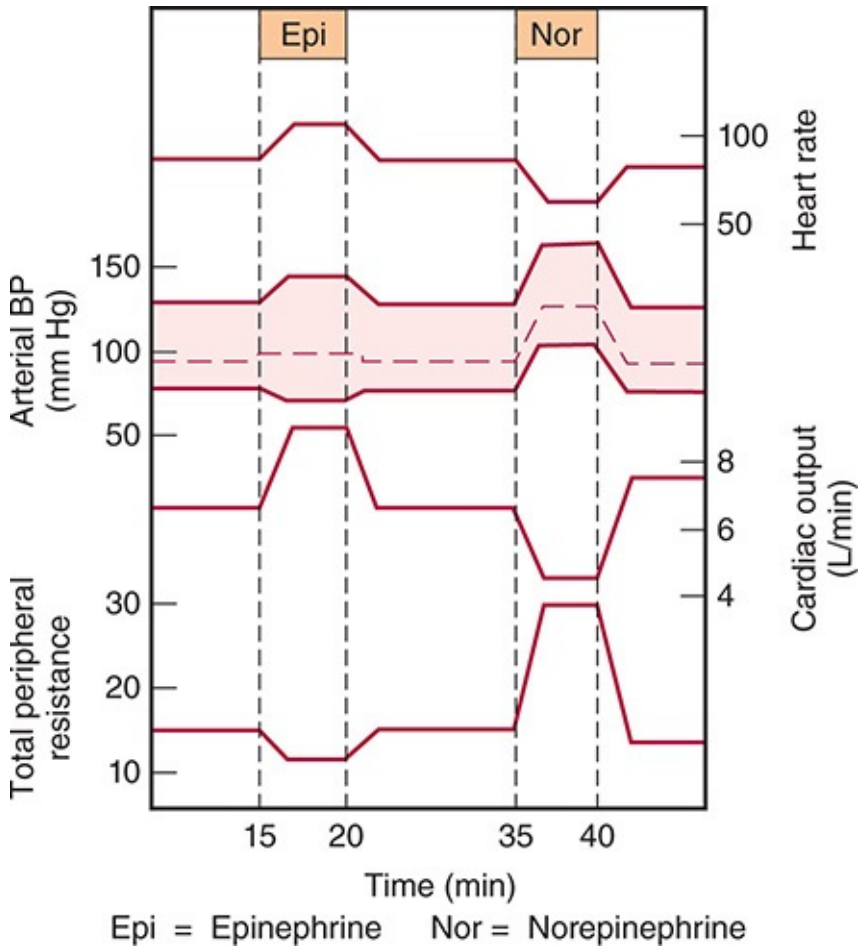


FIGURE 19–5 Circulatory changes produced in humans by the slow intravenous infusion of epinephrine and norepinephrine.

Catecholamines increase alertness (see Chapter 14). Epinephrine and norepinephrine are equally potent in this regard, although in humans epinephrine usually evokes more anxiety and fear.

The catecholamines have several different actions that affect blood glucose. Epinephrine and norepinephrine both cause glycogenolysis. They produce this effect via β-adrenergic receptors that increase cyclic adenosine monophosphate (cAMP), with activation of phosphorylase, and via α-adrenergic receptors that increase intracellular Ca^{2+} (see Chapter 7). In addition, the catecholamines

increase the secretion of insulin and glucagon via β-adrenergic mechanisms and inhibit the secretion of these hormones via α-adrenergic mechanisms.

Norepinephrine and epinephrine also produce a prompt rise in the metabolic rate that is independent of the liver and a smaller, delayed rise that is abolished by hepatectomy and coincides with the rise in blood lactate concentration. The initial rise in metabolic rate may be due to cutaneous vasoconstriction, which decreases heat loss and leads to a rise in body temperature, or to increased muscular activity, or both. The second rise is probably due to oxidation of lactate in the liver. Mice unable to make norepinephrine or epinephrine because their dopamine β-hydroxylase gene is knocked out are intolerant of cold, but surprisingly, their basal metabolic rate is elevated. The cause of this elevation is unknown.

When injected, epinephrine and norepinephrine cause an initial rise in plasma K^+ because of release of K^+ from the liver and then a prolonged fall in plasma K^+ because of an increased entry of K^+ into skeletal muscle that is mediated by β_2-adrenergic receptors. Some evidence suggests that activation of α-receptors opposes this effect.

The increases in plasma norepinephrine and epinephrine that are needed to produce the various effects listed above have been determined by infusion of catecholamines in resting humans. In general, the threshold for the cardiovascular and the metabolic effects of norepinephrine is about 1500 pg/mL, that is, about five times the resting value (Figure 19–4). Epinephrine, on the other hand, produces tachycardia when the plasma level is about 50 pg/mL, that is, about twice the resting value. The threshold for increased systolic blood pressure and lipolysis is about 75 pg/mL; the threshold for hyperglycemia, increased plasma lactate, and decreased diastolic blood pressure is about 150 pg/mL; and the threshold for the α-mediated decrease in insulin secretion is about 400 pg/mL. Plasma epinephrine often exceeds these thresholds. On the other hand, plasma norepinephrine rarely exceeds the threshold for its cardiovascular and metabolic effects, and most of its effects are due to its local release from postganglionic sympathetic neurons. Most adrenal medullary tumors **(pheochromocytomas)** secrete norepinephrine or epinephrine or both and produce sustained hypertension. However, 15% of epinephrine-secreting tumors secrete this catecholamine episodically, producing intermittent bouts of palpitations, headache, glycosuria, and extreme systolic hypertension. These same symptoms are produced by intravenous injection of a large dose of epinephrine.

EFFECTS OF DOPAMINE

The physiologic function of the dopamine in the circulation is unknown. However, injected dopamine produces renal vasodilation, probably by acting on a specific dopaminergic receptor. It also produces vasodilation in the mesentery. Elsewhere, it produces vasoconstriction, probably by releasing norepinephrine, and it has a positive inotropic effect on the heart by an action on β_1-adrenergic receptors. The net effect of moderate doses of dopamine is an increase in systolic pressure and no change in diastolic pressure. Because of these actions, dopamine is useful in the treatment of traumatic and cardiogenic shock (see Chapter 32).

Dopamine is made in the renal cortex. It causes natriuresis and may exert this effect by inhibiting renal Na, K, ATPase.

REGULATION OF ADRENAL MEDULLARY SECRETION

NEURAL CONTROL

Certain drugs act directly on the adrenal medulla, but physiologic stimuli affect medullary secretion through the nervous system. Catecholamine secretion is low in basal states, but the secretion of epinephrine and, to a lesser extent, that of norepinephrine is reduced even further during sleep.

Increased adrenal medullary secretion is part of the diffuse sympathetic discharge provoked in emergency situations, which Cannon called the "emergency function of the sympathoadrenal system." The ways in which this discharge prepares the individual for flight or fight are described in Chapter 13, and the increases in plasma catecholamines under various conditions are shown in Figure 19–4.

The metabolic effects of circulating catecholamines are probably important, especially in certain situations. The calorigenic action of catecholamines in animals exposed to cold is an example, and so is the glycogenolytic effect (see Chapter 24) in combating hypoglycemia.

SELECTIVE SECRETION

When adrenal medullary secretion is increased, the ratio of norepinephrine to epinephrine in the adrenal effluent is generally unchanged. However,

norepinephrine secretion tends to be selectively increased by emotional stresses with which the individual is familiar, whereas epinephrine secretion rises selectively in situations in which the individual does not know what to expect.

ADRENAL CORTEX: STRUCTURE & BIOSYNTHESIS OF ADRENOCORTICAL HORMONES

CLASSIFICATION & STRUCTURE

The hormones of the adrenal cortex are derivatives of cholesterol. Like cholesterol, bile acids, vitamin D, and ovarian and testicular steroids, they contain the **cyclopentanoperhydrophenanthrene nucleus** **(Figure 19–6)**. Gonadal and adrenocortical steroids are of three types: C_{21} steroids, which have a two-carbon side chain at position 17; C_{19} steroids, which have a keto or hydroxyl group at position 17; and C_{18} steroids, which, in addition to a 17-keto or hydroxyl group, have no angular methyl group attached to position 10. The adrenal cortex secretes primarily C_{21} and C_{19} steroids. Most of the C_{19} steroids have a keto group at position 17 and are therefore called **17-ketosteroids.** The C_{21} steroids that have a hydroxyl group at the 17 position in addition to the side chain are often called 17-hydroxycorticoids or 17-hydroxycorticosteroids.

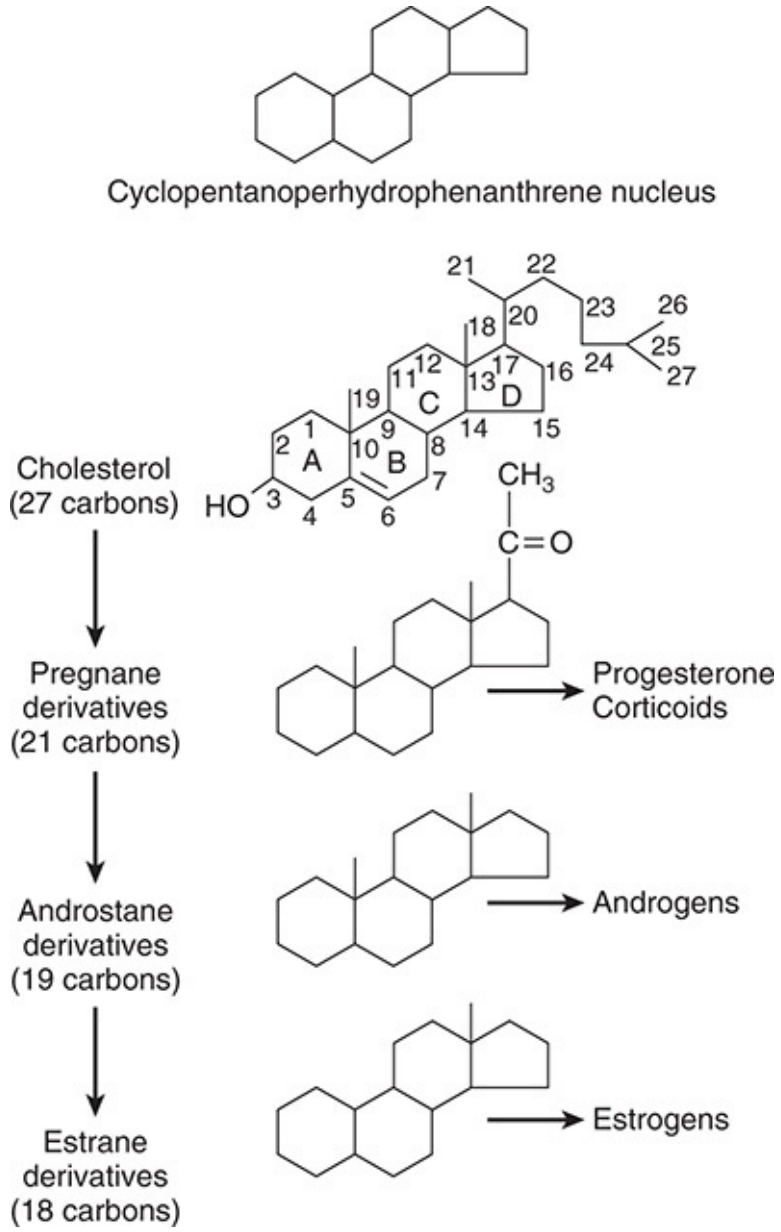


FIGURE 19–6 Basic structure of adrenocortical and gonadal steroids. The letters in the formula for cholesterol identify the four basic rings, and the numbers identify the positions in the molecule. As shown here, the angular methyl groups (positions 18 and 19) are usually indicated simply by straight lines.

The C_{19} steroids have androgenic activity. The C_{21} steroids are classified, using Selye's terminology, as mineralocorticoids or glucocorticoids. All secreted C_{21} steroids have both mineralocorticoid and glucocorticoid activity; **mineralocorticoids** are those in which effects on Na^+ and K^+ excretion

predominate and **glucocorticoids** are those in which effects on glucose and protein metabolism predominate.

The details of steroid nomenclature and isomerism can be found elsewhere. However, it is pertinent to mention that the Greek letter Δ indicates a double bond and that the groups that lie above the plane of each of the steroid rings are indicated by the Greek letter β and a solid line (—OH), whereas those that lie below the plane are indicated by α and a dashed line (- - -OH). Thus, the C_{21} steroids secreted by the adrenal have a Δ^4-3-keto configuration in the A ring. In most naturally occurring adrenal steroids, 17-hydroxy groups are in the α configuration, whereas 3-, 11-, and 21-hydroxy groups are in the β configuration. The 18-aldehyde configuration of naturally occurring aldosterone is the D form. L-aldosterone is physiologically inactive.

SECRETED STEROIDS

Innumerable steroids have been isolated from adrenal tissue, but the only steroids normally secreted in physiologically significant amounts are the mineralocorticoid **aldosterone,** the glucocorticoids **cortisol** and **corticosterone,** and the androgens **dehydroepiandrosterone (DHEA)** and **androstenedione.** The structures of these steroids are shown in **Figure 19–7** and **Figure 19–8**. **Deoxycorticosterone** is a mineralocorticoid that is normally secreted in about the same amount as aldosterone **(Table 19–1)** but has only 3% of the mineralocorticoid activity of aldosterone. Its effect on mineral metabolism is usually negligible, but in diseases in which its secretion is increased, its effect can be appreciable. Most of the estrogens that are not formed in the ovaries are produced in the circulation from adrenal androstenedione. Almost all the dehydroepiandrosterone is secreted conjugated with sulfate, although most if not all of the other steroids are secreted in the free, unconjugated form **(Clinical Box 19–1)**.

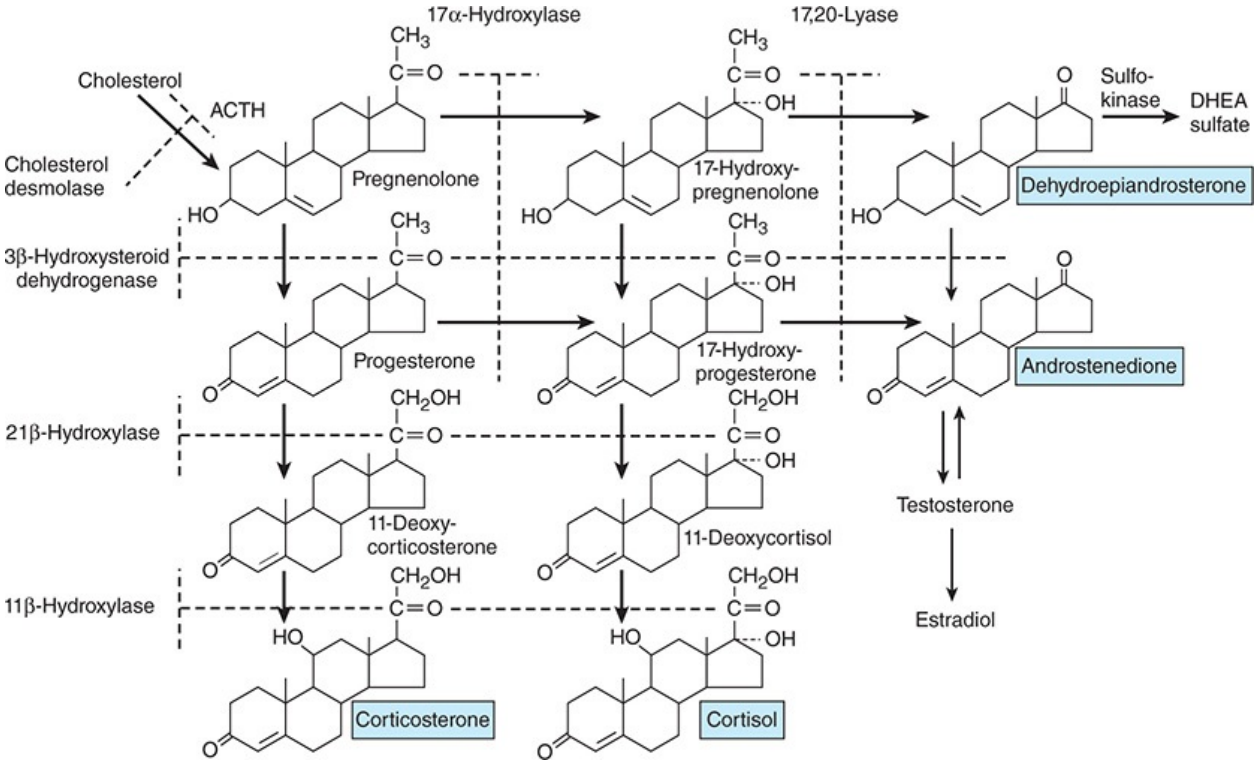


FIGURE 19–7 Outline of hormone biosynthesis in the zona fasciculata and zona reticularis of the adrenal cortex. The major secretory products are underlined. The enzymes for the reactions are shown on the left and at the top of the chart. When a particular enzyme is deficient, hormone production is blocked at the points indicated by the broken lines. ACTH, adrenocorticotropic hormone; DHEA, dehydroepiandrosterone.

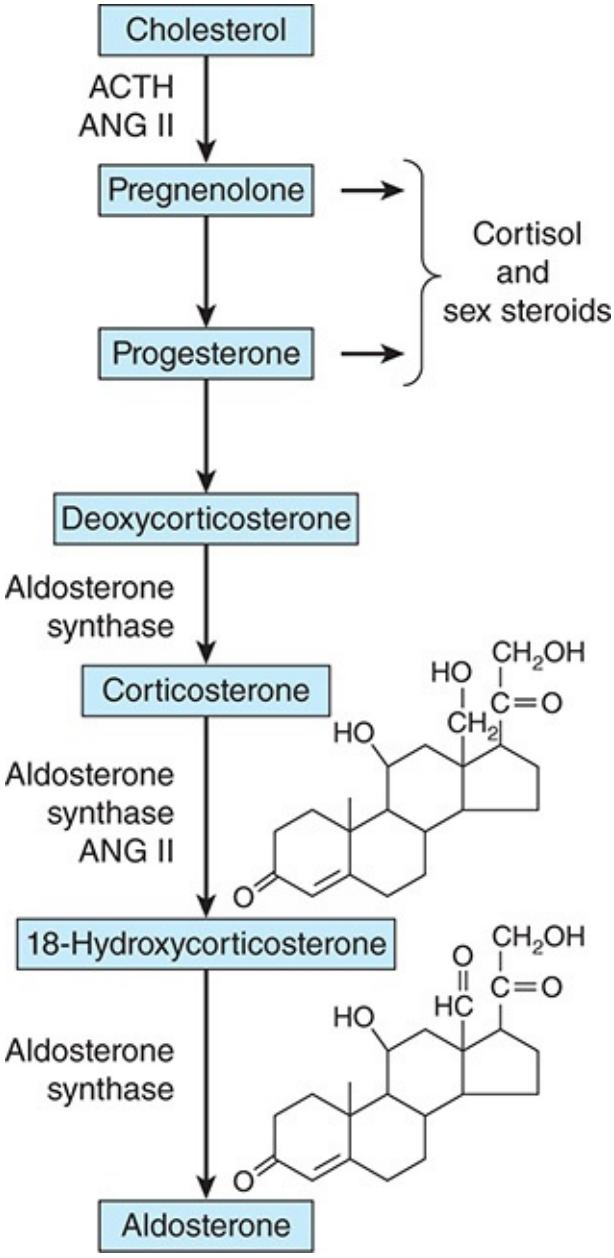


FIGURE 19–8 Hormone synthesis in the zona glomerulosa. The zona glomerulosa lacks 17α-hydroxylase activity, and only the zona glomerulosa can convert corticosterone to aldosterone because it is the only zone that normally contains aldosterone synthase. ACTH, adrenocorticotropic hormone; ANG II, angiotensin II.

TABLE 19–1 Principal adrenocortical hormones in adult humans.[a]

Name	Synonyms	Average Plasma Concentration (Free and Bound)[a] µg/dL)	Average Amount Secreted (mg/24 h)
Cortisol	Compound F, hydrocortisone	13.9	10
Corticosterone	Compound B	0.4	3
Aldosterone		0.0006	0.15
Deoxycorticosterone	DOC	0.0006	0.20
Dehydroepiandrosterone sulfate	DHEAS	175.0	20

[a]All plasma concentration values except DHEAS are fasting morning values after overnight recumbency.

CLINICAL BOX 19–1

Synthetic Steroids

As with many other naturally occurring substances, the activity of adrenocortical steroids can be increased by altering their structure. A number of synthetic steroids are available that have many times the activity of cortisol. The relative glucocorticoid and mineralocorticoid potencies of the natural steroids are compared with those of the synthetic steroids 9α-fluorocortisol, prednisolone, and dexamethasone in **Table 19–2**. The potency of dexamethasone is due to its high affinity for glucocorticoid receptors and its long half-life. Prednisolone also has a long half-life.

TABLE 19–2 Relative potencies of corticosteroids compared with cortisol.[a]

Steroid	Glucocorticoid Activity	Mineralocorticoid Activity
Cortisol	1.0	1.0
Corticosterone	0.3	15
Aldosterone	0.3	3000
Deoxycorticosterone	0.2	100
Costisone	0.7	0.8
Prednisolone	4	0.8
9α-Fluorocortisol	10	125
Dexamethasone	25	–0

[a]Values are approximations based on liver glycogen deposition or anti-inflammatory assays for glucocorticoid activity, and effect on urinary Na^+/K^+ or maintenance of adrenalectomized animals for mineralocorticoid activity. The last three steroids listed are synthetic compounds that do not occur naturally.

SPECIES DIFFERENCES

In all species from amphibia to humans, the major C_{21} steroid hormones secreted by adrenocortical tissue appear to be aldosterone, cortisol, and corticosterone, although the ratio of cortisol to corticosterone varies. Birds, mice, and rats secrete corticosterone almost exclusively; dogs secrete approximately equal amounts of the two glucocorticoids; and cats, sheep, monkeys, and humans secrete predominantly cortisol. In humans, the ratio of secreted cortisol to corticosterone is approximately 7:1.

STEROID BIOSYNTHESIS

The major paths by which the naturally occurring adrenocortical hormones are synthesized in the body are summarized in Figure 19–7 and 19–8. The precursor of all steroids is cholesterol. Some of the cholesterol is synthesized from acetate, but most of it is taken up from LDL in the circulation. LDL receptors are especially abundant in adrenocortical cells. The cholesterol is esterified and

stored in lipid droplets. **Cholesterol ester hydrolase** catalyzes the formation of free cholesterol in the lipid droplets **(Figure 19–9)**. The cholesterol is transported to mitochondria by a sterol carrier protein. In the mitochondria, it is converted to pregnenolone in a reaction catalyzed by an enzyme known as **cholesterol desmolase** or **side-chain cleavage enzyme.** This enzyme, like most of the enzymes involved in steroid biosynthesis, is a member of the cytochrome P450 superfamily and is also known as **P450scc** or **CYP11A1.** For convenience, the various names of the enzymes involved in adrenocortical steroid biosynthesis are summarized in **Table 19–3**.

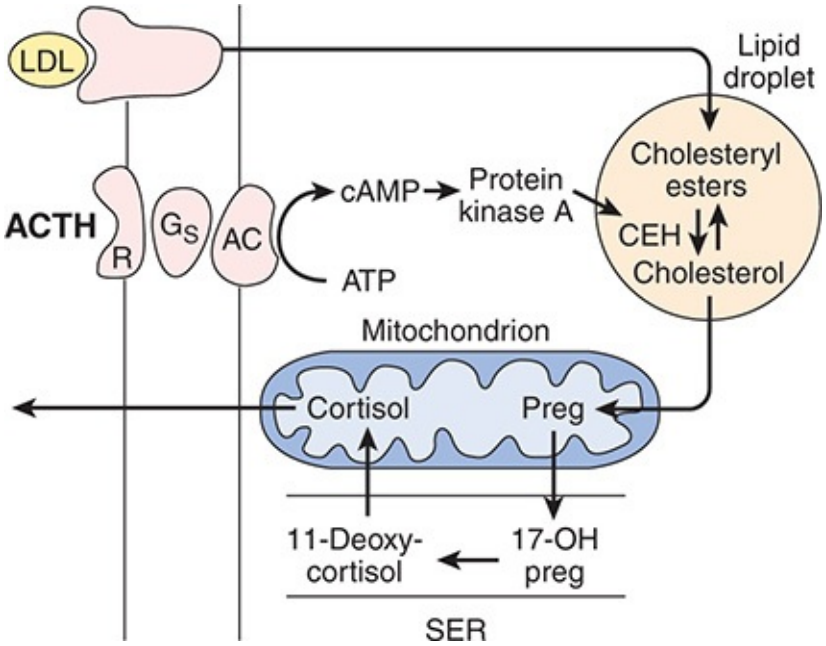


FIGURE 19–9 Mechanism of action of ACTH on cortisol-secreting cells in the inner two zones of the adrenal cortex. When ACTH binds to its receptor (R), adenylyl cyclase (AC) is activated via Gs. The resulting increase in cAMP activates protein kinase A, and the kinase phosphorylates cholesteryl ester hydrolase (CEH), increasing its activity. Consequently, more free cholesterol is formed and converted to pregnenolone. Note that in the subsequent steps in steroid biosynthesis, products are shuttled between the mitochondria and the smooth endoplasmic reticulum (SER). Corticosterone is also synthesized and secreted. ACTH, adrenocorticotropic hormone; ATP, adenosine triphosphate; cAMP, cyclic adenosine monophosphate.

TABLE 19–3 Nomenclature for adrenal steroidogenic enzymes and their location in adrenal cells.

Trivial Name	P450	CYP	Location
Cholesterol desmolase; side-chain cleavage enzyme	P450scc	CYP11A1	Mitochondria
3β-Hydroxysteroid dehydrogenase	. . .	. . .	SER
17α-Hydroxylase, 17,20-lyase	P450c17	CYP17	Mitochondria
21β-Hydroxylase	P450c21	CYP21A2	SER
11β-Hydroxylase	P450c11	CYP11B1	Mitochondria
Aldosterone synthase	P450c11AS	CYP11B2	Mitochondria

SER, smooth endoplasmic reticulum.

Pregnenolone moves to the smooth endoplasmic reticulum, where some of it is dehydrogenated to form progesterone in a reaction catalyzed by **3β-hydroxysteroid dehydrogenase.** This enzyme has a molecular weight of 46,000 and is not a cytochrome P450. It also catalyzes the conversion of 17α-hydroxypregnenolone to 17α-hydroxyprogesterone, and dehydroepiandrosterone to androstenedione (Figure 22–7) in the smooth endoplasmic reticulum. The 17α-hydroxypregnenolone and the 17α-hydroxyprogesterone are formed from pregnenolone and progesterone, respectively (Figure 19–7) by the action of **17α-hydroxylase.** This is another mitochondrial P450, and it is also known as **P450c17** or **CYP17.** Located in another part of the same enzyme is **17,20-lyase** activity that breaks the 17,20 bond, converting 17α-pregnenolone and 17α-progesterone to the C_{19} steroids dehydroepiandrosterone and androstenedione.

Hydroxylation of progesterone to 11-deoxycorticosterone and of 17α-hydroxyprogesterone to 11-deoxycortisol occurs in the smooth endoplasmic reticulum. These reactions are catalyzed by 21β-hydroxylase, a cytochrome P450 that is also known as **P450c21** or **CYP21A2.**

11-Deoxycorticosterone and the 11-deoxycortisol move back to the mitochondria, where they are 11-hydroxylated to form corticosterone and cortisol. These reactions occur in the zona fasciculata and zona reticularis and are catalyzed by 11β-hydroxylase, a cytochrome P450 also known as **P450c11** or

CYP11B1.

In the zona glomerulosa there is no 11β-hydroxylase, but a closely related enzyme called **aldosterone synthase** is present. This cytochrome P450 is 95% identical to 11β-hydroxylase and is also known as **P450c11AS** or **CYP11B2.** The genes that code CYP11B1 and CYP11B2 are both located on chromosome 8. However, aldosterone synthase is normally found only in the zona glomerulosa. The zona glomerulosa also lacks 17α-hydroxylase. This is why the zona glomerulosa makes aldosterone but fails to make cortisol or sex hormones.

Furthermore, subspecialization occurs within the inner two zones. The zona fasciculata has more 3β-hydroxysteroid dehydrogenase activity than the zona reticularis, and the zona reticularis has more of the cofactors required for the 17,20-lyase activity of 17α-hydroxylase. Therefore, the zona fasciculata makes more cortisol and corticosterone, and the zona reticularis makes more androgens. Most of the dehydroepiandrosterone that is formed is converted to dehydroepiandrosterone sulfate (DHEAS) by **adrenal sulfokinase,** and this enzyme is localized in the zona reticularis as well.

ACTION OF ACTH

ACTH binds to high-affinity receptors on the plasma membrane of adrenocortical cells. This activates adenylyl cyclase via G_s. The resulting reactions (Figure 19–9) lead to a prompt increase in the formation of pregnenolone and its derivatives, with secretion of the latter. Over longer periods, ACTH also increases the synthesis of the P450s involved in the synthesis of glucocorticoids.

ACTIONS OF ANGIOTENSIN II

Angiotensin II binds to AT_1 receptors (see Chapter 38) in the zona glomerulosa that act via a G-protein to activate phospholipase C. The resulting increase in protein kinase C fosters the conversion of cholesterol to pregnenolone (Figure 19–8) and facilitates the action of aldosterone synthase, resulting in increased secretion of aldosterone.

ENZYME DEFICIENCIES

The consequences of inhibiting any of the enzyme systems involved in steroid

biosynthesis can be predicted from Figures 19–7 and 19–8. Congenital defects in the enzymes lead to deficient cortisol secretion and the syndrome of **congenital adrenal hyperplasia.** The hyperplasia is due to increased ACTH secretion. Cholesterol desmolase deficiency is fatal in utero because it prevents the placenta from making the progesterone necessary for pregnancy to continue. A cause of severe congenital adrenal hyperplasia in newborns is a loss of function mutation of the gene for the **steroidogenic acute regulatory (StAR) protein.** This protein is essential in the adrenals and gonads but not in the placenta for the normal movement of cholesterol into the mitochondria to reach cholesterol desmolase, which is located on the matrix space side of the internal mitochondrial membrane (see Chapter 16, Figure 16-1). In its absence, only small amounts of steroids are formed. The degree of ACTH stimulation is marked, resulting eventually in accumulation of large numbers of lipoid droplets in the adrenal. For this reason, the condition is called **congenital lipoid adrenal hyperplasia.** Because androgens are not formed, female genitalia develop regardless of genetic sex (see Chapter 22). In 3β hydroxysteroid dehydrogenase deficiency, another rare condition, DHEA secretion is increased. This steroid is a weak androgen that can cause some masculinization in females with the disease, but it is not adequate to produce full masculinization of the genitalia in genetic males. Consequently, **hypospadias,** a condition where the opening of the urethra is on the underside of the penis rather than its tip, is common. In fully developed 17α-hydroxylase deficiency, a third rare condition due to a mutated gene for **CYP17,** no sex hormones are produced, so female external genitalia are present. However, the pathway leading to corticosterone and aldosterone is intact, and elevated levels of 11-deoxycorticosterone and other mineralocorticoids produce hypertension and hypokalemia. Cortisol is deficient, but this is partially compensated by the glucocorticoid activity of corticosterone.

Unlike the defects discussed in the preceding paragraph, 21β-hydroxylase deficiency is common, accounting for 90% or more of the enzyme deficiency cases. The 21β-hydroxylase gene, which is in the HLA complex of genes on the short arm of chromosome 6 (see Chapter 3) is one of the most polymorphic in the human genome. Mutations occur at many different sites in the gene, and the abnormalities that are produced therefore range from mild to severe. Production of cortisol and aldosterone are generally reduced, so ACTH secretion and consequently production of precursor steroids are increased. These steroids are converted to androgens, producing **virilization.** The characteristic pattern that develops in females in the absence of treatment is the **adrenogenital syndrome.** Masculization may not be marked until later in life and mild cases can be detected only by laboratory tests. In 75% of the cases, aldosterone deficiency

causes appreciable loss of Na^+ (**salt-losing form** of adrenal hyperplasia). The resulting hypovolemia can be severe.

In 11β-hydroxylase deficiency, virilization plus excess secretion of 11-deoxycortisol and 11-deoxycorticosterone take place. Because the former is an active mineralocorticoid, patients with this condition also have salt and water retention and, in two-thirds of the cases, hypertension (**hypertensive form** of congenital adrenal hyperplasia).

Glucocorticoid treatment is indicated in all of the virilizing forms of congenital adrenal hyperplasia because it repairs the glucocorticoid deficit and inhibits ACTH secretion, reducing the abnormal secretion of androgens and other steroids.

Expression of the cytochrome P450 enzymes responsible for steroid hormone biosynthesis depends on **steroid factor-1 (SF-1),** an orphan nuclear receptor. If *Ft2-F1*, the gene for SF-1, is knocked out, the gonads as well as adrenals fail to develop and additional abnormalities are present at the pituitary and hypothalamic level.

TRANSPORT, METABOLISM, & EXCRETION OF ADRENOCORTICAL HORMONES

GLUCOCORTICOID BINDING

Cortisol is bound in the circulation to an α globulin called **transcortin** or **corticosteroid-binding globulin (CBG).** A minor degree of binding to albumin also takes place. Corticosterone is similarly bound but to a lesser degree. The half-life of cortisol in the circulation is therefore longer (about 60–90 min) than that of corticosterone (50 min). Bound steroids are physiologically inactive (see Chapter 16). In addition, relatively little free cortisol and corticosterone are found in the urine because of protein binding.

The equilibrium between cortisol and its binding protein and the implications of binding in terms of tissue supplies and ACTH secretion are summarized in **Figure 19–10**. The bound cortisol functions as a circulating reservoir of hormone that keeps a supply of free cortisol available to the tissues. The relationship is similar to that of T_4 and its binding protein (see Chapter 20). At normal levels of total plasma cortisol (13.5 µg/dL or 375 nmol/L), very little free cortisol is present in the plasma, but the binding sites on CBG become saturated when the total plasma cortisol exceeds 20 µg/dL. At higher plasma levels,

binding to albumin increases, but the main increase is in the unbound fraction.

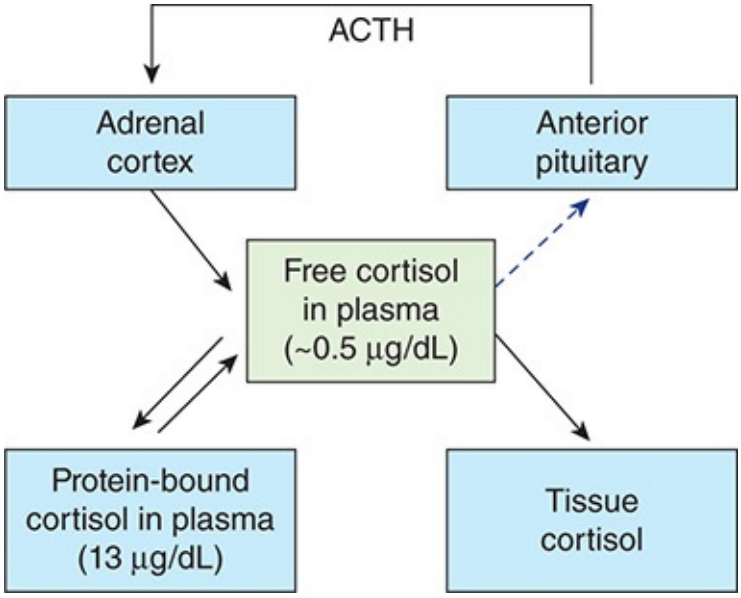


FIGURE 19–10 The interrelationships of free and bound cortisol. The dashed arrow indicates that cortisol inhibits ACTH secretion. The value for free cortisol is an approximation; in most studies, it is calculated by subtracting the protein-bound cortisol from the total plasma cortisol. ACTH, adrenocorticotropic hormone.

CBG is synthesized in the liver and its production is increased by estrogen. CBG levels are elevated during pregnancy and depressed in cirrhosis, nephrosis, and multiple myeloma. When the CBG level rises, more cortisol is bound, and initially the free cortisol level drops. This stimulates ACTH secretion, and more cortisol is secreted until a new equilibrium is reached at which the bound cortisol is elevated but the free cortisol is normal. Changes in the opposite direction occur when the CBG level falls. This explains why pregnant women have high total plasma cortisol levels without symptoms of glucocorticoid excess and, conversely, why some patients with nephrosis have low total plasma cortisol without symptoms of glucocorticoid deficiency.

METABOLISM & EXCRETION OF GLUCOCORTICOIDS

Cortisol is metabolized in the liver, which is the principal site of glucocorticoid catabolism. Most of the cortisol is reduced to dihydrocortisol and then to

tetrahydrocortisol, which is conjugated to glucuronic acid **(Figure 19–11)**. The glucuronyl transferase system responsible for this conversion also catalyzes the formation of the glucuronides of bilirubin (see Chapter 28) and a number of hormones and drugs. Competitive inhibition takes place between these substrates for the enzyme system.

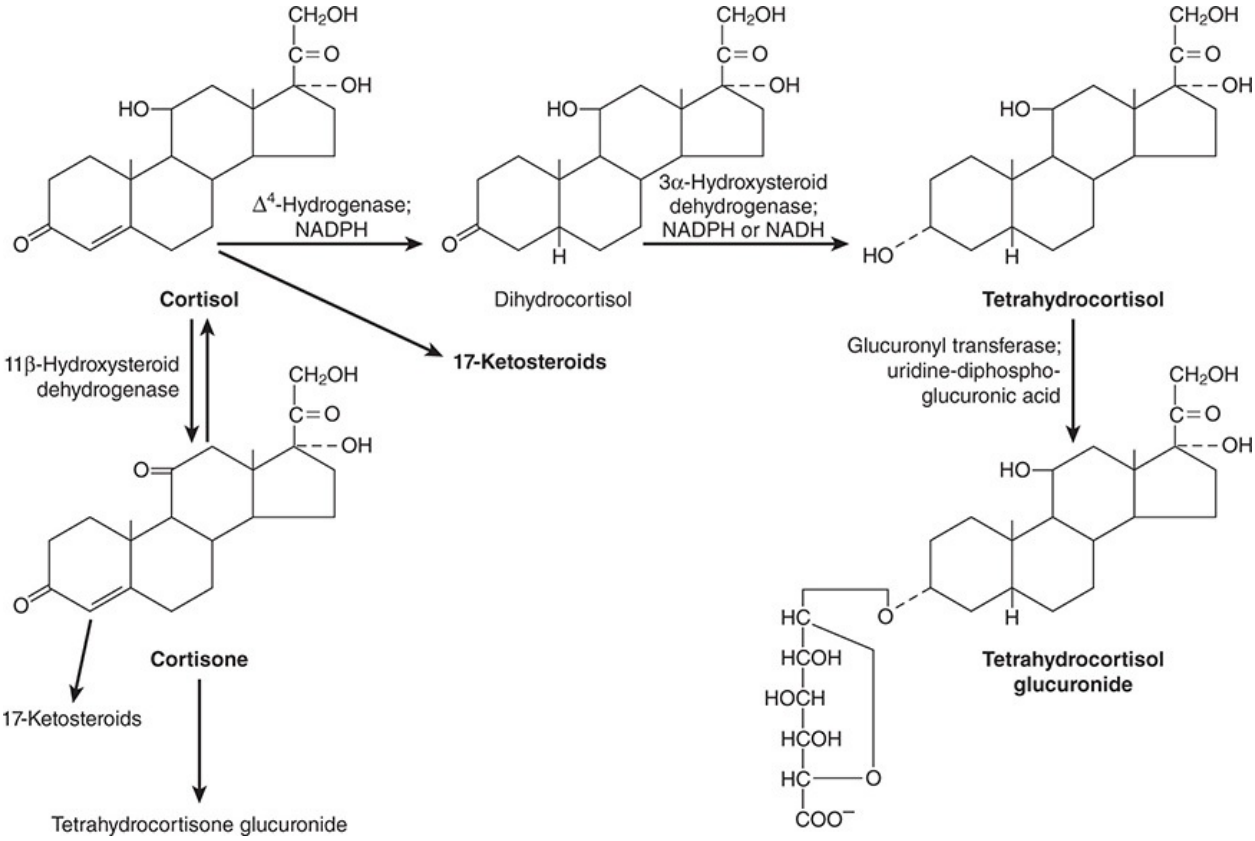


FIGURE 19–11 Outline of hepatic metabolism of cortisol.

CLINICAL BOX 19–2

Variations in the Rate of Hepatic Metabolism

The rate of hepatic inactivation of glucocorticoids is depressed in liver disease and, interestingly, during surgery and other stresses. Thus, in stressed humans, the plasma-free cortisol level rises higher than it does with maximal ACTH stimulation in the absence of stress.

The liver and other tissues contain the enzyme 11β hydroxysteroid

dehydrogenase. There are at least two forms of this enzyme. Type 1 catalyzes the conversion of cortisol to cortisone and the reverse reaction, though it functions primarily as a reductase, forming cortisol from corticosterone. Type 2 catalyzes almost exclusively the one-way conversion of cortisol to cortisone. Cortisone is an active glucocorticoid because it is converted to cortisol, and it is well known because of its extensive use in medicine. It is not secreted in appreciable quantities by the adrenal glands. Little, if any, of the cortisone formed in the liver enters the circulation, because it is promptly reduced and conjugated to form tetrahydrocortisone glucuronide. The tetrahydroglucuronide derivatives ("conjugates") of cortisol and corticosterone are freely soluble. They enter the circulation, where they do not become bound to protein. They are rapidly excreted in the urine.

About 10% of the secreted cortisol is converted in the liver to the 17-ketosteroid derivatives of cortisol and cortisone. The ketosteroids are conjugated for the most part to sulfate and then excreted in the urine. Other metabolites, including 20-hydroxy derivatives, are formed. There is an enterohepatic circulation of glucocorticoids and about 15% of the secreted cortisol is excreted in the stool. The metabolism of corticosterone is similar to that of cortisol, except that it does not form a 17-ketosteroid derivative **(see Clinical Box 19–2)**.

ALDOSTERONE

Aldosterone is bound to protein to only a slight extent, and its half-life is short (about 20 min). The amount secreted is small (Table 19–1), and the total plasma aldosterone level in humans is normally about 0.006 µg/dL (0.17 nmol/L), compared with a cortisol level (bound and free) of about 13.5 µg/dL (375 nmol/L). Much of the aldosterone is converted in the liver to the tetrahydroglucuronide derivative, but some is changed in the liver and in the kidneys to an 18-glucuronide. This glucuronide, which is unlike the breakdown products of other steroids, is converted to free aldosterone by hydrolysis at pH 1.0, and it is therefore often referred to as the "acid-labile conjugate." Less than 1% of the secreted aldosterone appears in the urine in the free form. Another 5% is in the form of the acid-labile conjugate, and up to 40% is in the form of the tetrahydroglucuronide.

17-KETOSTEROIDS

The major adrenal androgen is the 17-ketosteroid dehydroepiandrosterone, although androstenedione is also secreted. The 11-hydroxy derivative of androstenedione and the 17-ketosteroids formed from cortisol and cortisone by side chain cleavage in the liver are the only 17-ketosteroids that have an =O or an —OH group in the 11 position (“11-oxy-17-ketosteroids”). Testosterone is also converted to a 17-ketosteroid. Because the daily 17-ketosteroid excretion in normal adults is 15 mg in men and 10 mg in women, about two-thirds of the urinary ketosteroids in men are secreted by the adrenal or formed from cortisol in the liver and about one-third are of testicular origin.

Etiocholanolone, one of the metabolites of the adrenal androgens and testosterone, can cause fever when it is unconjugated (see Chapter 17). Certain individuals have episodic bouts of fever due to periodic accumulation in the blood of unconjugated etiocholanolone (“etiocholanolone fever”).

EFFECTS OF ADRENAL ANDROGENS & ESTROGENS

ANDROGENS

Androgens are the hormones that exert masculinizing effects and they promote protein anabolism and growth (see Chapter 23). Testosterone from the testes is the most active androgen and the adrenal androgens have less than 20% of its activity. Secretion of the adrenal androgens is controlled acutely by ACTH and not by gonadotropins. However, the concentration of DHEAS increases until it peaks at about 225 mg/dL in the early 20s, and then falls to very low values in old age **(Figure 19–12)**. These long-term changes are not due to changes in ACTH secretion and appear to be due instead to a rise and then a gradual fall in the lyase activity of 17α-hydroxylase.

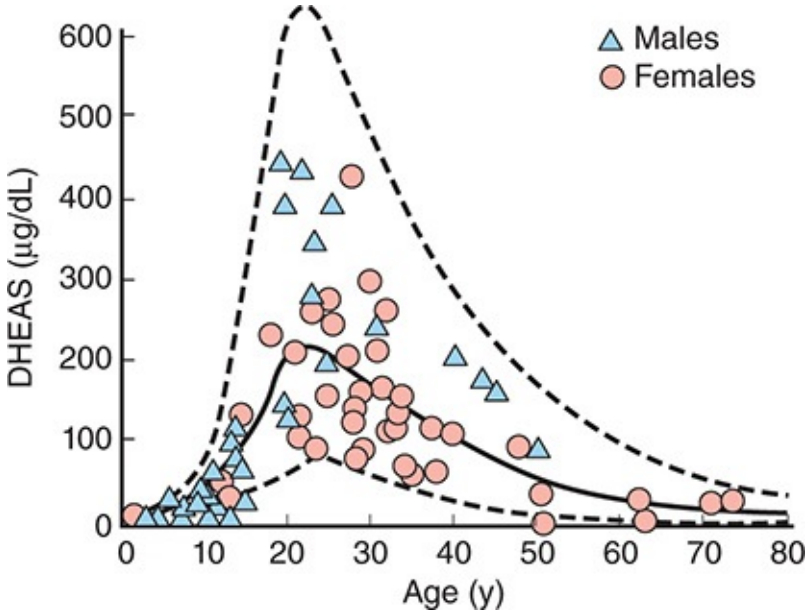


FIGURE 19–12 Change in serum dehydroepiandrosterone sulfate (DHEAS) with age. The middle line is the mean, and the dashed lines identify ±1.96 standard deviations. (Reproduced, with permission, from Smith MR, et al: A radioimmunoassay for the estimation of serum dehydroepiandrosterone sulfate in normal and pathological sera. Clin Chim Acta 1975; Nov 15; 65(1):5–13.)

All but about 0.3% of the circulating DHEA is conjugated to sulfate (DHEAS). The secretion of adrenal androgens is nearly as great in castrated males and females as it is in normal males, so it is clear that these hormones exert very little masculinizing effect when secreted in normal amounts. However, they can produce appreciable masculinization when secreted in excessive amounts. In adult males, excess adrenal androgens merely accentuate existing characteristics, but in prepubertal boys they can cause precocious development of the secondary sex characteristics without testicular growth **(precocious pseudopuberty).** In females they cause female pseudo-hermaphroditism and the adrenogenital syndrome. Some health practitioners recommend injections of dehydroepiandrosterone to combat the effects of aging (see Chapter 1), but results to date are controversial at best.

ESTROGENS

The adrenal androgen androstenedione is converted to testosterone and to estrogens (aromatized) in fat and other peripheral tissues. This is an important source of estrogens in postmenopausal women and men (see Chapters 22 and

23).

PHYSIOLOGIC EFFECTS OF GLUCOCORTICOIDS

ADRENAL INSUFFICIENCY

In untreated adrenal insufficiency, Na^+ loss and shock occurs due to the lack of mineralocorticoid activity, as well as abnormalities of water, carbohydrate, protein, and fat metabolism due to the lack of glucocorticoids. These metabolic abnormalities are eventually fatal despite mineralocorticoid treatment. Small amounts of glucocorticoids correct the metabolic abnormalities, in part directly and in part by permitting other reactions to occur. It is important to separate these physiologic actions of glucocorticoids from the quite different effects produced by large amounts of the hormones.

MECHANISM OF ACTION

The multiple effects of glucocorticoids are triggered by binding to glucocorticoid receptors, and the steroid–receptor complexes act as transcription factors that promote the transcription of certain segments of DNA (see Chapter 1). This, in turn, leads via the appropriate mRNAs to synthesis of enzymes that alter cell function. In addition, it seems likely that glucocorticoids have nongenomic actions.

EFFECTS ON INTERMEDIARY METABOLISM

The actions of glucocorticoids on the intermediary metabolism of carbohydrate, protein, and fat are discussed in Chapter 24. They include increased protein catabolism and increased hepatic glycogenesis and gluconeogenesis. Glucose- 6-phosphatase activity is increased, and the plasma glucose level rises. Glucocorticoids exert an anti-insulin action in peripheral tissues and make diabetes worse. However, the brain and the heart are spared, so the increase in plasma glucose provides extra glucose to these vital organs. In diabetics, glucocorticoids raise plasma lipid levels and increase ketone body formation, but in normal individuals, the increase in insulin secretion provoked by the rise in plasma glucose obscures these actions. In adrenal insufficiency, the plasma

glucose level is normal as long as an adequate caloric intake is maintained, but fasting causes hypoglycemia that can be fatal. The adrenal cortex is not essential for the ketogenic response to fasting.

PERMISSIVE ACTION

Small amounts of glucocorticoids must be present for a number of metabolic reactions to occur, although the glucocorticoids do not produce the reactions by themselves. This effect is called their **permissive action.** Permissive effects include the requirement for glucocorticoids to be present for glucagon and catecholamines to exert their calorigenic effects (see above and Chapter 24), for catecholamines to exert their lipolytic effects, and for catecholamines to produce pressor responses and bronchodilation.

EFFECTS ON ACTH SECRETION

Glucocorticoids inhibit ACTH secretion, which represents a negative feedback response on the pituitary. ACTH secretion is increased in adrenalectomized animals. The consequences of the negative feedback action of cortisol on ACTH secretion are discussed below in the section on regulation of glucocorticoid secretion.

VASCULAR REACTIVITY

In adrenally insufficient animals, vascular smooth muscle becomes unresponsive to norepinephrine and epinephrine. The capillaries dilate and, terminally, become permeable to colloidal dyes. Failure to respond to the norepinephrine liberated at noradrenergic nerve endings probably impairs vascular compensation for the hypovolemia of adrenal insufficiency and promotes vascular collapse. Glucocorticoids restore vascular reactivity.

EFFECTS ON THE NERVOUS SYSTEM

Changes in the nervous system in adrenal insufficiency that are reversed only by glucocorticoids include the appearance of electroencephalographic waves slower than the normal β rhythm, and personality changes. The latter, which are mild, include irritability, apprehension, and inability to concentrate.

EFFECTS ON WATER METABOLISM

Adrenal insufficiency is characterized by an inability to excrete a water load, causing the possibility of water intoxication. Only glucocorticoids repair this deficit. In patients with adrenal insufficiency who have not received glucocorticoids, glucose infusion may cause high fever (“glucose fever”) followed by collapse and death. Presumably, the glucose is metabolized, the water dilutes the plasma, and the resultant osmotic gradient between the plasma and the cells causes the cells of the thermoregulatory centers in the hypothalamus to swell to such an extent that their function is disrupted.

The cause of defective water excretion in adrenal insufficiency is unsettled. Plasma vasopressin levels are elevated in adrenal insufficiency and reduced by glucocorticoid treatment. The glomerular filtration rate is low, and this probably contributes to the reduction in water excretion. The selective effect of glucocorticoids on the abnormal water excretion is consistent with this possibility, because even though the mineralocorticoids improve filtration by restoring plasma volume, the glucocorticoids raise the glomerular filtration rate to a much greater degree.

EFFECTS ON THE BLOOD CELLS & LYMPHATIC ORGANS

Glucocorticoids decrease the number of circulating eosinophils by increasing their sequestration in the spleen and lungs. Glucocorticoids also lower the number of basophils in the circulation and increase the number of neutrophils, platelets, and red blood cells (**Table 19-4**).

TABLE 19–4 Typical effects of cortisol on the white and red blood cell counts in humans (cells/μL).

Cell	Normal	Cortisol-Treated
White blood cells		
Total	9000	10,000
PMNs	5760	8330
Lymphocytes	2370	1080
Eosinophils	270	20
Basophils	60	30
Monocytes	450	540
Red blood cells	5 million	5.2 million

Glucocorticoids decrease the circulating lymphocyte count and the size of the lymph nodes and thymus by inhibiting lymphocyte mitotic activity. They reduce secretion of cytokines by inhibiting the effect of NF-κB on the nucleus. The reduced secretion of the cytokine IL-2 leads to reduced proliferation of lymphocytes (see Chapter 3), and these cells undergo apoptosis.

RESISTANCE TO STRESS

The term **stress** as used in biology has been defined as any change in the environment that changes or threatens to change an existing optimal steady state. Most, if not all, of these stresses activate processes at the molecular, cellular, or systemic level that tend to restore the previous state, that is, they are homeostatic reactions. Some, but not all, of the stresses stimulate ACTH secretion. The increase in ACTH secretion is essential for survival when the stress is severe. If animals are then hypophysectomized, or adrenalectomized but treated with maintenance doses of glucocorticoids, they die when exposed to the same stress.

The reason an elevated circulating ACTH, and hence glucocorticoid level, is essential for resisting stress remains for the most part unknown. Most of the stressful stimuli that increase ACTH secretion also activate the sympathetic nervous system, and part of the function of circulating glucocorticoids may be maintenance of vascular reactivity to catecholamines. Glucocorticoids are also necessary for the catecholamines to exert their full FFA-mobilizing action, and the FFAs are an important emergency energy supply. However, sympathectomized animals tolerate a variety of stresses with relative impunity. Another theory holds that glucocorticoids prevent other stress-induced changes

from becoming excessive. At present, all that can be said is that stress causes increases in plasma glucocorticoids to high “pharmacologic” levels that in the short run are life-saving.

It should also be noted that the increase in ACTH, which is beneficial in the short term, becomes harmful and disruptive in the long term, causing among other things, the abnormalities of Cushing syndrome.

PHARMACOLOGIC & PATHOLOGIC EFFECTS OF GLUCOCORTICOIDS

CUSHING SYNDROME

The clinical picture produced by prolonged increases in plasma glucocorticoids was described by Harvey Cushing and is called **Cushing syndrome** **(Figure 19–13)**. It may be **ACTH-independent** or **ACTH-dependent.** The causes of ACTH-independent Cushing syndrome include glucocorticoid-secreting adrenal tumors, adrenal hyperplasia, and prolonged administration of exogenous glucocorticoids for diseases such as rheumatoid arthritis. Rare but interesting ACTH-independent cases have been reported in which adrenocortical cells abnormally express receptors for gastric inhibitory polypeptide (GIP) (see Chapter 25), vasopressin (see Chapter 38), β-adrenergic agonists, IL-1, or gonadotropin-releasing hormone (GnRH; see Chapter 22), causing these peptides to increase glucocorticoid secretion. The causes of ACTH-dependent Cushing syndrome include ACTH-secreting tumors of the anterior pituitary gland and tumors of other organs, usually the lungs, that secrete ACTH (ectopic ACTH syndrome) or corticotropin releasing hormone (CRH). Cushing syndrome due to anterior pituitary tumors is often called **Cushing disease** because these tumors were the cause of the cases described by Cushing. However, it is confusing to speak of Cushing disease as a subtype of Cushing syndrome, and the distinction seems to be of little more than historical value.